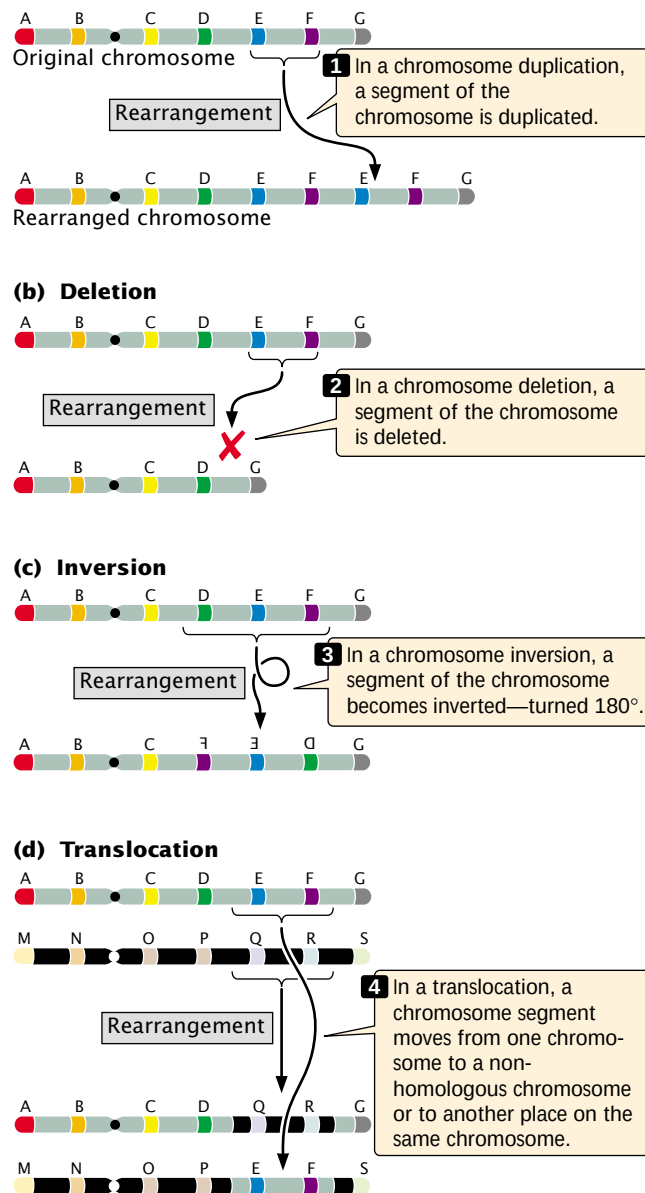


complete sets of chromosomes are added. Some organisms (such as yeast) possess a single chromosome set ($1n$) for most of their life cycles and are referred to as haploid, whereas others possess two chromosome sets and are referred to as diploid ($2n$). A polyploid is any organism that has more than two sets of chromosomes ($3n$, $4n$, $5n$, or more).

Chromosome Rearrangements

Chromosome rearrangements are mutations that change the structure of individual chromosomes. The four basic types of rearrangements are duplications, deletions, inversions, and translocations (FIGURE 9.5).

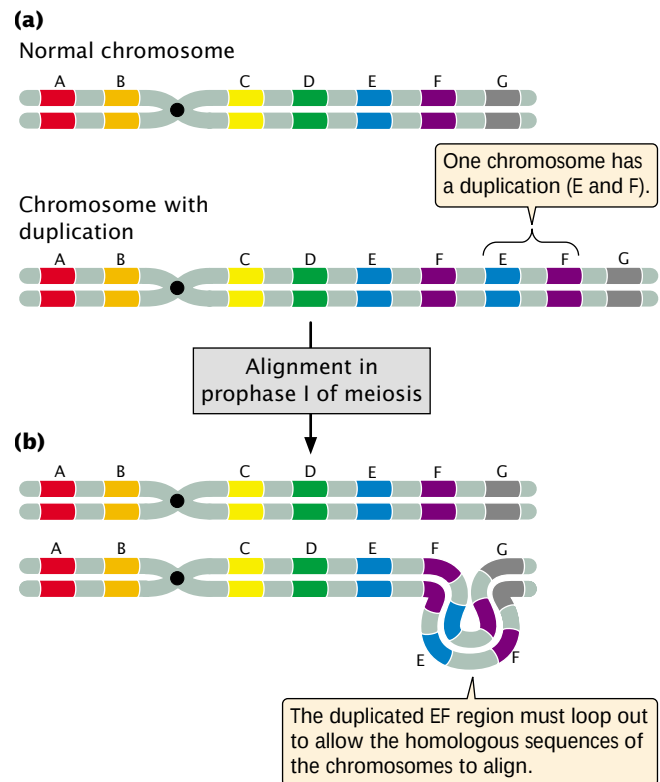


9.5 The four basic types of chromosome rearrangements are duplication, deletion, inversion, and translocation.

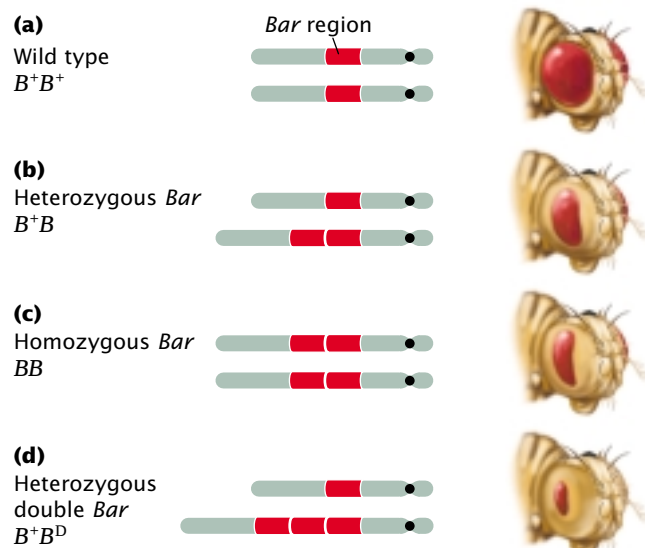
Duplications

A **chromosome duplication** is a mutation in which part of the chromosome has been doubled (see Figure 9.5a). Consider a chromosome with segments AB•CDEFG, in which • represents the centromere. A duplication might include the EF segments, giving rise to a chromosome with segments AB•CDEFEFG. This type of duplication, in which the duplicated region is immediately adjacent to the original segment, is called a **tandem duplication**. If the duplicated segment is located some distance from the original segment, either on the same chromosome or on a different one, this type is called a **displaced duplication**. An example of a displaced duplication would be AB•CDEFGEEF. A duplication can either be in the same orientation as the original sequence, as in the two preceding examples, or be inverted: AB•CDEFFEG. When the duplication is inverted, it is called a **reverse duplication**.

An individual homozygous for a rearrangement carries the rearrangement (the mutated sequence) on both homologous chromosomes, and an individual heterozygous for a rearrangement has one unmutated chromosome and one chromosome with the rearrangement. In the heterozygotes (FIGURE 9.6a), problems arise in chromosome pairing at prophase I of meiosis, because the two chromosomes are not homologous throughout their length. The homologous regions will pair and undergo synapsis, which often requires



9.6 In an individual heterozygous for a duplication, the duplicated chromosome loops out during pairing in prophase I.



9.7 The Bar phenotype in *Drosophila melanogaster* results from an X-linked duplication.

(a) Wild-type fruit flies have normal-size eyes. (b) Flies heterozygous and (c) homozygous for the *Bar* mutation have smaller, bar-shaped eyes. (d) Flies with double *Bar* have three copies of the duplication and much smaller bar-shaped eyes.

that one or both chromosomes loop and twist so that these regions are able to line up (FIGURE 9.6b). The appearance of this characteristic loop structure during meiosis is one way to detect duplications.

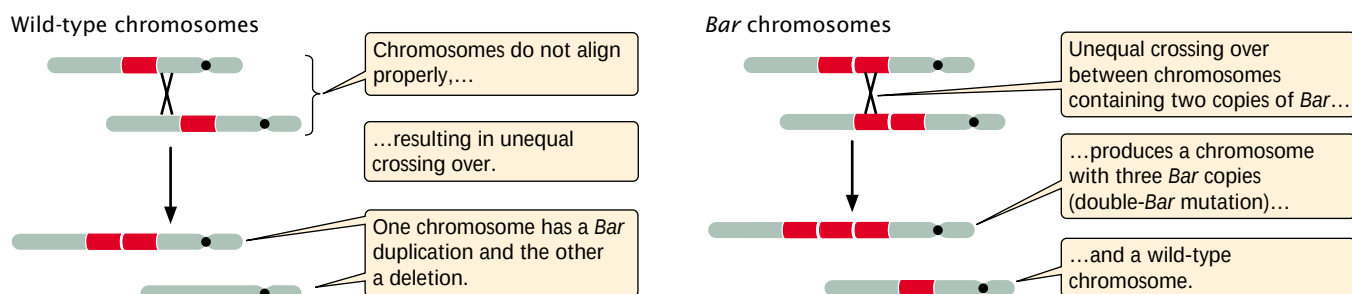
Duplications may have major effects on the phenotype. In *Drosophila melanogaster*, for example, a *Bar* mutant has a reduced number of facets in the eye, making the eye smaller and bar shaped instead of oval (FIGURE 9.7). The *Bar* mutant results from a small duplication on the X chromosome, which is inherited as an incompletely dominant, X-linked trait: heterozygous female flies have somewhat smaller eyes (the number of facets is reduced; see Figure 9.7b), whereas, in homozygous female and hemizygous male flies, the number of facets is greatly reduced (see Figure

9.7c). Occasionally, a fly carries three copies of the *Bar* duplication on its X chromosome; in such mutants, which are termed *double Bar*, the number of facets is extremely reduced (see Figure 9.7d). *Bar* arises from unequal crossing over, a duplication-generating process (FIGURE 9.8; see also Figure 17.17).

How does a chromosome duplication alter the phenotype? After all, gene sequences are not altered by duplications, and no genetic information is missing; the only change is the presence of additional copies of normal sequences. The answer to this question is not well understood, but the effects are most likely due to imbalances in the amounts of gene products (abnormal gene dosage). The amount of a particular protein synthesized by a cell is often directly related to the number of copies of its corresponding gene: an individual with three functional copies of a gene often produces 1.5 times as much of the protein encoded by that gene as that produced by an individual with two copies. Because developmental processes often require the interaction of many proteins, they may critically depend on the relative amounts of the proteins. If the amount of one protein increases while the amounts of others remain constant, problems can result (FIGURE 9.9). Although duplications can have severe consequences when the precise balance of a gene product is critical to cell function, duplications have arisen frequently throughout the evolution of many eukaryotic organisms and are a source of new genes that may provide novel functions. Human phenotypes associated with some duplications are summarized in Table 9.1.

Concepts

A chromosome duplication is a mutation that doubles part of a chromosome. In individuals heterozygous for a chromosome duplication, the duplicated region of the chromosome loops out when homologous chromosomes pair in prophase I of meiosis. Duplications often have major effects on the phenotype, possibly by altering gene dosage.



9.8 Unequal crossing over produces *Bar* and double-*Bar* mutations.

Table 9.1 Effects of some chromosome rearrangements in humans

Type of Rearrangement	Chromosome	Disorder	Symptoms
Duplication	4, short arm	—	Small head, short neck, low hairline, growth and mental retardation
Duplication	4, long arm	—	Small head, sloping forehead, hand abnormalities
Duplication	7, long arm	—	Delayed development, asymmetry of the head, fuzzy scalp, small nose, low-set ears
Duplication	9, short arm	—	Characteristic face, variable mental retardation, high and broad forehead, hand abnormalities
Deletion	5, short arm	<i>Cri-du-chat</i> syndrome	Small head, distinctive cry, widely spaced eyes, a round face, mental retardation
Deletion	4, short arm	Wolf-Hirschhorn syndrome	Small head with high forehead, wide nose, cleft lip and palate, severe mental retardation
Deletion	4, long arm	—	Small head, mild to moderate mental retardation, cleft lip and palate, hand and foot abnormalities
Deletion	15, long arm	Prader-Willi syndrome	Feeding difficulty at early age, but becoming obese after 1 year of age, mild to moderate mental retardation
Deletion	18, short arm	—	Round face, large low set-ears, mild to moderate mental retardation
Deletion	18, long arm	—	Distinctive mouth shape, small hands, small head, mental retardation

Deletions

A second type of chromosome rearrangement is a **deletion**, the loss of a chromosome segment (see Figure 9.5b). A chromosome with segments AB•CDEFG that undergoes a deletion of segment EF would generate the mutated chromosome AB•CDG.

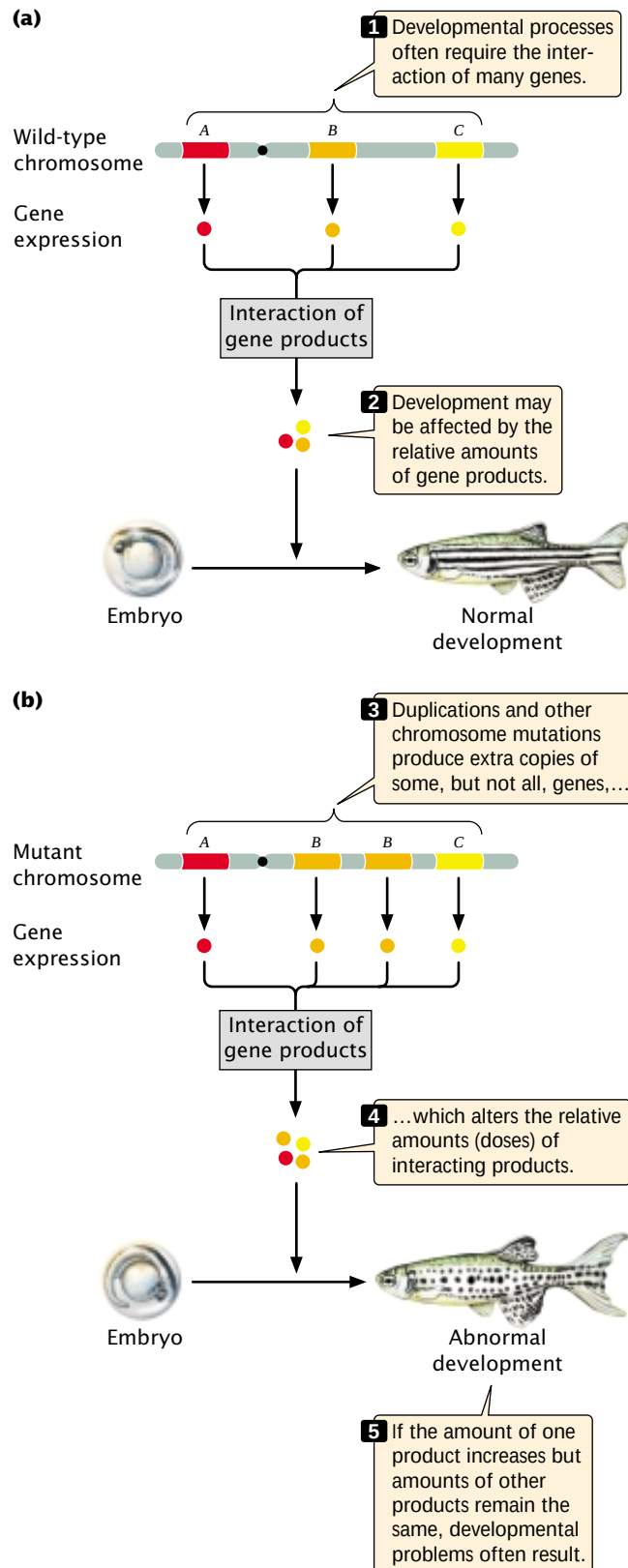
A large deletion can be easily detected because the chromosome is noticeably shortened. In individuals heterozygous for deletions, the normal chromosome must loop out during the pairing of homologs in prophase I of meiosis (FIGURE 9.10) to allow the homologous regions of the two chromosomes to align and undergo synapsis. This looping out generates a structure that looks very much like that seen in individuals heterozygous for duplications.

The phenotypic consequences of a deletion depend on which genes are located in the deleted region. If the deletion includes the centromere, the chromosome will not segregate in meiosis or mitosis and will usually be lost. Many deletions are lethal in the homozygous state because all copies of any essential genes located in the deleted region are missing. Even individuals heterozygous for a deletion may have multiple defects for three reasons.

First, the heterozygous condition may produce imbalances in the amounts of gene products, similar to the imbalances produced by extra gene copies. Second, deletions may

allow recessive mutations on the undeleted chromosome to be expressed (because there is no wild-type allele to mask their expression). This phenomenon is referred to as **pseudodominance**. The appearance of pseudodominance in otherwise recessive *alleles* is an indication that a deletion is present on one of the chromosomes. Third, some genes must be present in two copies for normal function. Such a gene is said to be **haploinsufficient**; loss of function mutations in haploinsufficient genes are dominant. *Notch* is a series of X-linked wing mutations in *Drosophila* that often result from chromosome deletions. *Notch* deletions behave as dominant mutations: when heterozygous for the *Notch* deletion, a fly has wings that are notched at the tips and along the edges (FIGURE 9.11). The *Notch* locus is therefore haploinsufficient—a single copy of the gene is not sufficient to produce a wild-type phenotype. Females that are homozygous for a *Notch* deletion (or males that are hemizygous) die early in embryonic development. The *Notch* gene codes for a receptor that normally transmits signals received from outside the cell to the cell's interior and is important in fly development. The deletion acts as a recessive lethal because loss of all copies of the *Notch* gene prevents normal development.

In humans, a deletion on the short arm of chromosome 5 is responsible for *cri-du-chat* syndrome. The name



9.9 Unbalanced gene dosage leads to developmental abnormalities.

(French for “cry of the cat”) derives from the peculiar, cat-like cry of infants with this syndrome. A child who is heterozygous for this deletion has a small head, widely spaced eyes, a round face, and mental retardation. Deletion of part of the short arm of chromosome 4 results in another human disorder—Wolf-Hirschhorn syndrome, which is characterized by seizures and by severe mental and growth retardation.

Concepts

A **chromosome deletion** is a mutation in which a part of the chromosome is lost. In individuals heterozygous for a deletion, the normal chromosome loops out during prophase I of meiosis. Deletions do not undergo reverse mutation. They cause recessive genes on the undeleted chromosome to be expressed and cause imbalances in gene products.

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Information on rare chromosome disorders

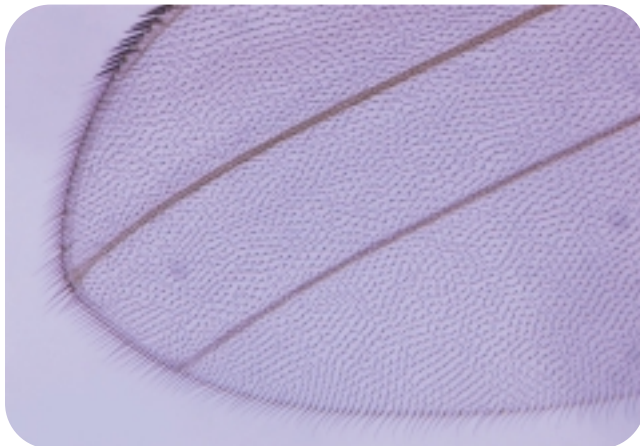
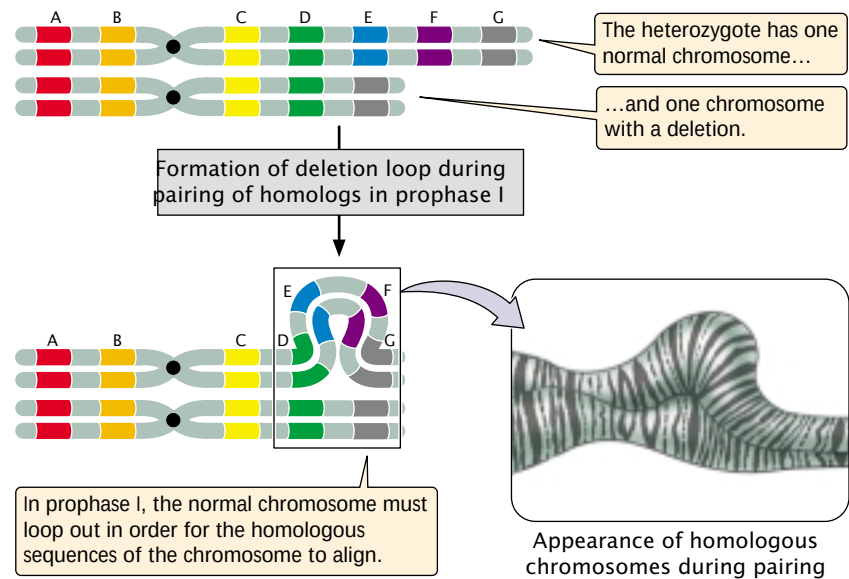
Inversions

A third type of chromosome rearrangement is a **chromosome inversion**, in which a chromosome segment is inverted—turned 180 degrees (see Figure 9.5c). If a chromosome originally had segments AB•CDEFG, then chromosome AB•CFEDG represents an inversion that includes segments DEF. For an inversion to take place, the chromosome must break in two places. Inversions that do not include the centromere, such as AB•CFEDG, are termed **paracentric inversions** (*para* meaning “next to”), whereas inversions that include the centromere, such as ADC•BEFG, are termed **pericentric inversions** (*peri* meaning “around”).

Individuals with inversions have neither lost nor gained any genetic material; just the gene order has been altered. Nevertheless, these mutations often have pronounced phenotypic effects. An inversion may break a gene into two parts, with one part moving to a new location and destroying the function of that gene. Even when the chromosome breaks are between genes, phenotypic effects may arise from the inverted gene order in an inversion. Many genes are regulated in a position-dependent manner; if their positions are altered by an inversion, they may be expressed at inappropriate times or in inappropriate tissues. This outcome is referred to as a **position effect**.

When an individual is homozygous for a particular inversion, no special problems arise in meiosis, and the two homologous chromosomes can pair and separate normally. When an individual is heterozygous for an inversion, however, the gene order of the two homologs differs, and the homologous sequences can align and pair only if

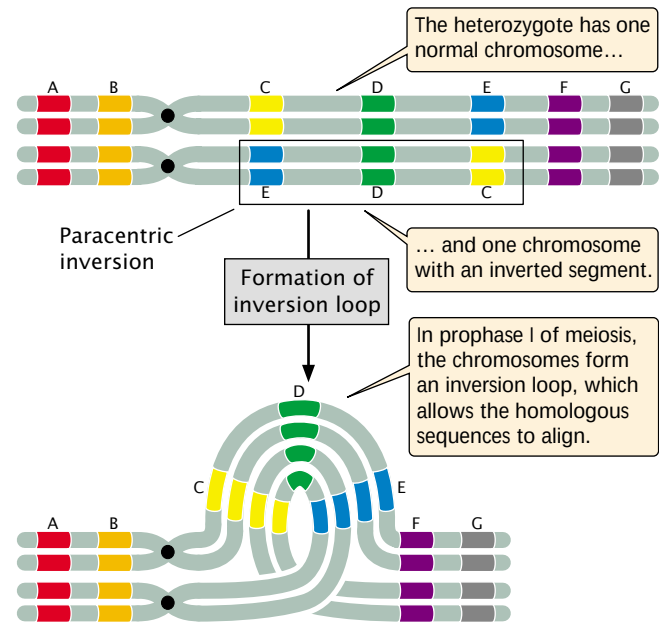
9.10 In an individual heterozygous for a deletion, the normal chromosome loops out during chromosome pairing in prophase I.



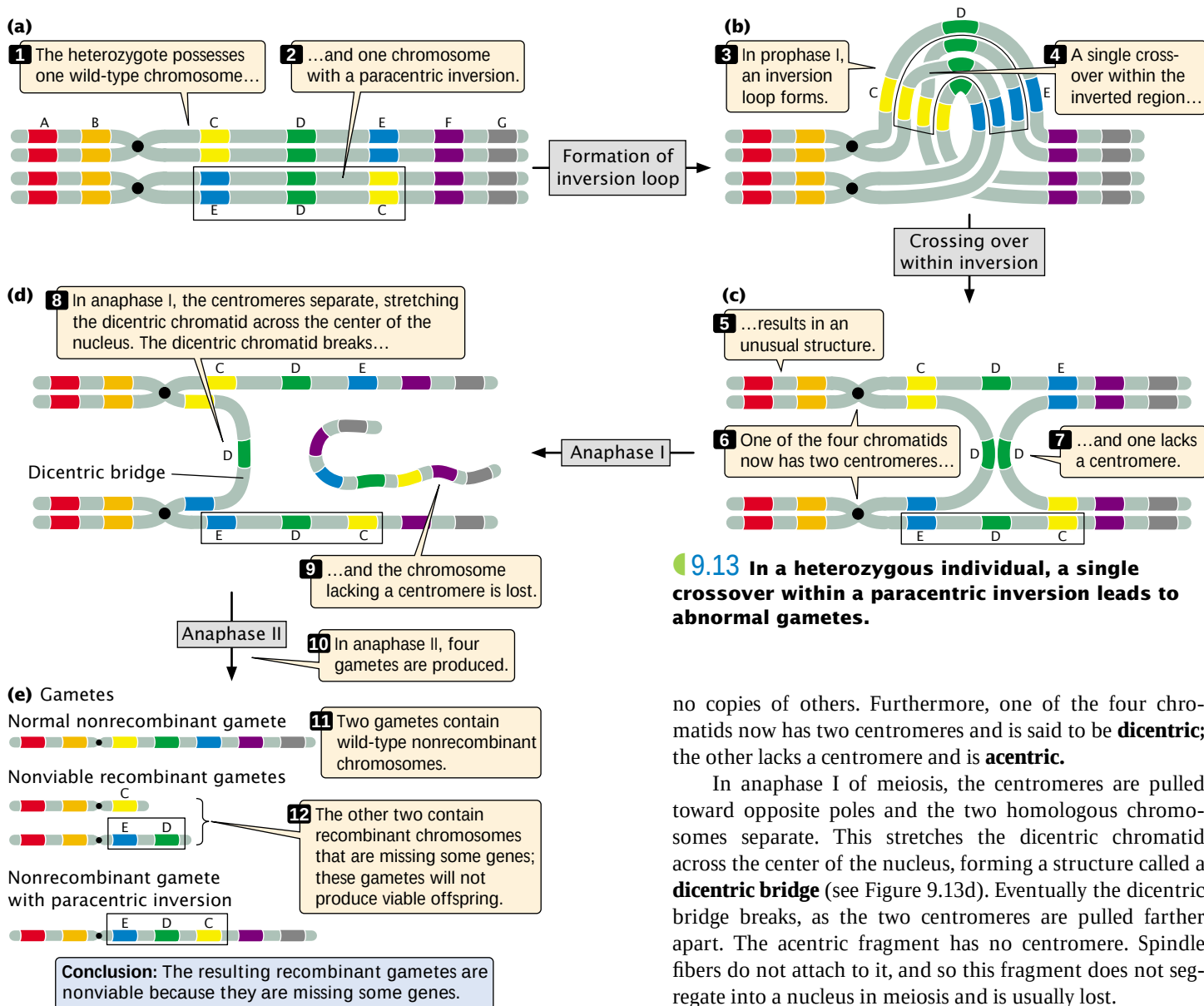
9.11 The Notch phenotype is produced by a chromosome deletion that includes the *Notch* gene. (top, normal wing venation; bottom, wing venation produced by *Notch* mutation. (Spyros Artavanis-Tsakonas, Kenji Matsuno, and Mark E. Fortini).

the two chromosomes form an inversion loop (FIGURE 9.12). The presence of an inversion loop in meiosis indicates that an inversion is present.

Individuals heterozygous for inversions also exhibit reduced recombination among genes located in the inverted region. The frequency of crossing over within the inversion is not actually diminished but, when crossing over does take place, the result is a tendency to produce gametes that are



9.12 In an individual heterozygous for a paracentric inversion, the chromosomes form an inversion loop during pairing in prophase I.



9.13 In a heterozygous individual, a single crossover within a paracentric inversion leads to abnormal gametes.

no copies of others. Furthermore, one of the four chromatids now has two centromeres and is said to be **dicentric**; the other lacks a centromere and is **acentric**.

In anaphase I of meiosis, the centromeres are pulled toward opposite poles and the two homologous chromosomes separate. This stretches the dicentric chromatid across the center of the nucleus, forming a structure called a **dicentric bridge** (see Figure 9.13d). Eventually the dicentric bridge breaks, as the two centromeres are pulled farther apart. The acentric fragment has no centromere. Spindle fibers do not attach to it, and so this fragment does not segregate into a nucleus in meiosis and is usually lost.

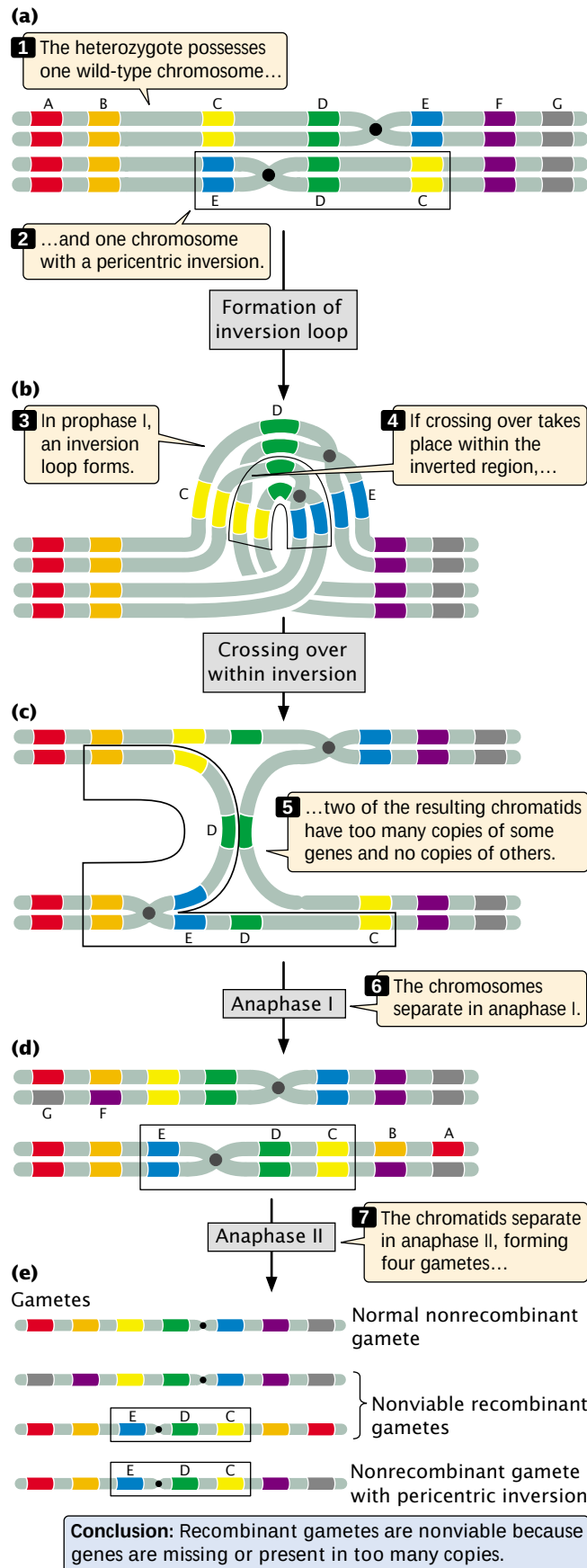
In the second division of meiosis, the chromatids separate and four gametes are produced (see Figure 9.11e). Two of the gametes contain the original, nonrecombinant chromosomes (AB•CDEFG and AB•EDCFG). The other two gametes contain recombinant chromosomes that are missing some genes; these gametes will not produce viable offspring. Thus, no recombinant progeny result when crossing over takes place within a paracentric inversion.

Recombination is also reduced within a pericentric inversion (Figure 9.14). No dicentric bridges or acentric fragments are produced, but the recombinant chromosomes have too many copies of some genes and no copies of others; so gametes that receive the recombinant chromosomes cannot produce viable progeny.

Figures 9.13 and 9.14 illustrate the results of single crossovers within inversions. Double crossovers, in which both crossovers are on the same two strands (two-strand,

not viable and thus no recombinant progeny are observed. Let's see why this occurs.

Figure 9.13 illustrates the results of crossing over within a paracentric inversion. The individual is heterozygous for an inversion (see Figure 9.13a), with one wild-type, unmutated chromosome (AB•CDEFG) and one inverted chromosome (AB•EDCFG). In prophase I of meiosis, an inversion loop forms, allowing the homologous sequences to pair up (see Figure 9.13b). If a single crossover takes place in the inverted region (between segments C and D in Figure 9.13), an unusual structure results (see Figure 9.13c). The two outer chromatids, which did not participate in crossing over, contain original, nonrecombinant gene sequences. The two inner chromatids, which did cross over, are highly abnormal: each has two copies of some genes and



9.14 In a heterozygous individual, a single crossover within a pericentric inversion leads to abnormal gametes.

double crossovers), result in functional, recombinant chromosomes. (Try drawing out the results of a double crossover.) Thus, even though the overall rate of recombination is reduced within an inversion, some viable recombinant progeny may still be produced through two-stranded double crossovers.

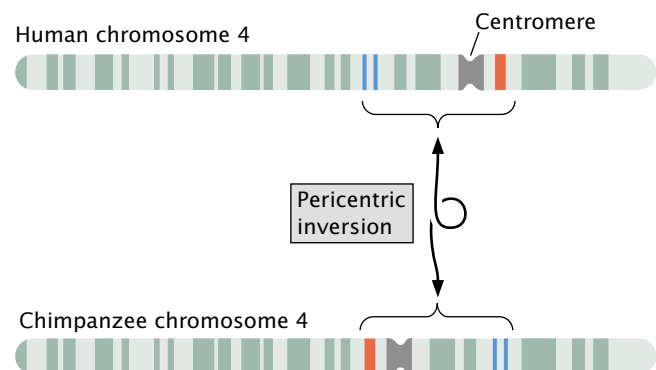
Inversion heterozygotes are common in many organisms, including a number of plants, some species of *Drosophila*, mosquitoes, and grasshoppers. Inversions may have played an important role in human evolution: G-banding patterns reveal that several human chromosomes differ from those of chimpanzees by only a pericentric inversion (FIGURE 9.15).

Concepts

In an inversion, a segment of a chromosome is inverted. Inversions cause breaks in some genes and may move others to new locations. In heterozygotes for a chromosome inversion, the chromosomes form loops in prophase I of meiosis. When crossing over takes place within the inverted region, nonviable gametes are usually produced, resulting in a depression in observed recombination frequencies.

Translocations

A **translocation** entails the movement of genetic material between nonhomologous chromosomes (see Figure 9.5d) or within the same chromosome. Translocation should not be confused with crossing over, in which there is an exchange of genetic material between *homologous* chromosomes.



9.15 Chromosome 4 differs in humans and chimpanzees in a pericentric inversion.

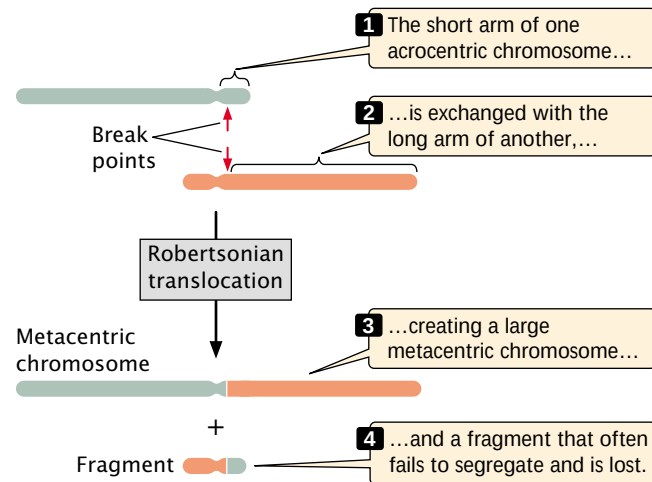
In **nonreciprocal translocations**, genetic material moves from one chromosome to another without any reciprocal exchange. Consider the following two nonhomologous chromosomes: $AB \bullet CDEFG$ and $MN \bullet OPQRS$. If chromosome segment EF moves from the first chromosome to the second without any transfer of segments from the second chromosome to the first, a nonreciprocal translocation has taken place, producing chromosomes $AB \bullet CDG$ and $MN \bullet OPEFQRS$. More commonly, there is a two-way exchange of segments between the chromosomes, resulting in a **reciprocal translocation**. A reciprocal translocation between chromosomes $AB \bullet CDEFG$ and $MN \bullet OPQRS$ might give rise to chromosomes $AB \bullet CDQRG$ and $MN \bullet OPEFS$.

Translocations can affect a phenotype in several ways. First, they may create new linkage relations that affect gene expression (a position effect): genes translocated to new locations may come under the control of different regulatory sequences or other genes that affect their expression—an example is found in Burkitt lymphoma, to be discussed later in this chapter.

Second, the chromosomal breaks that bring about translocations may take place within a gene and disrupt its function. Molecular geneticists have used these types of effects to map human genes. Neurofibromatosis is a genetic disease characterized by numerous fibrous tumors of the skin and nervous tissue; it results from an autosomal dominant mutation. Linkage studies first placed the locus for neurofibromatosis on chromosome 17. Geneticists later identified two patients with neurofibromatosis who possessed a translocation affecting chromosome 17. These patients were assumed to have developed neurofibromatosis because one of the chromosome breaks that occurred in the translocation disrupted a particular gene that causes neurofibromatosis. DNA from the regions around the breaks was sequenced and eventually led to the identification of the gene responsible for neurofibromatosis.

Deletions frequently accompany translocations. In a **Robertsonian translocation**, for example, the long arms of two acrocentric chromosomes become joined to a common centromere through a translocation, generating a metacentric chromosome with two long arms and another chromosome with two very short arms (FIGURE 9.16). The smaller chromosome often fails to segregate, leading to an overall reduction in chromosome number. As we will see, Robertsonian translocations are the cause of some cases of Down syndrome.

The effects of a translocation on chromosome segregation in meiosis depend on the nature of the translocation. Let us consider what happens in an individual heterozygous for a reciprocal translocation. Suppose that the original chromosome segments were $AB \bullet CDEFG$ and $MN \bullet OPQRS$ (designated N_1 and N_2), and a reciprocal translocation takes place, producing chromosomes $AB \bullet CDQRS$ and $MN \bullet OPEFG$ (designated T_1 and T_2). An individual heterozygous for this translocation would pos-

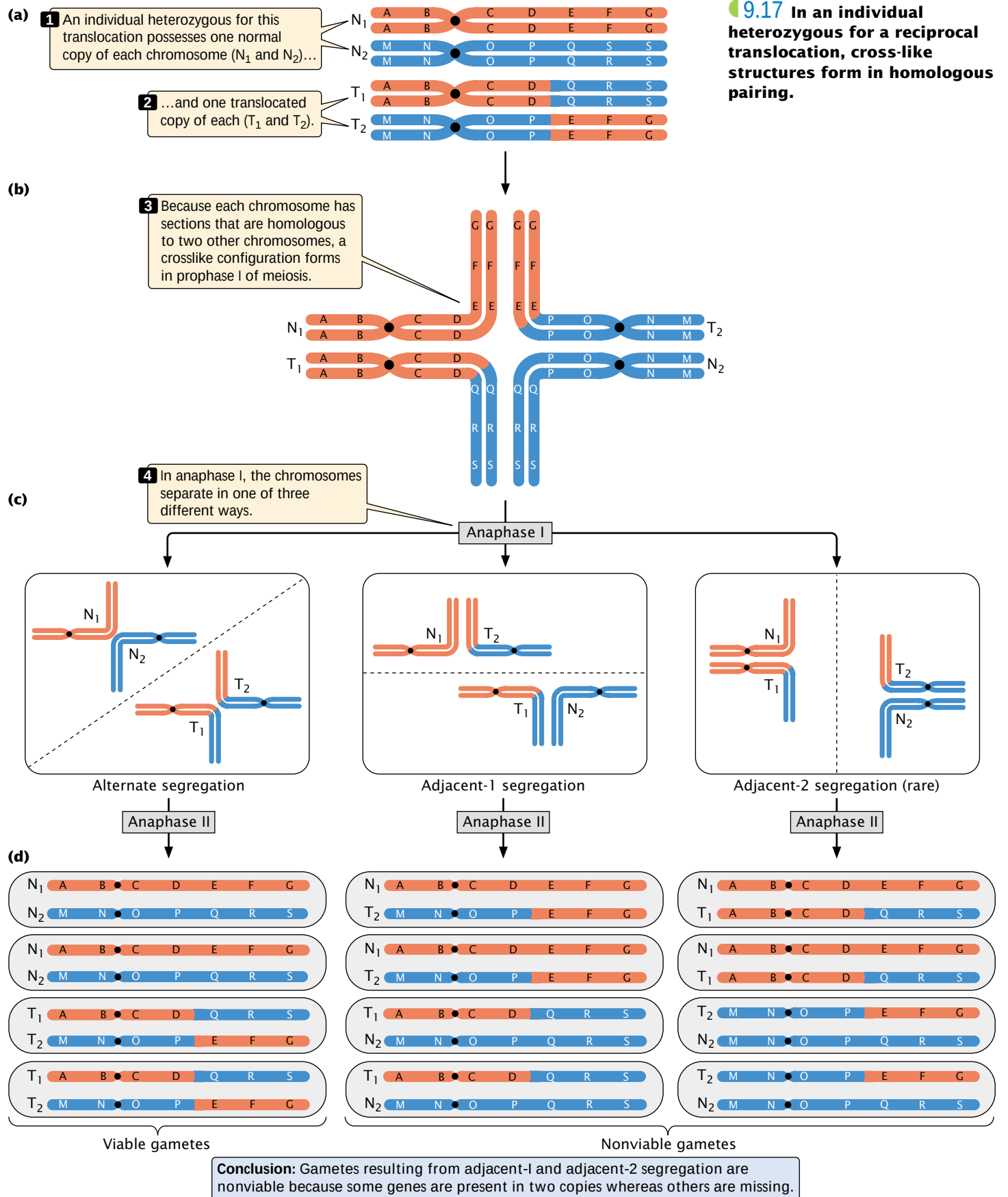


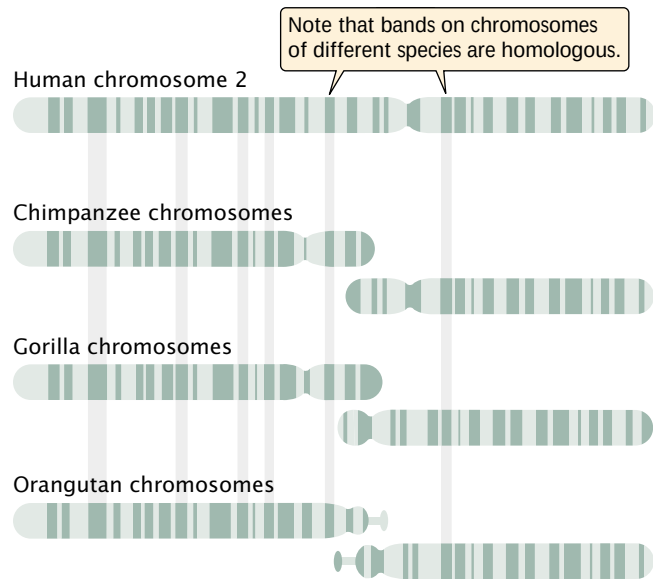
9.16 In a Robertsonian translocation, the short arm of one acrocentric chromosome is exchanged with the long arm of another.

sess one normal copy of each chromosome and one translocated copy (FIGURE 9.17a). Each of these chromosomes contains segments that are homologous to two other chromosomes. When the homologous sequences pair in prophase I of meiosis, crosslike configurations consisting of all four chromosomes (FIGURE 9.17b) form.

Notice that N_1 and T_1 have homologous centromeres (in both chromosomes the centromere is between segments B and C); similarly, N_2 and T_2 have homologous centromeres (between segments N and O). Normally, homologous centromeres separate and move toward opposite poles in anaphase I of meiosis. With a reciprocal translocation, the chromosomes may segregate in three different ways. In **alternate segregation** (FIGURE 9.17c), N_1 and N_2 move toward one pole and T_1 and T_2 move toward the opposite pole. In **adjacent-1 segregation**, N_1 and T_2 move toward one pole and T_1 and N_2 move toward the other pole. In both alternate and adjacent-1 segregation, homologous centromeres segregate toward opposite poles. **Adjacent-2 segregation**, in which N_1 and T_1 move toward one pole and T_2 and N_2 move toward the other, is rare.

The products of the three segregation patterns are illustrated in FIGURE 9.17d. As you can see, the gametes produced by alternate segregation possess one complete set of the chromosome segments. These gametes are therefore functional and can produce viable progeny. In contrast, gametes produced by adjacent-1 and adjacent-2 segregation are not viable, because some chromosome segments are present in two copies, whereas others are missing. Adjacent-2 segregation is rare, and so most gametes are produced by alternate and adjacent segregation. Therefore, approximately half of the gametes from an individual heterozygous for a reciprocal translocation are expected to be functional.





9.18 Human chromosome 2 contains a Robertsonian translocation that is not present in chimps, gorillas, or orangutans. G-banding reveals that a Robertsonian translocation in a human ancestor switched the long and short arms of the two acrocentric chromosomes that are still found in the other three primates. This translocation created the large metacentric human chromosome 2.

Translocations can play an important role in the evolution of karyotypes. Chimpanzees, gorillas, and orangutans all have 48 chromosomes, whereas humans have 46. Human chromosome 2 is a large, metacentric chromosome with G-banding patterns that match those found on two different acrocentric chromosomes of the apes (FIGURE 9.18). Apparently, a Robertsonian translocation took place in a human ancestor, creating a large metacentric chromosome from the two long arms of the ancestral acrocentric chromosomes and a small chromosome consisting of the two short arms. The small chromosome was subsequently lost, leading to the reduced human chromosome number.

Concepts

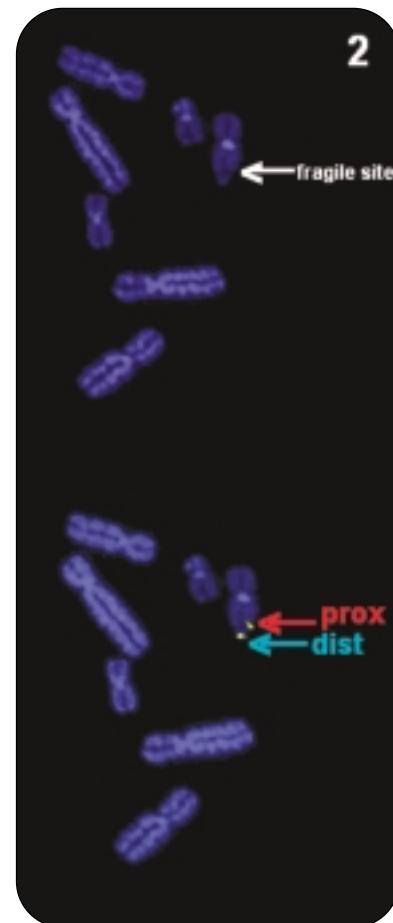
In translocations, parts of chromosomes move to other, nonhomologous chromosomes or other regions of the same chromosome. Translocations may affect the phenotype by causing genes to move to new locations, where they come under the influence of new regulatory sequences, or by breaking genes and disrupting their function.

www.whfreeman.com/pierce For gorilla and other great-ape chromosomes with a comparison of the human karyotype

and great-ape karyotypes; animation of the formation of a Robertsonian translocation and types of gametes produced by a translocation carrier; and pictures of karyotypes containing Robertsonian translocations

Fragile Sites

Chromosomes of cells grown in culture sometimes develop constrictions or gaps at particular locations called **fragile sites** (FIGURE 9.19) because they are prone to breakage under certain conditions. A number of fragile sites have been identified on human chromosomes. One of the most intensively studied is a fragile site on the human X chromosome that is associated with mental retardation, the fragile-X syndrome. Exhibiting X-linked inheritance and arising with a frequency of about 1 in 1250 male births, fragile-X syndrome has been shown to result from an increase in the number of repeats of a CGG trinucleotide (see Chapter 19). However, other common fragile sites do not consist of trinucleotide repeats, and their nature is still incompletely understood.



9.19 Fragile sites are chromosomal regions susceptible breakage under certain conditions.

(Erica Woollatt, Women's and Children's Hospital, Adelaide).

Ryan, age 4, is brought to the medical genetics clinic by his 27-year-old mother, Janet. Ryan is developmentally delayed and hyperactive and has undergone many tests, but all the results were normal. Janet and her husband, Terry, very much want another child. The family history is unremarkable with the exception of the 6-year-old son of one of Janet's cousins, who is apparently "slow." However, Janet reports that she does not get along with her siblings and that in fact she has little contact with any of the rest of her family. Both her parents are deceased. A friend told her recently that Janet's 25-year-old sister just found out that she is pregnant with her first child.

The cause of Ryan's delay is determined to be Fragile X-syndrome. As its name suggests, this condition is a sex-linked disorder carried by females and most seriously affecting males, in whom it can cause severe mental retardation. After describing the genetics of Fragile X-syndrome and its hereditary risks, the medical geneticist asks Janet to notify her sister of the information and alert her to the availability of prenatal testing. The next week, the geneticist calls Janet, who states that she has not called her sister and does not intend to.

Are there ways that the geneticist can persuade Janet to inform her sister? Failing that, can the physician alert Janet's sister to her risk without compromising the obligation to preserve the confidentiality of the relationship with Janet? If there is no other recourse, does the genetic professional have an ethical or legal right to breach confidentiality and inform Janet's sister—and perhaps others in the family—of the risk? Does the professional have a *duty* to do so? By law, a professional's ethical duty to a patient or client can be overridden only if (1) reasonable efforts to gain

consent to disclosure have failed; (2) there is a high probability of harm if information is withheld, and the information will be used to avert harm; (3) the harm that persons would suffer is serious; and (4) precautions are taken to ensure that only the genetic information needed for diagnosis or treatment or both is disclosed. However, who determines which genetic risks are among the "serious harms" that would permit breaching confidentiality in medical contexts?

Some might argue that all these questions miss the point: the familiar duties of doctor and patient don't apply in this case because, where genes are concerned, the patient is not the *individual*, but the *entire family* to which that patient belongs. Thus, the physician must do whatever best meets the needs of all members of the family. This line of reasoning may be increasingly popular as the powers of genetic medicine grow and physicians are more frequently asked to utilize and interpret genetic tests in the course of routine care. Nevertheless, we should be careful not to hastily discard the traditional ethical principle that the doctor's and medical team's first responsibility is to the presenting patient. Replacing it with a generalized responsibility to the whole family takes medical practice into uncharted territory and can impose serious new burdens on medical professionals.

In this case, the patient refuses to inform other family members who might benefit from prenatal testing—which would allow them to decide to continue or terminate a pregnancy or prepare for the birth of a child with a genetic disorder. But the same type of problem can arise in many other ways where genetic medicine or research is concerned. In some conditions, testing is aimed at determining whether an individual or family has a genetic

susceptibility to a disease, the knowledge of which can help them pursue preventative strategies. Current testing for known breast cancer mutations is an example. In these cases, whenever the specific genes involved have not yet been identified, researchers or clinicians must conduct extensive family linkage studies to determine the pattern of inheritance. One or more family members can block progress by refusing to participate in the study. The principle of respect for autonomy certainly supports such refusals, but should this principle trump research that is needed to improve the health of other members of the family? Sometimes, the reverse problem arises: some members demand participation by other relatives in ways that exert pressure on them.

Alternatively, in some cases one person's use of a genetic test may harm others in the family. This problem has arisen in connection with Huntington disease, a fatal, later-onset neurological disorder for which no treatment exists. There have been instances when one twin of a pair of identical twins has insisted on testing and the other twin, unwilling to be subjected to the fearful psychosocial harms that testing can bring, has objected.

Social workers, psychologists, genetic counselors, and others who work closely with families know that such disputes often reveal deep fault lines and sources of conflict within a family. When faced with these cases, they also recognize that it is not a matter of just solving an ethical problem, but of understanding and addressing the underlying problems that give rise to these conflicts. The familial nature of genetic information will undoubtedly increase the number and intensity of conflicts that come before caregivers or counseling professionals.

Aneuploidy

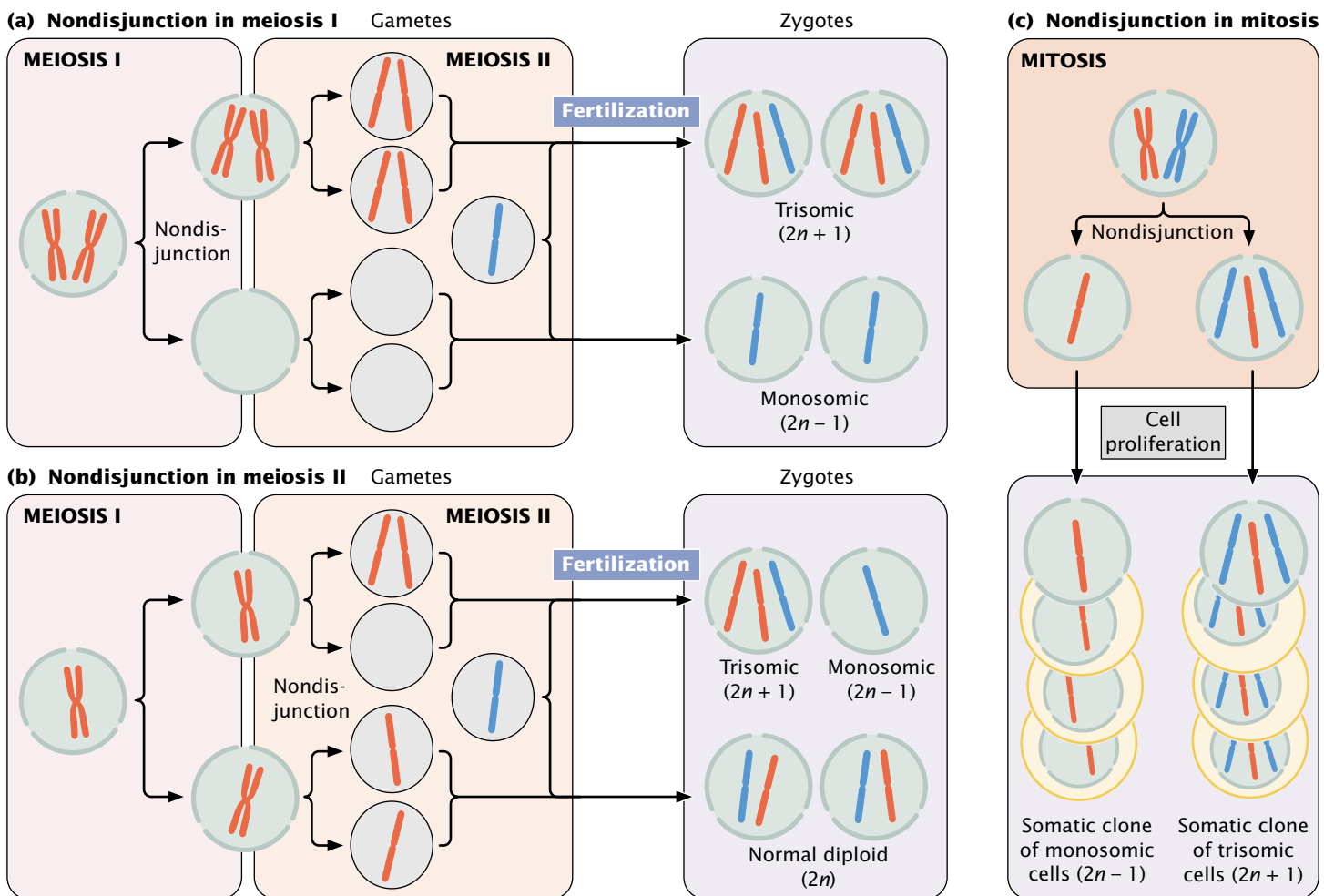
In addition to chromosome rearrangements, chromosome mutations also include changes in the *number* of chromosomes. Variations in chromosome number can be classified into two basic types: changes in the number of individual chromosomes (aneuploidy) and changes in the number of chromosome sets (polyploidy).

Aneuploidy can arise in several ways. First, a chromosome may be lost in the course of mitosis or meiosis if, for example, its centromere is deleted. Loss of the centromere prevents the spindle fibers from attaching; so the chromosome fails to move to the spindle pole and does not become incorporated into a nucleus after cell division. Second, the small chromosome generated by a Robertsonian translocation may be lost in mitosis or meiosis. Third, aneuploids may arise through nondisjunction, the failure of homologous chromosomes or sister chromatids to separate in meiosis or mitosis (see p. 000 in Chapter 4). Nondisjunction leads to some gametes or cells that contain an extra chromosome and others that are missing a chromosome (FIGURE 9.20).

Types of Aneuploidy

We will consider four types of relatively common aneuploid conditions in diploid individuals: nullisomy, monosomy, trisomy, and tetrasomy.

1. **Nullisomy** is the loss of both members of a homologous pair of chromosomes. It is represented as $2n - 2$, where n refers to the haploid number of chromosomes. Thus, among humans, who normally possess $2n = 46$ chromosomes, a nullisomic person has 44 chromosomes.
2. **Monosomy** is the loss of a single chromosome, represented as $2n - 1$. A monosomic person has 45 chromosomes.



9.20 Aneuploids can be produced through nondisjunction in (a) meiosis I, (b) meiosis II, and (c) mitosis. The gametes that result from meioses with nondisjunction combine with gamete (with blue chromosome) that results from normal meiosis to produce the zygotes.

3. **Trisomy** is the gain of a single chromosome, represented as $2n + 1$. A trisomic person has 47 chromosomes. The gain of a chromosome means that there are three homologous copies of one chromosome.
4. **Tetrasomy** is the gain of two homologous chromosomes, represented as $2n + 2$. A tetrasomic person has 48 chromosomes. Tetrasomy is not the gain of *any* two extra chromosomes, but rather the gain of two homologous chromosomes; so there will be four homologous copies of a particular chromosome.

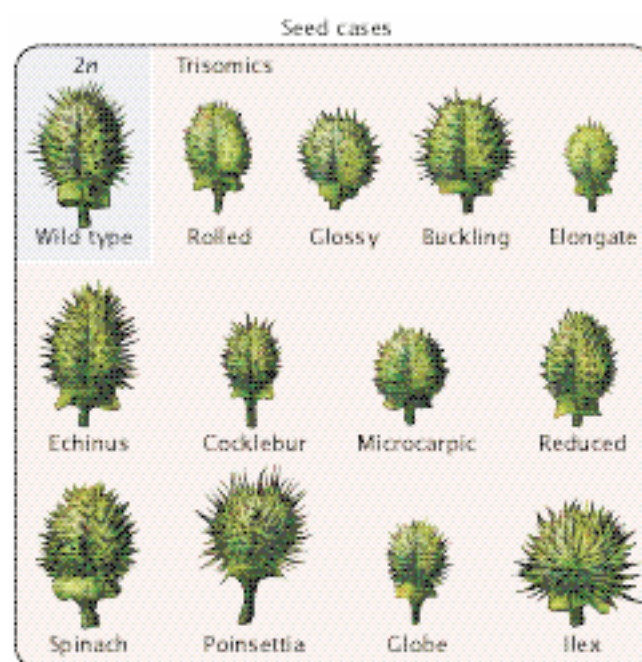
More than one aneuploid mutation may occur in the same individual. An individual that has an extra copy of two different (nonhomologous) chromosomes is referred to as being double trisomic and represented as $2n + 1 + 1$. Similarly, a double monosomic has two fewer nonhomologous chromosomes ($2n - 1 - 1$), and a double tetrasomic has two extra pairs of homologous chromosomes ($2n + 2 + 2$).

Effects of Aneuploidy

One of the first aneuploids to be recognized was a fruit fly with a single X chromosome and no Y chromosome, which was discovered by Calvin Bridges in 1913 (see p. 000 in Chapter 4). Another early study of aneuploidy focused on mutants in the Jimson weed, *Datura stramonium*. A. Francis Blakeslee began breeding this plant in 1913, and he observed that crosses with several Jimson mutants produced unusual ratios of progeny. For example, the *globe* mutant (having a seedcase globular in shape) was dominant but was inherited primarily from the female parent. When *globe* plants were self-fertilized, only 25% of the progeny had the globe phenotype, an unusual ratio for a dominant trait. Blakeslee isolated 12 different mutants (FIGURE 9.21) that also exhibited peculiar patterns of inheritance. Eventually, John Belling demonstrated that these 12 mutants are in fact trisomics. *Datura stramonium* has 12 pairs of chromosomes ($2n = 24$), and each of the 12 mutants is trisomic for a different chromosome pair. The aneuploid nature of the mutants explained the unusual ratios that Blakeslee had observed in the progeny. Many of the extra chromosomes in the trisomics were lost in meiosis, so fewer than 50% of the gametes carried the extra chromosome, and the proportion of trisomics in the progeny was low. Furthermore, the pollen containing an extra chromosome was not as successful in fertilization, and trisomic zygotes were less viable.

Aneuploidy usually alters the phenotype drastically. In most animals and many plants, aneuploid mutations are lethal. Because aneuploidy affects the number of gene copies but not their nucleotide sequences, the effects of aneuploidy are most likely due to abnormal gene dosage. Aneuploidy alters the dosage for some, but not all, genes, disrupting the relative concentrations of gene products and often interfering with normal development.

A major exception to the relation between gene number and protein dosage pertains to genes on the mammalian



9.21 Mutant capsules in Jimson weed (*Datura stramonium*) result from different trisomies. Each type of capsule is a phenotype that is trisomic for a different chromosome.

X chromosome. In mammals, X-chromosome inactivation ensures that males (who have a single X chromosome) and females (who have two X chromosomes) receive the same functional dosage for X-linked genes (see p. 000 in Chapter 4 for further discussion of X-chromosome inactivation). Extra X chromosomes in mammals are inactivated; so we might expect that aneuploidy of the sex chromosomes would be less detrimental in these animals. Indeed, this is the case for mice and humans, for whom aneuploids of the sex chromosomes are the most common form of aneuploidy seen in living organisms. Y-chromosome aneuploids are probably common because there is so little information on the Y-chromosome.

Concepts

Aneuploidy, the loss or gain of one or more individual chromosomes, may arise from the loss of a chromosome subsequent to translocation or from nondisjunction in meiosis or mitosis. It disrupts gene dosage and often has severe phenotypic effects.

Aneuploidy in Humans

Aneuploidy in humans usually produces serious developmental problems that lead to spontaneous abortion (miscarriage). In fact, as many as 50% of all spontaneously

aborted fetuses carry chromosome defects, and a third or more of all conceptions spontaneously abort, usually so early in development that the mother is not even aware of her pregnancy. Only about 2% of all fetuses with a chromosome defect survive to birth.

Sex-chromosome aneuploids The most common aneuploidy seen in living humans has to do with the sex chromosomes. As is true of all mammals, aneuploidy of the human sex chromosomes is better tolerated than aneuploidy of autosomal chromosomes. Turner syndrome and Klinefelter syndrome (see Figures 4.9 and 4.10) both result from aneuploidy of the sex chromosomes.

Autosomal aneuploids Autosomal aneuploids resulting in live births are less common than sex-chromosome aneuploids in humans, probably because there is no mechanism of dosage compensation for autosomal chromosomes. Most autosomal aneuploids are spontaneously aborted, with the exception of aneuploids of some of the small autosomes. Because these chromosomes are small and carry fewer genes, the presence of extra copies is less detrimental. For example, the most common autosomal aneuploidy in humans is **trisomy 21**, also called **Down syndrome**. The number of genes on different human chromosomes is not precisely known at the present time, but DNA sequence data indicate that chromosome 21 has fewer genes than any other autosome, with perhaps less than 300 genes of a total of 30,000 to 35,000 for the entire genome.

The incidence of Down syndrome in the United States is about 1 in 700 human births, although the incidence is higher among children born to older mothers. People with Down syndrome (FIGURE 9.22a) show variable degrees of

mental retardation, with an average IQ of about 50 (compared with an average IQ of 100 in the general population). Many people with Down syndrome also have characteristic facial features, some retardation of growth and development, and an increased incidence of heart defects, leukemia, and other abnormalities.

Approximately 92% of those who have Down syndrome have three full copies of chromosome 21 (and therefore a total of 47 chromosomes), a condition termed **primary Down syndrome** (FIGURE 9.22b). Primary Down syndrome usually arises from random nondisjunction in egg formation: about 75% of the nondisjunction events that cause Down syndrome are maternal in origin, and most arise in meiosis I. Most children with Down syndrome are born to normal parents, and the failure of the chromosomes to divide has little hereditary tendency. A couple who has conceived one child with primary Down syndrome has only a slightly higher risk of conceiving a second child with Down syndrome (compared with other couples of similar age who have not had any Down-syndrome children). Similarly, the couple's relatives are not more likely to have a child with primary Down syndrome.

Most cases of primary Down syndrome arise from maternal nondisjunction, and the frequency of this occurring correlates with maternal age (FIGURE 9.23). Although the underlying cause of the association between maternal age and nondisjunction remains obscure, recent studies have indicated a strong correlation between nondisjunction and aberrant meiotic recombination. Most chromosomes that failed to separate in meiosis I do not show any evidence of having recombined with one another. Conversely, chromosomes that appear to have failed to separate in meiosis II often show evidence of recombination in regions that do

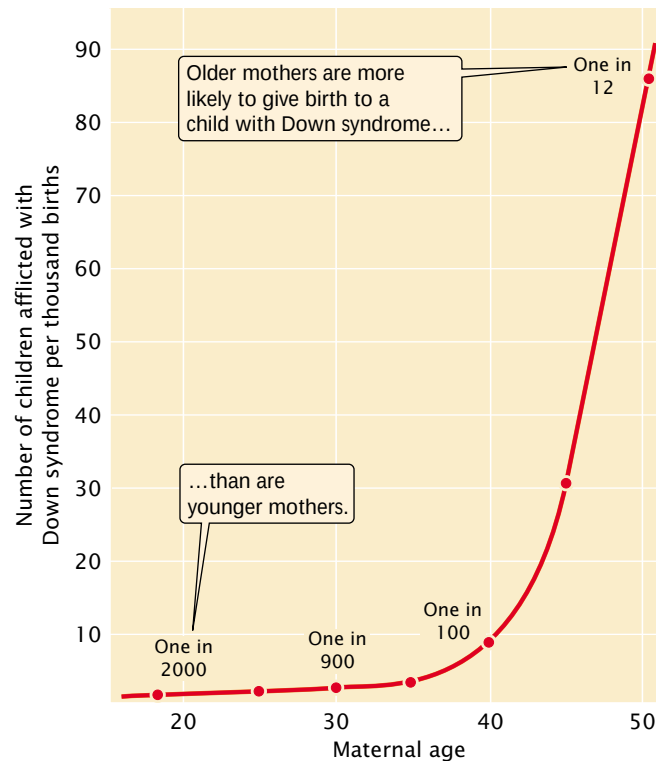
(a)



(b)



9.22 Primary Down syndrome is caused by the presence of three copies of chromosome 21. (a) A child who has Down syndrome. (b) Karyotype of a person who has primary Down syndrome. (Part a, Hattie Young/Science Photo Library/Photo Researchers; part b, L. Willatt. East Anglian Regional Genetics Service/Science Photo Library/Photo Researchers).



9.23 The incidence of primary Down syndrome increases with maternal age.

not normally recombine, most notably near the centromere. Although aberrant recombination appears to play a role in nondisjunction, the maternal age effect is more complex. In female mammals, prophase I begins in all oogonia during fetal development, and recombination is completed prior to birth. Meiosis then arrests in diplotene, and the primary oocytes remain suspended until just before ovulation. As each primary oocyte is ovulated, meiosis resumes and the first division is completed, producing a secondary oocyte. At this point, meiosis is suspended again, and remains so until the secondary oocyte is penetrated by a sperm. The second meiotic division takes place immediately before the nuclei of egg and sperm unite to form a zygote.

An explanation of the maternal age effect must take into account the aberrant recombination that occurs prenatally and the long suspension in prophase I. One theory is that the “best” oocytes are ovulated first, leaving those oocytes that had aberrant recombination to be used later in life. However, evidence indicates that the frequency of aberrant recombination is similar between oocytes that are ovulated in young women and those ovulated in older women. Another possible explanation is that aging of the cellular components needed for meiosis results in nondisjunction of chromosomes that are “at risk,” because they have failed to recombine or had some recombination defect. In younger oocytes, these chromosomes can still be segregated from one another, but in older oocytes, they are sensitive to other

perturbations in the meiotic machinery. In contrast, sperm are produced continually after puberty, with no long suspension of the meiotic divisions. This fundamental difference between the meiotic process in females and males may explain why most chromosome aneuploidy in humans is maternal in origin.

About 4% of people with Down syndrome have 46 chromosomes, but an extra copy of part of chromosome 21 is attached to another chromosome through a translocation (FIGURE 9.24). This condition is termed **familial Down syndrome** because it has a tendency to run in families. The phenotypic characteristics of familial Down syndrome are the same as those for primary Down syndrome.

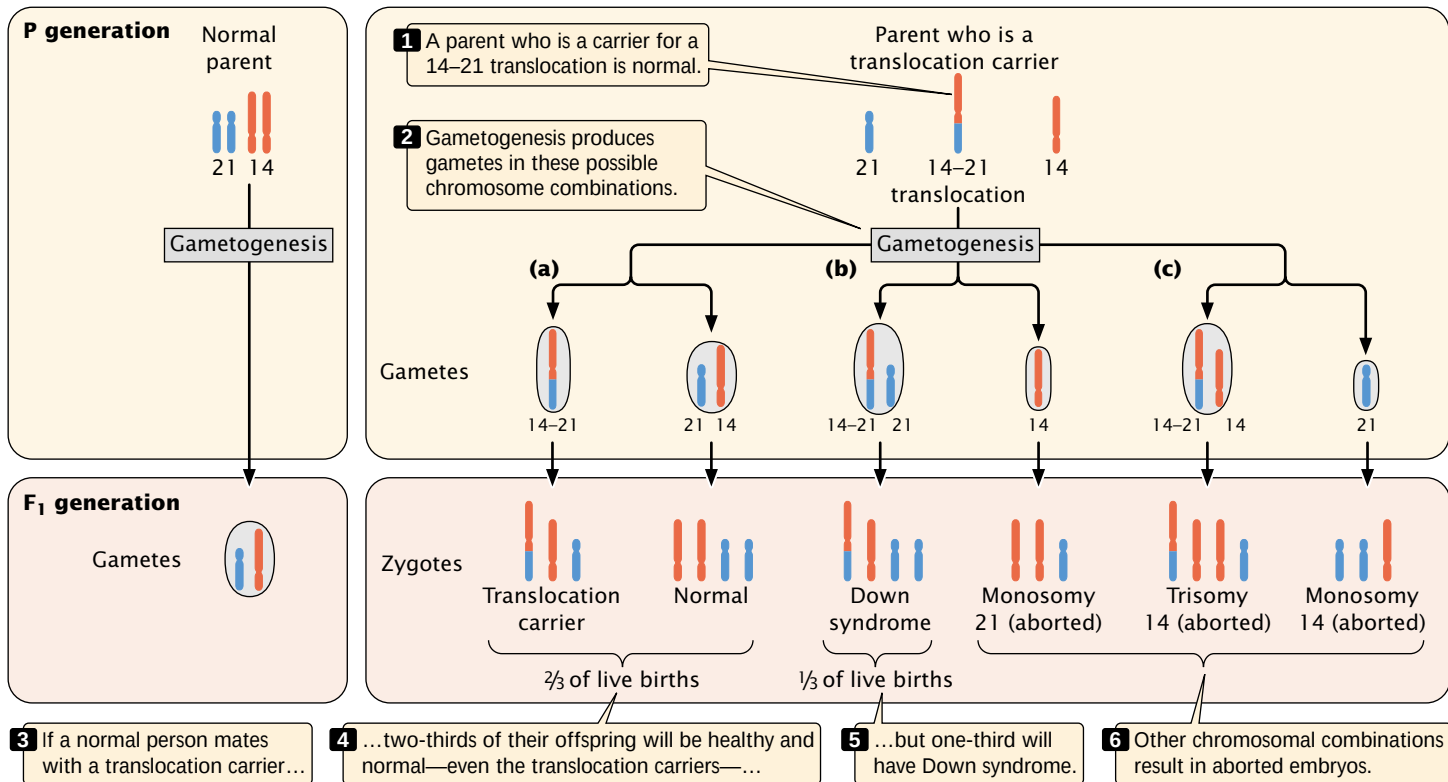
Familial Down syndrome arises in offspring whose parents are carriers of chromosomes that have undergone a Robertsonian translocation, most commonly between chromosome 21 and chromosome 14: the long arm of 21 and the short arm of 14 exchange places. This exchange produces a chromosome that includes the long arms of chromosomes 14 and 21, and a very small chromosome that consists of the short arms of chromosomes 21 and 14. The small chromosome is generally lost after several cell divisions.

Persons with the translocation, called **translocation carriers**, do not have Down syndrome. Although they possess only 45 chromosomes, their phenotypes are normal because they have two copies of the long arms of chromosomes 14 and 21, and apparently the short arms of these chromosomes (which are lost) carry no essential genetic information. Although translocation carriers are completely healthy, they have an increased chance of producing children with Down syndrome.

When a translocation carrier produces gametes, the translocation chromosome may segregate in three different



9.24 The translocation of chromosome 21 onto another chromosome results in familial Down syndrome. (Dr. Dorothy Warburton, HICC, Columbia University).



9.25 Translocation carriers are at increased risk for producing children with Down syndrome.

ways. First, it may separate from the normal chromosomes 14 and 21 in anaphase I of meiosis (FIGURE 9.25a). In this type of segregation, half of the gametes will have the translocation chromosome and no other copies of chromosomes 21 and 14; the fusion of such a gamete with a normal gamete will give rise to a translocation carrier. The other half of the gametes produced by this first type of segregation will be normal, each with a single copy of chromosomes 21 and 14, and will result in normal offspring.

Alternatively, the translocation chromosome may separate from chromosome 14 and pass into the same cell with the normal chromosome 21 (FIGURE 9.25b). This type of segregation produces all abnormal gametes; half will have two functional copies of chromosome 21 (one normal and one attached to chromosome 14) and the other half will lack chromosome 21. The gametes with the two functional copies of chromosome 21 will produce children with familial Down syndrome; the gametes lacking chromosome 21 will result in zygotes with monosomy 21 and will be spontaneously aborted.

In the third type of segregation, the translocation chromosome and the normal copy of chromosome 14 segregate together, and the normal chromosome 21 segregates by itself (FIGURE 9.25c). This pattern is presumably rare, because the two centromeres are both derived from chromosome 14 separately from each other. In any case, all the gametes produced by this process are abnormal: half result

in monosomy 14 and the other half result in trisomy 14—all are spontaneously aborted. Thus, only three of the six types of gametes that can be produced by a translocation carrier will result in the birth of a baby and, theoretically, these gametes should arise with equal frequency. One-third of the offspring of a translocation carrier should be translocation carriers like their parent, one-third should have familial Down syndrome, and one-third should be normal. In reality, however, fewer than one-third of the children born to translocation carriers have Down syndrome, which suggests that some of the embryos with Down syndrome are spontaneously aborted.

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Additional information on Down syndrome

Few autosomal aneuploids besides trisomy 21 result in human live births. **Trisomy 18**, also known as **Edward syndrome**, arises with a frequency of approximately 1 in 8000 live births. Babies with Edward syndrome are severely retarded and have low-set ears, a short neck, deformed feet, clenched fingers, heart problems, and other disabilities. Few live for more than a year after birth. **Trisomy 13** has a frequency of about 1 in 15,000 live births and produces features that are collectively known as **Patau syndrome**. Characteristics of this condition include severe mental retardation, a small head, sloping forehead, small eyes, cleft lip

and palate, extra fingers and toes, and numerous other problems. About half of children with trisomy 13 die within the first month of life, and 95% die by the age of 3. Rarer still is **trisomy 8**, which arises with a frequency of about 1 in 25,000 to 50,000 live births. This aneuploid is characterized by mental retardation, contracted fingers and toes, low-set malformed ears, and a prominent forehead. Many who have this condition have normal life expectancy.

Concepts

In humans, sex-chromosome aneuploids are more common than are autosomal aneuploids. X-chromosome inactivation prevents problems of gene dosage for X-linked genes. Down syndrome results from three functional copies of chromosome 21, either through trisomy (primary Down syndrome) or a Robertsonian translocation (familial Down syndrome).

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Additional information on trisomy 13 and trisomy 18

Uniparental Disomy

Normally, the two chromosomes of a homologous pair are inherited from different parents—one from the father and one from the mother. The development of molecular techniques that facilitate the identification of specific DNA sequences (see Chapter 18), has made it possible to determine the parental origins of chromosomes. Surprisingly, sometimes both chromosomes are inherited from the same parent, a condition termed **uniparental disomy**.

Uniparental disomy violates the rule that children affected with a recessive disorder appear only in families where both parents are carriers. For example, cystic fibrosis is an autosomal recessive disease; typically, both parents of an affected child are heterozygous for the cystic fibrosis mutation on chromosome 7. However, a small proportion of people with cystic fibrosis have only a single parent who is heterozygous for the cystic fibrosis gene. How can this be? These people must have inherited from the heterozygous parent two copies of the chromosome 7 that carries the defective cystic fibrosis allele and no copy of the normal allele from the other parent. Uniparental disomy has also been observed in Prader-Willi syndrome, a rare condition that arises when a paternal copy of a gene on chromosome 15 is missing. Although most cases of Prader-Willi syndrome result from a chromosome deletion that removes the paternal copy of the gene (see p. 000 in Chapter 4), from 20% to 30% arise when both copies of chromosome 15 are inherited from the mother and no copy is inherited from the father.

Many cases of uniparental disomy probably originate as a trisomy. Although most autosomal trisomies are lethal, a trisomic embryo can survive if one of the three chromosomes is lost early in development. If, just by chance, the

two remaining chromosomes are both from the same parent, uniparental disomy results.

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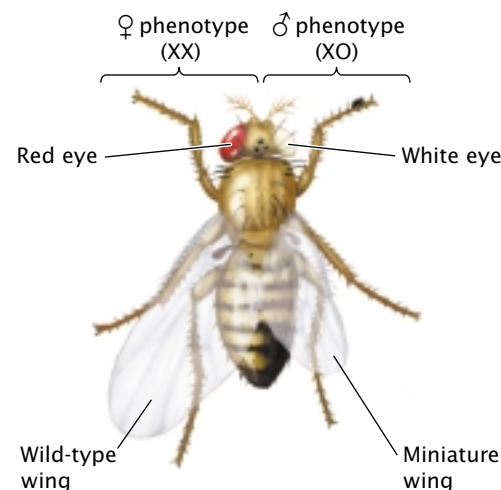
More on uniparental disomy or links to information on Prader-Willi syndrome

Mosaicism

Nondisjunction in a mitotic division may generate patches of cells in which every cell has a chromosome abnormality and other patches in which every cell has a normal karyotype. This type of nondisjunction leads to regions of tissue with different chromosome constitutions, a condition known as **mosaicism**. Growing evidence suggests that mosaicism is relatively common.

Only about 50% of those diagnosed with Turner syndrome have the 45,X karyotype (presence of a single X chromosome) in all their cells; most others are mosaics, possessing some 45,X cells and some normal 46,XX cells. A few may even be mosaics for two or more types of abnormal karyotypes. The 45,X/46,XX mosaic usually arises when an X chromosome is lost soon after fertilization in an XX embryo.

Fruit flies that are XX/XO mosaics (O designates the absence of a homologous chromosome; XO means the cell has a single X chromosome and no Y chromosome) develop a mixture of male and female traits, because the presence of two X chromosomes in fruit flies produces female traits and the presence of a single X chromosome produces male traits (FIGURE 9.26). Sex determination in fruit flies occurs



9.26 Mosaicism for the sex chromosomes produces a gynandromorph.

This XX/XO gynandromorph fruit fly carries one wild-type X chromosome and one X chromosome with recessive alleles for white eyes and miniature wings. The left side of the fly has a normal female phenotype, because the cells are XX and the recessive alleles on one X chromosome are masked by the presence of wild-type alleles on the other. The right side of the fly has a male phenotype with white eyes and miniature wing, because the cells are missing the wild-type X chromosome (are XO), allowing the white and miniature alleles to be expressed.

independently in each cell during development. Those cells that are XX express female traits; those that are XY express male traits. Such sexual mosaics are called **gynandromorphs**. Normally, X-linked recessive genes are masked in heterozygous females but, in XX/XO mosaics, any X-linked recessive genes present in the cells with a single X chromosome will be expressed.

Concepts

In uniparental disomy, an individual has two copies of a chromosome from one parent and no copy from the other. It may arise when a trisomic embryo loses one of the triplicate chromosomes early in development. In mosaicism, different cells within the same individual have different chromosome constitutions.

Polyploidy

Most eukaryotic organisms are diploid ($2n$) for most of their life cycles, possessing two sets of chromosomes. Occasionally, whole sets of chromosomes fail to separate in meiosis or mitosis, leading to polyploidy, the presence of more than two genomic sets of chromosomes. Polyploids include *triploids* ($3n$) *tetraploids* ($4n$), *pentaploids* ($5n$), and even higher numbers of chromosome sets.

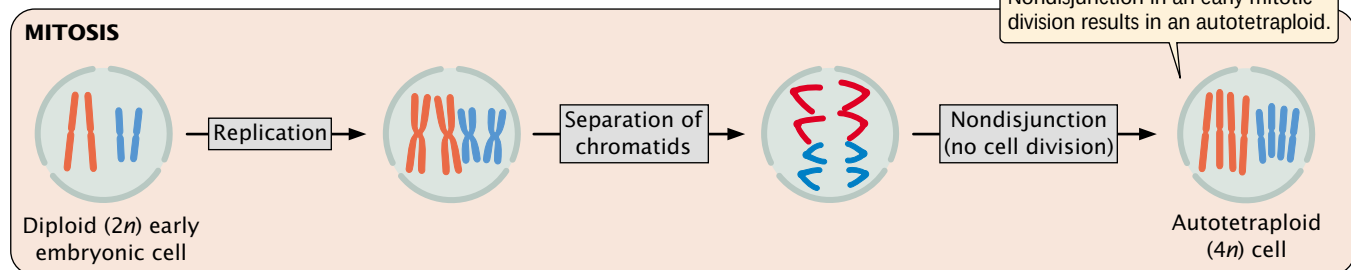
Polyploidy is common in plants and is a major mechanism by which new plant species have evolved. Approximately 40% of all flowering-plant species and from 70% to 80% of grasses are polyploids. They include a number of agriculturally important plants, such as wheat, oats, cotton, potatoes, and sugar cane. Polyploidy is less common in animals, but is found in some invertebrates, fishes, salamanders, frogs, and lizards. No naturally occurring, viable polyploids are known in birds, but at least one polyploid mammal—a rat from Argentina—has been reported.

We will consider two major types of polyploidy: **autopolyploidy**, in which all chromosome sets are from a single species; and **allopolyploidy**, in which chromosome sets are from two or more species.

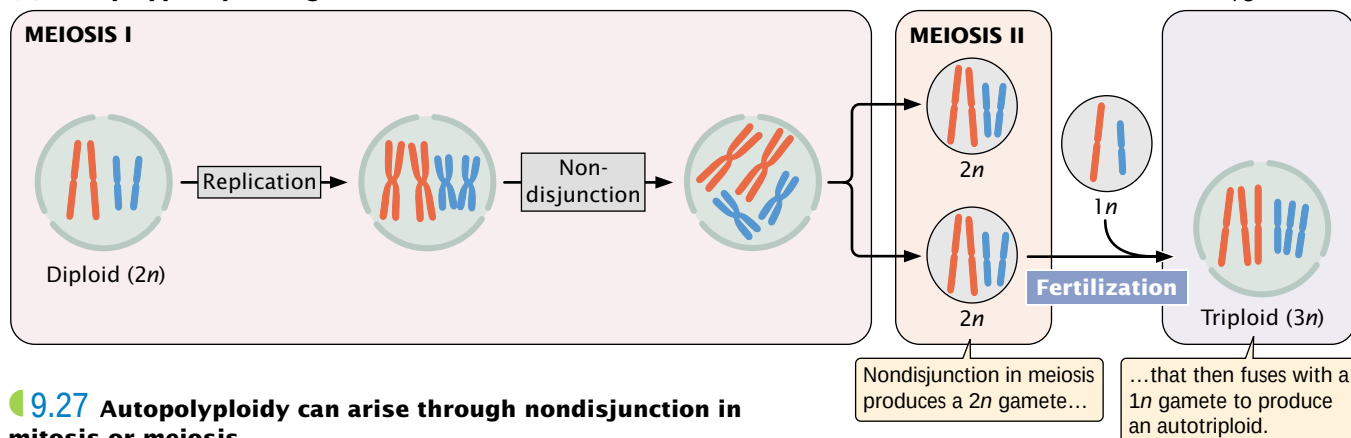
Autopolyploidy

Autopolyploidy results when accidents of meiosis or mitosis produce extra sets of chromosomes, all derived from a single species. Nondisjunction of all chromosomes in mitosis in an early $2n$ embryo, for example, doubles the chromosome number and produces an autotetraploid ($4n$) (FIGURE 9.27a). An autotriploid may arise when nondisjunction in meiosis produces a diploid gamete that then fuses with a normal haploid gamete to produce a triploid zygote (FIGURE 9.27b). Alternatively, triploids may arise from a cross between an autotetraploid that produces $2n$ gametes and a diploid that produces $1n$ gametes.

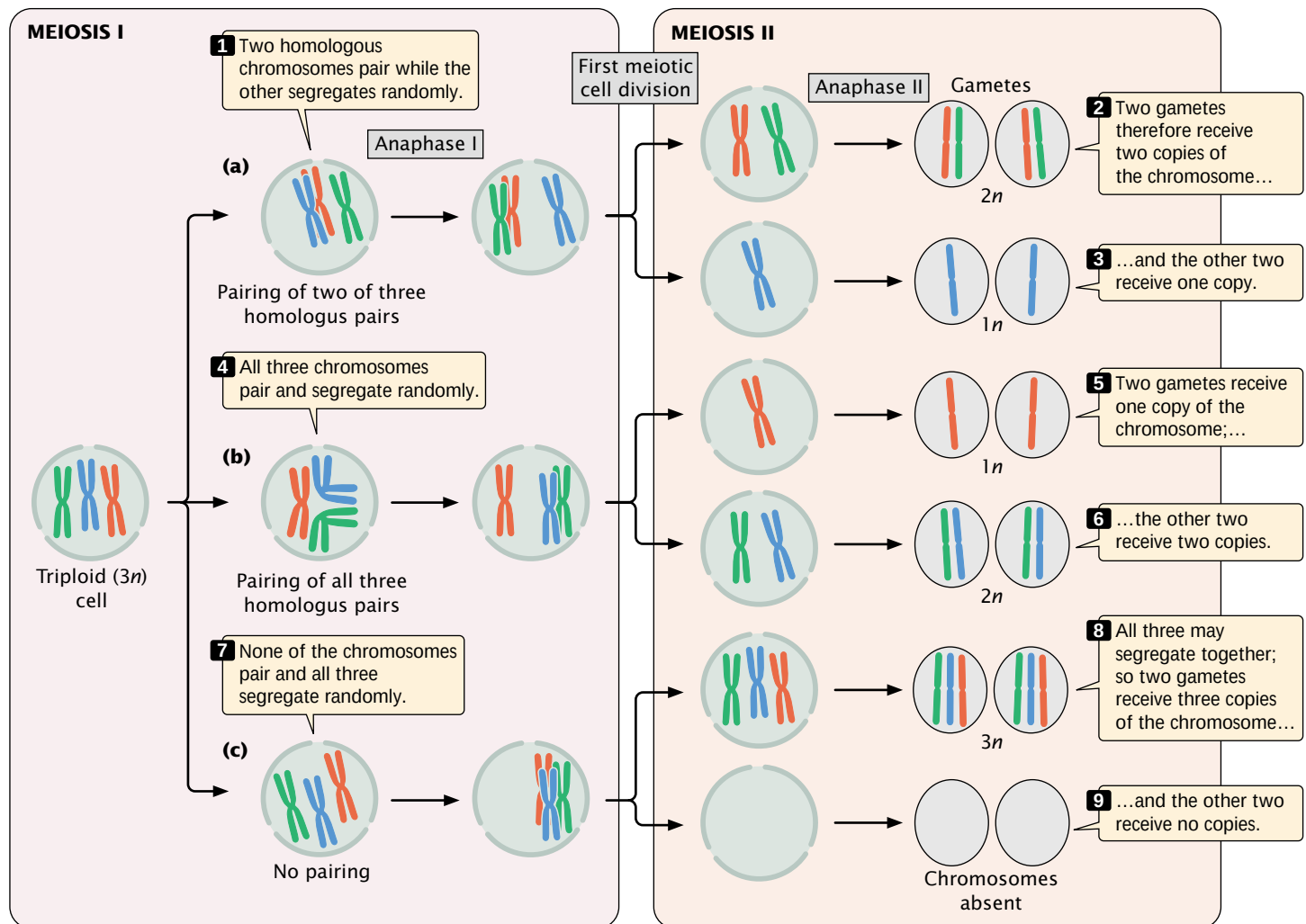
(a) Autopolyploidy through mitosis



(b) Autopolyploidy through meiosis



9.27 Autopolyploidy can arise through nondisjunction in mitosis or meiosis.



9.28 In meiosis of an autotriploid, homologous chromosomes can pair or not pair in three ways.

Because all the chromosome sets in autopolyploids are from the same species, they are homologous and attempt to align in prophase I of meiosis, which usually results in sterility. Consider meiosis in an autotriploid (FIGURE 9.28). In meiosis in a diploid cell, two chromosome homologs pair and align, but, in autotriploids, three homologs are present. One of the three homologs may fail to align with the other two, and this unaligned chromosome will segregate randomly (see Figure 9.28a). Which gamete gets the extra chromosome will be determined by chance and will differ for each homologous group of chromosomes. The resulting gametes will have two copies of some chromosomes and one copy of others. Even if all three chromosomes do align, two chromosomes must segregate to one gamete and one chromosome to the other (see Figure 9.28b). Occasionally, the presence of a third chromosome interferes with normal alignment, and all three chromosomes segregate to the same gamete (see Figure 9.28c).

No matter how the three homologous chromosomes align, their random segregation will create **unbalanced gametes**, with various numbers of chromosomes. A gamete produced by meiosis in such an autotriploid might receive, say, two copies of chromosome 1, one copy of chromosome 2, three copies of chromosome 3, and no copies of chromosome 4. When the unbalanced gamete fuses with a normal gamete (or with another unbalanced gamete), the resulting zygote has different numbers of the four types of chromosomes. This difference in number creates unbalanced gene dosage in the zygote, which is often lethal. For this reason, triploids do not usually produce viable offspring.

In even-numbered autopolyploids, such as autotetraploids, it is theoretically possible for the homologous chromosomes to form pairs and divide equally. However, this event rarely happens; so these types of autotetraploids also produce unbalanced gametes.

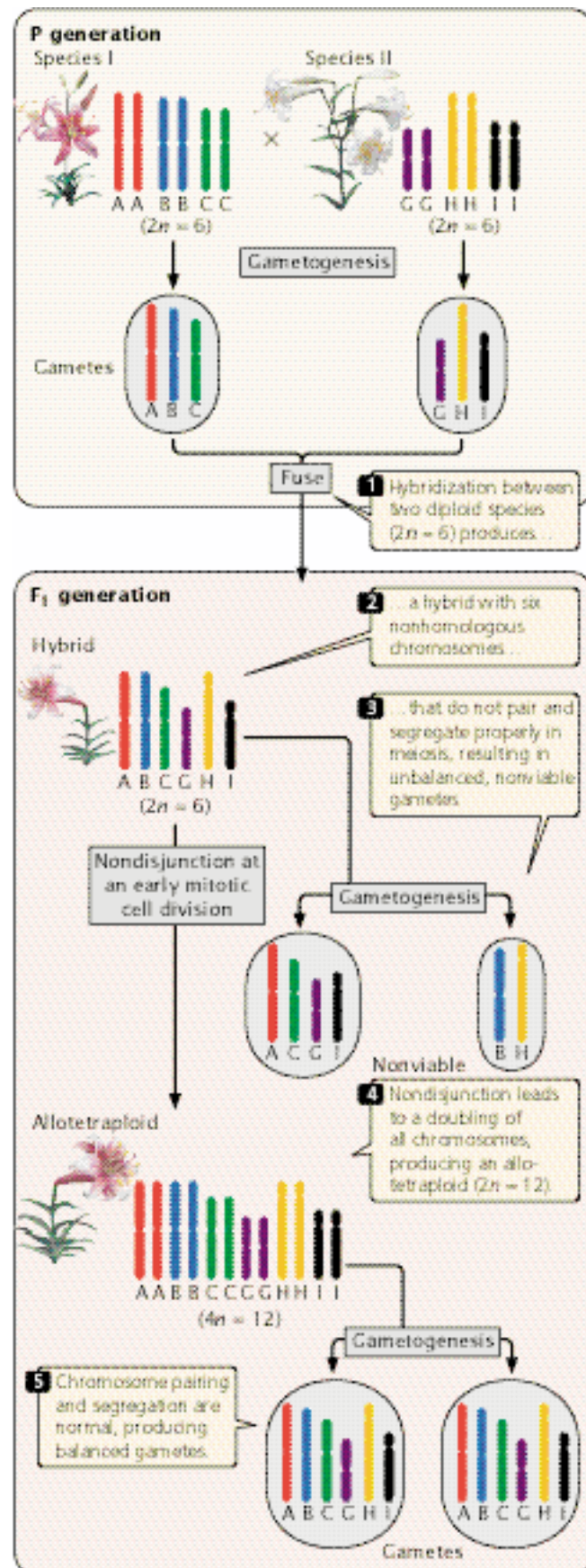
The sterility that usually accompanies autopolyploidy has been exploited in agriculture. Wild diploid bananas ($2n = 22$), for example, produce seeds that are hard and inedible, but triploid bananas ($3n = 33$) are sterile, and produce no seeds—they are the bananas sold commercially. Similarly, seedless triploid watermelons have been created and are now widely sold.

Allopolyploidy

Allopolyploidy arises from hybridization between two species; the resulting polyploid carries chromosome sets derived from two or more species. **FIGURE 9.29** shows how allopolyploidy can arise from two species that are sufficiently related that hybridization occurs between them. Species I (AABBCC, $2n = 6$) produces haploid gametes with chromosomes ABC, and species II (GGHHII, $2n = 6$) produces gametes with chromosomes GHI. If gametes from species I and II fuse, a hybrid with six chromosomes (ABCGHI) is created. The hybrid has the same chromosome number as that of both diploid species; so the hybrid is considered diploid. However, because the hybrid chromosomes are not homologous, they will not pair and segregate properly in meiosis; so this hybrid is functionally haploid and sterile.

The sterile hybrid is unable to produce viable gametes through meiosis, but it may be able to perpetuate itself through mitosis (asexual reproduction). On rare occasions, nondisjunction takes place in a mitotic division, which leads to a doubling of chromosome number and an allotetraploid, with chromosomes AABBCCGGHHII. This tetraploid is *functionally* diploid: every chromosome has one and only one homologous partner, which is exactly what meiosis requires for proper segregation. The allopolyploid can now undergo normal meiosis to produce balanced gametes having six chromosomes.

George Karpechenko created polyploids experimentally in the 1920s. Today, as well as in the early twentieth century, cabbage (*Brassica oleracea*, $2n = 18$) and radishes (*Raphanus sativa*, $2n = 18$) are agriculturally important plants, but only the leaves of the cabbage and the roots of the radish are normally consumed. Karpechenko wanted to produce a plant that had cabbage leaves and radish roots so that no part of the plant would go to waste. Because both cabbage and radish possess 18 chromosomes, Karpechenko was able to successfully cross them, producing a hybrid with $2n = 18$, but, unfortunately, the hybrid was sterile. After several crosses, Karpechenko noticed that one of his hybrid plants produced a few seeds. When planted, these seeds grew into plants that were viable and fertile. Analysis of their chromosomes revealed that the plants were allotetraploids, with $2n = 36$ chromo-



9.29 Allopolyploids usually arise from hybridization between two species followed by chromosome doubling.

somes. To Karpechenko's great disappointment, however, the new plants possessed the roots of a cabbage and the leaves of a radish.

The Significance of Polyploidy

In many organisms, cell volume is correlated with nuclear volume, which, in turn, is determined by genome size. Thus, the increase in chromosome number in polyploidy is often associated with an increase in cell size, and many polyploids are physically larger than diploids. Breeders have used this effect to produce plants with larger leaves, flowers, fruits, and seeds. The hexaploid ($6n = 42$) genome of wheat probably contains chromosomes derived from three different wild species (FIGURE 9.30). Many other cultivated plants also are polyploid (Table 9.2). (See next page).

Polyploidy is less common in animals than in plants for several reasons. As discussed, allopolyploids require hybridization between different species, which occurs less frequently in animals than in plants. Animal behavior often prevents interbreeding, and the complexity of animal development causes most interspecific hybrids to be non-viable. Many of the polyploid animals that do arise are in groups that reproduce through parthenogenesis (a type of reproduction in which individuals develop from unfertilized eggs). Thus asexual reproduction may facilitate the development of polyploids, perhaps because the perpetuation of hybrids through asexual reproduction provides greater opportunities for nondisjunction. Only a few human polyploid babies have been reported, and most died within a few days of birth. Polyploidy—usually triploidy—is seen in about 10% of all spontaneously aborted human fetuses. Different types of chromosome mutations are summarized in Table 9.3. (See next page).

Concepts

Polyploidy is the presence of extra chromosome sets: autopolyploids possess extra chromosome sets from the same species; allopolyploids possess extra chromosome sets from two or more species. Problems in chromosome pairing and segregation often lead to sterility in autopolyploids, but many allopolyploids are fertile.

9.30 Modern bread wheat, *Triticum aestivum*, is a hexaploid with genes derived from three different species. Two diploids species *T. monococcum* ($n = 14$) and probably *T. searsii* ($n = 14$) originally crossed to produce a diploid hybrid ($2n = 14$) that underwent chromosome doubling to create *T. turgidum* ($4n = 28$). A cross between *T. turgidum* and *T. tauschii* ($2n = 14$) produced a triploid hybrid ($3n = 21$) that then underwent chromosome doubling to produce *T. aestivum*, which is a hexaploid ($6n = 42$).

Note: Fig 9.30 to fill this column
See art page for exact additions.

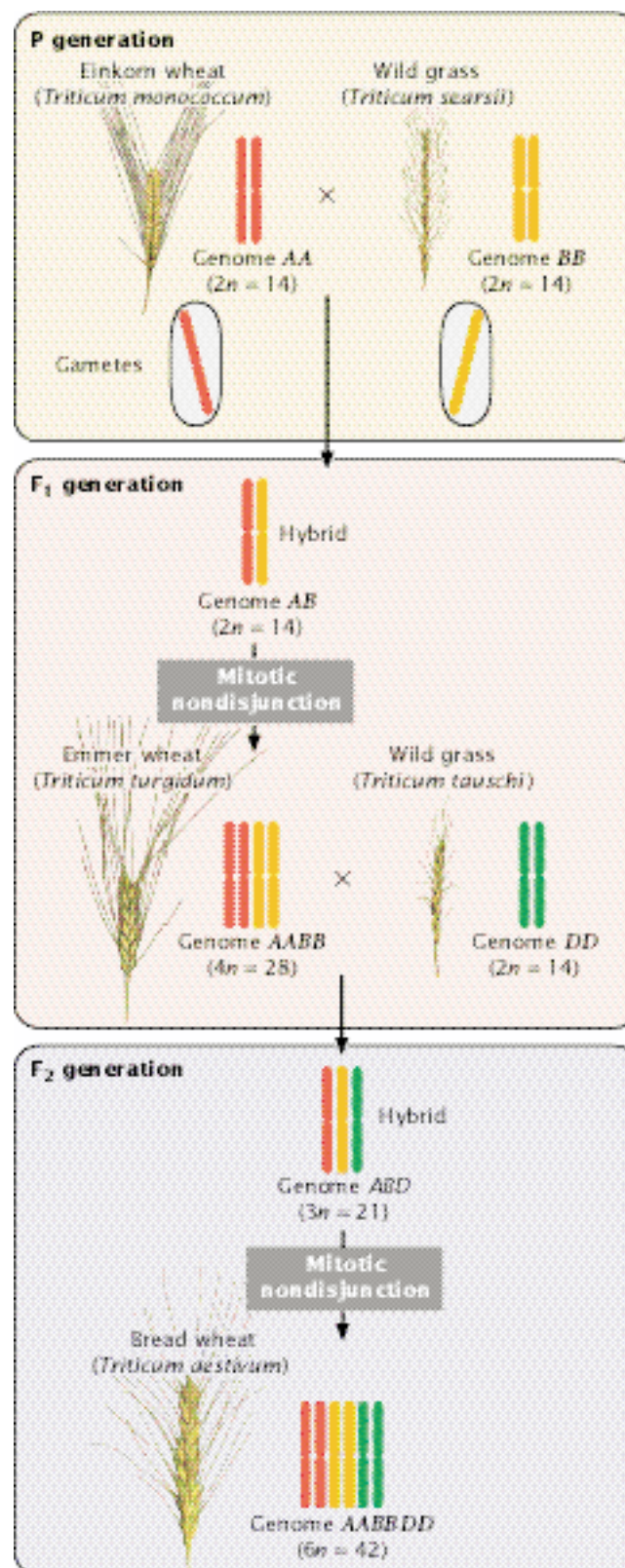


Table 9.2 Examples of polyploid crop plants

Plant	Type of Polyploidy	Ploidy	Chromosome Number
Potato	Autopolyploid	4 <i>n</i>	48
Banana	Autopolyploid	3 <i>n</i>	33
Peanut	Autopolyploid	4 <i>n</i>	40
Sweet potato	Autopolyploid	6 <i>n</i>	90
Tobacco	Allopolyploid	4 <i>n</i>	48
Cotton	Allopolyploid	4 <i>n</i>	52
Wheat	Allopolyploid	6 <i>n</i>	42
Oats	Allopolyploid	6 <i>n</i>	42
Sugar cane	Allopolyploid	8 <i>n</i>	80
Strawberry	Allopolyploid	8 <i>n</i>	56

Source: After F. C. Elliot, *Plant Breeding and Cytogenetics* (New York: McGraw-Hill, 1958), p. ***.

Table 9.3 Different types of chromosome mutations

Chromosome Mutation	Definition
Chromosome rearrangement	Change in chromosome structure
Chromosome duplication	Duplication of a chromosome segment
Chromosome	Deletion of a chromosome segment
Inversion	Chromosome segment inverted 180 degrees
Paracentric inversion	Inversion that does not include the centromere in the inverted region
Pericentric inversion	Inversion that includes the centromere in the inverted region
Translocation	Movement of a chromosome segment to a nonhomologous chromosome or region of the same chromosome
Nonreciprocal translocation	Movement of a chromosome segment to a nonhomologous chromosome or region of the same chromosome without reciprocal exchange
Reciprocal translocation	Exchange between segments of nonhomologous chromosomes or regions of the same chromosome
Aneuploidy	Change in number of individual chromosomes
Nullisomy	Loss of both members of a homologous pair
Monosomy	Loss of one member of a homologous pair
Trisomy	Gain of one chromosome, resulting in three homologous chromosomes
Tetrasomy	Gain of two homologous chromosomes, resulting in four homologous chromosomes
Polyploidy	Addition of entire chromosome sets
Autopolyploidy	Polyploidy in which extra chromosome sets are derived from the same species
Allopolyploidy	Polyploidy in which extra chromosome sets are derived from two or more species

Chromosome Mutations and Cancer

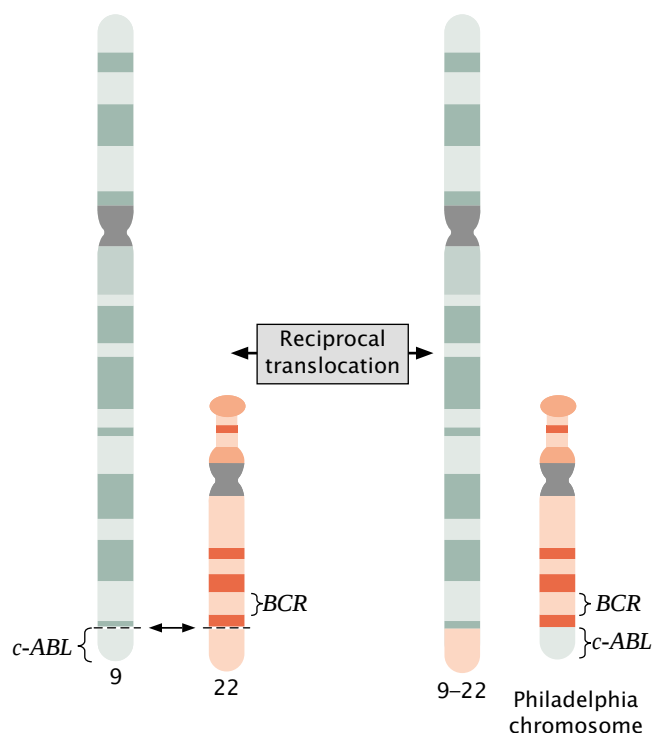
Most tumors contain cells with chromosome mutations. For many years, geneticists argued about whether these chromosome mutations were the cause or the result of can-

cer. Some types of tumors are consistently associated with *specific* chromosome mutations, suggesting that in these cases the specific chromosome mutation played a pivotal role in the development of the cancer. However, many cancers are not associated with specific types of chromosome

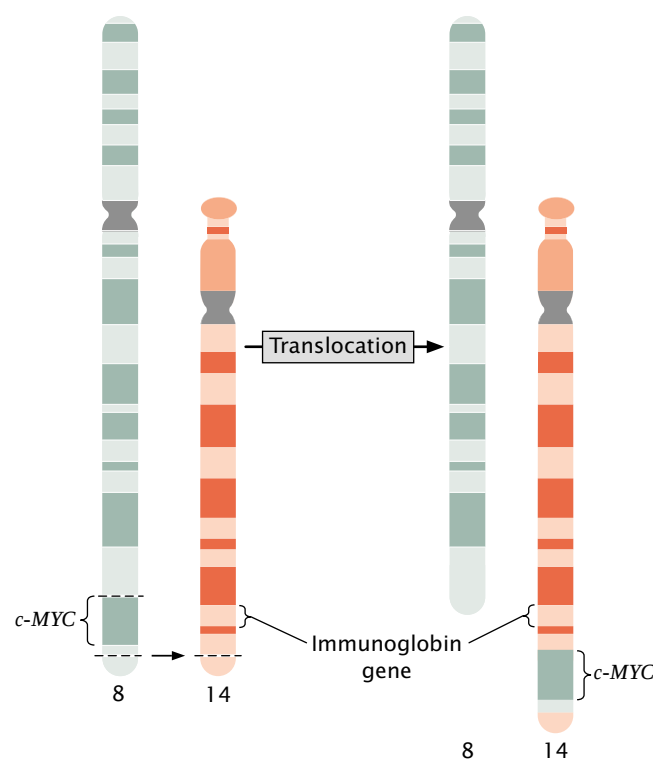
abnormalities, and individual *gene* mutations are now known to contribute to many types of cancer. Nevertheless, chromosome instability is a general feature of cancer cells, causing them to accumulate chromosome mutations, which then affect individual genes that contribute to the cancer process. Thus, chromosome mutations appear to both *cause* and *be a result of* cancer.

At least three types of chromosome rearrangements—deletions, inversions, and translocations—are associated with certain types of cancer. Deletions may result in the loss of one or more genes that normally hold cell division in check. When these so-called tumor-suppressor genes are lost, cell division is not regulated and cancer may result.

Inversions and translocations contribute to cancer in several ways. First, the chromosomal breakpoints that accompany these mutations may lie within tumor-suppressor genes, disrupting their function and leading to cell proliferation. Second, translocations and inversions may bring together sequences from two different genes, generating a fused protein that stimulates some aspect of the cancer process. Such fusions are seen in most cases of chronic myeloid leukemia, a fatal form of leukemia affecting bone-marrow cells. About 90% of patients with chronic myeloid leukemia have a reciprocal translocation between the long arm of chromosome 22 and the tip of the long arm of chromosome 9 (FIGURE 9.31). This translocation produces a



9.31 A reciprocal translocation between chromosomes 9 and 22 causes chronic myeloid leukemia.



9.32 A reciprocal translocation between chromosomes 8 and 14 causes Burkitt lymphoma.

shortened chromosome 22, called the Philadelphia chromosome because it was first discovered in Philadelphia. At the end of a normal chromosome 9 is a potential cancer-causing gene called *c-ABL*. As a result of the translocation, part of the *c-ABL* gene is fused with the *BCR* gene from chromosome 22. The protein produced by this *BCR*–*c-ABL* fusion gene is much more active than the protein produced by the normal *c-ABL* gene; the fusion protein stimulates increased, unregulated cell division and eventually leads to leukemia.

A third mechanism by which chromosome rearrangements may produce cancer is by the transfer of a potential cancer-causing gene to a new location, where it is activated by different regulatory sequences. Burkitt lymphoma is a cancer of the B cells, the lymphocytes that produce antibodies. Many people having Burkitt lymphoma possess a reciprocal translocation between chromosome 8 and chromosome 2, 14, or 22, each of which carries genes for immunological proteins (FIGURE 9.32). This translocation relocates a gene called *c-MYC* from the tip of chromosome 8 to a position in one of the aforementioned chromosomes that is next to a gene for one of the immunoglobulin proteins. At this new location, *c-MYC* comes under the control of regulatory sequences that normally activate the production of immunoglobulins, and *c-MYC* is expressed in B cells. The *c-MYC* protein stimulates the division of the B cells and leads to Burkitt lymphoma.

Concepts

Most tumors contain a variety of types of chromosome mutations. Some tumors are associated with specific deletions, inversions, and translocations. Deletions can eliminate or inactivate genes that control the cell cycle; inversions and translocations can cause breaks in genes that suppress tumors, fuse genes to produce cancer-causing proteins, or move genes to new locations, where they are under the influence of different regulatory sequences.

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myeloid leukemia

More information on chronic

Connecting Concepts Across Chapters

This chapter has focused on variations in the number and structure of chromosomes. Because these chromosome mutations affect many genes simultaneously, they have major effects on the phenotypes and often are not compatible with development. A major theme of this chapter has been that, even when the structure of a gene is not dis-

rupted, changes in gene number and position produced by chromosome mutations can have severe effects on gene expression.

Chromosome mutations most frequently arise through errors in mitosis and meiosis, and so a thorough understanding of these processes and chromosome structure (covered in Chapter 2) is essential for grasping the material in this chapter. The process of crossing over, discussed in Chapter 7, also is helpful for understanding the consequences of recombination in individuals heterozygous for chromosome rearrangements.

This chapter has provided a foundation for understanding the molecular nature of chromosome structure (discussed in Chapter 11). The movement of genes through a process called transposition often produces chromosome mutations, and so the current chapter is also relevant to the discussion of transposition in Chapter 11. The discussion in this chapter of chromosomes and cancer is closely linked to the more extended discussion of cancer genetics found in Chapter 21. Variation produced by chromosome mutations, along with gene mutations and recombination, provides the raw material for evolutionary change, which is covered in Chapters 22 and 23.

CONCEPTS SUMMARY

- Three basic types of chromosome mutations are: (1) chromosome rearrangements, which are changes in the structure of chromosomes; (2) aneuploidy, which is an increase or decrease in chromosome number; and (3) polyploidy, which is the presence of extra chromosome sets.
- Chromosome rearrangements include duplications, deletions, inversions, and translocations.
- Chromosome duplications arise when a chromosome segment is doubled. The segment may be adjacent to the original segment (a tandem duplication) or distant from the original segment (a displaced duplication). Reverse duplications have the duplicated sequence in the reverse order. In individuals heterozygous for a duplication, the duplicated region will form a loop when homologous chromosomes pair in meiosis. Duplications often have pronounced effects on the phenotype owing to unbalanced gene dosage.
- Chromosome deletion is the loss of part of a chromosome. In individuals heterozygous for a deletion, one of the chromosomes will loop out during pairing in meiosis. Many chromosome deletions are lethal in the homozygous state and cause deleterious effects in the heterozygous state, because of unbalanced gene dosage. Deletions may cause recessive alleles to be expressed.
- A chromosome inversion is the inversion of a chromosome segment. Pericentric inversions include the centromere; paracentric inversions do not. The phenotypic effects caused by inversions are due to the breaking of genes and their movement to new locations, where they may be influenced by different regulatory sequences. In individuals heterozygous for an inversion, the chromosomes form inversion loops in meiosis, with reduced recombination taking place within the inverted region.
- A translocation is the attachment of part of one chromosome to a nonhomologous chromosome. In translocation heterozygotes, the chromosomes form crosslike structures in meiosis, and the segregation of chromosomes produces unbalanced gametes.
- Fragile sites are constrictions or gaps that appear at particular regions on the chromosomes of cells grown in culture and are prone to breakage under certain conditions.
- Aneuploidy is the addition or loss of individual chromosomes. Nullisomy refers to the loss of two homologous chromosomes; monosomy is the loss of one homologous chromosome; trisomy is the addition of one homologous chromosome; tetrasomy is the addition of two homologous chromosomes.

- Aneuploidy usually causes drastic phenotypic effects because it leads to unbalanced gene dosage. In humans, sex-chromosome aneuploids are less detrimental than autosomal aneuploids because X-chromosome inactivation reduces the problems of unbalanced gene dosage.
- The most common autosomal aneuploid in living humans is trisomy 21, which results in Down syndrome. Primary Down syndrome is caused by the presence of three full copies of chromosome 21, whereas familial Down syndrome is caused by the presence of two normal copies of chromosome 21 and a third copy that is attached to another chromosome through a translocation.
- Uniparental disomy is the presence of two copies of a chromosome from one parent and no copy from the other.
- Mosaicism is caused by nondisjunction in an early mitotic division that leads to different chromosome constitutions in different cells of a single individual.
- Polyploidy is the presence of more than two full chromosome sets. In autopolyploidy, all the chromosomes derive from one species; in allopolyploidy, they come from two or more species.
- Autopolyploidy arises from nondisjunction in meiosis or mitosis. Here, problems with chromosome alignment and segregation frequently lead to the production of nonviable gametes.
- Allopolyploidy arises from nondisjunction that follows hybridization between two species. Allopolyploids are frequently fertile.
- Some types of cancer are associated with specific chromosome deletions, inversions, and translocations. Deletions may cause cancer by removing or disrupting genes that suppress tumors; inversions and translocations may break tumor-suppressing genes or they may move genes to positions next to different regulatory sequences, which alters their expression.

IMPORTANT TERMS

chromosome mutation (p. 000)	displaced duplication (p. 000)	reciprocal translocation (p. 000)	familial Down syndrome (p. 000)
metacentric chromosome (p. 000)	reverse duplication (p. 000)	robertsonian translocation (p. 000)	translocation carrier (p. 000)
submetacentric chromosome (p. 000)	chromosome deletion (p. 000)	alternate segregation (p. 000)	Edward syndrome (trisomy 18) (p. 000)
acrocentric chromosome (p. 000)	pseudodominance (p. 000)	adjacent-1 segregation (p. 000)	Patau syndrome (trisomy 13) (p. 000)
telocentric chromosome (p. 000)	haploinsufficient gene (p. 000)	adjacent-2 segregation (p. 000)	trisomy 8 (p. 000)
chromosome rearrangement (p. 000)	chromosome inversion (p. 000)	fragil site (p. 000)	uniparental disomy (p. 000)
aneuploidy (p. 000)	paracentric inversion (p. 000)	nullisomy (p. 000)	mosaicism (p. 000)
polyploidy (p. 000)	pericentric inversion (p. 000)	monosomy (p. 000)	gynandromorph (p. 000)
chromosome duplication (p. 000)	position effect (p. 000)	trisomy (p. 000)	autopolyploidy (p. 000)
tandem duplication (p. 000)	dicentric chromatid (p. 000)	tetrasomy (p. 000)	allopolyploidy (p. 000)
	acentric chromatid (p. 000)	down syndrome (trisomy 21) (p. 000)	unbalanced gametes (p. 000)
	dicentric bridge (p. 000)	primary Down syndrome (p. 000)	
	translocation (p. 000)		
	nonreciprocal translocation (p. 000)		

Worked Problems

1. A chromosome has the following segments, where • represents the centromere.

A B C D E • F G

What types of chromosome mutations are required to change this chromosome into each of the following chromosomes? (In some cases, more than one chromosome mutation may be required.)

(a) A B E • F G

(b) A E D C B • F G

(c) A B A B C D E • F G

(d) A F • E D C B G

(e) A B C D E E D C • F G

• Solution

The types of chromosome mutations are identified by comparing the mutated chromosome with the original, wild-type chromosome.

(a) The mutated chromosome (A B E • F G) is missing segment C D; so this mutation is a deletion.

(b) The mutated chromosome (A E D C B • F G) has one and only one copy of all the gene segments, but segment

B C D E has been inverted 180 degrees. Because the centromere has not changed location and is not in the inverted region, this chromosome mutation is a paracentric inversion.

(c) The mutated chromosome (A B A B C D E • F G) is longer than normal, and we see that segment A B has been duplicated. This mutation is a tandem duplication.

(d) The mutated chromosome (A F • E D C B G) is normal length, but the gene order and the location of the centromere have changed; this mutation is therefore a pericentric inversion of region (B C D E • F).

(e) The mutated chromosome (A B C D E E D C • F G) contains a duplication (C D E) that is also inverted; so this chromosome has undergone a duplication and a paracentric inversion.

2. Species I is diploid ($2n = 4$) with chromosomes AABB; related species II is diploid ($2n = 6$) with chromosomes MMNNOO. Give the chromosomes that would be found in individuals with the following chromosome mutations.

- Autotriploid for species I
- Allotetraploid including species I and II
- Monosomic in species I
- Trisomic in species II for chromosome M
- Tetrasomic in species I for chromosome A
- Allotriploid including species I and II
- Nullisomic in species II for chromosome N

• Solution

To work this problem, we should first determine the haploid genome complement for each species. For species I, $n = 2$ with

chromosomes AB and, for species II, $n = 3$ with chromosomes MNO.

(a) An autotriploid is $3n$, with all the chromosomes coming from a single species; so an autotriploid of species I will have chromosomes AAABBB ($3n = 6$).

(b) An allotetraploid is $4n$, with the chromosomes coming from more than one species. An allotetraploid could consist of $2n$ from species I and $2n$ from species II, giving the allotetraploid ($4n = 2 + 2 + 3 + 3 = 10$) chromosomes AABBMNNOO. An allotetraploid could also possess $3n$ from species I and $1n$ from species II ($4n = 2 + 2 + 2 + 3 = 9$; AAABBBMNO) or $1n$ from species I and $3n$ from species II ($4n = 2 + 3 + 3 + 3$; ABMMNNNOO).

(c) A monosomic is missing a single chromosome; so a monosomic for species I would be $2n - 1 = 4 - 1 = 3$. The monosomy might include either of the two chromosome pairs, with chromosomes ABB or AAB.

(d) Trisomy requires an extra chromosome; so a trisomic of species II for chromosome M would be $2n + 1 = 6 + 1 = 7$ (MMMNNNOO).

(e) A tetrasomic has two extra homologous chromosomes; so a tetrasomic of species I for chromosome A would be $2n + 2 = 4 + 2 = 6$ (AAAABB).

(f) An allotriploid is $3n$ with the chromosomes coming from two different species; so an allotriploid could be $3n = 2 + 2 + 3 = 7$ (AABBMNO) or $3n = 2 + 3 + 3 = 8$ (ABMMNNNOO).

(g) A nullisomic is missing both chromosomes of a homologous pair; so a nullisomic of species II for chromosome N would be $2n - 2 = 6 - 2 = 4$ (MMOO).

The New Genetics

MINING GENOMES

THE HUMAN GENOME PROJECT

The first successful efforts to clone single genes and to determine the sequence of DNA molecules began in the 1970s. Less than two decades later, techniques and strategies were being developed to organize and sequence clones that covered the entire human

genome. Today, the sequence of our genome is freshly available. This exercise will introduce you to some of the tools that are used to organize, retrieve, and understand information derived from the Human Genome Project.

COMPREHENSION QUESTIONS

- List the different types of chromosome mutations and define each one.
- Why do extra copies of genes sometimes cause drastic phenotypic effects?
- Draw a pair of chromosomes as they would appear during synapsis in prophase I of meiosis in an individual heterozygous for a chromosome duplication.
- How does a deletion cause pseudodominance?
- What is the difference between a paracentric and a pericentric inversion?
- How do inversions cause phenotypic effects?
- Draw a pair of chromosomes as they would appear during synapsis in prophase I of meiosis in an individual heterozygous for a paracentric inversion.

8. Explain why recombination is suppressed in individuals heterozygous for paracentric and pericentric inversions.
- * 9. How do translocations produce phenotypic effects?
10. Sketch the chromosome pairing and the different segregation patterns that can arise in an individual heterozygous for a reciprocal translocation.
11. What is a Robertsonian translocation?
12. List four major types of aneuploidy.
- *13. Why are sex-chromosome aneuploids more common in humans than autosomal aneuploids?
- *14. What is the difference between primary Down syndrome and familial Down syndrome? How does each arise?
- *15. What is uniparental disomy and how does it arise?
16. What is mosaicism and how does it arise?
- *17. What is the difference between autopolyploidy and allopolyploidy? How does each arise?

APPLICATION QUESTIONS AND PROBLEMS

- *18. Which types of chromosome mutations
 - (a) increase the amount of genetic material on a particular chromosome?
 - (b) increase the amount of genetic material for all chromosomes?
 - (c) decrease the amount of genetic material on a particular chromosome?
 - (d) change the position of DNA sequences on a single chromosome without changing the amount of genetic material?
 - (e) move DNA from one chromosome to a nonhomologous chromosome?
- *19. A chromosome has the following segments, where • represents the centromere:

A B • C D E F G

What types of chromosome mutations are required to change this chromosome into each of the following chromosomes? (In some cases, more than one chromosome mutation may be required.)

- (a) A B A B • C D E F G
 - (b) A B • C D E A B F G
 - (c) A B • C F E D G
 - (d) A • C D E F G
 - (e) A B • C D E
 - (f) A B • E D C F G
 - (g) C • B A D E F G
 - (h) A B • C F E D F E D G
 - (i) A B • C D E F C D F E G
- *20. A chromosome initially has the following segments:

A B • C D E F G

Draw and label the chromosome that would result from each of the following mutations.

- (a) Tandem duplication of DEF

- (b) Displaced duplication of DEF
- (c) Deletion of FG
- (e) Paracentric inversion that includes DEFG
- (f) Pericentric inversion of BCDE

21. The following diagram represents two nonhomologous chromosomes:

A B • C D E F G
R S • T U V W X

What type of chromosome mutation would produce the following chromosomes?

- (a) A B • C D
R S • T U V W X E F G
- (b) A U V B • C D E F G
R S • T W X
- (c) A B • T U V F G
R S • C D E W X
- (d) A B • C W G
R S • T U V D E F X

- *22. A species has $2n = 16$ chromosomes. How many chromosomes will be found per cell in each of the following mutants in this species?
 - (a) Monosomic
 - (b) Autotriploid
 - (c) Autotetraploid
 - (d) Trisomic
 - (e) Double monosomic
 - (f) Nullisomic
 - (g) Autopentaploid
 - (h) Tetrasomic
- *23. The *Notch* mutation is a deletion on the X chromosome of *Drosophila melanogaster*. Female flies heterozygous for *Notch* have an indentation on the margin of their wings; *Notch* is

lethal in the homozygous and hemizygous conditions. The *Notch* deletion covers the region of the X chromosome that contains the locus for white eyes, an X-linked recessive trait. Give the phenotypes and proportions of progeny produced in the following crosses.

- (a) A red-eyed, Notch female is mated with white-eyed male.
 (b) A white-eyed, Notch female is mated with a red-eyed male.
 (c) A white-eyed, Notch female is mated with a white-eyed male.
24. The green nose fly normally has six chromosomes, two metacentric and four acrocentric. A geneticist examines the chromosomes of an odd-looking green nose fly and discovers that it has only five chromosomes; three of them are metacentric and two are acrocentric. Explain how this change in chromosome number might have occurred.
25. Species I is diploid ($2n = 8$) with chromosomes AABBCDD; related species II is diploid ($2n = 8$) with chromosomes MMNNOOPP. Individuals with the following sets of chromosomes represent what types of chromosome mutations?
- (a) AAABBCDD
 (b) MMNNOOOOP
 (c) AABBCDD
 (d) AAABBBCCDDDD
 (e) AAABBCDDDD
 (f) AABBD
 (g) AABBCDDMMNNOOPP
 (h) AABBCDDMNOP
- *26. A wild-type chromosome has the following segments:
- A B C • D E F G H I
- An individual is heterozygous for the following chromosome mutations. For each mutation, sketch how the wild-type and mutated chromosomes would pair in prophase I of meiosis, showing all chromosome strands.
- (a) A B C • D E F D E F G H I
 (b) A B C • D H I
 (c) A B C • D G F E H I
 (d) A B E D • C F G H I
27. An individual that is heterozygous for a pericentric inversion has the following two chromosomes:
- A B C D • E F G H I
A B C F E • D G H I
- (a) Sketch the pairing of these two chromosomes in prophase I of meiosis, showing all four strands.
 (b) Draw the chromatids that would result from a single crossover between the E and F segments.
 (c) What will happen when the chromosomes separate in anaphase I of meiosis?

28. Answer part *b* of problem 28 for a two-strand double crossover between E and F.
- *29. An individual heterozygous for a reciprocal translocation possesses the following chromosomes:

A B • C D E F G
A B • C D V W X
R S • T U E F G
R S • T U V W X

- (a) Draw the pairing arrangement of these chromosomes in prophase I of meiosis.
 (b) Diagram the alternate, adjacent-1, and adjacent-2 segregation patterns in anaphase I of meiosis.
 (c) Give the products that result from alternate, adjacent-1, and adjacent-2 segregation.
30. Red-green color blindness is a human X-linked recessive disorder. A young man with a 47,XXY karyotype (Klinefelter syndrome) is color-blind. His 46,XY brother also is color-blind. Both parents have normal color vision. Where did the nondisjunction occur that gave rise to the young man with Klinefelter syndrome?
31. Some people with Turner syndrome are 45,X/46,XY mosaics. Explain how this mosaicism could arise.
- *32. Bill and Betty have had two children with Down syndrome. Bill's brother has Down syndrome and his sister has two children with Down syndrome. On the basis of these observations, which of the following statements is most likely correct? Explain your reasoning.
- (a) Bill has 47 chromosomes.
 (b) Betty has 47 chromosomes.
 (c) Bill and Betty's children each have 47 chromosomes.
 (d) Bill's sister has 45 chromosomes.
 (e) Bill has 46 chromosomes.
 (f) Betty has 45 chromosomes.
 (g) Bill's brother has 45 chromosomes.
- *33. Tay-Sachs disease is an autosomal recessive disease that causes blindness, deafness, brain enlargement, and premature death in children. It is possible to identify carriers for Tay-Sachs disease by means of a blood test. Mike and Sue have both been tested for the Tay-Sachs gene; Mike is a heterozygous carrier for Tay-Sachs, but Sue is homozygous for the normal allele. Mike and Sue's baby boy is completely normal at birth, but at age 2 develops Tay-Sachs disease. Assuming that a new mutation has not occurred, how could Mike and Sue's baby have inherited Tay-Sach's disease?
34. In mammals, sex-chromosome aneuploids are more common than autosomal aneuploids but, in fishes, sex-chromosome aneuploids and autosomal aneuploids are found with equal

frequency. Offer an explanation for these differences in mammals and fishes.

- *35. A young couple is planning to have children. Knowing that there have been a substantial number of stillbirths, miscarriages, and fertility problems on the husband's side of the family, they see a genetic counselor. A chromosome analysis reveals that, whereas the woman has a normal karyotype, the man possesses only 45 chromosomes and is a carrier for a Robertsonian translocation between chromosomes 22 and 13.

- (a) List all the different types of gametes that might be produced by the man.
- (b) What types of zygotes will develop when each of gametes produced by the man fuses with a normal gamete from the woman?
- (c) If trisomies and monosomies entailing chromosome 13 and 22 are lethal, what proportion of the surviving offspring will be carriers of the translocation?

CHALLENGE QUESTION

36. Red–green color blindness is a human X-linked recessive disorder. Jill has normal color vision, but her father is color-blind. Jill marries Tom, who also has normal color vision. Jill and Tom have a daughter who has Turner syndrome and is color-blind.

- (a) How did the daughter inherit color blindness?
- (b) Did the daughter inherit her X chromosome from Jill or from Tom?

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10

DNA: The Chemical Nature of the Gene



Ice Man is a 5300-year-old frozen corpse found in the Alps. Analysis of his mitochondrial DNA has revealed that he was a Neolithic hunter related to present-day Europeans living north of the Alps. (Brando Quilicia)

- The Elegantly Stable Double Helix: Ice Man's DNA
- Characteristics of Genetic Material
- The Molecular Basis of Heredity
 - Early Studies of DNA
 - DNA As the Source of Genetic Information
 - Watson and Crick's Discovery of the Three-Dimensional Structure of DNA
 - RNA as Genetic Material
- The Structure of DNA
 - The Primary Structure of DNA
 - Secondary Structures of DNA
- Special Structures in DNA and RNA
 - DNA Methylation Bends in DNA

The Elegantly Stable Double Helix: Ice Man's DNA

DNA, with its gentle double-stranded spiral, is among the most elegant of all biological molecules. But the double helix is not just a beautiful structure; it also gives DNA incredible stability and permanence, as illustrated by the story of Ice Man.

On September 19, 1991, German tourists hiking in the Tyrolean Alps near the border between Austria and Italy spotted a corpse trapped in glacial ice. A copper ax, dagger, bow, and quiver with 14 arrows were found alongside the body. Not realizing its antiquity, local residents made several crude and unsuccessful attempts to free the body from the ice. After 4 days, a team of forensic experts arrived to recover the body and transport it to the University of Innsbruck. There the mummified corpse, known as Ice Man, was refrozen and subjected to scientific study.

Radiocarbon dating indicates that Ice Man is approximately 5000 years old. Recent evidence from the South Tyrol Museum of Archeology has led to the conclusion that Ice Man was shot in the chest with an arrow and died soon thereafter. The body became dehydrated in the cold high-altitude air, was covered with snow that turned into ice, and remained frozen for the next 5000 years.

Some experts challenged Ice Man's origin, suggesting that he was a South American mummy who had been planted at the glacier site in an elaborate hoax. To establish his authenticity and ethnic origin, scientists removed eight samples of muscle, connective tissue, and bone from his left hip. Under sterile conditions, the investigators extracted DNA from the samples and used the polymerase chain reaction (see Chapter 18) to amplify a very small region of his mitochondrial DNA a millionfold. They determined the base sequence of this amplified DNA and compared it with mitochondrial sequences from present-day humans.

This analysis revealed that Ice Man's mitochondrial DNA sequences resemble those found in present-day Europeans living north of the Alps and are quite different from those of sub-Saharan Africans, Siberians, and Native Americans. Together, radiocarbon dating, the artifacts, and the DNA analysis all indicate that Ice Man was a Neolithic hunter who died while attempting to cross the Alps 5000 years ago. That some of Ice Man's DNA persists and faithfully carries his genetic instructions even after the passage of 5000 years is testimony to the remarkable stability of the double helix. Even more ancient DNA has been isolated from the fossilized bones of Neanderthals that are at least 30,000 years old.

This chapter focuses on how DNA was identified as the source of genetic information and how this elegant molecule encodes the genetic instructions. We begin by considering the basic requirements of the genetic material and the history of our understanding of DNA—how its relation to genes was uncovered and how its structure was determined. The history of DNA illustrates several important points about the nature of scientific research. As with so many important scientific advances, DNA's structure and its role as the genetic material were not discovered by any single person but were gradually revealed over a period of almost 100 years, thanks to the work of many investigators. Our understanding of the relation between DNA and genes was enormously enhanced in 1953, when James Watson and Francis Crick proposed a three-dimensional structure for DNA that brilliantly illuminated its role in genetics. As illustrated by Watson and Crick's discovery, major scientific advances are often achieved not through the collection of new data but through the interpretation of old data in new ways.

After reviewing the history of DNA, we will examine DNA structure. DNA structure is important in its own right, but the key genetic concept is the relation between the structure and the function of DNA—how its structure allows it to serve as the genetic material.

Characteristics of Genetic Material

Life is characterized by tremendous diversity, but the coding instructions of all living organisms are written in the same genetic language—that of nucleic acids. Surprisingly, the idea that genes are made of nucleic acids was not widely accepted until after 1950. This late recognition of the role of nucleic acids in genetics resulted principally from a lack of knowledge about the structure of deoxyribonucleic acid (DNA). Until the structure of DNA was fully elucidated, it wasn't clear how DNA could store and transmit genetic information. Even before nucleic acids were identified as the genetic material, biologists recognized that, whatever the nature of genetic material, it must possess three important characteristics.

1. Genetic material must contain complex information.

First and foremost, the genetic material must be capable of storing large amounts of information—instructions for all the traits and functions of an organism. This information must have the capacity to vary, because different species and even individual members of a species differ in their genetic makeup. At the same time, the genetic material must be stable, because most alterations to the genetic instructions (mutations) are likely to be detrimental.

2. Genetic material must replicate faithfully.

A second necessary feature is that genetic material must have the capacity to be copied accurately. Every organism begins life as a single cell, which must undergo billions of cell divisions to produce a complex, multicellular creature like yourself. At each cell division, the genetic instructions must be transmitted to descendent cells with great accuracy. When organisms reproduce and pass genes to their progeny, the coding instructions must be copied with fidelity.

3. Genetic material must encode phenotype.

The genetic material (the genotype) must have the capacity to “code for” (determine) traits (the phenotype). The product of a gene is often a protein; so there must be a mechanism for genetic instructions to be translated into the amino acid sequence of a protein.

Concepts

The genetic material must be capable of carrying large amounts information, replicating faithfully, and translating its coding instructions into phenotypes.

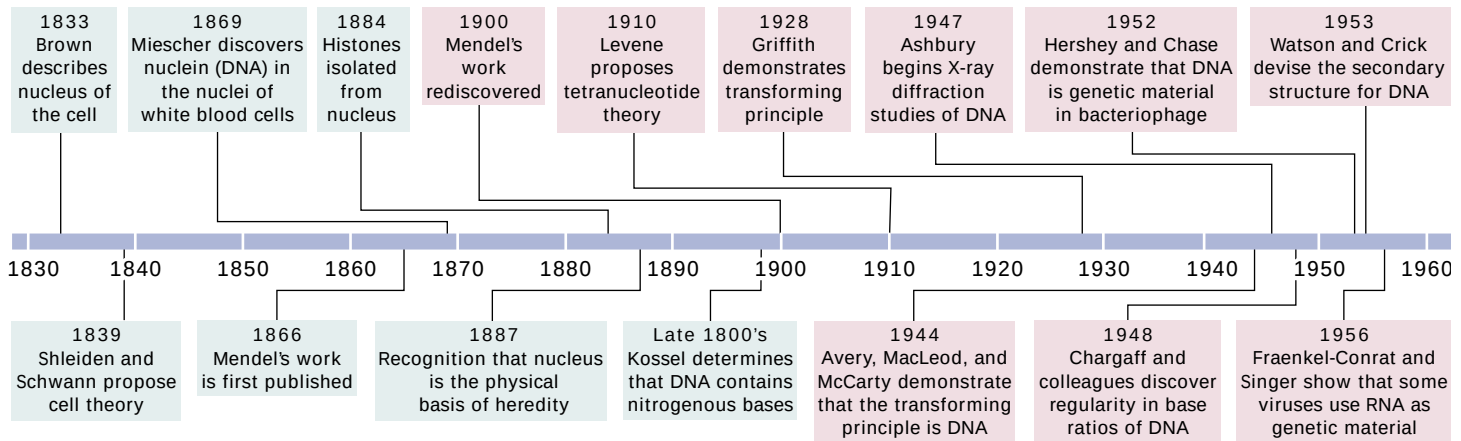


The Molecular Basis of Heredity

Although our understanding of how DNA encodes genetic information is relatively recent, the study of DNA structure stretches back 100 years.

Early Studies of DNA

In 1868, Johann Friedrich Miescher (**FIGURE 10.1**) graduated from medical school in Switzerland. Influenced by an uncle who believed that the key to understanding disease lay in the chemistry of tissues, Miescher traveled to Tübingen, Germany, to study under Ernst Felix Hoppe-Seyler, an early leader in the emerging field of biochemistry. Under Hoppe-Seyler's direction, Miescher turned his attention to the chemistry of pus, a substance of clear medical importance. Pus contains white blood cells with large nuclei; Miescher developed a method of isolating

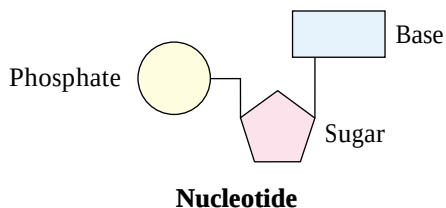


10.1 Many people have contributed to our understanding of the structure and function of DNA.

these nuclei. The minute amounts of nuclear material that he obtained were insufficient for a thorough chemical analysis, but he did establish that it contained a novel substance that was slightly acidic and high in phosphorus. This material, which consisted of DNA and protein, Miescher called *nuclein*. The substance was later renamed *nucleic acid* by one of his students.

By 1887, researchers had concluded that the physical basis of heredity lies in the nucleus. Chromatin was shown to consist of nucleic acid and proteins, but which of these substances is actually the genetic information was not clear. In the late 1800s, further work on the chemistry of DNA was carried out by Albrecht Kossel, who determined that DNA contains four nitrogenous bases: adenine, cytosine, guanine, and thymine (abbreviated A, C, G, and T).

In the early twentieth century, the Rockefeller Institute in New York City became a center for nucleic acid research. Phoebus Aaron Levene joined the Institute in 1905 and spent the next 40 years studying the chemistry of DNA. He discovered that DNA consists of a large number of linked, repeating units, each containing a sugar, a phosphate, and a base (together forming a **nucleotide**).



He incorrectly proposed that DNA consists of a series of four-nucleotide units, each unit containing all four bases—adenine, guanine, cytosine, and thymine—in a fixed sequence. This concept, known as the tetranucleotide theory, implied that the structure of DNA is too regular to serve as the genetic material. The tetranucleotide theory

contributed to the idea that *protein* is the genetic material because, with its 20 different amino acids, protein structure could be highly variable.

As additional studies of the chemistry of DNA were completed in the 1940s and 1950s, this notion of DNA as a simple, invariant molecule began to change. Erwin Chargaff and his colleagues carefully measured the amounts of the four bases in DNA from a variety of organisms and found that DNA from different organisms varies greatly in base composition. This finding disproved the tetranucleotide theory. They discovered that, within each species, there is some regularity in the ratios of the bases: the total amount of adenine is always equal to the amount of thymine ($A = T$), and the amount of guanine is always equal to the amount of cytosine ($G = C$; Table 10.1). These findings became known as **Chargaff's rules**.

Concepts

Details of the structure of DNA were worked out by a number of scientists. At first, DNA was interpreted as being too regular in structure to carry genetic information but, by the 1940s, DNA from different organisms was shown to vary in its base composition.

DNA As the Source of Genetic Information

While chemists were working out the structure of DNA, biologists were attempting to identify the source of genetic information. Two sets of experiments, one conducted on bacteria and the other on viruses, provided pivotal evidence that DNA, rather than protein, was the genetic material.

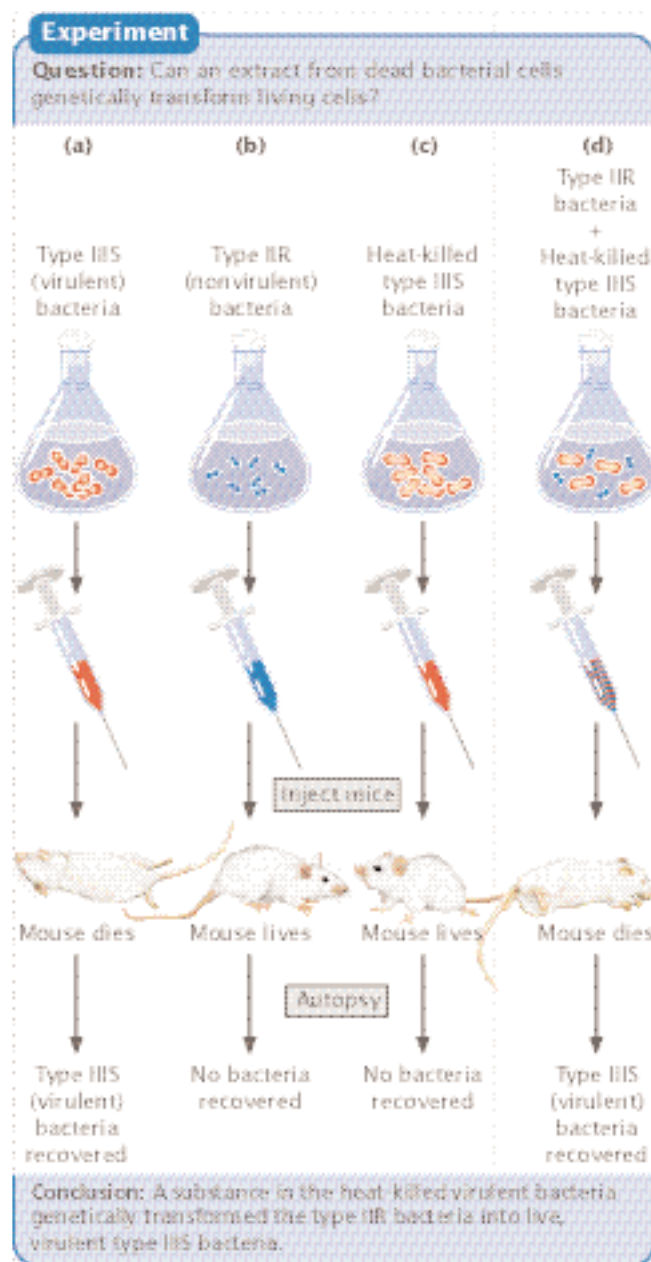
Table 10.1 Base composition of DNA from different sources and ratios of bases

Source of DNA	A	T	G	C	Ratio		
					A/T	G/C	A + G/T + C
<i>E. coli</i>	26.0	23.9	24.9	25.2	1.09	0.99	1.04
Yeast	31.3	32.9	18.7	17.1	.95	1.09	1.00
Sea urchin	32.8	32.1	17.7	18.4	1.02	.96	1.00
Rat	28.6	28.4	21.4	21.5	1.01	1.00	1.00
Human	30.3	30.3	19.5	19.9	1.00	0.98	0.99

The discovery of the transforming principle The first clue that DNA was the carrier of hereditary information came with the demonstration that DNA was responsible for a phenomenon called *transformation*. The phenomenon was first observed in 1928 by Fred Griffith, an English physician whose special interest was the bacterium that causes pneumonia, *Streptococcus pneumoniae*. Griffith had succeeded in isolating several different strains of *S. pneumoniae* (type I, II, III, and so forth). In the virulent (disease-causing) forms of a strain, each bacterium is surrounded by a polysaccharide coat, which makes the bacterial colony appear smooth when grown on an agar plate; these forms are referred to as S, for smooth. Griffith found that these virulent forms occasionally mutated to nonvirulent forms, which lack a polysaccharide coat and produce a rough-appearing colony on an agar plate; these forms are referred to as R, for rough.

Griffith was interested in the origins of the different strains of *S. pneumoniae* and why some types were virulent, whereas others were not. He observed that small amounts of living type IIIS bacteria injected into mice caused the mice to develop pneumonia and die; on autopsy, he found large amounts of type IIIS bacteria in the blood of the mice (FIGURE 10.2a). When Griffith injected type IIR bacteria into mice, the mice lived, and no bacteria were recovered from their blood (FIGURE 10.2b). Griffith knew that boiling killed all the bacteria and destroyed their virulence; when he injected large amounts of heat-killed type IIIS bacteria into mice, the mice lived and no type IIIS bacteria were recovered from their blood (FIGURE 10.2c).

The results of these experiments were not unusual. However, Griffith got a surprise when he infected his mice with a small amount of living type IIR bacteria, along with a large amount of heat-killed type IIIS bacteria. Because both the type IIR bacteria and the heat-killed type IIIS bacteria were nonvirulent, he expected these mice to live. Surprisingly, 5 days after the injections, the mice became infected with pneumonia and died (FIGURE 10.2d). When Griffith examined blood from the hearts of these mice, he observed live type IIIS bacteria. Furthermore, these bacteria retained their type IIIS characteristics through several generations; so the infectivity was heritable.



10.2 Griffith's experiments demonstrated transformation in bacteria.

Griffith's results had several possible interpretations, all of which he considered. First, it could have been the case that he had not sufficiently sterilized the type IIIS bacteria and thus a few live bacteria remained in the culture. Any live bacteria injected into the mice would have multiplied and caused pneumonia. Griffith knew that this possibility was unlikely, because he had used only heat-killed type IIIS bacteria in the control experiment, and they never produced pneumonia in the mice.

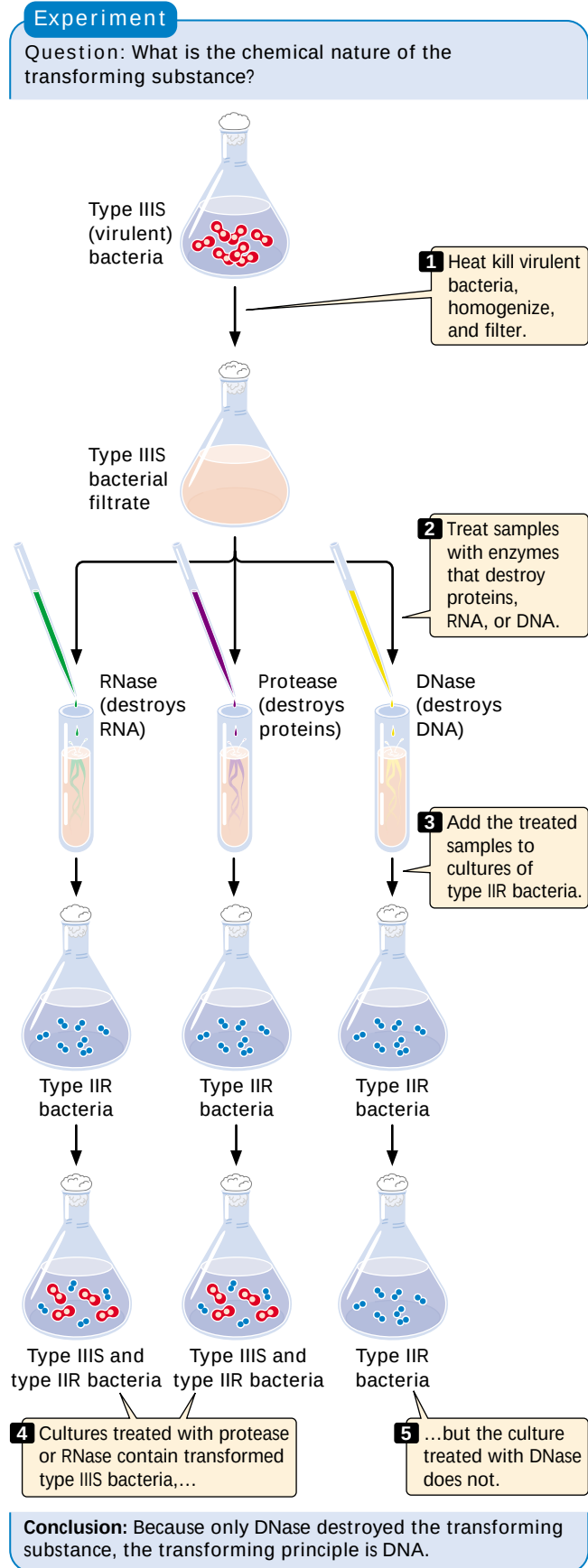
A second interpretation was that the live, type IIR bacteria had mutated to the virulent S form. Such a mutation would cause pneumonia in the mice, but it would produce type IIS bacteria, not the type IIIS that Griffith found in the dead mice. Many mutations would be required for type II bacteria to mutate to type III bacteria, and the chance of all the mutations occurring simultaneously was impossibly low.

Griffith finally concluded that the type IIR bacteria had somehow been *transformed*, acquiring the genetic virulence of the dead type IIIS bacteria. This transformation had produced a permanent, genetic change in the bacteria; though Griffith didn't understand the nature of transformation, he theorized that some substance in the polysaccharide coat of the dead bacteria might be responsible. He called this substance the **transforming principle**.

Identification of the transforming principle At the time of Griffith's report, Oswald Avery (see Figure 10.1) was a microbiologist at the Rockefeller Institute. At first Avery was skeptical but, after other microbiologists successfully repeated Griffith's experiments using other bacteria and showed that transformation took place, Avery set out to identify the nature of the transforming substance.

After 10 years of research, Avery, Colin MacLeod, and Maclyn McCarty succeeded in isolating and purifying the transforming substance. They showed that it had a chemical composition closely matching that of DNA and quite different from that of proteins. Enzymes such as trypsin and chymotrypsin, known to break down proteins, had no effect on the transforming substance. Ribonuclease, an enzyme that destroys RNA, also had no effect. Enzymes capable of destroying DNA, however, eliminated the biological activity of the transforming substance (FIGURE 10.3). Avery, MacLeod, and McCarty showed that purified transforming substance precipitated at about the same rate as purified DNA and that it absorbed ultraviolet light at the same wavelengths as does DNA. These results, published in 1944, provided compelling evidence that the transforming principle—and therefore genetic information—resides in DNA. Many biologists still refused to accept the idea, however, still preferring the hypothesis that the genetic material is protein.

10.3 Avery, MacLeod, and McCarty's experiment revealed the nature of the transforming principle.



Concepts

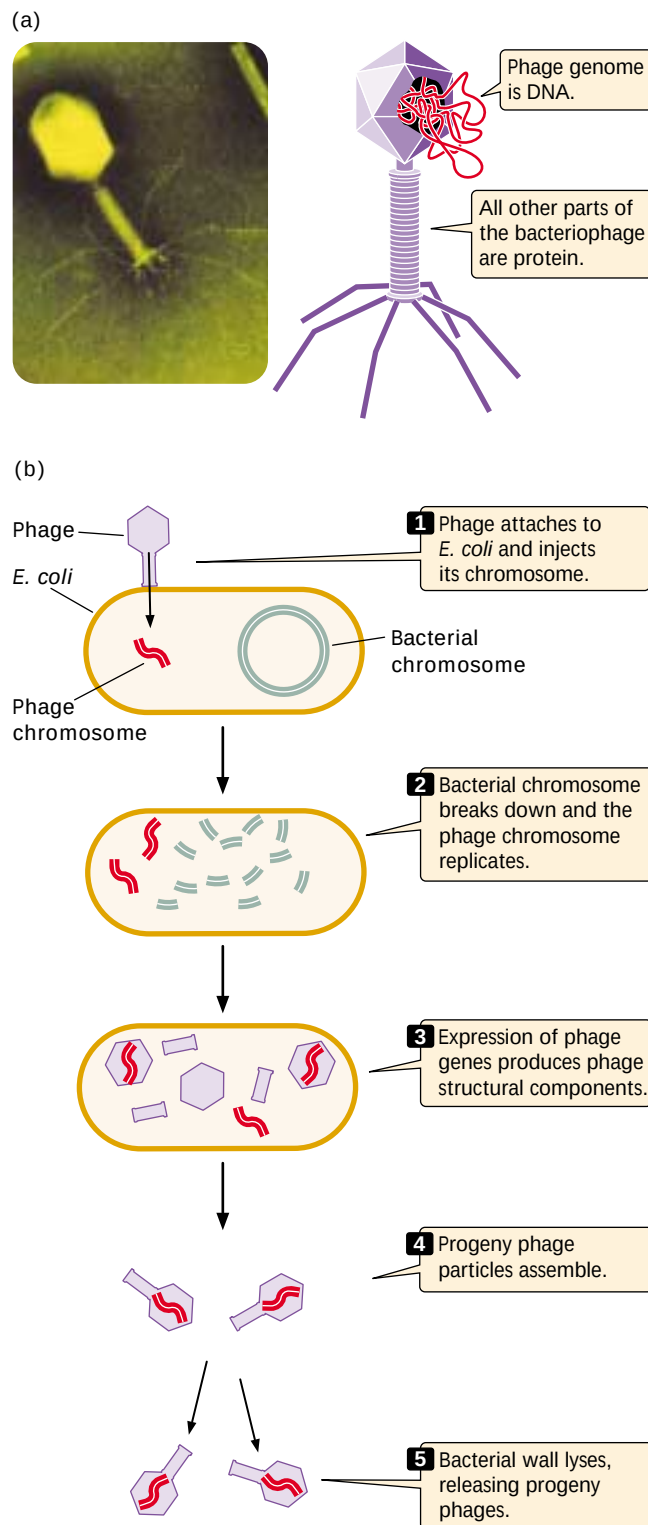
The process of transformation indicates that some substance—the transforming principle—is capable of genetically altering bacteria. Avery, MacLeod, and McCarty demonstrated that the transforming principle is DNA, providing the first evidence that DNA is the genetic material.

The Hershey-Chase experiment A second piece of evidence implicating DNA as the genetic material resulted from a study of the T2 virus conducted by Alfred Hershey and Martha Chase. T2 is a *bacteriophage* (phage) that infects the bacterium *Escherichia coli* (FIGURE 10.4a). As stated in Chapter 8, a phage reproduces by attaching to the outer wall of a bacterial cell and injecting its DNA into the cell, where it replicates and directs the cell to synthesize phage protein. The phage DNA becomes encapsulated within the proteins, producing progeny phages that lyse (break open) the cell and escape (FIGURE 10.4b).

At the time of the Hershey-Chase study (their paper was published in 1952), biologists did not understand exactly how phages reproduce. What they did know was that the T2 phage consists of approximately 50% protein and 50% nucleic acid, that a phage infects a cell by first attaching to the cell wall, and that progeny phages are ultimately produced within the cell. Because the progeny carried the same traits as the infecting phage, genetic material from the infecting phage must be transmitted to the progeny, but how this occurs was unknown.

Hershey and Chase designed a series of experiments to determine whether the phage *protein* or the phage *DNA* was transmitted in phage reproduction. To follow the fate of protein and DNA, they used radioactive forms (**isotopes**) of phosphorus and sulfur. A radioactive isotope can be used as a tracer to identify the location of a specific molecule, because any molecule containing the isotope will be radioactive and therefore easily detected. DNA contains phosphorus but not sulfur; so Hershey and Chase used ^{32}P to follow phage DNA during reproduction. Protein contains sulfur but not phosphorus; so they used ^{35}S to follow the protein.

First, Hershey and Chase grew *E. coli* in a medium containing ^{32}P and infected the bacteria with T2 so that all the new phages would have DNA labeled with ^{32}P (FIGURE 10.5). They grew a second batch of *E. coli* in a medium containing ^{35}S and infected these bacteria with T2 so that all these new phages would have protein labeled with ^{35}S . Hershey and Chase then infected separate batches of unlabeled *E. coli* with the ^{35}S - and ^{32}P -labeled phages. After allowing time for the phages to infect the cells, they placed the *E. coli* cells in a blender and sheared off the now-empty protein coats (ghosts) from the cell walls. They separated out the protein coats and cultured the infected bacterial

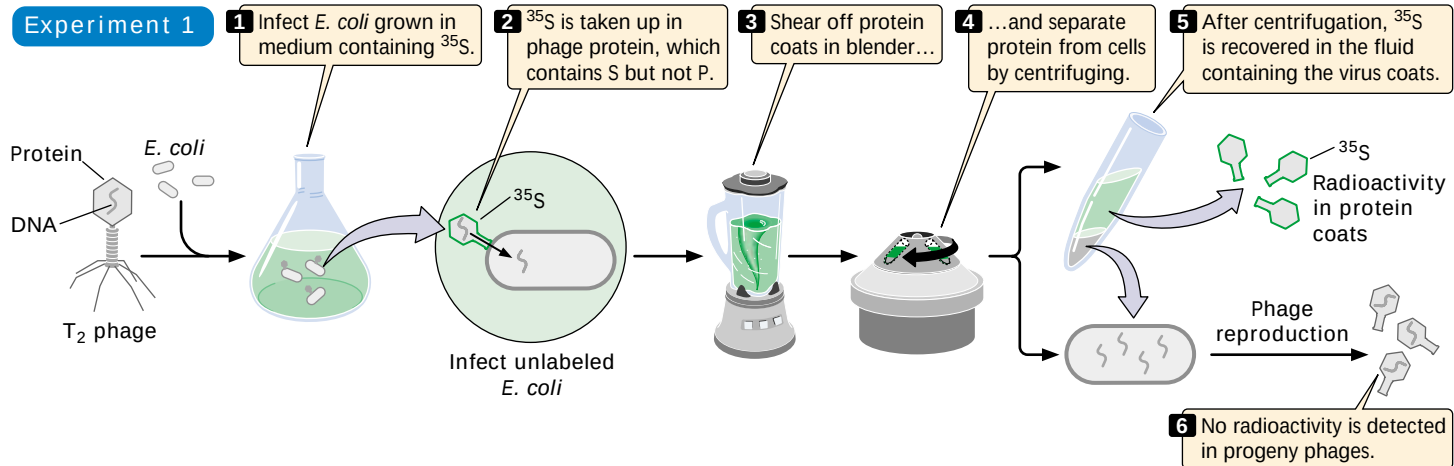


10.4 T2 is a bacteriophage that infects *E. coli*. (a) T2 phage. (b) Its life cycle. (Photo, Harold W. Fisher/Visuals Unlimited.)

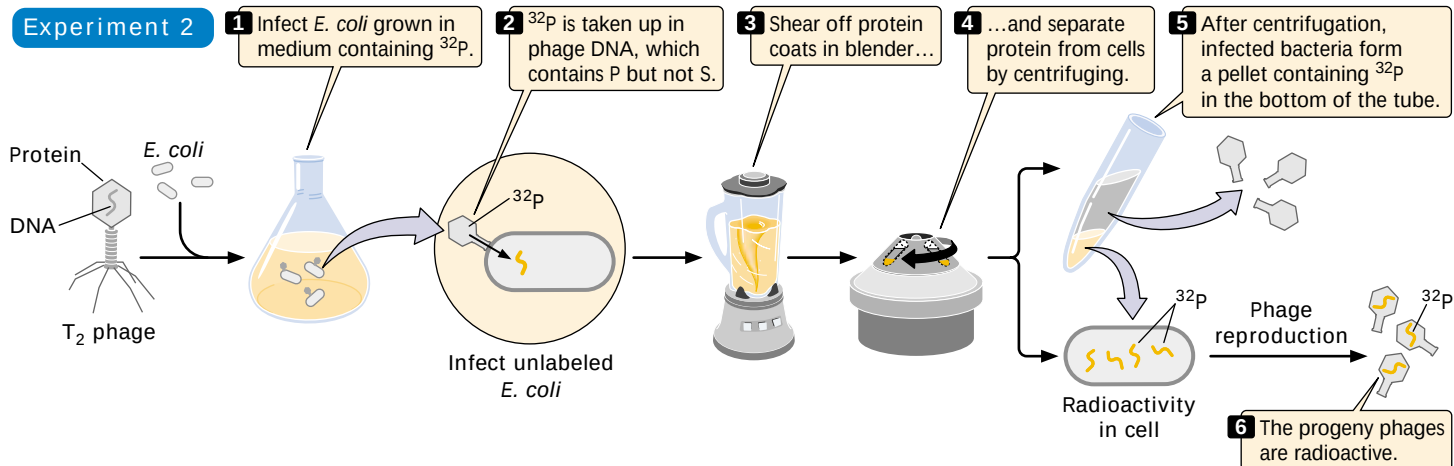
Experiment

Question: Which part of the phage—its DNA or its protein—serves as the genetic material and is transmitted to phage progeny?

Experiment 1



Experiment 2



Conclusion: DNA—not protein—is the genetic material in bacteriophages.

10.5 Hershey and Chase demonstrated that DNA carries the genetic information in bacteriophages.

cells. Eventually, the cells burst and new phage particles emerged.

When phages labeled with ^{35}S infected the bacteria, most of the radioactivity separated with the protein ghosts and little remained in the cells. Furthermore, when new phages emerged from the cell, they contained almost no radioactivity (see Figure 10.5). This result indicated that, although the protein component of a phage was necessary for infection, it didn't enter the cell and was not transmitted to progeny phages.

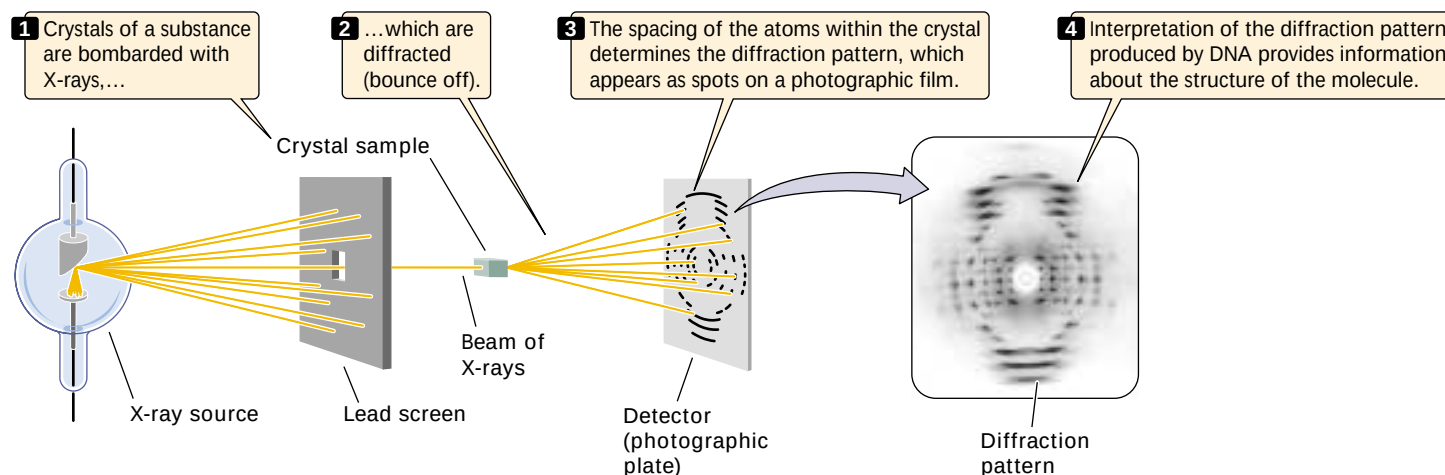
In contrast, when Hershey and Chase infected bacteria with ^{32}P -labeled phages and removed the protein ghosts, the bacteria were still radioactive. Most significantly, after the cells lysed and new progeny phages emerged, many of these phages emitted radioactivity from ^{32}P , demonstrating that DNA from the infecting phages had been passed on to the

progeny (see Figure 10.5). These results confirmed that DNA, not protein, is the genetic material of phages.

Concepts

Using radioactive isotopes, Hershey and Chase traced the movement of DNA and protein during phage infection. They demonstrated that DNA, not protein, enters the bacterial cell during phage reproduction and that only DNA is passed on to progeny phages.

www.whfreeman.com/pierce A discussion of the requirements of the genetic material and the history of our understanding of DNA structure and function.



10.6 X-ray diffraction provides information about the structures of molecules. (Photo from M. H. F. Wilkins, Department of Biophysics, King's College, University of London.)

Watson and Crick's Discovery of the Three-Dimensional Structure of DNA

The experiments on the nature of the genetic material set the stage for one of the most important advances in the history of biology—the discovery of the three-dimensional structure of DNA by James Watson and Francis Crick in 1953.

Watson had studied bacteriophage for his Ph.D.; he was familiar with Avery's work and thus understood the tremendous importance of DNA to genetics. Shortly after receiving his Ph.D., Watson went to the Cavendish Laboratory at Cambridge University in England, where a number of researchers were studying the three-dimensional structure of large molecules. Among these researchers was Francis Crick, who was still working on his Ph.D. Watson and Crick immediately became friends and colleagues.

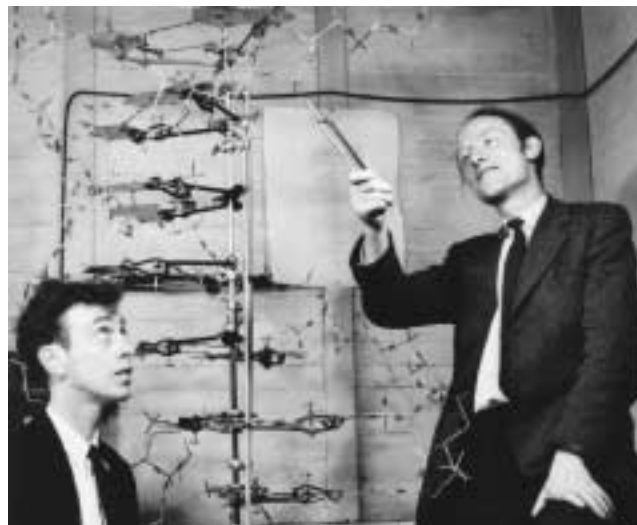
Much of the basic chemistry of DNA had already been determined by Miescher, Kossel, Levene, Chargaff, and others, who had established that DNA consisted of nucleotides, and that each nucleotide contained a sugar, base, and phosphate group. However, how the nucleotides fit together in the three-dimensional structure of the molecule was not at all clear.

In 1947, William Ashbury began studying the three-dimensional structure of DNA by using a technique called **X-ray diffraction** (see **FIGURE 10.6**), but his diffraction pictures did not provide enough resolution to reveal the structure. A research group at King's College in London, led by Maurice Wilkins and Rosalind Franklin, also was studying the structure of DNA by using X-ray diffraction and obtained strikingly better pictures of the molecule. Wilkins and Franklin, however, were unable to develop a complete structure of the molecule; their progress was impeded by personal discord that existed between them.

Watson and Crick investigated the structure of DNA, not by collecting new data but by using all available information about the chemistry of DNA to construct molecular

models (see **FIGURE 10.7**). By applying the laws of structural chemistry, they were able to limit the number of possible structures that DNA could assume. Watson and Crick tested various structures by building models made of wire and metal plates. With their models, they were able to see whether a structure was compatible with chemical principles and with the X-ray images.

The key to solving the structure came when Watson recognized that an adenine base could bond with a thymine base and that a guanine base could bond with a cytosine base; these pairings accounted for the base ratios that Chargaff had discovered earlier. The model developed by Watson and Crick showed that DNA consists of two strands of nucleotides wound around each other to form a right-



10.7 Watson and Crick provided a three-dimensional model of the structure of DNA. (A. Barrington Brown/Science Photo Library/Photo Researchers.)

handed helix, with the sugars and phosphates on the outside and the bases in the interior. They published an electrifying description of their model in *Nature* in 1953. At the same time, Wilkins and Franklin published their X-ray diffraction data, which demonstrated experimentally the theory that DNA was helical in structure.

Many have called the solving of DNA's structure the most important biological discovery of the twentieth century. For their discovery, Watson and Crick, along with Maurice Wilkins, were awarded a Nobel Prize in 1962. (Rosalind Franklin had died of cancer in 1957 and, thus, could not be considered a candidate for the shared prize.)

Concepts

By collecting existing information about the chemistry of DNA and building molecular models, Watson and Crick were able to discover the three-dimensional structure of the DNA molecule.

www.whfreeman.com/pierce A commentary on Watson and Crick's original paper describing the structure of DNA and more information about some of the key players in the discovery of DNA

RNA As Genetic Material

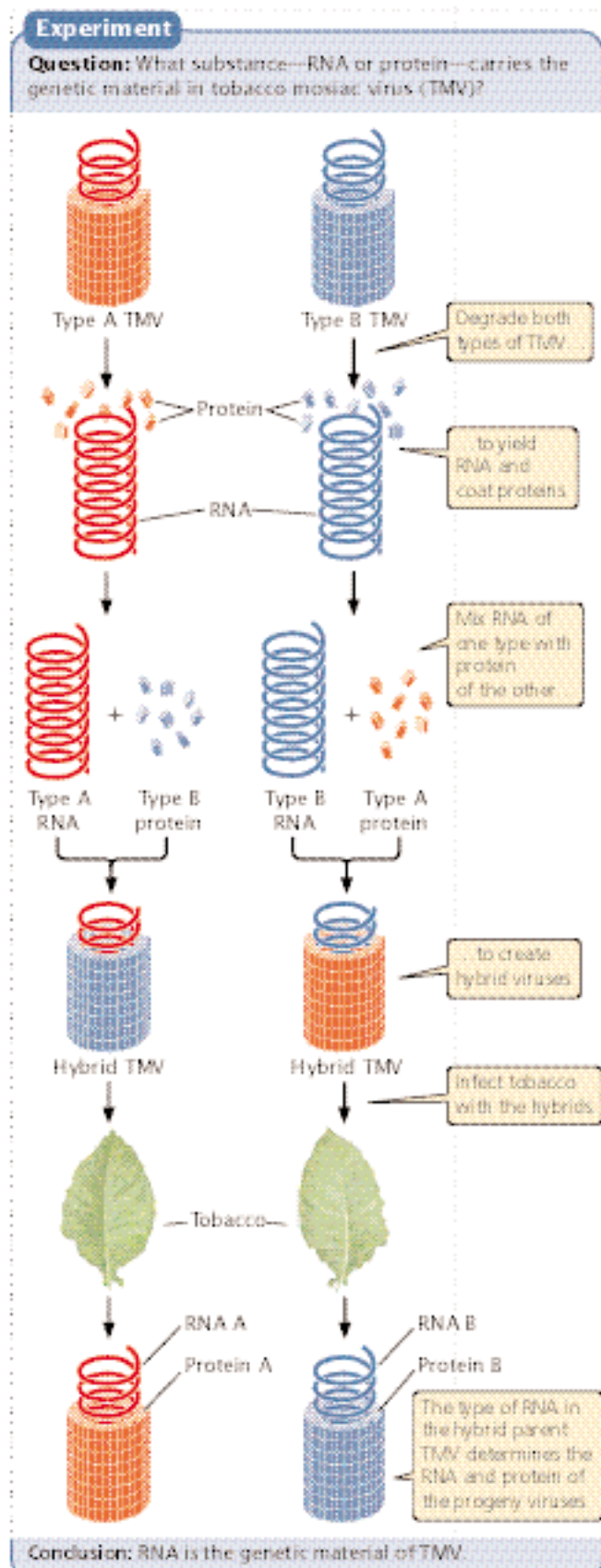
In most organisms, DNA carries the genetic information. However, a few viruses utilize RNA, not DNA, as their genetic material. This fact was demonstrated in 1956 by Heinz Fraenkel-Conrat and Bea Singer, who worked with tobacco mosaic virus (TMV), a virus that infects and causes disease in tobacco plants. The tobacco mosaic virus possesses a single molecule of RNA surrounded by a helically arranged cylinder of protein molecules. Fraenkel-Conrat found that, after separating the RNA and protein of TMV, he could remix them and obtain intact, infectious viral particles.

With Singer, Fraenkel-Conrat then created hybrid viruses by mixing RNA and protein from different strains of TMV (FIGURE 10.8). When these hybrid viruses infected tobacco leaves, new viral particles were produced. The new viral progeny were identical to the strain from which the RNA had been isolated and did not exhibit the characteristics of the strain that donated the protein. These results showed that RNA carries the genetic information in TMV.

Also in 1956, Alfred Gierer and Gerhard Schramm demonstrated that RNA isolated from TMV is sufficient to infect tobacco plants and direct the production of new TMV particles, confirming that RNA carries genetic instructions.

Concepts

RNA serves as the genetic material in some viruses.



10.8 Fraenkel-Conrat and Singer's experiment demonstrated that, in the tobacco mosaic virus, RNA carries the genetic information.

The Structure of DNA

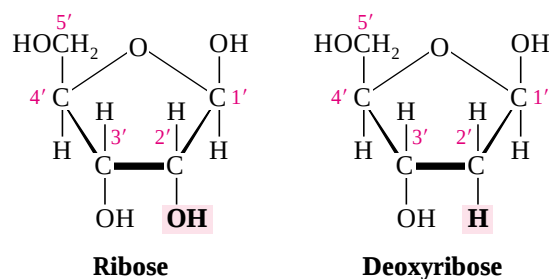
DNA, though relatively simple in structure, has an elegance and beauty unsurpassed by other large molecules. It is useful to consider the structure of DNA at three levels of increasing complexity, known as the primary, secondary, and tertiary structures of DNA. The primary structure of DNA refers to its nucleotide structure and how the nucleotides are joined together. The secondary structure refers to DNA's stable three-dimensional configuration, the helical structure worked out by Watson and Crick. In Chapter 11, we will consider DNA's tertiary structures, which are the complex packing arrangements of double-stranded DNA in chromosomes.

The Primary Structure of DNA

The primary structure of DNA consists of a string of nucleotides joined together by phosphodiester linkages.

Nucleotides DNA is typically a very long molecule and is therefore termed a macromolecule. For example, within each human chromosome is a single DNA molecule that, if stretched out straight, would be several centimeters in length. In spite of its large size, DNA has a relatively simple structure: it is a polymer, a chain made up of many repeating units linked together. As already mentioned, the repeating units of DNA are *nucleotides*, each comprising three parts: (1) a sugar, (2) a phosphate, and (3) a nitrogen-containing base.

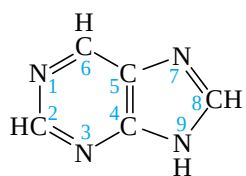
The sugars of nucleic acids—called pentose sugars—have five carbon atoms, numbered 1', 2', 3', and so forth (FIGURE 10.9). Four of the carbon atoms are joined by an oxygen atom to form a five-sided ring; the fifth (5') carbon atom projects upward from the ring. Hydrogen atoms or hydroxyl groups (OH) are attached to each carbon atom.



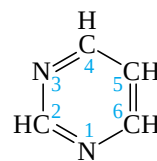
10.9 A nucleotide contains either a ribose sugar (in RNA) or a deoxyribose sugar (in DNA). The atoms of the five-sided ring are assigned primed numbers.

The sugars of DNA and RNA are slightly different in structure. RNA's **ribose** sugar has a hydroxyl group attached to the 2'-carbon atom, whereas DNA's sugar, called **deoxyribose**, has a hydrogen atom at this position and contains one oxygen atom fewer overall. This difference gives rise to the names ribonucleic acid (RNA) and deoxyribonucleic acid (DNA). This minor chemical difference is recognized by all the cellular enzymes that interact with DNA or RNA, thus yielding specific functions for each nucleic acid. Further, the additional oxygen atom in the RNA nucleotide makes it more reactive and less chemically stable than DNA. For this reason, DNA is better suited to serve as the long-term repository of genetic information.

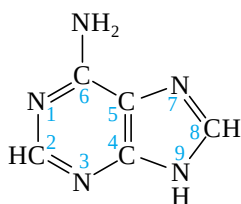
The second component of a nucleotide is its **nitrogenous base**, which may be of two types—a **purine** or a **pyrimidine** (FIGURE 10.10). Each purine consists of a six-sided ring attached to a five-sided ring, whereas each pyrimidine consists of a six-sided ring only. DNA and RNA both contain two purines, **adenine** and **guanine** (A



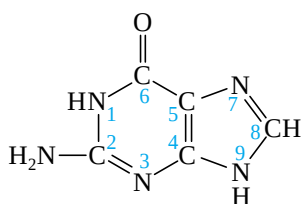
Purine
(basic structure)



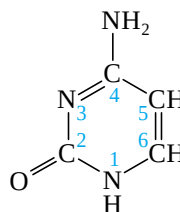
Pyrimidine
(basic structure)



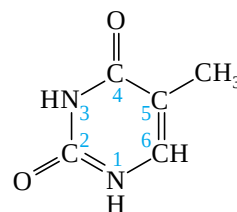
Adenine (A)



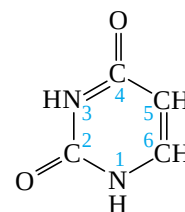
Guanine (G)



Cytosine (C)

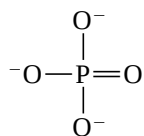


Thymine (T)
(present in DNA)



Uracil (U)
(present in RNA)

10.10 A nucleotide contains either a purine or a pyrimidine base. The atoms of the rings in the bases are assigned unprimed numbers.

**Phosphate**

10.11 A nucleotide contains a phosphate group.

and G), which differ in the positions of their double bonds and in the groups attached to the six-sided ring. There are three pyrimidines found in nucleic acids: **cytosine** (C), **thymine** (T), and **uracil** (U). Cytosine is present in both DNA and RNA; however, thymine is restricted to DNA, and uracil is found only in RNA. The three pyrimidines differ in the groups or atoms attached to the carbon atoms of the ring and in the number of double bonds in the ring. In a nucleotide, the nitrogenous base always forms a covalent bond with the 1'-carbon atom of the sugar (see Figure 10.9). A deoxyribose (or ribose) sugar and a base together are referred to as a **nucleoside**.

The third component of a nucleotide is the **phosphate group**, which consists of a phosphorus atom bonded to four oxygen atoms (10.11). Phosphate groups are found in every nucleotide and frequently carry a negative charge, which makes DNA acidic. The phosphate is always bonded to the 5'-carbon atom of the sugar in a nucleotide (see Figure 10.9).

The DNA nucleotides are properly known as **deoxyribonucleotides** or deoxyribonucleoside 5'-monophosphates. Because there are four types of bases, there are four different kinds of DNA nucleotides (10.12). The equivalent RNA nucleotides are termed **ribonucleotides** or ribonucleoside 5'-monophosphates. RNA molecules sometimes contain additional rare bases, which

are modified forms of the four common bases. These modified bases will be discussed in more detail when we examine the function of RNA molecules in Chapter 14.

Concepts

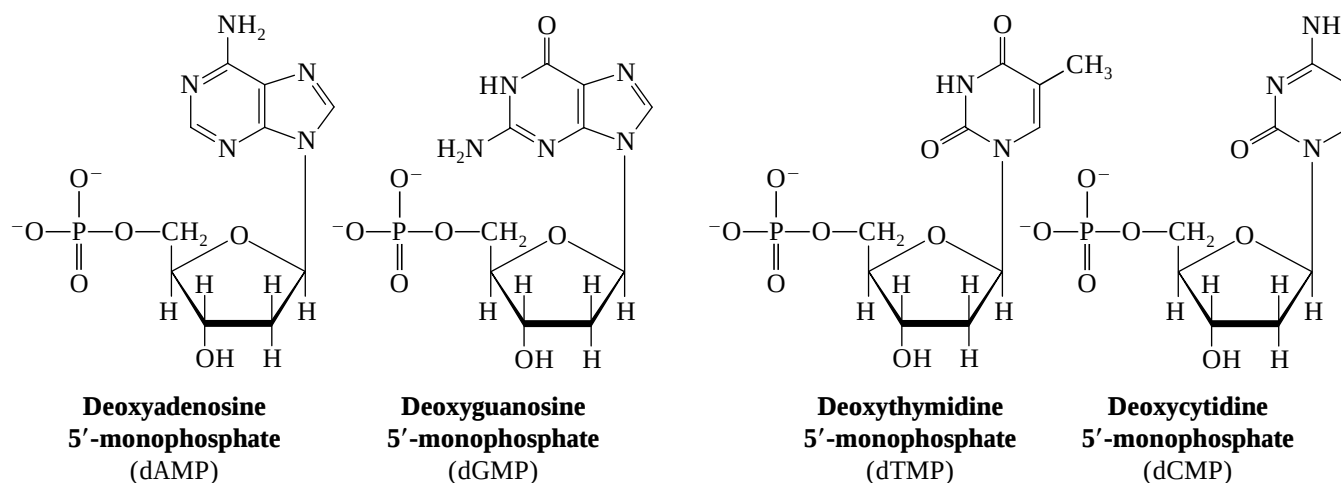


The primary structure of DNA consists of a string of nucleotides. Each nucleotide consists of a five-carbon sugar, a phosphate, and a base. There are two types of DNA bases: purines (adenine and guanine) and pyrimidines (thymine and cytosine).

Polynucleotide strands DNA is made up of many nucleotides connected by covalent bonds, which join the 5'-phosphate group of one nucleotide to the 3'-carbon atom of the next nucleotide (10.13). These bonds, called **phosphodiester linkages**, are relatively strong covalent bonds; a series of nucleotides linked in this way constitutes a **polynucleotide strand**. The backbone of the polynucleotide strand is composed of alternating sugars and phosphates; the bases project away from the long axis of the strand. The negative charges of the phosphate groups are frequently neutralized by the association of positive charges on proteins, metals, or other molecules.

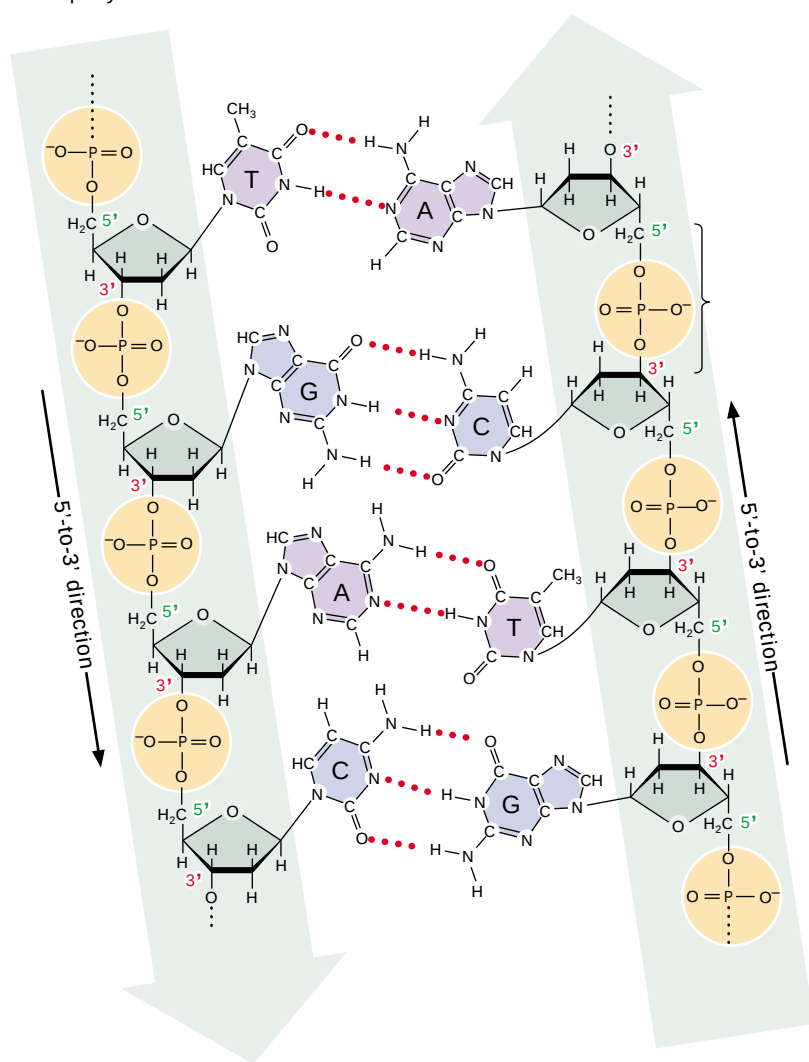
An important characteristic of the polynucleotide strand is its direction, or polarity. At one end of the strand a phosphate group is attached only to the 5'-carbon atom of the sugar in the nucleotide. This end of the strand is therefore referred to as the **5' end**. The other end of the strand, referred to as the **3' end**, has an OH group attached to the 3'-carbon atom of the sugar.

RNA nucleotides also are connected by phosphodiester linkages to form similar polynucleotide strands (see Figure 10.13).

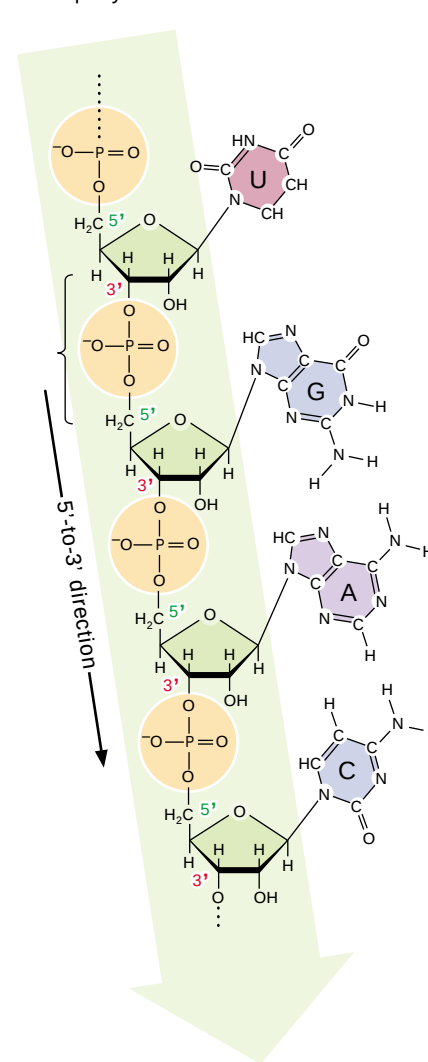


10.12 There are four types of DNA nucleotides.

DNA polynucleotide strand



RNA polynucleotide strand



10.13 DNA consists of two polynucleotide chains that are antiparallel and complementary, and RNA consists of a single nucleotide chain.

Concepts

The nucleotides of DNA are joined in polynucleotide strands by phosphodiester bonds that connect the 3' carbon atom of one nucleotide to the 5' phosphate group of the next. Each polynucleotide strand has polarity, with a 5' end and a 3' end.

Secondary Structures of DNA

The secondary structure of DNA refers to its three-dimensional configuration—its fundamental helical structure. DNA's secondary structure can assume a variety of configurations, depending on its base sequence and the conditions in which it is placed.

The double helix A fundamental characteristic of DNA's secondary structure is that it consists of two polynucleotide strands wound around each other—it's a double helix. The sugar-phosphate linkages are on the outside of the helix, and the bases are stacked in the interior of the molecule (see Figure 10.13). The two polynucleotide strands run in opposite directions—they are **antiparallel**, which means that the 5' end of one strand is opposite the 3' end of the second.

The strands are held together by two types of molecular forces. Hydrogen bonds link the bases on opposite strands (see Figure 10.13). These bonds are relatively weak compared with the covalent phosphodiester bonds that connect the sugar and phosphate groups of adjoining nucleotides. As we will see, several important functions of DNA require the separation of its two nucleotide strands, and this separation can be readily accomplished because

of the relative ease of breaking and reestablishing the hydrogen bonds.

The nature of the hydrogen bond imposes a limitation on the types of bases that can pair. Adenine normally pairs only with thymine through two hydrogen bonds, and cytosine normally pairs only with guanine through three hydrogen bonds (see Figure 10.13). Because three hydrogen bonds form between C and G and only two hydrogen bonds form between A and T, C–G pairing is stronger than A–T pairing. The specificity of the base pairing means that wherever there is an A on one strand, there must be a T in the corresponding position on the other strand, and wherever there is a G on one strand, a C must be on the other. The two polynucleotide strands of a DNA molecule are therefore not identical but are **complementary**.

The second force that holds the two DNA strands together is the interaction between the stacked base pairs. These stacking interactions contribute to the stability of the DNA molecule and do not require that any particular base follow another. Thus, the base sequence of the DNA molecule is free to vary, allowing DNA to carry genetic information.

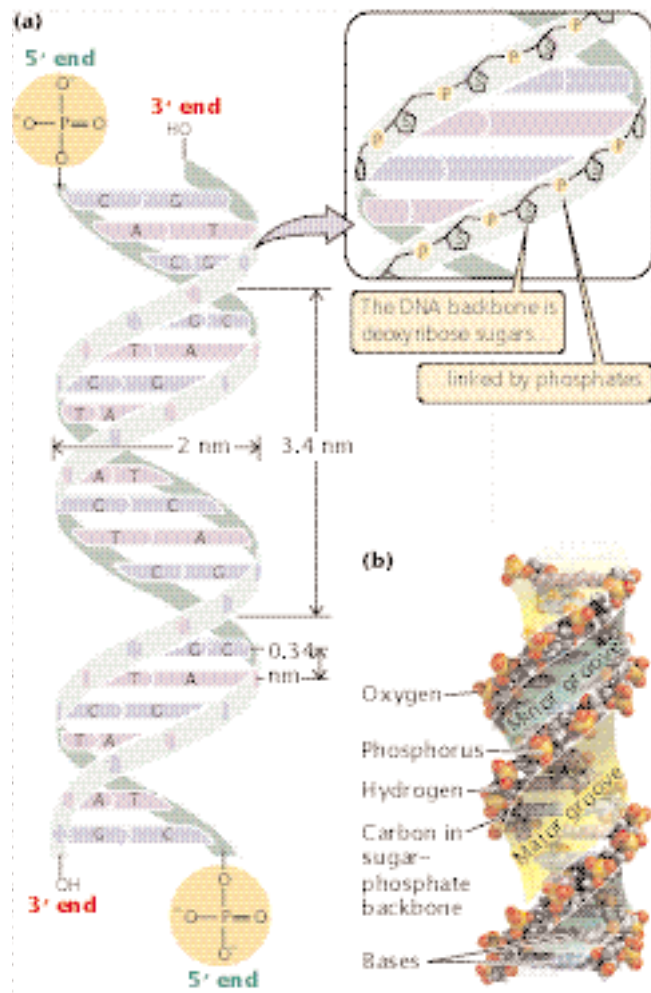
Concepts

DNA consists of two polynucleotide strands. The sugar–phosphate groups of each polynucleotide strand are on the outside of the molecule, and the bases are in the interior. Hydrogen bonding joins the bases of the two strands: guanine pairs with cytosine, and adenine pairs with thymine. The two polynucleotide strands of a DNA molecule are complementary and antiparallel.

Different secondary structures As we have seen, DNA normally consists of two polynucleotide strands that are antiparallel and complementary (exceptions are single-stranded DNA molecules in a few viruses). The precise three-dimensional shape of the molecule can vary, however, depending on the conditions in which the DNA is placed and, in some cases, on the base sequence itself.

The three-dimensional structure of DNA that Watson and Crick described is termed the **B-DNA** structure (FIGURE 10.14). This structure exists when plenty of water surrounds the molecule and there is no unusual base sequence in the DNA—conditions that are likely to be present in cells. The B-DNA structure is the most stable configuration for a random sequence of nucleotides under physiological conditions, and most evidence suggests that it is the predominate structure in the cell.

B-DNA is an alpha helix, meaning that it has a right-handed, or clockwise, spiral. It possesses approximately 10 base pairs (bp) per 360-degree rotation of the helix; so



10.14 B-DNA consists of an alpha helix with approximately 10 bases per turn. (a) Diagrammatic representation showing that the bases are 0.34 nanometer (nm) apart, that each rotation encompasses 3.4 nm, and that the diameter of the helix is 2 nm. (b) Space-filling model of B-DNA showing major and minor grooves.

each base pair is twisted 36 degrees relative to the adjacent bases (see Figure 10.14a). The base pairs are 0.34 nanometer (nm) apart; so each complete rotation of the molecule encompasses 3.4 nm. The diameter of the helix is 2 nm, and the bases are perpendicular to the long axis of the DNA molecule. A space-filling model shows that B-DNA has a relatively slim and elongated structure (see Figure 10.14b). Spiraling of the nucleotide strands creates major and minor grooves in the helix, features that are important for the binding of some DNA-binding proteins that regulate the expression of genetic information (Chapter 16). Some characteristics of the B-DNA structure, along with characteristics of other secondary structures that exist under certain conditions or with unusual base sequences, are given in Table 10.2.

Table 10.2 Characteristics of DNA secondary structures

Characteristic	A-DNA	B-DNA	Z-DNA
Conditions required to produce structure	75% H ₂ O	92% H ₂ O	Alternating purine and pyrimidine bases
Helix direction	Right-handed	Right-handed	Left-handed
Average base pairs per turn	11	10	12
Rotation per base pair	32.7°	36°–30°	
Distance between adjacent bases	0.26 nm	0.34 nm	0.37 nm
Diameter	2.3 nm	1.9 nm	1.8 nm
Overall shape	Short and wide	Long and narrow	Elongated and narrow

Note: Within each structure, the parameters may vary somewhat owing to local variation and method of analysis.

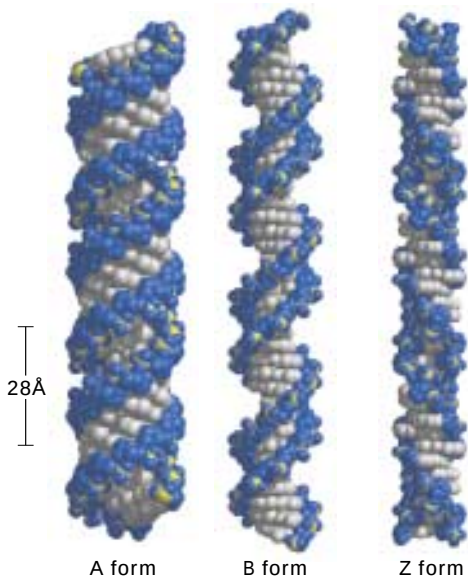
Another secondary structure that DNA can assume is the **A-DNA** structure, which exists when less water is present. Like B-DNA, A-DNA is an alpha (right-handed) helix (FIGURE 10.15a), but it is shorter and wider than B-DNA (FIGURE 10.15b) and its bases are tilted away from the main axis of the molecule. There is little evidence that A-DNA exists under physiological conditions.

A radically different secondary structure called **Z-DNA** (FIGURE 10.15c) forms a left-handed helix. In this form, the sugar–phosphate backbones zigzag back and forth, giving rise to the name Z-DNA (for zigzag). Z-DNA structures

can arise under physiological conditions when particular base sequences are present, such as stretches of alternating C and G sequences. Parts of some active genes form Z-DNA, suggesting that Z-DNA may play a role in regulating gene transcription.

Other secondary structures may exist under special conditions or with special base sequences, and characteristics of some of these structures are given in Table 10.2. Structures other than B-DNA exist rarely, if ever, within cells.

Local variation in secondary structures DNA is frequently presented as a static, rigid structure that is invariant in its secondary structure. In reality, the numbers describing the parameters for B-DNA in Figure 10.14 are average values, and the actual measurements vary slightly from one part of the molecule to another. The twist between base pairs within a single molecule of B-DNA, for example, can vary from 27 degrees to as high as 42 degrees. This **local variation** in DNA structure arises because of differences in local environmental conditions, such as the presence of proteins, metals, and ions that may bind to the DNA. The base sequence also influences DNA structure locally.



10.15 DNA can assume several different secondary structures. These structures depend on the base sequence of the DNA and the conditions under which it is placed. Lehninger, *Biochem* 3/e, p.338. Fig.10-19.

Concepts

DNA can assume different secondary structures, depending on the conditions in which it is placed and on its base sequence. B-DNA is thought to be the most common configuration in the cell. Local variation in DNA arises as a result of environmental factors and base sequence.

www.whfreeman.com/pierce More on DNA structure and some interesting images of DNA

Connecting Concepts

Genetic Implications of DNA Structure

After Oswald Avery and his colleagues demonstrated that the transforming principle is DNA, it was clear that the genotype resides within the chemical structure of DNA. Watson and Crick's great contribution was their elucidation of the genotype's chemical structure, making it possible for geneticists to begin to examine genes directly, instead of looking only at the phenotypic consequences of gene action. Determining the structure of DNA permitted the birth of molecular genetics—the study of the chemical and molecular nature of genetic information.

Watson and Crick's structure did more than just create the potential for molecular genetic studies; it was an immediate source of insight into key genetic processes. At the beginning of this chapter, three fundamental properties of the genetic material were identified. First, it must be capable of carrying large amounts of information; so it must vary in structure. Watson and Crick's model suggested that genetic instructions are encoded in the base sequence, the only variable part of the molecule. The sequence of the four bases—adenine, guanine, cytosine, and thymine—along the helix encodes the information that ultimately determines the phenotype. Watson and Crick were not sure *how* the base sequence of DNA determined the phenotype, but their structure clearly indicated that the genetic instructions were encoded in the bases.

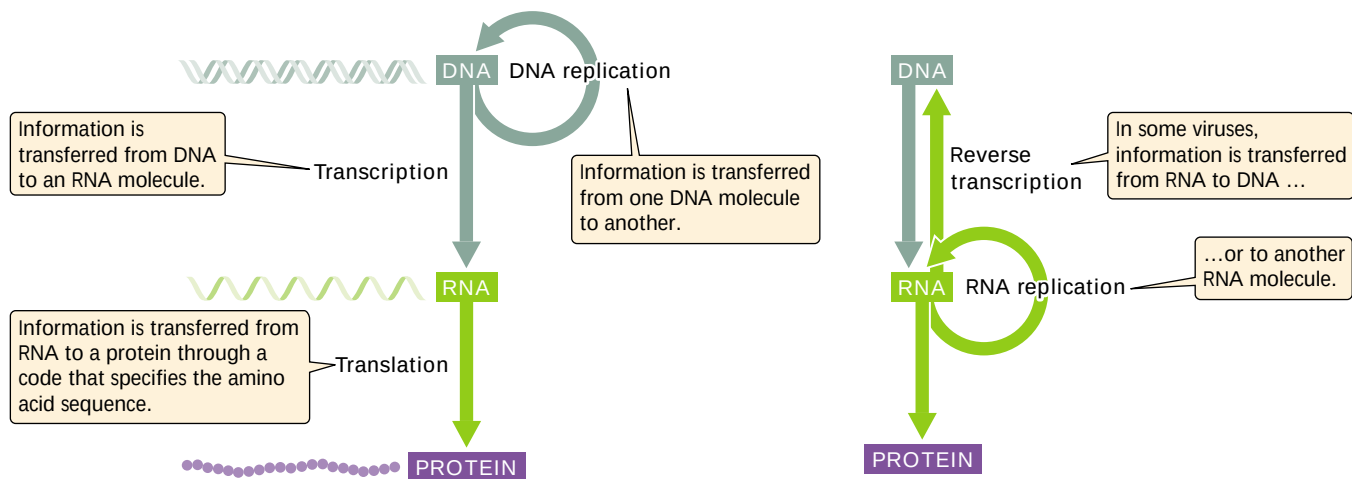
A second necessary property of genetic material is its ability to replicate faithfully. The complementary polynucleotide strands of DNA make this replication possible. Watson and Crick wrote, "It has not escaped our attention that the specific base pairing we have postulated immediately suggests a possible copying mechanism for the genetic material." They proposed that, in replication, the two

polynucleotide strands unzip, breaking the weak hydrogen bonds between the two strands, and each strand serves as a template on which a new strand is synthesized. The specificity of the base pairing means that only one possible sequence of bases—the complementary sequence—can be synthesized from each template. Newly replicated double-stranded DNA molecules will therefore be identical with the original double-stranded DNA molecule (see Chapter 12 on DNA replication).

The third essential property of genetic material is the ability to translate its instructions into the phenotype. For most traits, the immediate phenotype is production of a protein; so the genetic material must be capable of encoding proteins. Proteins, like DNA, are polymers, but their repeating units are amino acids, not nucleotides. A protein's function depends on its amino acid sequence; so the genetic material must be able to specify that sequence in a form that can be transferred in the course of protein synthesis.

DNA expresses its genetic instructions by first transferring its information to an RNA molecule, in a process termed **transcription** (see Chapter 13). The term *transcription* is appropriate because, although the information is transferred from DNA to RNA, the information remains in the language of nucleic acids. The RNA molecule then transfers the genetic information to a protein by specifying its amino acid sequence. This process is termed **translation** (see Chapter 15) because the information must be *translated* from the language of nucleotides into the language of amino acids.

We can now identify three major pathways of information flow in the cell (FIGURE 10.16): in **replication**, information passes from one DNA molecule to other DNA molecules; in transcription, information passes from DNA to RNA; and, in translation, information passes from RNA to protein. This concept of information flow was formalized by Francis Crick in a concept that he called the



10.16 The three major pathways of information transfer within the cell are replication, transcription, and translation.

central dogma of molecular biology. The central dogma states that genetic information passes from DNA to protein in a one-way information pathway. It indicates that genotype codes for phenotype but phenotype cannot code for genotype. We now realize, however, that the central dogma is an oversimplification. In addition to the three general information pathways of replication, transcription, and translation, other transfers may take place in certain organisms or under special circumstances, including the transfer of information from RNA to DNA, (in **reverse transcription**) and the transfer of information from RNA to RNA (in **RNA replication**; see Figure 10.16). Reverse transcription takes place in retroviruses and in some transposable elements; RNA replication takes place in some RNA viruses (see Chapter 8).

Concepts

The genetic information of DNA resides in the base sequence. When DNA replicates, the two strands separate, and each strand serves as a template on which a new strand is synthesized. Three principle pathways transfer genetic information: genetic information can pass from DNA to DNA through replication, from DNA to RNA through transcription, and from RNA to protein through translation.

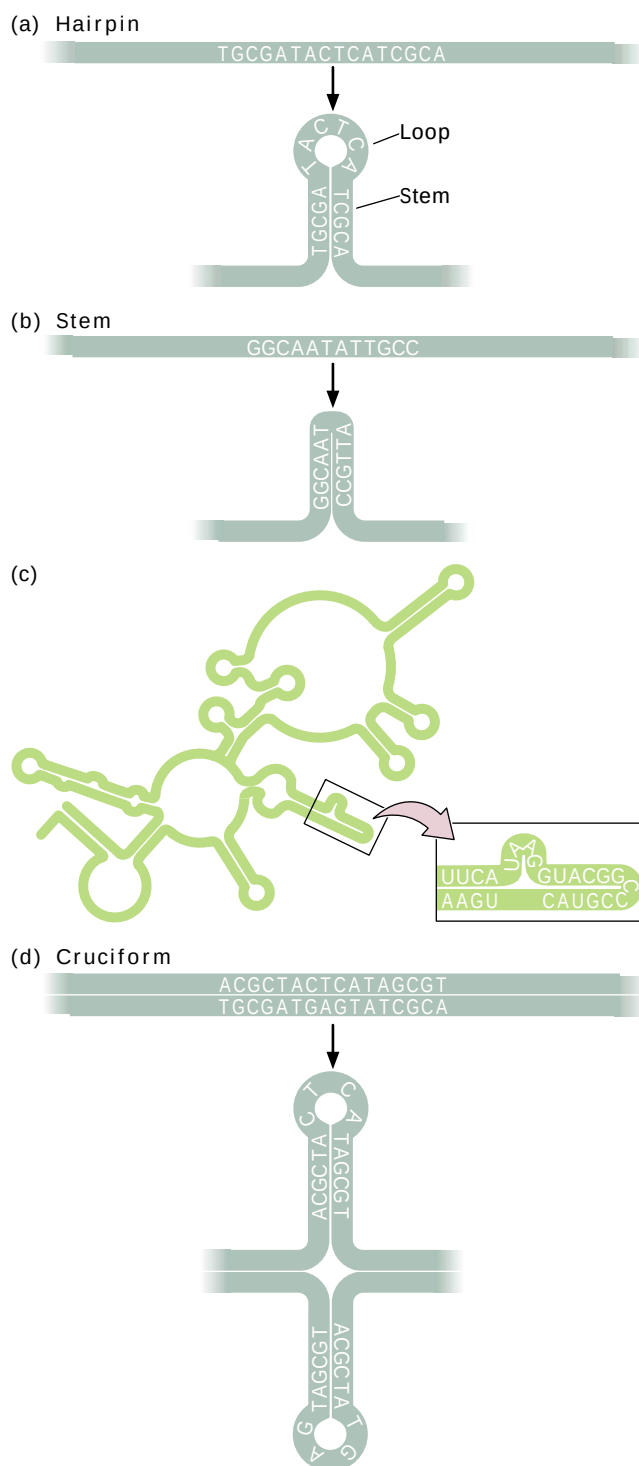
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More information on the central dogma

Special Structures in DNA and RNA

In double-stranded DNA, the pairing of bases on opposite nucleotide strands provides stability and produces the helical secondary structure of the molecule. Single-stranded DNA and RNA (the latter of which is almost always single stranded) lack the stabilizing influence of the paired nucleotide strands; so they exhibit no common secondary structure. Sequences *within* a single strand of nucleotides may be complementary to each other and can pair by forming hydrogen bonds, producing double-stranded regions (FIGURE 10.17). This internal base pairing imparts a secondary structure to a single-stranded molecule. In fact, internal base pairing within single strands of nucleotides can result in a great variety of secondary structures.

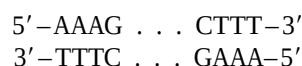
One common type of secondary structure found in single strands of nucleotides is a **hairpin**, which forms when sequences of nucleotides on the same strand are inverted complements. The sequence 5' TGCGAT 3' and 5' ATCGCA 3' are examples of inverted complements and, when these sequences are on the same nucleotide strand, they can pair



10.17 Both DNA and RNA can form special secondary structures. (a) A hairpin, consisting of a region of paired bases (which forms the stem) and a region of unpaired bases between the complementary sequences (which form a loop at the end of the stem). (b) A stem with no loop. (c) Secondary structure, showing many hairpins, of an RNA component of a riboprotein, commonly referred to as the enzyme RNase P of *E. coli*. (d) A cruciform structure.

and form a hairpin (see Figure 10.17a). A hairpin consists of a region of paired bases (the stem) and sometimes includes intervening unpaired bases (the loop). When the complementary sequences are contiguous, the hairpin has a stem but no loop (see Figure 10.17b). Hairpins frequently control aspects of information transfer. RNA molecules may contain numerous hairpins, allowing them to fold up into complex structures (see Figure 10.17c).

In double-stranded DNA, sequences that are inverted replicas of each other are called **inverted repeats**. The following double-stranded sequence is an example of inverted repeats:



Notice that the sequences on the two strands are the same when read from 5' to 3' but, because the polarities of the two strands are opposite, their sequences are reversed from left to right. An inverted repeat that is complementary to itself, such as:



is also a **palindrome**, defined as a word or sentence that reads the same forward and backward, such as “rotator.” Inverted repeats are palindromes because the sequences on the two strands are the same but in reverse orientation. When an inverted repeat forms a perfect palindrome, the double-stranded sequence reads the same forward and backward.

Another secondary structure, called a **cruciform**, can be made from an inverted repeat when a hairpin forms within each of the two single-stranded sequences. (see Figure 10.17d).

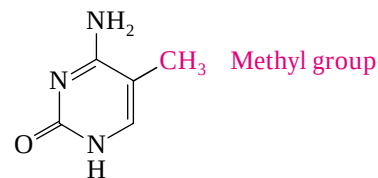
Concepts

In DNA and RNA, base pairing between nucleotides on the same strand produces special secondary structures such as hairpins and cruciforms.

DNA Methylation

The primary structure of DNA can be modified in various ways. These modifications are important in the expression of the genetic material, as we will see in the chapters to come. One such modification is **DNA methylation**, in which methyl groups ($-\text{CH}_3$) are added (by specific enzymes) to certain positions on the nucleotide bases.

In bacteria, adenine and cytosine are commonly methylated, whereas, in eukaryotes, cytosine is the most



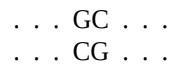
5-Methylcytosine

10.18 In eukaryotic DNA, cytosine bases are often methylated to form 5-methylcytosine.

commonly methylated base. Bacterial DNA is frequently methylated to distinguish it from foreign, unmethylated DNA that may be introduced by viruses; bacteria use proteins called restriction enzymes to cut up any unmethylated viral DNA (see Chapter 18).

In eukaryotic DNA, cytosine bases are often methylated to form **5-methylcytosine** (FIGURE 10.18). The extent of cytosine methylation varies; in most animal cells, about 5% of the cytosine bases are methylated, but more than 50% of the cytosine bases in some plants are methylated. On the other hand, no methylation of cytosine has been detected in yeast cells, and only very low levels of methylation (about 1 methylated cytosine base per 12,500 nucleotides) are found in *Drosophila*. Why eukaryotic organisms differ so widely in their degree of methylation is not clear.

Methylation is most frequent on cytosine nucleotides that sit next to guanine nucleotides on the same strand:



In eukaryotic cells, methylation is often related to gene expression. Sequences that are methylated typically show low levels of transcription while sequences lacking methylation are actively being transcribed (see Chapter 16). Methylation can also affect the three-dimensional structure of the DNA molecule.

Concepts

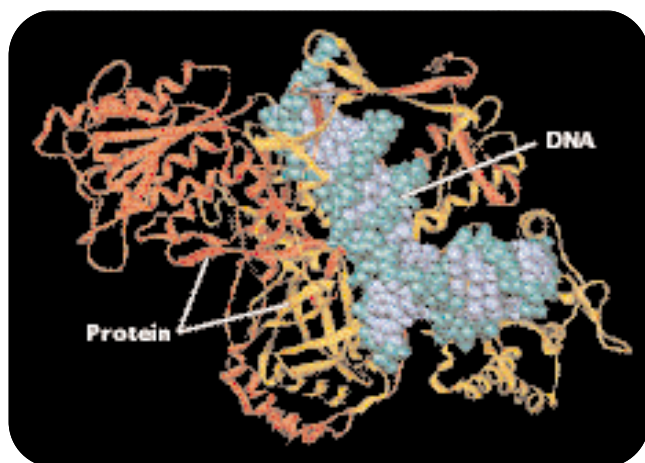
Methyl groups may be added to certain bases in DNA, depending on their positions in the molecule. Both prokaryotic and eukaryotic DNA can be methylated. In eukaryotes, cytosine bases are most often methylated to form 5-methylcytosine, and methylation is often related to gene expression.

www.whfreeman.com/pierce The latest on DNA methylation, at the Web site of the DNA Methylation Society

Bends in DNA

Some specific base sequences—such as a series of four or more adenine–thymine base pairs—cause the DNA double helix to bend. Bending affects how the DNA binds to certain proteins and may be important in controlling the transcription of some genes.

The DNA helix can also be made to bend by the binding of proteins to specific DNA sequences (FIGURE 10.19). The SRY protein, which is encoded by a Y-linked gene and is responsible for sex determination in mammals (see Chapter 4), binds to certain DNA sequences (along the minor groove) and activates nearby genes that encode male traits. When the SRY protein grips the DNA, it bends the molecule about 80 degrees. This distortion of the DNA helix apparently facilitates the binding of other proteins that activate the transcription of genes that encode male characteristics.



10.19 The DNA helix can be bent by the binding of proteins to the DNA molecule.

Connecting Concepts Across Chapters

This chapter has shifted the focus of our study to molecular genetics. The first nine chapters of this book examined various aspects of transmission genetics. In these chapters, the focus was on the individual: which phenotype was produced by an individual genotype, how the genes of an individual were transmitted to the next generation, and what types of offspring were produced when two individuals were crossed. In molecular genetics, our focus now shifts to genes: how they are encoded in DNA, how they are replicated, and how they are expressed.

Much of what follows in this book will depend on your knowledge of DNA. An understanding of all the major processes of information transfer—replication, transcription, and translation—requires an understanding of nucleic acid structure; discussions of recombinant DNA, mutation, gene expression, cancer genetics, and even population genetics are based on the assumption that you understand the basic structure and function of DNA. Thus the information in this chapter provides a critical foundation for much of the remainder of the book.

In this chapter, the history of how DNA's structure and function were unraveled has been strongly emphasized, because the DNA story illustrates how pivotal scientific discoveries are often made. No one scientist discovered the structure of DNA; rather, numerous persons, over a long period of time, made important contributions to our understanding of its structure. Watson and Crick's proposal for DNA's double-helical structure stands out as a singularly important contribution, because it combined many known facts about the structure into a new model that allowed important inferences about the fundamental nature of genes. The DNA story also illustrates the important lesson that science is a human enterprise, influenced by personalities, relations, and motivation.

CONCEPTS SUMMARY

- Genetic material must contain complex information, be replicated accurately, and have the capacity to be translated into the phenotype.
- Evidence that DNA is the source of genetic information came from the finding by Avery, MacLeod, and McCarty that transformation—the genetic alteration of bacteria—was dependent on DNA and from the demonstration by Hershey and Chase that viral DNA is passed on to progeny phages. The results of experiments with tobacco mosaic virus showed that RNA carries genetic information in some viruses.
- James Watson and Francis Crick proposed a new model for the three-dimensional structure of DNA in 1953.
- A DNA nucleotide consists of a deoxyribose sugar, a phosphate group, and a nitrogenous base. RNA consists of a ribose sugar, a phosphate group, and a nitrogenous base.
- The bases of a DNA nucleotide are of two types: purines (adenine and guanine) and pyrimidines (cytosine and thymine). RNA contains the pyrimidine uracil instead of thymine.
- Nucleotides are joined by phosphodiester linkages in a polynucleotide strand. Each polynucleotide strand has a 5' end with a phosphate and a 3' end with a hydroxyl group.
- DNA consists of two nucleotide strands that wind around each other to form a double helix. The sugars and phosphates lie on the outside of the helix, and the bases are stacked in

the interior. Bases from the two strands are joined by hydrogen bonding. The two strands are antiparallel and complementary.

- DNA molecules can form a number of different secondary structures, depending on the conditions in which the DNA is placed and on its base sequence. B-DNA, which consists of a right-handed helix with approximately 10 bases per turn, is the most common form of DNA in cells.
- The structure of DNA has several important genetic implications. Genetic information resides in the base sequence of DNA, which ultimately specifies the amino acid sequence of proteins. Complementarity of the bases on DNA's two strands allows genetic information to be replicated.
- Important pathways by which information passes from DNA to other molecules include: (1) replication, in which one molecule of DNA serves as a template for the synthesis of two new DNA molecules; (2) transcription, in which DNA

serves as a template for the synthesis of an RNA molecule; and (3) translation, in which RNA codes for protein.

- The central dogma of molecular biology proposes that information flows in a one-way direction, from DNA to RNA to protein. Clear exceptions to the central dogma are not known.
- Pairing between bases on the same nucleotide strand can lead to hairpins and other secondary structures. Inverted repeats are sequences on the same strand that are inverted and complementary; they can lead to cruciform structures.
- DNA methylation is the addition of methyl groups to the nucleotide bases. In bacteria, adenine and cytosine are commonly methylated. Among eukaryotes, cytosine bases are most commonly methylated to form 5-methylcytosine.
- Some sequences, such as a series of four or more adenine–thymine base pairs, can cause DNA to bend, which may affect gene expression.

IMPORTANT TERMS

nucleotide (p. 000)	guanine (G) (p. 000)	3' end (p. 000)	reverse transcription (p. 000)
Chargaff's rules (p. 000)	cytosine (C) (p. 000)	antiparallel (p. 000)	RNA replication (p. 000)
transforming principle (p. 000)	thymine (T) (p. 000)	complementary (p. 000)	hairpin (p. 000)
isotopes (p. 000)	uracil (U) (p. 000)	B-DNA (p. 000)	inverted repeats (p. 000)
X-ray diffraction (p. 000)	nucleoside (p. 000)	A-DNA (p. 000)	palindrome (p. 000)
ribose (p. 000)	phosphate group (p. 000)	Z-DNA (p. 000)	cruciform (p. 000)
deoxyribose (p. 000)	deoxyribonucleotide (p. 000)	local variation (p. 000)	DNA methylation (p. 000)
nitrogenous base (p. 000)	ribonucleotide (p. 000)	transcription (p. 000)	5-methylcytosine (p. 000)
purine (p. 000)	phosphodiester linkage (p. 000)	translation (p. 000)	
pyrimidine (p. 000)	polynucleotide strand (p. 000)	replication (p. 000)	
adenine (A) (p. 000)	5' end (p. 000)	central dogma (p. 000)	

Worked Problems

1. The percentage of cytosine in a double-stranded DNA molecule is 40%. What is the percentage of thymine?

• Solution

In double-stranded DNA, A pairs with T, whereas G pairs with C; so the percentage of A equals the percentage of T, and the percentage of G equals the percentage of C. If C = 40%, then G also must be 40%. The total percentage of C + G is therefore 40% + 40% = 80%. All the remaining bases must be either A or T; so the total percentage of A + T = 100% – 80% = 20%; because the percentage of A equals the percentage of T, the percentage of T is 20%/2 = 10%.

2. Which of the following relations will be true for the percentage of bases in double-stranded DNA?

(a) $C + T = A + G$ (b) $\frac{C}{A} = \frac{T}{G}$

• Solution

An easy way to determine whether the relations are true is to arbitrarily assign percentages to the bases, remembering that, in double-stranded DNA, A = T and G = C. For example, if the percentages of A and T are each 30%, then the percentages of G and C are each 20%. We can substitute these values into the equations to see if the relations are true.

- (a) $20 + 30 = 30 + 20$, so this relation is true.

(b) $\frac{20}{30} \neq \frac{30}{20}$; so this relation is not true.

The New Genetics

MINING GENOMES

INTRODUCTION TO GENBANK AND PUBMED

This exercise introduces you to some of the genetics databases that are most frequently used by contemporary researchers. You will explore some of the tools available at the National Center for

Biotechnology Information (NCBI), which is managed by the National Library of Medicine of the United States.

COMPREHENSION QUESTIONS

- * 1. What three general characteristics must the genetic material possess?
2. Briefly outline the history of our knowledge of the structure of DNA until the time of Watson and Crick. Which do you think were the principle contributions and developments?
- * 3. What experiments demonstrated that DNA is the genetic material?
4. What is transformation? How did Avery and his colleagues demonstrate that the transforming principle is DNA?
- * 5. How did Hershey and Chase show that DNA is passed to new phages in phage reproduction?
6. Why was Watson and Crick's discovery so important?
- * 7. Draw and label the three parts of a DNA nucleotide.
8. How does an RNA nucleotide differ from a DNA nucleotide?
9. How does a purine differ from a pyrimidine? What purines and pyrimidines are found in DNA and RNA?
- *10. Draw a short segment of a single polynucleotide strand, including at least three nucleotides. Indicate the polarity of the strand by labeling the 5' end and the 3' end.
11. Which bases are capable of forming hydrogen bonds with each other?
- *12. What is local variation in DNA structure and what causes it?
13. What are some of the important genetic implications of the DNA structure?
- *14. What are the major transfers of genetic information?
15. What are hairpins and how do they form?
16. What is DNA methylation?

APPLICATION QUESTIONS AND PROBLEMS

17. A student mixes some heat-killed type IIS *Streptococcus pneumoniae* bacteria with live type IIR bacteria and injects the mixture into a mouse. The mouse develops pneumonia and dies. The student recovers some type IIS bacteria from the dead mouse. It is the only experiment conducted by the student. Has the student demonstrated that transformation has taken place? What other explanations might explain the presence of the type IIS bacteria in the dead mouse?
- *18. (a) Why did Hershey and Chase choose ^{32}P and ^{35}S for use in their experiment? (b) Could they have used radioactive isotopes of carbon (C) and oxygen (O) instead? Why or why not?
19. What results would you expect if the Hershey and Chase experiment were conducted on tobacco mosaic virus?
- *20. Each nucleotide pair of a DNA double helix weighs about 1×10^{-21} g. The human body contains approximately 0.5 g of DNA. How many nucleotide pairs of DNA are in the human body? If you assume that all the DNA in human cells is in the B-DNA form, how far would the DNA reach if stretched end to end?
21. What aspects of its structure contribute to the stability of the DNA molecule? Why is RNA less stable than DNA?
- *22. Which of the following relations will be found in the percentages of bases of a double-stranded DNA molecule?

(a) $A + T = G + C$

(d) $\frac{A + T}{C + G} = 1.0$

(g) $\frac{A}{G} = \frac{T}{C}$

(b) $A + G = T + C$

(e) $\frac{A + G}{C + T} = 1.0$

(h) $\frac{A}{T} = \frac{G}{C}$

(c) $A + C = G + T$

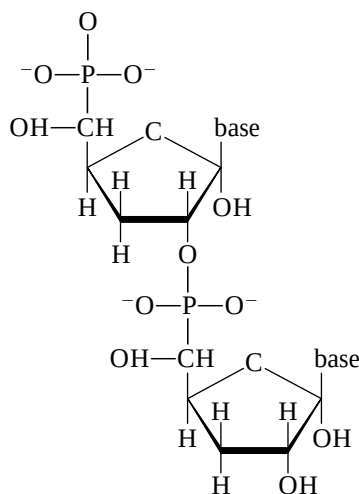
(f) $\frac{A}{C} = \frac{G}{T}$
- *23. If a double-stranded DNA molecule is 15% thymine, what are the percentages of all the other bases?
24. A virus contains 10% adenine, 24% thymine, 30% guanine, and 36% cytosine. Is the genetic material in this virus double-stranded DNA, single-stranded DNA, double-stranded RNA, or single-stranded RNA? Support your answer.
- *25. A B-DNA molecule has 1 million nucleotide pairs.

(a) How many complete turns are there in this molecule?
 (b) If this same molecule were in the Z-DNA configuration, how many complete turns would it have?

26. For entertainment on a Friday night, a genetics professor proposed that his children diagram a polynucleotide strand of DNA. Having learned about DNA in preschool, his 5-year-old daughter was able to draw a polynucleotide strand, but she made a few mistakes. The daughter's diagram (represented here) contained at least 10 mistakes.

(a) Make a list of all the mistakes in the structure of this DNA polynucleotide strand.

(b) Draw the correct structure for the polynucleotide strand.



- *27. Chapter 1 considered the theory of the inheritance of acquired characteristics and noted that this theory is no longer accepted. Is the central dogma consistent with the theory of the inheritance of acquired characteristics? Why or why not?
28. Write a sequence of bases in an RNA molecule that would produce a hairpin structure.
29. The following sequence is present in one strand of a DNA molecule:

5' –CATTGACCGA –3'

Write the sequence on the same strand that produces an inverted repeat and the sequence on the complementary strand.

CHALLENGE QUESTIONS

30. Suppose that an automated, unmanned probe is sent into deep space to search for extraterrestrial life. After wandering for many light-years among the far reaches of the universe, this probe arrives on a distant planet and detects life. The chemical composition of life on this planet is completely different from that of life on Earth, and its genetic material is not composed of nucleic acids. What predictions can you make about the chemical properties of the genetic material on this planet?
31. How might ^{32}P and ^{35}S be used to demonstrate that the transforming principle is DNA? Briefly outline an experiment that would show that DNA and not protein is the transforming principle.
32. Scientists have reportedly isolated short fragments of DNA from fossilized dinosaur bones hundreds of millions of years old. The technique used to isolate this DNA is the polymerase chain reaction (PCR), which is capable of amplifying very small amounts of DNA a millionfold (see Chapter 16). Critics have claimed that the DNA isolated from dinosaur bones is not of ancient origin but instead represents contamination of the samples with DNA from present-day organisms such as bacteria, mold, or humans. What precautions, analyses, and control experiments could be carried out to ensure that DNA recovered from fossils is truly of ancient origin?

SUGGESTED READINGS

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11

Chromosome Structure and Transposable Elements



The house mouse, *Mus musculus*, is one of the oldest and most important organisms used for genetic studies. Molecular techniques allow genes to be introduced into mice on yeast artificial chromosomes (YACs). Pigment in the brown mice is produced by a gene for tyrosinase that is carried on a YAC. The white mouse is a littermate, without the introduced gene. (Carolyn A. McKeone/Photo Researchers.)

YACs and the Common Mouse

The common house mouse, *Mus musculus*, is among the oldest and most valuable subjects for genetic study. It's an excellent genetic organism—small, prolific, and easy to keep, with a short generation time (about 3 months). It tolerates inbreeding well; so a large number of inbred strains have been developed through the years. Finally, being a mammal, the mouse is genetically and physiologically more similar to humans than are other organisms used in genetics studies, such as bacteria, yeast, corn, and fruit flies.

Powerful tools of molecular biology have enhanced the mouse's role in probing fundamental questions of heredity. New and altered genes can be added to the mouse genome

by injecting DNA directly into embryos that are implanted into surrogate mothers. The resulting **transgenic mice** can be bred to produce offspring carrying the new genes.

Today, it is possible to introduce not just individual genes, but entire chromosomes into mouse cells. In 1983, the first artificial chromosomes, made of parts culled from yeast and protozoans, were created for studying chromosome structure and segregation. In 1987, David Burke and Maynard Olson (at Washington University, St. Louis) used yeast to create much larger artificial chromosomes called yeast artificial chromosomes or YACs. Each YAC includes the three essential elements of a chromosome: a centromere, a pair of telomeres, and an origin of replication. These elements ensure that artificial chromosomes will segregate in

- YACs and the Common Mouse
- Packing DNA into Small Spaces
- The Bacterial Chromosome
- The Eukaryotic Chromosome
 - Chromatin Structure
 - Centromere Structure
 - Telomere Structure
- Variation in Eukaryotic DNA Sequences
 - Denaturation and Renaturation of DNA
 - Renaturation Reactions and C_0t Curves
 - Types of DNA sequence in Eukaryotes
- The Nature of Transposable Elements
 - General Characteristics of Transposable Elements
 - Transposition
 - Mechanisms of Transposition
 - The Mutagenic Effects of Transposition
 - The Regulation of Transposition
- The Structure of Transposable Elements
 - Transposable Elements in Bacteria
 - Transposable Elements in Eukaryotes
- The Evolution of Transposable Elements

mitosis and meiosis, will not be degraded, and will replicate successfully. Large chunks of extra DNA from any source can be added to a YAC, and the new artificial chromosome can be inserted into a cell. Eukaryotic centromeres, telomeres, and origins of replication are similar in different organisms; so YACs function well in almost any eukaryotic cell.

In 1993, molecular geneticists successfully modified YACs so that they could be transferred to mouse cells. Previously, transgenic mice could carry only relatively small pieces of DNA, usually no more than 50,000 bp. Now, large genes as well as the surrounding DNA, which may be important in the regulation of those genes, can be added to mouse-cell nuclei. Artificial chromosomes have also been made from chromosomal components of bacteria (BACs) and mammals (MACs).

The successful construction of YACs, BACs, and MACs illustrates the fundamental nature of eukaryotic chromosomes: huge amounts of DNA complexed with proteins and possessing telomeres, centromeres, and origins of replication. In this chapter, we explore the molecular nature of chromosomes, including details of the DNA–protein complex and the structure of telomeres and centromeres; origins of replication will be discussed in Chapter 12.

Much of this chapter focuses on a storage problem: how to cram tremendous amounts of DNA into the limited confines of a cell. Even in those organisms having the smallest amounts of DNA, the length of genetic material far exceeds the length of the cell. Thus, cellular DNA must be highly folded and tightly packed, but this packing creates problems—it renders the DNA inaccessible, unable to be copied or read. Functional DNA must be capable of partly unfolding and expanding so that individual genes can undergo replication and transcription. The flexible, dynamic nature of DNA packing will be a central theme of this chapter.

We begin this chapter by considering supercoiling, an important tertiary structure of DNA found in both prokaryotic and eukaryotic cells. After a brief look at the bacterial chromosome, we examine the structure of eukaryotic chromosomes. After considering chromosome structure, we pay special attention to the working parts of a chromosome, specifically centromeres and telomeres. We also consider the types of DNA sequences present in many eukaryotic chromosomes and how DNA sequences are analyzed.

The second part of this chapter focuses on genes that move. For many years, biologists viewed genes as static entities that occupied fixed positions on chromosomes. But we now recognize that many genetic elements do not occupy fixed positions. Genes that can move have been given a variety of names, including transposons, transposable genetic elements, mobile DNA, movable genes, controlling elements, and jumping genes. We will refer to mobile DNA sequences as **transposable elements**, and by this term we mean any DNA sequence that is capable of moving from one place to another place within the genome.

We begin the second part of the chapter by outlining some of the general features of transposable elements and the processes by which they move from place to place. We then consider several different types of transposable elements found in prokaryotic and eukaryotic genomes. Finally, we consider the evolutionary significance of transposable elements.

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More information about YACs and how genes are cloned into YACs and more information about mouse genetics

Packing DNA into Small Spaces

The packaging of tremendous amounts of genetic information into the small volume of a cell has been called the ultimate storage problem. Consider the chromosome of the bacterium *E. coli*, a single molecule of DNA with approximately 4.64 million base pairs. Stretched out straight, this DNA would be about 1000 times as long as the cell within which it resides (FIGURE 11.1). Human cells contain 6 billion base pairs of DNA, which would measure some 1.8 meters stretched end to end. Even DNA in the smallest human chromosome would stretch 14,000 times the length of the nucleus. Clearly, DNA molecules must be tightly packed to fit into such small spaces.

The structure of DNA can be considered at three hierarchical levels: the primary structure of DNA is its nucleotide sequence; the secondary structure is the double-stranded helix; and the tertiary structure refers to higher-order folding that allows DNA to be packed into the confined space of a cell.

Concepts

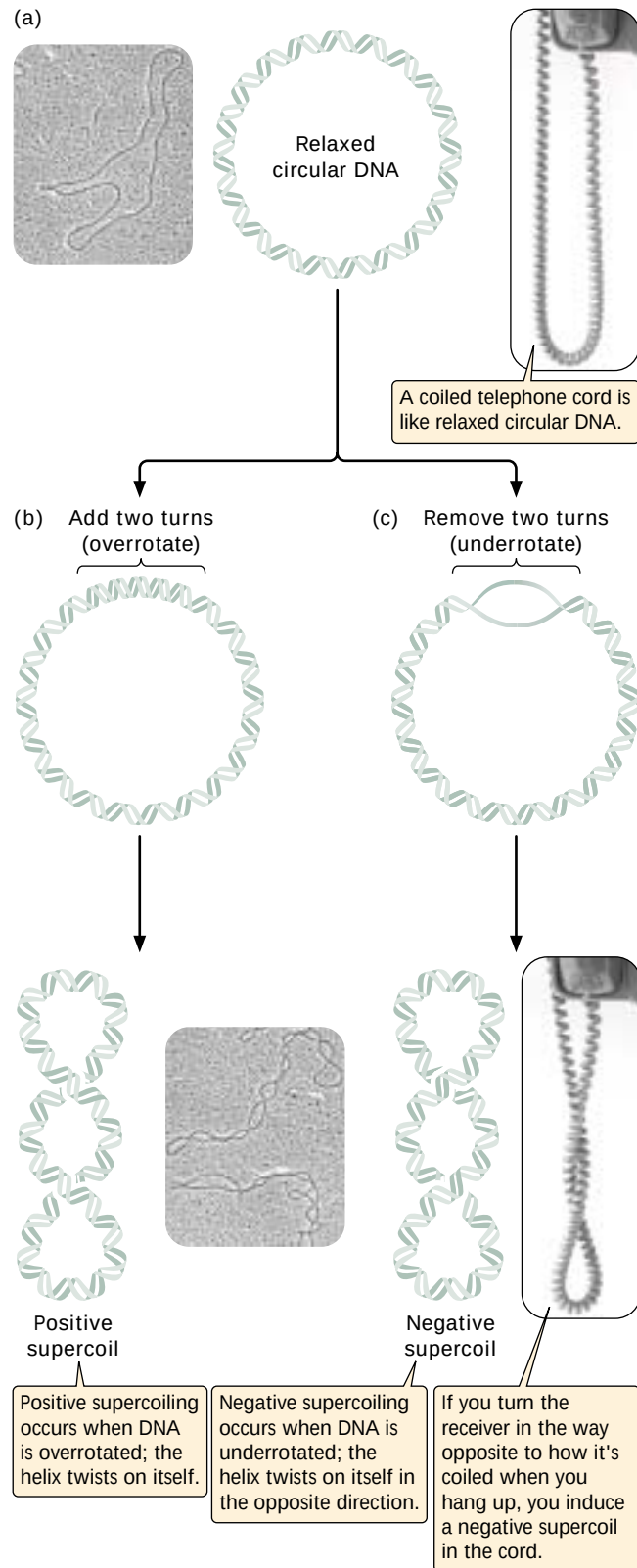
Chromosomal DNA exists in the form of very long molecules, which must be tightly packed to fit into the small confines of a cell.



One type of DNA tertiary structure is **supercoiling**, which occurs when the DNA helix is subjected to strain by being overwound or underwound. The lowest energy state for B-DNA is when it has approximately 10 bp per turn of its helix. In this **relaxed state**, a stretch of 100 bp of DNA would assume about 10 complete turns. (FIGURE 11.2a). If energy is used to add or remove any turns by rotating one strand around the other, strain is placed on the molecule, causing the helix to supercoil, or twist, on itself (FIGURE 11.2b and c).

Supercoiling is a natural consequence of the overrotating or underrotating of the helix; it occurs only when the

11.1 (overleaf, encircling pp. 4–5) The DNA in *E. coli* is about 1000 times as long as the cell itself.



11.2 Supercoiled DNA is overwound or underwound, causing it to twist on itself. Electron micrographs are of relaxed DNA (top) and supercoiled DNA (bottom). (Dr. Gopal Murti/Phototake.)

molecule is placed under strain. Molecules that are overrotated exhibit **positive supercoiling** (see Figure 11.2b). Underrotated molecules exhibit **negative supercoiling** (see Figure 11.2c), in which the direction of the supercoil is opposite that of the right-handed coil of the DNA helix.

Supercoiling occurs only if the two polynucleotide strands of the DNA double helix are unable to rotate about each other freely. If the chains *can* turn freely, their ends will simply turn as extra rotations are added or removed, and the molecule will spontaneously revert to the relaxed state. Supercoiling takes place when the strain of overrotating or underrotating cannot be compensated by the turning of the ends of the double helix, which is the case if the DNA is circular—that is, there are no free ends. Some viral chromosomes are in the form of simple circles and readily undergo supercoiling. Large molecules of bacterial DNA are typically a series of large loops, the ends of which are held together by proteins. Eukaryotic DNA is normally linear but also tends to fold into loops stabilized by proteins. In these chromosomes, the anchoring proteins prevent free rotation of the ends of the DNA; so supercoiling does take place.

Supercoiling relies on **topoisomerases**, enzymes that add or remove rotations from the DNA helix by temporarily breaking the nucleotide strands, rotating the ends around each other, and then rejoining the broken ends. The two classes of topoisomerases are: type I, which breaks only one of the nucleotide strands and reduces supercoiling by removing rotations; and type II, which adds or removes rotations by breaking both nucleotide strands.

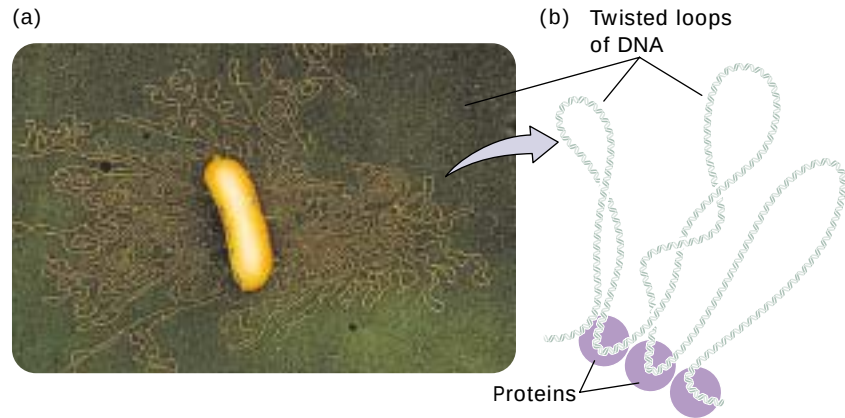
Most DNA found in cells is negatively supercoiled, which has two advantages for the cell. First, supercoiling makes the separation of the two strands of DNA easier during replication and transcription. Negatively supercoiled DNA is underrotated; so separation of the two strands during replication and transcription is more rapid and requires less energy. Second, supercoiled DNA can be packed into a smaller space because it occupies less volume than relaxed DNA.

Concepts

Overrotation or underrotation of a DNA double helix places strain on the molecule, causing it to supercoil. Supercoiling is controlled by topoisomerase enzymes. Most cellular DNA is negatively supercoiled, which eases the separation of nucleotide strands during replication and transcription and allows DNA to be packed into small spaces.

The Bacterial Chromosome

Most bacterial genomes consist of a single, circular DNA molecule, although linear DNA molecules have been found in a few species. In circular bacterial chromosomes, the



11.3 Bacterial DNA is highly folded into a series of twisted loops. (Part a, Dr. Gopal Murti/Photo Researchers.)

DNA does not exist in an open, relaxed circle; the 3 million to 4 million base pairs of DNA found in a typical bacterial genome would be much too large to fit into a bacterial cell (see Figure 11.1). Bacterial DNA is not attached to histone proteins (as is eukaryotic DNA, discussed later in the chapter). Consequently, for many years bacterial DNA was called “naked DNA.” However, this term is inaccurate, because bacterial DNA is complexed to a number of proteins that help compact it.

When a bacterial cell is viewed with the electron microscope, its DNA frequently appears as a distinct clump, the **nucleoid**, which is confined to a definite region of the cytoplasm. If a bacterial cell is broken open gently, its DNA spills out in a series of twisted loops (● **FIGURE 11.3a**). The ends of the loops are most likely held in place by proteins (● **FIGURE 11.3b**). Many bacteria contain additional DNA in the form of small circular molecules called plasmids, which replicate independently of the chromosome (see Chapter 8).

Concepts

The typical bacterial chromosome consists of a large, circular molecule of DNA that is a series of twisted loops. Bacterial DNA appears as a distinct clump, the nucleoid, within the bacterial cell.

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Information about the genome of the common bacterium *E. coli*

The Eukaryotic Chromosome

Individual eukaryotic chromosomes contain enormous amounts of DNA. Like bacterial chromosomes, each eukaryotic chromosome consists of a single, extremely long molecule of DNA. For all of this DNA to fit into the nucleus, tremendous packing and folding are required, the extent of which must change through time. The chromosomes are in an elongated, relatively uncondensed state during interphase of the cell cycle (see p. 000 in Chapter 2), but the term *relatively* is an important qualification here.

Although the DNA of interphase chromosomes is less tightly packed than DNA in mitotic chromosomes, it is still highly condensed; it's just *less* condensed. In the course of the cell cycle, the level of DNA packaging changes—chromosomes progress from a highly packed state to a state of extreme condensation. DNA packaging also changes locally in replication and transcription, when the two nucleotide strands must unwind so that particular base sequences are exposed. Thus, the packaging of eukaryotic DNA (its tertiary, chromosomal structure) is not static but changes regularly in response to cellular processes.

Chromatin Structure

As mentioned in Chapter 2, eukaryotic DNA is closely associated with proteins, creating *chromatin*. The two basic types of chromatin are: **euchromatin**, which undergoes the normal process of condensation and decondensation in the cell cycle, and **heterochromatin**, which remains in a highly condensed state throughout the cell cycle, even during interphase. Euchromatin constitutes the majority of the chromosomal material, whereas heterochromatin is found at the centromeres and telomeres of all chromosomes, at other specific places on some chromosomes, and along the entire inactive X chromosome in female mammals (see p. 000 in Chapter 4).

The most abundant proteins in chromatin are the *histones*, which are relatively small, positively charged proteins of five major types: H1, H2A, H2B, H3, and H4 (Table 11.1). All histones have a high percentage of arginine and lysine, positively charged amino acids that give them a net positive charge. The positive charges attract the negative charges on the phosphates of DNA and holds the DNA in contact with the histones.

A heterogeneous assortment of **nonhistone chromosomal proteins** make up about half of the protein mass of the chromosome. A fundamental problem in the study of these proteins is that the nucleus is full of all sorts of proteins; so, whenever chromatin is isolated from the nucleus, it may be contaminated by nonchromatin proteins. On the other hand, isolation procedures may also remove proteins that

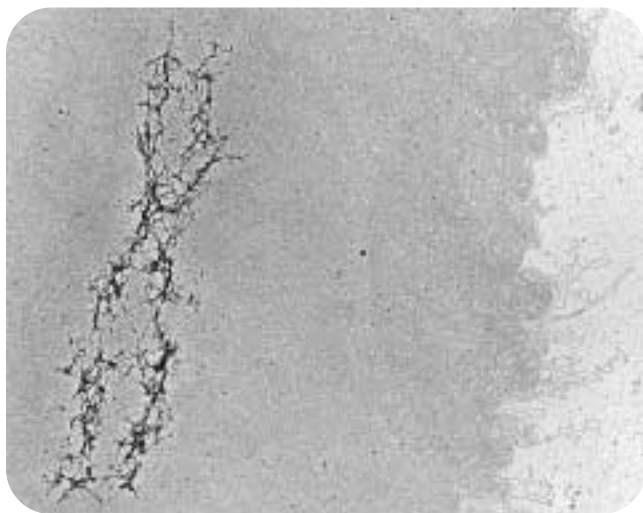
Table 11.1 Characteristics of histone proteins

Histone Protein	Molecular Weight	Number of Amino Acids
H1	21,130	223
H2A	13,960	129
H2B	13,774	125
H3	15,273	135
H4	11,236	102

Note: The sizes of H1, H2A, and H2B histones vary somewhat from species to species. The values given are for bovine histones. Source: Data are from A.L. Lehninger, D. L. Nelson, and M. M. Cox, *Principles of Biochemistry*, 3d ed. (New York: Worth Publishers, 1993), p. 924.

are associated with chromatin. In spite of these difficulties, we know that some groups of nonhistone proteins are clearly associated with chromatin.

Nonhistone chromosomal proteins may be broadly divided into those that serve structural roles and those that take part in genetic processes such as transcription and replication. **Chromosomal scaffold proteins** (FIGURE 11.4) are revealed when chromatin is treated with a concentrated salt solution, which removes histones and most other chromosomal proteins, leaving a chromosomal protein “skeleton” to which the DNA is attached. These scaffold proteins may play a role in the folding and packing of the chromosome. Other structural proteins make up the kinetochore, cap the chromosome ends by attaching to telomeres, and constitute the molecular motors that move chromosomes in mitosis and meiosis.



11.4 Scaffold proteins play a role in the folding and packing of chromosomes. (Professor U. Laemmli/Photo Researchers.)

Other types of nonhistone chromosomal proteins play a role in genetic processes. They are components of the replication machinery (DNA polymerases, helicases, primases; see Chapter 12) and proteins that carry out and regulate transcription (RNA polymerases, transcription factors, acetylases; see Chapter 13). **High-mobility-group proteins** are small, highly charged proteins that vary in amount and composition, depending on tissue type and stage of the cell cycle. Several of these proteins may play an important role in altering the packing of chromatin during transcription.

The highly organized structure of chromatin is best viewed from several levels. In the next sections, we will examine these levels of chromatin organization.

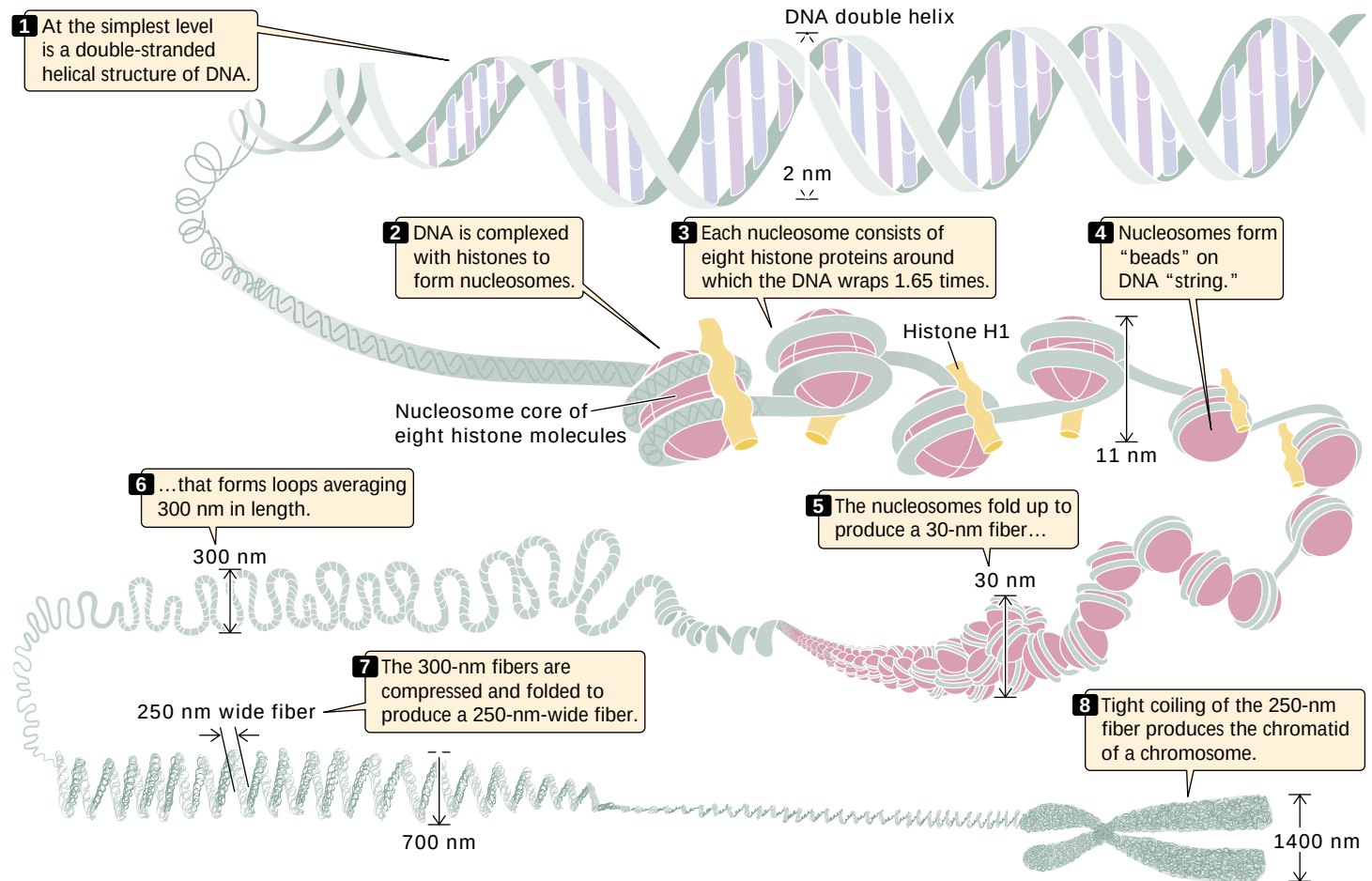
Concepts

Chromatin, which consists of DNA complexed to proteins, is the material that makes up eukaryotic chromosomes. The most abundant of these proteins are the five types of positively charged histone proteins: H1, H2A, H2B, H3, and H4.

The nucleosome Chromatin has a highly complex structure with several levels of organization. The simplest level (FIGURE 11.5) is the double helical structure of DNA discussed in Chapter 8. At a more complex level, the DNA molecule is associated with proteins and is highly folded to produce a chromosome.

When chromatin is isolated from the nucleus of a cell and viewed with an electron microscope, it frequently looks like beads on a string (FIGURE 11.6a on page 000). If a small amount of nuclease is added to this structure, the enzyme cleaves the string between the beads, leaving individual beads attached to about 200 bp of DNA (FIGURE 11.6b). If more nuclease is added, the enzyme chews up all of the DNA between the beads and leaves a core of proteins attached to a fragment of DNA (FIGURE 11.6c). Such experiments demonstrated that chromatin is not a random association of proteins and DNA but has a fundamental repeating structure.

The repeating core of protein and DNA produced by digestion with nuclease enzymes is the simplest level of chromatin structure, the **nucleosome** (see Figure 11.5). The nucleosome is a core particle consisting of DNA wrapped about two times around an octamer of eight histone proteins (two copies each of H2A, H2B, H3, and H4), much like thread wound around a spool (FIGURE 11.6d). The DNA in direct contact with the histone octamer is between 145 and 147 bp in length, coils around the histones in a left-handed direction, and is supercoiled. It does not wrap around the octamer smoothly; there are four bends,



11.5 Chromatin has a highly complex structure with several levels of organization.

or kinks, in its helical structure as it winds around the histones.

The fifth type of histone, H1, is not a part of the core particle but plays an important role in the nucleosome structure. The precise location of H1 with respect to the core particle is still uncertain. The traditional view is that H1 sits outside the octamer and binds to the DNA where the DNA joins and leaves the octamer (see Figure 11.5). However, the results of recent experiments suggest that the H1 histone sits *inside* the coils of the nucleosome. Regardless of its position, H1 helps to lock the DNA into place, acting as a clamp around the nucleosome octamer.

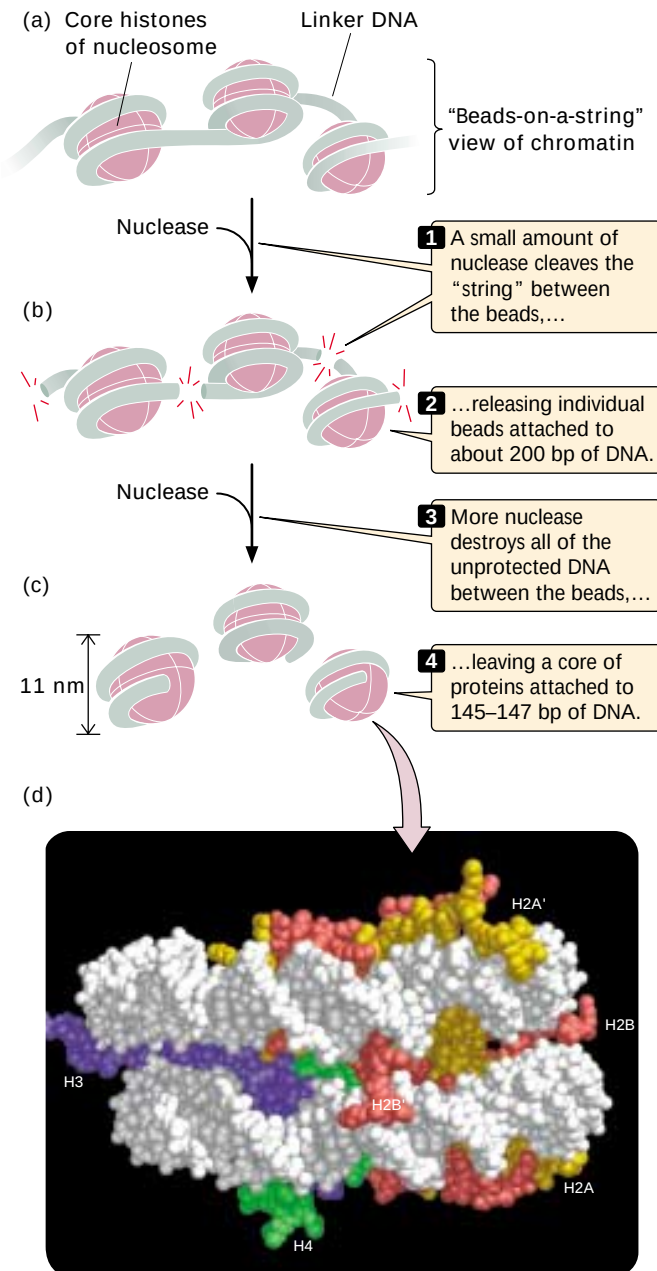
Together, the core particle and its associated H1 histone are called the **chromatosome**, the next level of chromatin organization. The H1 protein is attached to between 20 and 22 bp of DNA, and the nucleosome encompasses an additional 145 to 147 bp of DNA; so about 167 bp of DNA are held within the chromatosome. Chromatosomes are located at regular intervals along the DNA molecule and are separated from one another by

linker DNA, which varies in size among cell types—most cells have from about 30 bp to 40 bp of linker DNA. Nonhistone chromosomal proteins may be associated with this linker DNA, and a few also appear to bind directly to the core particle.

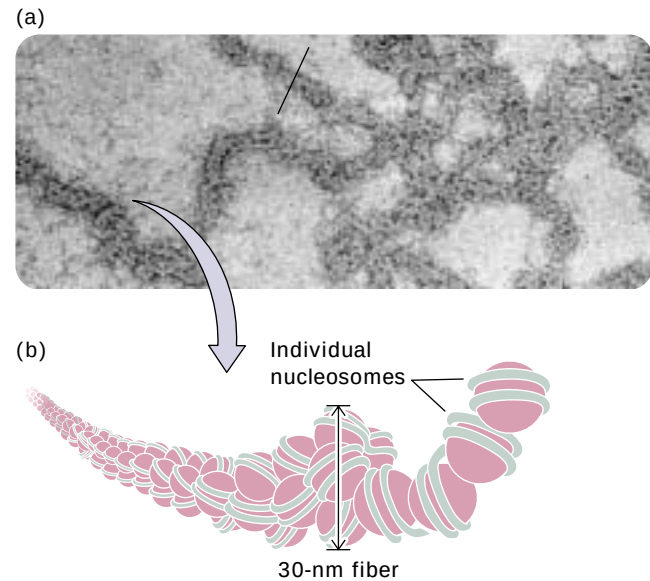
Higher-order chromatin structure In chromosomes, adjacent nucleosomes are not separated by space equal to the length of the linker DNA; rather, nucleosomes fold on themselves to form a dense, tightly packed structure (see Figure 11.5). This structure is revealed when nuclei are gently broken open and their contents are examined with the use of an electron microscope; much of the chromatin that spills out appears as a fiber with a diameter of about 30 nm (FIGURE 11.7a). A model of how this 30-nm fiber forms is shown in FIGURE 11.7b.

The next-higher level of chromatin structure is a series of loops of 30-nm fibers, each anchored at its base by proteins in the nuclear scaffold (see Figure 11.5). On average, each loop encompasses some 20,000 to 100,000 bp of

DNA and is about 300 nm in length, but the individual loops vary considerably. The 300-nm fibers are packed and folded to produce a 250-nm-wide fiber. Tight helical coiling of the 250-nm fiber, in turn, produces the structure that appears in metaphase: an individual chromatid approximately 700 nm in width.



11.6 The nucleosome is the fundamental repeating unit of chromatin. The space-filling model shows that the nucleosome core particle consists of two copies each of H2A, H2B, H3, and H4, around which DNA (white) coils. (Part d, from K. Luger et al., 1997, *Nature* 389:251; courtesy of T. H. Richmond.)



11.7 Adjacent nucleosomes pack together to form a 30-nm fiber. (Part a, Barbara Hamkalo, Molecular Biology and Biochemistry, University of California at Irvine.)

Concepts

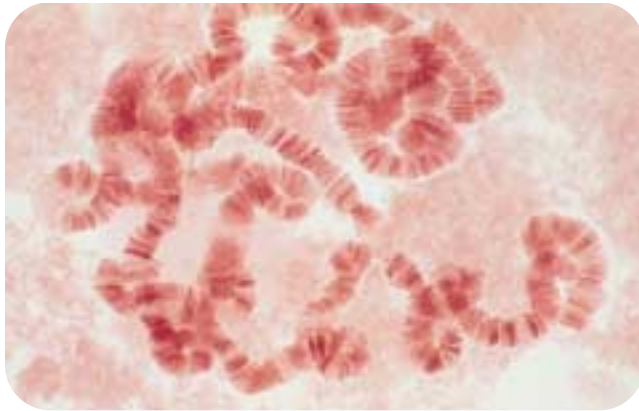
The nucleosome consists of a core particle of eight histone proteins and DNA, about 146 bp in length, that wraps around the core. Chromosomes, each including the core particle plus an H1 histone, are separated by linker DNA. Nucleosomes fold up to form a 30-nm chromatin fiber, which appears as a series of loops that pack to create a 250-nm-wide fiber. Helical coiling of the 250-nm fiber produces a 700-nm-wide chromatid.

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A virtual tour of nucleosome and chromatin structure and links to additional sites on chromatin structure and research

Changes in chromatin structure Although eukaryotic DNA must be tightly packed to fit into the cell nucleus, it must also periodically unwind to undergo transcription and replication. Evidence of the changing nature of chromatin structure is seen in the puffs of polytene chromosomes and in the sensitivity of genes to digestion by DNase I.

Polytene chromosomes are giant chromosomes found in certain tissues of *Drosophila* and some other organisms (**FIGURE 11.8**). These large, unusual chromosomes arise when repeated rounds of DNA replication take place without accompanying cell divisions, producing thousands of copies of DNA that lie side by side. When polytene chromosomes are stained with dyes, numerous bands are revealed. Under certain conditions, the bands may exhibit **chromosomal puffs**—localized swellings of the chromosome. Each puff is a region of the chromatin that has relaxed its



11.8 Polytene chromosomes are giant chromosomes isolated from the salivary glands of larval *Drosophila*. Each puff represents a region of relaxed chromatin where transcription is taking place. (Andrew Syred/Science Photo Library/Photo Researchers.)

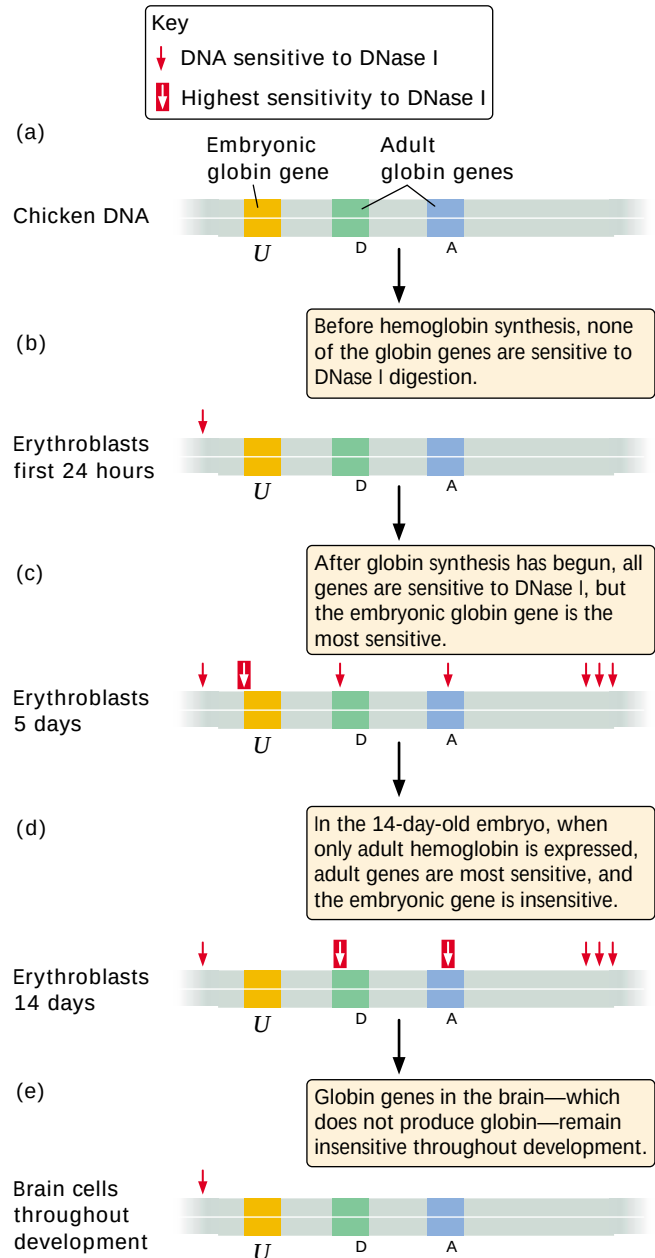
structure, assuming a more open state. If radioactively labeled uridine (a precursor to RNA) is briefly added to a *Drosophila* larva, radioactivity accumulates in chromosomal puffs, indicating that they are regions of active transcription. Additionally, the appearance of puffs at particular locations on the chromosome can be stimulated by exposure to hormones and other compounds that are known to induce the transcription of genes at those locations. This correlation between the occurrence of transcription and the relaxation of chromatin at a puff site indicates that chromatin structure undergoes dynamic change associated with gene activity.

A second piece of evidence indicating that chromatin structure changes with gene activity is sensitivity to DNase I, an enzyme that digests DNA. The ability of this enzyme to digest DNA depends on chromatin structure: when DNA is tightly bound to histone proteins, it is less sensitive to digestion by DNase I, whereas unbound DNA is more sensitive to digestion by DNase I. The results of experiments that examine the effect of DNase I on specific genes show that DNase sensitivity is correlated with gene activity. For example, globin genes code for hemoglobin in the erythroblasts (precursors of red blood cells) of chickens. The forms of hemoglobin produced in chick embryos and chickens are different and are encoded by different genes (FIGURE 11.9a). However, no hemoglobin is synthesized in chick embryos in the first 24 hours after fertilization. If DNase I is applied to chromatin from chick erythroblasts in this first 24-hour period, all the globin genes are insensitive to digestion (FIGURE 11.9b). From day 2 to day 6 after fertilization, after hemoglobin synthesis has begun, the globin genes become sensitive to DNase I, and the genes that code for embryonic hemoglobin are the most sensitive (FIGURE 11.9c). After 14 days of development, embryonic hemoglobin is replaced by the adult forms of hemoglobin. The most

Experiment

Question: Is chromatin structure altered during transcription?

Method: Sensitivity to DNase I was tested on different tissues and at different times in development.



Conclusion: Sensitivity of DNA to digestion by DNase I is correlated with gene expression, suggesting that chromatin structure changes during transcription.

11.9 DNase I sensitivity is correlated with the transcription of globin genes in erythroblasts of chick embryos. The U gene codes for embryonic hemoglobin; the α^D and α^A genes code for adult hemoglobin.

sensitive regions now lie near the genes that produce the adult hemoglobins (FIGURE 11.9d). DNA from brain cells, which produce no hemoglobin, remains insensitive to DNase digestion throughout development (FIGURE 11.9e). In summary, when genes become transcriptionally active, they also become sensitive to DNase I, indicating that the chromatin structure is more exposed during transcription.

What is the nature of the change in chromatin structure that produces chromosome puffs and DNase I sensitivity? In both cases, the chromatin relaxes; presumably the histones loosen their grip on the DNA. One process that appears to be implicated in changing chromatin structure is acetylation, a reaction that adds chemical groups called acetyls to the histone proteins. Enzymes called acetyltransferases attach acetyl groups to lysine amino acids at one end (called a tail) of the histone protein. This modification reduces the positive charges that normally exist on lysine and destabilizes the nucleosome structure, and so the histones hold the DNA less tightly. Proteins taking part in transcription can then bind more easily to the DNA and carry out transcription.

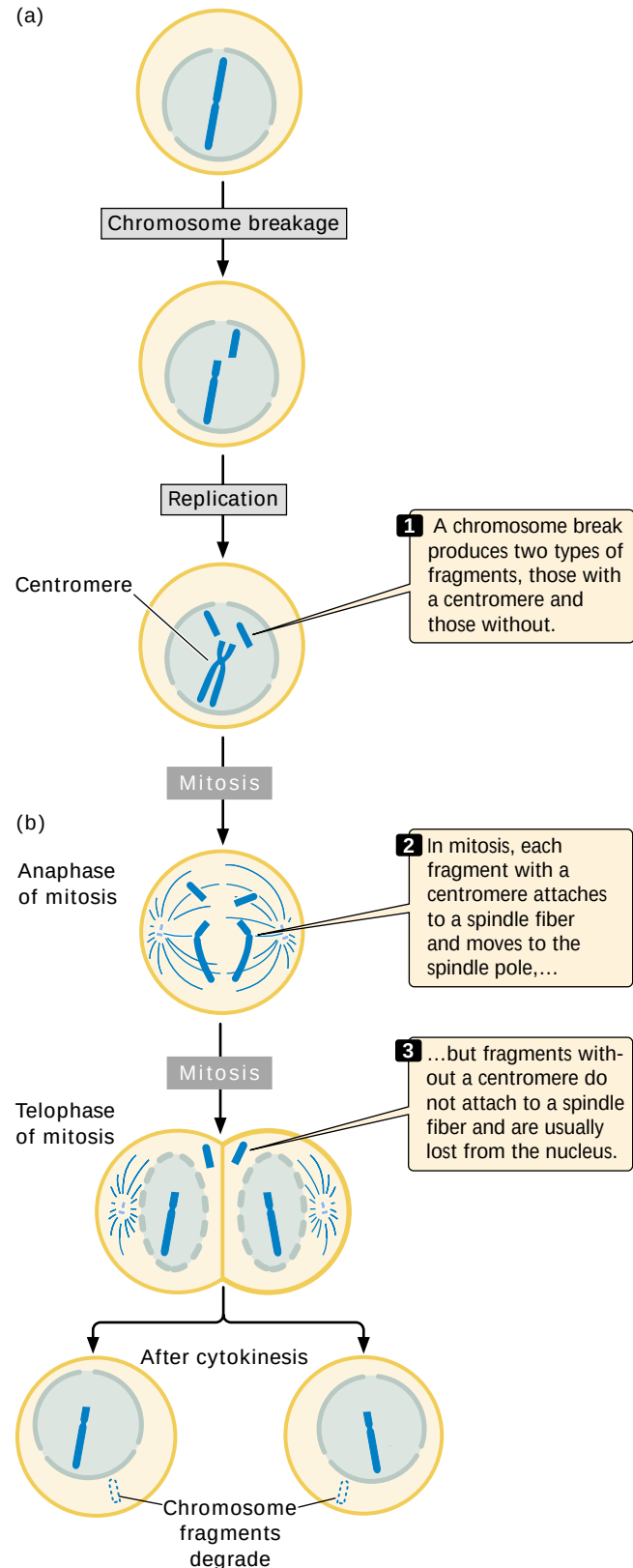
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Images of polytene chromosomes

Centromere Structure

The centromere is a constricted region of the chromosome where spindle fibers attach and is essential for proper movement of the chromosome in mitosis and meiosis (Chapter 2). The essential role of the centromere in chromosome movement was recognized by early geneticists, who observed what happens when a chromosome breaks in two. A chromosome break produces two fragments, one with a centromere and one without (FIGURE 11.10a). In mitosis, the chromosome fragment containing the centromere attaches to spindle fibers and moves to the spindle pole, whereas the fragment lacking a centromere never connects to a spindle fiber and is usually lost because it fails to move into the nucleus of a daughter cell (FIGURE 11.10b).

Although the centromere's role in chromosome movement has been recognized for some time, its molecular nature has only recently been revealed. The first centromeres to be isolated and studied at the molecular level came from yeast, which have small, linear chromosomes. When molecular biologists attached sequences from yeast centromeres to plasmids (small circular DNA molecules that don't have centromeres), the plasmids behaved in mitosis as if they were eukaryotic chromosomes. This finding indicated that the sequences from yeast, called **centromeric sequences** (FIGURE 11.11), contain a functional centromere that allows segregation to take place. Centromeric sequences are the binding sites for proteins that function as the *kinetochore*, a complex that assembles on the centromere and to which the spindle fibers attach.



11.10 Chromosome fragments that lack a centromere are lost in mitosis.

Chromosomes in these cells don't progressively shorten and self-destruct, because the cells possess an enzyme called telomerase that replicates the telomeres. The ability of telomerase to replicate a chromosome end depends on the unique molecular structure of the telomere. We will examine this mechanism of replication in Chapter 12.

Telomeres were first isolated from the protozoan *Tetrahymena thermophila* and were found to possess multiple copies of the sequence:



Telomeres have now been isolated from protozoans, plants, humans, and other organisms; most are similar in structure (Table 11.2). These **telomeric sequences** usually consist of a series of cytosine nucleotides followed by several adenine or thymine nucleotides or both, taking the form 5'–C_n(A or T)_m–3', where *n* is 2 or greater and *m* is from 1 to 4. For example, the repeating unit in human telomeres is CCCTAA, which may be repeated from 250 to 1500 times. The sequence is always oriented with the string of Cs andGs toward the end of the chromosome, as shown here:

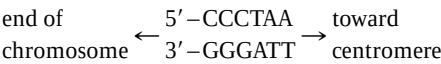
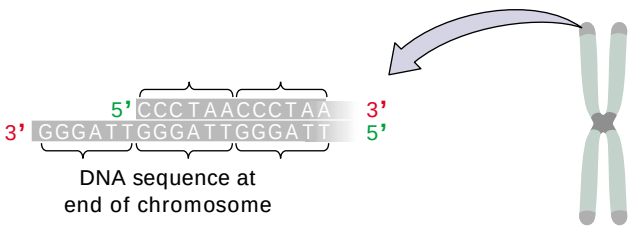


Table 11.2 DNA sequences typically found in telomeres of various organisms	
Organism	Sequence
<i>Tetrahymena</i> (protozoan)	5'–CCCCAA–3' 3'–GGGGTT–5'
<i>Oxytricha</i> (protozoan)	5'–CCCCAAAA–3' 3'–GGGGTTTT–5'
<i>Trypanosoma</i> (protozoan)	5'–CCCTAA–3' 3'–GGGATT–5'
<i>Saccharomyces</i> (yeast)	5'–C ₂₋₃ ACA ₁₋₆ –3' 3'–G ₂₋₃ TGT ₁₋₆ –5'
<i>Neurospora</i> (fungus)	5'–CCCTAA–3' 3'–GGGATT–5'
<i>Caenorhabditis</i> (nematode)	5'–GCCTAA–3' 3'–CGGATT–5'
<i>Bombyx</i> (insect)	5'–CCTAA–3' 3'–GGATT–5'
Vertebrate	5'–CCCTAA–3' 3'–GGGATT–5'
<i>Arabidopsis</i> (plant)	5'–CCCTAAA–3' 3'–GGGATTT–5'

Source: V. A. Zakian, *Science* 270(1995): 1602.



11.12 DNA at the ends of eukaryotic chromosomes consists of telomeric sequences.

The G-rich strand often protrudes beyond the complementary C-rich strand at the end of the chromosome (FIGURE 11.12). The length of the telomeric sequence varies from chromosome to chromosome and from cell to cell, suggesting that each telomere is a dynamic structure that actively grows and shrinks. The telomeres of *Drosophila* chromosomes are different in structure. They consist of multiple copies of the two different retrotransposons (discussed later in this chapter), *Het-A* and *Tart*, arranged in tandem repeats. Apparently, in *Drosophila*, loss of telomere sequences during replication is balanced by transposition of additional copies of the *Het-A* and *Tart* elements.

Farther away from the end of the chromosome, from several thousand to hundreds of thousands of base pairs form **telomere-associated sequences**. They, too, contain repeated sequences, but the repeats are longer, more varied, and more complex than those found in telomeric sequences.

Concepts

A telomere is the stabilizing end of a chromosome. At the end of each telomere are many short telomeric sequences. Longer, more complex telomere-associated sequences are found adjacent to the telomeric sequences.

www.whfreeman.com/pierce More detailed information on telomeres

Variation in Eukaryotic DNA Sequences

Prokaryotic and eukaryotic cells differ dramatically in the amount of DNA per cell, a quantity termed an organism's **C value** (Table 11.3). Each cell of a fruit fly, for example, contains 35 times the amount of DNA found in a cell of the bacterium *E. coli*. In general, eukaryotic cells contain more DNA than that of prokaryotes, but variability in the C values of different eukaryotes is huge. Human cells contain more than 10 times the amount of DNA found in *Drosophila* cells, whereas some salamander cells contain 20

Table 11.3 Genome sizes of various organisms

Organism	Approximate Genome Size (bp)
λ (bacteriophage)	50,000
<i>E. coli</i> (bacterium)	4,600,000
<i>Saccharomyces cerevisiae</i> (yeast)	13,500,000
<i>Arabidopsis thaliana</i> (plant)	100,000,000
<i>Drosophila melanogaster</i> (insect)	140,000,000
<i>Homo sapiens</i> (human)	3,000,000,000
<i>Zea mays</i> (corn)	4,500,000,000
<i>Amphiuma</i> (salamander)	765,000,000,000

times as much DNA as that of human cells. Clearly, these differences in C value cannot be explained simply by differences in organismal complexity. So what is all this extra DNA in eukaryotic cells doing? We do not yet have a complete answer to this question, but examination of DNA sequences has revealed that eukaryotic DNA has complexity that is absent from prokaryotic DNA.

Denaturation and Renaturation of DNA

The first clue that the DNA of eukaryotes contains several types of sequences came from the results of studies in which double-stranded DNA was separated and then allowed to reassociate. When double-stranded DNA in solution is heated, the hydrogen bonds that hold the two strands together are weakened and, with enough heat, the two nucleotide strands separate completely, a process called **denaturation** or melting (FIGURE 11.13). DNA is typically denatured within a narrow temperature range. The midpoint of this range, the **melting temperature** (T_m), depends on the base sequence of a particular sample of DNA: G–C base pairs have three hydrogen bonds, whereas A–T base

pairs only have two; so the separation of G–C pairs requires more energy than does the separation of A–T pairs. A DNA molecule with a higher percentage of G–C pairs will therefore have a higher T_m than that of DNA with more A–T pairs.

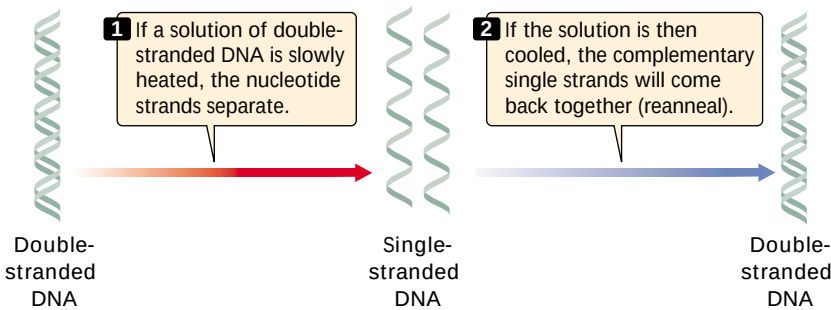
The denaturation of DNA by heating is reversible; if single-stranded DNA is slowly cooled, single strands will collide and hydrogen bonds will again form between complementary base pairs, producing double-stranded DNA (see Figure 11.13). This reaction, called **renaturation** or reannealing, takes place in two steps. First, single strands in solution collide randomly with their complementary strands. Second, hydrogen bonds form between complementary bases.

Two single-stranded molecules of DNA from different sources will anneal if they are complementary, a process termed **hybridization**. For hybridization to take place, the two strands do not have to be complementary at all their bases—just at enough bases to hold the two strands together. The extent of hybridization can be used to measure the similarity of nucleic acids from two different sources and is a common tool for assessing evolutionary relationships. The rate at which hybridization takes place also provides information about the sequence complexity of DNA (see next subsection).

Renaturation Reactions and C_0t Curves

In a typical renaturation reaction, DNA molecules are first sheared into fragments several hundred base pairs in length. Next, the fragments are heated to about 100°C, which causes the DNA to denature. The solution is then cooled slowly, and the amount of renaturation is measured by observing optical absorbance. Double-stranded DNA absorbs less UV light than does single-stranded DNA; so the amount of renaturation can be monitored by shining a UV light through the solution and measuring the amount of the light absorbed.

The amount of renaturation depends on two critical factors: (1) initial concentration of single-stranded DNA (C_0) and (2) amount of time allowed for renaturation (t). Other things being equal, there will be more renaturation



11.13 The slow heating of DNA causes the two strands to separate (denature).

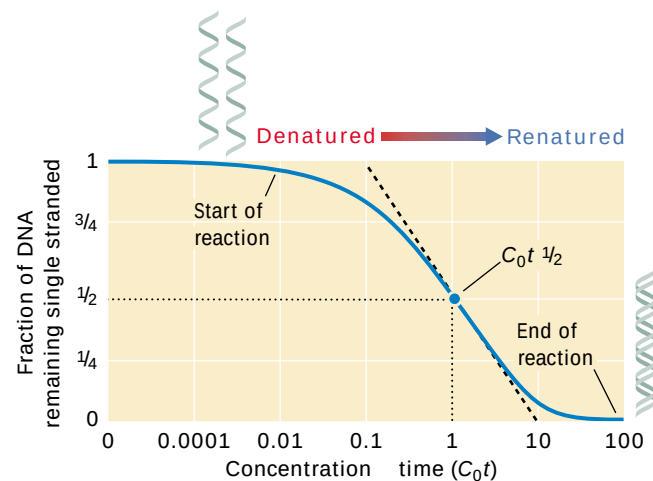
at higher concentrations of DNA, because high concentrations increase the likelihood that the two complementary strands will collide. There will also be more renaturation with increasing time, because there are more opportunities for two complementary sequences to collide. These two factors together form a parameter called C_0t , which equals the initial concentration multiplied by the renaturation time ($C_0 \times t = C_0t$).

A plot of the fraction of single-stranded DNA as a function of C_0t during a renaturation reaction is called a C_0t curve. A typical C_0t curve for a prokaryotic organism is shown in **FIGURE 11.14**. The upper left-hand side of the curve represents the start of the renaturation reaction, when all of the DNA is single stranded, and so the proportion of single-stranded DNA is 1. As the reaction proceeds, single-stranded DNA pairs to form double-stranded DNA, represented by the decreasing fraction of single-stranded DNA. At the end of the reaction, the proportion of single-stranded DNA is 0, because all of the DNA is now double stranded. The value at which half of the DNA is reannealed is called $C_0t^{1/2}$.

The rate of renaturation also depends on the size and complexity of the DNA molecules used. Consider the following analogy. Suppose we distribute 100 cards equally among the students in a class. We ask each student to write his or her name on the cards, and we put all the cards in a hat. We then randomly draw two cards from the hat and see if the names on the two cards match. If they don't match, we put them back in the hat; if they do match, we remove them, and we continue drawing until all the cards have been removed. If there are only four students in the class, each student will receive 25 cards. Because each student's name is on 25 cards, the chance of drawing two cards that match is high, and we will quickly empty the hat. If we do the same exercise in another class with 50 students, again using 100 cards, each student's name will appear on only two cards, and the chance of removing two cards with the same name is much lower. Thus, it will take longer to empty the hat.

This exercise resembles what occurs in the renaturation reaction. If we start with the same total amount of DNA, but there are only a few different sequences in the DNA, a chance collision between two complementary fragments is more likely to occur than if there were many different sequences. Therefore DNA from organisms with larger genomes will have a larger $C_0t^{1/2}$ value.

Thus far, we have considered renaturation reactions in which each DNA sequence is present only once in each molecule. If some sequences are present in multiple copies, these sequences will be more likely to collide with a complementary copy, and renaturation of these sequences will be rapid. Think about our analogy of drawing names from a hat. Imagine that we have 50 students and 100 cards; each student gets two cards. This time, the students write only their first names on the cards. Again, we place the cards in the hat and draw out two cards at random. If



11.14 A C_0t curve represents the fraction of DNA remaining single stranded in a renaturation reaction, plotted as a function of DNA concentration $\times$ time (C_0t). This graph is a typical C_0t curve for a prokaryotic organism.

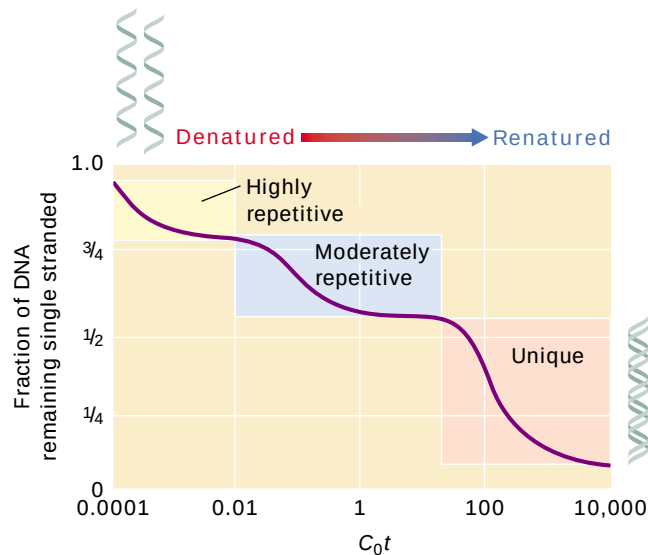
there are five students in the class named Scott, this name will appear on ten cards; so the chance of drawing out two cards at random bearing the name Scott is fairly high. On the other hand, if there is only one Susan in the class, this name will appear on only two cards, and the chance of drawing out two cards with the name Susan is low. The cards with Scott match up more quickly than the cards with Susan, because there are more copies with the name Scott. Similarly, in a renaturation reaction, if some sequences of DNA are present in multiple copies, they will renature more quickly.

Concepts

When double-stranded DNA is heated, it denatures, separating into single-stranded molecules. On cooling, these single-stranded molecules pair and re-form double-stranded DNA, a process called renaturation. A C_0t curve is a plot of a renaturation reaction.

Types of DNA Sequences in Eukaryotes

For most eukaryotic organisms, C_0t curves similar to the one presented in **FIGURE 11.15** are produced and indicate that eukaryotic DNA consists of at least three types of sequences. Slowly renaturing DNA consists of sequences that are present only once, or at most a few times, in the genome. This nonrepetitive, **unique-sequence DNA** includes sequences that code for proteins, as well as a great deal of DNA whose function is unknown. The more rapidly renaturing DNA represents two kinds of **repetitive DNA**—



11.15 A typical C_0t curve for a eukaryotic organism contains several steps. The first step in the curve represents DNA renaturing at very low C_0t values, because these sequences are present in many copies (highly repetitive). The second step represents DNA renaturing at intermediate C_0t values; these sequences are present in an intermediate number of copies (moderately repetitive). The last step represents DNA that renatures slowly; these sequences are present singly or in few copies (unique).

DNA sequences that exist in multiple copies. Although not identical, these copies are similar enough to reanneal.

Moderately repetitive DNA typically consists of sequences from 150 to 300 bp in length (although they may be longer) that are repeated many thousands of times. Some of these sequences perform important functions for the cell; for example, the genes for ribosomal RNAs (rRNAs) and transfer RNAs (tRNAs) make up a part of the moderately repetitive DNA. However, much of the moderately repetitive DNA has no known function in the cell. Moderately repetitive DNA itself is of two types of repeats. **Tandem repeat sequences** appear one after another and tend to be clustered at a few locations on the chromosomes. **Interspersed repeat sequences** are scattered throughout the genome. An example of an interspersed repeat is the *Alu* sequence, each of which consists of about 200 bp. The *Alu* sequence is present more than a million times in the human genome and makes up about 11% of each person's DNA. Short repeats, such as the *Alu* sequences, are called **SINEs (short interspersed elements)**. Longer interspersed repeats consisting of several thousand base pairs are called **LINEs (long interspersed elements)**. Most interspersed repeats are transposable genetic elements, sequences that can multiply and move (see next section).

The other major class of repetitive DNA is **highly repetitive DNA**. These short sequences, often less than 10 bp in

length, are present in hundreds of thousands to millions of copies that are repeated in tandem and clustered in certain regions of the chromosome, especially at centromeres and telomeres. Highly repetitive DNA is sometimes called satellite DNA, because it has a different base composition from those of the other DNA sequences and separates as a satellite fraction when centrifuged at high speeds. Highly repetitive DNA is rarely transcribed into RNA. Although these sequences may contribute to centromere and telomere function, most highly repetitive DNA has no known function.

Concepts

Eukaryotic DNA comprises three major classes: unique-sequence DNA, moderately repetitive DNA, and highly repetitive DNA. Unique-sequence DNA consists of sequences that exist in one or only a few copies; moderately repetitive DNA consists of sequences that may be several hundred base pairs in length and is present in thousands to hundreds of thousands of copies. Highly repetitive DNA consists of very short sequences repeated in tandem and present in hundreds of thousands to millions of copies.

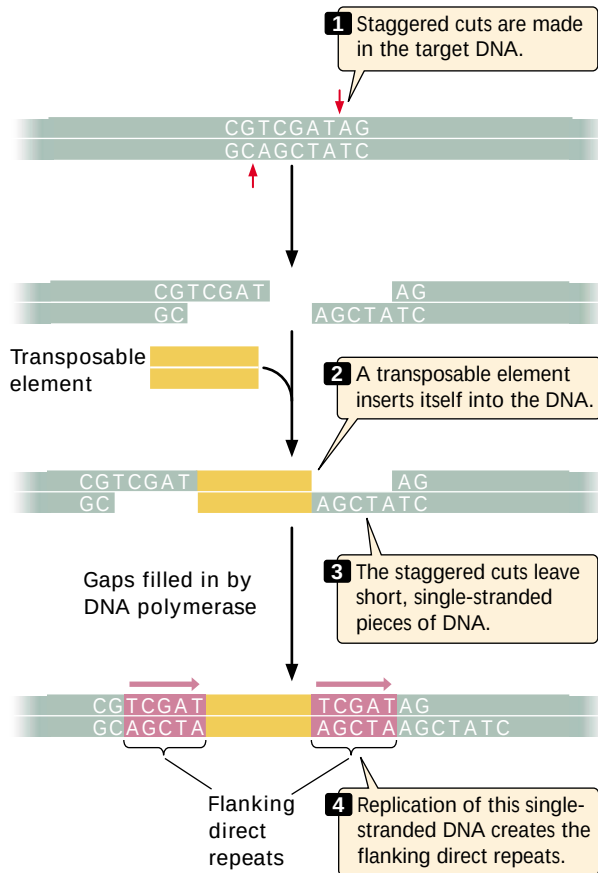
The Nature of Transposable Elements

Transposable elements are mobile DNA sequences found in the genomes of all organisms. In many genomes, they are quite abundant: for example, they make up at least 50% of human DNA. Most transposable elements are able to insert at many different locations, relying on mechanisms that are distinct from homologous recombination. They often cause mutations, either by inserting into another gene and disrupting it or by promoting DNA rearrangements such as deletions, duplications, and inversions (see Chapter 9).

General Characteristics of Transposable Elements

There are many different types of transposable elements: some have simple structures, encompassing only those sequences necessary for their own transposition (movement), whereas others have complex structures and encode a number of functions not directly related to transposition. Despite this variation, many transposable elements have certain features in common.

Short, **flanking direct repeats** of 3 to 12 base pairs are present on both sides of most transposable elements. They are not a part of a transposable element and do not travel with it. Rather, they are generated in the process of transposition, at the point of insertion. The sequences of these repeats vary, but the length is constant for each type of transposable element.



11.16 Flanking direct repeats are generated when a transposable element inserts into DNA.

The presence of flanking direct repeats indicates that staggered cuts are made in the target DNA when a transposable element inserts itself, as shown in **FIGURE 11.16**. The staggered cuts leave short, single-stranded pieces of DNA on either side of the transposable element. Replication of the single-stranded DNA then creates the flanking direct repeats.

At the ends of many, but not all, transposable elements are **terminal inverted repeats**, which are sequences from 9 to 40 bp in length that are inverted complements of one another. For example, the following sequences are inverted repeats:

5'–ACAGTTCAG ... CTGAACTGT–3'
3'–TGTC AAGTC ... GACTTGACA–5'

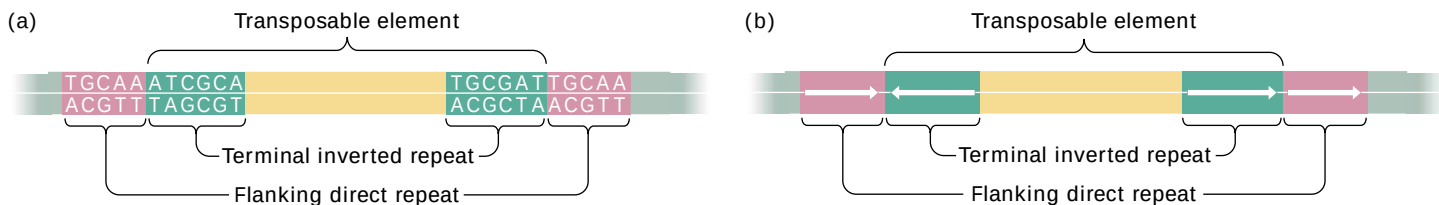
On the same strand, the two sequences are not simple inversions, as their name might imply; rather, they are both inverted and complementary. (Notice that the sequence from left to right in the top strand is the same as the sequence from right to left in the bottom strand.) Terminal inverted repeats are recognized by enzymes that carry out transposition and are required for transposition to take place. **FIGURE 11.17** summarizes the general characteristics of transposable elements.

Concepts

Transposable elements are mobile DNA sequences that often cause mutations. There are many different types of transposable elements; most generate short, flanking direct repeats at the target site as they insert. Many transposable elements also possess short terminal inverted repeats.

Transposition

Transposition is the movement of a transposable element from one location to another. Although our understanding of transposition is still incomplete, it's clear that, rather than a single mechanism, several different mechanisms are required for transposition in both prokaryotic and eukaryotic cells. Nevertheless, all types of transposition have several features in common: (1) staggered breaks are made in the target DNA (see Figure 11.16); (2) the



11.17 Many transposable elements have common characteristics. (a) Most transposable elements generate flanking direct repeats on each side of the point of insertion into target DNA. Many transposable elements also possess terminal inverted repeats. (b) These representations of direct and indirect repeats are used in illustrations throughout this chapter.

transposable element is joined to single-stranded ends of the target DNA; and (3) DNA is replicated at the single-strand gaps.

Mechanisms of Transposition

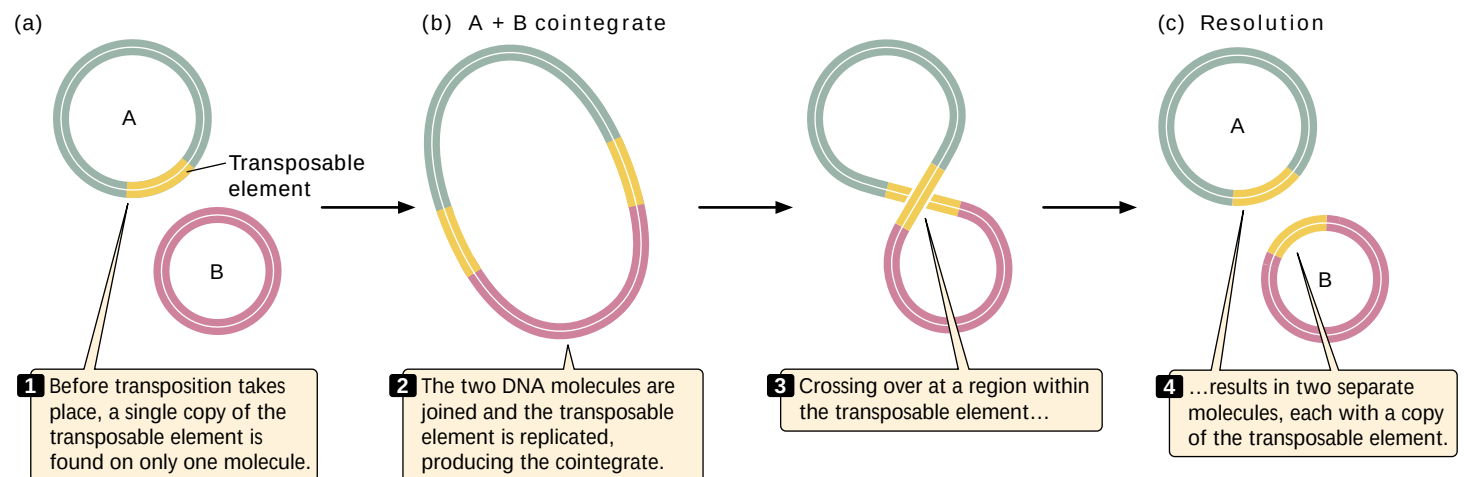
Some transposable elements transpose through DNA intermediates, whereas others use RNA intermediates. Among those that transpose through DNA, transposition may be replicative or nonreplicative. In **replicative transposition**, a new copy of the transposable element is introduced at a new site while the old copy remains behind at the original site; the number of copies of the transposable element increases. In **nonreplicative transposition**, the transposable element excises from the old site and inserts at a new site without any increase in the number of its copies. Nonreplicative transposition requires replication of only the few nucleotides that constitute the direct repeats.

Replicative transposition Replicative transposition, sometimes called copy-and-paste transposition, can be either between two different DNA molecules or between two parts of the same DNA molecule. **FIGURE 11.18** summarizes the steps of transposition between two circular DNA molecules. Before transposition (see Figure 11.18a), the transposable element is on one molecule. In the first step, the two DNA molecules are joined, and the transposable element is replicated, producing the **cointegrate structure** that consists of molecules A + B fused together with two copies of the transposable element (see Figure 11.18b). In a moment, we'll see how the copy is produced, but let's first look at the second step of the replicative transposition process. After the cointegrate has formed, crossing over at regions within the transposable elements produces two molecules, each with a copy of the transposable element

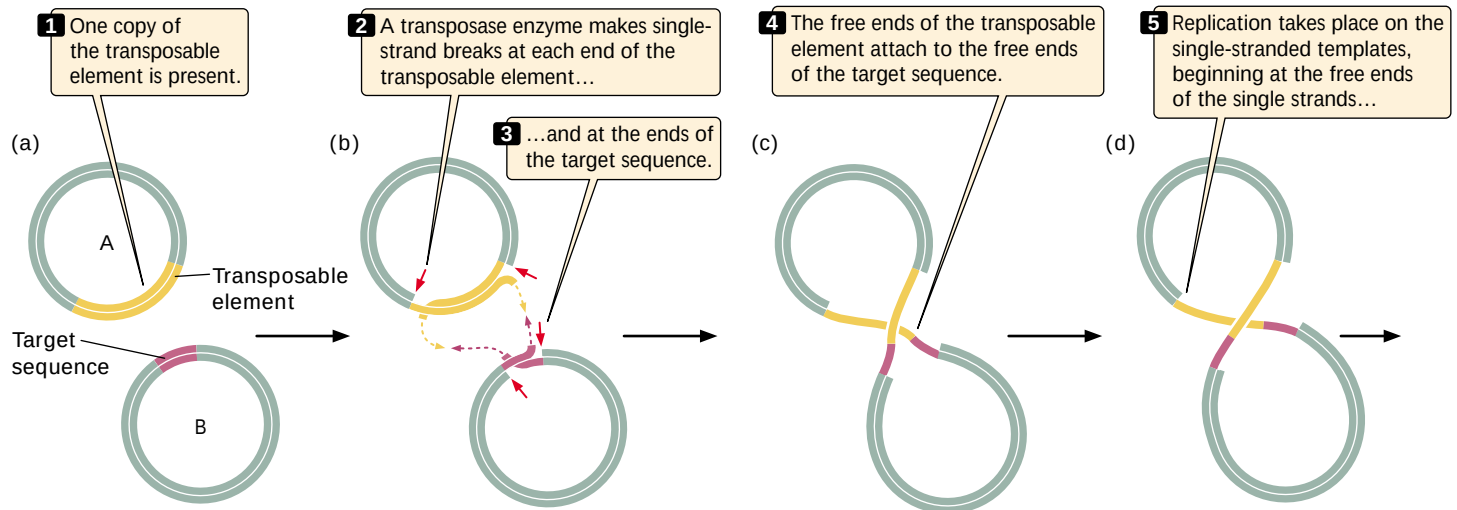
(see Figure 11.18c). This second step is known as resolution of the cointegrate.

How are the steps of replicative transposition (cointegrate formation and resolution) brought about? Cointegrate formation requires four events. First, a **transposase** enzyme (often encoded by the transposable element) makes single-strand breaks at each end of the transposable element and on either side of the target sequence where the element inserts (**FIGURE 11.19 a and b**). Second, the free ends of the transposable element attach to the free ends of the target sequence (**FIGURE 11.19c**). Third, replication takes place on the single-stranded templates, beginning at the 3' ends of the single strands and proceeding through the transposable element (**FIGURE 11.19d and e**). This replication creates the cointegrate, with its two copies of both the transposable element and the sequence at the target site, which is now on one side of each copy (**FIGURE 11.19f**). The enzymes that perform the replication and ligation functions are cellular enzymes that function in replication and DNA repair. Fourth, after the cointegrate has formed, it undergoes resolution, which requires crossing over between sites located within the transposon. Resolution gives rise to two copies of the transposable element (**FIGURE 11.19g**). The resolution step is brought about by **resolvase** enzymes (encoded in some cases by the transposable element and in other cases by a cellular gene) that function in homologous recombination.

Nonreplicative transposition In nonreplicative transposition, the transposable element moves from one site to another without replication of the entire transposable element, although short sequences in the target DNA are replicated, generating flanking direct repeats. Sometimes referred to as cut-and-paste transposition, nonreplicative transposition requires only that the transposable element



11.18 Replicative transposition increases the number of copies of the transposable element.



11.19 Replicative transposition requires single-strand breaks, replication, and resolution.

and the target DNA be cleaved and joined together. Cleavage requires a transposase enzyme produced by the transposable element. The joining of the transposable element and target DNA is probably carried out by normal replication and repair enzymes. If a transposable element moves by nonreplicative transposition, how does it increase in copy number in the genome? The answer comes from examining the fate of the original site of the element. After excision, a break will be left at the original insertion site. Such breaks are harmful to the cell, and so they are repaired efficiently (see Chapter 17). One common method of repair is to copy sequence information from a homologous template; the sister chromatid is the preferred template for this type of repair. Before transposition, both sisters will have a copy of the transposable element. After excision from one chromatid, repair of the break can result in copying the transposable element sequence off the sister chromatid. Thus, the transposable element is moved from the original site to a new site, but a copy is restored to the original site by DNA repair mechanisms.

Transposition through an RNA intermediate Eukaryotic transposable elements that transpose through RNA intermediates are called **retrotransposons**. A retrotransposon in DNA (FIGURE 11.20a) is first transcribed into an RNA sequence (FIGURE 11.20b), which may be processed. The processed RNA undergoes reverse transcription by a reverse transcriptase enzyme to produce a double-stranded DNA copy of the RNA (FIGURE 11.20c). Staggered cuts are made in the target DNA (FIGURE 11.20d), and the DNA copy of the retrotransposon inserts into the genome (FIGURE 11.20e). Replication fills in the short gaps produced by the staggered cuts, generating flanking direct repeats on both sides of the retrotransposon.

Concepts

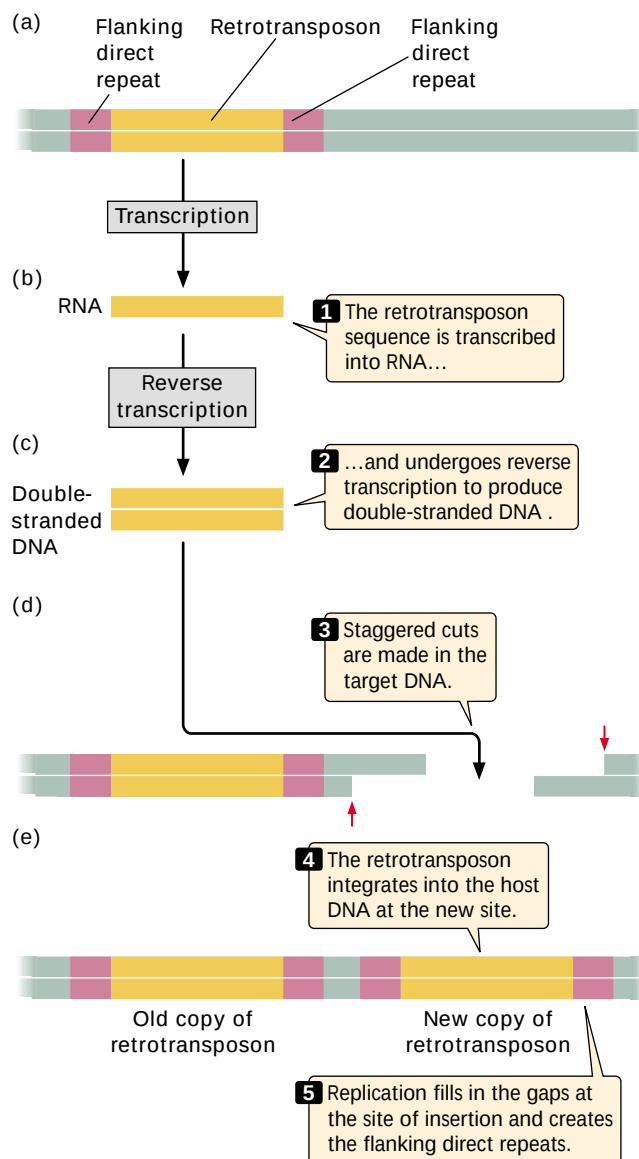
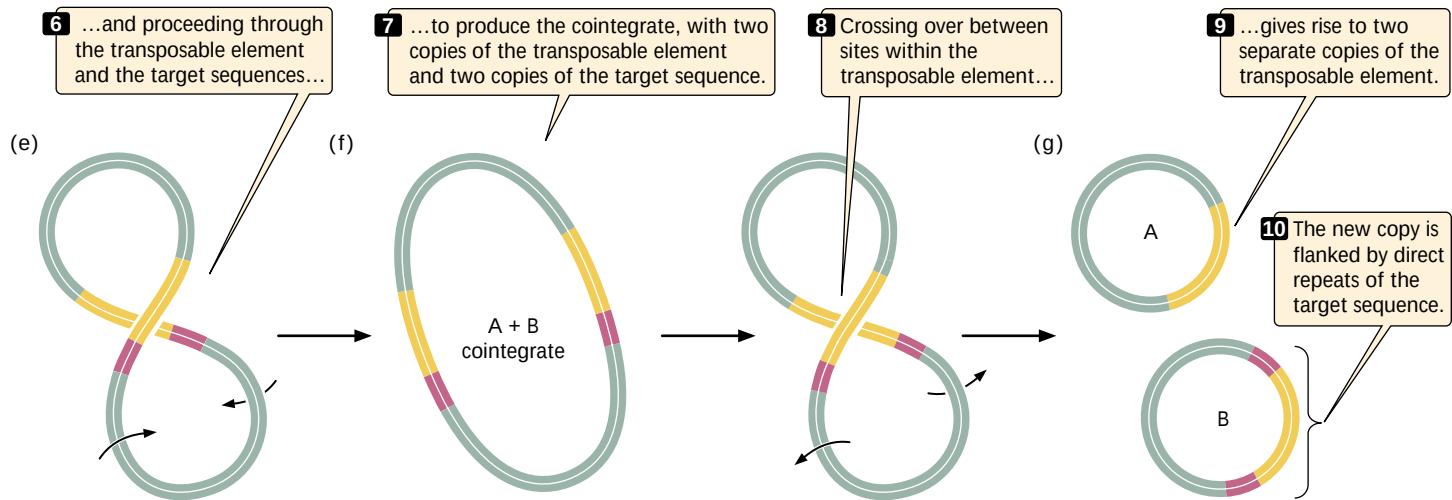
Transposition may be through either a DNA or an RNA intermediate. In replicative transposition, a new copy of the transposable element inserts in a new location and the old copy stays behind; in nonreplicative transposition, the old copy excises from the old site and moves to a new site. Transposition through an RNA intermediate requires reverse transcription, in which a retrotransposon is transcribed into RNA, the RNA is copied into DNA, and the new DNA copy is integrated into the target site.

The Mutagenic Effects of Transposition

Because transposable elements may insert into other genes and disrupt their function, transposition is generally mutagenic. In fact, more than half of all spontaneously occurring mutations in *Drosophila* result from the insertion of a transposable element in or near a functional gene. Although most of these mutations are detrimental, transposition may occasionally activate a gene or change the phenotype of the cell in a beneficial way. Additionally, a transposable element may carry information that benefits the cell, such as antibiotic resistance conferred by genes carried on bacterial transposable elements.

In 1991, Francis Collins and his colleagues discovered a 31-year-old man with neurofibromatosis caused by a transposition of the *Alu* sequence. Neurofibromatosis is a disease

11.20 (opposite page) Retrotransposons transpose through RNA intermediates.



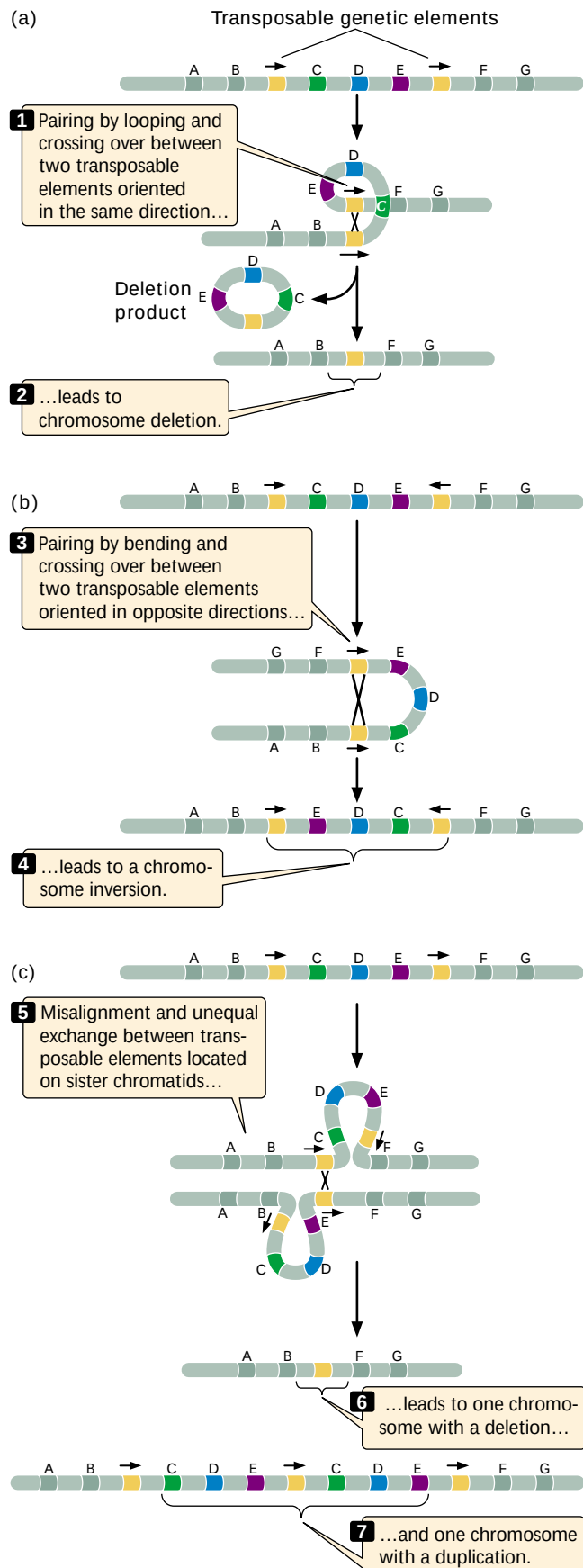
that produces numerous tumors of the skin and nerves; it results from mutations in a gene called *NF1*. Collins and his colleagues found a copy of *Alu* in one of the introns of this man's *NF1* gene. The *Alu* had caused an RNA splicing error, with the result that one of the exons was left out of the *NF1* mRNA. The absence of the exon caused a shift in the reading frame and resulted in an abnormal protein, which eventually caused the neurofibromatosis. Examination of DNA from the man's mother and father revealed that the *Alu* sequence was not present in their *NF1* genes—the insertion was new. Cases of hemophilia and muscular dystrophy also have been traced to mutations caused by transposition.

Because transposition entails the exchange of DNA sequences and recombination, it often leads to DNA rearrangements. Homologous recombination between multiple copies of transposons also leads to duplications, deletions, and inversions, as shown in [FIGURE 11.21](#). The Bar mutation in *Drosophila* (see Figures 9.7 and 9.8) is a tandem duplication thought to have arisen through homologous recombination between two copies of a transposable element present in different locations on the X chromosome.

DNA rearrangements can also be caused by excision of transposable elements in a cut-and-paste transposition. If the broken DNA is not repaired properly, a chromosome rearrangement can be generated. If it is not repaired at all, the acentric fragment will be lost, resulting in a deletion. This type of chromosome breakage led to the first discovery of transposable elements by Barbara McClintock (described below). She named the gene that appeared at these sites *Dissociation* because of the tendency for it to cause chromosome breakage and loss of a fragment.

The Regulation of Transposition

Many transposable elements move through replicative transposition and increase in number with each transposition. As the number of copies of the transposon increases, the rate of transposition increases because the concentration



11.21 Chromosome rearrangements are often generated by transposition.

of transposase in the cell becomes greater (remember that transposase is produced by the transposon). In the absence of mechanisms to restrict transposition, the number of copies of transposable elements would increase continuously, and the host DNA would be harmed by the resulting high rate of mutation (caused by frequent insertion of transposable elements). Furthermore, large amounts of energy and resources would be required to replicate the “extra” DNA in the proliferating transposable elements. For these reasons, it isn’t surprising that cells have evolved mechanisms to regulate transposition, just as they have mechanisms to regulate gene expression (see Chapter 16).

When a transposable element first enters a cell that possesses no other copies of that element, transposition is frequent. As the number of copies of the transposable element increases, the frequency of transposition diminishes until a steady-state number of transposable elements is reached. This regulation of transposition means that most cells have a characteristic number of copies of a particular transposable element.

Many transposable elements regulate transposition by limiting the production of the transposase enzyme required for movement. In some cases, *transcription* of the transposase gene is regulated but, more frequently, *translation* of the transposase mRNA is controlled (see p. 000 in Chapter 16). Other regulatory mechanisms do not affect the level of transposase; rather, they directly inhibit the transposition event.

Concepts

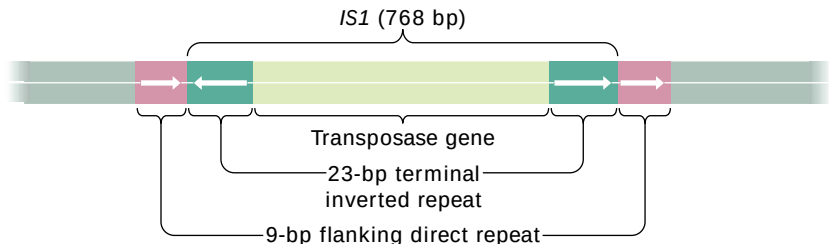
Transposable elements frequently cause mutations and DNA rearrangements. Many transposable elements regulate their own transposition, either by controlling the amount of transposase produced or by direct inhibition of the transposition event.

The Structure of Transposable Elements

Bacteria and eukaryotic organisms possess a number of different types of transposable elements, the structures of which vary extensively. In this section, we consider the structures of representative types of transposable elements.

Transposable Elements in Bacteria

The two major groups of bacterial transposable elements are: (1) simple transposable elements that carry only the information required for movement and (2) more-complex



11.22 Insertion sequences are simple transposable elements found in bacteria.

transposable elements that contain DNA sequences not directly related to transposition.

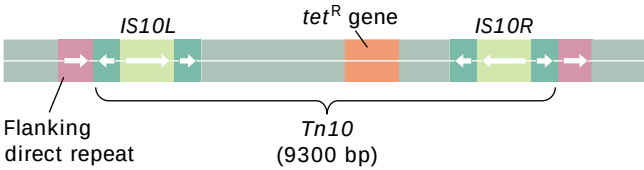
Insertion sequences The simplest type of transposable element in bacterial chromosomes and plasmids is an **insertion sequence (IS)**. This type of element carries only the genetic information necessary for its movement. Insertion sequences are common constituents of bacteria and plasmids. They are designated by IS, followed by an identifying number. For example, *IS1* is a common insertion sequence found in *E. coli*.

Insertion sequences are typically from 800 to 2000 bp in length and possess the two hallmarks of transposable elements: terminal inverted repeats and the generation of flanking direct repeats at the site of insertion. Most insertion sequences contain one or two genes that code for transposase. *IS1*, a typical insertion sequence, is 768 nucleotide pairs long and has terminal inverted repeats of 23 bp at each end (Figure 11.22). The flanking direct repeats created by *IS1* are each 9 bp long—the most common length for flanking direct repeats. Table 11.4 summarizes these features for several bacterial insertion sequences.

Composite transposons Any segment of DNA that becomes flanked by two copies of an insertion sequence may itself transpose and is called a **composite transposon**. Each type of composite transposon is designated by the abbreviation *Tn*, followed by a number. *Tn10* is a

composite transposon of about 9300 bp that carries a gene (about 6500 bp) for tetracycline resistance between two *IS10* insertion sequences (Figure 11.23). The insertion sequences have terminal inverted repeats; so the composite transposon also ends in inverted repeats. Composite transposons also generate flanking direct repeats at their sites of insertion (see Figure 11.23). The insertion sequences at the ends of a composite transposon may be in the same orientation or they may be inverted relative to one other (as in *Tn10*).

The insertion sequences at the ends of a composite transposon are responsible for transposition. The DNA between the insertion sequences is not required for movement and may carry additional information (such as antibiotic resistance). Presumably, composite transposons evolve when one insertion sequence transposes to a location close to another of the same type. The transposase produced by one of the IS sequences catalyzes the transposition of both insertion sequences, allowing them to move together and carry along the DNA that lies between



11.23 *Tn10* is a composite transposon in bacteria.

Table 11.4 Structures of some common insertion sequences

Insertion Sequence	Total Length (bp)	Length of	
		Inverted Repeats (bp)	Flanking Direct Repeats (bp)
<i>IS1</i>	768	23	9
<i>IS2</i>	1327	41	5
<i>IS4</i>	1428	18	11 or 12
<i>IS5</i>	1195	16	4

Source: B. Lewin, *Genes*, 3d ed. (New York: Wiley, 1987), p. 591.

Table 11.5 Characteristics of several composite transposons			
Composite Transposon	Total Length (bp)	Associated IS Elements	Other Genes Within the Transposon
<i>Tn9</i>	2500	<i>IS1</i>	Chloramphenicol resistance
<i>Tn10</i>	9300	<i>IS10</i>	Tetracycline resistance
<i>Tn5</i>	5700	<i>IS50</i>	Kanamycin resistance
<i>Tn903</i>	3100	<i>IS903</i>	Kanamycin resistance

them. In some composite transposons (such as *Tn10*), one of the insertion sequences may be defective; so its movement depends on the transposase produced by the other. Characteristics of several composite transposons are listed in Table 11.5.

Noncomposite transposons As already stated, insertion sequences carry only information for their own movement, whereas bacterial transposons are more complex. Some transposable elements in bacteria lack insertion sequences and are referred to as noncomposite transposons. For instance, *Tn3* is a noncomposite transposon that is about 5000 bp long, possesses terminal inverted repeats of 38 bp, and generates flanking direct repeats that are 5 bp in length. *Tn3* carries genes for transposase and resolvase (mentioned earlier in this chapter), plus a gene that codes for the enzyme β -lactamase, which provides resistance to ampicillin.

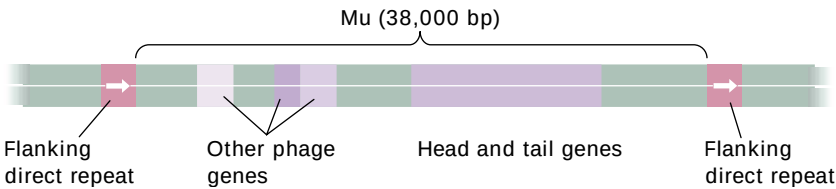
A few bacteriophage genomes reproduce by transposition and use transposition to insert themselves into a bacterial chromosome in their lysogenic cycle; the best studied of these transposing bacteriophages is Mu (FIGURE 11.24). Although Mu does not possess terminal inverted repeats, it does generate short (5-bp) flanking direct repeats when it inserts randomly into DNA. Mu replicates through transposition and causes mutations at the site of insertion, properties characteristic of transposable elements.

Concepts

Insertion sequences are prokaryotic transposable elements that carry only the information needed for transposition. A composite transposon is a more complex element that consists of two insertion sequences plus intervening DNA. Noncomposite transposons in bacteria lack insertion sequences but have terminal inverted repeats and carry information not related to transposition. All of these transposable elements generate flanking direct repeats at their points of insertion.

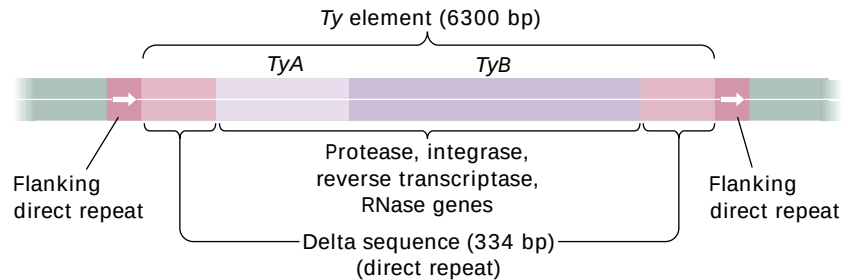
Transposable Elements in Eukaryotes

Eukaryotic transposable elements can be divided into two groups. One group is structurally similar to transposable elements found in bacteria, typically ending in short inverted repeats and transposing through DNA intermediates. The other group comprises retrotransposons (see Figure 11.20); they use RNA intermediates and are similar in structure and movement to retroviruses (see p. 000 in Chapter 8). On the basis of their structure, function, and genomic sequences, it is clear that some retrotransposons are evolutionarily related to retroviruses. Although their mechanism of movement is fundamentally different from that of other transposable elements, retrotransposons



11.24 Mu is a transposing bacteriophage.

11.25 *Ty* is a transposable element of yeast.



also generate direct repeats at the point of insertion. Retrotransposons include the *Ty* elements in yeast, the *copia* elements in *Drosophila*, and the *Alu* sequences in humans.

***Ty* elements in yeast** *Ty* (for transposon yeast) elements are a family of common transposable elements found in yeast; many yeast cells have 30 copies of *Ty* elements. These elements are retrotransposons that are about 6300 nucleotide pairs in length and generate 5-bp flanking direct repeats when they insert into DNA (FIGURE 11.25). At each end of a *Ty* element are direct repeats called **delta sequences**, which are 334 bp long. The delta sequences are analogous to the long terminal repeats found in retroviruses (see p. 000 in Chapter 8). These delta sequences contain promoters required for the transcription of *Ty* genes, and the promoters may also stimulate the transcription of genes that lie downstream of the *Ty* element. Between the delta sequences at each end of a *Ty* element are two genes (*TyA* and *TyB*, which encode several enzymes) that are related to the *gag* and *pol* genes found in retroviruses (see p. 000 in Chapter 8). Many *Ty* elements are defective and no longer capable of undergoing transposition.

***Ac* and *Ds* elements in maize** Transposable elements were first identified in maize (corn), more than 50 years ago by Barbara McClintock (FIGURE 11.26). McClintock spent much of her long career studying their properties, and her work stands among the landmark discoveries of genetics. Her results, however, were misunderstood and ignored for many years. Not until molecular techniques were developed in the late 1960s and 1970s did the importance of transposable elements become widely accepted.

Born in 1902, Barbara McClintock attended Cornell University as an undergraduate and, later, as a graduate student. She was especially interested in genetics, but the subject was taught in the department of plant breeding, which did not accept women students. So she registered for botany instead and studied maize chromosomes for her Ph.D. dissertation.

After receiving her degree, McClintock remained at Cornell, continuing her cytogenetic analysis of maize chromosomes. Her discoveries in the next 10 years included the identification of all the chromosomes in maize, the assignment of linkage groups to chromosomes, proof of crossing over, mapping genes to chromosomes by using rearrangements, and associating chromosome elements with the nucleolus.



11.26 Barbara McClintock was the first to discover transposable elements. (CSHL Archives/Peter Arnold.)

McClintock's discovery of transposable elements had its genesis in the early work of Rollins A. Emerson on the maize genes that caused variegated (multicolored) kernels. Most corn kernels are either wholly pigmented or colorless (yellow), but Emerson noted that some yellow kernels had spots or streaks of color (FIGURE 11.27). He proposed that these kernels resulted from an unstable mutation: a muta-



11.27 Variegated (spotted) kernels in corn are caused by mobile genes. The study of variegated corn led Barbara McClintock to discover transposable elements. (Matt Meadows/Peter Arnold.)

tion in the wild-type gene for pigment produced a colorless kernel; but, in some cells, the mutation reverted back to the wild type, causing a spot of pigment. However, Emerson didn't know why these mutations were unstable.

McClintock discovered that the cause of the unstable mutation was a gene that moved. She noticed that chromosome breakage in maize often occurred at a locus that she called *Dissociation* (*Ds*) but only if another gene, the *Activator* (*Ac*), also was present. *Ds* and *Ac* exhibited unusual patterns of inheritance; occasionally, the genes moved together. McClintock called these moving genes controlling elements, because they controlled the expression of other genes.

McClintock published her conclusion that controlling elements moved in 1948. Although her results were not disputed, they were neither understood nor recognized by most geneticists. Of her work, Alfred Sturtevant, then a prominent geneticist remarked, "I didn't understand one word she said, but if she says it is so, it must be so!" He expressed what seems to have been the attitude of many geneticists at the time. McClintock was frustrated by the genetics community's reaction to her research, but she continued to pursue it nonetheless. In the 1960s, bacteria and bacteriophages were shown to possess transposable elements, and the development of recombinant DNA techniques in the 1970s and 1980s demonstrated that transposable elements exist in all organisms. The significance of McClintock's early discoveries was finally recognized in 1983, when she was awarded the Nobel Prize in Physiology or Medicine.

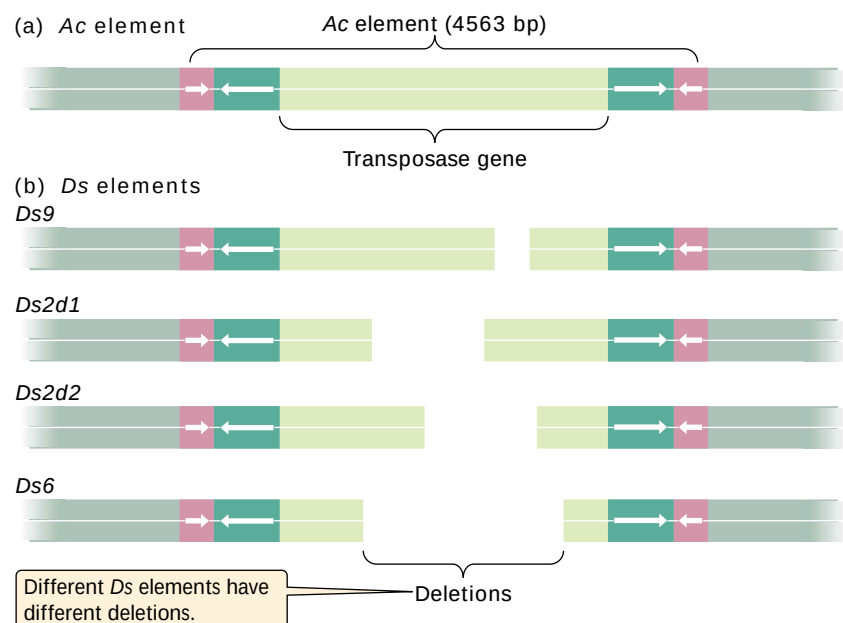
www.whfreeman.com/pierce A series of links to Barbara McClintock and her work on transposable elements

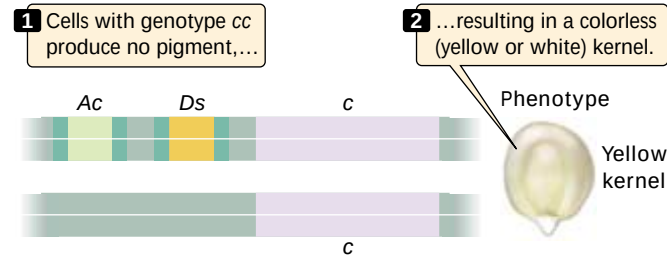
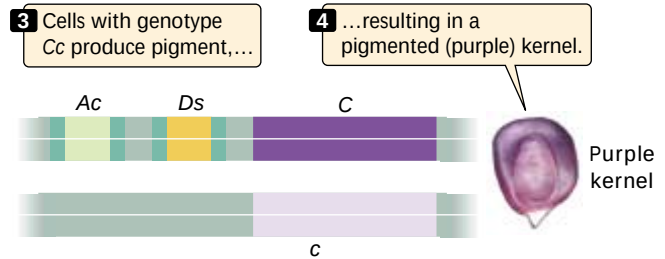
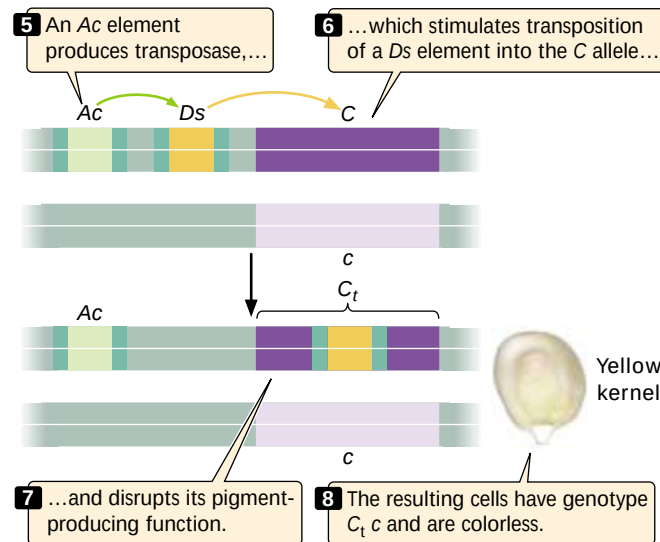
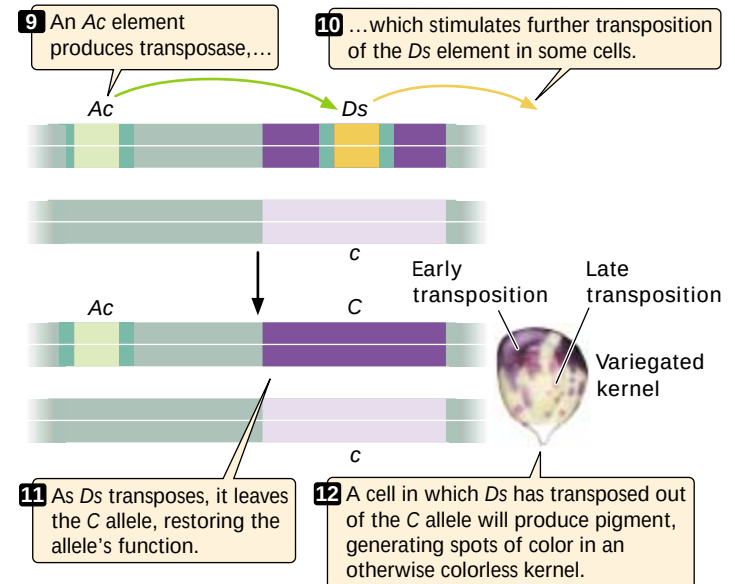
Ac and *Ds* elements in maize have now been examined in detail, and their structure and function are similar to those of transposable elements found in bacteria: they possess terminal inverted repeats and generate flanking direct repeats at the points of insertion. *Ac* elements are about 4500 bp long, including terminal inverted repeats of 11 bp, and the flanking direct repeats that they generate are 8 bp in length (FIGURE 11.28a). Each *Ac* element contains a single gene that encodes a transposase enzyme. Thus *Ac* elements are *autonomous*—that is, able to transpose. *Ds* elements are *Ac* elements with one or more deletions that have inactivated the transposase gene (FIGURE 11.28b). Unable to transpose on their own, (*nonautonomous*), *Ds* elements can transpose in the presence of *Ac* elements because they still possess terminal inverted repeats recognized by *Ac* transposase.

Each kernel in an ear of corn is a separate individual, originating as an ovule fertilized by a pollen grain. A kernel's pigment pattern is determined by several loci. A pigment-encoding allele at one of these loci can be designated *C*, and an allele at the same locus that does not confer pigment is designated as *c*. A kernel with genotype *cc* will be colorless—that is, yellow or white (FIGURE 11.29a); a kernel with genotype *CC* or *Cc* will produce pigment and be purple (FIGURE 11.29b).

A *Ds* element, transposing under the influence of a nearby *Ac* element, may insert into the *C* allele, destroying its ability to produce pigment (FIGURE 11.29c). An allele inactivated by a transposable element is designated with a subscript "t"; so in this case it would be designated *C_t*. After the transposition of *Ds* into the *C* allele, the kernel cell has genotype *C_tc*. This kernel will be colorless (white or yellow), because neither the *C_t* nor the *c* allele confers pigment.

11.28 *Ac* and *Ds* are transposable elements in maize.



(a) Genotype cc : no transposition(b) Genotype Cc : no transposition(c) Genotype $Cc \rightarrow C_t c$: transposition(d) Genotype $C_t c \rightarrow C_t c/cc$: mosaic (transposition during development)

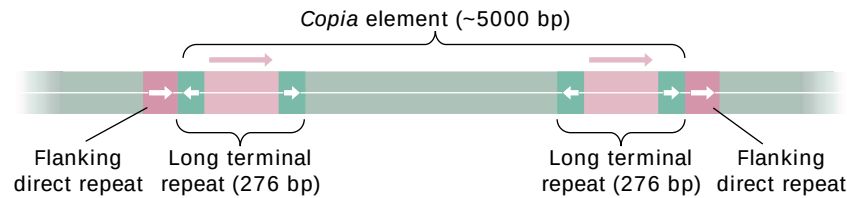
Conclusion: Variegated corn kernels result from excision of Ds elements from genes controlling pigment production during development.

11.29 Transposition results in variegated maize kernels.

As development takes place and the original one-celled maize embryo divides by mitosis, additional transpositions may take place in some cells. In any cell in which the transposable element excises from the C_t allele and moves to a new location, the C allele is rendered functional again: all cells derived from those in which this event has taken place will have the genotype Cc and be purple. The presence of these pigmented cells, surrounded by the colorless ($C_t c$) cells, produces a purple spot or streak (called a sector) in the otherwise yellow kernel (FIGURE 11.29d). The size of the sector varies, depending on when the excision of the transposable element from the $C_t c$ allele occurred. If excision occurred early in development, then many cells will contain the functional C allele and the pigmented sector will be large; if excision occurred late in development, few cells will have the functional C allele and the pigmented sector will be small.

Transposable elements in *Drosophila* A number of different transposable elements are found in *Drosophila*. One of the best studied is *copia*, a retrotransposon about 5000 bp long (FIGURE 11.30). *Copia* has direct (i.e., not inverted) repeats of 276 bp at each end, and within each direct repeat are terminal inverted repeats. When *copia* transposes, it generates flanking direct repeats that are 5 bp long at the site of insertion. Like *Ty* elements, *copia* contains sequences similar to those found in the *gag* and *pol* genes of retroviruses (see Figure 8.36). The number of *copia* elements in a typical fruit fly genome varies from 20 to 60.

Another family of transposable elements found in *Drosophila* are the P elements. Most functional P elements are about 2900 bp long, although shorter P elements with deletions also exist. Each P element possesses terminal inverted repeats that are 31 bp long and generates flanking direct repeats of 8 bp at the site of insertion. Like transpos-



11.30 Copia is a transposable element of *Drosophila*.

able elements in bacteria, *P* elements transpose through DNA intermediates. Each element encodes both a transposase and a repressor of transposition.

The role of this repressor in controlling transposition is demonstrated dramatically in **hybrid dysgenesis**, which is the sudden appearance of numerous mutations, chromosome aberrations, and sterility in the offspring of a cross between a P^+ male fly (with *P* elements) and a P^- female fly (without them). The reciprocal cross between a P^+ female and a P^- male produces normal offspring.

Hybrid dysgenesis arises from a burst of transposition that takes place when *P* elements are introduced into a cell that does not possess them. A cell that contains *P* elements produces the repressor in the cytoplasm that inhibits transposition. When a P^+ female produces eggs, the repressor protein is incorporated into the egg cytoplasm, which prevents further transposition in the embryo and thus prevents mutations from arising. The resulting offspring are fertile as adults (FIGURE 11.31a). However, a P^- female does not produce the repressor; so none is stored in the cytoplasm of her eggs. When her eggs are fertilized by sperm from a P^+ male, the absence of repression allows the *P* elements contributed by the sperm to undergo rapid transposition in the embryo, causing hybrid dysgenesis (FIGURE 11.31b).

P elements appear to have invaded *D. melanogaster* within the past 50 years. Today, almost all *D. melanogaster* caught in the wild possess *P* elements, but these transposable elements are uncommon in laboratory colonies of flies that were established more than 30 years ago. In fact, no strain of *D. melanogaster* collected before 1945 possesses them, suggesting that *P* elements have recently invaded *D. melanogaster* and have spread rapidly throughout the species.

Because *P* elements are not present in most laboratory stocks, they have been useful experimentally as vectors for introducing modified or foreign DNA into the *Drosophila* genome. *P* elements have been extensively manipulated and engineered for a variety of uses.

If *P* elements are a recent addition to the genome of *D. melanogaster*, where did they come from? A likely source is *Drosophila willistoni*, another fruit fly species. *D. willistoni* appears to have long possessed *P* elements that are virtually identical with those now found in *D. melanogaster*. Researchers Marilyn Houck and Margaret Kidwell proposed that the *P* elements made the leap from *D. willistoni* to *D. melanogaster* by hitching a ride on a mite.

All fruit flies are infected with a variety of mites. One mite species, *Proctolaelaps regalis*, infests both *D. willistoni* and *D. melanogaster*. This mite has needlelike mouth parts that allow it to pierce and feed on the eggs and larvae of the flies. Houck and Kidwell suggest that, while feeding on *D. willistoni*, a mite picked up fruit fly DNA with *P* elements, which it later injected into a developing *D. melanogaster*. This hypothesis is supported by the finding that mites do pick up *P* element DNA from P^+ fruit flies.

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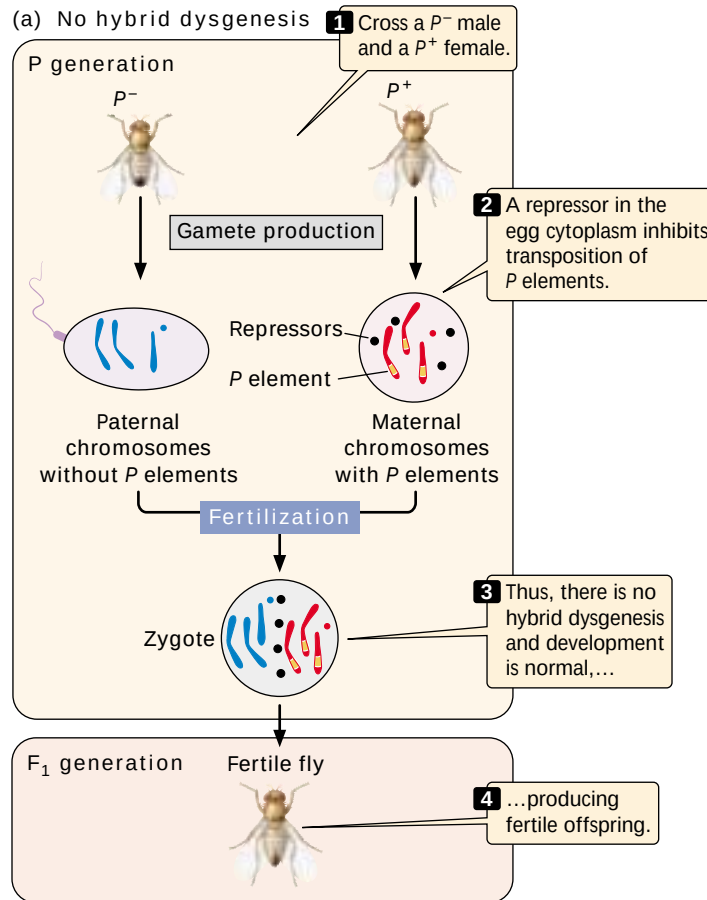
More information on *copia*

and *P* elements

Transposable elements in humans Almost 50% of the human genome consists of sequences derived from transposable elements, although most of these elements are now inactive and no longer capable of transposing. One of the most common transposable elements in the human genome is *Alu*, named after a restriction enzyme (*AluI*), which cleaves the element into two parts. Every human cell contains more than 1 million related, but not identical, copies of *Alu* in its chromosomes. Unlike the retrotransposons we have described earlier (*Ty* elements from yeast and *copia* elements from *Drosophila*), *Alu* sequences are not similar to retroviruses. They do not have genes resembling *gag* and *pol*, and are therefore nonautonomous. Rather, *Alu* sequences are similar to the gene that encodes the 7S RNA molecule, which transports newly synthesized proteins across the endoplasmic reticulum. *Alu* sequences create short flanking direct repeats when they insert into DNA and have characteristics that suggest that they have transposed through an RNA intermediate.

Alu belongs to a class of repetitive sequences found frequently in mammalian and some other genomes. These sequences are collectively referred to as SINEs, (short interspersed sequences). The human genome also has many LINES (long interspersed sequences), which are somewhat more similar in structure to retroviruses, but not as similar as *Ty* or *copia*.

The human genome contains evidence for several classes of transposable elements that transpose through a DNA intermediate, by the cut-and-paste mechanism. However, these all appear to have been inactive for about 50 million years; the nonfunctional sequences that remain have been referred to as DNA fossils.



11.31 Hybrid dysgenesis in *Drosophila* is caused by the transposition of P elements.

Concepts

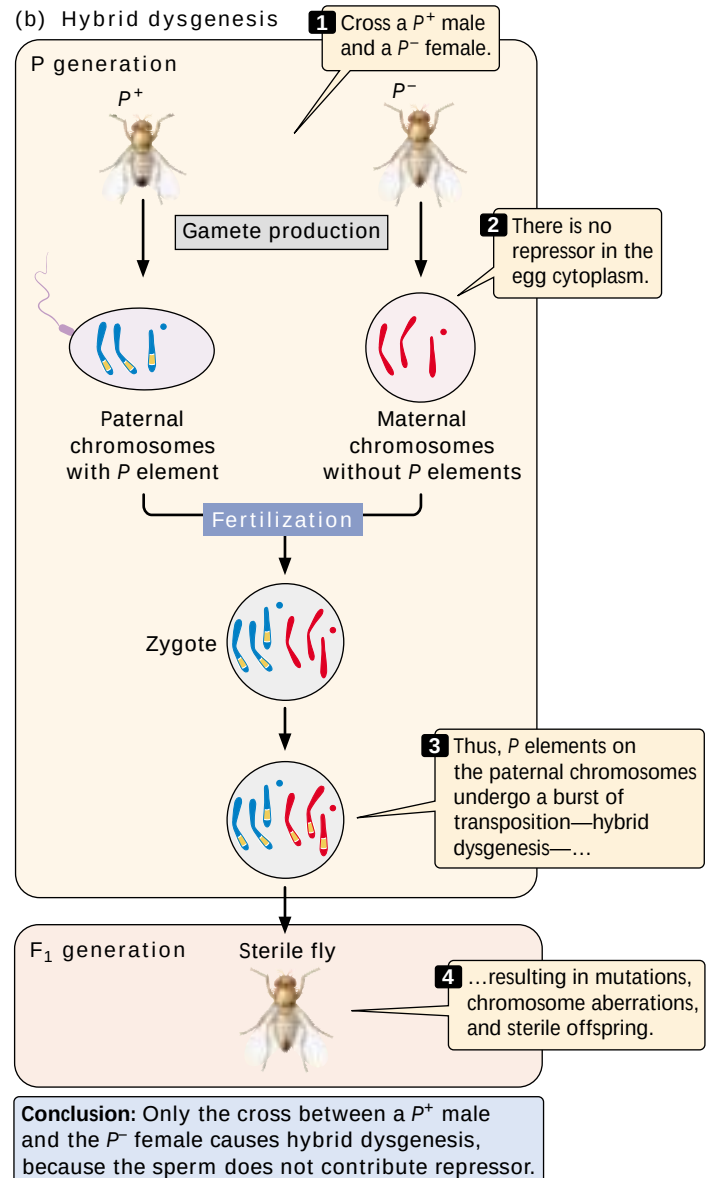
A great variety of transposable elements exist in eukaryotes. Some resemble transposable elements in prokaryotes, having terminal inverted repeats, and transpose through a DNA intermediate. Others are retrotransposons with long direct repeats at their ends and transpose through an RNA intermediate.

Connecting Concepts

Classes of Transposable Elements

Now that we have examined the process of transposition, let us review the major classes of transposable elements (Table 11.6).

Transposable elements can be divided into two major classes on the basis of structure and movement. The first class consists of elements that possess terminal inverted repeats and transpose through DNA intermediates. They all generate flanking direct repeats at their points of



insertion into DNA. All active forms of these transposable elements encode transposase, which is required for their movement. Some also encode resolvase, repressors, and other proteins. Their transposition may be replicative or nonreplicative, but they never use RNA intermediates. Examples of transposable elements in this first class include insertion sequences and all complex transposons in bacteria, the *Ac* and *Ds* elements of maize, and the *P* elements of *Drosophila*.

The second class of transposable elements are the retrotransposons, which transpose through RNA intermediates. They generate flanking direct repeats at their points of insertion when they transpose into DNA. Retrotransposons do not encode transposase, but some types are similar in structure to retroviruses and carry sequences that produce reverse transcriptase. Transposition

Table 11.6 Characteristics of two major classes of transposable genetic elements

Transposable Genetic Element	Structure	Genes Encoded	Transposition	Examples
Class I	Short, terminal inverted repeats; short flanking direct repeats at target site	Transposase gene (and sometimes others)	By DNA intermediate (replicative or nonreplicative)	<i>IS1</i> (<i>E. coli</i>) <i>Tn3</i> (<i>E. coli</i>) <i>Ac</i> , <i>Ds</i> (maize) <i>P</i> elements (<i>Drosophila</i>)
Class II (retrotransposon)	Long, terminal direct repeats; short flanking direct repeats at target site	Reverse transcriptase gene (and sometimes others)	By RNA intermediate	<i>Ty</i> (yeast) <i>copia</i> (<i>Drosophila</i>) <i>Alu</i> (human)

takes place when transcription produces an RNA intermediate, which is then transcribed into DNA by reverse transcriptase and inserted into the target site. Examples of retrotransposons in this class include *Ty* elements in yeast, *copia* elements in *Drosophila*, and *Alu* sequences in humans. Retrotransposons are not found in prokaryotes.

The Evolution of Transposable Elements

As mentioned earlier, transposable elements exist in all organisms, often in large numbers. Why are they so common? Three principal hypotheses have been proposed to explain their widespread occurrence.

The *cellular function hypothesis* proposes that transposable elements serve a valuable function within the cell, such as the control of gene expression or the regulation of development. Although the insertion of a transposable element can alter gene expression, there are few data to suggest that transposition plays a routine role in either of these or any other cellular processes.

The *genetic variation hypothesis* proposes that transposable elements exist because of their mutagenic activity. It suggests that a certain amount of genetic variation is useful because it allows a species to adapt to environmental change. Although some mutations caused by transposable elements may allow species to evolve beneficial traits, the vast majority of mutations generated by random transposition have deleterious effects. Thus, although mutations produced by transposable elements may be useful in the future, their immediate effect is usually deleterious and they will be selected against. The fact that many organisms have evolved mechanisms to regulate transposition suggests that there is selective pressure to limit the extent of transposition. In fact, if their only effect were to generate mutations, transposable genetic elements could be expected to disappear in time.

The *selfish DNA hypothesis* asserts that transposable elements serve no purpose for the cell; they exist simply because

they are capable of replicating and spreading. They can be thought of as “selfish” parasites of DNA that provide no benefit to the cell and may even be somewhat detrimental. Their capacity to reproduce and spread is what makes them common.

Which, if any, of these hypotheses is the correct explanation for the existence of transposable elements is not known. These hypotheses are not mutually exclusive, and all may contribute to the existence of mobile genes. Regardless of the evolutionary forces responsible for their existence, transposable elements have clearly played an important role in shaping the genomes of many organisms. In some cases, they have even been adopted for useful purposes by their host cells. One example is the mechanism that generates antibody diversity in the immune systems of vertebrates.

As will be discussed in Chapter 21, the ability of the immune system to recognize and attack foreign substances (antigens) depends on a mechanism whereby lymphocytes join several DNA segments that code for antigen-recognition proteins. Three DNA segments, called V, D, and J, exist in multiple forms within each cell. In the development of a lymphocyte, particular V, D, and J segments are randomly joined to produce a protein that recognizes a specific antigen. Within different lymphocytes, different V, D, and J segments are joined together in different combinations. The variety of combinations provides a large array of cells, each of which recognizes a particular antigen. Close examination of the V, D, and J joining process reveals that its mechanism is the same as that for transposition. The genes—designated *RAG1* and *RAG2*—participating in bringing about V, D, and J joining may have at one time been transposable elements that inserted into the germ line of a vertebrate ancestor, some 450 million years ago.

Another cellular function that may have originated as the result of a transposable element is the process that maintains the ends of chromosomes in eukaryotic organisms. As mentioned earlier in this chapter, DNA polymerases are unable to replicate the ends of chromosomes. In germ cells and single-celled eukaryotic organisms, chromosome length is maintained by telomerase, an enzyme

that extends the chromosome ends by copying repeated DNA sequences from an RNA template that is a part of the telomerase enzyme. The mechanism used by the telomerase enzyme is similar to the reverse transcription process used in retrotransposition, and telomerase is evolutionarily related to the reverse transcriptases encoded by certain retrotransposons.

These findings suggest that an invading retrotransposon in an ancestral eukaryotic cell may have provided the ability to copy the ends of chromosomes and eventually evolved into the gene that encodes the modern telomerase enzyme. *Drosophila* lacks the telomerase enzyme; retrotransposons appear to have resumed the role of telomere maintenance in this case.

Connecting Concepts Across Chapters



The material covered in this chapter has important connections to several topics already covered and to others in chapters yet to come. In Chapter 2, the gross structure of chromosomes and their behavior during mitosis and meiosis were introduced. The present chapter has built on that introduction by examining the molecular details of chromosome structure and the higher-level folding and packing of DNA that allows these very large molecules to maintain their functionality and still fit into the confined space of the cell. The solution to this cellular storage problem and the essential elements of eukaryotic chromosomes have been major themes of this chapter, completing the story of DNA structure introduced in Chapter 8.

Transposable genetic elements, DNA sequences that move, are a part of chromosome structure. Earlier chapters dealt with crossing over, in which homologous DNA sequences switch places, and chromosome rearrangements, in which the breakage and rejoining of chromosome segments moves blocks of genes to new locations. The movement of transposable elements is fundamentally different

from these other mechanisms of gene movement because transposable elements possess sequences that facilitate their movement. Understanding the structure of transposable genetic elements requires a basic knowledge of DNA structure and sequence, topics covered in Chapter 10.

Transposable elements violate a basic premise of classical genetics—that genes have a particular fixed location on a chromosome. This departure from a long-held view helps to explain why the discovery of transposable elements by Barbara McClintock was ignored for many years. A common theme in the history of genetics is that fundamental discoveries are often overlooked or unrecognized, because they require a radical rethinking of basic principles. Transposable elements today are recognized as ubiquitous DNA sequences with important implications for medicine, recombinant DNA technology, and evolution, but the reason for their widespread occurrence is still not completely understood.

This chapter has provided a foundation for topics introduced in several later chapters of the book. Transposition requires the replication of DNA (Chapter 12) or reverse transcription (Chapter 14) and generates gene mutations (Chapter 17). In Chapter 16, we explore the control of gene expression, which requires changes in chromatin structure. Condensed chromatin structure tends to inhibit the transcription of genetic information; some of the proteins that take part in activating and repressing transcription are known to affect the binding of DNA to histones. The regulation of transposition is by some of the same mechanisms that regulate the expression of other genes, also discussed in Chapter 16. Additional topics covered in more detail in later chapters include the origins of replication (Chapter 12) and the application of repetitive sequences to DNA fingerprinting (Chapter 18). Transposable elements are important in the generation of immune-system diversity (Chapter 21) and in molecular evolution (Chapter 23).

CONCEPTS SUMMARY

- Chromosomes contain very long DNA molecules that are tightly packed. Packing is accomplished through tertiary structures and the binding of DNA to proteins.
- Supercoiling results from strain produced when rotations are added or removed from a relaxed DNA molecule. Overrotation produces positive supercoiling; underrotation produces negative supercoiling.
- Topoisomerases control the degree of supercoiling by adding or removing rotations to DNA.
- A bacterial chromosome consists of a single, circular DNA molecule that is bound to proteins and exists as a series of large loops. It usually appears in the cell as a distinct clump known as the nucleoid.
- Each eukaryotic chromosome contains a single, very long linear DNA molecule that is bound to histone and nonhistone chromosomal proteins. Euchromatin undergoes the normal cycle of decondensation and condensation in the cell cycle. Heterochromatin remains highly condensed throughout the cell cycle.
- The nucleosome is a core of eight histone proteins (two each of H2A, H2B, H3, and H4) and DNA (145–147 bp) that wraps around it. The H1 protein holds DNA onto the histone core.
- Nucleosomes are folded into a 30-nm fiber that forms a series of 300-nm-long loops; these loops are anchored at their bases by proteins associated with the nuclear scaffold. The 300-nm loops are condensed to form a

- fiber that is 250 nm in diameter, which is itself tightly coiled to produce a 700-nm-wide chromatid.
- Chromosomal puffs are regions of localized unpacking of the DNA that are associated with regions of active transcription. Chromosome regions that are undergoing active transcription are relatively sensitive to digestion by DNase I, indicating that DNA unfolds during transcription.
 - Centromeres are chromosomal regions where spindle fibers attach; chromosomes without centromeres are usually lost in the course of cell division. Centromeres play an important role in the regulation of the cell cycle.
 - Telomeres stabilize the ends of chromosomes. Telomeric sequences consist of many copies of short sequences, which usually consist of a series of cytosine nucleotides followed by several adenine nucleotides. Longer telomere-associated sequences are found adjacent to the telomeric sequences.
 - The C value is the amount of DNA in an organism's genome. Eukaryotic organisms exhibit much variation in C value owing to differences in sequence complexity, which can be measured by observing the time required for denatured DNA to reanneal in a hybridization reaction, as plotted by a C_0t curve.
 - Eukaryotic DNA exhibits three classes of sequences. Unique-sequence DNA exists in very few copies. Moderately repetitive DNA consists of moderately long sequences that are repeated from hundreds to thousands of times. Highly repetitive DNA consists of very short sequences that are repeated in tandem from many thousands to millions of times.
 - Transposable elements are mobile DNA sequences that insert into many locations within a genome and often cause mutations and DNA rearrangements.
 - Most transposable elements have two common characteristics: terminal inverted repeats and the generation of short direct repeats in DNA at the point of insertion.
 - Transposition may take place through a DNA molecule or through the production of an RNA molecule that is then reverse transcribed into DNA. Transposition may be replicative, in which the transposable element is copied and the copy moves to a new site, or nonreplicative, in which the transposable element excises from the old site and moves to a new site.
 - Retrotransposons transpose through RNA molecules that undergo reverse transcription to produce DNA.
 - In many transposable elements, transposition is tightly regulated.
 - Insertion sequences are small bacterial transposable elements that carry only the information needed for their own movement. Composite transposons in bacteria are more complex elements that consist of DNA between two insertion sequences. Some complex transposable elements in bacteria do not contain insertion sequences.
 - Some transposable elements in eukaryotic cells are similar to those found in bacteria, ending in short inverted repeats and producing flanking direct repeats at the point of insertion. Others are retrotransposons, similar in structure to retroviruses and transposing through RNA intermediates.
 - Hybrid dysgenesis is the appearance of numerous mutations, chromosome rearrangements, and sterility when transposable *P* elements undergo a burst of transposition in *Drosophila*.
 - The evolutionary significance of transposable elements is unknown, but three hypotheses have been proposed to explain their common occurrence. The cellular function hypothesis suggests that transposable elements provide some important function for the cell; the genetic variation hypothesis proposes that transposable elements provide evolutionary flexibility by inducing mutations; and the selfish DNA hypothesis suggests that transposable elements do not benefit the cell but are widespread because they can replicate and spread.

IMPORTANT TERMS

transgenic mouse (p. 000)	nucleosome (p. 000)	unique-sequence DNA (p. 000)	transposition (p. 000)
transposable element (p. 000)	chromatosome (p. 000)	repetitive DNA (p. 000)	replicative transposition (p. 000)
supercoiling (p. 000)	linker DNA (p. 000)	moderately repetitive DNA (p. 000)	nonreplicative transposition (p. 000)
relaxed state of DNA (p. 000)	polytene chromosome (p. 000)	tandem repeat sequence (p. 000)	cointegrate structure (p. 000)
positive supercoiling (p. 000)	chromosomal puff (p. 000)	interspersed repeat sequences (p. 000)	transposase (p. 000)
negative supercoiling (p. 000)	centromeric sequence (p. 000)	short interspersed element (SINE) (p. 000)	resolvase (p. 000)
topoisomerase (p. 000)	telomeric sequence (p. 000)	long interspersed element (LINE) (p. 000)	retrotransposon (p. 000)
nucleoid (p. 000)	telomere-associated sequence (p. 000)	highly repetitive DNA (p. 000)	insertion sequence (p. 000)
euchromatin (p. 000)	C value (p. 000)	flanking direct repeat (p. 000)	composite transposon (p. 000)
heterochromatin (p. 000)	denaturation (melting) (p. 000)	terminal inverted repeats (p. 000)	delta sequence (p. 000)
nonhistone chromosomal proteins (p. 000)	melting temperature (T_m) (p. 000)		hybrid dysgenesis (p. 000)
chromosomal scaffold protein (p. 000)	renaturation (reannealing) (p. 000)		
high-mobility-group proteins (p. 000)	hybridization (p. 000)		

Worked Problems

1. A diploid plant cell contains 2 billion base pairs of DNA.

(a) How many nucleosomes are present in the cell?

(b) Give the numbers of molecules of each type of histone protein associated with the genomic DNA.

• Solution

Each nucleosome encompasses about 200 bp of DNA: from 144 to 147 bp of DNA wrapped twice around the histone core, from 20 to 22 bp of DNA associated with the H1 protein, and another 30 to 40 bp of linker DNA.

(a) To determine how many nucleosomes are present in the cell, we simply divide the total number of base pairs of DNA (2×10^9 bp) by the number of base pairs per nucleosome:

$$\frac{2 \times 10^9 \text{ nucleotides}}{2 \times 10^2 \text{ nucleotides per nucleosome}} = 1 \times 10^7 \text{ nucleosomes}$$

Thus, there are approximately 10 million nucleosomes in the cell.

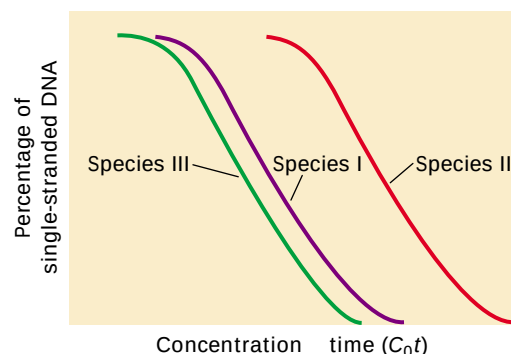
(b) Each nucleosome includes two molecules each of H2A, H2B, H3, and H4 histones. Therefore, there are 2×10^7 molecules each of H2A, H2B, H3, and H4 histones. Each nucleosome has associated with it one copy of the H1 histone; so there are 1×10^7 molecules of H1.

2. A renaturation reaction is carried out on the genomic DNA from three different bacterial species. Species I has a genome size of 2×10^6 bp, species II has a genome size of 1×10^8 bp, and species III has a genome size of 1×10^6 bp. Assume that the same total amount of DNA is used in each renaturation reaction and draw a C_0t curve for each species, showing the relative positions of each species on the same graph.

• Solution

Because this DNA is from bacteria, which contain only unique-sequence DNA, the complexity of renaturation with repetitive DNA can be ignored. If the total amount of DNA is the same for all three bacterial species, the number of copies of each sequence will depend on the genome size: there are fewer different sequences in organisms with small genomes, and so a chance collision is more likely to be between two complementary sequences that will anneal. Consequently, renaturation will proceed more rapidly in organisms with smaller genomes. (Recall the analogy of drawing cards from a hat; when there are only a few different names on the cards, the hat empties more quickly.)

At the start of the reaction, all the DNA is single stranded; so the proportion of single-stranded DNA is 1. As the reaction proceeds, single-stranded DNA pairs to form double-stranded DNA; so the proportion of single-stranded DNA decreases. This decrease will occur at a low C_0t in the organisms with a smaller genome, as shown in the following graph.



3. Genomic DNAs from species I, II, and III have the following base compositions:

Species	%			
	A	G	T	C
I	10	40	10	40
II	27	23	27	23
III	46	4	46	4

DNA from which species has a higher T_m ? Explain your reasoning.

• Solution

The melting temperature (T_m) of DNA depends on its base sequence. The three hydrogen bonds of a G–C base pair require more energy to break than do the two hydrogen bonds of an A–T pair; so a molecule with a higher percentage of G–C pairs will have a higher T_m . Species I has the highest G–C content of the three species; so it should exhibit the highest T_m .

4. Certain repeated sequences in eukaryotes are flanked by short direct repeats, suggesting that they originated as transposable elements. These same sequences lack introns and possess a string of thymine nucleotides at their 3' ends. Have these elements transposed through DNA or RNA sequences? Explain your reasoning.

• Solution

The absence of introns and the string of thymine nucleotides (which would be complementary to adenine nucleotides in RNA) at the 3' end are characteristics of processed RNA. These similarities to RNA suggest that the element was originally transcribed into mRNA, processed to remove the introns and to add a poly(A) tail, and then reverse transcribed into a complementary DNA that was inserted into the chromosome.

5. Which of the following pairs of sequences might be found at the ends of an insertion sequence?

- (a) 5'–TAAGGCCG–3' and 5'–TAAGGCCG–3'
- (b) 5'–AAAGGGCTA–3' and 5'–ATCGGGAAA–3'
- (c) 5'–GATCCAGTT–3' and 5'–CTAGGGTCAA–3'

(d) 5'–GATCCAGGT–3' and 5'–ACCTGGATC–3'

(e) 5'–AAAATTTT–3' and 5'–TTTTAAAA–3'

(f) 5'–AAAATTTT–3' and 5'–AAAATTTT–3'

• Solution

The correct answer is *d* and *f*. The ends of insertion sequences always have inverted repeats, which are sequences on the same strand that are inverted and complementary. The sequences in

part *a* are direct repeats, which are generated on the outside of an insertion sequence but are not part of the transposable element itself. The sequences in part *b* are inverted but not complementary. The sequences in part *c* are complementary but not inverted. The sequences in part *d* are both inverted and complementary. The sequences in part *e* are complementary but not inverted. Interestingly, the sequences in part *f* are both inverted complements and direct repeats.

The New Genetics

MINING GENOMES

INTRODUCTION TO BLAST AND BLAST SEARCHING

This exercise casts you in the role of biological detective, trying to figure out the functions of newly discovered genes. The simplest way to determine what is encoded by new sequences is to compare them with information already in the databases by using

BLAST (Basic Local Alignment Search Tools). You will use the National Center for Biotechnology Information (NCBI) Web site to explore some of the strengths and weaknesses of this powerful approach.

COMPREHENSION QUESTIONS

- * 1. How does supercoiling arise? What is the difference between positive and negative supercoiling?
2. What functions does supercoiling serve for the cell?
- * 3. Describe the composition and structure of the nucleosome. How do core particles differ from chromatosomes?
4. Describe in steps how the double helix of DNA, which is 2 nm in width, gives rise to a chromosome that is 700 nm in width.
5. What are polytene chromosomes and chromosomal puffs?
- * 6. Describe the function and molecular structure of the centromere.
- * 7. Describe the function and molecular structure of a telomere.
8. What is the C value of an organism?
9. What is a C_0t curve? Explain how C_0t curves of DNA provide evidence for the existence of repetitive DNA in eukaryotic cells.
- *10. Describe the different types of DNA sequences that exist in eukaryotes.
- *11. What general characteristics are found in many transposable elements? Describe the differences between replicative and nonreplicative transposition.
- *12. What is a retrotransposon and how does it move?
- *13. Describe the process of replicative transposition through DNA intermediates. What enzymes are required?
- *14. Draw and label the structure of a typical insertion sequence.
15. Draw and label the structure of a typical composite transposon in bacteria.
16. How are composite transposons and retrotransposons alike and how are they different?
17. Explain how *Ac* and *Ds* elements produce variegated corn kernels.
18. Briefly explain hybrid dysgenesis and how *P* elements lead to hybrid dysgenesis.
- *19. Briefly summarize three hypotheses for the widespread occurrence of transposable elements.

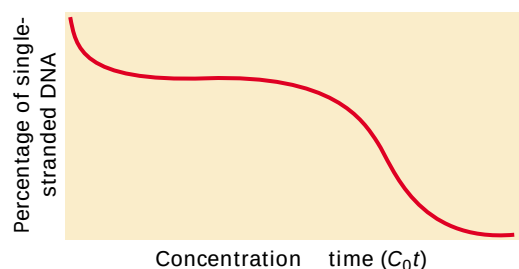
APPLICATION QUESTIONS AND PROBLEMS

- *20. Compare and contrast prokaryotic and eukaryotic chromosomes. How are they alike and how do they differ?
21. (a) In a typical eukaryotic cell, would you expect to find more molecules of the H1 histone or more molecules of the H2A histone? Explain your reasoning. (b) Would you expect to find more molecules of H2A or more molecules of H3? Explain your reasoning.
22. Suppose you examined polytene chromosomes from the salivary glands of fruit fly larvae and counted the number of chromosomal puffs observed in different regions of DNA.
 - (a) Would you expect to observe more puffs from euchromatin or from heterochromatin? Explain your answer.
 - (b) Would you expect to observe more puffs in unique-sequence DNA, moderately repetitive DNA, or repetitive DNA? Why?

- *23. A diploid human cell contains approximately 6 billion base pairs of DNA.
- How many nucleosomes are present in such a cell? (Assume that the linker DNA encompasses 40 bp.)
 - How many histone proteins are complexed to this DNA?
- *24. Would you expect to see more or less acetylation in regions of DNA that are sensitive to digestion by DNase I? Why?
25. A YAC that contains only highly repetitive, nonessential DNA is added to mouse cells that are growing in culture. The cells are then divided into two groups, A and B. A laser is then used to damage the centromere on the YACs in cells of group A. The centromeres on the YACs of group B are not damaged. In spite of the fact that the YACs contain no essential DNA, the cells in group A divide more slowly than those in group B. Provide a possible explanation.
26. Species A possesses only unique-sequence DNA. Species B possesses unique-sequence DNA and highly repetitive DNA. Species C possesses only moderately repetitive DNA. The genomes of all three species are similar in size. A student performs typical renaturation reactions with DNA from each species and plots a C_0t curve for each. Draw a C_0t curve for the renaturation reaction of each species.
- *27. Which of the following two molecules of DNA has the lower melting temperature? Why?

AGTTACTAAAGCAATACATC
TCAATGATTTCGTTATGTAG
AGGCGGGTAGGCACCCTTA
TCCGCCCATCCGTGGAAT

28. DNA was isolated from a newly discovered worm collected near a deep-sea vent in the Pacific Ocean. This DNA was sheared into pieces, heated to melting, and then cooled slowly. The amount of renaturation was measured with optical absorbance, and the following results were obtained.



What conclusions can you draw about the type of sequences found in this DNA?

29. Which of the following pairs of sequences might be found at the ends of an insertion sequence?
- 5'–GGGCCAATT–3' and 5'–CCCGGTTAA–3'
 - 5'–AAACCCTTT–3' and 5'–AAAGGGTTT–3'

- 5'–TTTCGAC–3' and 5'–CAGCTTT–3'
- 5'–ACGTACG–3' and 5'–CGTACGT–3'
- 5'–GCCCCAT–3' and 5'–GCCCCAT–3'

- *30. A particular transposable element generates flanking direct repeats that are 4 bp long. Give the sequence that will be found on both sides of the transposable element if this transposable element inserts at the position indicated on each of the following sequences.

- (a) Transposable element

5'–ATTTCGAACTGACCGATCA–3'

- (b) Transposable element

5'–ATTTCGAACTGACCGATCA–3'

- *31. White eyes in *Drosophila melanogaster* result from an X-linked recessive mutation. Occasionally, white-eyed mutants give rise to offspring that possess white eyes with small red spots. The number, distribution, and size of the red spots are variable. Explain how a transposable element could be responsible for this spotting phenomenon.
- *32. An insertion sequence contains a large deletion in its transposase gene. Under what circumstances would this insertion sequence be able to transpose?
- *33. What factor do you think determines the length of the flanking direct repeats that are produced in transposition?
34. A transposable element is found to encode a transposase enzyme. On the basis of this information, what conclusions can you make about the likely structure and method of transposition of this element?
35. A transposable element is found to encode a reverse transcriptase enzyme. On the basis of this information, what conclusions can you make about the likely structure and method of transposition of this element?
36. Transposition often produces chromosome rearrangements, such as deletions, inversions, and translocations. Can you suggest a reason why transposition leads to these chromosome mutations?
37. A geneticist studying the DNA of the Japanese bottle fly finds many copies of a particular sequence that appears similar to the *copia* transposable element in *Drosophila*. Using recombinant DNA techniques, the geneticist places an intron into a copy of this DNA sequence and inserts it into the genome of a Japanese bottle fly. If the sequence is a transposable element similar to *copia*, what prediction would you make concerning the fate of the introduced sequence in the genomes of offspring of the fly receiving it?

CHALLENGE QUESTIONS

38. An explorer discovers a strange new species of plant and sends some of the plant tissue to a geneticist to study. The geneticist isolates chromatin from the plant and examines it with the electron microscope. She observes what appear to be beads on a string. She then adds a small amount of nuclease, which cleaves the string into individual beads that each contain 280 bp of DNA. After digestion with more nuclease, she finds that a 120-bp fragment of DNA remains attached to a core of histone proteins. Analysis of the histone core reveals histones in the following proportions:

H1	12.5%
H2A	25%
H2B	25%
H3	0%
H4	25%
H7 (a new histone)	12.5%

On the basis of these observations, what conclusions could the geneticist make about the probable structure of the nucleosome in the chromatin of this plant?

39. Although highly repetitive DNA is common in eukaryotic chromosomes, it does not code for proteins; in fact, it is probably never transcribed into RNA. If highly repetitive DNA does not code for RNA or proteins, why is it present in eukaryotic genomes? Suggest some possible reasons for the widespread presence of highly repetitive DNA.
40. As discussed in the chapter, *Alu* sequences are retrotransposons that are common in the human genome. *Alu* sequences are thought to have evolved from the 7S RNA gene, which encodes an RNA molecule that takes part in transporting newly synthesized proteins across the endoplasmic reticulum. The 7S RNA gene is transcribed by RNA polymerase III, which uses an internal promoter (see Chapter 13). How might this observation explain the large number of copies of *Alu* sequences?
41. Houck and Kidwell proposed that *P* elements were carried from *Drosophila willistoni* to *D. melanogaster* by mites that fed on fruit flies. What evidence do you think would be required to demonstrate that *D. melanogaster* acquired *P* elements in this way? Propose a series of experiments to provide such evidence.

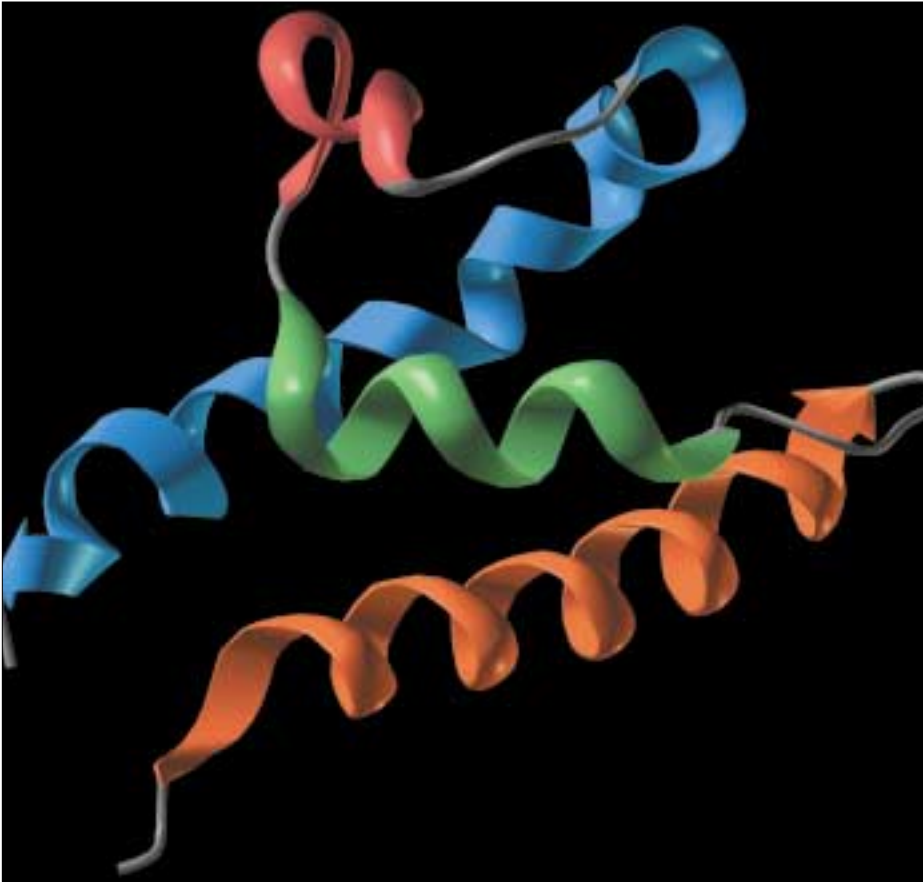
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12

DNA Replication and Recombination



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The Cause of Bloom Syndrome

Tommy was a full-term baby but weighed only 4.5 pounds (2 kg) at birth. At about 9 months of age, an unusual and persistent rash appeared on his face, and he frequently caught colds and infections. The illnesses caused no serious problems; so his parents were not concerned. Throughout childhood, Tommy remained small; by age 18, he was only 4 feet 6 inches (137 cm) in height.

Tommy's first major health problem arose shortly after he turned 22—he was diagnosed with intestinal cancer. The

tumor was surgically removed but additional, unrelated tumors appeared spontaneously over the next 10 years. Their appearance startled Tommy's doctors; the chance of multiple, independent cancers arising in the same person is generally remote. The propensity of Tommy's cells to become cancerous hinted at a high mutation rate in his genes. Indeed, when pathologists studied Tommy's chromosomes, they observed a wide range of abnormalities. Tommy had inherited Bloom syndrome.

Bloom syndrome is a rare autosomal recessive condition characterized by short stature, a facial rash induced by

- The Cause of Bloom Syndrome
- The Central Problem of Replication
- Semiconservative Replication
 - Meselson and Stahl's Experiment
 - Modes of Replication
 - Requirements of Replication
 - Direction of Replication
- The Mechanism of Replication
 - Bacterial DNA Replication
 - Eukaryotic DNA Replication
 - Replication in Archaea
- The Molecular Basis of Recombination
 - The Holliday Model
 - The Double-Strand-Break Model
 - Enzymes Required for Recombination

sun exposure, a small narrow head, and a predisposition to cancers of all types. The disorder is extremely rare; only several hundred cases have been reported worldwide. Cells from persons with Bloom syndrome exhibit excessive mutations in all genes, and numerous gaps and breaks occur in chromosomes that lead to extensive genetic exchange in cell division. Rates of DNA synthesis are retarded.

The characteristics of Bloom syndrome suggest that its underlying cause is a defect in DNA replication. In 1995, researchers at the New York Blood Center traced Bloom syndrome to a gene on chromosome 15 that encodes an enzyme called DNA helicase. A variety of helicase enzymes are responsible for unwinding double-stranded DNA during replication and repair. The cells of a person with Bloom syndrome carry two mutated copies of the gene and possess little or no activity for a particular helicase. Normal DNA replication is disrupted, leading to chromosome breaks and numerous mutations. The genetic damage resulting from faulty DNA replication leads to tumors. It is not yet clear whether the basic defect in DNA synthesis is associated with replication or DNA repair or both.

Rapid and accurate DNA replication is fundamental to normal cell function and health. Replication is a complex process in which dozens of proteins, enzymes, and DNA structures take part; a single defective component, such as DNA helicase, can disrupt the whole process.

This chapter deals with DNA replication, the process whereby a cell doubles its DNA before division. We begin with the basic mechanism of replication that emerged from the Watson and Crick structure of DNA. We then examine several different modes of replication, the requirements of replication, and the universal direction of DNA synthesis. We examine the enzymes and proteins that participate in the process and conclude the chapter by considering the molecular details of recombination, which is closely related to replication and is essential for the segregation of homologous chromosomes in meiosis, production of genetic variation, and for DNA repair.

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More information about the symptoms and genetics of Bloom syndrome

The Central Problem of Replication

In a schoolyard game, a verbal message, such as “John’s brown dog ran away from home,” is whispered to a child, who runs to a second child and repeats the message. The message is relayed from child to child around the schoolyard until it returns to the original sender. Inevitably, the last child returns with an amazingly transformed message, such as “Joe Brown has a pig living under his porch.” The more children playing the game, the more garbled the message becomes. This game illustrates an important principle: errors arise whenever information is copied; the more times it is copied, the greater the number of errors.

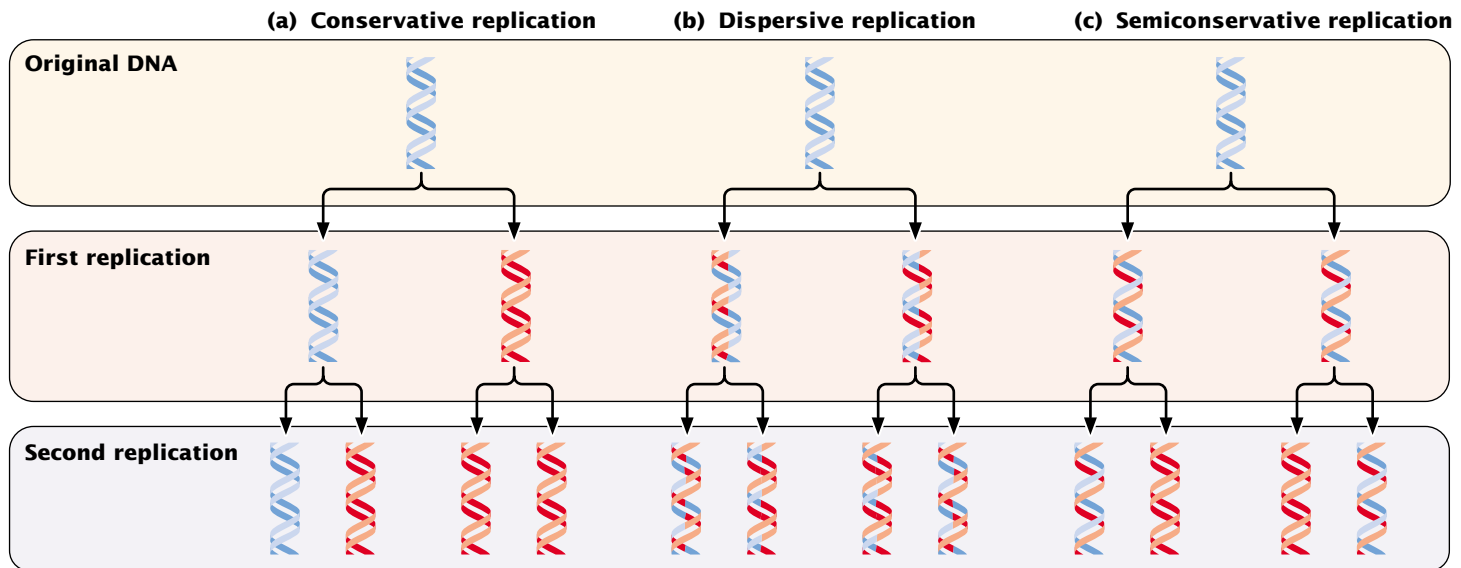
A complex, multicellular organism faces a problem similar to that of the children in the schoolyard game: how to faithfully transmit genetic instructions each time its cells divide. The solution to this problem is central to replication. A huge amount of genetic information and an enormous number of cell divisions are required to produce a multicellular adult organism; even a low rate of error during copying would be catastrophic. A single-celled human zygote contains 6 billion base pairs of DNA. If a copying error occurred only once per million base pairs, 6000 mistakes would be made every time a cell divided—errors that would be compounded at each of the millions of cell divisions that take place in human development.

Not only must the copying of DNA be astoundingly accurate, it must also take place at breakneck speed. The single, circular chromosome of *E. coli* contains about 4.7 million base pairs. At a rate of more than 1000 nucleotides per minute, replication of the entire chromosome would require almost 3 days. Yet, these bacteria are capable of dividing every 20 minutes. *E. coli* actually replicates its DNA at a rate of 1000 nucleotides per *second*, with fewer than one in a billion errors. How is this extraordinarily accurate and rapid process accomplished?

Semiconservative Replication

From the three-dimensional structure of DNA that Watson and Crick proposed in 1953 (see Figure 10.8), several important genetic implications were immediately apparent. The complementary nature of the two nucleotide strands in a DNA molecule suggested that, during replication, each strand can serve as a template for the synthesis of a new strand. The specificity of base pairing (adenine with thymine; guanine with cytosine) implied that only one sequence of bases can be specified by each template, and so two DNA molecules built on the pair of templates will be identical with the original. This process is called **semiconservative replication**, because each of the original nucleotide strands remains intact (conserved), despite no longer being combined in the same molecule; the original DNA molecule is half (semi) conserved during replication.

Initially, three alternative models were proposed for DNA replication. In conservative replication (● **FIGURE 12.1a**), the entire double-stranded DNA molecule serves as a template for a whole new molecule of DNA, and the original DNA molecule is *fully* conserved during replication. In dispersive replication (● **FIGURE 12.1b**), both nucleotide strands break down (disperse) into fragments, which serve as templates for the synthesis of new DNA fragments, and then somehow reassemble into two complete DNA molecules. In this model, each resulting DNA molecule is interspersed with fragments of old and new DNA; none of the original molecule is conserved. Semiconservative replication (● **FIGURE 12.1c**) is intermediate between these two models; the two nucleotide strands unwind and each serves as a template for a new DNA molecule.

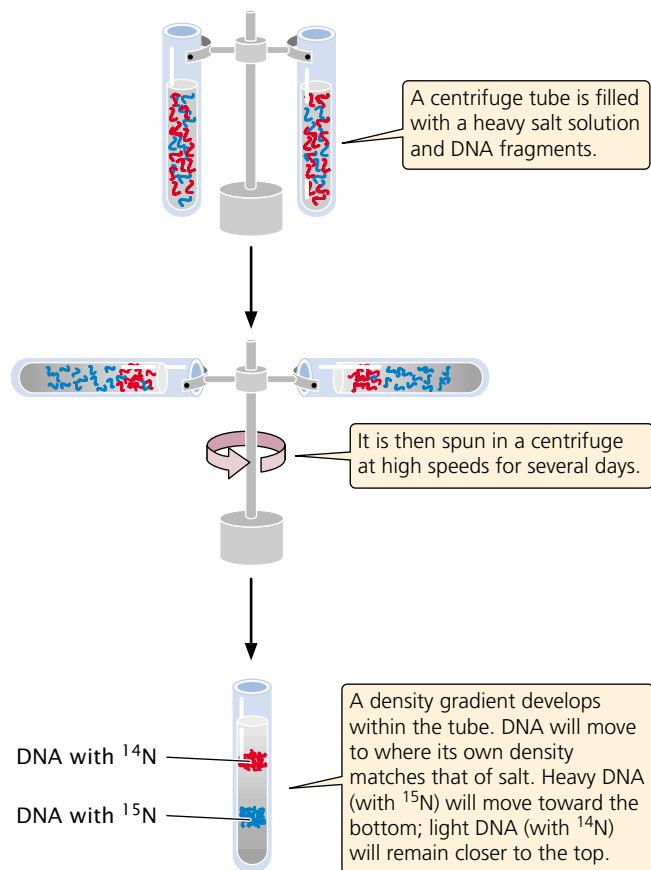


12.1 Three proposed models of replication are conservative replication, dispersive replication, and semiconservative replication.

These three models allow different predictions to be made about the distribution of original DNA and newly synthesized DNA after replication. With conservative replication, after one round of replication, 50% of the molecules would consist entirely of the original DNA and 50% would consist entirely of new DNA. After a second round of replication, 25% of the molecules would consist entirely of the original DNA and 75% would consist entirely of new DNA. With each additional round of replication, the proportion of molecules with new DNA would increase, although the number of molecules with the original DNA would remain constant. Dispersive replication would always produce hybrid molecules, containing some original and some new DNA, but the proportion of new DNA within the molecules would increase with each replication event. In contrast, with semiconservative replication, one round of replication would produce two hybrid molecules, each consisting of half original DNA and half new DNA. After a second round of replication, half the molecules would be hybrid, and the other half would consist of new DNA only. Additional rounds of replication would produce more and more molecules consisting entirely of new DNA, and a few hybrid molecules would persist.

Meselson and Stahl's Experiment

To determine which of the three models of replication applied to *E. coli* cells, Matthew Meselson and Franklin Stahl needed a way to distinguish old and new DNA. They did so by using two isotopes of nitrogen, ^{14}N (the common form) and ^{15}N (a rare, heavy form). Meselson and Stahl grew a culture of *E. coli* in a medium that contained ^{15}N as the sole nitrogen source; after many generations, all



12.2 Meselson and Stahl used equilibrium density gradient centrifugation to distinguish between heavy, ^{15}N -laden DNA and lighter, ^{14}N -laden DNA.

the *E. coli* cells had ^{15}N incorporated into the purine and pyrimidine bases of DNA (see Figure 10.10). Meselson and Stahl took a sample of these bacteria, switched the rest of the bacteria to a medium that contained only ^{14}N , and then took additional samples of bacteria over the next few cellular generations. In each sample, the bacterial DNA that was synthesized before the change in medium contained ^{15}N and was relatively heavy, whereas any DNA synthesized after the switch contained ^{14}N and was relatively light.

Meselson and Stahl distinguished between the heavy ^{15}N -laden DNA and the light ^{14}N -containing DNA with the use of **equilibrium density gradient centrifugation** (FIGURE 12.2). In this technique, a centrifuge tube is filled with a heavy salt solution and a substance whose density is to be measured—in this case, DNA fragments. The tube is then spun in a centrifuge at high speeds. After several days of spinning, a gradient of density develops within the tube, with high density at the bottom and low density at the top.

The density of the DNA fragments matches that of the salt: light molecules rise and heavy molecules sink.

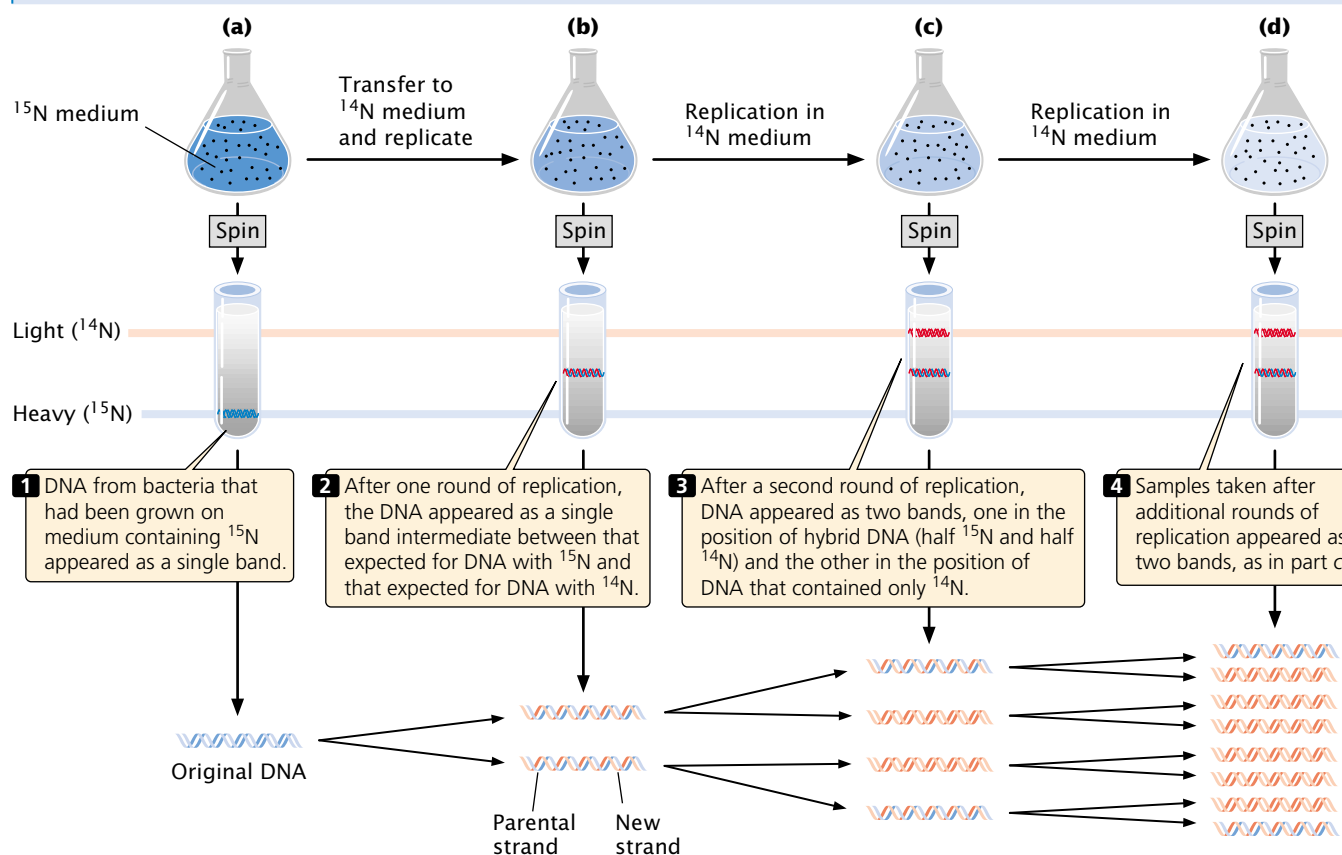
Meselson and Stahl found that DNA from bacteria grown only on medium containing ^{15}N produced a single band at the position expected of DNA containing only ^{15}N (FIGURE 12.3a). DNA from bacteria transferred to the medium with ^{14}N and allowed one round of replication also produced a single band, but at a position intermediate between that expected of DNA with only ^{15}N and that expected of DNA with only ^{14}N (FIGURE 12.3b).^{*} After a second round of replication in medium with ^{14}N , two bands of equal intensity appeared, one in the intermediate posi-

^{*}This result is inconsistent with the conservative replication model, which predicts one heavy band (the original DNA molecules) and one light band (the new DNA molecules). A single band of intermediate density is predicted by both the semiconservative and dispersive models.

To distinguish between these two models, Meselson and Stahl grew the bacteria in medium containing ^{14}N for a second generation.

Experiment

Question: Which model of DNA replication—conservative, dispersive or semiconservative—applies to *E. coli*?



Conclusion: DNA replication in *E. coli* is semiconservative.

12.3 Meselson and Stahl demonstrated that DNA replication is semiconservative.

tion and the other at the position expected of DNA that contained only ^{14}N (FIGURE 12.3c). All samples taken after additional rounds of replication produced two bands, and the band representing light DNA became progressively stronger (FIGURE 12.3d). Meselson and Stahl's results were exactly as expected for semiconservative replication and are incompatible with those predicated for both conservative and dispersive replication.

Concepts

Replication is semiconservative: each DNA strand serves as a template for the synthesis of a new DNA molecule. Meselson and Stahl convincingly demonstrated that replication in *E. coli* is semiconservative.

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A summary of Meselson and Stahl's experiment

A summary of Meselson and Stahl's experiment

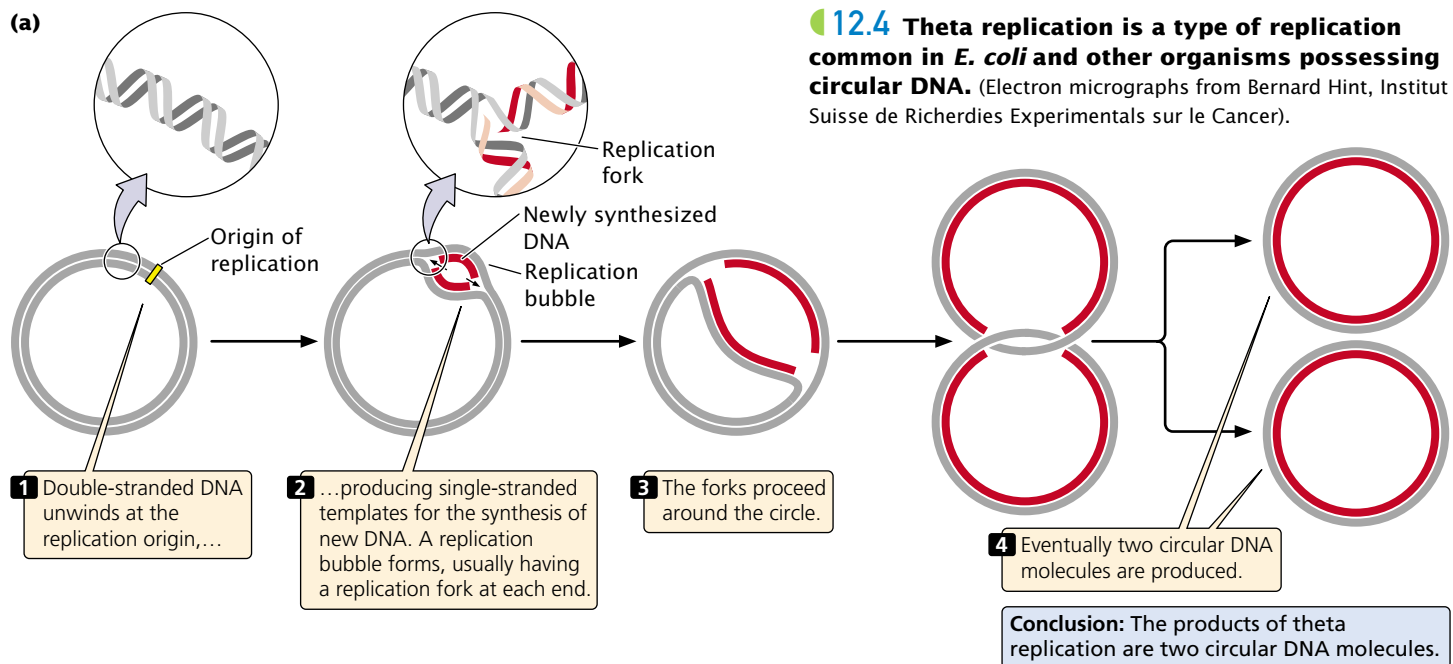
Modes of Replication

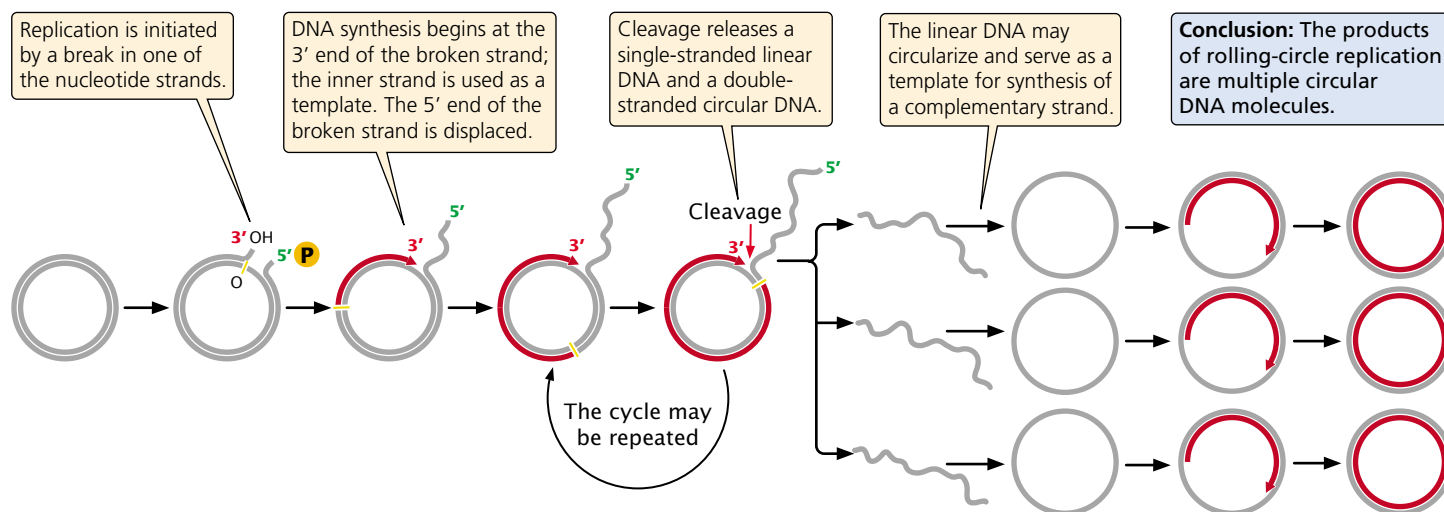
Following Meselson and Stahl's work, investigators confirmed that other organisms also use semiconservative replication. No evidence was found for conservative or dispersive replication. There are, however, several different ways that semiconservative replication can take place, differing principally in the nature of the template DNA—whether it is linear or circular—and in the number of replication forks.

Individual units of replication are called **replicons**, each of which contains a **replication origin**. Replication starts at the origin and continues until the entire replicon has been replicated. Bacterial chromosomes have a single replication origin, whereas eukaryotic chromosomes contain many.

Theta replication A common type of replication that takes place in circular DNA, such as that found in *E. coli* and other bacteria, is called **theta replication** (FIGURE 12.4a), because it generates a structure that resembles the Greek letter theta (θ). In theta replication, double-stranded DNA begins to unwind at the replication origin, producing single-stranded

12.4 Theta replication is a type of replication common in *E. coli* and other organisms possessing circular DNA. (Electron micrographs from Bernard Hint, Institut Suisse de Richerches Experimentales sur le Cancer).





12.5 Rolling-circle replication takes place in some viruses and in the F factor of *E. coli*.

nucleotide strands that then serve as templates on which new DNA can be synthesized. The unwinding of the double helix generates a loop, termed a **replication bubble**. Unwinding may be at one or both ends of the bubble, making it progressively larger. DNA replication on both of the template strands is simultaneous with unwinding. The point of unwinding, where the two single nucleotide strands separate from the double-stranded DNA helix, is called a **replication fork**.

If there are two replication forks, one at each end of the replication bubble, the forks proceed outward in both directions in a process called **bidirectional replication**, simultaneously unwinding and replicating the DNA until they eventually meet. If a single replication fork is present, it proceeds around the entire circle to produce two complete circular DNA molecules, each consisting of one old and one new nucleotide strand.

John Cairns provided the first visible evidence of theta replication in 1963 by growing bacteria in the presence of radioactive nucleotides. After replication, each DNA molecule consisted of one "hot" (radioactive) strand and one "cold" (nonradioactive) strand. Cairns isolated DNA from the bacteria after replication and placed it on an electron microscope grid, which was then covered with a photographic emulsion. Radioactivity present in the sample exposes the emulsion and produces a picture of the molecule (called an autoradiograph), similar to the way that light exposes a photographic film. Because the newly synthesized DNA contained radioactive nucleotides, Cairns was able to produce an electron micrograph of the replication process, similar to those shown in [FIGURE 12.4b](#).

Rolling-circle replication Another form of replication, called **rolling-circle replication** ([FIGURE 12.5](#)), takes place in some viruses and in the F factor (a small circle of extra-chromosomal DNA that controls mating, discussed in

Chapter 8) of *E. coli*. This form of replication is initiated by a break in one of the nucleotide strands that creates a 3'-OH group and a 5'-phosphate group. New nucleotides are added to the 3' end of the broken strand, with the inner (unbroken) strand used as a template. As new nucleotides are added to the 3' end, the 5' end of the broken strand is displaced from the template, rolling out like thread being pulled off a spool. The 3' end grows around the circle, giving rise to the name rolling-circle model.

The replication fork may continue around the circle a number of times, producing several linked copies of the same sequence. With each revolution around the circle, the growing 3' end displaces the nucleotide strand synthesized in the preceding revolution. Eventually, the linear DNA molecule is cleaved from the circle, resulting in a double-stranded circular DNA molecule and a single-stranded linear DNA molecule. The linear molecule circularizes either before or after serving as a template for the synthesis of a complementary strand.

Linear eukaryotic replication Circular DNA molecules that undergo theta or rolling-circle replication have a single origin of replication. Because of the limited size of these DNA molecules, replication starting from one origin can traverse the entire chromosome in a reasonable amount of time. The large linear chromosomes in eukaryotic cells, however, contain far too much DNA to be replicated speedily from a single origin. Eukaryotic replication proceeds at a rate ranging from 500 to 5000 nucleotides per minute at each replication fork (considerably slower than bacterial replication). Even at 5000 nucleotides per minute at each fork, DNA synthesis starting from a single origin would require 7 days to replicate a typical human chromosome consisting of 100 million base pairs of DNA. The replication of eukaryotic chromosomes actually occurs in a matter of minutes or hours, not

Table 12.1 Number and length of replicons

Organism	Number of Replication Origins	Average Length of Replicon (bp)
<i>Escherichia coli</i> (bacterium)	1	4,200,000
<i>Saccharomyces cerevisiae</i> (yeast)	500	40,000
<i>Drosophila melanogaster</i> (fruit fly)	3,500	40,000
<i>Xenopus laevis</i> (toad)	15,000	200,000
<i>Mus musculus</i> (mouse)	25,000	150,000

Source: Data from B. L. Lewin, *Genes V* (Oxford: Oxford University Press, 1994), p. 536.

days. This rate is possible because replication takes place simultaneously from thousands of origins.

Typical eukaryotic replicons are from 20,000 to 300,000 base pairs in length (Table 12.1). At each replication origin, the DNA unwinds and produces a replication bubble. Replication takes place on both strands at each end of the bubble, with the two replication forks spreading outward. Eventually, replication forks of adjacent replicons run into each other, and the replicons fuse to form long stretches of newly synthesized DNA (FIGURE 12.6). Replication and fusion of all the replicons leads to two identical DNA molecules. Important features of theta replication, rolling-circle replication, and linear eukaryotic replication are summarized in Table 12.2.

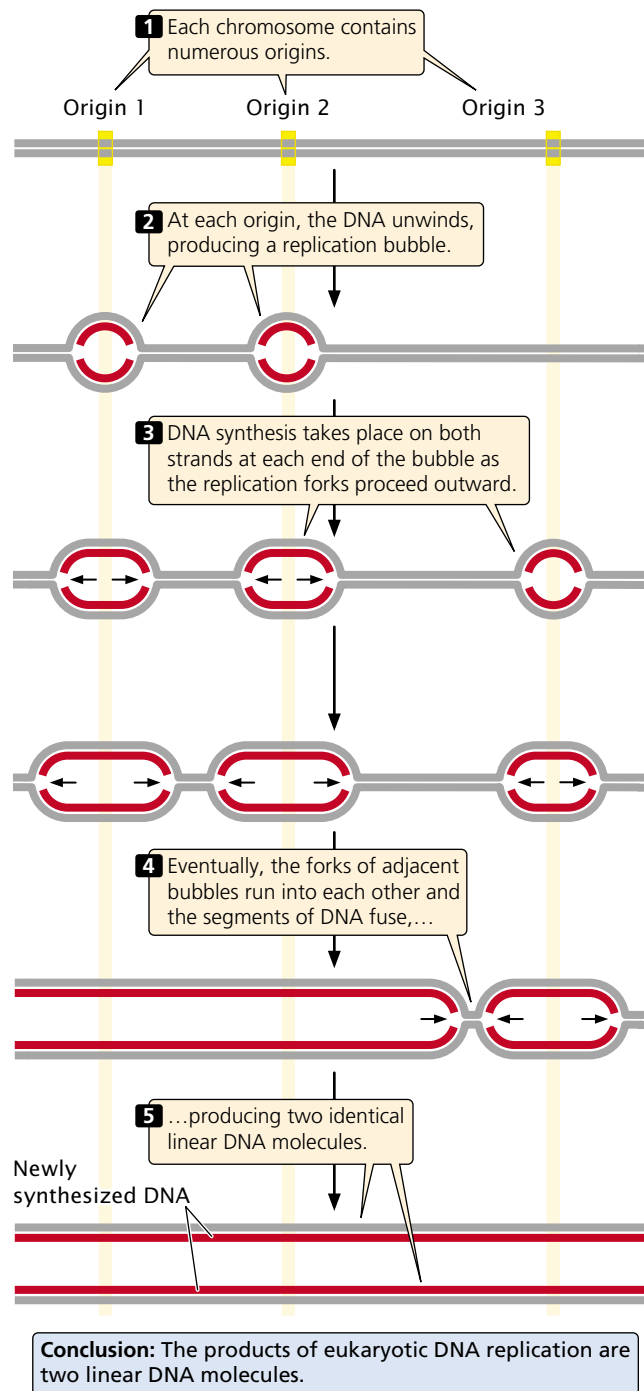
Concepts

Theta replication, rolling-circle replication, and linear replication differ with respect to the initiation and progression of replication, but all produce new DNA molecules by semiconservative replication.

Requirements of Replication

Although the process of replication includes many components, they can be combined into three major groups:

1. a template consisting of single-stranded DNA,
2. raw materials (substrates) to be assembled into a new nucleotide strand, and
3. enzymes and other proteins that “read” the template and assemble the substrates into a DNA molecule.



12.6 Linear DNA replication takes place in eukaryotic chromosomes.

Because of the semiconservative nature of DNA replication, a double-stranded DNA molecule must unwind to expose the bases that act as a template for the assembly of new polynucleotide strands, which are made complementary and antiparallel to the template strands. The raw materials from which new DNA molecules are synthesized are deoxyribonucleoside triphosphates (dNTPs), each consisting of a deoxyribose sugar and a base (a nucleoside)

proteins and enzymes as we consider the replication process in more detail.

Concepts

DNA synthesis requires a single-stranded DNA template, deoxyribonucleoside triphosphates, a growing nucleotide strand, and a group of enzymes and proteins.

Direction of Replication

In DNA synthesis, new nucleotides are joined one at a time to the 3' end of the newly synthesized strand. **DNA polymerases**, the enzymes that synthesize DNA, can add nucleotides *only* to the 3' end of the growing strand (not the 5' end), so new DNA strands always elongate in the same 5'→3' direction (5'→3'). Because the two single-stranded DNA templates are antiparallel and strand elongation is always 5'→3', if synthesis on one template proceeds from, say, right to left, then synthesis on the other template must proceed in the opposite direction, from left to right (FIGURE 12.8). As DNA unwinds during replication, the antiparallel nature of the two DNA strands means that one template is exposed in the 5'→3' direction and the other template is exposed in the 3'→5' direction (see Figure 12.8); so how can synthesis take place simultaneously on both strands at the fork?

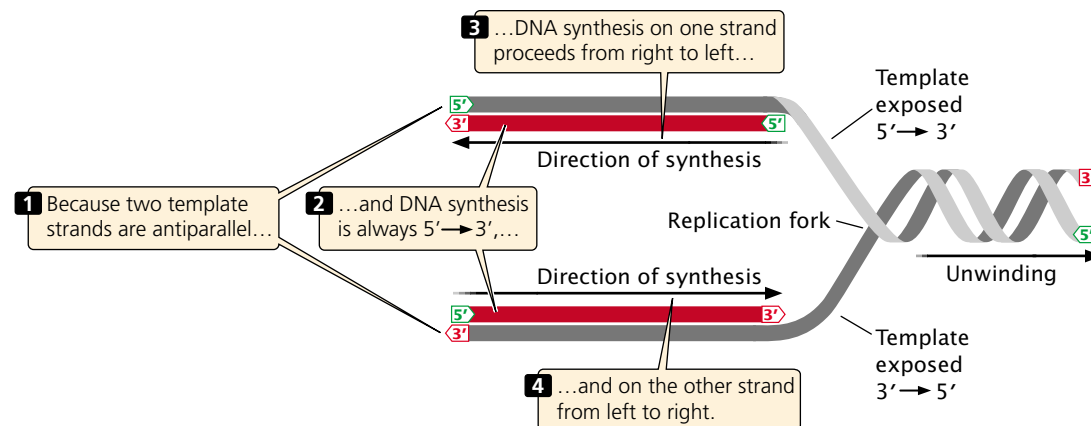
As the DNA unwinds, the template strand that is exposed in the 3'→5' direction (the lower strand in Figures 12.8 and 12.9) allows the new strand to be synthesized continuously, in the 5'→3' direction. This new strand, which undergoes **continuous replication**, is called the **leading strand**.

The other template strand is exposed in the 5'→3' direction (the upper strand in Figures 12.8 and 12.9). After

a short length of the DNA has been unwound, synthesis must proceed 5'→3'; that is, in the direction *opposite* that of unwinding (FIGURE 12.9). Because only a short length of DNA needs to be unwound before synthesis on this strand gets started, the replication machinery soon runs out of template. By that time, more DNA has unwound, providing new template at the 5' end of the new strand. DNA synthesis must start anew at the replication fork and proceed in the direction opposite that of the movement of the fork until it runs into the previously replicated segment of DNA. This process is repeated again and again, so synthesis of this strand is in short, discontinuous bursts. The newly made strand that undergoes **discontinuous replication** is called the **lagging strand**.

The short lengths of DNA produced by discontinuous replication of the lagging strand are called **Okazaki fragments**, after Reiji Okazaki, who discovered them. In bacterial cells, each Okazaki fragment ranges in length from about 1000 to 2000 nucleotides; in eukaryotic cells, they are about 100 to 200 nucleotides long. Okazaki fragments on the lagging strand are linked together to create a continuous new DNA molecule.

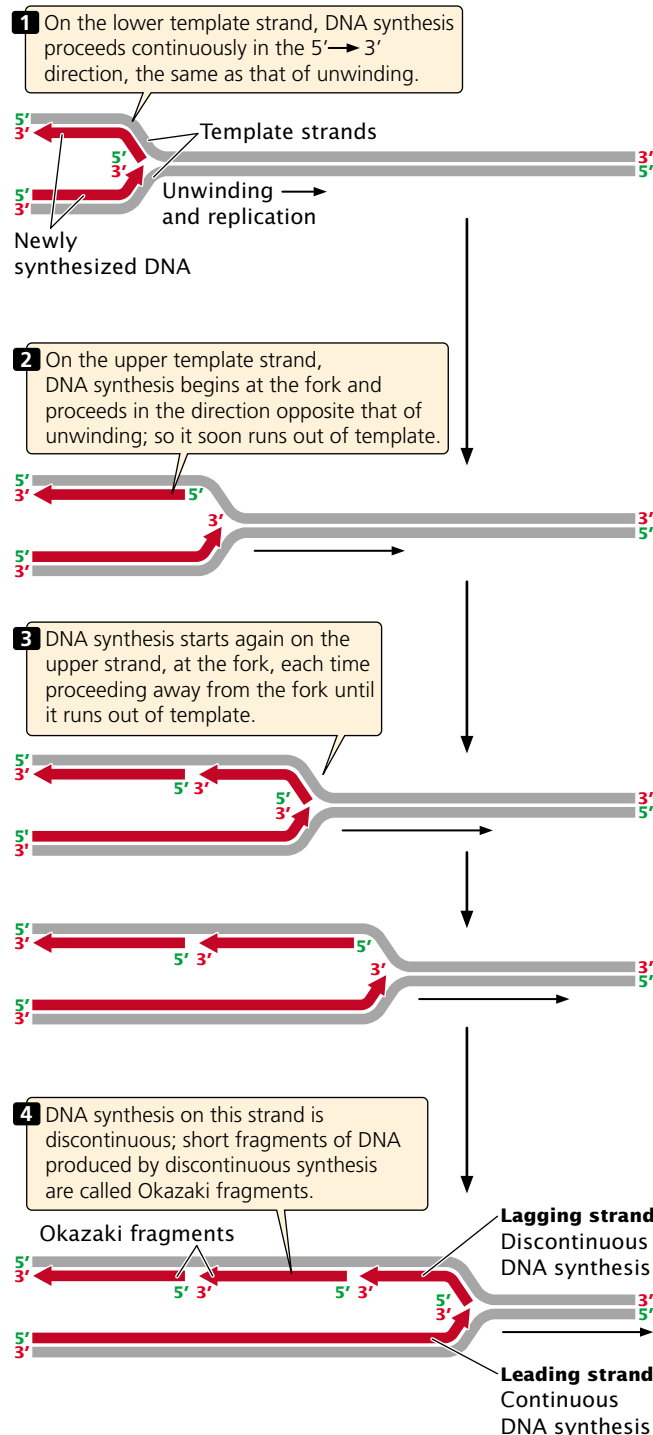
Let's relate the direction of DNA synthesis to the modes of replication examined earlier. In the theta model (FIGURE 12.10a), the DNA unwinds at one particular location, the origin, and a replication bubble is formed. If the bubble has two forks, one at each end, synthesis takes place simultaneously at both forks (bidirectional replication). At each fork, synthesis on one of the template strands proceeds in the same direction as that of unwinding; the newly replicated strand is the leading strand with continuous replication. On the other template strand, synthesis is proceeding in the direction opposite that of unwinding; this newly synthesized strand is the lagging strand with discontinuous replication. Focus on just one of the template strands within the bubble. Notice that synthesis on this template



12.8 DNA synthesis takes place simultaneously but in opposite directions on the two DNA template strands. DNA replication at a single replication fork begins when a double-stranded DNA molecule unwinds to provide two single-strand templates.

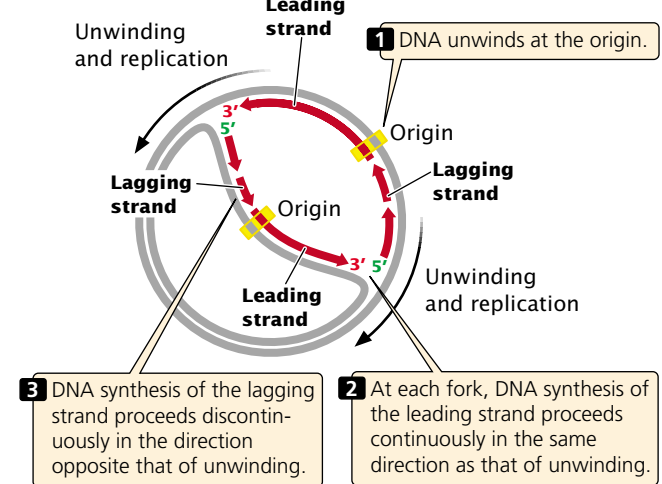
strand is continuous at one fork but discontinuous at the other. This difference arises because DNA synthesis is always in the same direction ($5' \rightarrow 3'$), but the two forks are moving in opposite directions.

Replication in the rolling-circle model (FIGURE 12.10b) is somewhat different, because there is no replication bubble.

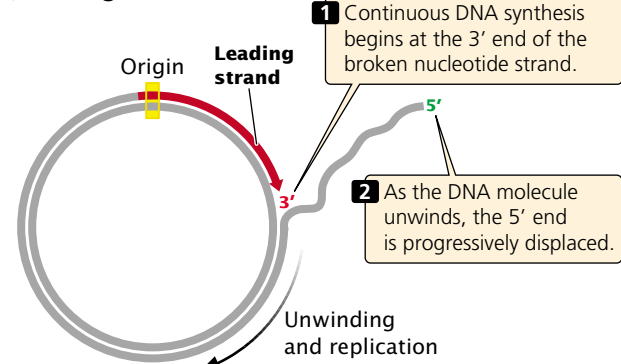


12.9 DNA synthesis is continuous on one template strand of DNA and discontinuous on the other.

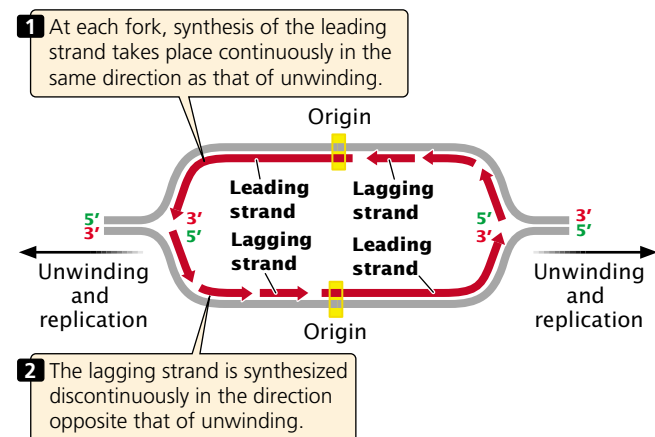
(a) Theta model



(b) Rolling-circle model



(c) Linear eukaryotic replication



12.10 The process of replication differs in theta replication, rolling-circle replication, and linear replication.

Replication begins at the $3'$ end of the broken nucleotide strand. Continuous replication takes place on the circular template as new nucleotides are added to this $3'$ end.

The replication of linear molecules of DNA, such as those found in eukaryotic cells, produces a series of replica-

tion bubbles (FIGURE 12.10c). DNA synthesis in these bubbles is the same as that in the single replication bubble of the theta model; it begins at the center of each replication bubble and proceeds at two forks, one at each end of the bubble. At both forks, synthesis of the leading strand proceeds in the same direction as that of unwinding, whereas synthesis of the lagging strand proceeds in the direction opposite that of unwinding.

Concepts

All DNA synthesis is $5' \rightarrow 3'$, meaning that new nucleotides are always added to the $3'$ end of the growing nucleotide strand. At each replication fork, synthesis of the leading strand proceeds continuously and that of the lagging strand proceeds discontinuously.

The Mechanism of Replication

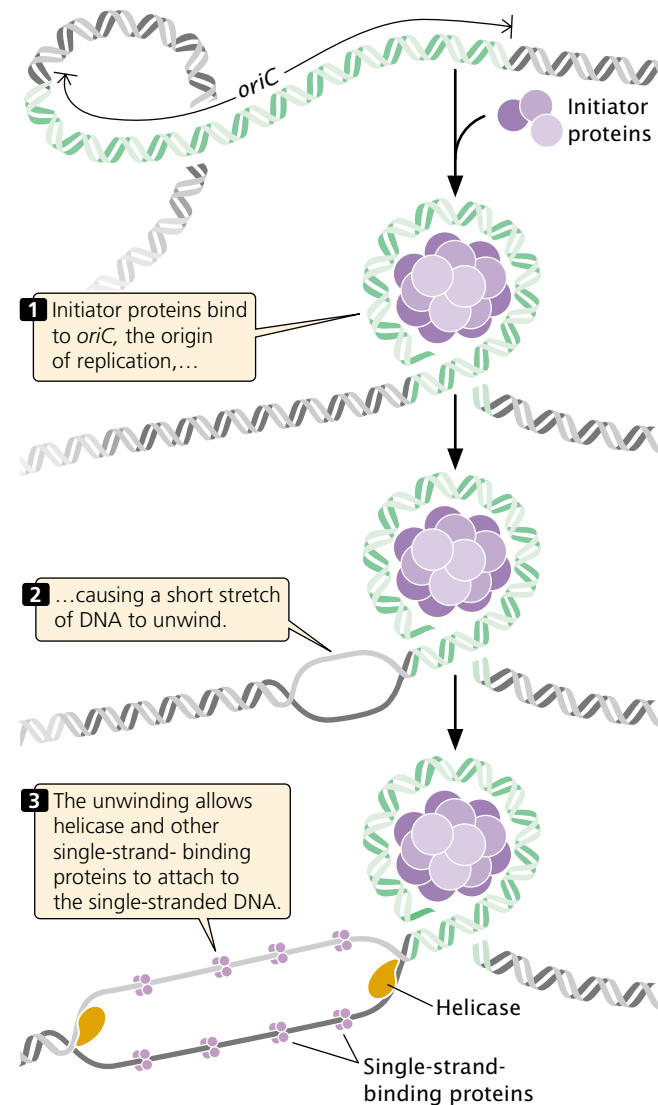
Replication takes place in four stages: initiation, unwinding, elongation, and termination.

Bacterial DNA Replication

The following discussion of the process of replication will focus on bacterial systems, where replication has been most thoroughly studied and is best understood. Although many aspects of replication in eukaryotic cells are similar to those of prokaryotic cells, there are some important differences. We will compare bacterial and eukaryotic replication later in the chapter.

Initiation The circular chromosome of *E. coli* has a single replication origin (*oriC*). The minimal sequence required for *oriC* to function consists of 245 bp that contain several critical sites. **Initiator proteins** bind to *oriC* and cause a short section of DNA to unwind. This unwinding allows helicase and other single-strand-binding proteins to attach to the polynucleotide strand (FIGURE 12.11).

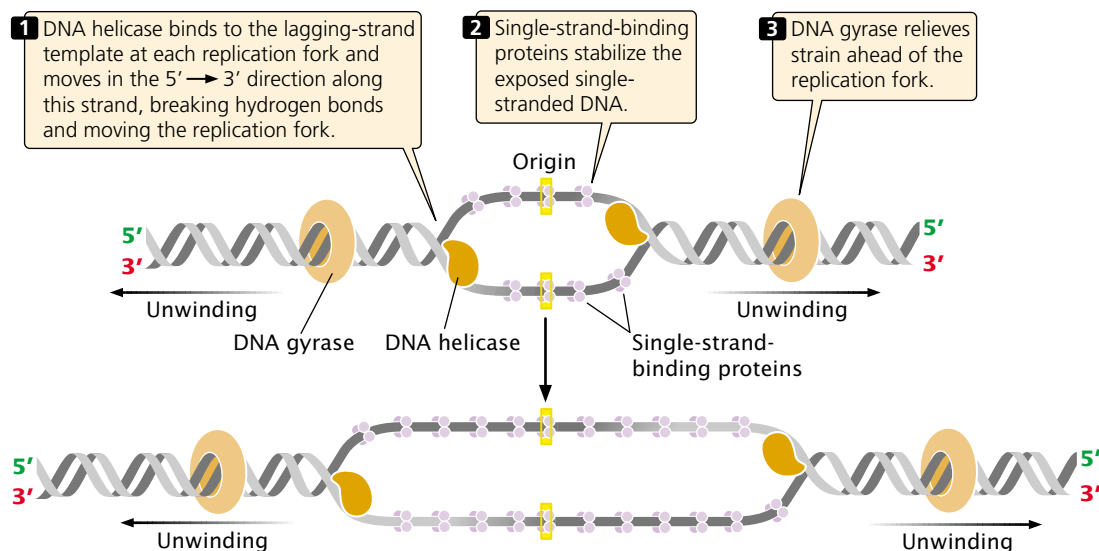
Unwinding Because DNA synthesis requires a single-stranded template and double-stranded DNA must be unwound before DNA synthesis can take place, the cell relies on several proteins and enzymes to accomplish the unwinding. **DNA helicases** break the hydrogen bonds that exist between the bases of the two nucleotide strands of a DNA molecule. Helicases cannot *initiate* the unwinding of double-stranded DNA; the initiator proteins first separate DNA strands at the origin, providing a short stretch of single-stranded DNA to which a helicase binds. Helicases bind to the lagging-strand template at each replication fork and move in the $5' \rightarrow 3'$ direction along this strand, thus also moving the replication fork (FIGURE 12.12).



12.11 *E. coli* DNA replication begins when initiator proteins bind to *oriC*, the origin of replication, causing a short stretch of DNA to unwind.

After DNA has been unwound by helicase, the single-stranded nucleotide chains have a tendency to form hydrogen bonds and reanneal (stick back together). Secondary structures, such as hairpins (see Figure 8.21), also may form between complementary nucleotides on the same strand. To stabilize the single-stranded DNA long enough for replication to take place, **single-strand-binding (SSB) proteins** attach tightly to the exposed single-stranded DNA (see Figure 12.12). Unlike many DNA-binding proteins, SSBs are indifferent to base sequence—they will bind to any single-stranded DNA. Single-strand-binding proteins form tetramers (groups of four) that together cover from 35 to 65 nucleotides.

Another protein essential for the unwinding process is the enzyme **DNA gyrase**, a topoisomerase. As discussed in



12.12 DNA helicase unwinds DNA by binding to the lagging-strand template at each replication fork and moving in the 5' → 3' direction along the strand.

Chapter 11, topoisomerases control the supercoiling of DNA. In replication, DNA gyrase reduces torsional strain (torque) that builds up ahead of the replication fork as a result of unwinding (see Figure 12.12). It reduces torque by making a double-stranded break in one segment of the DNA helix, passing another segment of the helix through the break, and then resealing the broken ends of the DNA. This action removes a twist in the DNA and reduces the supercoiling.

Concepts

Replication is initiated at a replication origin, where an initiator protein binds and causes a short stretch of DNA to unwind. DNA helicase breaks hydrogen bonds at a replication fork, and single-strand-binding proteins stabilize the separated strands. DNA gyrase reduces torsional strain that develops as the two strands of double-helical DNA unwind.

Primers All DNA polymerases require a nucleotide with a 3'-OH group to which a new nucleotide can be added. Because of this requirement, DNA polymerases cannot initiate DNA synthesis on a bare template; rather, they require a primer—an existing 3'-OH group—to get started. How, then, does DNA synthesis begin?

An enzyme called **primase** synthesizes short stretches of nucleotides (**primers**) to get DNA replication started. Primase synthesizes a short stretch of RNA nucleotides (about 10–12 nucleotides long), which provides a 3'-OH

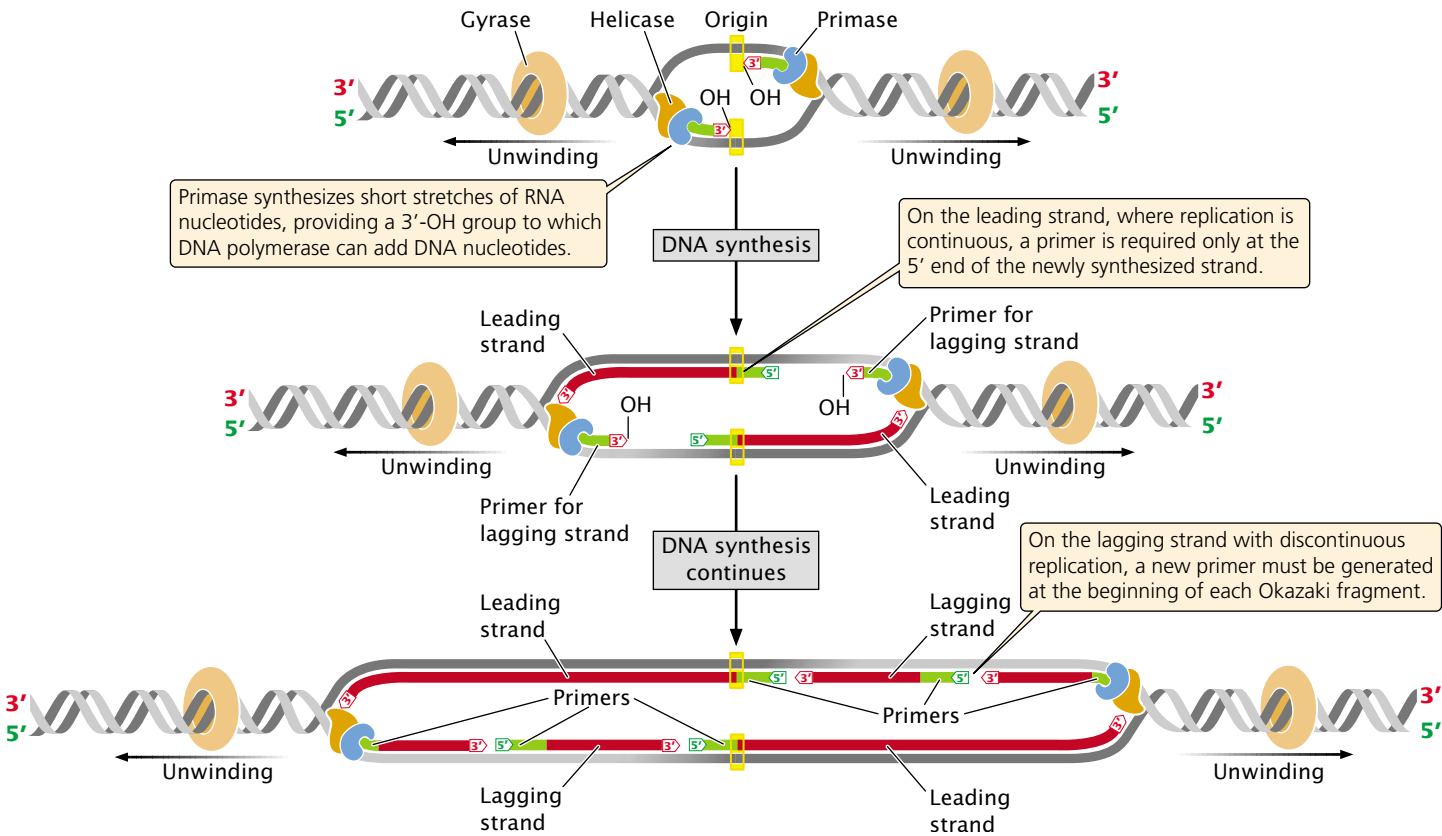
group to which DNA polymerase can attach DNA nucleotides. (Because primase is an RNA polymerase, it does not require an existing 3'-OH group to which nucleotides can be added.) All DNA molecules initially have short RNA primers imbedded within them; these primers are later removed and replaced by DNA nucleotides.

On the leading strand, where DNA synthesis is continuous, a primer is required only at the 5' end of the newly synthesized strand. On the lagging strand, where replication is discontinuous, a new primer must be generated at the beginning of each Okazaki fragment (FIGURE 12.13). Primase forms a complex with helicase at the replication fork and moves along the template of the lagging strand. The single primer on the leading strand is probably synthesized by the primase–helicase complex on the template of the lagging strand of the *other* replication fork, at the opposite end of the replication bubble.

Concepts

Primase synthesizes a short stretch of RNA nucleotides (primers), which provides a 3'-OH group for the attachment of DNA nucleotides to start DNA synthesis.

Elongation After DNA is unwound and a primer has been added, DNA polymerases elongate the polynucleotide strand by catalyzing DNA polymerization. The best-studied polymerases are those of *E. coli*, which has at least five different DNA polymerases. Two of them, DNA polymerase I and DNA polymerase III, carry out DNA synthesis associ-



12.13 Primase synthesizes short stretches of RNA nucleotides, providing a 3'-OH group to which DNA polymerase can add DNA nucleotides.

ated with replication; the other three have specialized functions in DNA repair (Table 12.3).

DNA polymerase III is a large multiprotein complex that acts as the main workhorse of replication. DNA polymerase III synthesizes nucleotide strands by adding new nucleotides to the 3' end of growing DNA molecules. This enzyme has two enzymatic activities (Table 12.3). Its 5'→3'

polymerase activity allows it to add new nucleotides in the 5'→3' direction. Its 3'→5' exonuclease activity allows it to remove nucleotides in the 3'→5' direction, enabling it to correct errors. If a nucleotide having an incorrect base is inserted into the growing DNA molecule, DNA polymerase III uses its 3'→5' exonuclease activity to back up and remove the incorrect nucleotide. It then resumes its 5'→3'

Table 12.3 Characteristics of DNA Polymerases in <i>E. coli</i>				
DNA Polymerase	5'→3' Polymerization	3'→5' Exonuclease	5'→3' Exonuclease	Function
I	Yes	Yes	Yes	Removes and replaces primers
II	Yes	Yes	No	DNA repair; restarts replication after damaged DNA halts synthesis
III	Yes	Yes	No	Elongates DNA
IV	Yes	No	No	DNA repair
V	Yes	No	No	DNA repair; translesion DNA synthesis

polymerase activity. These two functions together allow DNA polymerase III to efficiently and accurately synthesize new DNA molecules.

The first *E. coli* polymerase to be discovered, **DNA polymerase I**, also has 5'→3' polymerase and 3'→5' exonuclease activities (see Table 12.3), permitting the enzyme to synthesize DNA and to correct errors. Unlike DNA polymerase III, however, DNA polymerase I also possesses 5'→3' exonuclease activity, which is used to remove the primers laid down by primase and to replace them with DNA nucleotides by moving in a 5'→3' direction. The removal and replacement of primers appear to constitute the main function of DNA polymerase I. DNA polymerases IV and V function in DNA repair.

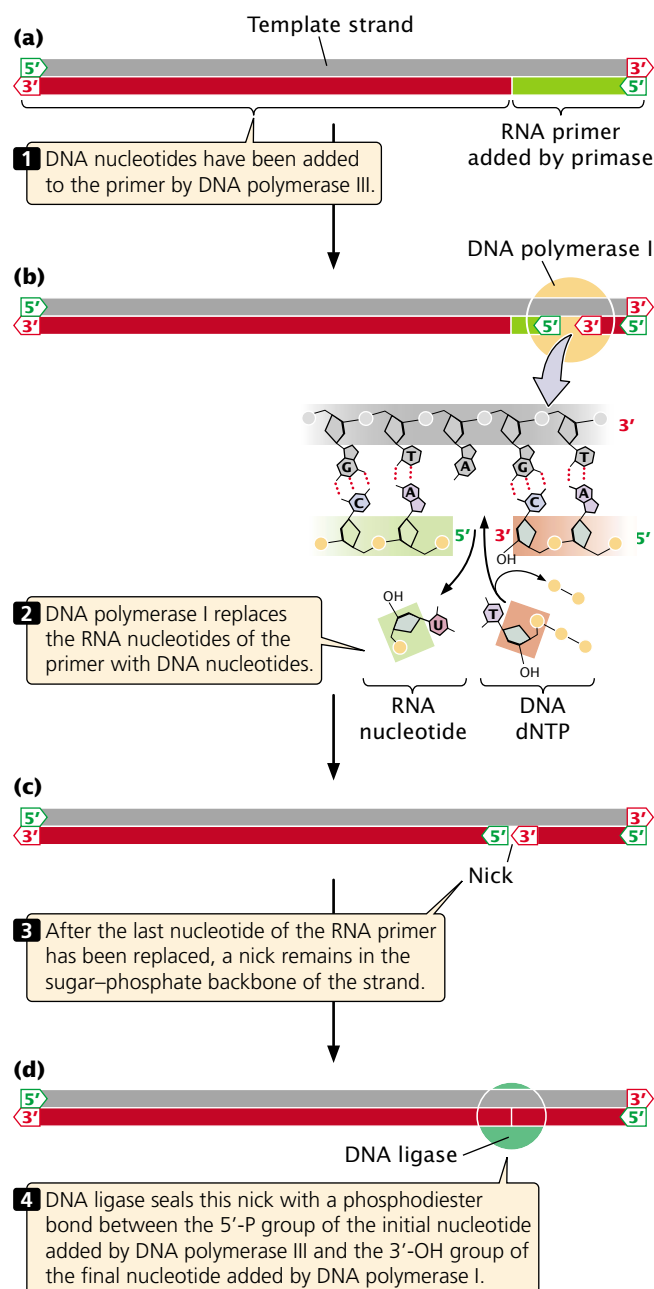
Despite their differences, all of *E. coli*'s DNA polymerases

1. synthesize any sequence specified by the template strand;
2. synthesize in the 5'→3' direction by adding nucleotides to a 3'-OH group;
3. use dNTPs to synthesize new DNA;
4. require a primer to initiate synthesis;
5. catalyze the formation of a phosphodiester bond by joining the 5' phosphate group of the incoming nucleotide to the 3'-OH group of the preceding nucleotide on the growing strand, cleaving off two phosphates in the process;
6. produce newly synthesized strands that are complementary and antiparallel to the template strands; and
7. are associated with a number of other proteins.

Concepts

DNA polymerases synthesize DNA in the 5'→3' direction by adding new nucleotides to the 3' end of a growing nucleotide strand.

DNA ligase After DNA polymerase III attaches a DNA nucleotide to the 3'-OH group on the last nucleotide of the RNA primer, each new DNA nucleotide then provides the 3'-OH group needed for the next DNA nucleotide to be added. This process continues as long as template is available (● **FIGURE 12.14a**). DNA polymerase I follows DNA polymerase III and, using its 5'→3' exonuclease activity, removes the RNA primer. It then uses its 5'→3' polymerase activity to replace the RNA nucleotides with DNA nucleotides. DNA polymerase I attaches the first nucleotide to the OH group at the 3' end of the preceding Okazaki fragment and then continues, in the 5'→3' direction along the nucleotide strand, removing and replacing, one at a time, the RNA nucleotides of the primer (● **FIGURE 12.14b**).



12.14 DNA ligase seals the nick left by DNA polymerase I in the sugar-phosphate backbone after the polymerase has added the final nucleotide.

After polymerase I has replaced the last nucleotide of the RNA primer with a DNA nucleotide, a nick remains in the sugar-phosphate backbone of the new DNA strand. The 3'-OH group of the last nucleotide to have been added by DNA polymerase I is not attached to the 5'-phosphate group of the first nucleotide added by DNA polymerase III (● **FIGURE 12.14c**). This nick is sealed by the enzyme **DNA ligase**, which catalyzes the formation of a

Table 12.4 Components required for replication in bacterial cells

Component	Function
Initiator protein	Binds to origin and separates strands of DNA to initiate replication
DNA helicase	Unwinds DNA at replication fork
Single-strand-binding proteins	Attach to single-stranded DNA and prevent reannealing
DNA gyrase	Moves ahead of the replication fork, making and resealing breaks in the double-helical DNA to release torque that builds up as a result of unwinding at the replication fork
DNA primase	Synthesizes short RNA primers to provide a 3'-OH group for attachment of DNA nucleotides
DNA polymerase III	Elongates a new nucleotide strand from the 3'-OH group provided by the primer
DNA polymerase I	Removes RNA primers and replaces them with DNA
DNA ligase	Joins Okazaki fragments by sealing nicks in the sugar-phosphate backbone of newly synthesized DNA

phosphodiester bond without adding another nucleotide to the strand (◀ **FIGURE 12.14d**). Some of the major enzymes and proteins required for replication are summarized in Table 12.4.

Concepts

After primers are removed and replaced, the nick in the sugar-phosphate linkage is sealed by DNA ligase.



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The replication fork Now that the major enzymatic components of elongation—DNA polymerases, helicase, primase, and ligase—have been introduced, let’s consider how these components interact at the replication fork. Because the synthesis of both strands takes place simultaneously, two units of DNA polymerase III must be present at the replication fork, one for each strand. In one model of the replication process (◀ **FIGURE 12.15**), the two units of DNA polymerase III are connected, and the lagging-strand template loops around so that, as the DNA polymerase III complex moves along the helix, the two antiparallel strands can undergo 5’→3’ replication simultaneously.

In summary, each active replication fork requires five basic components:

1. helicase to unwind the DNA,
2. single-strand-binding proteins to keep the nucleotide strands separate long enough to allow replication,

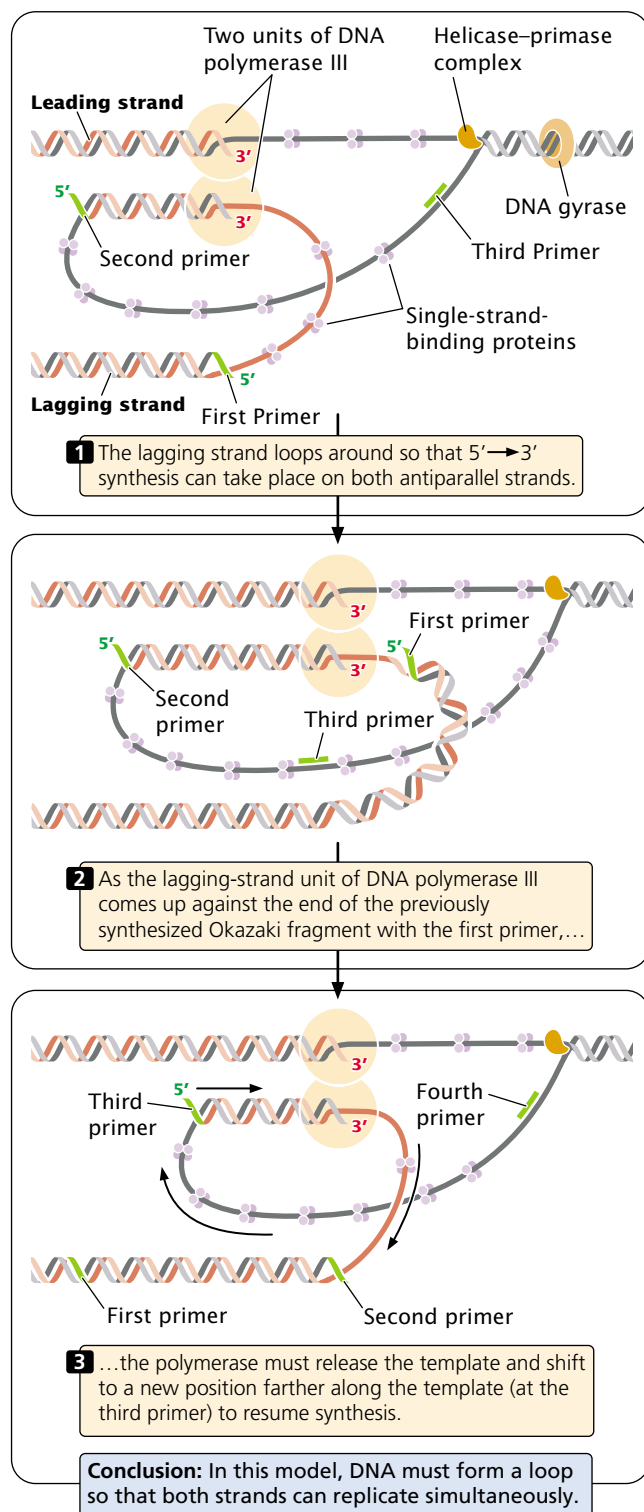
3. the topoisomerase gyrase to remove strain ahead of the replication fork,
4. primase to synthesize primers with a 3'-OH group at the beginning of each DNA fragment, and
5. DNA polymerase to synthesize the leading and lagging nucleotide strands.

www.whfreeman.com/pierce Additional information about the mechanism of replication and an animation of a replication fork

Termination In some DNA molecules, replication is terminated whenever two replication forks meet. In others, specific termination sequences block further replication. A termination protein, called Tus in *E. coli*, binds to these sequences. Tus blocks the movement of helicase, thus stalling the replication fork and preventing further DNA replication.

The fidelity of DNA replication Overall, replication results in an error rate of less than one mistake per billion nucleotides. How is this incredible accuracy achieved? No single process could produce this level of accuracy; a series of processes are required, each catching errors missed by the preceding ones (◀ **FIGURE 12.16**).

DNA polymerases are very particular in pairing nucleotides with their complements on the template strand. Errors in nucleotide selection by DNA polymerase arise only about once per 100,000 nucleotides. Most of the errors that do arise in nucleotide selection are corrected in a second process called **proofreading**. When a DNA polymerase inserts an incorrect nucleotide into the growing strand, the 3'-OH group of the mispaired nucleotide is not correctly positioned for accepting the next nucleotide. The incorrect positioning stalls the poly-



12.15 In one model of DNA replication in *E. coli*, the two units of DNA polymerase III are connected, and the lagging-strand template forms a loop so that replication can take place on the two anti-parallel DNA strands. Components of the replication machinery at the replication fork are shown at the top.

merization reaction, and the 3'→5' exonuclease activity of DNA polymerase removes the incorrectly paired nucleotide. DNA polymerase then inserts the correct nucleotide. Together, proofreading and nucleotide selection result in an error rate of only one in 10 million nucleotides.

A third process, called **mismatch repair** (discussed further in Chapter 17), corrects errors after replication is complete. Any incorrectly paired nucleotides remaining after replication produce a deformity in the secondary structure of the DNA; the deformity is recognized by enzymes that excise an incorrectly paired nucleotide and use the original nucleotide strand as a template to replace the incorrect nucleotide. Mismatch repair requires the ability to distinguish between the old and the new strands of DNA, because the enzymes need some way of determining which of the two incorrectly paired bases to remove. In *E. coli*, methyl groups (–CH₃) are added to particular nucleotide sequences, but only *after* replication. Thus, methylation lags behind replication: so, immediately after DNA synthesis, only the old DNA strand is methylated. Therefore it can be distinguished from the newly synthesized strand, and mismatch repair takes place preferentially on the unmethylated nucleotide strand.

Concepts

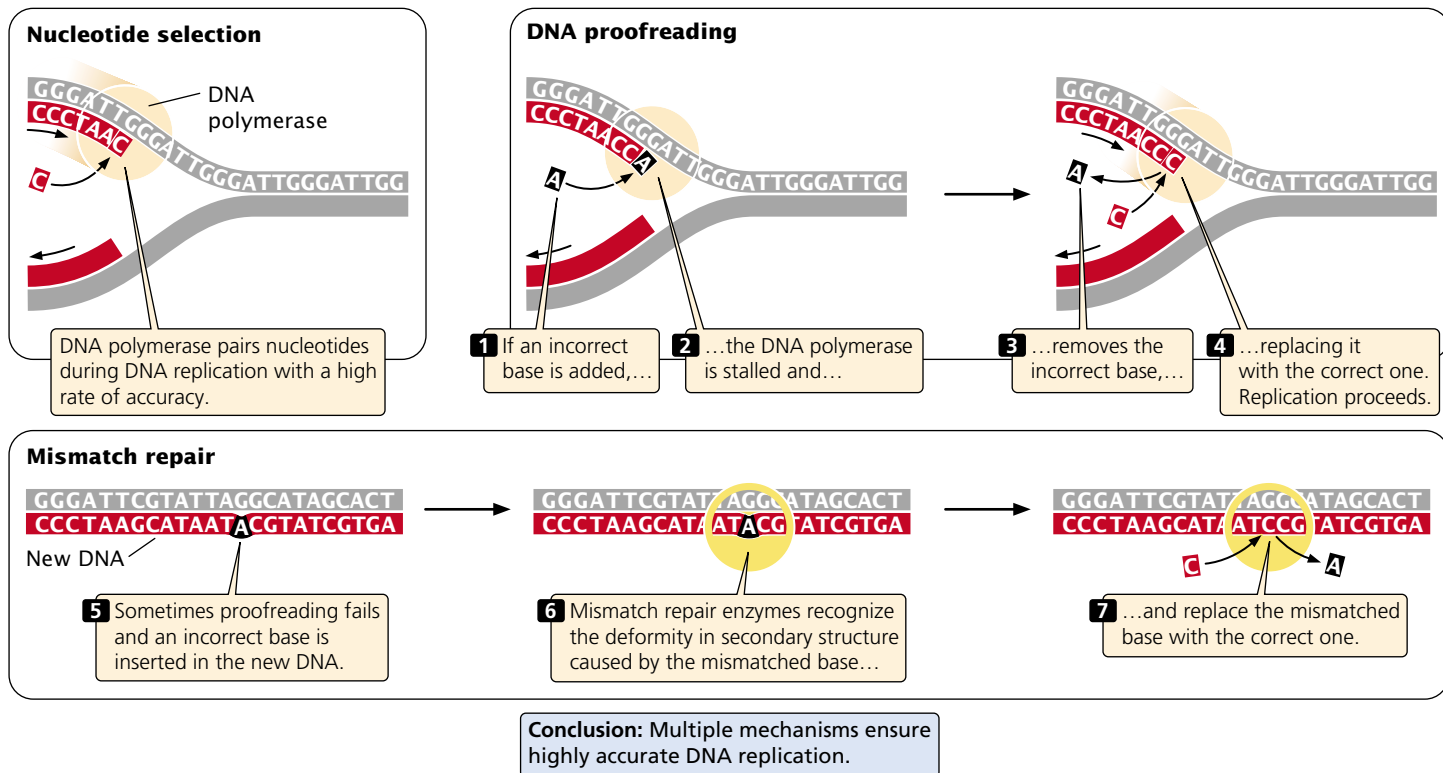
Replication is extremely accurate, with less than one error per billion nucleotides. This accuracy results from the processes of nucleotide selection, proofreading, and mismatch repair.

Connecting Concepts

The Basic Rules of Replication

Bacterial replication requires a number of enzymes, proteins, and DNA sequences that function together to synthesize a new DNA molecule. These components are important, but it is critical that we not become so immersed in the details of the process that we lose sight of general principles of replication.

1. Replication is always semiconservative.
2. Replication begins at sequences called origins.
3. DNA synthesis is initiated by short segments of RNA called primers.
4. The elongation of DNA strands is always in the 5'→3' direction.
5. New DNA is synthesized from dNTPs; in the polymerization of DNA, two phosphates are cleaved from a dNTP and the resulting nucleotide is added to the 3'-OH group of the growing nucleotide strand.



12.16 A series of processes are required to ensure the incredible accuracy of DNA replication. Among these processes are DNA selection, proofreading, and mismatch repair.

6. Replication is continuous on the leading strand and discontinuous on the lagging strand.
7. New nucleotide strands are made complementary and antiparallel to their template strands.
8. Replication takes place at very high rates and is astonishingly accurate, thanks to precise nucleotide selection, proofreading, and repair mechanisms.

Eukaryotic DNA Replication

Although not as well understood, eukaryotic replication resembles bacterial replication in many respects. The most obvious differences are that eukaryotes have: (1) multiple replication origins in their chromosomes; (2) more types of DNA polymerases, with different functions; and (3) nucleosome assembly immediately following DNA replication.

Eukaryotic origins Researchers first isolated eukaryotic origins of replication from yeast cells by demonstrating that certain DNA sequences confer the ability to replicate when transferred from a yeast chromosome to small circular pieces of DNA (plasmids). These **autonomously replicating sequences** (ARSs) enabled any DNA to which they were attached to replicate. They were subsequently shown to be the origins of replication in yeast chromosomes.

Yeast ARSs typically consist of 100 to 120 bp of DNA. A multiprotein complex, the origin recognition complex (ORC), binds to the ARS and probably unwinds the DNA in this region. Interestingly, ORCs also function in regulating transcription.

Concepts

Eukaryotic DNA contains many origins of replication. At each origin, a multiprotein origin recognition complex binds to initiate the unwinding of the DNA.

Licensing of DNA replication Eukaryotic cells utilize thousands of origins, and so the entire genome can be replicated in a timely manner. The use of multiple origins, however, creates a special problem in the timing of replication: the entire genome must be precisely replicated once and only once in each cell cycle so that no genes are left unreplicated and no genes are replicated more than once. How does a cell ensure that replication is initiated at thousands of origins only once per cell cycle?

The precise replication of DNA is accomplished by the separation of the initiation of replication into two distinct steps. In the first step, the origins are licensed, meaning that they are approved for replication. This step is early in the cell

cycle when a **replication licensing factor** attaches to an origin. In the second step, initiator proteins cause the separation of DNA strands and the initiation of replication at each licensed origin. The key is that initiator proteins function only at *licensed* origins. As the replication forks move away from the origin, the licensing factor is removed, leaving the origin in an unlicensed state, where replication cannot be initiated again until the license is renewed. To ensure that replication takes place only once each cell cycle, the licensing factor is active only after the cell has completed mitosis and before the initiator proteins become active.

Unwinding Several helicases that separate double-stranded DNA have been isolated from eukaryotic cells, as have single-strand-binding proteins and topoisomerases (which have a function equivalent to the DNA gyrase in bacterial cells). These enzymes and proteins are assumed to function in unwinding eukaryotic DNA in much the same way as unwinding in bacterial cells.

Eukaryotic DNA polymerases A significant difference in the processes of bacterial and eukaryotic replication is in the number and functions of DNA polymerases. Eukaryotic cells contain a number of different DNA polymerases that function in replication, recombination, and DNA repair

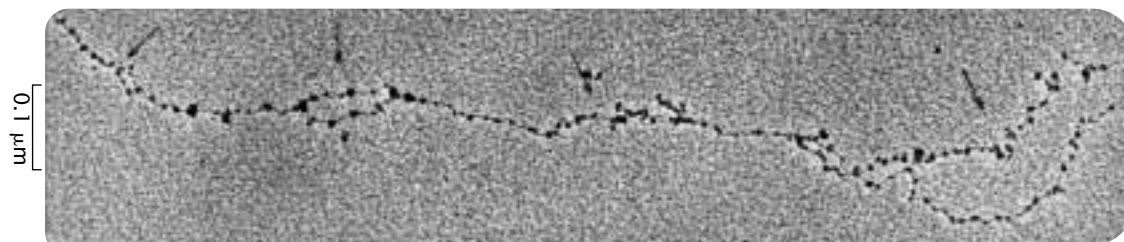
(Table 12.5). **DNA polymerase α** , which contains primase activity, initiates nuclear DNA synthesis by synthesizing an RNA primer, followed by a short string of DNA nucleotides. After DNA polymerase α has laid down from 30 to 40 nucleotides, **DNA polymerase δ** completes replication on the leading and lagging strands. **DNA polymerase β** does not participate in replication but is associated with the repair and recombination of nuclear DNA. **DNA polymerase γ** replicates mitochondrial DNA; a γ -like polymerase also replicates chloroplast DNA. Similar in structure and function to DNA polymerase δ , **DNA polymerase ϵ** appears to take part in nuclear replication of both the leading and the lagging strands, but its precise role is not yet clear. Other DNA polymerases (ζ , η , θ , κ , λ , μ) allow replication to bypass damaged DNA (called translesion replication) or play a role in DNA repair. Many of the DNA polymerases have multiple roles in replication and DNA repair (see Table 12.5).

Concepts

There are at least thirteen different DNA polymerases in eukaryotic cells. DNA polymerases α and δ carry out replication on the leading and lagging strands.

Table 12.5 DNA polymerases in eukaryotic cells

DNA Polymerase	5' $\rightarrow$ 3' Polymerase Activity	3' $\rightarrow$ 5' Exonuclease Activity	Cellular Function
α (alpha)	Yes	No	Initiation of nuclear DNA synthesis and DNA repair
β (beta)	Yes	No	DNA repair and recombination of nuclear DNA
γ (gamma)	Yes	Yes	Replication of mitochondrial DNA
δ (delta)	Yes	Yes	Leading- and lagging-strand synthesis of nuclear DNA, DNA repair, and translesion DNA synthesis
ϵ (epsilon)	Yes	Yes	Unknown; probably repair and replication of nuclear DNA
ζ (zeta)	Yes	No	Translesion DNA synthesis
η (eta)	Yes	No	Translesion DNA synthesis
θ (theta)	Yes	No	DNA repair
ι (iota)	Yes	No	Translesion DNA synthesis
κ (kappa)	Yes	No	Translesion DNA synthesis
λ (lambda)	Yes	No	DNA repair
μ (mu)	Yes	No	DNA repair
σ (sigma)	Yes	No	Nuclear DNA replication (possibly), DNA repair, and sister-chromatid cohesion



12.17 This electron micrograph of eukaryotic DNA in the process of replication clearly shows that newly replicated DNA is already covered with nucleosomes (dark circles). (Victoria Foe).

Nucleosome assembly Eukaryotic DNA is complexed to histone proteins in nucleosome structures that contribute to the stability and packing of the DNA molecule (see Figure 11.6). The disassembly and reassembly of nucleosomes on newly synthesized DNA probably takes place in replication, but the precise mechanism for these processes has not yet been determined. The unwinding of double-stranded DNA and the assembly of the replication enzymes on the single-stranded templates probably require the disassembly of the nucleosome structure. Electron micrographs of eukaryotic DNA show recently replicated DNA already covered with nucleosomes (FIGURE 12.17), indicating that nucleosome structure is reassembled quickly.

Before replication, a single DNA molecule is associated with histone proteins. After replication and nucleosome assembly, two DNA molecules are associated with histone proteins. Do the original histones remain together, attached to one of the new DNA molecules, or do they disassemble and mix with new histones on both DNA molecules?

Techniques similar to those employed by Meselson and Stahl to determine the mode of DNA replication were used to address this question. Cells were cultivated for several generations in a medium containing amino acids labeled with a heavy isotope. The histone proteins incorporated these heavy amino acids and were relatively dense (FIGURE 12.18). The cells were then transferred to a culture medium that contained amino acids with a light isotope. Histones assembled after the transfer possessed the new, relatively light amino acids and were less dense.

After allowing replication to take place, the histone octamers were isolated and centrifuged in a density gradient. Results show that, after replication, the octamers were in a continuous band between high density (representing old octamers) and low density (representing new octamers). This finding suggests that newly assembled octamers consist of a random mixture of old and new histones.

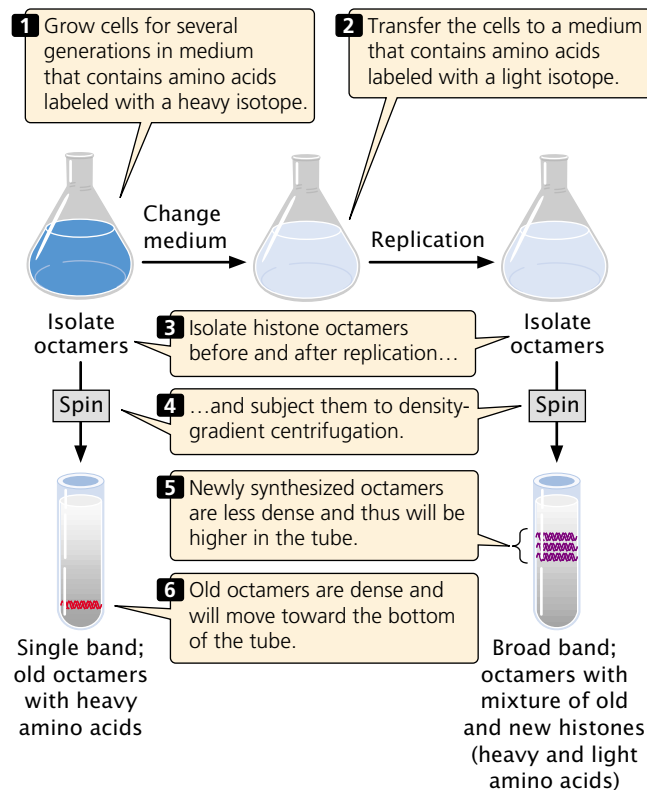
Concepts

After DNA replication, new nucleosomes quickly reassemble on the molecules of DNA. Nucleosomes apparently break down in the course of replication and reassemble from a random mixture of old and new histones.

Location of replication within the nucleus The DNA polymerases that carry out replication are frequently depicted as moving down the DNA template, much as a locomotive travels along a train track. Recent evidence suggests that this view is incorrect. A more accurate view is that the polymerase is fixed in location, and template DNA is threaded through it, with newly synthesized DNA molecules emerging from the other end.

Experiment

Question: What happens to histones during eukaryotic DNA replication?



Conclusion: After DNA replication, the new octamers are a random mixture of old and new histones.

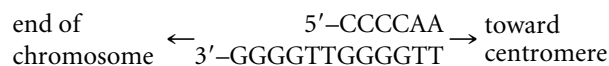
12.18 Experimental procedure for studying how nucleosomes dissociate and reassociate during replication.

Techniques of fluorescence microscopy, which are capable of revealing active sites of DNA synthesis, show that most replication in the nucleus of a eukaryotic cell takes place at a limited number of fixed sites, often referred to as replication factories. Time-lapse micrographs reveal that newly duplicated DNA is extruded from these particular sites. Similar results have also been obtained with bacterial cells.

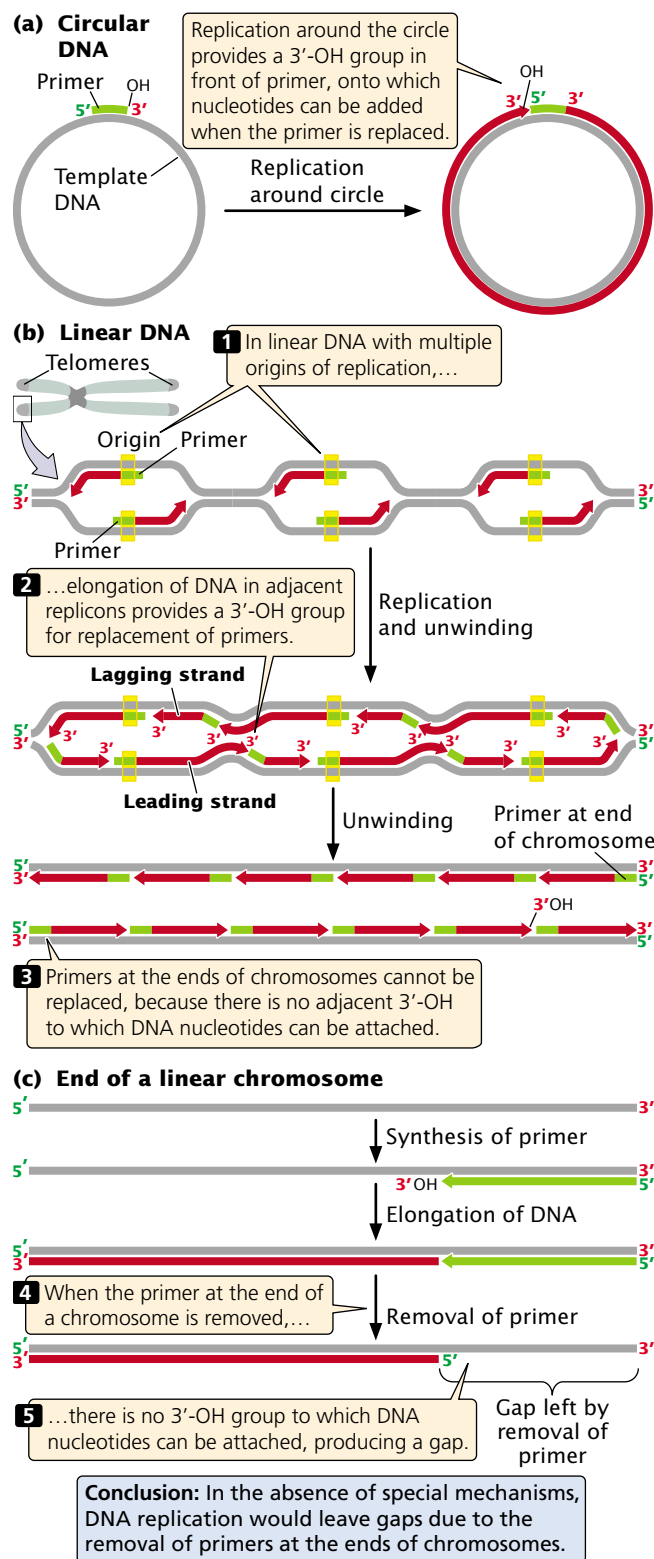
DNA synthesis at the ends of chromosomes A fundamental difference between eukaryotic and bacterial replication arises because eukaryotic chromosomes are linear and thus have ends. As already stated, the 3'-OH group needed for replication by DNA polymerases is provided at the initiation of replication by RNA primers that are synthesized by primase. This solution is temporary, because eventually the primers must be removed and replaced by DNA nucleotides. In a circular DNA molecule, elongation around the circle eventually provides a 3'-OH group immediately in front of the primer (● **FIGURE 12.19a**). After the primer has been removed, the replacement DNA nucleotides can be added to this 3'-OH group.

In linear chromosomes with multiple origins, the elongation of DNA in adjacent replicons also provides a 3'-OH group preceding each primer (● **FIGURE 12.19b**). At the very end of a linear chromosome, however, there is no adjacent stretch of replicated DNA to provide this crucial 3'-OH group. Once the primer at the end of the chromosome has been removed, it cannot be replaced with DNA nucleotides, which produces a gap at the end of the chromosome (● **FIGURE 12.19c**), suggesting that the chromosome should become progressively shorter with each round of replication. The chromosome would be shortened each generation, leading to the eventual elimination of the entire telomere, destabilization of the chromosome, and cell death. But chromosomes don't become shorter each generation and destabilize; so how are the ends of linear chromosomes replicated?

The ends of chromosomes—the telomeres—possess several unique features, one of which is the presence of many copies of a short repeated sequence. In the protozoan *Tetrahymena*, this telomeric repeat is CCCCAG (see Table 11.2), with the G-rich strand typically protruding beyond the C-rich strand (● **FIGURE 12.20a**):

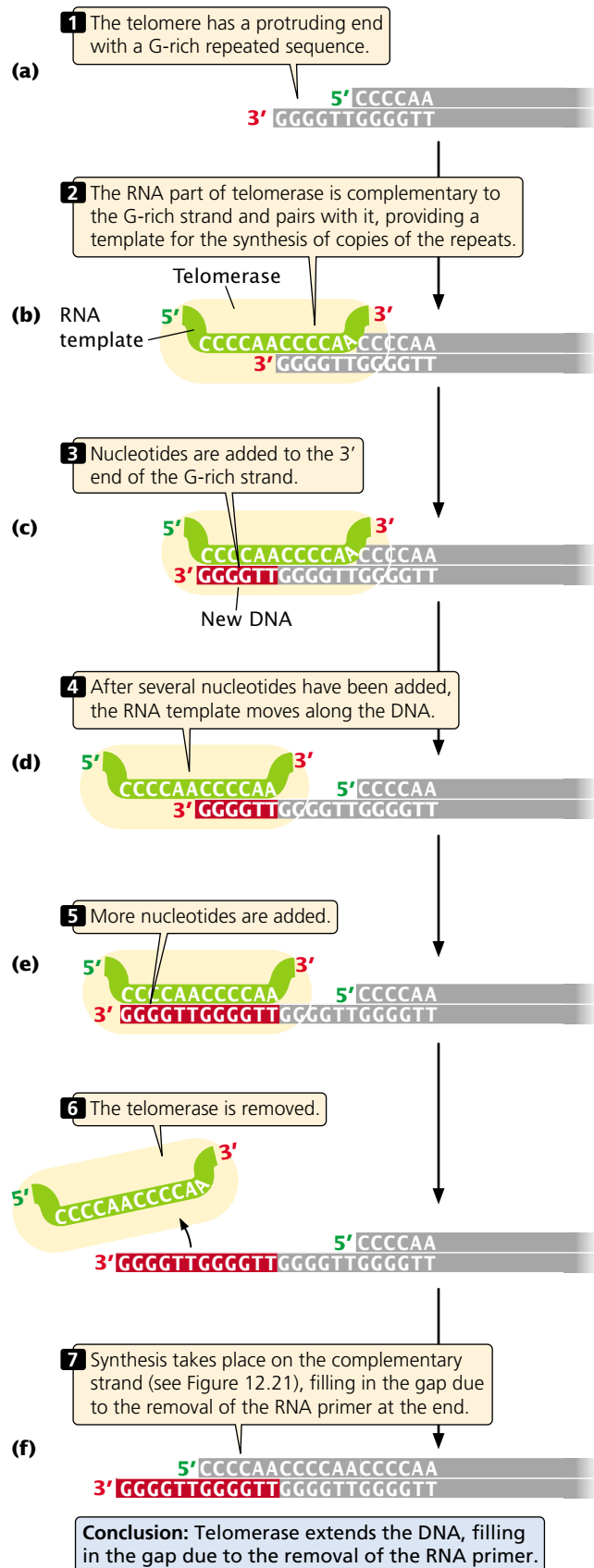


The single-stranded protruding end of the telomere can be extended by **telomerase**, an enzyme with both a protein and an RNA component (also known as a ribonucleoprotein). The RNA part of the enzyme contains from 15 to 22 nucleotides that are complementary to the sequence on the G-rich strand. This sequence pairs with the overhanging 3' end of the DNA (● **FIGURE 12.20b**) and provides a template for the synthesis of additional DNA copies of the repeats. DNA nucleotides are added to the 3' end of the



12.19 DNA synthesis must differ at the ends of circular and linear chromosomes.

strand one at a time (● **FIGURE 12.20c**) and, after several nucleotides have been added, the RNA template moves down the DNA and more nucleotides are added to the 3' end



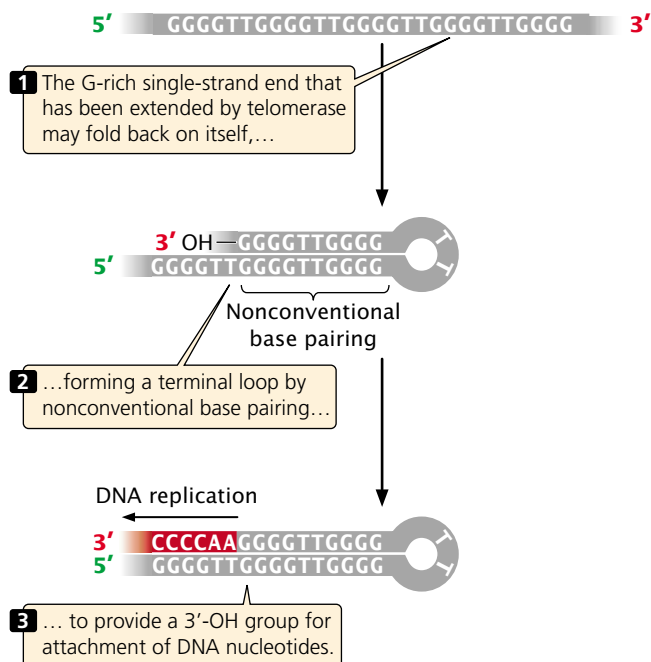
12.20 The enzyme telomerase is responsible for the replication of chromosome ends.

(**FIGURE 12.20d**). Usually, from 14 to 16 nucleotides are added to the 3' end of the G-rich strand.

In this way, the telomerase can extend the 3' end of the chromosome without the use of a complementary DNA template (**FIGURE 12.20e**). How the complementary C-rich strand is synthesized (**FIGURE 12.20f**) is not yet clear. It may be synthesized by conventional replication, with primase synthesizing an RNA primer on the 5' end of the extended (G-rich) template. The removal of this primer once again leaves a gap at the 5' end of the chromosome, but this gap does not matter, because the end of the chromosome is extended at each replication by telomerase; no genetic information is lost, and the chromosome does not become shorter overall. The extended single-strand end may fold back on itself, forming a terminal loop by nonconventional pairing of bases (**FIGURE 12.21**). This loop could provide a 3'-OH group for the attachment of DNA nucleotides along the C-rich strand.

Telomerase is present in single-celled organisms, germ cells, early embryonic cells, and certain proliferative somatic cells (such as bone-marrow cells and cells lining the intestine), all of which must undergo continuous cell division. Most somatic cells have little or no telomerase activity, and chromosomes in these cells progressively shorten with each cell division. These cells are capable of only a limited number of divisions; once the telomeres shorten beyond a critical point, a chromosome becomes unstable, has a tendency to undergo rearrangements, and is degraded. These events lead to cell death.

The shortening of telomeres may contribute to the process of aging. Genetically engineered mice that lack a functional telomerase gene (and therefore do not express



12.21 The complementary G-rich strand at the end of the telomere must be primed before the extension of the 3' end of the chromosome by telomerase.

telomerase in somatic or germ cells) experience progressive shortening of their telomeres in successive generations. After several generations, these mice show some signs of premature aging, such as graying, hair loss, and delayed wound healing. Through genetic engineering, it is also possible to create somatic cells that express telomerase. In these cells, telomeres do not shorten, cell aging is inhibited, and the cells will divide indefinitely.

Telomerase also appears to play a role in cancer. Cancer tumor cells have the capacity to divide indefinitely, and many tumor cells express the telomerase enzyme. As will be discussed in Chapter 21, cancer is a complex, multistep process that usually requires mutations in at least several genes. Telomerase activation alone does not lead to cancerous growth in most cells, but it does appear to be required along with other mutations for cancer to develop.

Concepts

The ends of eukaryotic chromosomes are replicated by an RNA–protein enzyme called telomerase. This enzyme adds extra nucleotides to the G-rich DNA strand of the telomere.

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More on telomerase, including an animated cartoon that illustrates the process of replication by telomerase

Replication in Archaea

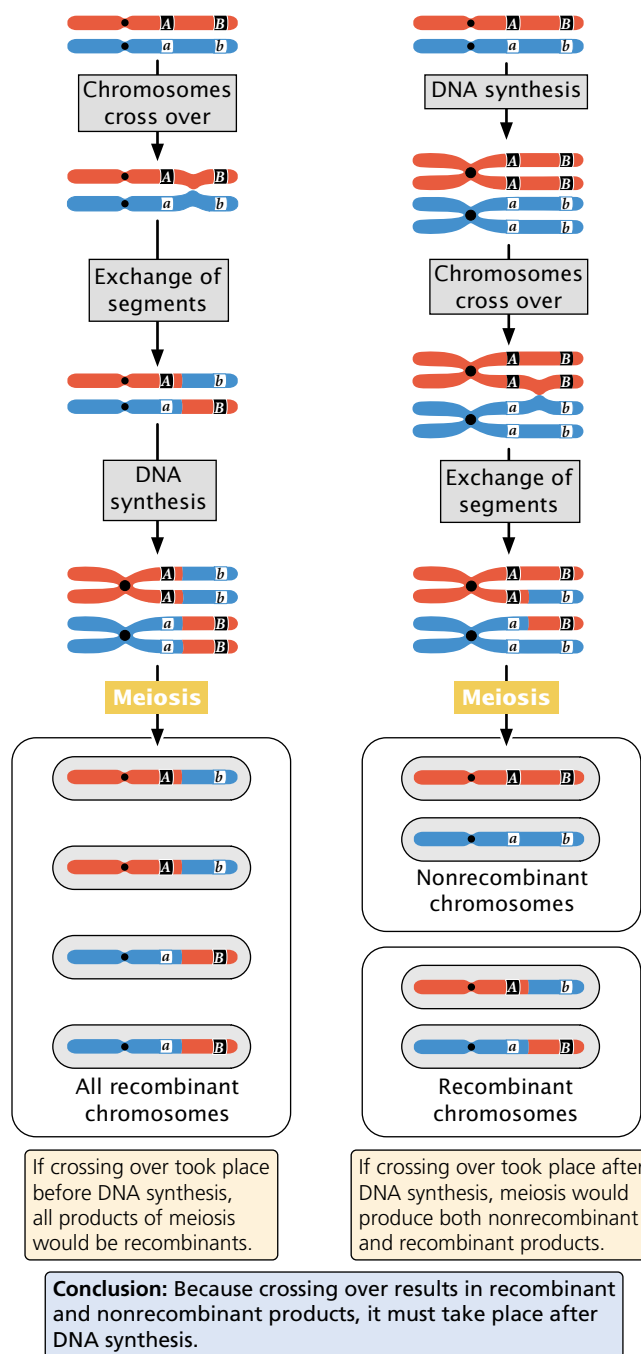
The process of replication in archaeobacteria has a number of features in common with replication in eukaryotic cells—many of the proteins taking part are more similar to those in eukaryotic cells than to those in to those in eubacteria. Although some archaea have a single origin of replication, as do eubacteria, this origin does not contain the typical sequences recognized by bacterial initiator proteins but instead has sequences that are similar to those found in eukaryotic origins. These similarities in replication between archaeal and eukaryotic cells reinforce the conclusion that the archaea are more closely related to eukaryotic cells than to the prokaryotic eubacteria.

The Molecular Basis of Recombination

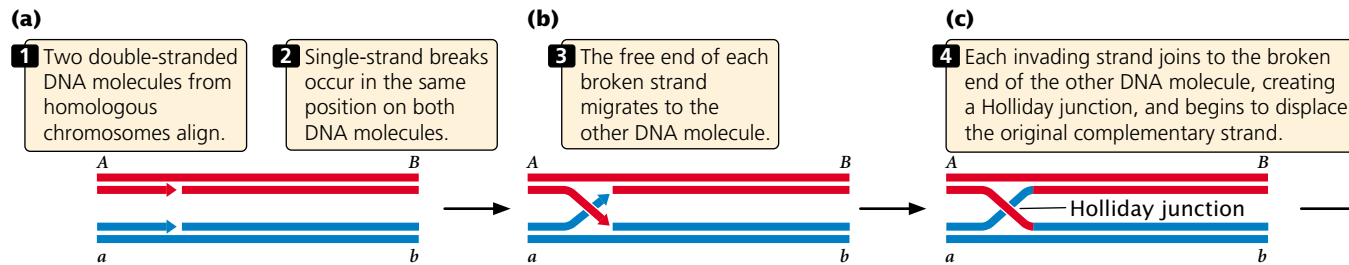
Recombination is the exchange of genetic information between DNA molecules; when the exchange is between homologous DNA molecules, it is called **homologous recombination**. This process takes place in crossing over, in which homologous regions of chromosomes are exchanged (see Figure 2.17) and genes are shuffled into new combinations. Recombination is an extremely important genetic process because it increases genetic variation. Rates of recombination provide important information about linkage relations among genes, which is used to create genetic maps (see Figures 7.12 through 7.14). Recombination is also essen-

tial for some types of DNA repair (as will be discussed in Chapter 17).

Homologous recombination is a remarkable process: a nucleotide strand of one chromosome aligns precisely with a nucleotide strand of the homologous chromosome, breaks arise in corresponding regions of different DNA molecules, parts of the molecules precisely change place, and then the pieces are correctly joined. In this complicated series of events, no genetic information is lost or gained. Although the precise



12.22 Genetic evidence suggests that crossing over takes place after DNA synthesis.



12.23 In the Holliday model, homologous recombination is accomplished through a single-strand break in each DNA duplex, strand displacement, branch migration, and resolution of a single Holliday junction.

molecular mechanism of homologous recombination is still poorly known, the exchange is probably accomplished through the pairing of complementary bases. A single-stranded DNA molecule of one chromosome pairs with a single-stranded DNA molecule of another, forming **heteroduplex DNA**.

In meiosis, homologous recombination (crossing over) could theoretically take place before, during, or after DNA synthesis. Cytological, biochemical, and genetic evidence indicates that it takes place in prophase I of meiosis, whereas DNA replication takes place earlier, in interphase. Thus, crossing over must entail the breaking and rejoining of chromatids when homologous chromosomes are at the four-strand stage (FIGURE 12.22). This section explores some theories about how the process of recombination takes place.

The Holliday Model

One model of homologous recombination, the **Holliday junction**, states that the process is initiated by single-strand breaks in the DNA molecule. This model begins with double-stranded DNA molecules from two homologous chromosomes that carry identical (or nearly identical) nucleotide sequences. These two DNA molecules align precisely, and so their homologous sequences sit side by side (FIGURE 12.23a). Single-strand breaks in the same position on both DNA molecules allow the free ends of the strands to move to the other DNA molecule (FIGURE 12.23b and c). Each invading strand joins to the broken end of the other homologous DNA molecule and begins to displace the original complementary strand, taking its place by hydrogen bonding to the original strand. The invasion and joining take place on both DNA molecules, creating two heteroduplex DNAs, each consisting of one original strand plus one new strand from the other DNA molecule. The point at which nucleotide strands pass from one DNA molecule to the other is the cross bridge. In the Holliday model of recombination, there is a single cross bridge. As the two nucleotide strands exchange positions, the cross bridge moves along the molecules in a process called **branch migration** (FIGURE 12.23d). The exchange

of nucleotide strands and branch migration create two duplex molecules connected by the cross bridge. This structure is termed the **Holliday intermediate**. Holliday intermediates in *E. coli* and yeast have been observed with electron microscopy.

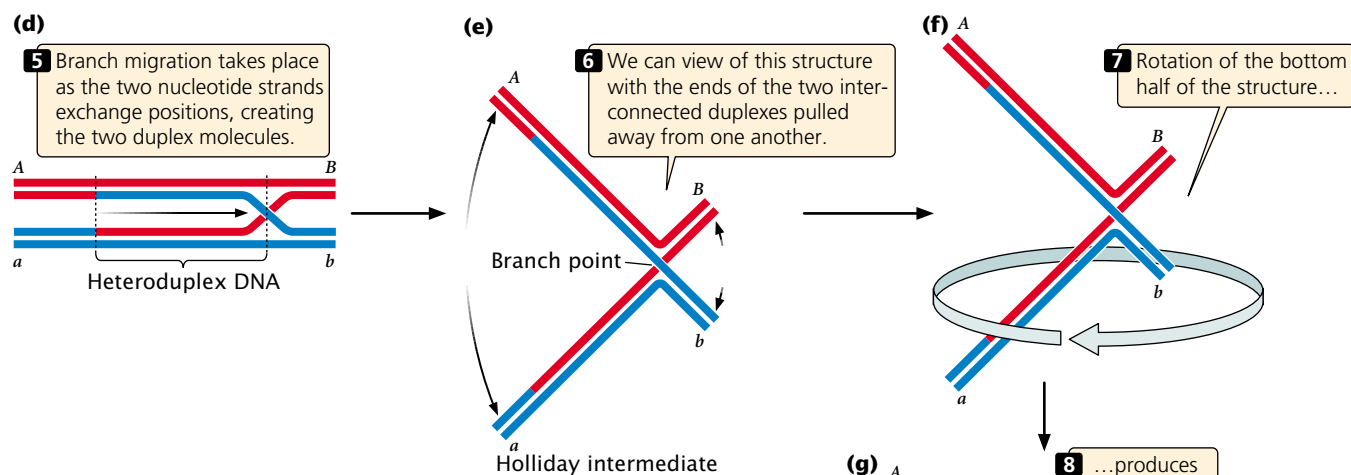
If the ends of the two interconnected duplexes illustrated in Figure 12.23d are pulled away from one another, we obtain the structure illustrated in FIGURE 12.23e. If you carefully compare parts d and e, you will see that the structures in each are the same; the only difference is that, in part e, the ends of the molecules have been pulled apart.

The next step in the Holliday model is easier to visualize if we rotate the bottom half of the Holliday intermediate by 180 degrees, producing the structure shown in FIGURE 12.23f. These interconnected DNA duplexes are then separated by additional cleavage and reunion of the nucleotide strands. The duplexes can be cleaved in one of two ways, as shown by two different pathways in Figure 12.23. Cleavage may be in the horizontal plane (FIGURE 12.23g), in which case the nucleotide strands are rejoined as shown in FIGURE 12.23h, and two DNA molecules are produced. Although both resulting DNA molecules contain a patch of heteroduplex DNA, the genes on either end of the molecules are identical with those originally present (gene A with B, and gene a with b). These DNA molecules are called **patched recombinants** (FIGURE 12.23i).

On the other hand, cleavage of the Holliday structure in the vertical plane and rejoining of the nucleotide strands (FIGURE 12.23j) produces **spliced recombinants** (FIGURE 12.23k). In these recombinants, *both* resulting DNA molecules are heteroduplex, and recombination has taken place between loci at the ends of the molecules; now gene A is paired with b, and gene a is paired with B. Recombination is equally likely to produce patched and spliced recombinants.

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An animated cartoon that will help you visualize the Holliday structure and its resolution



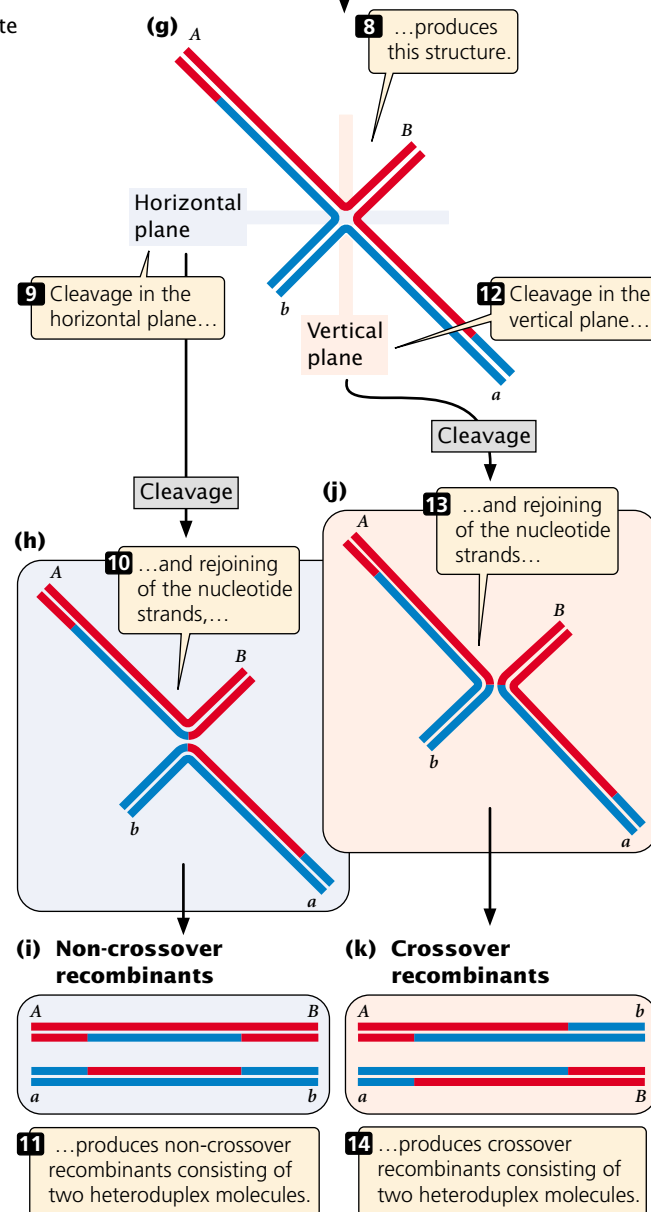
The Double-Strand-Break Model

In the Holliday model, recombination starts with single-strand breaks at the same positions in two homologous DNA molecules. The **double-strand-break model**, in contrast, begins with double-strand breaks in one of the two aligned DNA molecules (FIGURE 12.24a). On both sides of the break, an enzyme nibbles away nucleotides, producing a gap in the DNA, with some single-stranded DNA on each side (FIGURE 12.24b). A free 3' end then invades the other unbroken DNA molecule and displaces the homologous strand (FIGURE 12.24c). The 3' end of the invading strand is elongated by DNA synthesis, which further displaces the original strand of the unbroken molecule (FIGURE 12.24d).

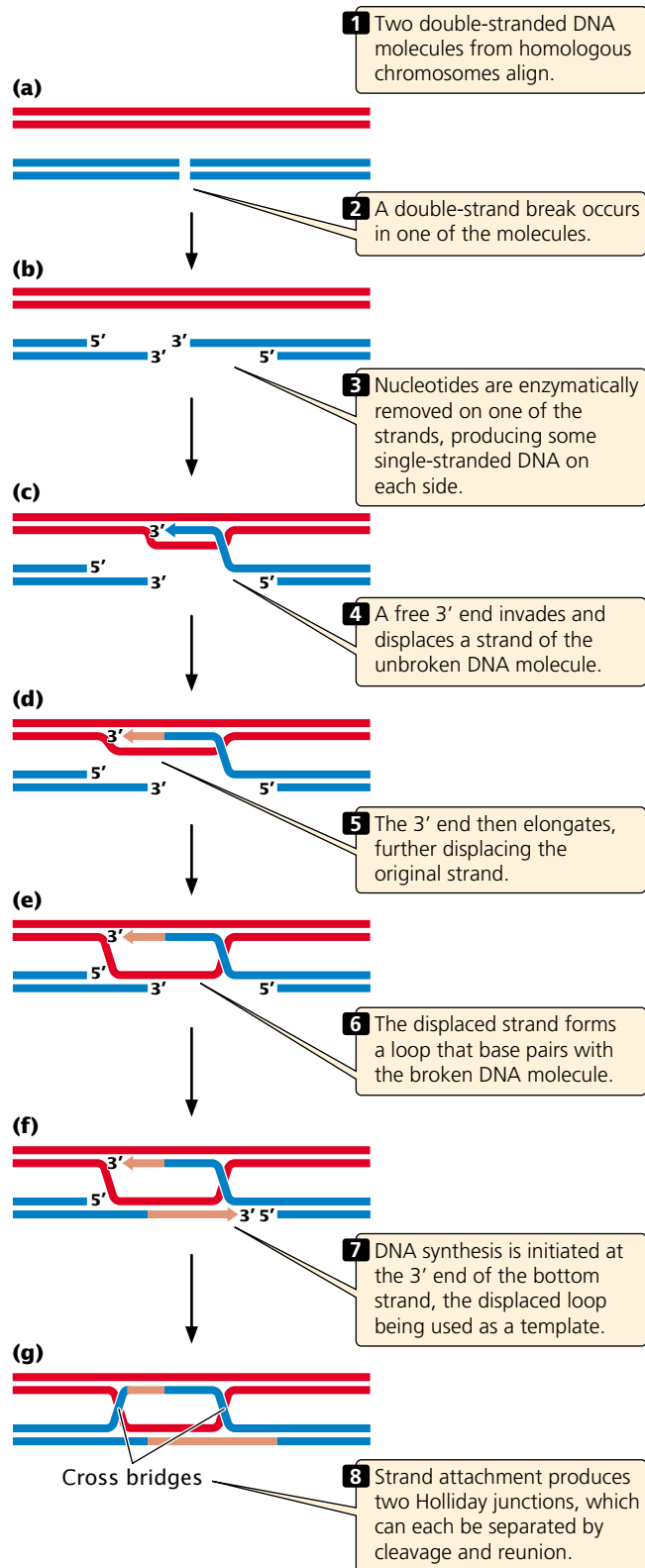
The displaced strand forms a loop that fills the gap in the broken DNA molecule (FIGURE 12.24e) and serves as a template for the synthesis of a complementary DNA strand (FIGURE 12.24f). The result is that two heteroduplex DNA molecules are joined by two cross bridges (FIGURE 12.24g), in contrast with the single cross bridge produced in the Holliday single-strand-break model.

The interconnected molecules produced in the double-strand-break model can be separated by further cleavage and reunion of the nucleotide strands, in the same way that the Holliday intermediate is separated in the Holliday single-strand-break model (see Figure 12.23g–k). Patched or spliced recombinant products can be produced, depending on whether cleavage is in the vertical or the horizontal plane.

Evidence for the double-strand-break model originally came from results of genetic crosses in yeast that could not be explained by the Holliday model. Subsequent observations in yeast showed that double-strand breaks appear in meiosis during prophase I when crossing over occurs and that mutant strains that are unable to form double-strand breaks do not exhibit meiotic recombination. Although considerable evidence supports the double-strand-break model in yeast, the extent to which it applies to other organisms is not known.



Conclusion: The Holliday model predicts non-crossover or crossover recombinant DNA, depending on whether cleavage is in the horizontal or the vertical plane.



12.24 In the double-strand-break model, recombination is accomplished through a double-strand break in one DNA duplex, strand displacement, DNA synthesis, and resolution of two Holliday junctions.

Concepts

Homologous recombination requires the formation of heteroduplex DNA consisting of one nucleotide strand from each of two different chromosomes. In the Holliday model, homologous recombination is accomplished through a single-strand break in the DNA, strand displacement, and branch migration. In the double-strand-break model, recombination is accomplished through double-strand breaks, strand displacement, and branch migration.

Enzymes Required for Recombination

Recombination between DNA molecules requires the unwinding of DNA helices, the cleavage of nucleotide strands, strand invasion, and branch migration, followed by further strand cleavage and union to remove cross bridges. Much of what we know about these processes arises from studies of gene exchange in *E. coli*. Although bacteria do not undergo meiosis, they do have a type of sexual reproduction (conjugation), in which one bacterium donates its chromosome to another (discussed more fully in Chapter 8). Subsequent to conjugation, the recipient bacterium has two chromosomes, which may undergo homologous recombination. Geneticists have isolated mutant strains of *E. coli* that are deficient in recombination; the study of these strains has resulted in the identification of genes and proteins that play a role in bacterial recombination, revealing several different pathways by which it can take place.

Three genes that play a pivotal role in *E. coli* recombination are *recB*, *recC*, and *recD*, which encode three polypeptides that together form the RecBCD protein. This protein unwinds double-stranded DNA and is capable of cleaving nucleotide strands. The *recA* gene encodes the RecA protein that allows a single strand to invade a DNA helix and the subsequent displacement of one of the original strands. Thus invasion and displacement are necessary for both the single-strand- and the double-strand-break models of homologous recombination.

The *ruvA* and *ruvB* genes encode proteins that catalyze branch migration, and the *ruvC* gene produces a protein, called resolvase, that cleaves Holliday structures. Single-strand-binding proteins, DNA ligase, DNA polymerases, and DNA gyrase also play roles in various types of recombination, in addition to their functions in DNA replication.

Concepts

A number of proteins have roles in recombination, including RecA, RecBCD, RuvA, RuvB, resolvase, single-strand-binding proteins, ligase, DNA polymerases, and gyrase.

Connecting Concepts Across Chapters



This chapter has built on a central concept introduced in Chapter 2, that cell division is preceded by replication of the genetic material. In Chapter 2, we saw that DNA replication takes place in the S phase of the cell cycle and that several checkpoints ensure that division does not take place in the absence of DNA replication. The current chapter examined the process of DNA synthesis.

DNA is sometimes said to be a self-replicating molecule, but nothing could be farther from the truth. Replication requires much more than a DNA template; a large number of proteins and enzymes also are necessary. Despite this complexity, a few rules summarize the process: (1) all replication is semiconservative, (2) new DNA molecules always elongate at the 3' end (replication is 5' → 3'), (3) replication begins at sequences called origins and requires RNA primers for initiation, (4) DNA synthesis takes place continuously on one strand and discontinuously on the other, and (5) newly synthesized nucleotide strands are antiparallel and complementary to their template strands.

As we have seen, replication takes place with a high degree of accuracy; this accuracy is essential to maintain the integrity of genetic information as DNA molecules are copied again and again. The accuracy of replication is maintained by several different mechanisms, including precision in nucleotide selection, the ability of DNA polymerases to proofread and correct mistakes, and the detection and repair of residual mismatches after replication (mismatch repair).

An understanding of DNA replication provides a foundation for several topics that will be introduced in later chapters of this book. Chapter 18 (on recombinant DNA technology) examines the polymerase chain reaction and other techniques (DNA sequence analysis and cloning) that require an understanding of DNA synthesis. In Chapter 17 (on gene mutation and DNA repair), we learn that, in spite of the accuracy of DNA synthesis, errors do arise and sometimes lead to mutations. These errors are addressed by mechanisms of DNA repair, many of which require DNA synthesis. The movement of transposable genetic elements (Chapter 11) also requires DNA synthesis.

CONCEPTS SUMMARY

- Replication is semiconservative: DNA's two nucleotide strands separate and each serves as a template on which a new strand is synthesized
- A replicon is a unit of replication that contains an origin of replication.
- In theta replication of DNA, the two nucleotide strands of a circular DNA molecule unwind, creating a replication bubble; within each replication bubble, DNA is normally synthesized on both strands and at both replication forks, producing two circular DNA molecules.
- Rolling-circle replication is initiated by a nick in one strand of circular DNA, which produces a 3'-OH group to which new nucleotides are added while the 5' end of the broken strand is displaced from the circle. Replication proceeds around the circle, producing a circular DNA molecule and a single-stranded linear molecule.
- Linear eukaryotic DNA contains many origins of replication. At each origin, the DNA unwinds, producing two nucleotide strands that serve as templates. Unwinding and replication take place on both templates at both ends of the replication bubble until adjacent replicons meet, resulting in two linear DNA molecules.
- DNA synthesis requires a single-stranded DNA template, deoxyribonucleoside triphosphates; and a group of enzymes and proteins that carry out replication.
- All DNA synthesis is in the 5' → 3' direction. Because the two nucleotide strands of DNA are antiparallel, replication takes place continuously on one strand (the leading strand) and discontinuously on the other (the lagging strand).
- Replication begins when an initiator protein binds to a replication origin and unwinds a short stretch of DNA, to which DNA helicase attaches. DNA helicase unwinds the DNA at the replication fork, single-strand-binding proteins bind to single nucleotide strands to prevent them from reannealing, and DNA gyrase (a topoisomerase) removes the strain ahead of the replication fork that is generated by unwinding.
- During replication, primase synthesizes short primers of RNA nucleotides, providing a 3'-OH group to which DNA polymerase can add DNA nucleotides.
- DNA polymerase adds new nucleotides to the 3' end of a growing polynucleotide strand. Bacteria have two DNA polymerases that have primary roles in replication: DNA polymerase III, which synthesizes new DNA on the leading and lagging strands; and DNA polymerase I, which removes and replaces primers.
- DNA ligase seals nicks that remain in the sugar-phosphate backbones when the RNA primers are replaced by DNA nucleotides.
- Several mechanisms ensure the high rate of accuracy in replication, including precise nucleotide selection, proofreading, and mismatch repair.

- Eukaryotic replication is similar to bacterial replication, although eukaryotes have multiple origins of replication and different DNA polymerases.
- Precise replication at multiple origins is ensured by a licensing factor that must attach to an origin before replication can begin. The licensing factor is removed after replication is initiated and renewed after cell division.
- Eukaryotic nucleosomes are quickly assembled on new molecules of DNA; newly assembled nucleosomes consist of a random mixture of old and new histone proteins.
- The ends of linear eukaryotic DNA molecules are replicated by the enzyme telomerase.
- Replication in archaeal bacteria has a number of features in common with eukaryotic replication.
- Homologous recombination takes place through the exchange of genetic material between homologous DNA molecules. In the Holliday model, homologous recombination begins with single-strand breaks in both DNA molecules, followed by strand displacement, branch migration, and Holliday junction resolution. In the double-strand break model, it begins with a double-strand-break, followed by strand displacement, DNA synthesis, and resolution of two Holliday junctions.
- Homologous recombination in *E. coli* requires a number of enzymes, including RecA, RecBCD, resolvase, single-strand-binding proteins, ligase, DNA polymerases, and gyrase.

IMPORTANT TERMS

semiconservative replication (p. 000)

equilibrium density gradient centrifugation (p. 000)

replicon (p. 000)

replication origin (p. 000)

theta replication (p. 000)

replication bubble (p. 000)

replication fork (p. 000)

bidirectional replication (p. 000)

rolling-circle replication (p. 000)

DNA polymerase (p. 000)

continuous replication (p. 000)

leading strand (p. 000)

discontinuous replication (p. 000)

lagging strand (p. 000)

Okazaki fragments (p. 000)

initiator protein (p. 000)

DNA helicase (p. 000)

single-strand-binding protein (SSB) (p. 000)

DNA gyrase (p. 000)

primase (p. 000)

primer (p. 000)

DNA polymerase III (p. 000)

DNA polymerase I (p. 000)

DNA ligase (p. 000)

proofreading (p. 000)

mismatch repair (p. 000)

autonomously replicating sequence (p. 000)

replication licensing factor (p. 000)

DNA polymerase α (p. 000)

DNA polymerase δ (p. 000)

DNA polymerase β (p. 000)

DNA polymerase γ (p. 000)

DNA polymerase ϵ (p. 000)

telomerase (p. 000)

homologous recombination (p. 000)

heteroduplex DNA (p. 000)

Holliday junction (p. 000)

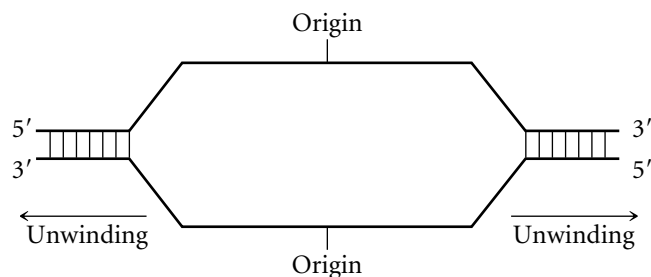
branch migration (p. 000)

Holliday intermediate (p. 000)

double-strand-break model (p. 000)

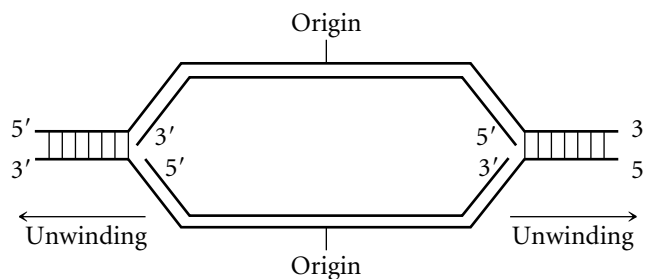
Worked Problems

1. The following diagram (below) represents the template strands of a replication bubble in a DNA molecule. Draw in the newly synthesized strands and label the leading and lagging strands.

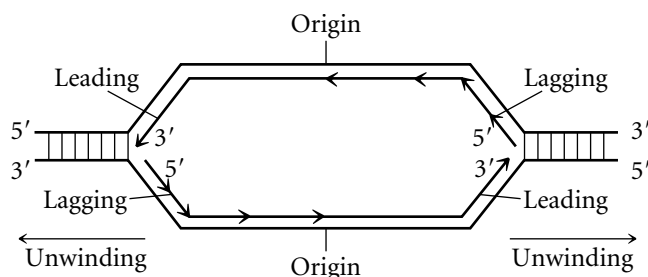


• Solution

To determine the leading and lagging strands, first note which end of each template strand is 5' and which end is 3'. With a pencil, draw in the strands being synthesized on these templates, and label their 5' and 3' ends, recalling that the newly synthesized strands must be antiparallel to the templates.



Next, determine the direction of replication for each new strand, which must be 5' $\rightarrow$ 3'. You might draw arrows on the new strands to indicate the direction of replication. After you have established the direction of replication for each strand, look at each fork and determine whether the direction of replication for a strand is the same as the direction of unwinding. The strand on which replication is in the same direction as unwinding is the leading strand. The strand on which replication is in the direction opposite that of unwinding is the lagging strand. Make sure that you have one leading strand and one lagging strand for each fork.

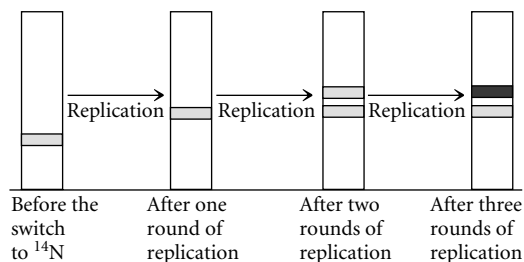


2. Consider the experiment conducted by Meselson and Stahl in which they used ^{14}N and ^{15}N in cultures of *E. coli* and equilibrium density gradient centrifugation. Draw pictures to represent the bands produced by bacterial DNA in the density-gradient tube before the switch to medium containing ^{14}N and after one, two, and three rounds of replication after the switch to the medium containing ^{14}N . Use a separate set of drawings to show the bands that would appear if replication were (a) semiconservative; (b) conservative; (c) dispersive.

• Solution

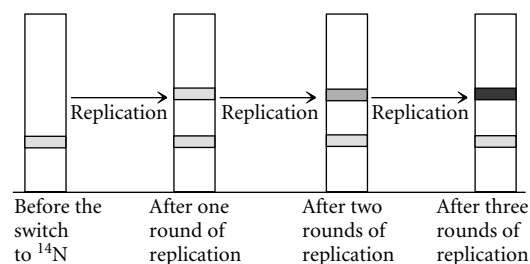
DNA labeled with ^{15}N will be denser than DNA labeled with ^{14}N ; therefore ^{15}N -labeled DNA will sink lower in the density-gradient tube. Before the switch to medium containing ^{14}N , all DNA in the bacteria will contain ^{15}N and will produce a single band in the lower end of the tube.

(a) With semiconservative replication, the two strands separate, and each serves as a template on which a new strand is synthesized. After one round of replication, the original template strand of each molecule will contain ^{15}N and the new strand of each molecule will contain ^{14}N ; so a single band will appear in the density gradient halfway between the positions expected of DNA with ^{15}N and of DNA with ^{14}N . In the next round of replication, the two strands again separate and serve as templates for new strands. Each of the new strands contains only ^{14}N , thus some DNA molecules will contain one strand with the original ^{15}N and one strand with new ^{14}N , whereas the other molecules will contain two strands with ^{14}N . This labeling will produce two bands, one at the intermediate position and one at a higher position in the tube. Additional rounds of replication should produce increasing amounts of DNA that contains only ^{14}N ; so the higher band will get darker.

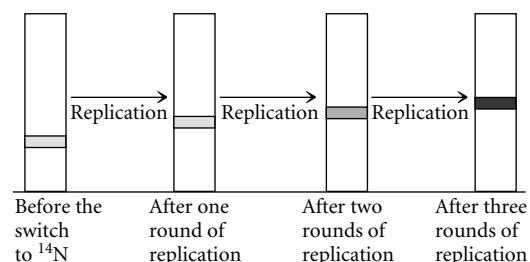


(b) With conservative replication, the entire molecule serves as a template. After one round of replication, some molecules will consist entirely of ^{15}N , and others will consist entirely of ^{14}N ; so

two bands should be present. Subsequent rounds of replication will increase the fraction of DNA consisting entirely of new ^{14}N , thus the upper band will get darker. However, the original DNA with ^{15}N will remain, so two bands will be present.



(c) In dispersive replication, both nucleotide strands break down into fragments that serve as templates for the synthesis of new DNA. The fragments then reassemble into DNA molecules. After one round of replication, all DNA should contain approximately half ^{15}N and half ^{14}N , producing a single band that is halfway between the positions expected of DNA labeled with ^{15}N and of DNA labeled with ^{14}N . With further rounds of replication, the proportion of ^{14}N in each molecule increases; so a single hybrid band remains, but its position in the density gradient will move upward. The band is also expected to get darker as the total amount of DNA increases.



3. The *E. coli* chromosome contains 4.7 million base pairs of DNA. If synthesis at each replication fork occurs at a rate of 1000 nucleotides per second, how long will it take to completely replicate the *E. coli* chromosome with theta replication?

• Solution

Bacterial chromosomes contain a single origin of replication, and theta replication usually employs two replication forks, which proceed around the chromosome in opposite directions. Thus, the overall rate of replication for the whole chromosome is 2000 nucleotides per second. With a total of 4.7 million base pairs of DNA, the entire chromosome will be replicated in:

$$4,700,000 \text{ bp} \times \frac{1 \text{ second}}{2000 \text{ bp}} = 2350 \text{ seconds} \times \frac{1 \text{ minute}}{60 \text{ seconds}} = 39.17 \text{ minutes}$$

At the beginning of this chapter it was stated that *E. coli* is capable of dividing every 20 minutes. How is this possible if it

takes almost twice as long to replicate its genome? The answer is that a second round of replication begins before the first round has finished. Thus, when an *E. coli* cell divides, the chromosomes that are passed on to the daughter cells are

already partially replicated. This is in contrast to eukaryotic cells, which replicate their entire genome once, and only once, during each cell cycle.

COMPREHENSION QUESTIONS

1. What is semiconservative replication?
- * 2. How did Meselson and Stahl demonstrate that replication in *E. coli* takes place in a semiconservative manner?
- * 3. Draw a molecule of DNA undergoing theta replication. On your drawing, identify (1) origin, (2) polarity (5' and 3' ends) of all template strands and newly synthesized strands, (3) leading and lagging strands, (4) Okazaki fragments, and (5) location of primers.
4. Draw a molecule of DNA undergoing rolling-circle replication. On your drawing, identify (1) origin, (2) polarity (5' and 3' ends) of all template and newly synthesized strands, (3) leading and lagging strands, (4) Okazaki fragments, and (5) location of primers.
5. Draw a molecule of DNA undergoing eukaryotic linear replication. On your drawing, identify (1) origin; (2) polarity (5' and 3' ends) of all template and newly synthesized strands, (3) leading and lagging strands, (4) Okazaki fragments, and (5) location of primers.
6. What are three major requirements of replication?
- * 7. What substrates are used in the DNA synthesis reaction?
8. List the different proteins and enzymes taking part in bacterial replication. Give the function of each in the replication process.
9. What similarities and differences exist in the enzymatic activities of DNA polymerases I, II, and III? What is the function of each type of DNA polymerase in bacterial cells?
- * 10. Why is primase required for replication?
11. What three mechanisms ensure the accuracy of replication in bacteria?
12. How does replication licensing ensure that DNA is replicated only once at each origin per cell cycle?
- * 13. In what ways is eukaryotic replication similar to bacterial replication, and in what ways is it different?
14. Outline in words and pictures how telomeres at the end of eukaryotic chromosomes are replicated.
15. Briefly outline with diagrams the Holliday model of homologous recombination.
- * 16. What are some of the enzymes taking part in recombination in *E. coli* and what roles do they play?

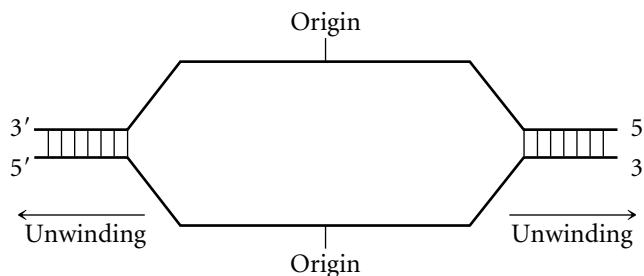
APPLICATION QUESTIONS AND PROBLEMS

- * 17. Suppose a future scientist explores a distant planet and discovers a novel form of double-stranded nucleic acid. When this nucleic acid is exposed to DNA polymerases from *E. coli*, replication takes place continuously on both strands. What conclusion can you make about the structure of this novel nucleic acid?
- * 18. Phosphorus is required to synthesize the deoxyribonucleoside triphosphates used in DNA replication. A geneticist grows some *E. coli* in a medium containing nonradioactive phosphorus for many generations. A sample of the bacteria is then transferred to a medium that contains a radioactive isotope of phosphorus (^{32}P). Samples of the bacteria are removed immediately after the transfer and after one and two rounds of replication. What will be the distribution of radioactivity in the DNA of the bacteria in each sample? Will radioactivity be detected in neither, one, or both strands of the DNA?
19. A line of mouse cells is grown for many generations in a medium with ^{15}N . Cells in G_1 are then switched to a new medium that contains ^{14}N . Draw a pair of homologous chromosomes from these cells at the following stages, showing the two strands of DNA molecules found in the chromosomes. Use different colors to represent strands with ^{14}N and ^{15}N .
 - (a) Cells in G_1 , before switching to medium with ^{14}N
 - (b) Cells in G_2 , after switching to medium with ^{14}N
 - (c) Cells in anaphase of mitosis, after switching to medium with ^{14}N
 - (d) Cells in metaphase I of meiosis, after switching to medium with ^{14}N
 - (e) Cells in anaphase II of meiosis, after switching to medium with ^{14}N
- * 20. A circular molecule of DNA contains 1 million base pairs. If DNA synthesis at a replication fork occurs at a rate of 100,000 nucleotides per minute, how long will theta replication require to completely replicate the molecule, assuming that theta replication is bidirectional? How long

will replication of this circular chromosome take by rolling-circle replication? Ignore replication of the displaced strand in rolling-circle replication.

21. A bacterium synthesizes DNA at each replication fork at a rate of 1000 nucleotides per second. If this bacterium completely replicates its circular chromosome by theta replication in 30 minutes, how many base pairs of DNA will its chromosome contain?
- *22. The following diagram represents a DNA molecule that is undergoing replication. Draw in the strands of newly synthesized DNA and identify the following:
- Polarity of newly synthesized strands
 - Leading and lagging strands
 - Okazaki fragments
 - RNA primers

- *23. What would be the effect on DNA replication of mutations that destroyed each of the following activities in DNA polymerase I?



- $3' \rightarrow 5'$ exonuclease activity
- $5' \rightarrow 3'$ exonuclease activity
- $5' \rightarrow 3'$ polymerase activity

CHALLENGE QUESTIONS

24. Conditional mutations express their mutant phenotype only under certain conditions (the restrictive conditions) and express the normal phenotype under other conditions (the permissive conditions). One type of conditional mutation is a temperature-sensitive mutation, which expresses the mutant phenotype only at certain temperatures.

Strains of *E. coli* have been isolated that contain temperature-sensitive mutations in the genes encoding different components of the replication machinery. In each of these strains, the protein produced by the mutated gene is nonfunctional under the restrictive conditions. These strains are grown under permissive conditions and then abruptly switched to the restrictive condition. After one

round of replication under the restrictive condition, the DNA from each strain is isolated and analyzed. What would you predict to see in the DNA isolated from each strain in the following list?

Temperature-sensitive mutation in gene encoding:

- DNA ligase
- DNA polymerase I
- DNA polymerase III
- Primase
- Initiator protein

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13

Transcription



Molecular image of the hammerhead ribozyme (in blue) bound to RNA (in orange). Ribozymes are catalytic RNA molecules that may have been the first carriers of genetic information. (K. Eward/Biografx/Photo Researchers.)

- RNA in the Primeval World
- RNA Molecules
 - The Structure of RNA
 - Classes of RNA
- Transcription: Synthesizing RNA from a DNA Template
 - The Template
 - The Substrate for Transcription
 - The Transcription Apparatus
- The Process of Bacterial Transcription
 - Initiation
 - Elongation
 - Termination
- The Process of Eukaryotic Transcription
 - Transcription and Nucleosome Structure
 - Transcription Initiation
 - RNA Polymerase II Promoters
 - RNA Polymerase I Promoters
 - RNA Polymerase III Promoters
 - Evolutionary Relationships and the TATA-Binding Protein
 - Termination

RNA in the Primeval World

Life requires two basic functions. First, living organisms must be able to store and faithfully transmit genetic information during reproduction. Second, they must have the ability to catalyze chemical transformations, to fire the reactions that drive life processes. It was long believed that the functions of information storage and chemical transformation are handled by two entirely different types of molecules. Genetic information is stored in nucleic acids. Catalysis of chemical transformations was held to be the exclusive domain of certain proteins that serve as biological catalysts or enzymes, making reactions take place rapidly within the cell. This biochemical dichotomy—nucleic acid

for information, proteins for catalysts—revealed a dilemma in our understanding of the early stages in the evolution of life. Which came first: proteins or nucleic acids? If nucleic acids carry the coding instructions for proteins, how could proteins be generated without them? Because nucleic acids are unable to copy themselves, how could they be generated without proteins? If DNA and proteins each require the other, how could life begin?

This apparent paradox disappeared in 1981 when Thomas Cech and his colleagues discovered that RNA can serve as a biological catalyst. They found that RNA from the protozoan *Tetrahymena thermophila* can excise 400 nucleotides from its RNA in the absence of any protein. Other examples of catalytic RNAs have now been discovered in

different types of cells. Called **ribozymes**, these RNA molecules can cut out parts of their own sequences, connect some RNA molecules together, replicate others, and even catalyze the formation of peptide bonds between amino acids. The discovery of ribozymes complements other evidence suggesting that the original genetic material was RNA.

Ribozymes that were self-replicating probably first arose between 3.5 billion and 4 billion years ago and may have begun the evolution of life on Earth. Early life was an RNA world, with RNA molecules serving both as carriers of genetic information and as catalysts that drove the chemical reactions needed to sustain and perpetuate life. These catalytic RNAs may have acquired the ability to synthesize protein-based enzymes, which are more efficient catalysts; with enzymes taking over more and more of the catalytic functions, RNA probably became relegated to the role of information storage and transfer. DNA, with its chemical stability and faithful replication, eventually replaced RNA as the primary carrier of genetic information. In modern cells, RNA still plays a vital role in both DNA replication and protein synthesis.

Transcription is the synthesis of RNA molecules, with DNA as a template, and it is the first step in the transfer of genetic information from genotype to phenotype. The process is complex, and requires a number of protein components. As we examine the stages of transcription, try to keep all the detail in perspective; focus on understanding how the details relate to the overall purpose of transcription—the selective synthesis of an RNA molecule.

This chapter begins with a brief review of RNA structure and a discussion of the different classes of RNA. We then consider the major components required for transcription. Finally, we explore the process of transcription in eubacteria and eukaryotic cells. At several points in the text, we'll pause to absorb some general principles that emerge.

www.whfreeman.com/pierce Current research on ribozymes

RNA Molecules

Before we begin our study of transcription, let's review the structure of RNA and consider the different types of RNA molecules.

The Structure of RNA

RNA, like DNA, is a polymer consisting of nucleotides joined together by phosphodiester bonds (see Chapter 10 for a discussion of RNA structure). However, there are several important differences in the structures of DNA and RNA. Whereas DNA nucleotides contain deoxyribose sugars, RNA nucleotides have ribose sugars ( **FIGURE 13.1a**). With a free hydroxyl group on the 2'-carbon atom of the ribose sugar, RNA is degraded rapidly under alkaline condi-

tions. The deoxyribose sugar of DNA lacks this free hydroxyl group; so DNA is a more stable molecule. Another important difference is that thymine, one of the two pyrimidines found in DNA, is replaced by uracil in RNA.

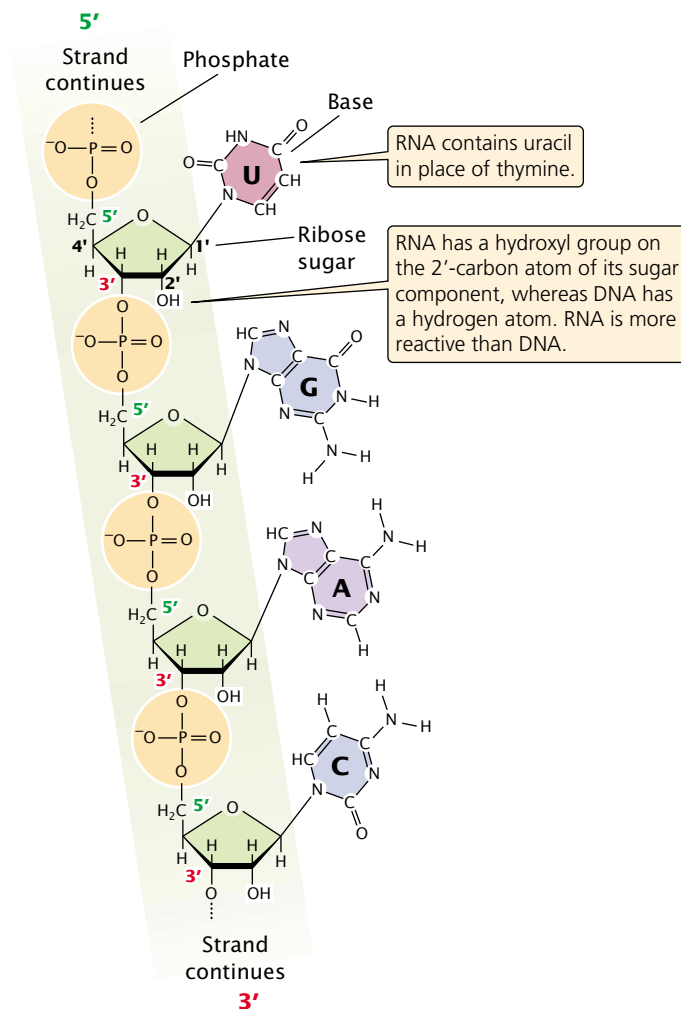
A final difference in the structures of DNA and RNA is that RNA is usually single stranded, consisting of a single polynucleotide strand ( **FIGURE 13.1b**), whereas DNA normally consists of two polynucleotide strands joined by hydrogen bonding between complementary bases. Some viruses contain double-stranded RNA genomes, as discussed in Chapter 8. Although RNA is usually single stranded, short complementary regions within a nucleotide strand can pair and form secondary structures (see Figure 13.1b). These RNA secondary structures are often called hairpin-loops or stem-loop structures. When two regions within a single RNA molecule pair up, the strands in those regions must be antiparallel, with pairing between cytosine and guanine and between adenine and uracil (although occasionally guanine pairs with uracil).

The formation of secondary structures plays an important role in RNA function. Secondary structure is determined by the base sequence of the nucleotide strand; so different RNA molecules can assume different structures. Because their structure determines their function, RNA molecules have the potential for tremendous variation in function. With its two complementary strands forming a helix, DNA is much more restricted in the range of secondary structures that it can assume, and it serves fewer functional roles in the cell. Similarities and differences in DNA and RNA structures are summarized in Table 13.1.

Table 13.1 The structures of DNA and RNA compared

Characteristic	DNA	RNA
Composed of nucleotides	Yes	Yes
Type of sugar	Deoxyribose	Ribose
Presence of 2'-OH group	No	Yes
Bases	A, G, C, T	A, G, C, U
Nucleotides joined by phosphodiester bonds	Yes	Yes
Double or single stranded	Usually double	Usually single
Secondary structure	Double helix	Many types
Stability	Quite stable	Easily degraded

(a)



(b) Primary structure

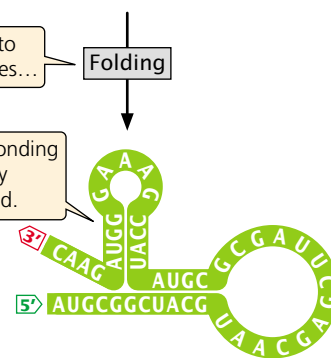
5' AUGCGGCUACGUAACGAGCUUAGCGCGUAUACCGAAAGGGUAGAAC 3'

An RNA molecule folds to form secondary structures...

Folding

...owing to hydrogen bonding between complementary bases on the same strand.

Secondary structure



13.1 RNA has a primary and a secondary structure.

Classes of RNA

RNA molecules perform a variety of functions in the cell. **Ribosomal RNA (rRNA)**, along with ribosomal protein subunits, makes up the ribosome, the site of protein assembly. We'll take a more detailed look at the ribosome in Chapter 14. **Messenger RNA (mRNA)** carries the coding instructions for polypeptide chains from DNA to the ribosome. After attaching to a ribosome, an mRNA molecule specifies the sequence of the amino acids in a polypeptide chain and provides a template for joining amino acids. Large precursor molecules, which are termed **pre-messenger RNAs (pre-mRNAs)**, are the immediate products of transcription in eukaryotic cells. Pre-mRNAs are modified extensively before they exit the nucleus for translation into protein. Bacterial cells do not possess pre-mRNA; in these cells, transcription takes place concurrently with translation.

Transfer RNA (tRNA) serves as the link between the coding sequence of nucleotides in the mRNA and the amino acid sequence of a polypeptide chain. Each tRNA attaches to one particular type of amino acid and helps to incorporate that amino acid into a polypeptide chain (discussed in Chapter 15).

Additional classes of RNA molecules are found in the nuclei of eukaryotic cells. **Small nuclear RNAs (snRNAs)** combine with small nuclear protein subunits to form **small nuclear ribonucleoproteins (snRNPs)**, affectionately known as "snurps"). The snRNPs are analogous to ribosomes in structure, only smaller, and they typically contain a single RNA molecule combined with approximately 10 small nuclear protein subunits. Some snRNAs participate in the processing of RNA, converting pre-mRNA into mRNA. **Small nucleolar RNAs (snoRNAs)** take part in the processing of rRNA. Small

Table 13.2 Locations and functions of different classes of RNA molecules

Class of RNA	Cell Type	Location of Function* in Eukaryotic Cells	Function
Ribosomal RNA (rRNA)	Bacterial and eukaryotic	Cytoplasm	Structural and functional components of the ribosome
Messenger RNA (mRNA)	Bacterial and eukaryotic	Nucleus and cytoplasm	Carries genetic code for proteins
Transfer RNA (tRNA)	Bacterial and eukaryotic	Cytoplasm	Helps incorporate amino acids into polypeptide chain
Small nuclear RNA (snRNA)	Eukaryotic	Nucleus	Processing of pre-mRNA
Small nucleolar RNA (snoRNA)	Eukaryotic	Nucleus	Processing and assembly of rRNA
Small cytoplasmic RNA (scRNA)	Eukaryotic	Cytoplasm	Variable

*All eukaryotic RNAs are transcribed in the nucleus.

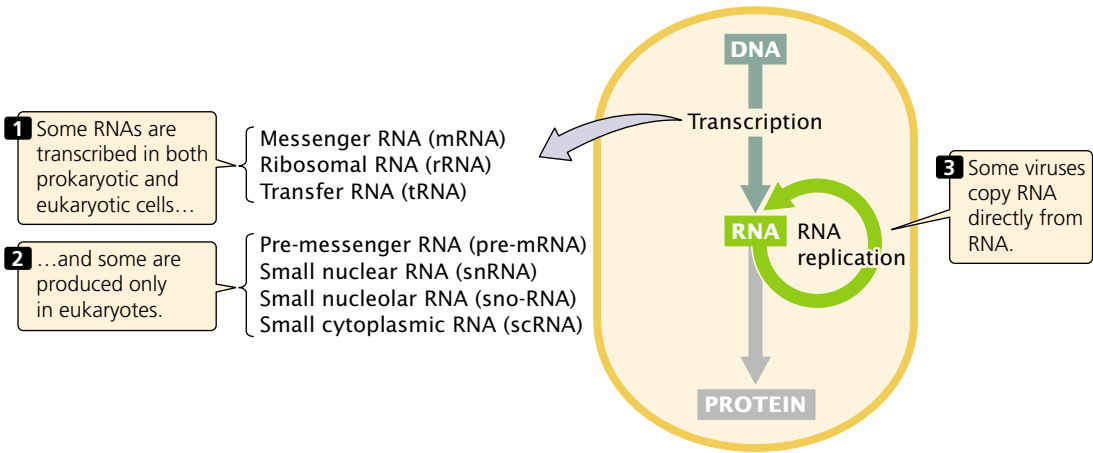
RNA molecules also are found in the cytoplasm of eukaryotic cells; these molecules are called **small cytoplasmic RNAs** (scRNAs). The different classes of RNA molecules are summarized in Table 13.2.

Concepts

RNA differs from DNA in that it possesses a hydroxyl group on the 2'-carbon atom of its sugar, contains uracil instead of thymine, and is normally single stranded. Several classes of RNA exist within bacterial and eukaryotic cells.

Transcription: Synthesizing RNA from a DNA Template

All cellular RNAs are synthesized from a DNA template through the process of transcription (FIGURE 13.2). Transcription is in many ways similar to the process of replication, but one fundamental difference relates to the length of the template used. During replication, all the nucleotides in the DNA template are copied, but, during transcription, only small parts of the DNA molecule—usually a single gene or, at most, a few genes—are transcribed into RNA. Because not all gene products are needed at the same time or in the same cell, it would be highly



13.2 All cellular types of RNA are transcribed from DNA.

inefficient for a cell to constantly transcribe all of its genes. Furthermore, much of the DNA does not code for a functional product, and transcription of such sequences would be pointless. Transcription is, in fact, a highly selective process—individual genes are transcribed only as their products are needed. But this selectivity imposes a fundamental problem on the cell—the problem of how to recognize individual genes and transcribe them at the proper time and place.

Like replication, transcription requires three major components:

1. a DNA template;
2. the raw materials (substrates) needed to build a new RNA molecule; and
3. the transcription apparatus, consisting of the proteins necessary to catalyze the synthesis of RNA.

The Template

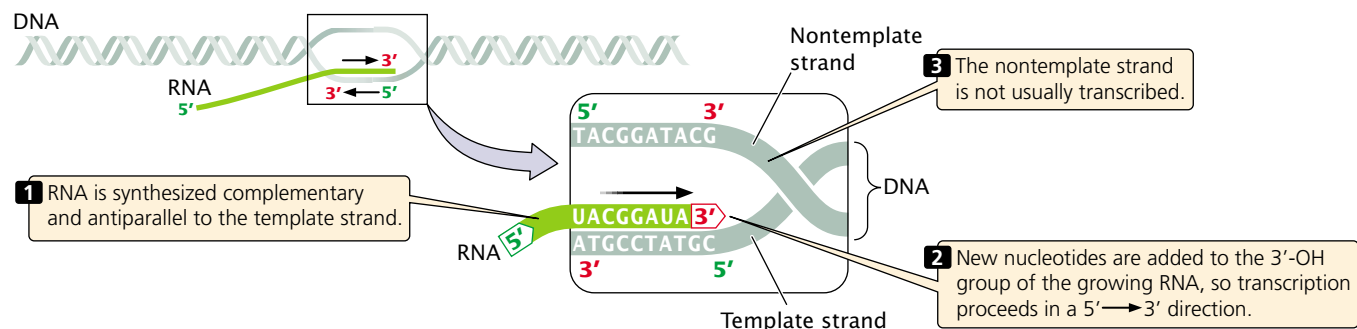
In 1970, Oscar Miller, Jr., Barbara Hamkalo, and Charles Thomas used electron microscopy to examine cellular contents and demonstrate that RNA is transcribed from a DNA template. The results of this study revealed within the cell the presence of Christmas-tree-like structures: thin central fibers (the trunk of the tree), to which were attached strings (the branches) with granules (FIGURE 13.3). The addition of deoxyribonuclease (an enzyme that degrades DNA) caused the central fibers to disappear, indicating that the “tree trunks” were DNA molecules. Ribonuclease (an enzyme that degrades RNA) removed the granular strings, indicating that the branches were RNA. Their conclusion was that each Christmas tree represented a gene undergoing transcription. The transcription of each gene begins at the top of the tree; there, little of the DNA has been transcribed and the RNA branches are short. As the transcription apparatus moves down the tree, transcribing more of the template, the RNA molecules lengthen, producing the long branches at the bottom.



13.3 Under the electron microscope, DNA molecules undergoing transcription exhibit Christmas-tree-like structures. The trunk of each “Christmas tree” (a transcription unit) represents a DNA molecule; the tree branches (granular strings attached to the DNA) are RNA molecules that have been transcribed from the DNA. As the transcription apparatus moves down the DNA, transcribing more of the template, the RNA molecules become longer and longer. (O. L. Miller, B. R. Beatty, D. W. Fawcett/Visuals Unlimited.)

The transcribed strand The template for RNA synthesis, as for DNA synthesis, is a single strand of the DNA double helix. Unlike replication, however, transcription typically takes place on only one of the two nucleotide strands of DNA (FIGURE 13.4). The nucleotide strand used for transcription is termed the **template strand**. The other strand, called the **nontemplate strand**, is not ordinarily transcribed. Thus, in any one section of DNA, only one of the nucleotide strands normally carries the genetic information that is transcribed into RNA (there *are* some exceptions to this rule).

Evidence that only one DNA strand serves as a template came from several experiments carried out by Julius



13.4 RNA molecules are synthesized that are complementary and antiparallel to one of the two nucleotide strands of DNA, the template strand.

Marmur and his colleagues in 1963 on the DNA of bacteriophage SP8, which infects the bacterium *Bacillus subtilis*. This phage carries its genetic information in the form of a double-stranded DNA molecule. The two strands have different base compositions and therefore different densities, which permits the separation of the strands by equilibrium density gradient centrifugation (see Figure 12.2) into “heavy” and “light” DNA strands.

Marmur and his colleagues placed some *B. subtilis* in a medium that contained a radioactively labeled precursor of RNA (FIGURE 13.5). They infected the bacteria with SP8, and the phage injected their DNA into the bacterial cells. Transcription of the phage DNA within the cells incorporated the radioactive precursor into the newly synthesized RNA, producing radioactively labeled RNA complementary to the phage DNA (step 2), which was then isolated from the cells (step 3).

The DNA of another culture of SP8 phage was isolated (step 4) and the heavy and light strands of the DNA were separated (step 5). When the radioactively labeled RNA (obtained in steps 1 through 3 of Figure 13.5) was com-

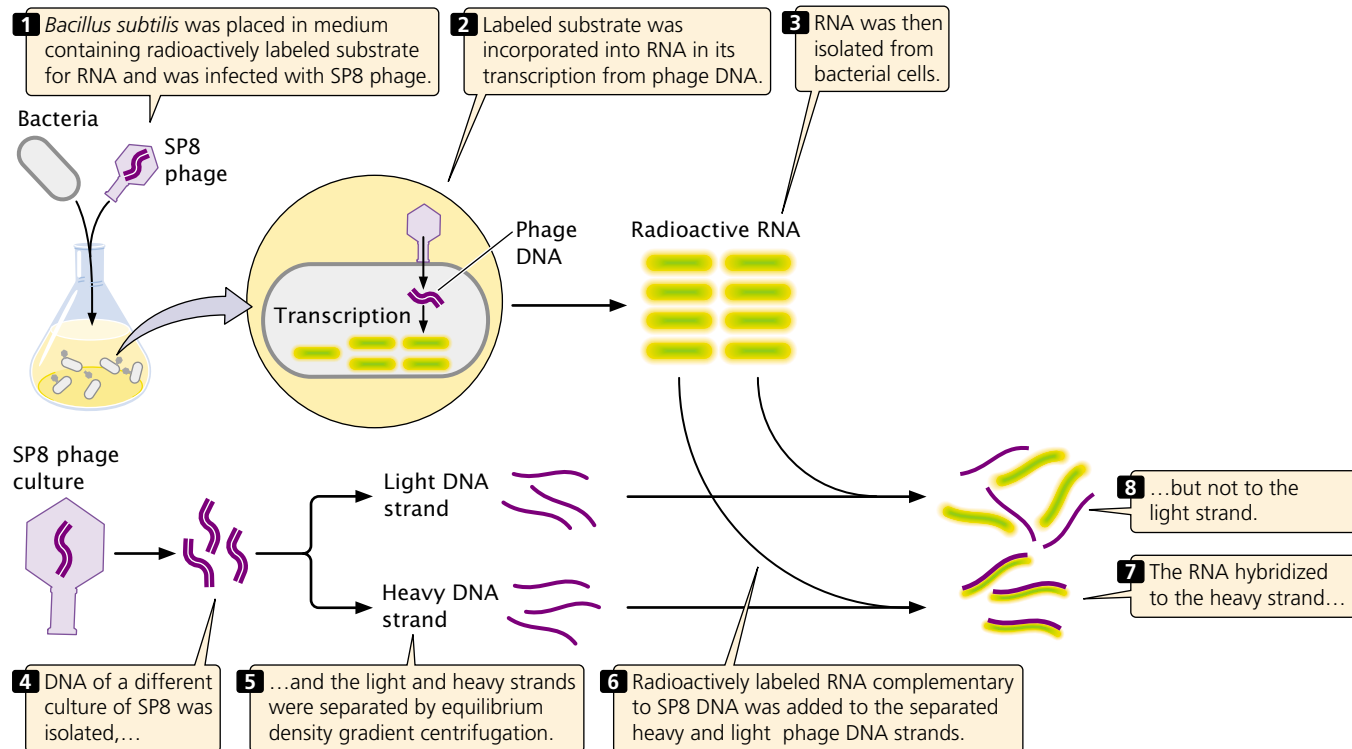
bined with the heavy strand (step 6), the RNA hybridized to it, indicating that the RNA and DNA were complementary (step 7). However, when radioactively labeled RNA was added to the light strand (step 8), no hybridization took place. These findings led Marmur and his colleagues to conclude that RNA is transcribed from only one of the DNA strands in SP8—in this case, the heavy strand.

SP8 is unusual in that all of its genes are transcribed from the same strand. In most organisms, each gene is transcribed from a single strand, but different genes may be transcribed from different strands (FIGURE 13.6). Notice that one of the strands in Figure 13.6 is identified as plus (+) and the other as minus (−). The plus strand is the template for genes *a* and *c*, and the minus strand is the template for gene *b*.

During transcription, an RNA molecule is synthesized that is complementary and antiparallel to the DNA template strand (see Figure 13.4). The RNA transcript has the same polarity and base sequence as does the nontemplate strand, with the exception that U in RNA substitutes for T in DNA.

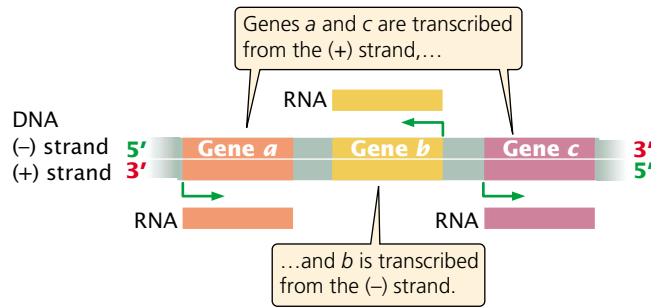
Experiment

Question: Do both strands of DNA serve as templates for RNA synthesis?



Conclusion: The fact that RNA hybridized to only one of the DNA strands indicates that it was transcribed only from that strand.

13.5 Marmur and colleagues showed that only one DNA strand serves as template during transcription.



13.6 RNA is transcribed from one DNA strand. In most organisms, each gene is transcribed from a single DNA strand, but different genes may be transcribed from one or the other of the two DNA strands.

Concepts

Within a single gene, only one of the two DNA strands, the template strand, is generally transcribed into RNA.

The transcription unit A **transcription unit** is a stretch of DNA that codes for an RNA molecule and the sequences necessary for its transcription. In eukaryotes, as discussed in Chapter 14, alternative RNA molecules can be produced from each transcription unit. How does the complex of enzymes and proteins that performs transcription—the transcription apparatus—recognize a transcription unit? How does it know which DNA strand to read, and where to start and stop? This information is encoded by the DNA sequence.

Included within a transcription unit are three critical regions: a promoter, an RNA coding sequence, and a terminator (FIGURE 13.7). The **promoter** is a DNA sequence that the transcription apparatus recognizes and binds. It indicates which of the two DNA strands is to be read as the template and the direction of transcription. The promoter also determines the transcription start site, the first nucleotide that will be transcribed into RNA. In most transcription units, the promoter is located next to the transcription start site but is not, itself, transcribed.

The second critical region of the transcription unit is the **RNA-coding region**, a sequence of DNA nucleotides

that is copied into an RNA molecule. A third component of the transcription unit is the **terminator**, a sequence of nucleotides that signals where transcription is to end. Terminators are usually part of the coding sequence; that is, transcription stops only after the terminator has been copied into RNA.

Molecular biologists often use the terms *upstream* and *downstream* to refer to the direction of transcription and the location of nucleotide sequences surrounding the RNA coding sequence. The transcription apparatus is said to move downstream during transcription: it binds to the promoter (which is usually upstream of the start site) and moves toward the terminator (which is downstream of the start site).

When DNA sequences are written out, often the sequence of only one of the two strands is listed. Molecular biologists typically write the sequence of the nontemplate strand, because it will be the same as the sequence of the RNA transcribed from the template (with the exception that U in RNA replaces T in DNA). By convention, the sequence on the nontemplate strand is written with the 5' end on the left and the 3' end on the right. The first nucleotide transcribed (the transcription start site) is numbered +1; nucleotides downstream of the start site are assigned positive numbers, and nucleotides upstream of the start site are assigned negative numbers. So, nucleotide +34 would be 34 nucleotides downstream of the start site, whereas nucleotide -75 would be 75 nucleotides upstream of the start site.

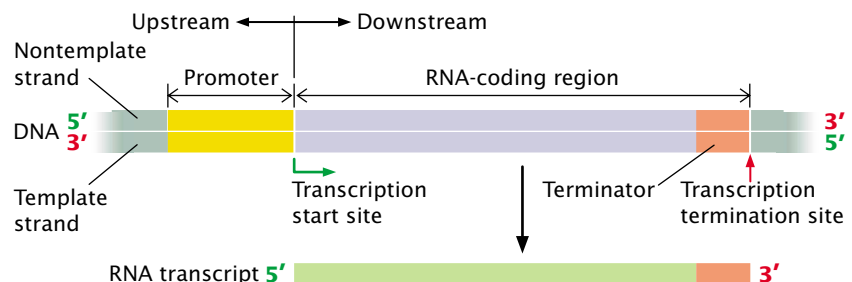
Concepts

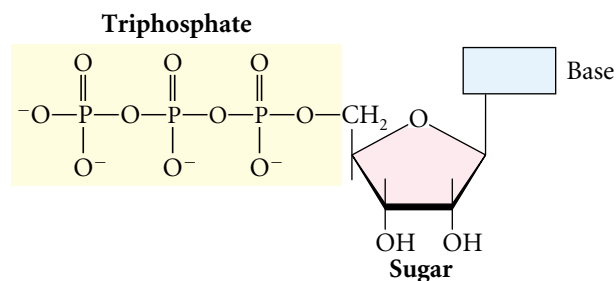
A transcription unit is a piece of DNA that encodes an RNA molecule and the sequences necessary for its proper transcription. Each transcription unit includes a promoter, an RNA-coding region, and a terminator.

The Substrate for Transcription

RNA is synthesized from **ribonucleoside triphosphates** (rNTPs) (FIGURE 13.8). In synthesis, nucleotides are added one at a time to the 3'-OH group of the growing RNA molecule. Two phosphates are cleaved from the

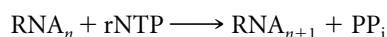
13.7 A transcription unit includes a promoter, an RNA-coding region, and a terminator.





13.8 Ribonucleoside triphosphates are substrates used in RNA synthesis.

incoming ribonucleoside triphosphate; the remaining phosphate participates in a phosphodiester bond that connects the nucleotide to the growing RNA molecule. The overall chemical reaction for the addition of each nucleotide is:



where PP_i represents two atoms of inorganic phosphorus. Nucleotides are always added to the 3' end of the RNA molecule, and the direction of transcription is therefore $5' \rightarrow 3'$ (FIGURE 13.9), the same as the direction of DNA synthesis during replication. RNA is made complementary and antiparallel to one of the DNA strands (the template strand).

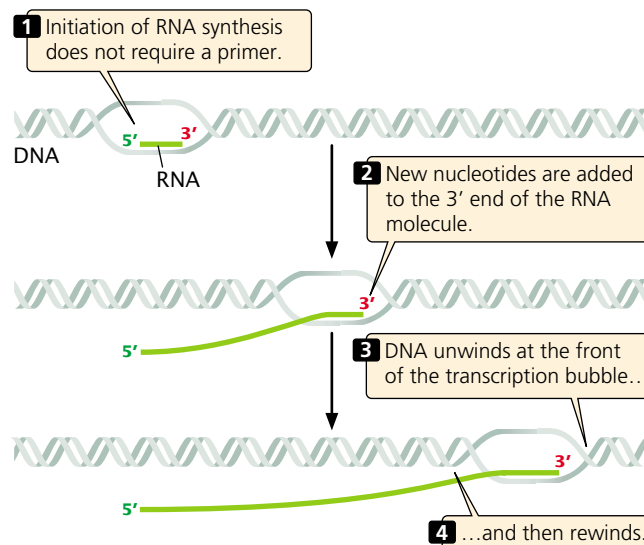
Concepts

RNA is synthesized from ribonucleoside triphosphates. Transcription is $5' \rightarrow 3'$: each new nucleotide is joined to the 3'-OH group of the last nucleotide added to the growing RNA molecule. RNA synthesis does not require a primer.

The Transcription Apparatus

Recall that, in replication, a number of different enzymes and proteins are required to bring about DNA synthesis. Although transcription might initially appear to be quite different, because a single enzyme—**RNA polymerase**—carries out all the required steps of transcription, on closer inspection, the processes are actually similar. The action of RNA polymerase is enhanced by a number of accessory proteins that join and leave the polymerase at different stages of the process. Each accessory protein is responsible for providing or regulating a special function. Thus, transcription, like replication, requires an array of proteins.

Bacterial RNA polymerase Bacterial cells typically possess only one type of RNA polymerase, which catalyzes the synthesis of all classes of bacterial RNA: mRNA, tRNA, and



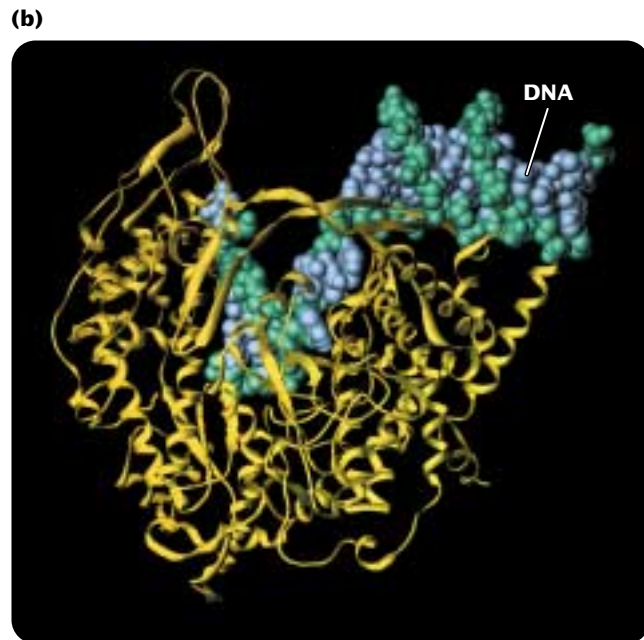
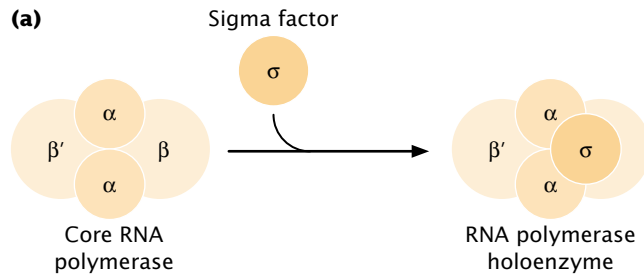
13.9 In transcription, nucleotides are always added to the 3' end of the RNA molecule.

rRNA. Bacterial RNA polymerase is a large, multimeric enzyme (meaning that it consists of several polypeptide chains).

At the heart of bacterial RNA polymerase are four subunits (individual polypeptide chains) that make up the **core enzyme**: two copies of a subunit called alpha (α), a single copy of beta (β), and single copy of beta prime (β') (FIGURE 13.10). The core enzyme catalyzes the elongation of the RNA molecule by the addition of RNA nucleotides. Other functional subunits join and leave the core enzyme at particular stages of the transcription process. The **sigma (σ) factor** controls the binding of the RNA polymerase to the promoter. Without sigma, RNA polymerase will initiate transcription at a random point along the DNA. After sigma has associated with the core enzyme (forming a **holoenzyme**), RNA polymerase binds stably only to the promoter region and initiates transcription at the proper start site. Sigma is required only for promoter binding and initiation; when a few RNA nucleotides have been joined together, sigma detaches from the core enzyme.

Many bacteria possess multiple types of sigma. *E. coli*, for example, possesses sigma 28 (σ^{28}), sigma 32 (σ^{32}), sigma 54 (σ^{54}), and sigma 70 (σ^{70}), named on the basis of their molecular weights. Each type of sigma initiates the binding of RNA polymerase to a particular set of promoters. For example, σ^{32} binds to promoters of genes that protect against environmental stress, σ^{54} binds to promoters of genes used during nitrogen starvation, and σ^{70} binds to many different promoters.

Other subunits provide the core RNA polymerase with additional functions. Rho (ρ) and NusA, for example, facilitate the termination of transcription.



13.10 In bacterial RNA polymerase, the core enzyme consists of four subunits: two copies of alpha (α), a single copy of beta (β), and single copy of beta prime (β'). The core enzyme catalyzes the elongation of the RNA molecule by the addition of RNA nucleotides. (a) The sigma factor (σ) joins the core to form the holoenzyme, which is capable of binding to a promoter and initiating transcription. (b) The molecular model shows RNA polymerase (shown in yellow) binding DNA.

Eukaryotic RNA polymerases Eukaryotic cells possess three distinct types of RNA polymerase, each of which is responsible for transcribing a different class of RNA: **RNA polymerase I** transcribes rRNA; **RNA polymerase II** transcribes pre-mRNAs, snoRNAs, and some snRNAs; and **RNA polymerase III** transcribes small RNA molecules—specifically tRNAs, small rRNA, and some snRNAs (Table 13.3). All three eukaryotic polymerases are large, multimeric enzymes, typically consisting of more than a dozen subunits. Some subunits are common to all three RNA polymerases, whereas others are limited to one of the polymerases. As in bacterial cells, a number of accessory proteins bind to the core enzyme and affect its function.

Table 13.3 Eukaryotic RNA polymerases

Type	Transcribes
RNA polymerase I	Large rRNAs
RNA polymerase II	Pre-mRNA, some snRNAs, snoRNAs
RNA polymerase III	tRNAs, small rRNA, snRNAs

Concepts

Bacterial cells possess a single type of RNA polymerase, consisting of a core enzyme and other subunits that participate in various stages of transcription. Eukaryotic cells possess three distinct types of RNA polymerase: RNA polymerase I transcribes rRNA; RNA polymerase II transcribes pre-mRNA, snoRNAs, and some snRNAs; and RNA polymerase III transcribes tRNAs, small rRNAs, and some snRNAs.

The Process of Bacterial Transcription

Now that we've considered some of the major components of transcription, we're ready to take a detailed look at the process. Transcription can be conveniently divided into three stages:

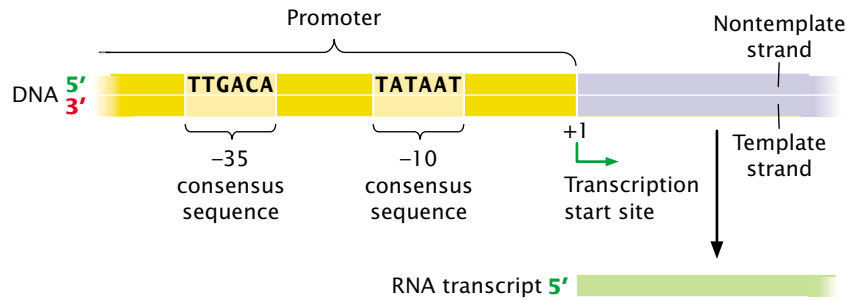
1. initiation, in which the transcription apparatus assembles on the promoter and begins the synthesis of RNA;
2. elongation, in which RNA polymerase moves along the DNA, unwinding it and adding new nucleotides, one at a time, to the 3' end of the growing RNA strand; and
3. termination, the recognition of the end of the transcription unit and the separation of the RNA molecule from the DNA template.

We will first examine each of these steps in bacterial cells, where the process is best understood; then we will consider eukaryotic transcription.

Initiation

Initiation includes all the steps necessary to begin RNA synthesis, including (1) promoter recognition, (2) formation of the transcription bubble, (3) creation of the first bonds between rNTPs, and (4) escape of the transcription apparatus from the promoter.

Transcription initiation requires that the transcription apparatus recognize and bind to the promoter. At this step, the selectivity of transcription is enforced; the binding of RNA polymerase to the promoter determines which parts of the DNA template are to be transcribed and how often.



13.11 In bacterial promoters, consensus sequences are found upstream of the start site, approximately at positions -10 and -35.

Different genes are transcribed with different frequencies, and promoter binding is primarily responsible for determining the frequency of transcription for a particular gene. Promoters also have different affinities for RNA polymerase. Even within a single promoter, the affinity can vary over time, depending on its interaction with RNA polymerase and a number of other factors.

Bacterial promoters Essential information for the transcription unit—where it will start transcribing, which strand is to be read, and in what direction the RNA polymerase will move—is imbedded in the nucleotide sequence of the promoter. Promoters are sequences in the DNA that are recognized by the transcription apparatus and are required for transcription to take place. In bacterial cells, promoters are usually adjacent to an RNA coding sequence. The examination of many promoters in *E. coli* and other bacteria reveals a general feature: although most of the nucleotides within the promoters vary in sequence, short stretches of nucleotides are common to many. Furthermore, the spacing and location of these nucleotides relative to the transcription start site are similar in most promoters. These short stretches of common nucleotides are called **consensus sequences**.

The term “consensus sequence” refers to sequences that possess considerable similarity or consensus. By definition, the consensus sequence comprises the most commonly encountered nucleotides found at a specific location. For example, consider the following nucleotides found near the transcription start site of four prokaryotic genes.

5' - A A T A A A - 3'
 5' - T T T A A T - 3'
 5' - T A T T T T - 3'
 5' - T A A A A T - 3'

Consensus sequence = 5' - T A T A A T - 3'

If two bases are equally frequent, they are designated by listing both bases separated by a line or a slash, as in 5' - T A T A A A/T - 3'. Purines can be indicated by the abbreviation R, pyrimidines by Y, and any nucleotide by N. For example, the consensus sequence 5' - T A Y A R N A - 3'

means that the third nucleotide in the consensus sequence (Y) is usually a pyrimidine, but either pyrimidine is equally likely. Similarly, the fifth nucleotide in the sequence (R) is most likely one of the purines, but both are equally frequent. In the sixth position (N), no particular base is more common than any other. The presence of consensus in a set of nucleotides usually implies that the sequence is associated with an important function. Consensus exists in a sequence because natural selection has favored a restricted set of nucleotides in that position.

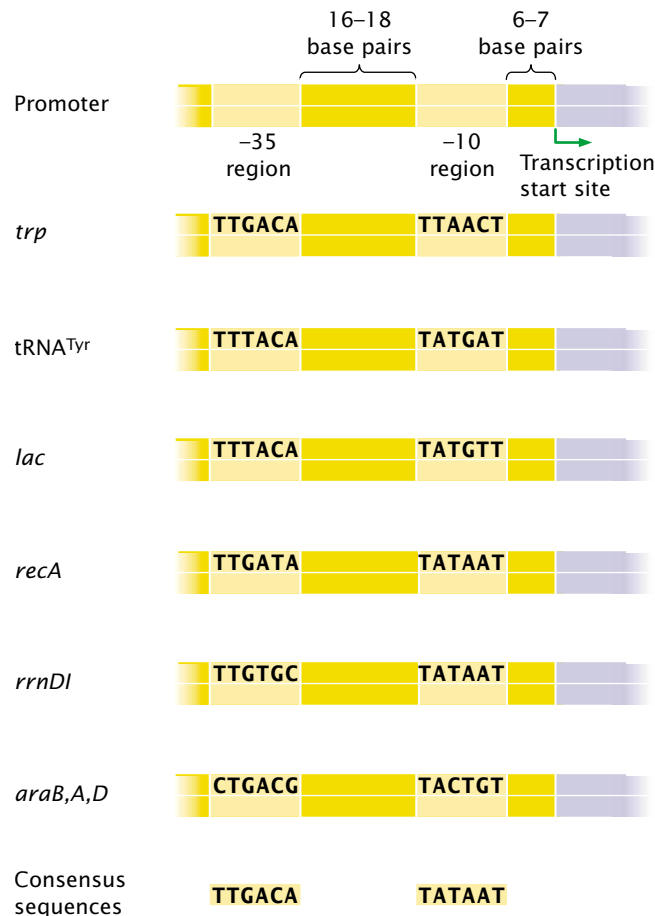
The most commonly encountered consensus sequence, found in almost all bacterial promoters, is located just upstream of the start site, centered on position -10. Called the **-10 consensus sequence** or, sometimes, the Pribnow box, its sequence is

5' TATAAT 3'
 3' ATATTA 5'

often written simply as TATAAT (FIGURE 13.11). Remember that TATAAT is just the *consensus* sequence—representing the most commonly encountered nucleotides at each of these positions. In most prokaryotic promoters, the actual sequence is not TATAAT (FIGURE 13.12).

Another consensus sequence common to most bacterial promoters is TTGACA, which lies approximately 35 nucleotides upstream of the start site and is termed the **-35 consensus sequence** (see Figure 13.11). The nucleotides on either side of the -10 and -35 consensus sequences and those between them vary greatly from promoter to promoter, suggesting that they are relatively unimportant in promoter recognition.

The function of these consensus sequences in bacterial promoters has been studied by inducing mutations at various positions within the consensus sequences and observing the effect of the changes on transcription. The results of these studies reveal that most base substitutions within the -10 and -35 consensus sequences reduce the rate of transcription; these substitutions are termed *down mutations* because they slow down the rate of transcription. Occasionally, a particular change in a consensus sequence increases the rate of transcription; such a change is called an *up mutation*.



13.12 In most prokaryotic promoters, the actual sequence is not TATAAT. The sequences shown are found in five *E. coli* promoters, including those of genes for tryptophan biosynthesis (*trp*), tyrosine tRNA (*tRNA^{Tyr}*), lactose metabolism (*lac*), a recombination protein (*recA*), and arabinose metabolism (*araB, A, D*). These sequences are on the nontemplate strand and read 5' → 3', left to right.

The sigma factor associates with the core enzyme (FIGURE 13.13a) to form a holoenzyme, which binds to the -35 and -10 consensus sequences in the DNA promoter (FIGURE 13.13b). Although it binds only the nucleotides of consensus sequences, the enzyme extends from -50 to +20 when bound to the promoter. The holoenzyme initially binds weakly to the promoter but then undergoes a change in structure that allows it to bind more tightly and unwind the double-stranded DNA (FIGURE 13.13c). Unwinding begins within the -10 consensus sequence and extends downstream for about 17 nucleotides, including the start site.

Some bacterial promoters contain a third consensus sequence that also takes part in the initiation of transcription. Called the **upstream element**, this sequence contains a number of A-T pairs and is found at about -40 to -60. The alpha subunit of the RNA polymerase interacts directly

with this upstream element, greatly enhancing the rate of transcription in those bacterial promoters that possess it. A number of other proteins may bind to sequences in and near the promoter; some stimulate the rate of transcription and others repress it; we will consider the proteins that regulate gene expression in Chapter 16.

Concepts

A promoter is a DNA sequence that is adjacent to a gene and required for transcription. Promoters contain short consensus sequences that are important in the initiation of transcription.

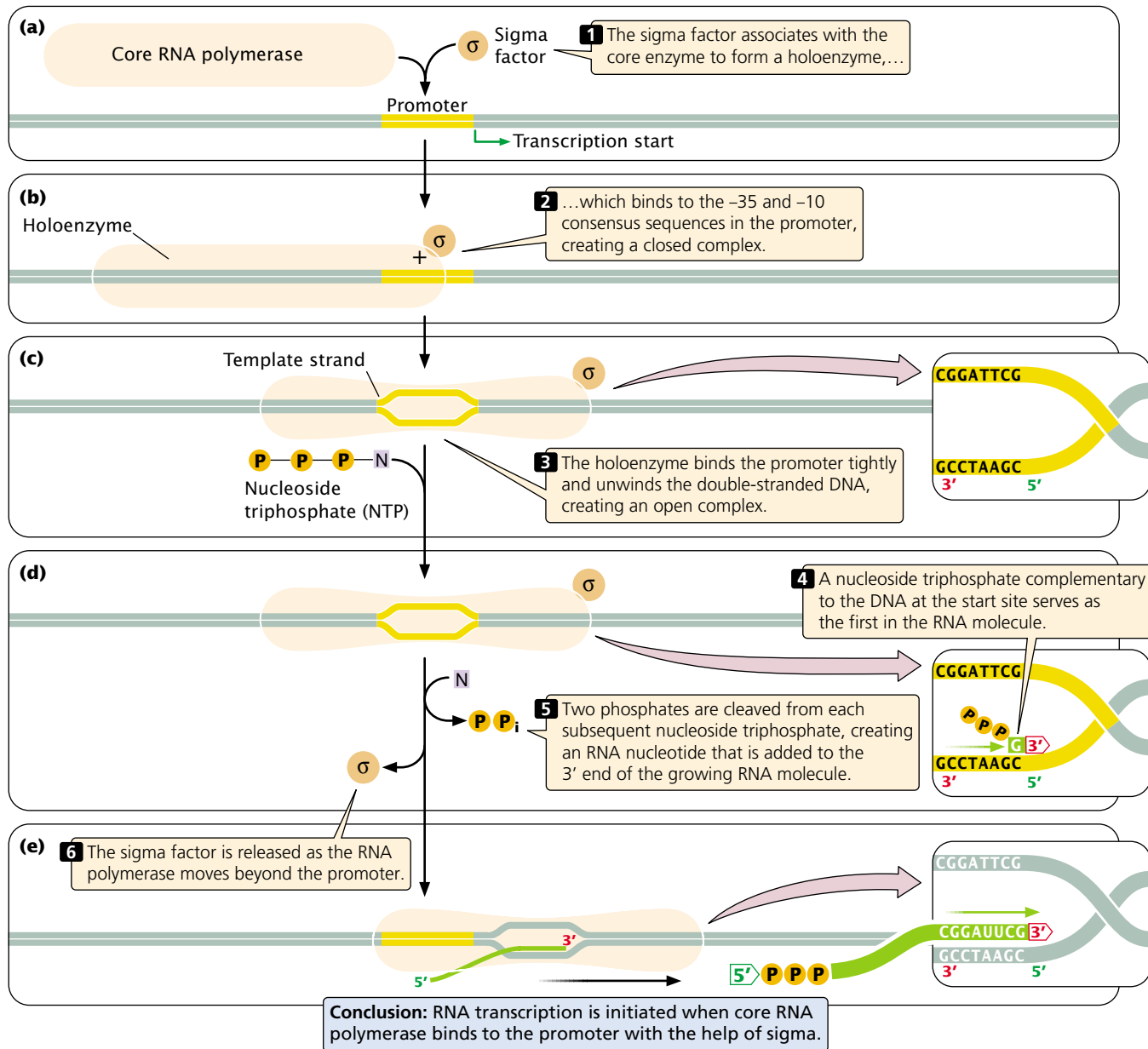
Initial RNA synthesis After the holoenzyme has attached to the promoter, RNA polymerase is positioned over the start site for transcription (at position +1) and has unwound the DNA to produce a single-stranded template. The orientation and spacing of consensus sequences on a DNA strand determine which strand will be the template for transcription, and thereby determine the direction of transcription.

The start site itself is not marked by a consensus sequence but often has the sequence CAT, with the start site at the A. The position of the start site is determined not by the sequences located there but by the location of the consensus sequences, which positions RNA polymerase so that the enzyme's active site is aligned for initiation of transcription at +1. If the consensus sequences are artificially moved upstream or downstream, the location of the starting point of transcription correspondingly changes.

To begin the synthesis of an RNA molecule, RNA polymerase pairs the base on a ribonucleoside triphosphate with its complementary base at the start site on the DNA template strand (FIGURE 13.13d). No primer is required to initiate the synthesis of the 5' end of the RNA molecule. Two of the three phosphates are cleaved from the ribonucleoside triphosphate as the nucleotide is added to the 3' end of the growing RNA molecule. However, because the 5' end of the first ribonucleoside triphosphate does not take part in the formation of a phosphodiester bond, all three of its phosphates remain. An RNA molecule therefore possesses, at least initially, three phosphates at its 5' end (FIGURE 13.13e).

Elongation

After initiation, RNA polymerase moves downstream along the template, progressively unwinding the DNA at the leading (downstream) edge of the transcription bubble, joining nucleotides to the RNA molecule according to the sequence on the template, and rewinding the DNA at the trailing (upstream) edge of the bubble. In bacterial



13.13 Transcription in bacteria is carried out by RNA polymerase, which must bind to the sigma factor to initiate transcription.

cells at 37°C, about 40 nucleotides are added per second. This rate of RNA synthesis is much lower than that of DNA synthesis, which is more than 1500 nucleotides per second in bacterial cells.

Transcription takes place within a short stretch of about 18 nucleotides of unwound DNA—the transcription bubble. Within this region, RNA is continuously synthesized, with single-stranded DNA used as a template. About 8 nucleotides of newly synthesized RNA are paired with the DNA-template nucleotides at any one time. As the transcription apparatus moves down the DNA template, it

generates positive supercoiling ahead of the transcription bubble and negative supercoiling behind it. Topoisomerase enzymes probably relieve the stress associated with the unwinding and rewinding of DNA in transcription, as they do in DNA replication.

Concepts

Transcription is initiated at the start site, which, in bacterial cells, is set by the binding of RNA polymerase to the consensus sequences of the



promoter. Transcription takes place within the transcription bubble. DNA is unwound ahead of the bubble and rewound behind it.

Termination

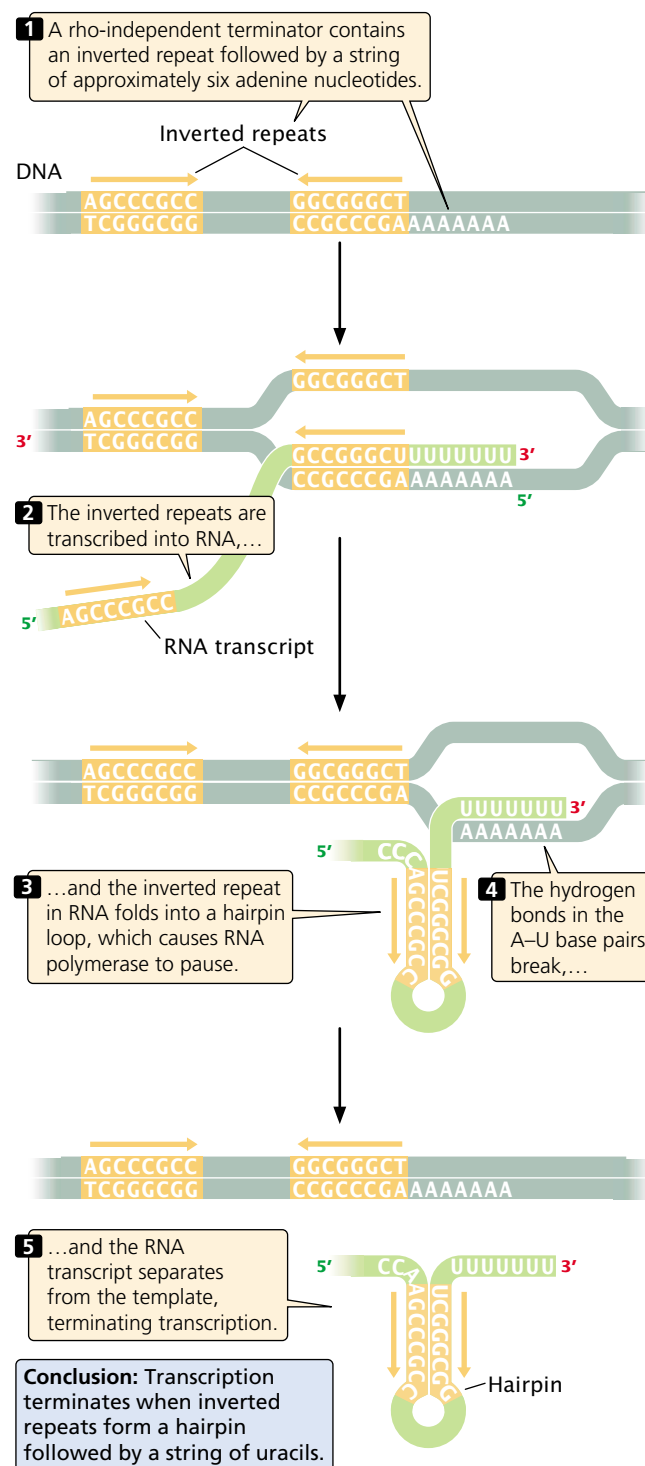
RNA polymerase moves along the template, adding nucleotides to the 3' end of the growing RNA molecule until it transcribes a terminator. Most terminators are found upstream of the point of termination. Transcription therefore does not suddenly end when polymerase reaches a terminator, like a car stopping in front of a stop sign. Rather, transcription ends after the terminator has been transcribed, like a car that stops only after running over a speed bump. At the terminator, several overlapping events are needed to bring an end to transcription: RNA polymerase must stop synthesizing RNA, the RNA molecule must be released from RNA polymerase, the newly made RNA molecule must dissociate fully from the DNA, and RNA polymerase must detach from the DNA template.

Bacterial cells possess two major types of terminators. **Rho-dependent terminators** are able to cause the termination of transcription only in the presence of an ancillary protein called the **rho factor**. **Rho-independent terminators** are able to cause the end of transcription in the absence of rho.

Rho-independent terminators have two common features. First, they contain inverted repeats (sequences of nucleotides on one strand that are inverted and complementary). When inverted repeats have been transcribed into RNA, a hairpin secondary structure forms (▶ **FIGURE 13.14**). Second, in rho-independent terminators, a string of approximately six adenine nucleotides follows the second inverted repeat in the template DNA. Their transcription produces a string of uracil nucleotides after the hairpin in the transcribed RNA.

The presence of a hairpin in an RNA transcript causes RNA polymerase to slow down or pause, which creates an opportunity for termination. The adenine–uracil base pairings downstream of the hairpin are relatively unstable compared with other base pairings, and the formation of the hairpin may itself destabilize the DNA–RNA pairing, causing the RNA molecule to separate from its DNA template. When the RNA transcript has separated from the template, RNA synthesis can no longer continue (see **Figure 13.14**).

Rho-dependent terminators have two features: (1) DNA sequences that produce a pause in transcription; and (2) a DNA sequence that encodes a stretch of RNA upstream of the terminator that is devoid of any secondary structures. This unstructured RNA serves as binding site for the rho protein, which binds the RNA and moves toward its 3' end, following the RNA polymerase (▶ **FIGURE 13.15**). When RNA polymerase encounters the terminator, it pauses, allowing



13.14 Termination by bacterial rho-independent terminators is a multistep process.

rho to catch up. The rho protein has helicase activity, which it uses to unwind the RNA–DNA hybrid in the transcription bubble, bringing an end to transcription.

Concepts

Transcription ends after RNA polymerase transcribes a terminator. Bacterial cells possess two types of terminator: a rho-independent terminator, which RNA polymerase can recognize by itself; and a rho-dependent terminator, which RNA polymerase can recognize only with the help of the rho protein.

Connecting Concepts

The Basic Rules of Transcription

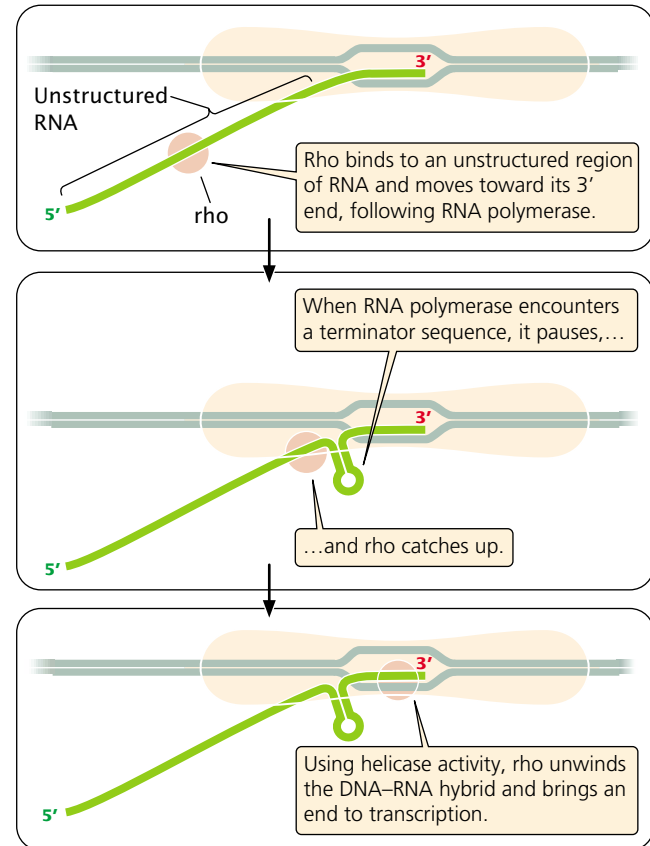
Before we examine the process of eukaryotic transcription, let's pause to summarize some of the general principles of bacterial transcription.

The Basic Rules of Transcription

1. Transcription is a selective process; only certain parts of the DNA are transcribed.
2. RNA is transcribed from single-stranded DNA. Normally, only one of the two DNA strands—the template strand—is copied into RNA.
3. Ribonucleoside triphosphates are used as the substrates in RNA synthesis. Two phosphates are cleaved from a ribonucleoside triphosphate, and the resulting nucleotide is joined to the 3'-OH group of the growing RNA strand.
4. RNA molecules are antiparallel and complementary to the DNA template strand. Transcription is always in the 5' → 3' direction, meaning that the RNA molecule grows at the 3' end.
5. Transcription depends on RNA polymerase—a complex, multimeric enzyme. RNA polymerase consists of a core enzyme, which is capable of synthesizing RNA, and other subunits that may join transiently to perform additional functions.
6. The core enzyme of RNA polymerase requires a sigma factor in order to bind to a promoter and initiate transcription.
7. Promoters contain short sequences crucial in the binding of RNA polymerase to DNA; these consensus sequences are interspersed with nucleotides that play no known role in transcription.
8. RNA polymerase binds to DNA at a promoter, begins transcribing at the start site of the gene, and ends transcription after a terminator has been transcribed.

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of transcription

A brief overview of the process



13.15 The termination of transcription in some bacterial genes requires the presence of the rho protein.

The Process of Eukaryotic Transcription

The process of eukaryotic transcription is similar to that of bacterial transcription. Eukaryotic transcription also includes initiation, elongation, and termination, and the basic principles of transcription already outlined apply to eukaryotic transcription. However, there are some important differences. Eukaryotic cells possess three different RNA polymerases, each of which transcribes a different class of RNA and recognizes a different type of promoter. Thus, a generic promoter cannot be described for eukaryotic cells, as was done for bacterial cells; rather, a promoter's description depends on whether the promoter is recognized by RNA polymerase I, II, or III. Another difference is in the nature of promoter recognition and initiation. Many proteins take part in the binding of eukaryotic RNA polymerases to DNA templates, and the different types of promoters require different proteins.

Transcription and Nucleosome Structure

Transcription requires that sequences on DNA are accessible to RNA polymerase and other proteins. However, in

eukaryotic cells, DNA is complexed with histone proteins in highly compressed chromatin (see Figure 11.5). How can the proteins necessary for transcription gain access to eukaryotic DNA when it is complexed with histones?

The answer to this question is that, before transcription, the chromatin structure is modified so that the DNA is in a more open configuration and is more accessible to the transcription machinery. Several types of proteins have roles in chromatin modification. Acetyltransferases add acetyl groups to amino acids at the ends of the histone proteins, which destabilizes the nucleosome structure and makes the DNA more accessible. Other types of histone modification also can affect chromatin packing. In addition, proteins called chromatin-remodeling proteins may bind to the chromatin and displace nucleosomes from promoters and other regions important for transcription. We will take a closer look at the role of changes to chromatin structure associated with gene expression in Chapter 16.

Transcription Initiation

The initiation of transcription is a complex process in eukaryotic cells because of the variety of initiation sequences and because numerous proteins bind to these sequences. Two broad classes of DNA sequences are important for the initiation of transcription: promoters and enhancers. A promoter is always found adjacent to (or sometimes within) the gene that it regulates and has a fixed location with regard to the transcription start point. An enhancer, in contrast, need not be adjacent to the gene; enhancers can affect the transcription of genes that are thousands of nucleotides away, and their positions relative to start sites can vary.

A significant difference between bacterial and eukaryotic transcription is the existence of three different eukaryotic RNA polymerases, which recognize different types of promoters. In bacterial cells, the holoenzyme (RNA polymerase plus sigma) recognizes and binds directly to sequences in the promoter. In eukaryotic cells, promoter recognition is carried out by accessory proteins that bind to the promoter and then recruit a specific RNA polymerase (I, II, or III) to the promoter.

One class of accessory proteins comprises **general transcription factors**, which, along with RNA polymerase, form the **basal transcription apparatus** that assembles near the

start site and is sufficient to initiate minimal levels of transcription. Another class of accessory proteins consists of **transcriptional activator proteins**, which bind to specific DNA sequences and bring about higher levels of transcription by stimulating the assembly of the basal transcription apparatus at the start site.

Concepts

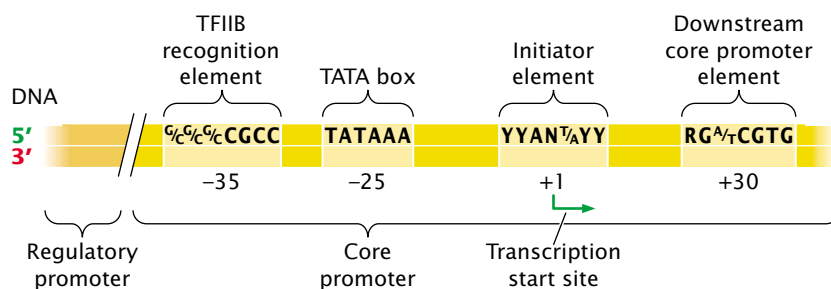
Two classes of DNA sequences in eukaryotic cells affect transcription: enhancers and promoters. A promoter is near the gene and has a fixed position relative to the start site of transcription. An enhancer can be distant from the gene and variable in location.

RNA Polymerase II Promoters

We will focus most of our attention on promoters recognized by RNA polymerase II, which transcribes the genes that encode proteins. A promoter for a gene transcribed by RNA polymerase II typically consists of two primary parts: the core promoter and the regulatory promoter.

Core promoter The **core promoter** is located immediately upstream of the gene (FIGURE 13.16) and typically includes one or more consensus sequences. The most common of these consensus sequences is the **TATA box**, which has the consensus sequence TATAAA and is located from –25 to –30 bp upstream of the start site. Mutations in the sequence of the TATA box affect the rate of transcription, and changing its position alters the location of the transcription start site.

Another common consensus sequence in the core promoter is the TFIIB recognition element (BRE), which has the consensus sequence G/C G/C G/C C G C C and is located from –32 to –38 bp upstream of the start site. (TFIIB is the abbreviation for a transcription factor that binds to this element; see next subsection). Instead of a TATA box, some core promoters have an initiator element (Inr) that directly overlaps the start site and has the consensus Y Y A N T/A Y Y. Another consensus sequence called the downstream core promoter element (DPE) is found



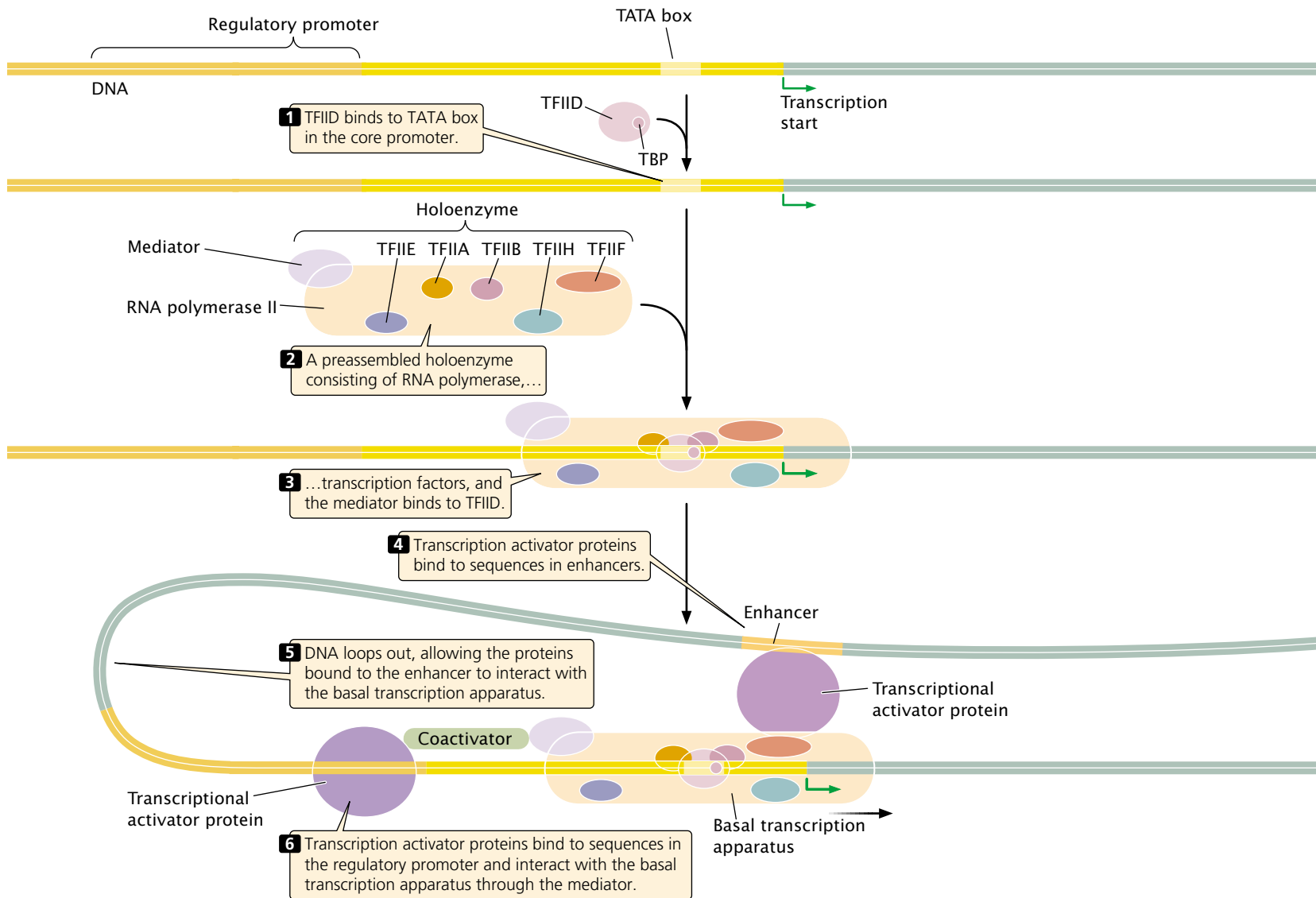
13.16 The promoters of genes transcribed by RNA polymerase II consist of a core promoter and a regulatory promoter that contain consensus sequences. Not all the consensus sequences shown are found in all promoters.

approximately +30 bp downstream of the start site in many promoters that also have Inr; the consensus sequence of DPE is R G A/T C G T G. All of these consensus sequences in the core promoter are recognized by transcription factors that bind to them and serve as a platform for the assembly of the basal transcription apparatus.

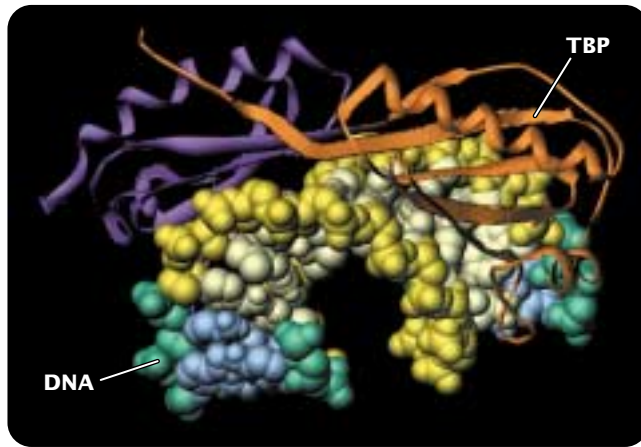
Assembly of the basal transcription apparatus The basic transcriptional machinery that binds to DNA at the start site is called the basal transcription apparatus and is required to initiate minimal levels of transcription. It consists of RNA polymerase, a series of general transcription factors, and a complex of proteins known as the mediator

(**FIGURE 13.17**). The general transcription factors include TFIIA, TFIIB, TFIID, TFII E, TFII F, and TFII H, in which TFII stands for transcription factor for RNA polymerase II and the letter designates the individual factor.

TFIID binds to the TATA box and positions the active site of RNA polymerase II so that it begins transcription at the correct place. TFIID consists of at least nine polypeptides. One of them is the **TATA-binding protein (TBP)**, which recognizes and binds to the TATA box on the DNA template. The TATA-binding protein binds to the minor groove and straddles the DNA as a molecular saddle (**FIGURE 13.18**), bending the DNA and partly unwinding it. Other proteins, called TBP-associated factors (TAFs),



13.17 Transcription is initiated at RNA polymerase II promoters when the TFIID transcription factor binds to the TATA box, followed by the binding of a preassembled holoenzyme containing general transcription factors, RNA polymerase II, and the mediator.



13.18 The TATA-binding protein (TBP) binds to the minor groove of DNA, straddling the double helix of DNA like a saddle.

combine with TBP to form the complete TFIID transcription factor.

The large holoenzyme consisting of RNA polymerase, additional transcription factors, and the mediator are thought to preassemble and bind as a unit to TFIID. The other transcription factors provide additional functions: TFIIA helps to stabilize the interaction between TBP and DNA, TFIIB plays a role in the selection of the start site, and TFIIH has helicase activity and unwinds the DNA during transcription. The mediator plays a role in communication between the basal transcription apparatus and transcriptional activator proteins (see next subsection).

Regulatory promoter The **regulatory promoter** is located immediately upstream of the core promoter. A variety of different consensus sequences may be found in the regulatory promoters, and they can be mixed and matched in different combinations (FIGURE 13.19). Transcriptional activator proteins bind to these sequences and, either directly or indirectly (through the mediation of coactivator proteins), make contact with the mediator in the basal transcription apparatus and affect the rate at which transcription is initiated. Some regulatory promoters also contain repressing sequences, which are bound by proteins that lower the rate of transcription through inhibitory interactions with the mediator.

Enhancers DNA sequences that increase the rate of transcription at distant genes are called **enhancers**. Furthermore, the precise position of an enhancer relative to a gene's transcriptional start site is not critical; most enhancers can stimulate any promoter in their vicinities, and an enhancer may be upstream or downstream from the affected gene or, in some cases, within an intron of the gene itself.

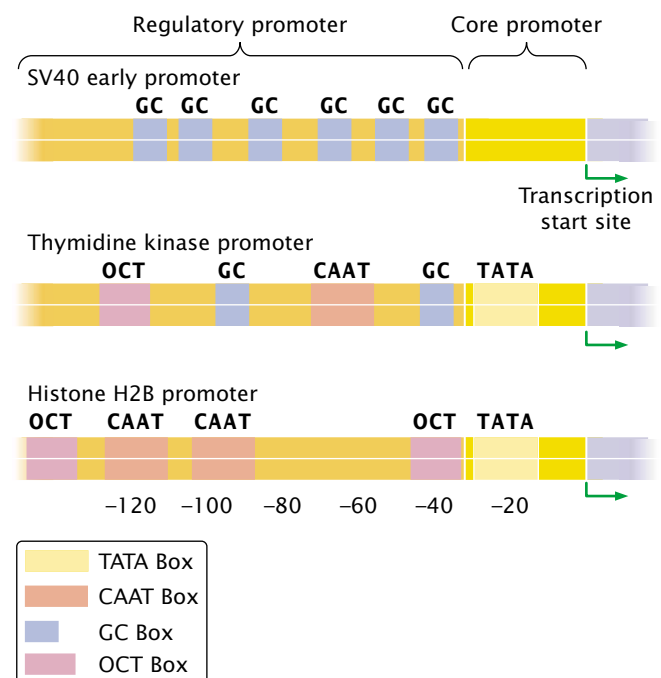
Enhancers also contain sequences that are recognized by transcriptional activator proteins. How does the binding

of a transcriptional activator protein to an enhancer affect the initiation of transcription at a gene thousands of nucleotides away? The answer is that the DNA between the enhancer and the promoter loops out, allowing the enhancer and the promoter to lie close to each other. Transcriptional activator proteins bound to the enhancer interact with proteins bound to the promoter and stimulate the transcription of the adjacent gene (see Figure 13.17). The looping of DNA between the enhancer and the promoter explains how the position of an enhancer can vary with regard to the start site—enhancers that are farther from the start site simply cause a longer length of DNA to loop out.

Sequences having many of the properties possessed by enhancers sometimes take part in *repressing* transcription instead of enhancing it; such sequences are called **silencers**. Although enhancers and silencers are characteristic of eukaryotic DNA, some enhancer-like sequences have been found in bacterial cells.

Concepts

General transcription factors assemble into the basal transcription apparatus, which binds to DNA near the start site and is necessary for transcription to take place at minimal levels. Additional proteins called transcriptional activators bind to other consensus sequences in promoters and enhancers, and affect the rate of transcription.



13.19 The consensus sequences in promoters of three eukaryotic genes illustrate the principle that different sequences can be mixed and matched to yield a functional promoter.

RNA Polymerase I Promoters

RNA polymerase I promoters have two functional sequences near the start site (◀ **FIGURE 13.20**). A core element surrounds the start site, extending from -45 to $+20$, and is needed to initiate transcription. An upstream control element extends from -180 to -107 and increases the efficiency of the core element. The DNA sequences of the core element and the upstream control element are rich in guanine and cytosine nucleotides and are similar in sequence.

RNA polymerase I requires two proteins to initiate transcription: SL1 and UBF. SL1 is made up of four subunits, one of which is TBP—the same protein that binds the TATA box in RNA polymerase II promoters (see Figure 13.17). The other protein, UBF, binds to both the core element and the upstream control element (see Figure 13.20) and enables SL1 to bind to the promoter. SL1 then recruits RNA polymerase I to the promoter. Thus, RNA polymerase I promoters function much as RNA polymerase II promoters do: transcription factors bind to a consensus sequence in the promoter and recruit RNA polymerase to the start site.

Concepts

RNA polymerase I promoters have two key components: (1) the core element, which surrounds the start site and is sufficient to initiate transcription, and (2) the upstream control sequence, which increases the efficiency of the core promoter.

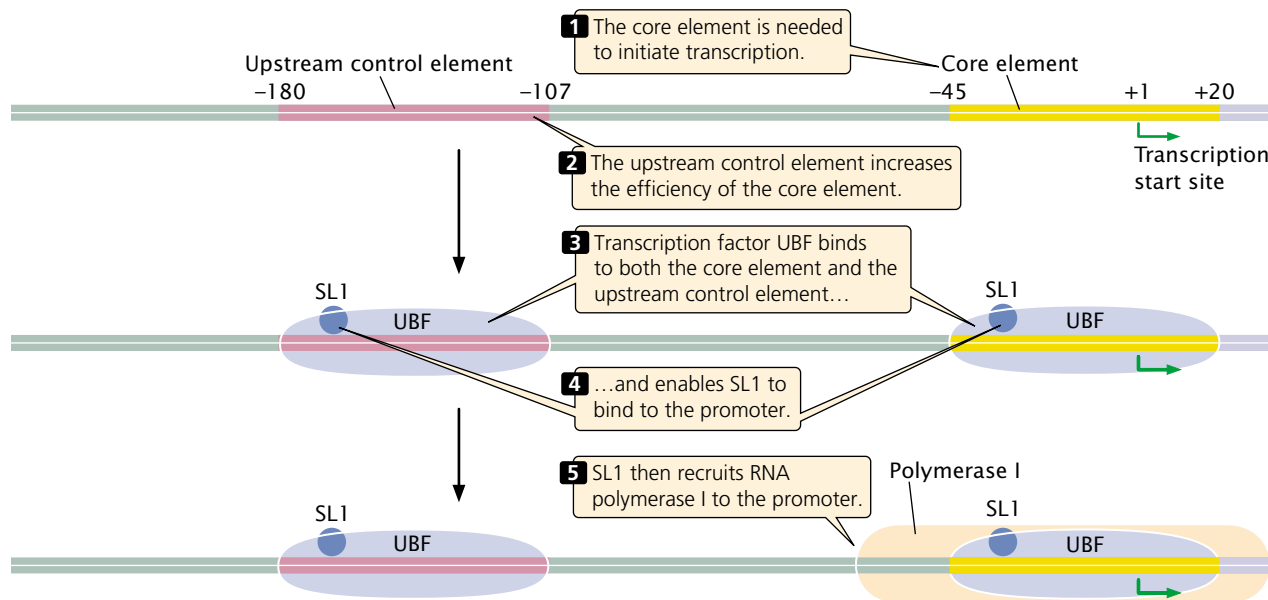
RNA Polymerase III Promoters

RNA polymerase III transcribes small rRNA, tRNAs, and some snRNAs (see Table 13.3) and recognizes several distinct types of promoters. The promoters of snRNA genes transcribed by RNA polymerase III contain several consensus sequences that are also found in some promoters transcribed by RNA polymerase II (◀ **FIGURE 13.21a**). These consensus sequences include the TATA box, which is recognized by a transcription factor that contains TBP. As in other types of eukaryotic promoters, TBP positions the active site of RNA polymerase over the start site for transcription in these promoters.

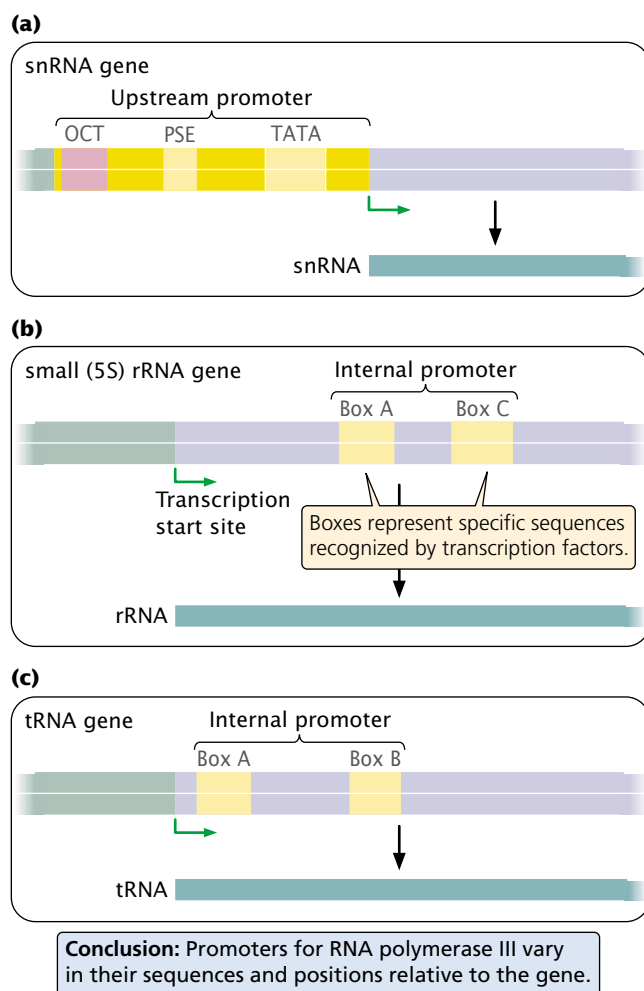
Promoters for small rRNA and tRNA genes, also transcribed by RNA polymerase III, contain **internal promoters** that are *downstream* of the start site and are actually transcribed into the RNA (◀ **FIGURE 13.21 b AND c**). These promoters contain critical sets of nucleotides (boxes A, B, and C) that also are recognized by transcription factors. One of these transcription factors includes TBP, which, again, positions the active site of RNA polymerase III over the upstream start site and ensures that the enzyme initiates transcription at the correct location. Additional transcription factors then bind to the DNA-binding factor and recruit RNA polymerase to the initiation complex.

Concepts

Some RNA polymerase III promoters are upstream of the start site and contain a TATA box. Others are internal, imbedded within the transcribed sequence downstream of the start site.



13.20 The basal transcription apparatus assembles at RNA polymerase I promoters.



13.21 RNA polymerase III recognizes several different types of promoters. OCT and PSE are consensus sequences that may also be present in RNA polymerase II promoters.

Connecting Concepts

Characteristics of Eukaryotic Promoters and Transcription Factors

Mastering the details of eukaryotic promoters and their associated transcription factors is a daunting task even for experienced researchers, never mind the beginning genetics student. Let's step back from the detail for a moment and identify some general principles of eukaryotic promoters and transcription factors:

1. Several types of DNA sequences take part in the initiation of transcription in eukaryotic cells. These sequences generally serve as the binding sites for proteins that interact with RNA polymerase and influence the initiation of transcription.

2. Some sequences that affect transcription, called promoters, are adjacent to or within the RNA coding region and are relatively fixed with regard to the start site of transcription. Promoters consist of a core promoter located adjacent to the gene and a regulatory promoter located farther upstream.
3. Other sequences, called enhancers, are distant from the gene and function independently of position and direction. Enhancers stimulate transcription.
4. General transcription factors bind to the core promoter near the start site and, with RNA polymerase, assemble into a basal transcription apparatus. The TATA-binding protein (TBP) is a critical transcription factor that positions the active site of RNA polymerase over the start site.
5. Transcriptional activator proteins bind to sequences in the regulatory promoter and enhancers and affect transcription by interacting with the basal transcription apparatus.
6. Proteins binding to enhancers interact with the basal transcription apparatus by causing the DNA between the promoter and the enhancer to loop out, bringing the enhancer into close proximity to the promoter.

Evolutionary Relationships and the TATA-Binding Protein

Some 2 billion to 3 billion years ago, life diverged into three lines of evolutionary descent: the eubacteria, the archaea, and the eukaryotes (see Chapter 2). Although eubacteria and archaea are superficially similar—both are unicellular and lack a nucleus—the results of studies of their DNA sequences and other biochemical properties indicate that they are distantly related. The evolutionary distinction between archaea, eubacteria, and eukaryotes is clear: however, did eukaryotes first diverge from an ancestral prokaryote, with the later separation of prokaryotes into eubacteria and archaea, or did the archaea and the eubacteria split first, with the eukaryotes later evolving from one of these groups?

Studies of transcription in eubacteria, archaea, and eukaryotes have yielded important findings about the evolutionary relationships of these organisms. The results of studies in 1994 demonstrated that archaea possess a TATA-binding protein, a critical transcription factor in all three of the eukaryotic polymerases. The binding of TBP to DNA is the first step in the assembly of the eukaryotic transcription apparatus. In earlier studies, TATA-like sequences were found in eukaryotic cells, but no such sequences have been found in eubacteria. TBP binds the TATA box in archaea with the help of another transcription factor, TFIIB, which is also found in eukaryotes but not in eubacteria.

Together these findings indicate that transcription, one of the most basic of life processes, has strong similarities in

eukaryotes and archaea, suggesting that these two groups are more closely related to each other than either is to the eubacteria. This conclusion is supported by other data, including those obtained from a comparison of gene sequences.

Termination

The termination of transcription in eukaryotic genes is less well understood than in bacterial genes. The three eukaryotic RNA polymerases use different mechanisms for termination. RNA polymerase I requires a termination factor, like the rho factor utilized in termination of some bacterial genes. Unlike rho, which binds to the newly transcribed RNA molecule, the termination factor for RNA polymerase I binds to a DNA sequence downstream of the termination site.

RNA polymerase III ends transcription after transcribing a terminator sequence that produces a string of Us in the RNA molecule, like that produced by the rho-independent terminators of bacteria. Unlike rho-independent terminators in bacterial cells, RNA polymerase III does not require that a hairpin structure precede the string of Us.

In many of the genes transcribed by RNA polymerase II, transcription can end at multiple sites located within a span of hundreds or thousands of base pairs. As we will see in Chapter 14, the transcription of these genes continues well beyond the coding sequence necessary to produce the mRNA. After transcription, the 3' end of pre-mRNA is cleaved at a specific site, designated by a consensus sequence, producing the mature mRNA. Research findings suggest that termination is coupled to cleavage, which is carried out by a cleavage complex that probably associates with the RNA polymerase. This complex may suppress termination until the consensus sequence that marks the cleavage site is encountered. The 3' end of the pre-mRNA is then cleaved by the complex, and transcription is terminated downstream.

Concepts

The different eukaryotic RNA polymerases utilize different mechanisms of termination.



Connecting Concepts Across Chapters



This chapter has focused on the process of transcription, during which an RNA molecule that is complementary and antiparallel to a DNA template is synthesized. Transcription is the first step in gene expression, the transfer of genetic information from genotype to phenotype and, as we will see in Chapter 16, is an important point at which gene expression is regulated. Transcription is similar in many respects to replication—it utilizes a DNA template, takes place in the 5' → 3' direction, synthesizes a molecule that is antiparallel and complementary to the template, and utilizes nucleoside triphosphates as substrates. But there are important differences as well: only one strand is typically transcribed, each gene is transcribed separately, and the process is subject to numerous regulatory mechanisms.

This chapter has provided important links to topics discussed in several other chapters of the book. Transcription is the first step in the molecular transfer of genetic information from the genotype to the phenotype and is therefore the starting point for discussions of RNA processing in Chapter 14 and translation in Chapter 15. Knowledge of the details of transcription is also essential for understanding gene regulation (Chapter 16), because transcription is an important point at which the expression of many genes is controlled. Additionally, because transcription factors play an important role in some types of cancer, the information in this chapter will be useful when we consider the molecular basis of cancer in Chapter 21.

CONCEPTS SUMMARY

- RNA molecules can function as biological catalysts and may have been the first carriers of genetic information.
- RNA is a polymer, consisting of nucleotides joined together by phosphodiester bonds. Each RNA nucleotide consists of a ribose sugar, a phosphate, and a base. RNA contains the base uracil; it is usually single stranded, which allows it to form secondary structures.
- Ribosomal RNA is a component of the ribosome, messenger RNA carries coding instructions for proteins, and transfer RNA helps incorporate the amino acids into a polypeptide chain. Other RNA molecules found in eukaryotic cells include pre-mRNAs, the precursor of mRNA; snRNAs, which function in the processing of pre-mRNAs; snoRNAs, which process rRNA; and scRNAs, which exist in the cytoplasm.
- The template for RNA synthesis is single-stranded DNA. In transcription, RNA synthesis is complementary and antiparallel to the DNA template strand.
- A transcription unit consists of a promoter, an RNA-coding region, and a terminator.
- The substrates for RNA synthesis are ribonucleoside triphosphates. In transcription, two phosphates are cleaved from a ribonucleoside triphosphate and the remaining phosphate takes part in a phosphodiester bond with the 3'-OH group at the growing end of the RNA molecule.
- RNA polymerase in bacterial cells consists of a core enzyme, which catalyzes the addition of nucleotides to an RNA molecule, and other subunits, which join the core enzyme to provide additional functions. The sigma factor controls the

binding of the core enzyme to the promoter; rho and NusA assist in the termination of transcription.

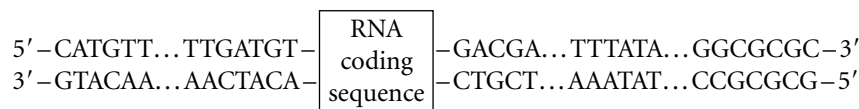
- Eukaryotic cells contain three RNA polymerases: RNA polymerase I, which transcribes rRNA; RNA polymerase II, which transcribes pre-mRNA and some snRNAs; and RNA polymerase III, which transcribes tRNAs, small rRNA, and some snRNAs.
- The process of transcription consists of three stages: initiation, elongation, and termination.
- Promoters are recognized by the transcription apparatus and are required for transcription. They contain short consensus sequences imbedded within longer stretches of DNA.
- Transcription begins at the start site, which is determined by the consensus sequences. A short stretch of DNA is unwound near the start site, RNA is synthesized from a single strand of DNA as a template, and the DNA is rewound at the lagging end of the transcription bubble.
- Terminators consist of sequences within the RNA coding region; RNA synthesis ceases after the terminator has been transcribed. Bacterial cells have two types of terminators: rho-independent terminators, which RNA polymerase can recognize by itself, and rho-dependent terminators, which RNA polymerase can recognize only with the help of the rho protein.
- In eukaryotic cells, DNA is complexed to histone proteins, which interfere with the binding of transcription factors and RNA polymerase. Chromatin may be modified by acetylation, chromatin-remodeling proteins, and other factors, allowing transcription factors and RNA polymerase to bind to the DNA.
- Two classes of sequences affect transcription in eukaryotic cells: promoters, which are adjacent to genes, and enhancers, which may be distant to the genes that they affect.
- A promoter for RNA polymerase II consists of a core promoter, which is required for minimal levels of transcription, and a regulatory promoter, which affects the rate of transcription.
- General transcription factors bind to the core promoter and are part of the basal transcription apparatus. Transcriptional activator proteins bind to sequences in regulatory promoters and enhancers and interact with the basal transcription apparatus at the core promoter.
- The three types of RNA polymerase in eukaryotic cells recognize different types of promoters, all of which have consensus sequences that serve as binding sites for transcription factors.
- The three RNA polymerases found in eukaryotic cells use different mechanisms of termination.

IMPORTANT TERMS

ribozyme (p. 354)	small cytoplasmic RNA (scRNA) (p. 356)	RNA polymerase II (p. 361)	basal transcription apparatus (p. 367)
ribosomal RNA (rRNA) (p. 355)	template strand (p. 357)	RNA polymerase III (p. 361)	transcriptional activator protein (p. 367)
messenger RNA (mRNA) (p. 355)	nontemplate strand (p. 357)	consensus sequence (p. 362)	core promoter (p. 367)
pre-messenger RNA (pre-mRNA) (p. 355)	transcription unit (p. 359)	–10 consensus sequence (Pribnow box) (p. 362)	TATA box (p. 367)
transfer RNA (tRNA) (p. 355)	promoter (p. 359)	–35 consensus sequence (p. 362)	TATA-binding protein (TBP) (p. 368)
small nuclear RNA (snRNA) (p. 355)	RNA-coding region (p. 359)	upstream element (p. 363)	regulatory promoter (p. 369)
small nuclear ribonucleoprotein (snRNP) (p. 355)	terminator (p. 359)	rho-dependent terminator (p. 365)	enhancer (p. 369)
small nucleolar RNA (snoRNA) (p. 355)	ribonucleoside triphosphate (rNTP) (p. 359)	rho factor (p. 365)	silencer (p. 369)
	RNA polymerase (p. 360)	rho-independent terminator (p. 365)	internal promoter (p. 370)
	core enzyme (p. 360)	general transcription factor (p. 367)	
	sigma factor (p. 360)		
	holoenzyme (p. 360)		
	RNA polymerase I (p. 361)		

Worked Problems

1. The following diagram represents a sequence of nucleotides surrounding an RNA coding sequence.



(a) Is the RNA coding sequence likely to be from a bacterial cell or from a eukaryotic cell? How can you tell?

(b) Which DNA strand will serve as the template strand during transcription of the RNA coding sequence?

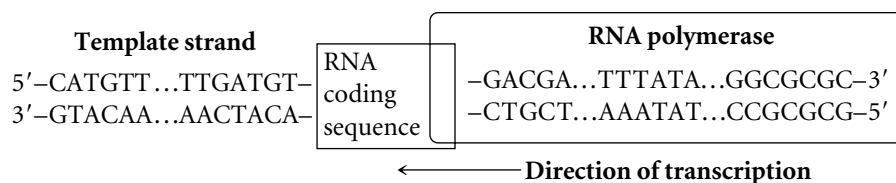
• Solution

(a) Bacterial and eukaryotic cells use the same DNA bases (A, T, G, and C); so the bases themselves provide no clue to the origin of the sequence. The RNA coding sequence must have a promoter, and bacterial and eukaryotic cells do differ in the consensus sequences found in their promoters; so we should examine the sequences for the presence of familiar consensus sequences. On the bottom strand to the right of the RNA coding sequence we find AAATAT, which, written in the conventional manner (5' on the left), is 5'–TATAAA–3'. This sequence is the TATA box found in most eukaryotic promoters. However, the sequence is also quite similar to the –10 consensus sequence (5'–TATAAT–3') found in bacterial promoters.

Farther to the right on the bottom strand, we also see 5'–GCGCGCC–3', which is the TFIIB recognition element

(BRE) in eukaryotic RNA polymerase II promoters. No similar consensus sequence is found in bacterial promoters; so we can be fairly certain that this sequence is a eukaryotic promoter and RNA coding sequence.

(b) The TATA box and BRE of RNA polymerase II promoters are upstream of the RNA coding sequences; so RNA polymerase must bind to these sequences and then proceed downstream, transcribing the RNA coding sequence. Thus RNA polymerase must proceed from right (upstream) to left (downstream). The RNA molecule is always synthesized in the 5'→3' direction and is antiparallel to the DNA template strand; so the template strand must be read 3'→5'. If the enzyme proceeds from right to left and reads the template in the 3'→5' direction, the upper strand must be the template, as shown here.



2. Suppose that a consensus sequence in the regulatory promoter of a gene that encodes enzyme A were deleted. Which of the following effects would result from this deletion?

- (a) Enzyme A would have a different amino acid sequence.
- (b) The mRNA for enzyme A would be abnormally short.
- (c) Enzyme A would be missing some amino acids.
- (d) The mRNA for enzyme A would be transcribed but not translated.
- (e) The amount of mRNA transcribed would be affected.

Explain your reasoning.

• Solution

The correct answer is part *e*. The regulatory protein contains binding sites for transcriptional activator proteins. These sequences are not part of the RNA coding sequence for enzyme A; so the mutation would have no effect on the length or amino acid sequence of the enzyme, eliminating answers *a*, *b*, and *c*. The TATA box is the binding site for the basal transcription apparatus. Transcriptional activator proteins bind to the regulatory promoter and affect the amount of transcription that takes place through interactions with the basal transcription apparatus at the core promoter.

COMPREHENSION QUESTIONS

- * 1. Draw an RNA nucleotide and a DNA nucleotide, highlighting the differences. How is the structure of RNA similar to that of DNA? How is it different?
2. What are the major classes of cellular RNA? Where would you expect to find each class of RNA within eukaryotic cells?
- * 3. What parts of DNA make up a transcription unit? Draw and label a typical transcription unit in a bacterial cell.
4. What is the substrate for RNA synthesis? How is this substrate modified and joined together to produce an RNA molecule?
5. Describe the structure of bacterial RNA polymerase.
- * 6. Give the names of the three RNA polymerases found in eukaryotic cells and the types of RNA that they transcribe.
7. What are the four basic stages of transcription? Describe what happens at each stage.
- * 8. Draw and label a typical bacterial promoter. Include any common consensus sequences.
9. What are the two basic types of terminators found in bacterial cells? Describe the structure of each.
10. How is the process of transcription in eukaryotic cells different from that in bacterial cells?
- * 11. How are promoters and enhancers similar? How are they different?
12. How can an enhancer affect the transcription of a gene that is thousands of nucleotides away?
13. Compare the roles of general transcription factors and transcriptional activator proteins.
14. What are some of the common consensus sequences found in RNA polymerase II promoters?
- * 15. What protein associated with a transcription factor is common to all eukaryotic promoters? What is its function in transcription?
- * 16. Compare and contrast transcription and replication. How are these processes similar and how are they different?

APPLICATION QUESTIONS AND PROBLEMS

17. Write the consensus sequence for the following set of nucleotide sequences.

AGGAGTT
AGCTATT
TGCAATA
ACGAAAA
TCCTAAT
TGCAATT

- *18. List at least five properties that DNA polymerases and RNA polymerases have in common. List at least three differences.
19. RNA molecules have *three* phosphates at the 5' end, but DNA molecules never do. Explain this difference.
20. An RNA molecule has the following percentages of bases: A = 23%, U = 42%, C = 21%, and G = 14%.
- (a) Is this RNA single stranded or double stranded? How can you tell?
- (b) What would be the percentages of bases in the template strand of the DNA that contains the gene for this RNA?
- *21. The following diagram represents DNA that is part of the RNA-coding sequence of a transcription unit. The bottom strand is the template strand. Give the sequence found on the RNA molecule transcribed from this DNA and label the 5' and 3' ends of the RNA.

5'–ATAGGCGATGCCA–3'
3'–TATCCGCTACGGT–5' ← Template strand

22. The following sequence of nucleotides is found in a single-stranded DNA template:

ATTGCCAGATCATCCCAATAGAT

Assume that RNA polymerase proceeds along this template from left to right.

- (a) Which end of the DNA template is 5' and which end is 3'?
- (b) Give the sequence and label the 5' and 3' ends of the RNA copied from this template.
23. Write out a hypothetical sequence of bases that might be found in the first 20 nucleotides of a promoter of a bacterial gene. Include both strands of DNA and label the 5' and 3' ends of both strands. Be sure to include the start site for transcription and any consensus sequences found in the promoter.
24. The following diagram represents a transcription unit in a hypothetical DNA molecule.

5' . . . TTGACA . . . TATAAT . . . 3'
3' . . . AACTGT . . . ATATTA . . . 5'

- (a) On the basis of the information given, is this DNA from a bacterium or from a eukaryotic organism?

(b) If this DNA molecule is transcribed, which strand will be the template strand and which will be the nontemplate strand?

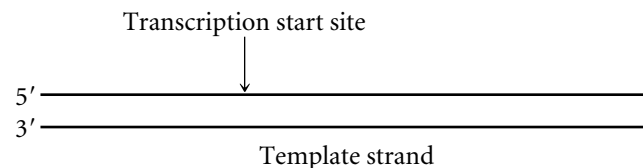
(c) Where, approximately, will the start site of transcription be?

- *25. What would be the most likely effect of a mutation at the following locations in *E. coli* gene?

(a) –8 (c) –20
(b) –35 (d) Start site

26. A strain of bacteria possesses a temperature-sensitive mutation in the gene that encodes the sigma factor. At elevated temperatures, the mutant bacteria produce a sigma factor that is unable to bind to RNA polymerase. What effect will this mutation have on the process of transcription when the bacteria are raised at elevated temperatures?

- *27. The following diagram represents a transcription unit on a DNA molecule.



- (a) Assume that this DNA molecule is from a bacterial cell. Draw in the approximate location of the promoter and terminator for this transcription unit.
- (b) Assume that this DNA molecule is from a eukaryotic cell. Draw in the approximate location of an RNA polymerase II promoter.
- (c) Assume that this DNA molecule is from a eukaryotic cell. Draw in the approximate location of an internal RNA polymerase III promoter.
28. The following DNA nucleotides are found near the end of a bacterial transcription unit. Find the terminator in this sequence.

3'–AGCATACAGCAGACCGTTGGTCTGAAAAAAGCATACA–5'

- (a) Mark the point at which transcription will terminate.
- (b) Is this terminator rho independent or rho dependent?
- (c) Draw a diagram of the RNA that will be transcribed from this DNA, including its nucleotide sequence and any secondary structures that form.
- *29. A strain of bacteria possesses a temperature-sensitive mutation in the gene that encodes the rho subunit of RNA polymerase. At high temperatures, rho is not functional. When these bacteria are raised at elevated temperatures, which of the following effects would you expect to see?
- (a) Transcription does not take place.
- (b) All RNA molecules are shorter than normal.

- (c) All RNA molecules are longer than normal.
- (d) Some RNA molecules are longer than normal.
- (e) RNA is copied from both DNA strands.

Explain your reasoning for accepting or rejecting each of these five options.

30. Suppose that the string of As following the inverted repeat in a rho-independent terminator were deleted, but the inverted repeat were left intact. How would this deletion affect termination? What would happen when RNA polymerase reached this region?

CHALLENGE QUESTIONS

33. Enhancers are sequences that affect the initiation of the transcription of genes that are hundreds or thousands of nucleotides away. Enhancer-binding proteins usually interact directly with transcription factors at promoters by causing the intervening DNA to loop out. An enhancer of bacteriophage T4 does not function by looping of the DNA (D. R. Herendeen, et al., 1992, *Science* 256:1298–1303). Propose some additional mechanisms (other than DNA looping) by which this enhancer might affect transcription at a gene thousands of nucleotides away.

*34. The location of the TATA box in two species of yeast, *Saccharomyces pombe* and *S. cerevisiae*, differs dramatically. The TATA box of *S. pombe* is about 30 nucleotides upstream of the start site, similar to the location for most other eukaryotic cells. However, the TATA box of *S. cerevisiae* can be as many as 120 nucleotides upstream of the start site. To understand how the TATA box functions in these two species, a series of experiments was conducted to determine which components of the transcription apparatus of these two species could be interchanged. In these experiments, different components of the transcription apparatus were switched in *S. pombe* and *S. cerevisiae*, and the effects of the switch on the level of RNA synthesis and on the start point of transcription were observed. TFIID from *S. pombe* could be used in *S. cerevisiae* cells and vice versa, without any effect on the transcription start site in either cell type. Switching TFIIB, TFIIE, or RNA polymerase did alter the level of transcription. However, the following pairs of components could be exchanged without affecting transcription: TFIIE together with TFIIH; and TFIIB together with RNA polymerase. The exchange of TFIIE–TFIIH did not alter the start point, but the exchange of TFIIB–RNA polymerase did shift it. (Y. Li, P. M. Flanagan, H. Tschochner, and R. D. Kornberg, 1994, *Science* 263:805–807.)

On the basis of these results, what conclusions can you draw about how the different components of the transcription apparatus interact and which components are

- *31. Through genetic engineering, a geneticist mutates the gene that encodes TBP in cultured human cells. This mutation destroys the ability of TBP to bind to the TATA box. Predict the effect of this mutation on cells that possess it.
- 32. Elaborate repair mechanisms are associated with replication to prevent permanent mutations in DNA, yet no similar repair is associated with transcription. Can you think of a reason for these differences in replication and transcription? (Hint: Think about the relative effects of a permanent mutation in a DNA molecule compared with one in an RNA molecule.)

responsible for setting the start site? Propose a mechanism for the determination of the start site in eukaryotic RNA polymerase II promoters.

- 35. The relation between chromatin structure and transcription activity has been the focus of recent research. In one set of experiments, the level of in vitro transcription of a *Drosophila* gene by RNA polymerase II was studied with the use of DNA and various combinations of histone proteins. First, the level of transcription was measured for naked DNA with no associated histone proteins. Then, the level of transcription was measured after nucleosome octamers (without H1) were added to the DNA. The addition of the octamers caused the level of transcription to drop by 50%. When both nucleosome octamers and H1 proteins were added to the DNA, transcription was greatly repressed, dropping to less than 1% of that obtained with naked DNA (see the table below).

GAL4-VP16 is a protein that binds to the DNA of certain eukaryotic genes. When GAL4-VP16 is added to DNA, the level of RNA polymerase II transcription is greatly elevated. Even in the presence of the H1 protein, GAL4-VP16 stimulates high levels of transcription.

Propose a mechanism for how the H1 protein represses transcription and how GAL4-VP16 overcomes this repression. Explain how your proposed mechanism would produce the results obtained in these experiments.

Treatment	Relative amount of transcription
Naked DNA	100
DNA + octamers	50
DNA + octamers + H1	<1
DNA + GAL4-VP16	1000
DNA + octamers + GAL4-VP16	1000
DNA + octamers + H1 + GAL4-VP16	1000

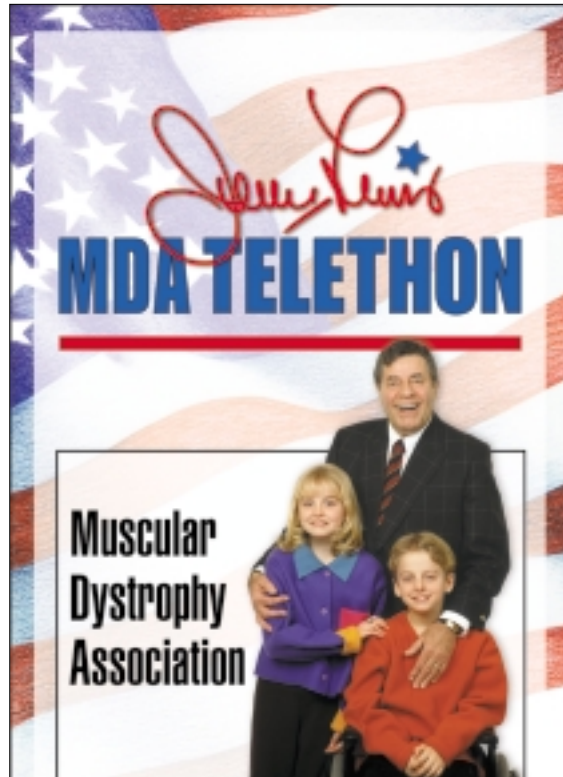
(Based on experiments reported in an article by G. E. Croston et al., 1991, *Science* 251:643–649.)

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14

RNA Molecules and RNA Processing



This is Chapter 14 Opener photo legend to position here. (Courtesy of the Muscular Dystrophy Association.)

- The Immense Dystrophin Gene
- Gene Structure
 - Gene Organization
 - Introns
 - The Concept of the Gene Revisited
- Messenger RNA
 - The Structure of Messenger RNA
 - Pre-mRNA Processing
 - The Addition of the 5' Cap
 - The Addition of the Poly(A) Tail
 - RNA Splicing
 - Alternative Processing Pathways
 - RNA Editing
- Transfer RNA
 - The Structure of Transfer RNA
 - Transfer RNA Gene Structure and Processing
- Ribosomal RNA
 - The Structure of the Ribosome
 - Ribosomal RNA Gene Structure and Processing

The Immense Dystrophin Gene

The most common and devastating of the muscular dystrophies is Duchenne muscular dystrophy, a fatal disease that strikes nearly 1 in 3500 males. At birth, affected boys appear normal. The first symptom is mild muscle weakness appearing between 3 and 5 years of age: the child stumbles frequently, has difficulty climbing stairs, and is unable to rise from a sitting position. In time, the arm and leg muscles become progressively weaker. By age 11, those affected are usually confined to a wheel chair and, by age 20, most persons with Duchenne muscular dystrophy have died. At present, there is no cure for the disease.

Duchenne muscular dystrophy was first recognized in 1852, and the disease was fully described in 1861 by Benjamin A. Duchenne, a Paris physician. Even before Mendel's laws were discovered, physicians noticed its X-linked

pattern of inheritance, remarking that the disease developed almost exclusively in males and seemed to be inherited through unaffected mothers. In spite of this early recognition of its hereditary basis, the biochemical cause of Duchenne muscular dystrophy remained a mystery until 1987.

In 1985, Louis Kunkel and his colleagues at Harvard Medical School observed a boy with Duchenne muscular dystrophy whose X chromosome had a visible deletion on the short arm. Reasoning that this boy's disease was caused by the absence of a gene within the deletion, they recognized that the deletion pointed to the location on the X chromosome of the gene responsible for Duchenne muscular dystrophy. Kunkel and his colleagues located and cloned the piece of DNA responsible for the disease. Shortly thereafter, the sequence of the gene was determined, and the protein that it encodes was isolated. This large protein, called dystrophin, consists of nearly 4000 amino acids and

is an integral component of muscle cells. Persons with Duchenne muscular dystrophy lack functional dystrophin.

The dystrophin gene is among the most remarkable of all genes yet examined. It's *huge*, encompassing more than 2 million nucleotides of DNA. However, only about 12,000 of its nucleotides encode its amino acids. Why is the dystrophin gene so large? What are all those other nucleotides doing?

The unusual properties of the dystrophin gene make sense only in the context of RNA processing—the alteration of RNA after it has been transcribed. Dystrophin mRNA, like many eukaryotic RNAs, undergoes extensive processing after transcription, including the removal of large sections that are not required for translation. Chapter 13 focused on transcription—the process of RNA synthesis. In this chapter, we will examine the function and processing of RNA.

We begin by taking a careful look at the nature of the gene. Next, we examine messenger RNA, its structure, and how it is modified in eukaryotes after transcription. We'll also see how, through alternative pathways of RNA modification, one gene can produce several different proteins. Then, we turn to transfer RNA, the adapter molecule that forms the interface between amino acids and mRNA in protein synthesis. Finally, we examine ribosomal RNA, the structure and organization of rRNA genes, and how rRNAs are processed.

As we explore the world of RNA and its role in gene function, we will see evidence of two important characteristics of this nucleic acid. First, RNA is extremely versatile, both structurally and biochemically. It can assume a number of different secondary structures, which provide the basis for its functional diversity. Second, RNA processing and function frequently include interactions between two or more RNA molecules.

www.whfreeman.com/pierce More information about Duchenne muscular dystrophy

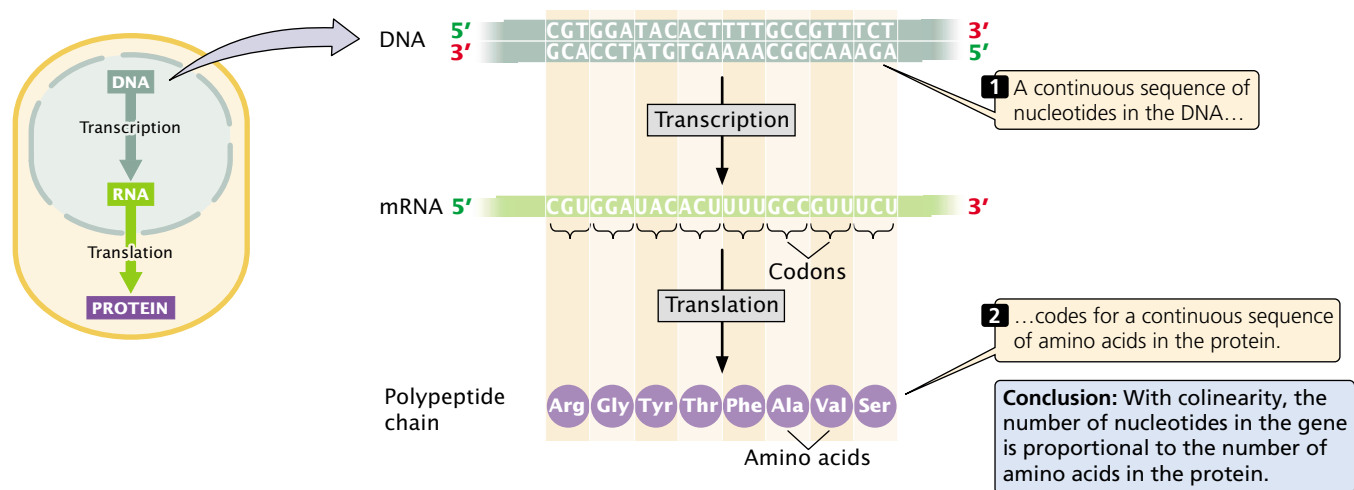
Gene Structure

What is a gene? In Chapter 3, it was noted that the definition of *gene* would appear to change as we explored different aspects of heredity. A gene was defined as an inherited factor that determined a trait. This definition may have seemed vague, because it says nothing about what a gene is, only what it does. Nevertheless, this definition was appropriate for our purposes at the time, because our focus was on how genes influence the inheritance of traits. It wasn't necessary to consider the physical nature of the gene in learning the rules of inheritance.

Knowing something about the chemical structure of DNA and the process of transcription enables us to be more precise about what a gene is. Chapter 10 described how genetic information is encoded in the base sequence of DNA; so a gene consists of a set of DNA nucleotides. But how many nucleotides are encompassed in a gene, and how is the information in these nucleotides organized? In 1902, Archibald Garrod suggested, correctly, that genes code for proteins (see p. 000). Proteins are made of amino acids; so a gene contains the nucleotides that specify the amino acids of a protein. We could, then, define a gene as a set of nucleotides that specifies the amino acid sequence of a protein, which indeed was for many years the working definition of a gene. As geneticists learned more about the structure of genes, however, it became clear that this concept of a gene was an oversimplification.

Gene Organization

Early work on gene structure was carried out largely through the examination of mutations in bacteria and viruses. This research led Francis Crick in 1958 to propose that genes and proteins are **colinear**—that there is a direct correspondence between the nucleotide sequence of DNA and the amino acid sequence of a protein (FIGURE 14.1).



14.1 The concept of colinearity suggests that a continuous sequence of nucleotides in DNA encodes a continuous sequence of amino acids in a protein.

The concept of colinearity suggests that the number of nucleotides in a gene should be proportional to the number of amino acids in the protein encoded by that gene. In a general sense, this concept is true for genes found in bacterial cells and many viruses, although these genes are slightly longer than expected if colinearity is strictly applied (the mRNAs encoded by the genes contain sequences at their ends that do not specify amino acids). At first, eukaryotic genes and proteins also were generally assumed to be colinear, but there were hints that eukaryotic gene structure was fundamentally different. Eukaryotic cells contain far more DNA than is required to encode proteins (the so-called C-value paradox; see Chapter 11). Furthermore, many large RNA molecules observed in the nucleus were absent from the cytoplasm, suggesting that nuclear RNAs undergo some type of change before they are exported to the cytoplasm.

Most geneticists were nevertheless surprised by the announcement in the 1970s that four coding sequences in a gene from a eukaryotic virus were interrupted by nucleotides that did not specify amino acids. This discovery was made when the viral DNA was hybridized with the mRNA transcribed from it, and the hybridized structure was examined with the use of an electron microscope (FIGURE 14.2). The DNA was clearly much longer than the mRNA, because regions of DNA looped out from the hybridized molecules. These regions contained nucleotides in the DNA that were absent from the coding nucleotides in the mRNA. Many other examples of interrupted genes were subsequently discovered; it quickly became apparent that most eukaryotic genes comprise stretches of coding and noncoding nucleotides.

Concepts

When a continuous sequence of nucleotides in DNA encodes a continuous sequence of amino acids in a protein, the two are said to be colinear. The discovery of coding and noncoding regions within eukaryotic genes shows that not all genes are colinear with the proteins that they encode.

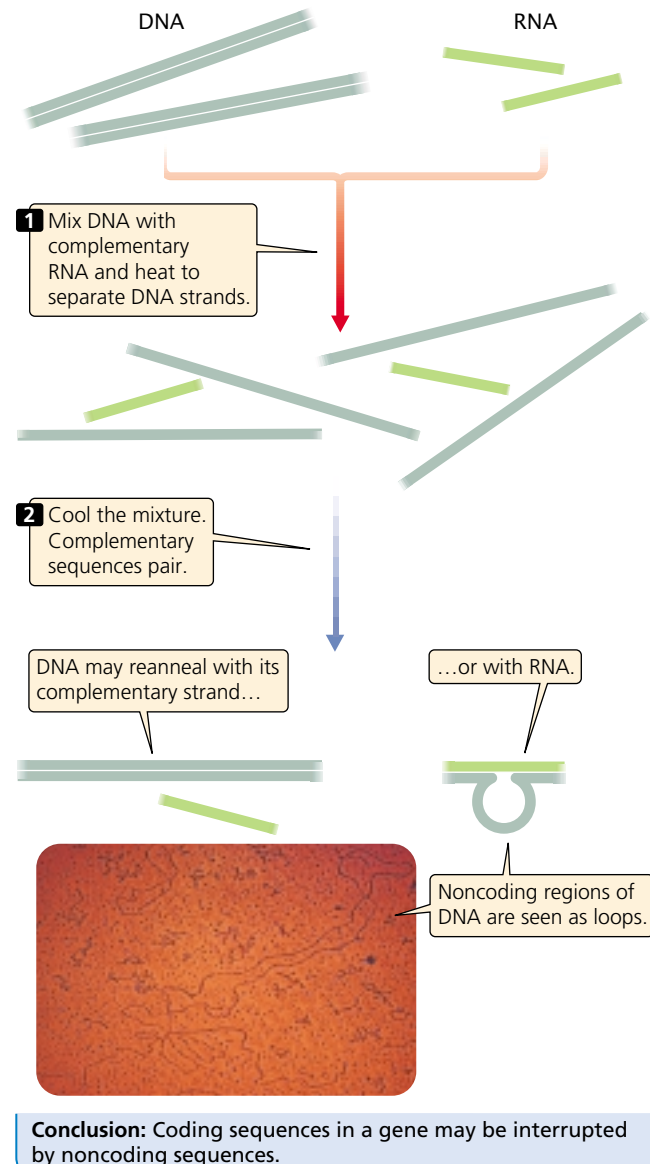
Introns

Many eukaryotic genes contain coding regions called **exons** and noncoding regions called intervening sequences or **introns**. For example, the ovalbumin gene has eight exons and seven introns; the gene for cytochrome *b* has five exons and four introns (FIGURE 14.3). All the introns and the exons are initially transcribed into RNA but, after transcription, the introns are removed by splicing and the exons are joined to yield the mature RNA.

Introns are common in eukaryotic genes but are rare in bacterial genes. For a number of years after their discovery,

Experiment

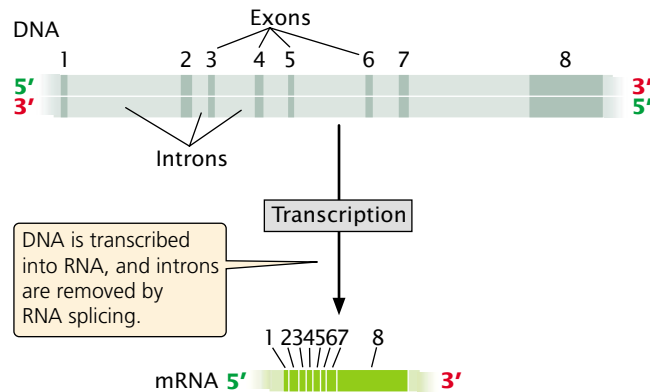
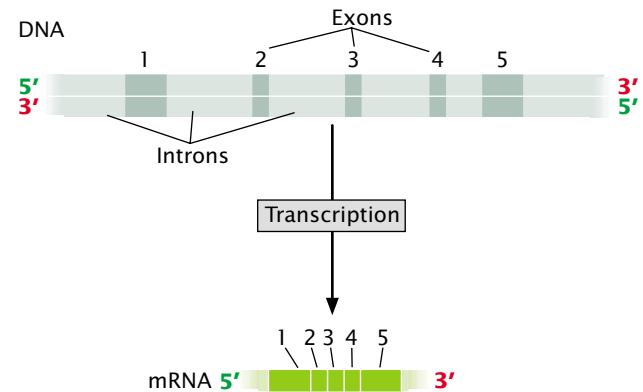
Question: Is the coding sequence in a gene always continuous?



Conclusion: Coding sequences in a gene may be interrupted by noncoding sequences.

14.2 The noncolinearity of eukaryotic genes was discovered by hybridizing DNA and mRNA. (Electromicrograph from O.L. Miller, B.R. Beatty, D.W. Fawcett/Visuals Unlimited.)

introns were thought to be entirely absent from prokaryotic genomes, but they have now been observed in archaea, bacteriophages, and even some eubacteria. Introns are present in mitochondrial and chloroplast genes, as well as nuclear genes. In eukaryotic genomes, the size and number of introns appear to be directly related to increasing organismal complexity. Yeast genes contain only a few short introns; *Drosophila* introns are longer and more numerous; and most vertebrate genes are interrupted by long introns. All

Ovalbumin gene**Cytochrome *b* gene**

14.3 The coding sequences of many eukaryotic genes are disrupted by noncoding introns.

classes of genes—those that code for rRNA, tRNA, and proteins—may contain introns. The number and size of introns vary widely: some eukaryotic genes have no introns, whereas others may have more than 60; intron length varies from fewer than 200 nucleotides to more than 50,000. Introns tend to be longer than exons, and most eukaryotic genes contain more noncoding nucleotides than coding nucleotides. Finally, most introns do not encode proteins (an intron of one gene is not usually an exon for another), although there are exceptions.

There are four major types of introns (Table 14.1). **Group I introns**, found in some rRNA genes, are self-splicing—they can catalyze their own removal. **Group II introns** are present in some protein-encoding genes of mitochondria, chloroplasts, and a few eubacteria; they also are self-splicing, but their mechanism of splicing differs from that of the group I introns. **Nuclear pre-mRNA introns** are the best studied; they include introns located in the protein-encoding genes of the nucleus. The splicing mechanism by

which these introns are removed is similar to that of the group II introns, but nuclear introns are not self-splicing; their removal requires snRNAs (discussed later) and a number of proteins. **Transfer RNA introns**, found in tRNA genes, utilize yet another splicing mechanism that relies on enzymes to cut and reseal the RNA. In addition to these major groups, there are several other types of introns.

We'll take a detailed look at the chemistry and mechanics of RNA splicing later in the chapter. For now, we should keep in mind two general characteristics of the splicing process: (1) the splicing of all pre-mRNA introns takes place in the nucleus and is probably required for RNA to move to the cytoplasm; and (2) the order of exons in DNA is usually maintained in the spliced RNA—the coding sequences of a gene may be split up, but they are not usually jumbled up.

Table 14.1 Major types of introns

Type of Intron	Location	Type of Splicing
Group I	Some rRNA genes	Self-splicing
Group II	Protein-encoding genes in mitochondria and chloroplasts	Self-splicing
Nuclear pre-mRNA	Protein-encoding genes in the nucleus	Spliceosomal
tRNA	tRNA genes	Enzymatic

Note: There are also several types of minor introns, including group III introns, twintrons, and archaeal introns.

Concepts

Many eukaryotic genes contain exons and introns, both of which are transcribed into RNA, but introns are later removed by RNA processing. The number and size of introns vary from gene to gene; they are common in many eukaryotic genes but uncommon in bacterial genes.

The Concept of the Gene Revisited

How does the presence of introns affect our concept of a gene? It no longer seems appropriate to define a gene as a sequence of nucleotides that codes for amino acids in a protein, because this definition excludes from the gene those sequences in introns that don't specify amino acids. This definition also excludes nucleotides that code for the 5' and 3' ends of a mRNA molecule, which are required for translation but do not code for amino acids. And defining a gene

in these terms also excludes sequences that encode rRNA, tRNA, and other RNAs that do not encode proteins. In view of our current understanding of DNA structure and function, we need a more satisfactory definition of gene.

Many geneticists have broadened the concept of a gene to include all sequences in the DNA that are transcribed into a single RNA molecule. Defined in this way, a gene includes all exons, introns, and those sequences at the beginning and end of the RNA that are not translated into a protein. This definition also includes DNA sequences that code for rRNAs, tRNAs, and other types of non-messenger RNA. Many geneticists have expanded the definition of a gene even further, to include the entire transcription unit—the promoter, the RNA coding sequence, and the terminator.

Concepts

The discovery of introns forced a reevaluation of the definition of the gene. Today, a gene is often defined as a DNA sequence that codes for an RNA molecule.

Messenger RNA

As soon as DNA was identified as the source of genetic information, it became clear that DNA could not directly encode proteins. In eukaryotic cells, DNA resides in the nucleus, yet most protein synthesis takes place in the cytoplasm. Geneticists recognized that an additional molecule must take part in the transfer of genetic information.

The results of studies of bacteriophage infection conducted in the late 1950s and early 1960s pointed to RNA as a likely candidate for this transport function. Bacteriophages inject their DNA into bacterial cells, where the DNA is replicated, and large amounts of phage protein are produced on the bacterial ribosomes. As early as 1953, Alfred Hershey discovered a type of RNA that was synthesized rapidly after bacteriophage infection. Findings from later studies showed that the bacteriophage T2 produced short-lived RNA having a nucleotide composition similar to that of phage DNA but quite different from that of the bacterial RNA. These observations were consistent with the idea that RNA was copied from DNA and that this RNA then directed the synthesis of proteins.

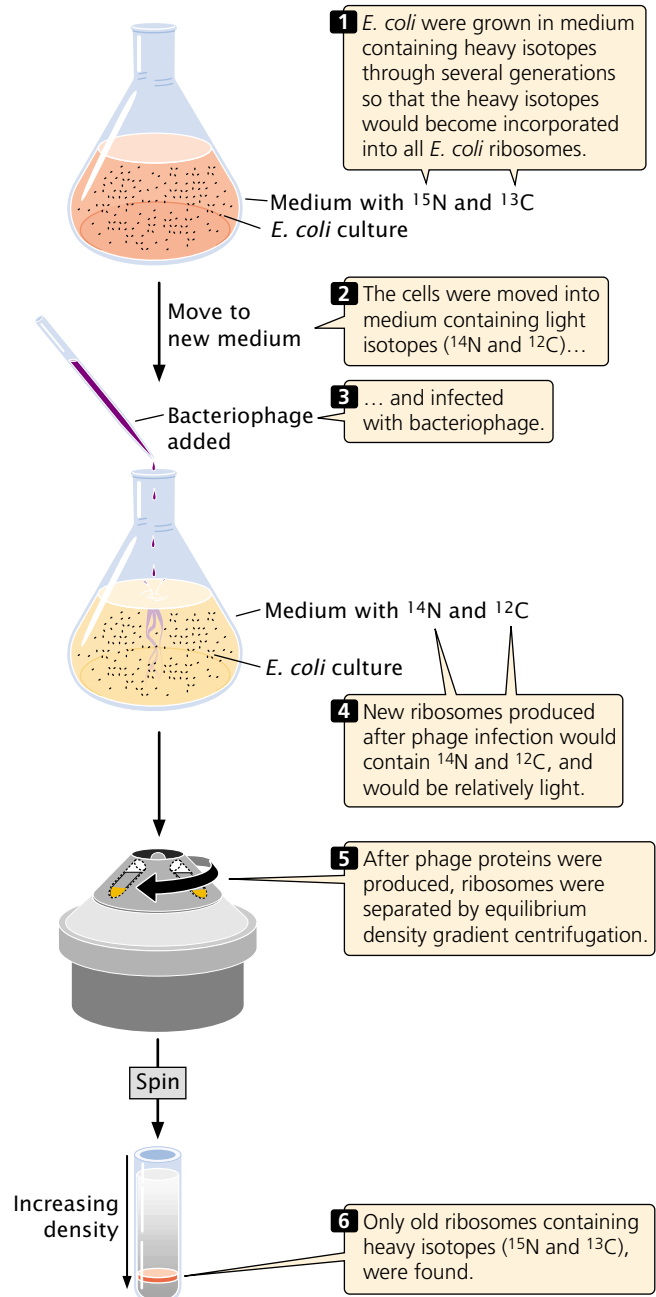
At the time, ribosomes were known to be *somehow* implicated in protein synthesis, and much of the RNA in a cell was known to be in the form of ribosomes. Each gene was thought to direct the synthesis of a special type of ribosome in the nucleus, which then moved to the cytoplasm and produced a specific protein. Using equilibrium density-gradient centrifugation (see Figure 10.2), Sydney Brenner, François Jacob, and Matthew Meselson demonstrated in 1961 that new ribosomes are *not* produced during the burst of protein synthesis that accompanies phage infection

(**FIGURE 14.4**). The genetic information needed to produce new phage proteins was not carried by the ribosomes.

In a related experiment, François Gros and his colleagues infected *E. coli* cells with bacteriophages while radioactively

Experiment

Question: Do ribosomes carry genetic information?



Conclusion: Ribosomes are not produced during phage reproduction.

14.4 Brenner, Jacob, and Meselson demonstrated that ribosomes do not carry genetic information.

labeled (“hot”) uracil was added to the medium (which would become incorporated into newly produced phage RNA). After a few minutes, they transferred the cells to a medium that contained unlabeled (“cold”) uracil. This type of experiment is called a pulse–chase experiment: the cells are exposed to a brief pulse of label, which is then “chased” by cold, unlabeled precursor. Pulse–chase experiments make it possible to follow, by tracking the presence of the radioactivity, products of short-term biochemical events, such as RNA synthesis immediately following phage infection.

Gros and his coworkers found that the newly produced phage RNA was short lived, lasting only a few minutes, and was associated with ribosomes but was distinct from them. They concluded that newly synthesized, short-lived RNA carries the genetic information for protein structure to the ribosome. The term *messenger RNA* was coined for this carrier.

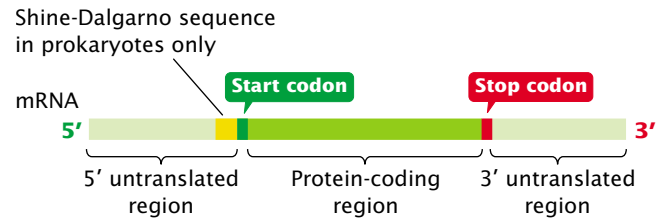
The Structure of Messenger RNA

Messenger RNA functions as the template for protein synthesis; it carries genetic information from DNA to a ribosome and helps to assemble amino acids in their correct order. Each amino acid in a protein is specified by a set of three nucleotides in the mRNA, called a **codon**. Both prokaryotic and eukaryotic mRNAs contain three primary regions (FIGURE 14.5). The **5′ untranslated region** (5′ UTR; sometimes call the leader) is a sequence of nucleotides at the 5′ end of the mRNA that does not code for the amino acid sequence of a protein. In bacterial mRNA, this region contains a consensus sequence called the **Shine-Dalgarno sequence**, which serves as the ribosome-binding site during translation; it is found approximately seven nucleotides upstream of the first codon translated into an amino acid (called the start codon). Eukaryotic mRNA has no equivalent consensus sequence in its 5′ untranslated region. In eukaryotic cells, ribosomes bind to a modified 5′ end of mRNA, as discussed later in the chapter.

The next section of mRNA is the **protein-coding region**, which comprises the codons that specify the amino acid sequence of the protein. The protein-coding region begins with a start codon and ends with a stop codon. The last region of mRNA is the **3′ untranslated region** (3′ UTR), a sequence of nucleotides at the 3′ end of mRNA that is not translated into protein. The 3′ untranslated region affects the stability of mRNA and the translation of the mRNA protein-coding sequence.

Concepts

Messenger RNA molecules contain three main regions: a 5′ untranslated region, a protein-coding region, and a 3′ untranslated region. The 5′ and 3′ untranslated regions do not code for the amino acids of a protein.



14.5 Three primary regions of mature mRNA are the 5′ untranslated region, the protein-coding region, and the 3′ untranslated region.

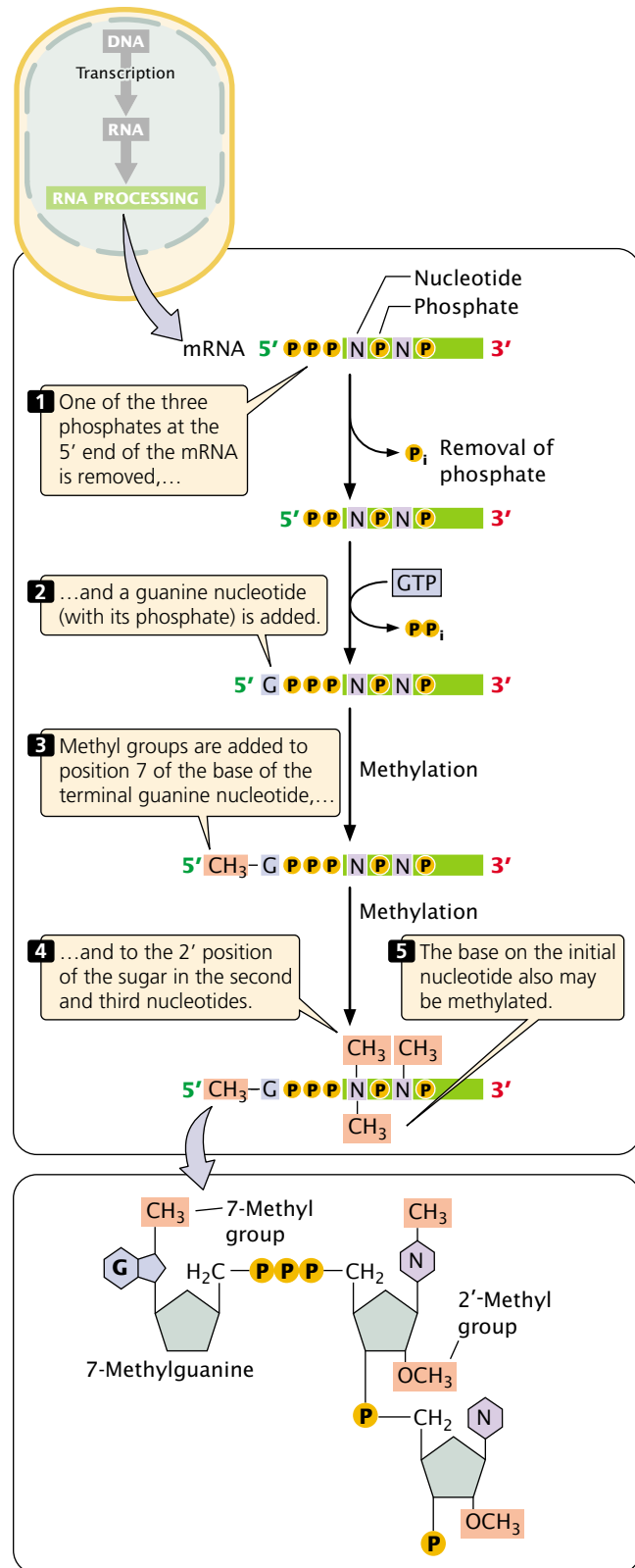
Pre-mRNA Processing

In bacterial cells, transcription and translation take place simultaneously; while the 3′ end of an mRNA is undergoing transcription, ribosomes attach to the Shine-Dalgarno sequence near the 5′ end and begin translation. Because transcription and translation are coupled, there is little opportunity for the bacterial mRNA to be modified before protein synthesis. In contrast, transcription and translation are separated in both time and space in eukaryotic cells. Transcription takes place in the nucleus, whereas most translation takes place in the cytoplasm; this separation provides an opportunity for eukaryotic RNA to be modified before it is translated. Indeed, eukaryotic mRNA is extensively altered after transcription. Changes are made to the 5′ end, the 3′ end, and the protein-coding section of the RNA molecule. The initial transcript of protein-encoding genes of eukaryotic cells is called pre-mRNA, whereas the mature, processed transcript is mRNA. We will reserve the term mRNA for RNA molecules that have been completely processed and are ready to undergo translation.

The Addition of the 5′ Cap

Almost all eukaryotic pre-mRNAs are modified at their 5′ ends by the addition of a structure called a **5′ cap**. This capping consists of the addition of an extra nucleotide at the 5′ end of the mRNA and methylation by the addition of a methyl group (CH_3) to the base in the newly added nucleotide and to the 2′-OH group of the sugar of one or more nucleotides at the 5′ end (FIGURE 14.6). Capping takes place rapidly after the initiation of transcription and, as will be discussed in more depth in Chapter 15, the 5′ cap functions in the initiation of translation. Capping proteins recognize the cap and attach to it; a ribosome then binds to these proteins and moves downstream along the mRNA until the start codon is reached and translation begins. The presence of a 5′ cap also increases the stability of mRNA and influences the removal of introns.

In the discussion of transcription in Chapter 13, it was noted that three phosphates are present at the 5′ end of all RNA molecules, because phosphates are not cleaved



14.6 Most eukaryotic mRNAs have a 5' cap.

The cap consists of a nucleotide with 7-methyl guanine attached to the pre-mRNA by a unique 5'–5' bond (shown in detail in the bottom box). The cap is added shortly after the initiation of transcription. A methyl group is added to position 7 of the guanine base of the newly added (now the terminal) nucleotide and to the 2' position of each sugar of the next two nucleotides.

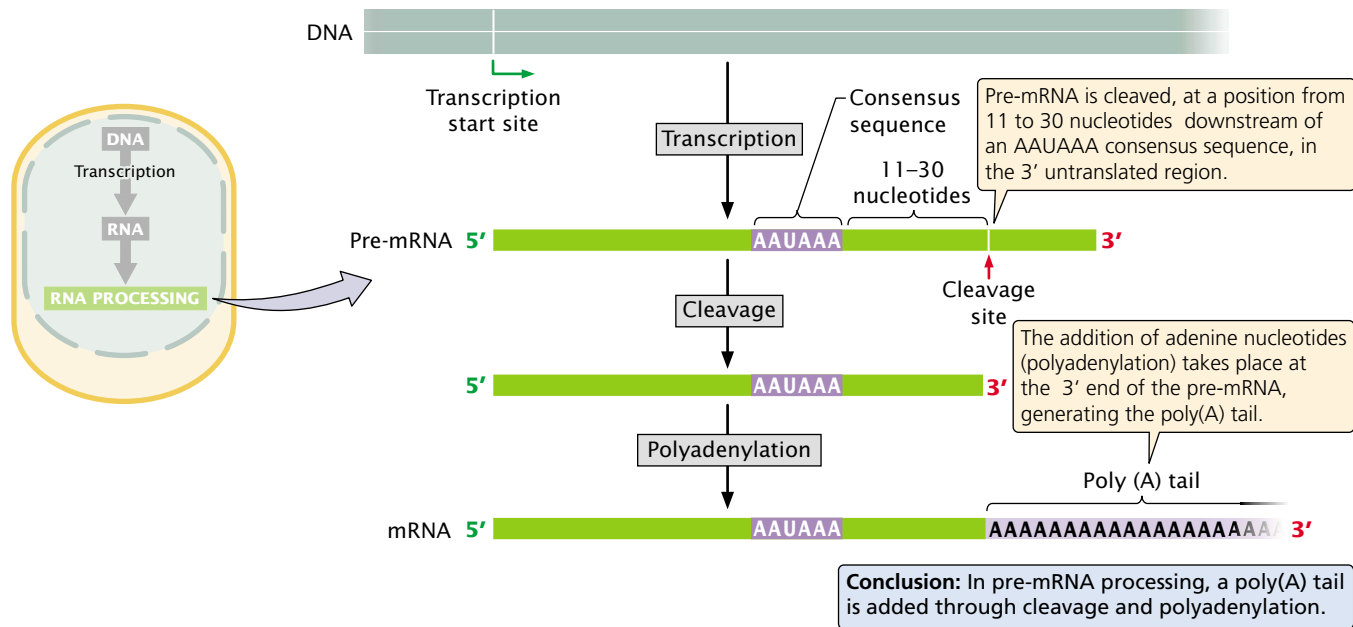
from the first ribonucleoside triphosphate in the transcription reaction. The 5' end of pre-mRNA can be represented as 5'–pppNpNpN..., in which the letter N represents a ribonucleotide and p represents a phosphate. Shortly after the initiation of transcription, one of these phosphates is removed and a guanine nucleotide is added (see Figure 14.6). This guanine nucleotide is attached to the pre-mRNA by a unique 5'–5' bond, which is quite different from the usual 5'–3' phosphodiester bond that joins all the other nucleotides in RNA. One or more methyl groups are then added to the 5' end; the first of these methyl groups is added to position 7 of the base of the terminal guanine nucleotide, making the base 7-methylguanine. Next, a methyl group may be added to the 2' position of the sugar in the second and third nucleotides, as shown in Figure 14.6. Rarely, additional methyl groups may be attached to the bases of the second and third nucleotides of the pre-mRNA.

The Addition of the Poly(A) Tail

Most mature eukaryotic mRNAs have from 50 to 250 adenine nucleotides at the 3' end (a **poly(A) tail**). These nucleotides are not encoded in the DNA but are added after transcription (see Figure 14.7) in a process termed polyadenylation. Many eukaryotic genes transcribed by RNA polymerase II are transcribed well beyond the end of the coding sequence (see Chapter 13); the extra material at the 3' end is then cleaved and the poly(A) tail is added. For some pre-mRNA molecules, more than 1000 nucleotides may be cleaved from the 3' end.

Processing of the 3' end of pre-mRNA requires sequences both upstream and downstream of the cleavage site (see Figure 14.8a). The consensus sequence AAUAAA is usually from 11 to 30 nucleotides upstream of the cleavage site (see Figure 14.7) and determines the point at which cleavage will take place. A sequence rich in Us (or Gs and Us) is typically downstream of the cleavage site.

In mammals, 3' cleavage and the addition of the poly(A) tail requires a complex consisting of several proteins: cleavage and polyadenylation specificity factor (CPSF); cleavage stimulation factor (CstF); at least two cleavage factors (CFI and CFII); and polyadenylate polymerase (PAP). CPSF binds to the upstream AAUAAA consensus sequence, whereas CstF binds to the downstream sequence (see Figure 14.8b).



14.7 Most eukaryotic mRNAs have a 3' poly(A) tail.

The pre-mRNA is cleaved, and CstF and the cleavage factors leave the complex; the cleaved 3' end of the pre-mRNA is then degraded (FIGURE 14.8c). CFSF and PAP remain bound to the pre-mRNA and carry out polyadenylation (FIGURE 14.8d). After the addition of approximately 10 adenine nucleotides, a poly(A)-binding protein (PABII) attaches to the poly(A) tail and increases the rate of polyadenylation (FIGURE 14.8e). As more of the tail is synthesized, additional molecules of PABII attach to it (FIGURE 14.8f).

The poly(A) tail confers stability on many mRNAs, increasing the time during which the mRNA remains intact and available for translation before it is degraded by cellular enzymes. The stability conferred by the poly(A) tail is dependent on the proteins that attached to the tail.

Eukaryotic mRNAs that lack a poly(A) tail depend on a different mechanism for 3' cleavage that requires the formation of a hairpin structure in the pre-mRNA and a small ribonucleoprotein particle (snRNP) called U7 (FIGURE 14.9). U7 contains an snRNA with nucleotides that are complementary to a sequence on the pre-mRNA just downstream of the cleavage site, and U7 most likely binds to this sequence. A hairpin-binding protein binds to the hairpin structure and stabilizes the binding of U7 to the complementary sequence on the pre-mRNA.

a ribosome, increases the stability of the mRNA, and may affect the removal of introns. Processing at the 3' end includes cleavage downstream of an AAUAAA consensus sequence and the addition of a poly(A) tail.

RNA Splicing

The other major type of modification that takes place in eukaryotic pre-mRNA is the removal of introns by **RNA splicing**. This occurs in the nucleus following transcription but before the RNA moves to the cytoplasm.

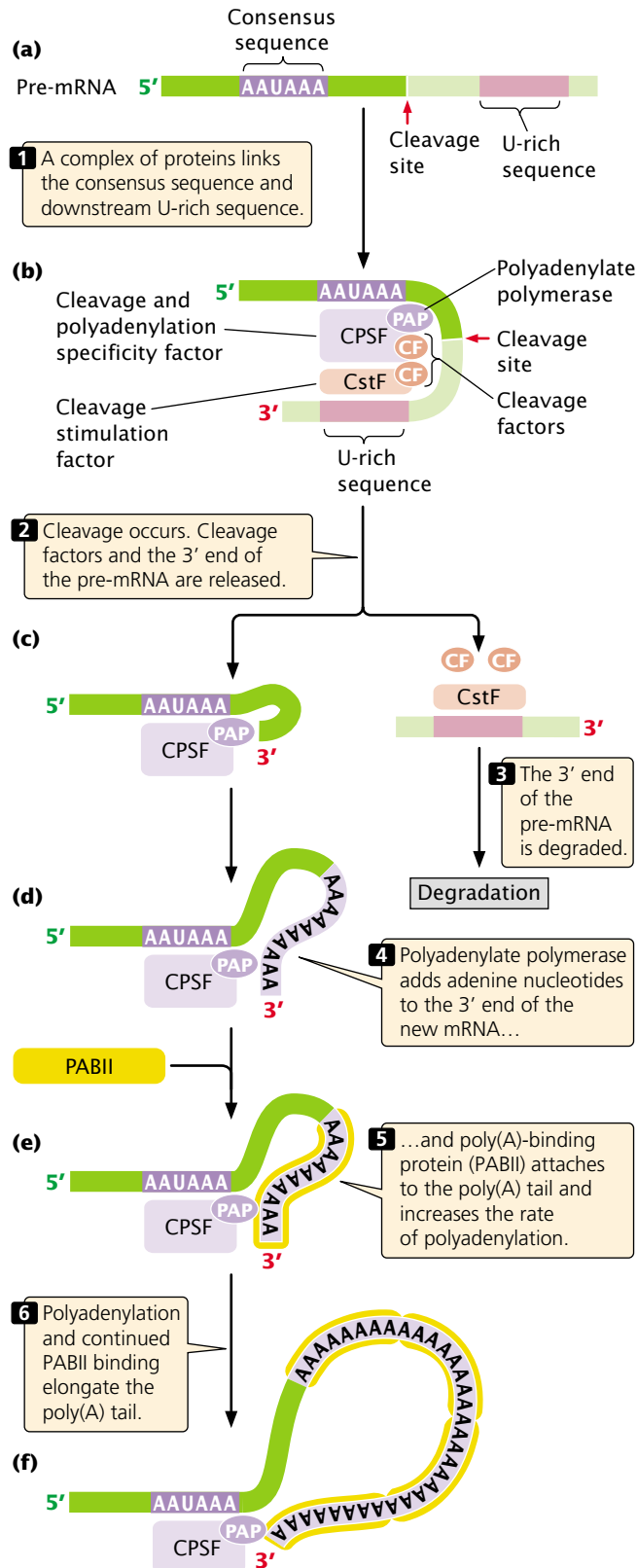
Consensus sequences and the spliceosome Splicing requires the presence of three sequences in the intron. One end of the intron is referred to as the **5' splice site**, and the other end is the **3' splice site** (FIGURE 14.10); these splice sites possess short consensus sequences. Most introns in pre-mRNA begin with GU and end with AG, suggesting that these sequences play a crucial role in splicing. Changing a single nucleotide at either of these sites does indeed prevent splicing. A few introns in pre-mRNA begin with AU and end with AC. These introns are spliced by a process that is similar to that seen in GU...AG introns but utilizes a different set of splicing factors. This discussion will focus on splicing of the more common GU...AG introns.

The third sequence important for splicing is at the so-called **branch point**, which is an adenine nucleotide that lies from 18 to 40 nucleotides upstream of the 3' splice site (see Figure 14.10). The sequence surrounding the branch point does not have a strong consensus but usually takes the

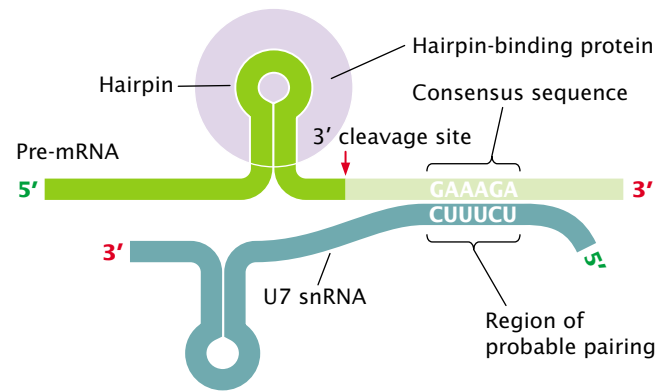
Concepts

Eukaryotic pre-mRNAs are processed at their 5' and 3' ends. A cap, consisting of a modified nucleotide and several methyl groups, is added to the 5' end. The cap facilitates the binding of





14.8 Processing of the 3' end of pre-mRNA requires a consensus sequence and several factors.



14.9 Eukaryotic mRNAs that lack a poly(A) tail depend on a different mechanism for 3' cleavage.

Cleavage requires the presence of U7 snRNA, which has bases complementary to a consensus sequence downstream of the 3' cleavage site. Cleavage depends on the formation of a hairpin structure near the 3' end of the pre-mRNA; base pairing probably takes place between the complementary regions of the pre-mRNA and the U7 snRNA.

form YNYYRAY (Y is any pyrimidine, N is any base, R is any purine, and A is adenine). The deletion or mutation of the adenine nucleotide at the branch point prevents splicing.

Splicing takes place within a large complex called the **spliceosome**, which consists of several RNA molecules and many proteins. The RNA components are small nuclear RNAs (Chapter 13); these snRNAs associate with proteins to form small ribonucleoprotein particles. Each snRNP contains a single snRNA molecule and multiple proteins. The spliceosome is composed of five snRNPs, named for the snRNAs that they contain (U1, U2, U4, U5, and U6), and some proteins not associated with an snRNA.

Concepts

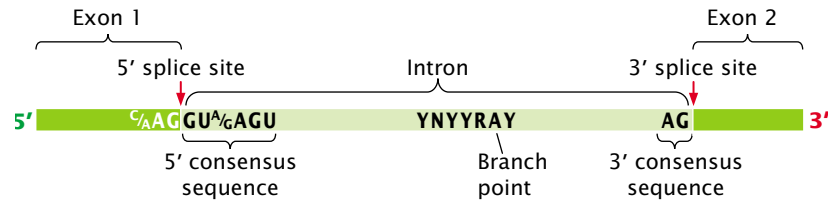
Introns in nuclear genes contain three consensus sequences critical to splicing: a 5' splice site, a 3' splice site, and a branch point. Splicing of pre-mRNA takes place within a large complex called the spliceosome, which consists of snRNAs and proteins.

The process of splicing To illustrate the process of RNA splicing, we'll first consider the chemical reactions that take place. Then we'll see how these splicing reactions constitute a set of coordinated processes within the context of the spliceosome.

Before splicing takes place, an upstream exon (exon 1) and a downstream exon (exon 2) are separated by an intron (FIGURE 14.11). Pre-mRNA is spliced in two distinct steps.

14.10 Splicing of pre-mRNA

requires consensus sequences. In the consensus sequence surrounding the branch point (YNYRAY) Y is any pyrimidine, R is any purine, A is adenine, and N is any base.



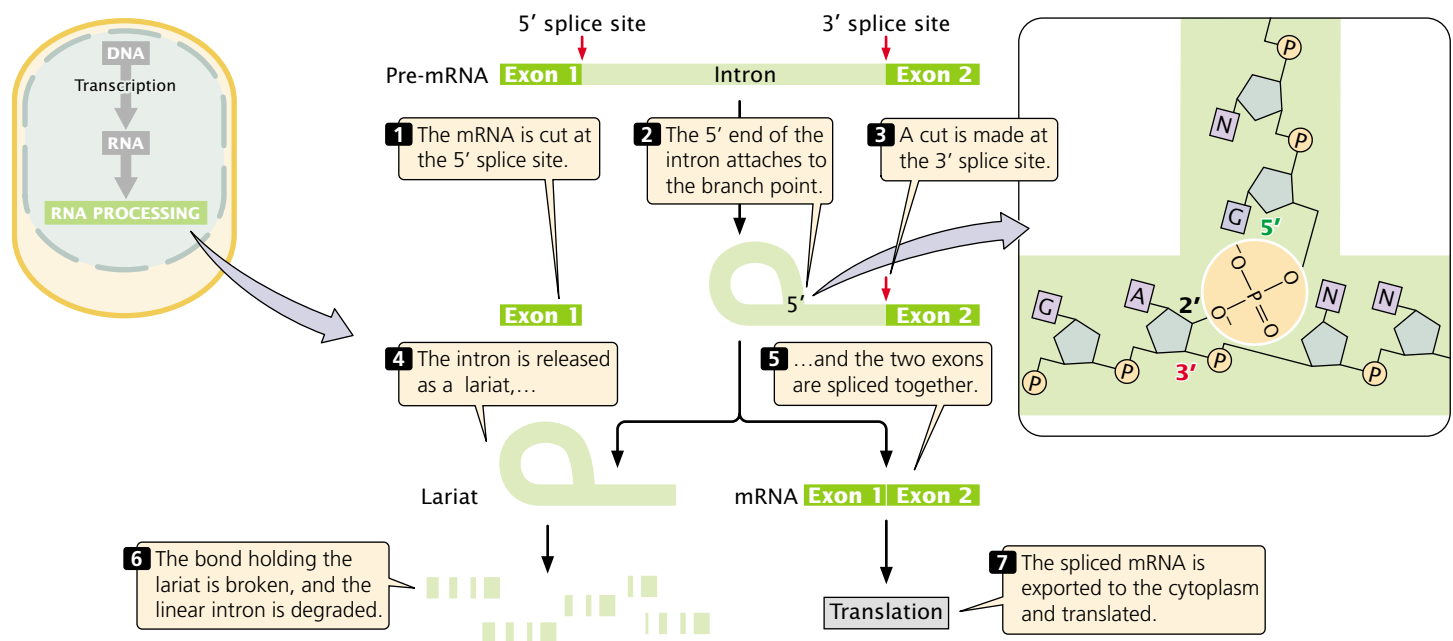
Conclusion: Critical consensus sequences are present at the 5' splice site, the branch point, and the 3' splice site.

In the first step, the pre-mRNA is cut at the 5' splice site. This cut frees exon 1 from the intron, and the 5' end of the intron attaches to the branch point; that is, the intron folds back on itself, forming a structure called a **lariat**. The guanine nucleotide in the consensus sequence at the 5' splice site bonds with the adenine nucleotide at the branch point. This bonding is accomplished through **transesterification**, a chemical reaction in which the OH group on the 2'-carbon atom of the adenine nucleotide at the branch point attacks the 5' phosphodiester bond of the guanine nucleotide at the 5' splice site, cleaving it and forming a new 5'-2' phosphodiester bond between the guanine and adenine nucleotides.

In the second step of RNA splicing, a cut is made at the 3' splice site and, simultaneously, the 3' end of exon

1 becomes covalently attached (spliced) to the 5' end of exon 2. This bond also forms through a transesterification reaction, in which the 3'-OH group attached to the end of exon 1 attacks the phosphodiester bond at the 3' splice site, cleaving it and forming a new phosphodiester bond between the 3' end of exon 1 and the 5' end of exon 2; the intron is released as a lariat. The intron becomes linear when the bond breaks at the branch point and is then rapidly degraded by nuclear enzymes. The mature mRNA consisting of the exons spliced together is exported to the cytoplasm where it is translated.

Although splicing is illustrated in Figure 14.11 as a two-step process, the reactions are in fact coordinated within the spliceosome. A key feature of the spliceosome is a series of interactions between the mRNA and snRNAs and



14.11 The splicing of nuclear introns requires a two-step process.

First, cleavage takes place at the 5' splice site, and a lariat is formed by the attachment of the 5' end of the intron to the branch point. Second, cleavage takes place at the 3' splice site, and two exons are spliced together.

Table 14.2 RNA–RNA interactions in pre-mRNA splicing

Interaction	Function
U1 with 5' splice site	U1 attaches to 5' end of intron; commits intron to splicing; no direct role in splicing
U2 with branch point	Positions 5' end of intron near branch point for lariat formation
U2 with U6	Holds 5' end of intron near branch point
U6 with 5' splice site	Positions 5' end of intron near branch point
U5 with 3' end of first exon	Anchors first exon to spliceosome subsequent to cleavage; juxtaposes two ends of exon for splicing
U5 with 3' end of one exon and 5' end of the other	Juxtaposes two ends of exon for splicing
U4 with U6	Delivers U6 to intron; no direct role in splicing

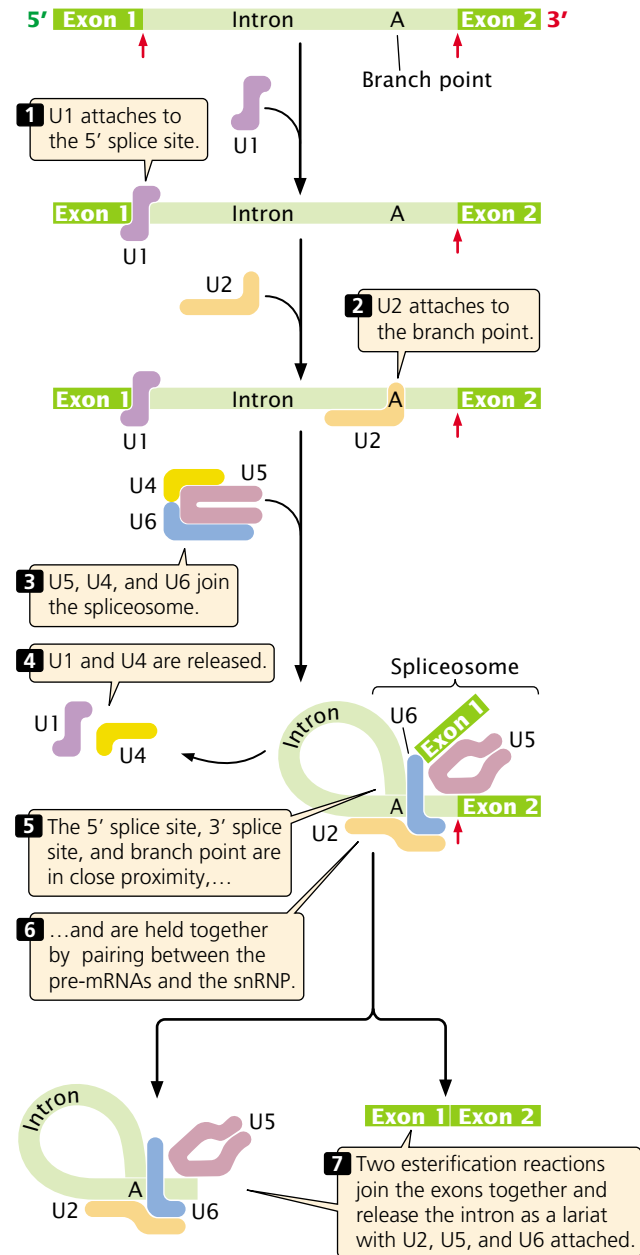
between different snRNAs (summarized in Table 14.2). These interactions depend on complementary base pairing between the different RNA molecules and bring the essential components of the pre-mRNA transcript and the spliceosome close together, which makes splicing possible.

The spliceosome is assembled on the pre-mRNA transcript in a step-by-step fashion (FIGURE 14.12). First, snRNP U1 attaches to the 5' splice site, and then U2 attaches to the branch point. A complex consisting of U5 and U4–U6 (which form a single snRNP) joins the spliceosome. At this point, the intron loops over and the 5' splice site is brought close to the branch point. U1 and U4 dissociate from the spliceosome. The 5' splice site, 3' splice site, and branch point are in close proximity, held together by the spliceosome. The two transesterification reactions take place, joining the two exons together and releasing the intron as a lariat.

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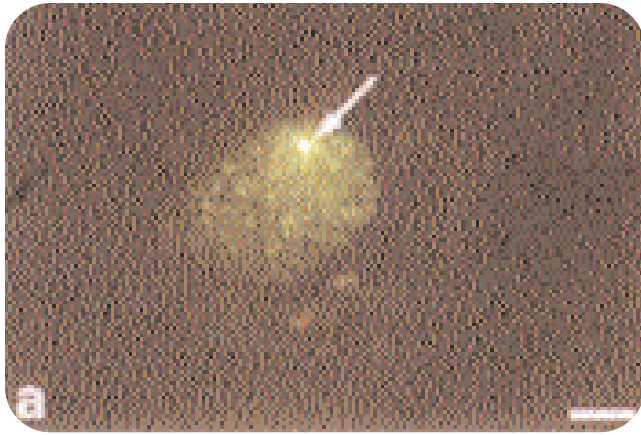
An animation of the splicing process

Nuclear organization RNA splicing takes place in the nucleus and must occur before the RNA can move into the cytoplasm. For many years, the nucleus was viewed as a biochemical soup, in which components such as the spliceosome diffused and reacted randomly. Now, the nucleus is believed to have a highly ordered internal structure, with transcription and RNA processing taking place



14.12 RNA splicing takes place within the spliceosome.

at particular locations within it. By attaching fluorescent tags to pre-mRNA and using special imaging techniques, researchers have been able to observe the location of pre-mRNA as it is transcribed and processed. The results of these studies revealed that intron removal and other processing reactions take place at the same sites as those of transcription (FIGURE 14.13), suggesting that these processes may be physically coupled. This suggestion is supported by the observation that part of RNA polymerase II is also required for the splicing and 3' processing of pre-mRNA.

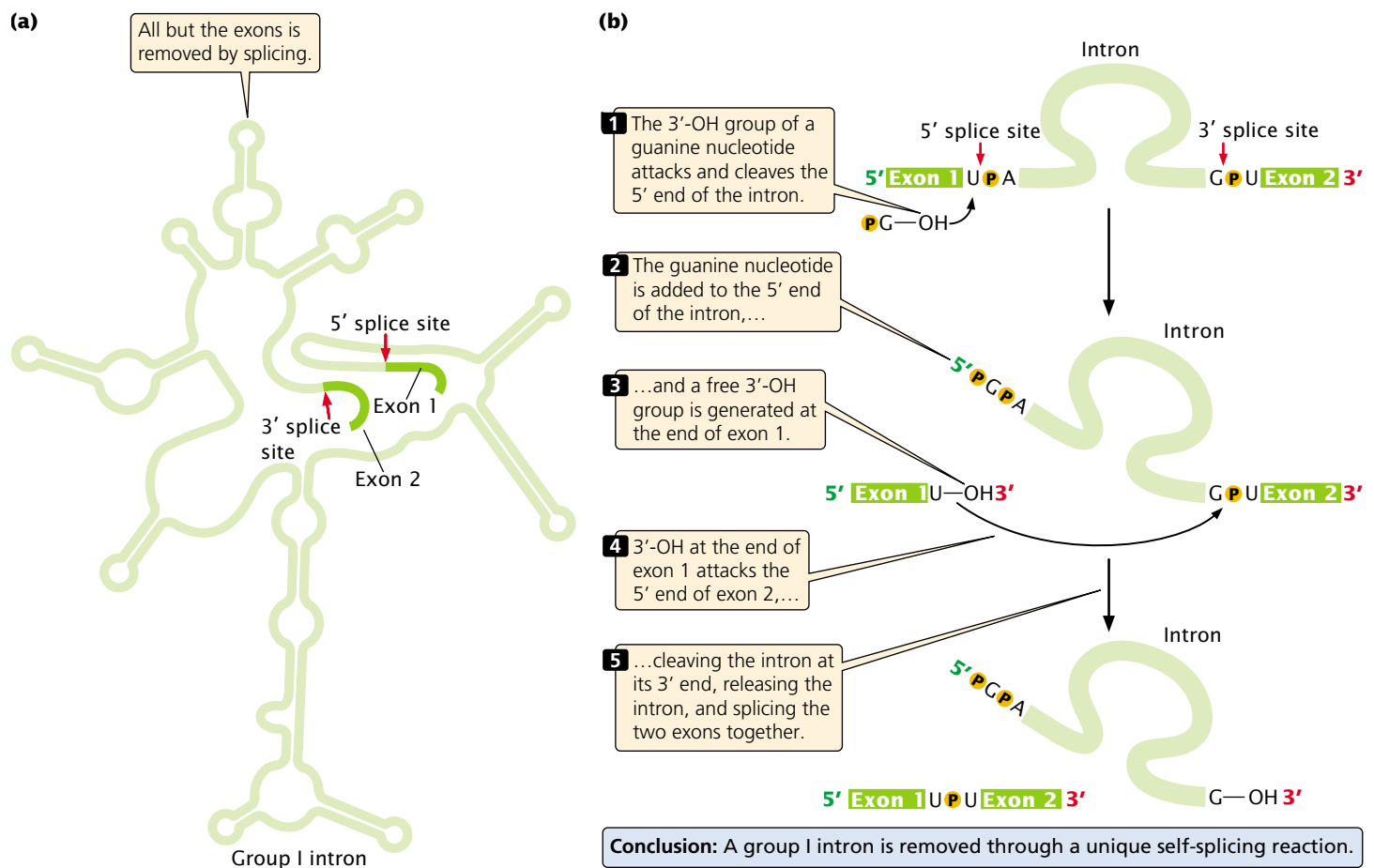


14.13 Intron removal, processing, and transcription take place at the same site. RNA tracks can be seen in the nucleus of a eukaryotic cell. Fluorescent tags were attached to DNA (red) and RNA (green). Transcribed RNA does not disperse; rather, it accumulates near the site of synthesis and follows a defined track during processing. (Credit for Fig 14.13 allowing additional line for photo credit.)

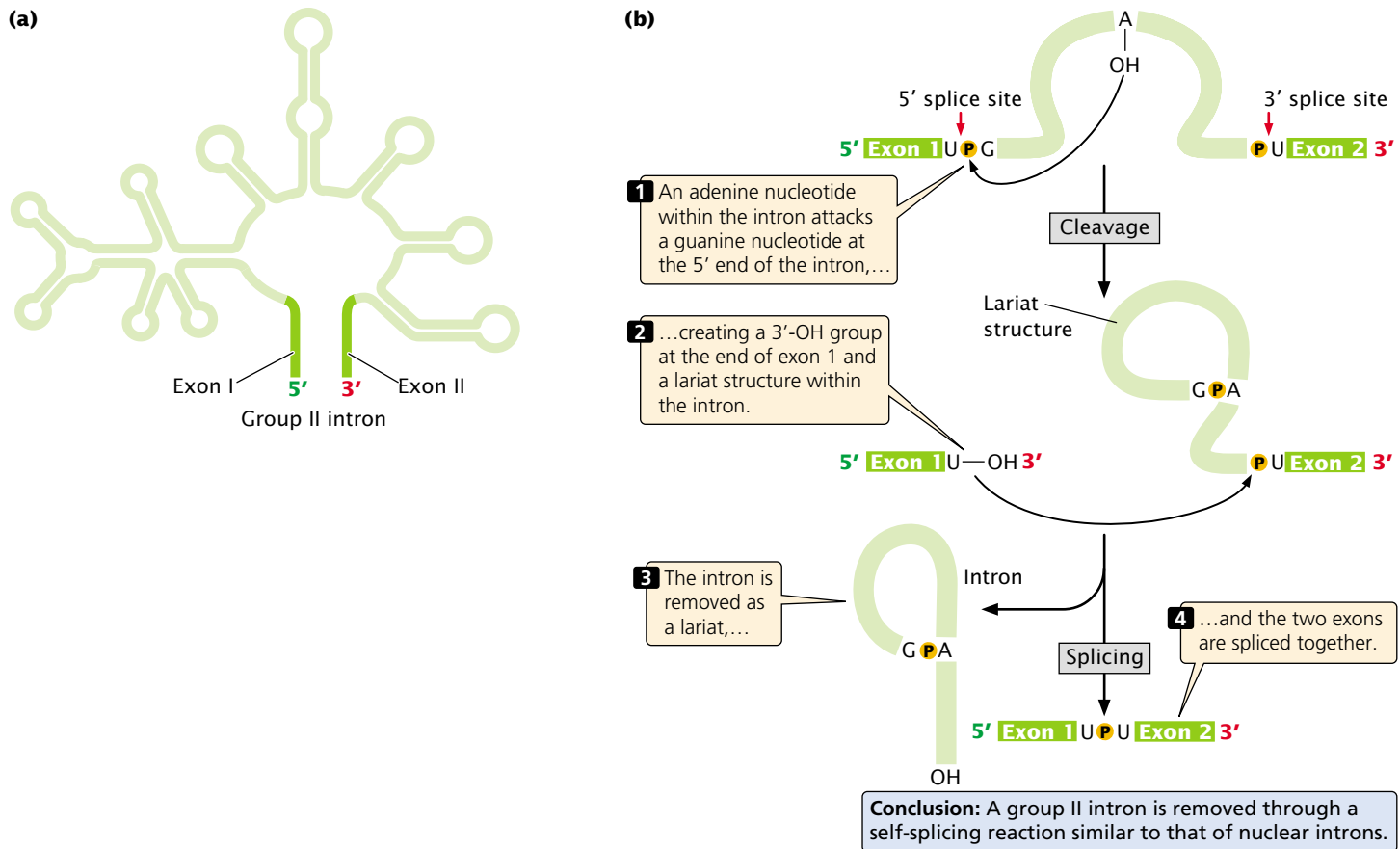
Concepts

Intron splicing of nuclear genes is a two-step process: (1) the 5' end of the intron is cleaved and attached to the branch point to form a lariat and (2) the 3' end of the intron is cleaved and the two ends of the exon are spliced together. These reactions take place within the spliceosome.

Self-splicing introns Some introns are self-splicing, meaning that they possess the ability to remove themselves from an RNA molecule. These self-splicing introns fall into two major categories. Group I introns are found in a variety of genes, including some rRNA genes in prokaryotes, some mitochondrial genes in fungi, and even some bacteriophage genes. Although the lengths of group I introns vary, all of them fold into a common secondary structure with nine looped stems (FIGURE 14.14a), which are necessary for splicing. Transesterification reactions are required for the splicing of group I introns (FIGURE 14.14b).



14.14 Group I introns undergo self-splicing. (a) Secondary structure of a group I intron. (b) Self-splicing of a group I intron.



14.15 Group II introns undergo self-splicing by a different mechanism from that for group I introns. (a) Secondary structure of a group II intron. (b) Self-splicing of group II introns, which is similar to the splicing of nuclear introns.

Group II introns, present in some mitochondrial genes, also have the ability to self-splice. All group II introns fold into similar secondary structures (FIGURE 14.15a). The splicing of group II introns is accomplished by a mechanism that has some similarities to the spliceosomal-mediated splicing of nuclear genes; splicing takes place through two transesterification reactions that generate a lariat structure (FIGURE 14.15b). Because of these similarities, group II introns and nuclear pre-mRNA introns have been suggested to be evolutionarily related—perhaps the nuclear introns evolved from self-splicing group II introns and later adopted the proteins and snRNAs of the spliceosome to carry out the splicing reaction.

Concepts

Self-splicing introns are of two types: group I introns and group II introns. These introns have complex secondary structures that enable them to catalyze their excision from RNA molecules without the aid of enzymes or other proteins.

Alternative Processing Pathways

Another finding that complicates the view of a gene as a sequence of nucleotides that specifies the amino acid sequence of a protein is the existence of **alternative processing pathways**, in which a single pre-mRNA is processed in different ways to produce alternative types of mRNA, resulting in the production of different proteins from the same DNA sequence.

One type of alternative processing is **alternative splicing**, in which the same pre-mRNA can be spliced in more than one way to yield multiple mRNAs that are translated into proteins with different amino acid sequences (FIGURE 14.16a). Another type of alternative processing requires the use of **multiple 3' cleavage sites** (FIGURE 14.16b); two or more potential sites for cleavage and polyadenylation are present in the pre-mRNA. In our example, cleavage at the first site produces a relatively short mRNA, compared with the mRNAs produced through cleavage at other sites.

Both alternative splicing and multiple 3' cleavage sites can exist in the same pre-mRNA transcript; an example is seen in the mammalian calcitonin gene, which contains six

exons and five introns (FIGURE 14.17a). The entire gene is transcribed into pre-mRNA (FIGURE 14.17b). There are two possible 3' cleavage sites. In cells of the thyroid gland, 3' cleavage and polyadenylation take place after the fourth exon, and the first three introns are then removed to produce a mature mRNA consisting of exons 1, 2, 3, and 4 (FIGURE 14.17c). This mRNA is translated into the hormone calcitonin. In brain cells, the *identical* pre-mRNA is transcribed from DNA, but it is processed differently. Cleavage and polyadenylation take place after the sixth exon, yielding an initial transcript that includes all six exons. During splicing, exon 4 (part of the calcitonin mRNA) is removed, along with all the introns; so only exons 1, 2, 3, 5, and 6 are present in the mature mRNA (FIGURE 14.17d). When translated, this mRNA produces a protein called calcitonin-gene-related peptide (CGRP), which has an amino acid sequence quite different from that of calcitonin. Alternative splicing may produce different combinations of

exons in the mRNA, but the order of the exons is not usually changed. Different processing pathways contribute to gene regulation, as discussed in Chapter 16.

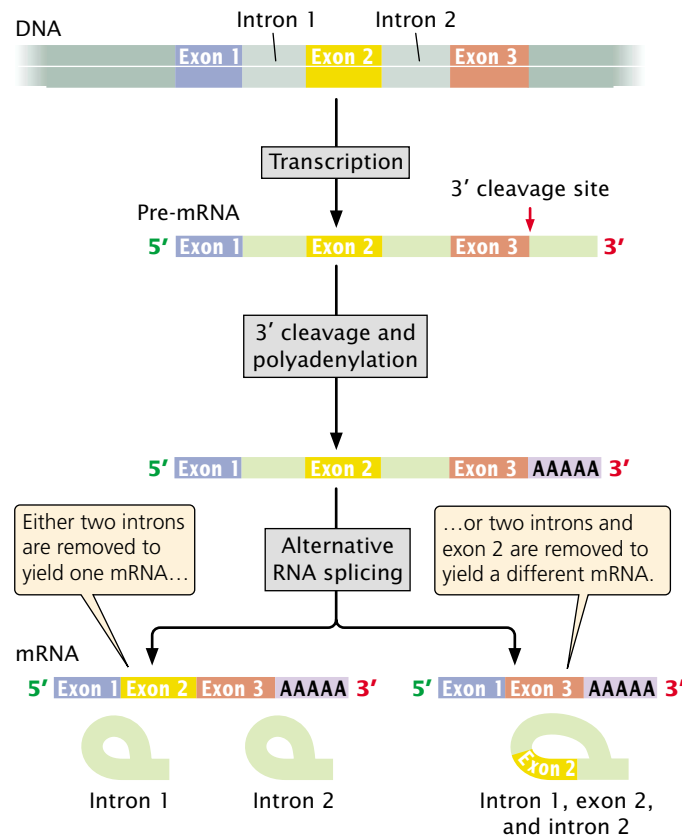
Concepts

Alternative splicing enables exons to be spliced together in different combinations to yield mRNAs that encode different proteins. Alternative 3' cleavage sites allow pre-mRNA to be cleaved at different sites to produce mRNAs of different lengths.

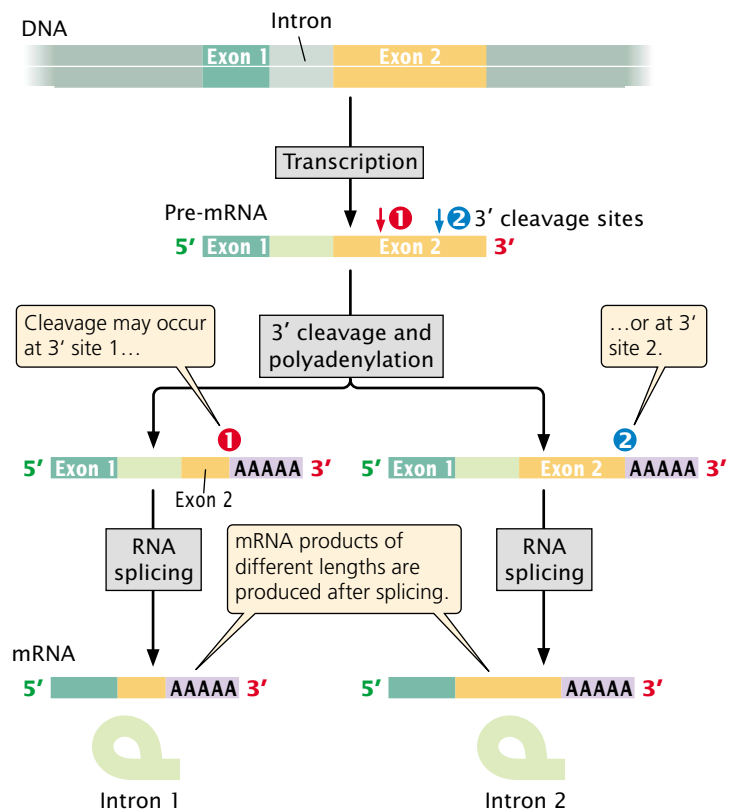
RNA Editing

A long-standing principle of molecular genetics is that genetic information ultimately resides in the nucleotide sequence of DNA, except in RNA viruses (Chapter 10). This

(a) Alternative splicing

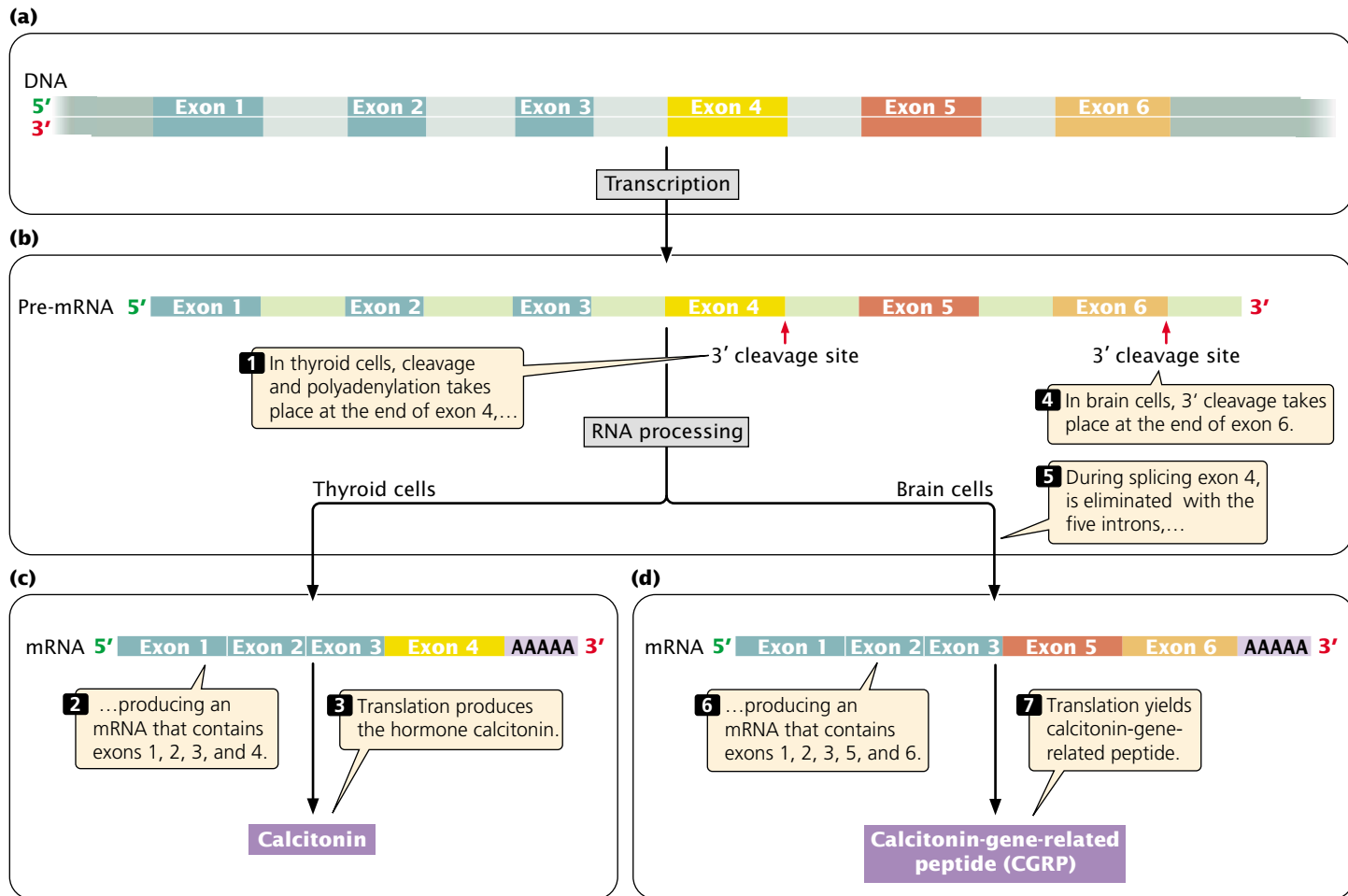


(b) Multiple 3' cleavage sites



Conclusion: Both alternate splicing and multiple 3' cleavage sites produce different mRNAs from a single pre-mRNA.

14.16 Eukaryotic cells have alternative pathways for processing pre-mRNA. (a) With alternative splicing; pre-mRNA can be spliced in different ways to produce different mRNAs. (b) With multiple 3' cleavage sites, there are two or more potential sites for cleavage and polyadenylation; use of the different sites produces mRNAs of different lengths.



14.17 Pre-mRNA encoded by the calcitonin gene undergoes alternative processing.

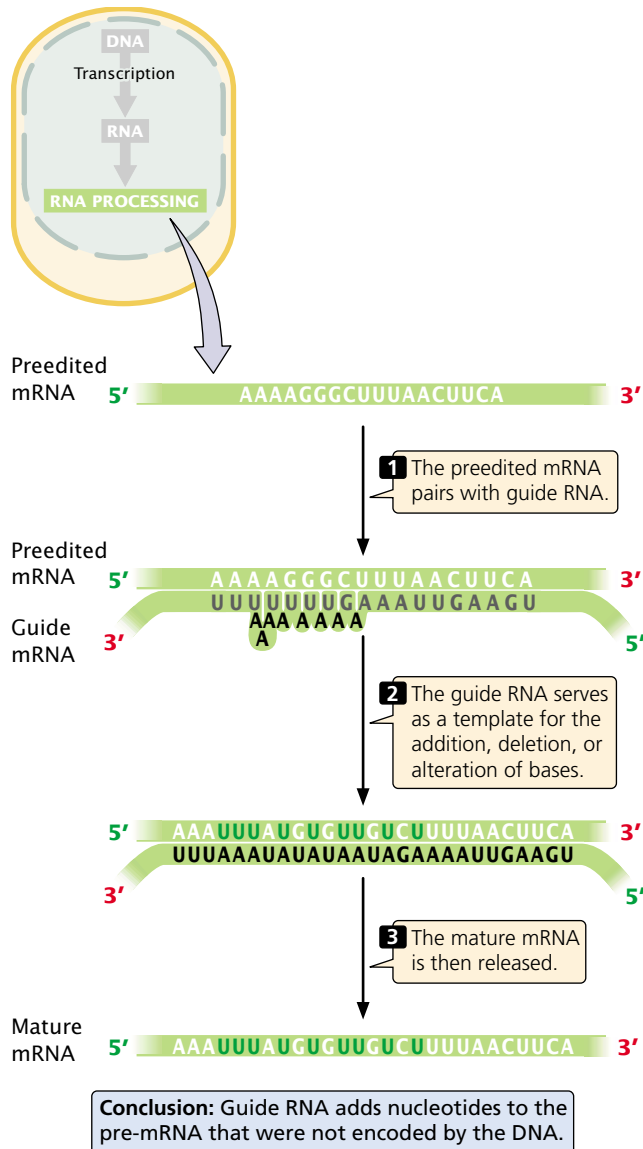
information is transcribed into mRNA, and mRNA is then translated into a protein. The assumption that all information about the amino acid sequence of a protein resides in DNA is violated by a process called **RNA editing**. In RNA editing, the coding sequence of an mRNA molecule is altered after transcription, and so the protein has an amino acid sequence that differs from that encoded by the gene.

RNA editing was first detected in 1986 when the coding sequences of mRNAs were compared with the coding sequences of the DNAs from which they had been transcribed. Discrepancies were found for some nuclear genes in mammalian cells and for mitochondrial genes in plant cells. In these cases, substitutions had occurred in some of the nucleotides of the mRNA. More extensive RNA editing has been found in the mRNA for some mitochondrial genes in trypanosome parasites (which cause African sleeping sickness). In some mRNAs of these organisms, more than 60% of the sequence is determined

by RNA editing. Different types of RNA editing have now been observed in mRNAs, tRNAs, and rRNAs from a wide range of organisms; they include the insertion and the deletion of nucleotides and the conversion of one base into another.

If the modified sequence in edited RNA molecules doesn't come from a DNA template, then how is it specified? There are a variety of mechanisms that may bring about changes in RNA sequences. In some cases, molecules called **guide RNAs** (gRNAs) play a crucial role. The gRNAs contain sequences that are partly complementary to segments of the preedited RNA, and the two molecules undergo base pairing in these sequences (FIGURE 14.18). After the mRNA is anchored to the gRNA, the mRNA undergoes cleavage and nucleotides are added, deleted, or altered according to the template provided by gRNA. The ends of the mRNA are then joined together.

In other cases, enzymes bring about base conversion. In humans, for example, a gene is transcribed into



14.18 RNA editing is carried out by guide RNAs.

mRNA that codes for a lipid-transporting polypeptide called apolipoprotein-B100, which has 4563 amino acids and is synthesized in liver cells. A truncated form of the protein called apolipoprotein-B48—with only 2153 amino acids—is synthesized in intestinal cells. The truncated protein is produced from an edited version of the same mRNA that codes for apolipoprotein-B100. In editing, an enzyme deaminates a cytosine base, converting it into uracil. This conversion changes a codon that specifies the amino acid glutamine into a stop codon that prematurely terminates translation, resulting in the shortened protein.

Concepts

Individual nucleotides in the interior of pre-mRNA may be changed, added, or deleted by RNA editing. The amino acid sequence produced by the edited mRNA is not the same as that encoded by DNA.

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More information on RNA editing and a database of guide RNA sequences

Connecting Concepts

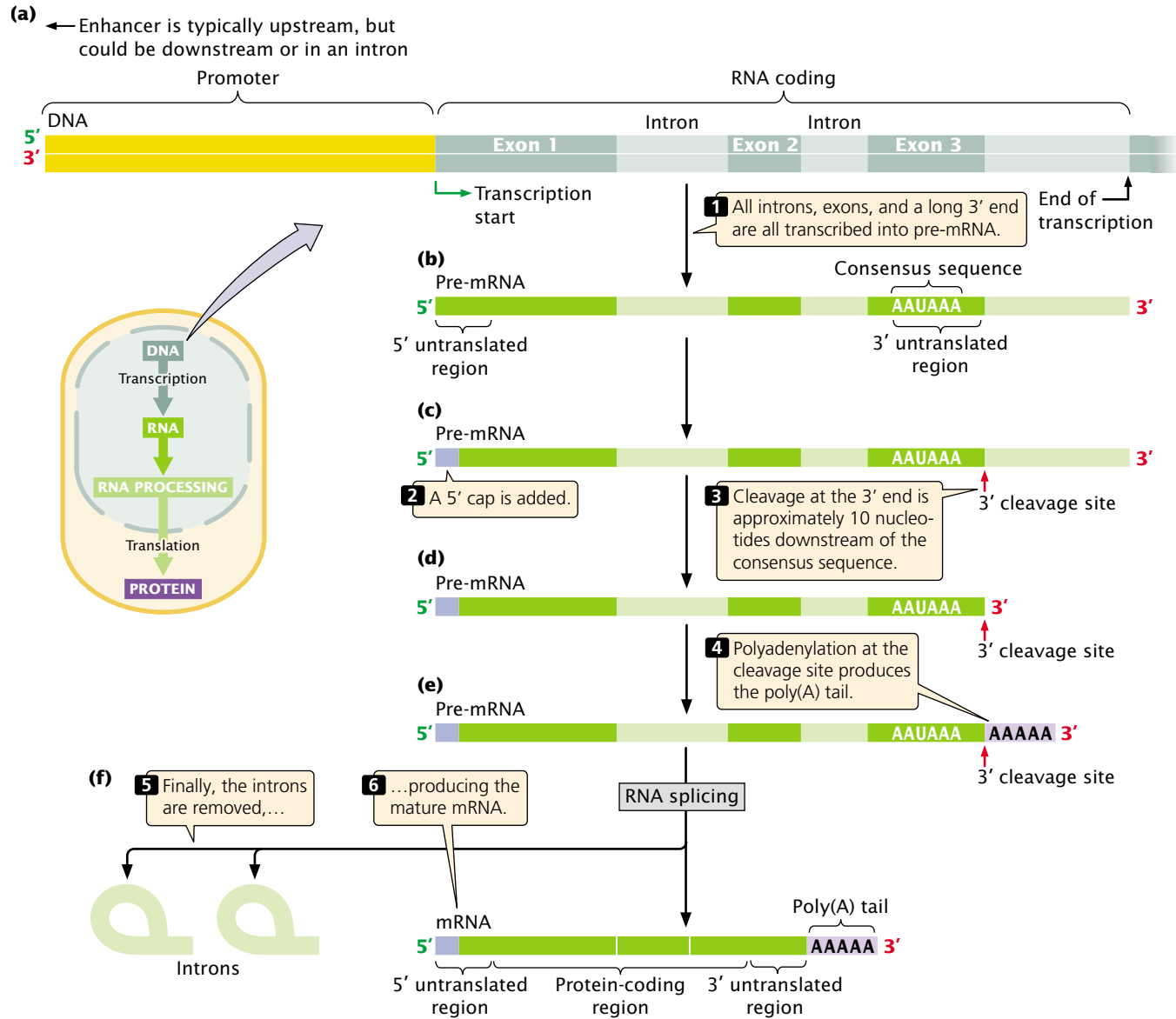
Eukaryotic Gene Structure and Pre-mRNA Processing

Chapters 13 and 14 have introduced a number of different components of genes and RNA molecules, including promoters, 5' untranslated regions, coding sequences, introns, 3' untranslated regions, poly(A) tails, and caps. Let's see how some of these components are combined to create a typical eukaryotic gene and how a mature mRNA is produced from them.

The promoter, which typically encompasses about 100 nucleotides upstream of the transcription start site, is necessary for transcription to take place but is itself not usually transcribed when protein-encoding genes are transcribed by RNA polymerase II (FIGURE 14.19a). Farther upstream or downstream of the start site, there may be enhancers that also regulate transcription.

In transcription, all the nucleotides between the transcription start site and the stop site are transcribed into pre-mRNA, including exons, introns, and a long 3' end that is later cleaved from the transcript (FIGURE 14.19b). Notice that the 5' end of the first exon contains the sequence that codes for the 5' untranslated region, and the 3' end of the last exon contains the sequence that codes for the 3' untranslated region.

The pre-mRNA is then processed to yield a mature mRNA. The first step in this processing is the addition of a cap to the 5' end of the pre-mRNA (FIGURE 14.19c). Next, the 3' end is cleaved at a site downstream of the AAUAAA consensus sequence in the last exon (FIGURE 14.19d). Immediately after cleavage, a poly(A) tail is added to the 3' end (FIGURE 14.19e). Finally, the introns are removed to yield the mature mRNA (FIGURE 14.19f). The mRNA now contains 5' and 3' untranslated regions, which are not translated into amino acids, and the nucleotides that carry the protein-coding sequences. The nu-



14.19 Mature eukaryotic mRNA is produced when pre-mRNA is transcribed and undergoes several types of processing.

cleotide sequence of a small gene, with these components labeled, is presented in (FIGURE 14.20).

Transfer RNA

In 1956, Francis Crick proposed the idea of a molecule that transported amino acids to the ribosome and interacted with codons in mRNA, placing amino acids in their proper order in protein synthesis. By 1963, the existence of such an adapter molecule, called transfer RNA, had been confirmed. Transfer RNA (tRNA) serves as a link between the genetic code in mRNA and the amino acids that make up a protein. Each tRNA attaches to a particular amino acid and carries it

to the ribosome, where the tRNA adds its amino acid to the growing polypeptide chain at the position specified by the genetic instructions in the mRNA. We'll take a closer look at the mechanism of this process in Chapter 15.

Each tRNA is capable of attaching to only one type of amino acid. The complex of tRNA plus its amino acid can be written in abbreviated form by adding a three-letter superscript representing the amino acid to the term tRNA. For example, a tRNA that attaches to the amino acid alanine is written as tRNA^{Ala}. Because 20 different amino acids are found in proteins, there must be a minimum of 20 different types of tRNA. In fact, most organisms possess from at least 30 to 40 different types of tRNA, each



14.20 This representation of the nucleotide sequence of the human interleukin 2 gene includes the TATA box, transcription start site, start and stop codons, introns, exons, poly(A) consensus sequence, and the 3' cleavage site.

encoded by a different gene (or, in some cases, multiple copies of a gene) in DNA.

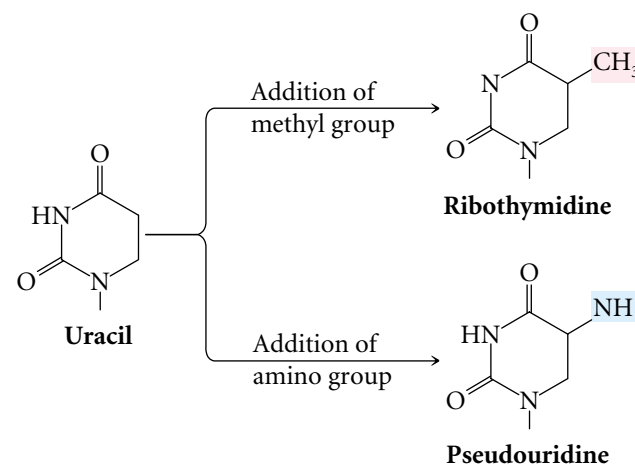
The Structure of Transfer RNA

A unique feature of tRNA is the occurrence of rare, **modified bases**. All RNAs have the four standard bases (adenine, cytosine, guanine, and uracil) specified by DNA, but tRNAs have additional bases, including ribothymine, pseudourasil (which is also occasionally present in snRNAs and rRNA), and dozens of others. The structures of some of these modified bases are shown in (FIGURE 14.21).

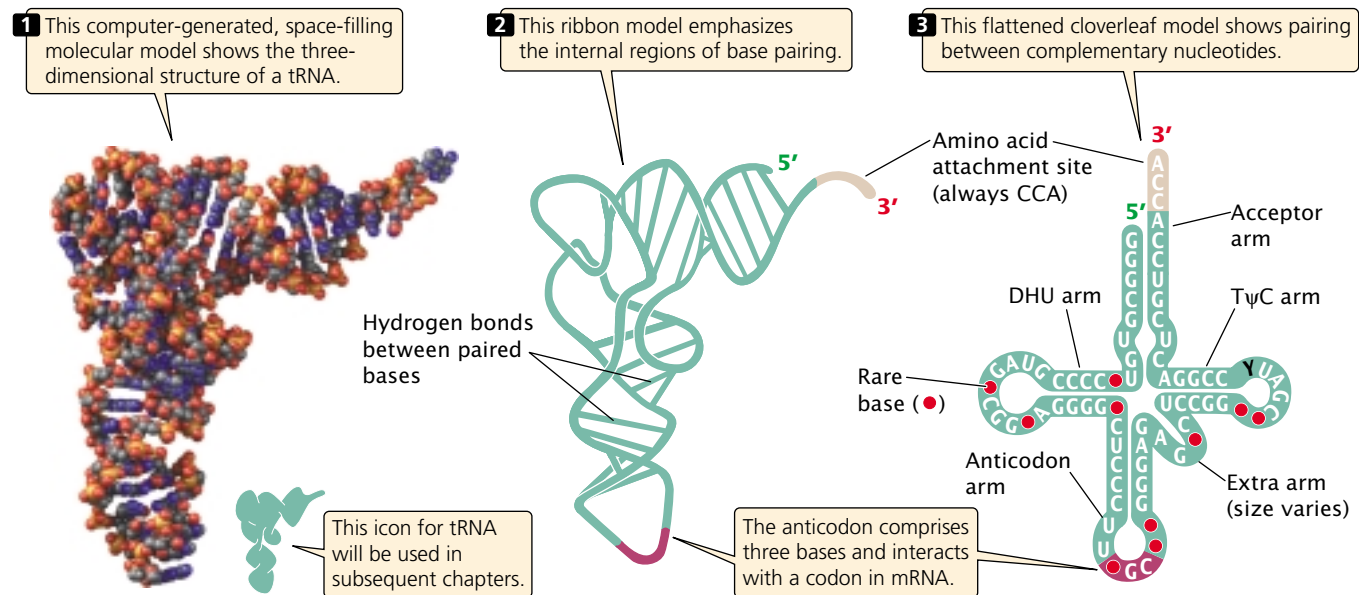
If there are only four bases in DNA, and all RNA molecules are transcribed from DNA, how do tRNAs acquire these additional bases? Modified bases arise from chemical changes made to the four standard bases after transcription. These changes are carried out by special **tRNA-modifying enzymes**. For example, the addition of a methyl group to uracil creates the modified base ribothymine.

The structures of all tRNAs are similar, a feature critical to tRNA function. Most tRNAs contain between 74 and 95 nucleotides, some of which are complementary to each other and form intramolecular hydrogen bonds. As a result, each tRNA has a **cloverleaf structure** (FIGURE 14.22). The cloverleaf has four major arms, each consisting of a stem and a loop. The stem is formed by the pairing of comple-

mentary nucleotides, and the loop lies at the terminus of the stem, where there is no nucleotide pairing. If we start at the top and proceed clockwise around the tRNA shown at the right in Figure 14.22, the four major arms are the acceptor arm, the T ψ C arm, the anticodon arm, and the DHU arm.



14.21 Modified bases are found in tRNAs. All the modified bases are produced by the chemical alteration of the four standard RNA bases.



14.22 All tRNAs possess a common secondary structure, the cloverleaf structure. The base sequence in the flattened model is for tRNA^{Ala}. (Credit for Fig 14.22 allowing rest of line for photo credit here.)

The acceptor arm has no loop but contains the 5' and 3' ends of the tRNA molecule. All tRNAs have the same sequence (CCA) at the 3' end, where the amino acid attaches to the tRNA; so clearly this sequence is not responsible for specifying which amino acid will attach to the tRNA.

The T ψ C arm is named for the bases of three nucleotides in the loop of this arm: thymine (T), pseudouracil (ψ), and cytosine (C). The anticodon arm lies at the bottom of the tRNA. Three nucleotides at the end of this arm make up the **anticodon**, which pairs with the corresponding codon on mRNA to ensure that the amino acids link in the correct order. The DHU arm is so named because it often contains the modified base dihydrouridine.

Although each tRNA molecule folds into a cloverleaf owing to the complementary pairing of bases, the cloverleaf is not the three-dimensional (tertiary) structure of tRNAs found in the cell. The results of X-ray crystallographic studies have shown that the cloverleaf folds upon itself to form an L-shaped structure, as illustrated by the space-filling and ribbon models in Figure 14.22. Notice that the acceptor stem is at one end of the tertiary structure and the anticodon is at the other end.

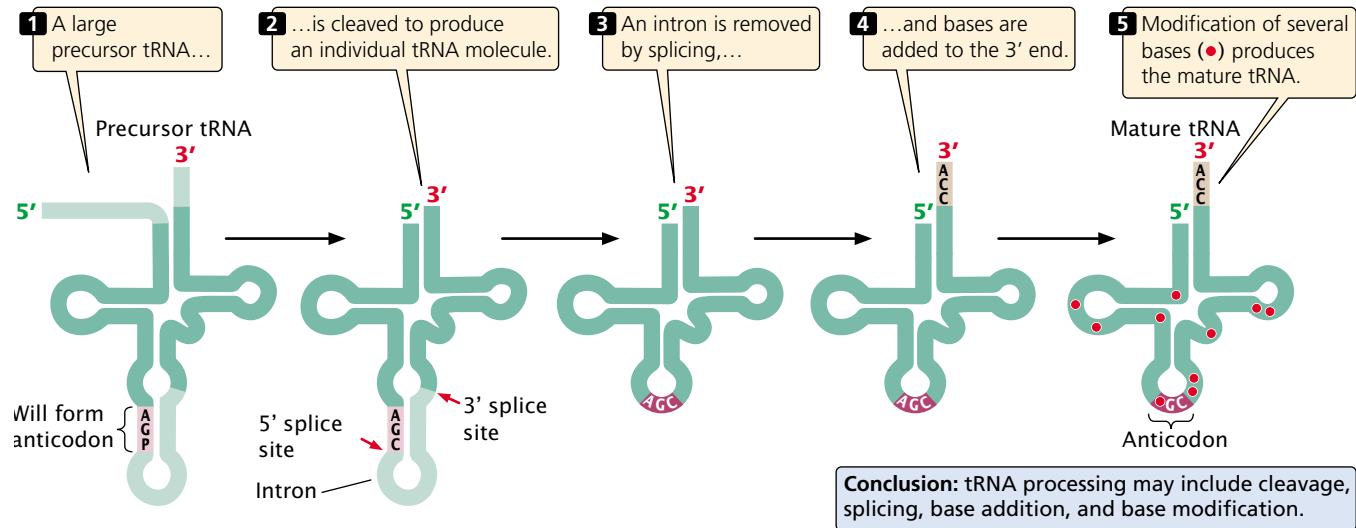
Transfer RNA Gene Structure and Processing

The genes that produce tRNAs may be scattered about the genome or may be in clusters. In *E. coli*, the genes for some tRNAs are present in a single copy, whereas the genes for

other tRNAs are present in several copies; eukaryotic cells usually have many copies of each tRNA gene. All tRNA molecules in both bacterial and eukaryotic cells undergo processing after transcription.

In *E. coli*, several tRNAs are usually transcribed together as one large precursor tRNA, which is then cut up into pieces, each containing a single tRNA. Additional nucleotides may then be removed one at a time from the 5' and 3' ends of the tRNA in a process known as trimming. Base-modifying enzymes may then change some of the standard bases into modified bases, and additional bases (such as CCA at the 3' end) may be added (Figure 14.23). Different tRNAs are processed in different ways; so it is not possible to outline a generic processing pathway for all tRNAs. Eukaryotic tRNAs are processed in a manner similar to that for bacterial tRNAs: most are transcribed as larger precursors that are then cleaved, trimmed, and modified to produce mature tRNAs.

Some eukaryotic tRNA genes possess introns of variable length that must be removed in processing. For example, about 40 of the 400 tRNA genes in yeast contain a single intron that is always found adjacent to the 3' side of the anticodon. The splicing process for tRNA genes (see Figure 14.23) is quite different from the spliceosome-mediated reactions that remove introns from protein-encoding genes. The intron in the precursor tRNA is cut at both ends by an endonuclease enzyme, which releases the linear intron from the rest of the tRNA. The two pieces of tRNA, which are held together by intramolecular bonding, are then folded and ligated to produce the mature tRNA.



14.23 Transfer RNAs are processed in both bacterial and eukaryotic cells. Different tRNAs are modified in different ways. One example is shown here.

Concepts

All tRNAs are similar in size and have a common secondary structure known as the cloverleaf. Transfer RNAs contain modified bases and are extensively processed after transcription in both bacterial and eukaryotic cells.

Ribosomal RNA

Within ribosomes, the genetic instructions contained in mRNA are translated into the amino acid sequences of polypeptides. Thus, ribosomes play an integral part in the transfer of genetic information from genotype to phenotype. We will examine the role of ribosomes in the process of translation in Chapter 15. Here, we will consider ribo-

some structure and examine how ribosomes are processed before becoming functional.

The Structure of the Ribosome

The ribosome is one of the most abundant organelles in the cell: a single bacterial cell may contain as many as 20,000 ribosomes, and eukaryotic cells possess even more. Ribosomes typically contain about 80% of the total cellular RNA. They are complex organelles, each consisting of more than 50 different proteins and RNA molecules (Table 14.3). A functional ribosome consists of two subunits, a **large subunit** and a **small subunit**, each of which consists of one or more pieces of RNA and a number of proteins. The sizes of the ribosomes and their RNA components are given in Svedberg (S) units (a measure of how rapidly an object sediments in a centrifugal field). It is important to note that Svedberg units are not additive; in other words, combining a 10S

Table 14.3 Composition of ribosomes in bacterial and eukaryotic cells

Cell Type	Ribosome Size	Subunit	rRNA Component	Proteins
Bacterial	70S	Large (50S)	23S (2900 nucleotides)	31
		Small (30S)	16S (1500 nucleotides)	
Eukaryotic	80S	Large (60S)	28S (4700 nucleotides)	49
			5.8S (160 nucleotides)	
			5S (120 nucleotides)	
		Small (40S)	18S (1900 nucleotides)	33

Note: The letter S stands for Svedberg unit.

structure and a 20S structure does not necessarily produce a 30S structure, because the sedimentation rate is affected by the three-dimensional structure as well as the mass.

Ribosomal RNA Gene Structure and Processing

The genes for rRNA, like those for tRNA, can be present in multiple copies, and the numbers vary among species (Table 14.4); all copies of the rRNA gene in a species are identical or nearly identical. In bacteria, rRNA genes are dispersed, but, in eukaryotic cells, they are clustered, with the genes arrayed in tandem, one after another.

Eukaryotic cells possess two types of rRNA genes: a large one that encodes 18S rRNA, 28S rRNA, and 5.8S rRNA, and a small one that encodes the 5S rRNA. All three bacterial rRNAs (23S rRNA, 16S rRNA, and 5S rRNA) are encoded by a single type of gene.

Ribosomal RNA is processed in both bacterial and eukaryotic cells. In *E. coli*, the immediate product of transcription is a 30S rRNA precursor (FIGURE 14.24a). Methyl groups (CH₃) are added to specific bases and to the 2' carbon of some of the ribose sugars of this 30S precursor, which is then cleaved into several pieces and trimmed to produce 16S rRNA, 23S rRNA, and 5S rRNA, along with one or more tRNAs. All rRNA genes in *E. coli* produce the same three rRNA molecules, but the number and location of these rRNAs within the 30S rRNA transcript differ among genes.

Eukaryotic rRNAs undergo similar processing (FIGURE 14.24b). Small nucleolar RNAs help to cleave and

Table 14.4 Number of rRNA genes in different organisms

Species	Copies of rRNA Genes or Genome
<i>Escherichia coli</i>	1
Yeast	100–200
Human	280
Frog	450

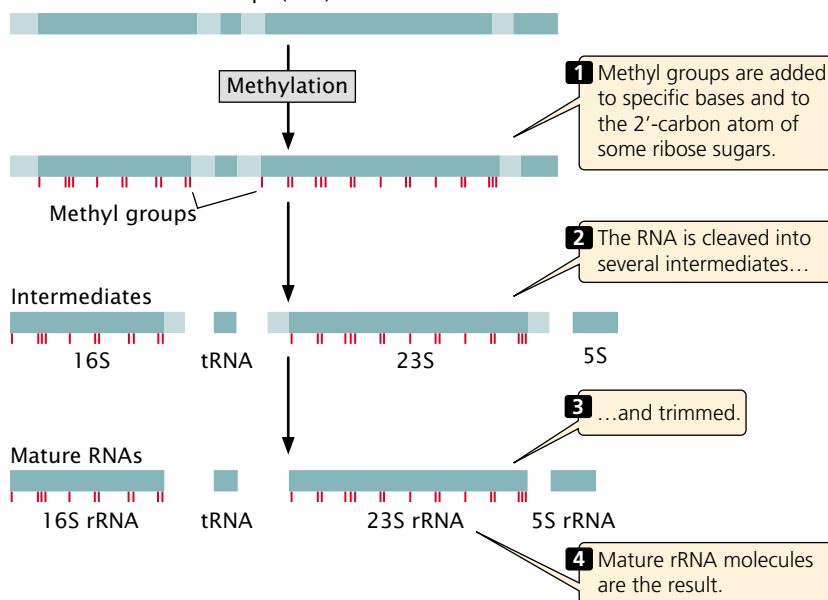
modify eukaryotic rRNAs (as well as some archaeal rRNAs), and help to assemble the processed rRNAs into mature ribosomes. The snoRNAs have extensive complementarity to the rRNA sequences where modification takes place. Interestingly, some snoRNAs are encoded by sequences in the introns of other protein-encoding genes.

Concepts

A ribosome is a complex organelle consisting of several rRNA molecules and many proteins. Each functional ribosome consists of a large and a small subunit. rRNAs in both bacterial and eukaryotic cells are modified after transcription. In eukaryotes, rRNA processing is carried out by small nucleolar RNAs (snoRNAs).

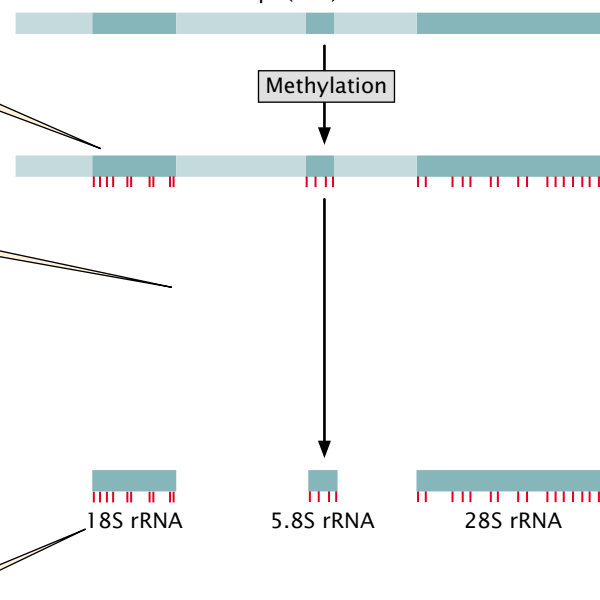
(a) Prokaryotic rRNAs

Precursor rRNA transcript (30S)



(b) Eukaryotic rRNAs

Precursor rRNA transcript (45S)



14.24 Ribosomal RNA is processed after transcription. Note that eukaryotic rRNA does not undergo trimming and that 5S rRNA is transcribed separately from the small eukaryotic rRNA gene.

Connecting Concepts Across Chapters



Because it is single stranded and can form hydrogen bonds between complementary bases on the same strand, RNA is capable of assuming a number of secondary structures. This ability gives RNA functional flexibility, and it assumes a number of important roles in information transfer within the cell.

A central theme in this chapter has been the nature of the gene. The concept of a gene has changed with time and even today depends on the particular question that is being addressed. A modern definition used by many geneticists is: a gene is a sequence of nucleotides in DNA that is transcribed into a single RNA molecule.

The details of RNA function and processing covered in this chapter are important for understanding the process of protein synthesis, which is the focus of Chapter 15. Knowledge of the structure of the ribosome and tRNAs will be important for understanding how amino acids are assembled into a protein. In eukaryotic cells, features [such as the 5' cap and the poly(A) tail] that are added to pre-mRNA and those removed (introns) from it are essential for translation to proceed properly. These features of processed mRNA also play an important role in eukaryotic gene regulation, a subject to be addressed in Chapter 16.

CONCEPTS SUMMARY

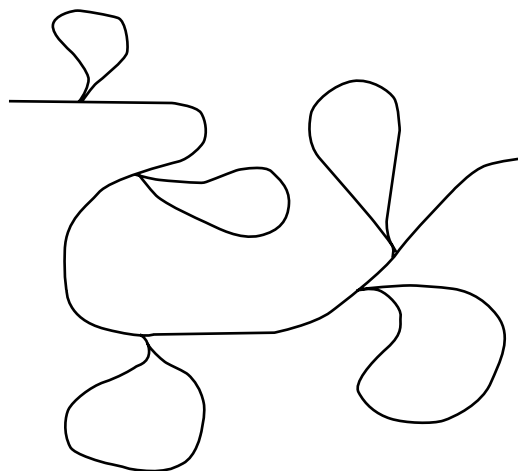
- The discovery of introns in eukaryotic genes forced the redefinition of the gene at the molecular level. Today, a gene is often defined as a sequence of DNA nucleotides that is transcribed into a single RNA molecule.
- Introns are noncoding sequences that interrupt the coding sequences (exons) of genes. Common in eukaryotic cells but rare in bacterial cells, introns exist in all types of genes and vary in size and number. They comprise four major types: group I introns, group II introns, nuclear pre-mRNA introns, and tRNA introns.
- The results of experiments in the late 1950s and early 1960s suggested that genetic information is carried from DNA to ribosomes by short-lived RNA molecules called messenger RNA. An mRNA molecule has three primary parts: a 5' untranslated region, a protein-coding sequence, and a 3' untranslated region.
- Bacterial mRNA is translated immediately after transcription and undergoes little processing.
- The primary transcript (pre-mRNA) of a eukaryotic protein-encoding gene is extensively processed: a modified nucleotide and methyl groups, collectively termed the cap, are added to the 5' end of pre-mRNA; the 3' end is cleaved and a poly(A) tail is added; and introns are removed.
- The process of RNA splicing takes place within a structure called the spliceosome, which is composed of several small nuclear RNAs and proteins. RNA splicing takes place in a two-step process that entails RNA–RNA interactions among snRNAs of the spliceosome and the pre-mRNA.
- Some introns found in rRNA genes and mitochondrial genes are self-splicing.
- Some pre-mRNAs undergo alternative splicing, in which different combinations of exons are spliced together or different 3' cleavage sites are used.
- Messenger RNAs may also be altered by the addition, deletion, or modification of nucleotides in the coding sequence, a process called RNA editing.
- Transfer RNA serves as a bridge between amino acids and the genetic information carried in mRNA. Transfer RNAs are relatively short molecules that assume a common secondary structure and contain modified bases. Most organisms have multiple copies of tRNA genes; the tRNAs transcribed from these genes are extensively processed in bacterial and eukaryotic cells.
- Ribosomes are the sites of protein synthesis in the cell. Each ribosome is composed of several rRNA molecules and a number of proteins that form a large and a small subunit. Genes for rRNA exist in multiple copies; the primary transcripts from these genes are extensively modified after transcription in bacterial and eukaryotic cells. In eukaryotic cells, rRNA processing is carried out by small nucleolar RNAs.

IMPORTANT TERMS

colinearity (p. 000)	Shine-Dalgarno sequence (p. 000)	spliceosome (p. 000)	modified base (p. 000)
exon (p. 000)	protein-coding region (p. 000)	lariat (p. 000)	tRNA-modifying enzyme (p. 000)
intron (p. 000)	3' untranslated region (p. 000)	transesterification (p. 000)	cloverleaf structure (p. 000)
group I intron (p. 000)	5' cap (p. 000)	alternative processing pathway (p. 000)	anticodon (p. 000)
group II intron (p. 000)	poly(A) tail (p. 000)	alternative splicing (p. 000)	large ribosomal subunit (p. 000)
nuclear pre-mRNA intron (p. 000)	RNA splicing (p. 000)	multiple 3' cleavage site (p. 000)	small ribosomal subunit (p. 000)
tRNA intron (p. 000)	5' splice site (p. 000)	RNA editing (p. 000)	
codon (p. 000)	3' splice site (p. 000)	guide RNA (p. 000)	
5' untranslated region (p. 000)	branch point (p. 000)		

Worked Problems

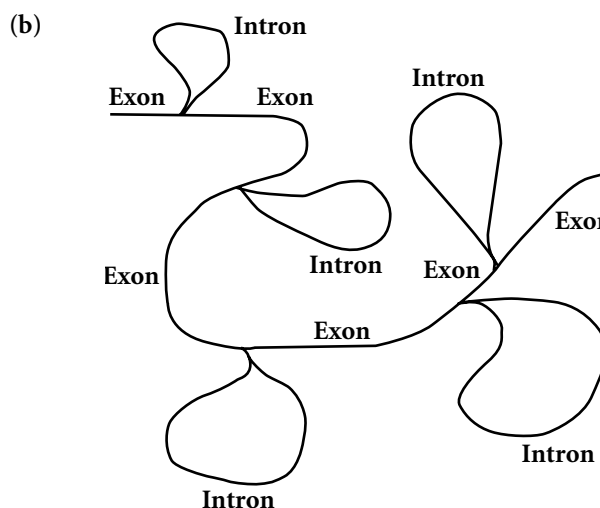
1. DNA from a eukaryotic gene was isolated, denatured, and hybridized to the mRNA transcribed from the gene; the hybridized structure was then observed with the use of an electron microscope. The following structure was observed.



- (a) How many introns and exons are there in this gene? Explain your answer.
 (b) Identify the exons and introns in this hybridized structure.

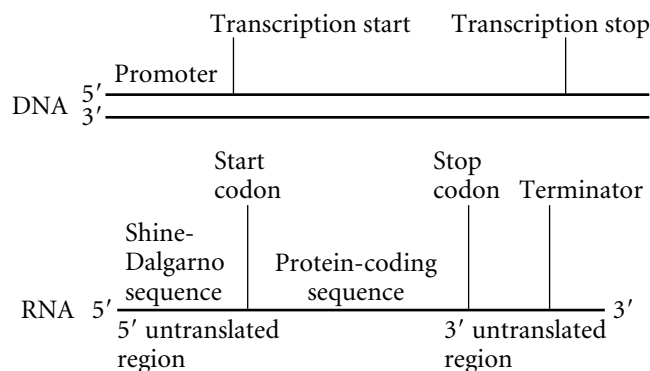
• Solution

(a) Each of the loops represents a region where there are sequences in the DNA that do not have corresponding sequences in the RNA; these regions are introns. There are five loops in the hybridized structure; so there must be five introns in the DNA.
 (b) Identify the exons and introns in this hybridized structure.



2. Draw a typical bacterial mRNA and the gene from which it was transcribed. Label the 5' and 3' ends of the RNA and DNA molecules, and identify the following regions or sequences:

- (a) Promoter (e) Transcription start site
 (b) 5' untranslated region (f) Terminator
 (c) 3' untranslated region (g) Shine-Dalgarno sequence
 (d) Protein-coding sequence



• **Solution**

3. A test-tube splicing system has been developed that contains all the components (snRNAs, proteins, splicing factors) necessary for the splicing of nuclear genes. When a piece of RNA containing an intron and two exons is added to the system, the intron is removed as a lariat and the exons are spliced together. If the RNA molecule added to the system has the following mutations, what intermediate products of the splicing reactions will accumulate? Explain your answer.

- (a) GT at the 5' splice site is deleted.
- (b) A at the branch point is deleted.
- (c) AG at the 3' splice site is deleted.

• **Solution**

(a) The GT sequence at the 5' splice site is required for the attachment of the U1 snRNP and the first cleavage reaction. If this sequence is mutated, cleavage will not take place. Thus, the original pre-mRNA with the intron will accumulate.

(b) After cleavage at the 5' splice site, the 5' end of the intron attaches to the A at the branch point in a transesterification reaction. If the A at the branch point is deleted, no lariat structure will form. The separated first exon and the intron attached to the second exon will accumulate as intermediate products.

(c) The AG sequence at the 3' splice site is required for cleavage at the 3' splice site. If this sequence is mutated, accumulated intermediate products will be: (1) the separated first exon and (2) the intron attached to the second exon, with the 5' end of the intron attached to the branch point to form a lariat structure.

The New Genetics

MINING GENOMES

RIBOSOMAL RNA STUDIES

Ribosomal RNA is the most plentiful nucleic acid in cells and is widely exploited for a variety of genetic studies. This exercise introduces you to some of the ways that ribosomal RNA

sequences are used and to the collection of tools at the Ribosomal Database Project II, managed by Michigan State University's Center for Microbial Ecology.

COMPREHENSION QUESTIONS

- * 1. What is the concept of colinearity? In what way is this concept fulfilled in bacterial and eukaryotic cells?
- 2. What are some characteristics of introns?
- * 3. What are the four basic types of introns? In which genes are they found?
- * 4. What are the three principal elements in mRNA sequences in bacterial cells?
- 5. What is the function of the Shine-Dalgarno consensus sequence?
- * 6. (a) What is the 5' cap? (b) How is the 5' cap added to eukaryotic pre-mRNA? (c) What is the function of the 5' cap?
- 7. How is the poly(A) tail added to pre-mRNA? What is the purpose of the poly(A) tail?
- * 8. What makes up the spliceosome? What is the function of the spliceosome?
- 9. Explain the process of pre-mRNA splicing in nuclear genes.
- 10. Describe two types of alternative processing pathways. How do they lead to the production of multiple proteins from a single gene?
- *11. What is RNA editing? Explain the role of guide RNAs in RNA editing.
- *12. Summarize the different types of processing that can take place in pre-mRNA.
- *13. What are some of the modifications in tRNA processing?
- 14. Describe the basic structure of ribosomes in bacterial and eukaryotic cells
- *15. Explain how rRNA is processed.

APPLICATION QUESTIONS AND PROBLEMS

- *16. At the beginning of the chapter, we considered Duchenne muscular dystrophy and the dystrophin gene. We learned that the gene causing Duchenne muscular dystrophy encompasses more than 2 million nucleotides, but less than 1% of the gene encodes the protein dystrophin. On the basis of what you now know about gene structure and RNA processing in eukaryotic cells, provide a possible explanation for the large size of the dystrophin gene.
- 17. How do the mRNA of bacterial cells and the pre-mRNA of eukaryotic cells differ? How do the mature mRNAs of bacterial and eukaryotic cells differ?

- *18. Draw a typical eukaryotic gene and the pre-mRNA and mRNA derived from it. Assume that the gene contains three exons. Identify the following items and, for each item, give a brief description of its function:
- | | |
|-------------------------------|----------------------------|
| (a) 5' untranslated region | (e) 3' untranslated region |
| (b) Promoter | (f) Introns |
| (c) AAUAAA consensus sequence | (g) Exons |
| (d) Transcription start site | (h) Poly(A) tail |
| | (i) 5' cap |
19. How would the deletion of the Shine-Dalgarno sequence affect a bacterial mRNA?
- *20. How would the deletion of the following sequences or features most likely affect a eukaryotic pre-mRNA?
- | |
|-------------------------------|
| (a) AAUAAA consensus sequence |
| (b) 5' cap |
| (c) Poly(A) tail |
21. What would be the most likely effect on the amino acid sequence of a protein of a mutation that occurred in an intron of the gene encoding the protein? Explain your answer.
22. A geneticist induces a mutation in the gene that codes for cleavage and polyadenylation specificity factor (CPSF) in a line of cells growing in the laboratory. What would be the immediate effect of this mutation on RNA molecules in the cultured cells?
- *23. A geneticist mutates the gene for proteins that bind to the poly(A) tail in a line of cells growing in the laboratory. What would be the immediate effect of this mutation in the cultured cells?
- *24. An in vitro (within a test tube) splicing system has been developed that contains all the components (snRNAs, proteins, splicing factors) necessary for the splicing of nuclear pre-mRNA genes. When a piece of RNA containing an intron and two exons is added to the system, the intron is removed as a lariat and the exons are spliced together. What intermediate products of the splicing reaction would accumulate if the following components were omitted from the splicing system? Explain your reasoning.
- | | | |
|--------|--------|--------|
| (a) U1 | (c) U6 | (e) U4 |
| (b) U2 | (d) U5 | |
25. The splicing system introduced in Problem 24 is used to splice an RNA molecule containing two exons and one intron. This time, however, the U2 snRNA used in the splicing reaction contains several mutations in the sequence that pairs with the U6 snRNA. What would be the effect of these mutations on the splicing process?
26. A geneticist isolates a gene that contains five exons. He then isolates the mature mRNA produced by this gene. After making the DNA single stranded, he mixes the single-stranded DNA and RNA. Some of the single-stranded DNA hybridizes (pairs) with the complementary mRNA. Draw a picture of what the DNA–RNA hybrids would look like under the electron microscope.
27. The chemical reagent psoralen can be used to elucidate nucleic acid structure. This chemical attaches itself to nucleic acids and, on exposure to UV light, forms covalent bonds between closely associated nucleotide sequences. Such cross-links provide information about the proximity of RNA molecules to one another in complex structures.
- Psoralen cross-linking has been used to examine the structure of the spliceosome. In one study, the following cross-linked structures were obtained during splicing. U1, U2, U5, and U6 became cross-linked to pre-mRNA. U2 was cross-linked to U6 and to pre-mRNA. The U1, U5, and U6 cross-links with pre-mRNA were mapped to sequences near the 5' splice site, whereas the U2 snRNA cross-links with pre-mRNA were mapped to the branch site. After splicing, U2, U5, and U6 were cross-linked to the excised lariat.
- Explain these results in regard to what is known about the structure of the spliceosome and how it functions in RNA splicing. (Based on D. A. Wassarman, and J. A. Steitz, 1992, Interactions of small nuclear RNAs with precursor messenger RNA during in vitro splicing, *Science* 257:1918–1925.)

CHALLENGE QUESTIONS

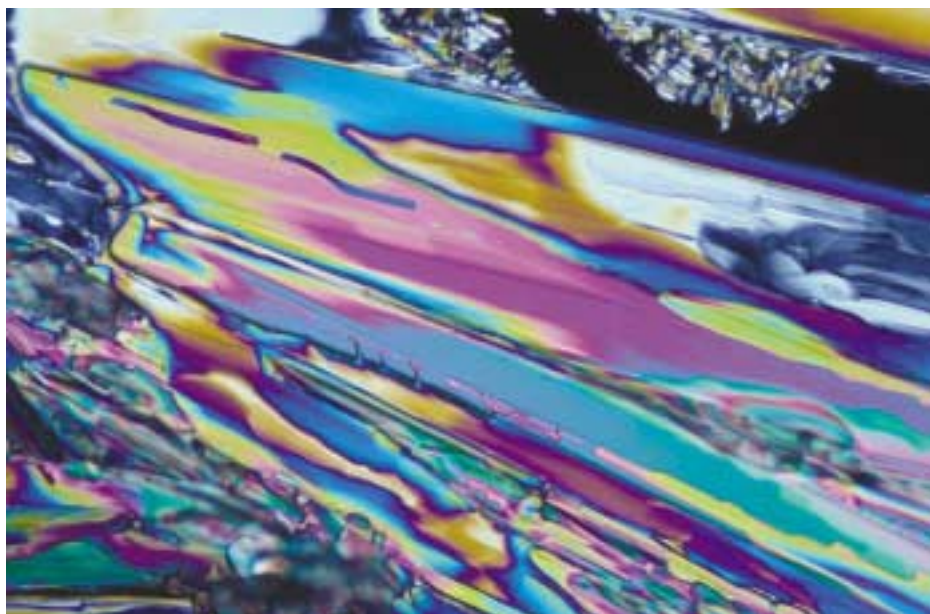
28. In addition to snRNAs, the spliceosome contains a number of proteins. Some of these proteins are associated with the snRNAs to form snRNPs. Other proteins are associated with the spliceosome but are not associated with any specific snRNA.
- One group of spliceosomal proteins comprises the precursor RNA-processing (PRP) proteins. Three PRP proteins that directly take part in splicing are PRP2, PRP16, and PRP22. The results of studies have shown that PRP2 is required for the first step of the splicing reaction, PRP16 acts at the second step, and PRP22 is required for the release of the mRNA from the spliceosome. Other studies have found that these PRP proteins have amino acid sequences similar to the sequences found in RNA helicase enzymes—enzymes that are capable of unwinding two paired RNA molecules. On the basis of this information, propose a functional role for PRP2, PRP16, and PRP22 in RNA splicing.
29. Propose a scenario by which spliceosomal-mediated splicing might have evolved from the splicing of group II introns.

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15

The Genetic Code and Translation



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(Phil A. Harrington/Peter Arnold).

Lesch-Nyhan Syndrome and the Relation Between Genotype and Phenotype

In 1962, Dr. William Nyhan and his student Michael Lesch examined a seriously ill boy with a strange combination of symptoms. The boy had blood in his urine, high concentrations of uric acid in his blood, and uncontrollable spasms in his arms and legs. He was mentally retarded and self-destructively bit his fingers and lips. After carefully studying the boy, Nyhan and Lesch came to the conclusion that he was afflicted by an undescribed disease. Soon other patients with similar symptoms were reported, and the disease became known as the Lesch-Nyhan syndrome.

- Lesch-Nyhan Syndrome and the Relation Between Genotype and Phenotype
 - The Molecular Relation Between Genotype and Phenotype
 - The One Gene, One Enzyme Hypothesis
 - The Structure and Function of Proteins
- The Genetic Code
 - Breaking the Genetic Code
 - The Degeneracy of the Code
 - The Reading Frame and Initiation Codons
 - Termination Codons
 - The Universality of the Code
- The Process of Translation
 - The Binding of Amino Acids to Transfer RNAs
 - The Initiation of Translation
 - Elongation
 - Termination
 - The Energy of Protein Connecting Concepts
 - RNA-RNA Interactions in Translation
 - Polyribosomes
 - The Posttranslational Modifications of Proteins
 - Translation and Antibiotics

One of the earliest symptoms of Lesche-Nyhan syndrome is the appearance of orange “sand”—actually uric acid crystals—in diapers a few weeks after birth. Within a year, the child begins to exhibit writhing movements of the hands and feet and involuntary spasms. About half of the children have seizures. After 2 or 3 years, some of the children exhibit compulsive self-mutilation—biting fingers, lips, the tongue, and the insides of the mouth.

Lesch-Nyhan syndrome develops almost exclusively in boys, and the trait is inherited as an X-linked recessive disorder; the presence of a defective gene on a male’s single X chromosome causes the disease. In 1967, a team of scientists at the National Institutes of Health determined that the disease results from a defective copy of the gene that normally encodes the enzyme hypoxanthine-guanine phos-

phoribosyl transferase (HGPRT). As DNA and RNA are degraded in the cell, purines are liberated, and HGPRT salvages these purines and uses them to synthesize new RNA and DNA nucleotides. In people who have Lesch-Nyhan syndrome, a mutation in the gene for HGPRT changes the amino acid sequence of the enzyme, rendering it nonfunctional. The result is that purines are not recycled; they accumulate and are converted into uric acid. High levels of uric acid produce the symptoms of the disease.

Lesch-Nyhan syndrome illustrates the link between genotype and phenotype: a mutation in a gene affects a protein, which then produces the symptoms of the disease. Preceding chapters described how DNA encodes genetic information and how that information is transferred from DNA to RNA. In this chapter, we examine translation, the process by which the nucleotide sequence in mRNA specifies the amino acid sequence of a protein.

We begin by examining the molecular relation between genotype and phenotype. As with Lesch-Nyhan syndrome, the final phenotype may be complex, including biochemical, physiological, and behavioral traits, but it is ultimately caused by the protein that the gene encodes. We next study the genetic code—the instructions that specify the amino acid sequence of a protein—and then examine the mechanism of protein synthesis. Our primary focus will be on protein synthesis in bacterial cells, but we will highlight some of the differences in eukaryotic cells.

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More information on

Lesch-Nyhan syndrome

The Molecular Relation Between Genotype and Phenotype

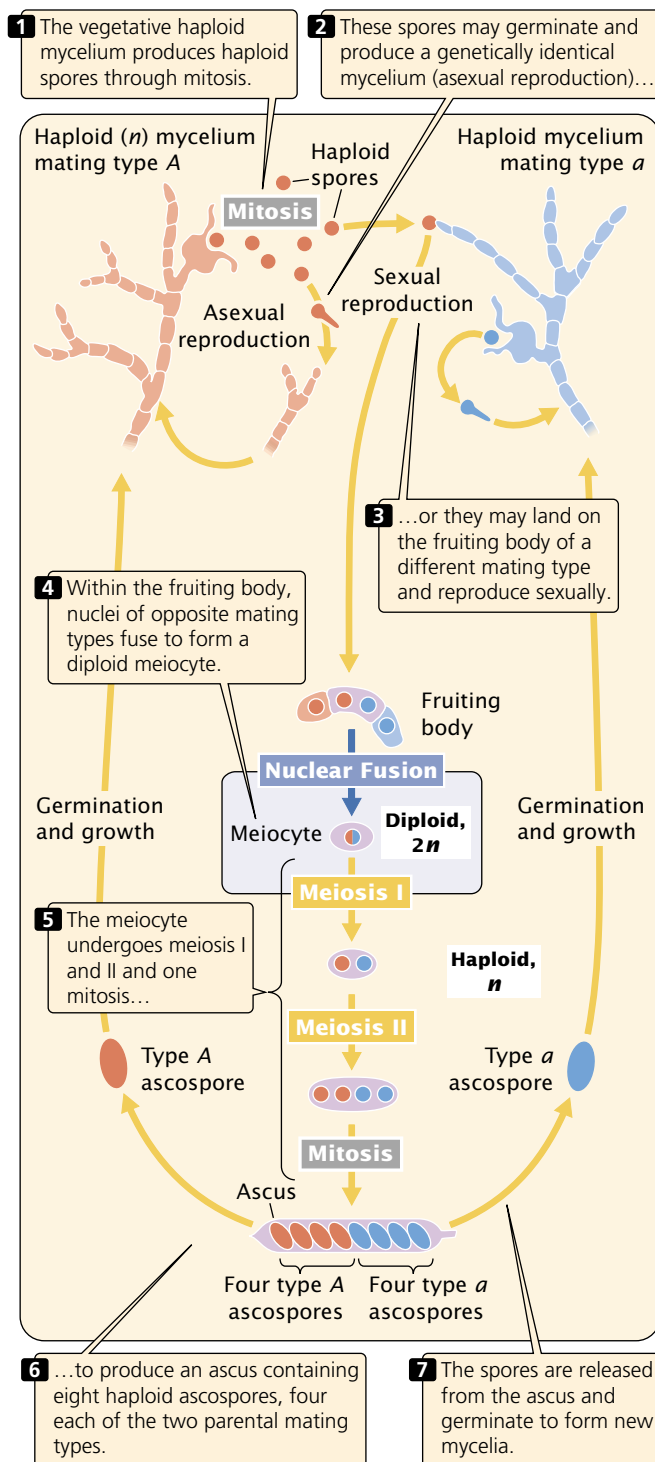
The first person to suggest the existence of a relation between genotype and proteins was Archibald Garrod. In 1908, Garrod correctly proposed that genes encode enzymes (see p. 000), but, unfortunately, his theory made little impression on his contemporaries. Not until the 1940s, when George Beadle and Edward Tatum examined the genetic basis of biochemical pathways in *Neurospora*, did the relation between genes and proteins become widely accepted.

The One Gene, One Enzyme Hypothesis

Beadle and Tatum used the bread mold *Neurospora* to study the biochemical result of mutations. *Neurospora* is a good model organism because it is easy to cultivate in the laboratory and because the main vegetative part of the fungus is haploid; the haploid state allows the effects of recessive mutations to be easily observed (FIGURE 15.1).

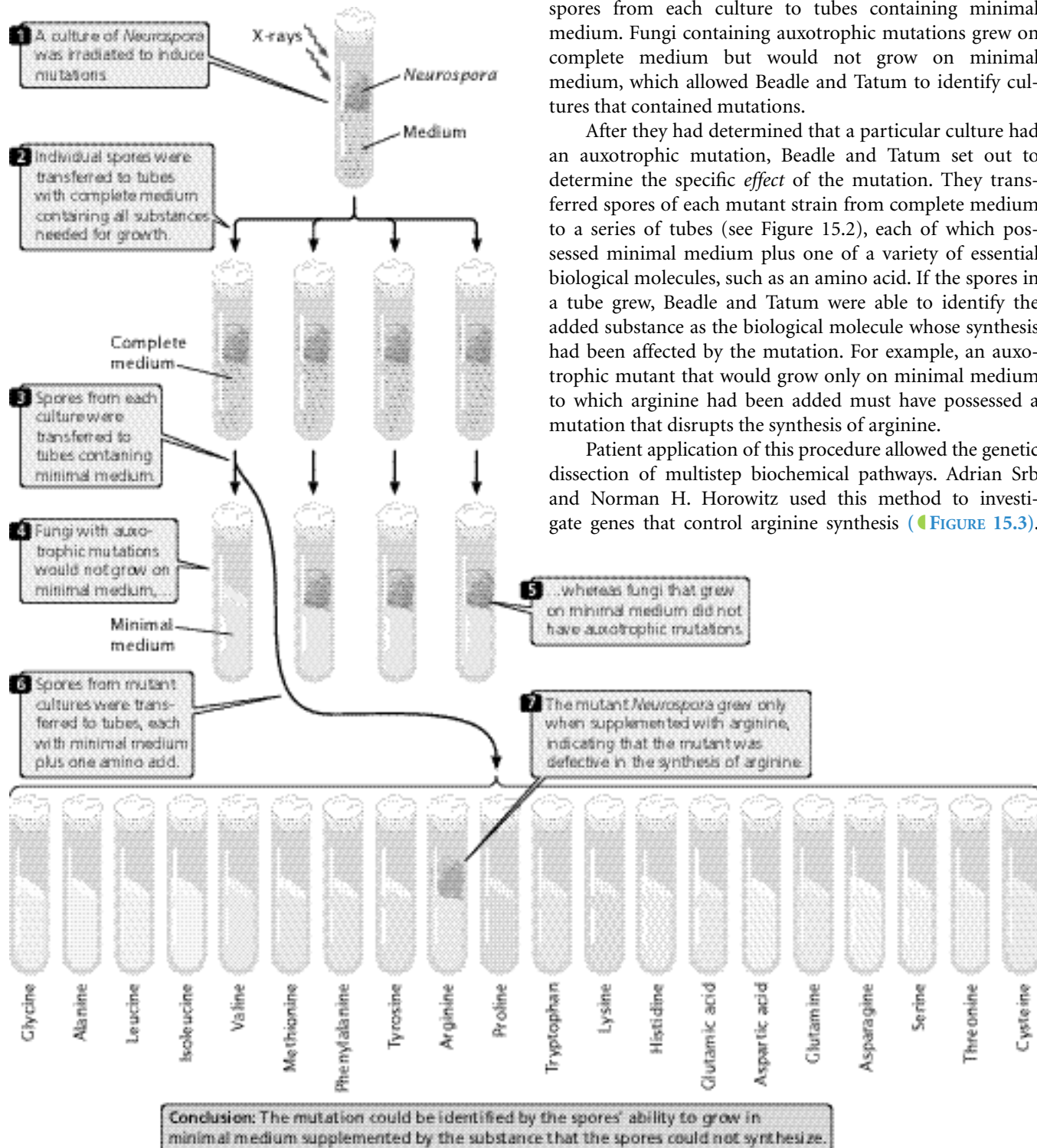
Wild-type *Neurospora* grows on minimal medium, which contains only inorganic salts, nitrogen, a carbon source such as sucrose, and the vitamin biotin. The fungus

can synthesize all the biological molecules that it needs from these basic compounds. However, mutations may arise that disrupt fungal growth by destroying the fungus's ability to synthesize one or more essential biological molecules.



15.1 Beadle and Tatum used the fungus *Neurospora*, which has a complex life cycle, to work out the relation of genes to proteins.

These nutritionally deficient mutants, termed **auxotrophs**, will not grow on minimal medium, but they *can* grow on medium that contains the substance that they are no longer able to synthesize.



Beadle and Tatum first irradiated spores of *Neurospora* to induce mutations (FIGURE 15.2). After irradiation, each spore was placed into a different culture tube with complete medium (medium containing all the biological substances needed for growth (see Figure 15.2). Next, they transferred spores from each culture to tubes containing minimal medium. Fungi containing auxotrophic mutations grew on complete medium but would not grow on minimal medium, which allowed Beadle and Tatum to identify cultures that contained mutations.

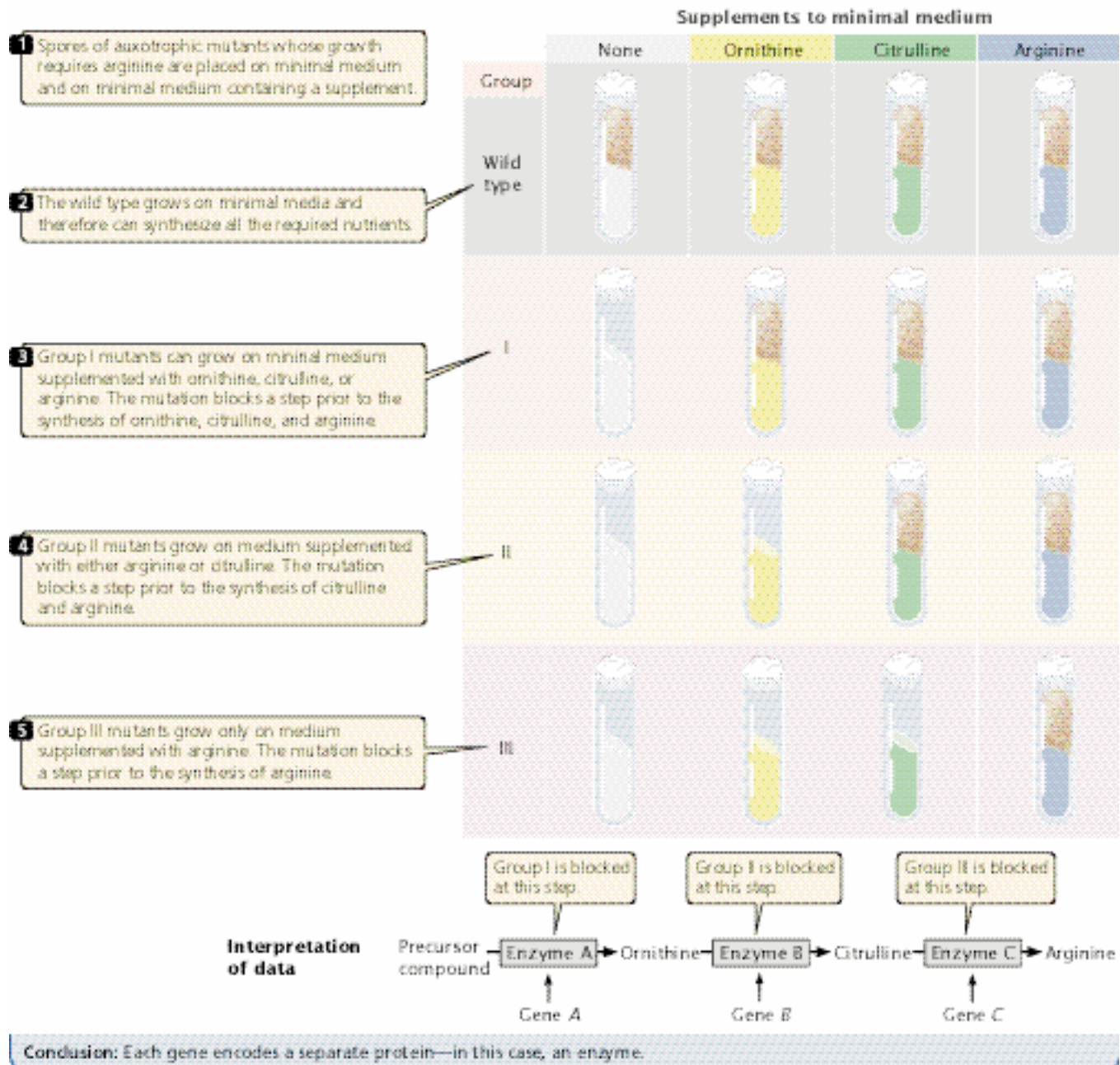
After they had determined that a particular culture had an auxotrophic mutation, Beadle and Tatum set out to determine the specific *effect* of the mutation. They transferred spores of each mutant strain from complete medium to a series of tubes (see Figure 15.2), each of which possessed minimal medium plus one of a variety of essential biological molecules, such as an amino acid. If the spores in a tube grew, Beadle and Tatum were able to identify the added substance as the biological molecule whose synthesis had been affected by the mutation. For example, an auxotrophic mutant that would grow only on minimal medium to which arginine had been added must have possessed a mutation that disrupts the synthesis of arginine.

Patient application of this procedure allowed the genetic dissection of multistep biochemical pathways. Adrian Srb and Norman H. Horowitz used this method to investigate genes that control arginine synthesis (FIGURE 15.3).

15.2 Beadle and Tatum developed a method for isolating auxotrophic mutants in *Neurospora*.

Experiment

Question: What do the effects of genetic mutation on a biochemical pathway tell us about the gene-protein relation?



15.3 Beadle and Tatum established that each step in a pathway is controlled by a different enzyme. This biochemical pathway leads to the synthesis of arginine in *Neurospora*. Steps in the pathway are catalyzed by enzymes affected by mutants.

They first isolated a series of auxotrophic mutants whose growth required arginine. They then tested these mutants for their ability to grow on minimal medium supplemented with three compounds: ornithine, citrulline, and arginine. From the results, they were able to place the mutants

into three groups (Table 15.1) based on which of the substances allowed growth. Group I mutants grew on minimal medium supplemented with ornithine, citrulline, or arginine. Group II mutants grew on minimal medium supplemented with either arginine or citrulline but did not grow on

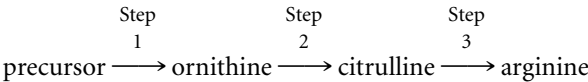
Table 15.1 Growth of arginine auxotrophic mutants on minimal medium with various supplements

Mutant Strain Number	Ornithine	Citrulline	Arginine
Group I	+	+	+
Group II	–	+	+
Group III	–	–	+

Note: + indicates growth; – indicates no growth.

medium supplemented only with ornithine. Finally, group III mutants grew only on medium supplemented with arginine.

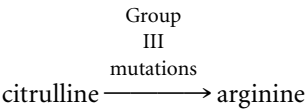
Srb and Horowitz therefore proposed that the biochemical pathway leading to the amino acid arginine has at least three steps:



They concluded that the mutations in group I affected step 1 of this pathway, mutations in group II affected step 2, and mutations in group III affected step 3. But how did they know that the order of the compounds in the biochemical pathway was correct?

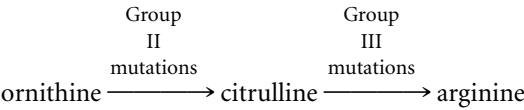
Notice that, if step 1 is blocked by a mutation, then the addition of either ornithine or citrulline allows growth, because these compounds can still be converted into arginine (see Figure 15.3). Similarly, if step 2 is blocked, the addition of citrulline allows growth, but the addition of ornithine has no effect. If step 3 is blocked, the spores will grow only if arginine is added to the medium. The underlying principle is that an auxotrophic mutant cannot synthesize any compound that comes after the step blocked by a mutation.

Using this reasoning with the information in Table 15.1, we can see that the addition of arginine to the medium allows all three groups of mutants to grow. Therefore, biochemical steps affected by all the mutants precede the step that results in arginine. The addition of citrulline allows group I and group II mutants to grow but not group III mutants; therefore, group III mutations must affect a biochemical step that takes place after the production of citrulline but before the production of arginine.

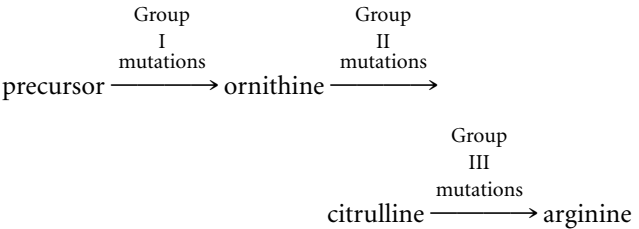


The addition of ornithine allows the growth of group I mutants but not group II or group III mutants; thus, muta-

tions in groups II and III affect steps that come after the production of ornithine. We've already established that group II mutations affect a step *before* the production of citrulline; so group II mutations must block the conversion of ornithine into citrulline.



Because group I mutations affect some step before the production of ornithine, we can conclude that they must affect the conversion of some precursor into ornithine. We can now outline the biochemical pathway yielding ornithine, citrulline, and arginine.



It is important to note that this procedure does not necessarily detect all steps in a pathway; rather, it detects only the steps producing the compounds tested.

Using mutations and this type of reasoning, Beadle and Tatum were able to identify genes that control several biosynthetic pathways in *Neurospora*. They established that each step in a pathway is controlled by a different enzyme, as shown in Figure 15.3 for the arginine pathway. The results of genetic crosses and mapping studies demonstrated that mutations affecting any one step in a pathway always map to the same chromosomal location. Beadle and Tatum reasoned that mutations affecting a particular biochemical step occurred at a single locus that encoded a particular enzyme. This idea became known as the **one gene, one enzyme hypothesis**: genes function by encoding enzymes, and each gene encodes a separate enzyme. When research showed that some proteins are composed of more than one polypeptide chain and that different polypeptide chains are encoded by separate genes, this model was modified to become the **one gene, one polypeptide hypothesis**.

Concepts

Beadle and Tatum's studies of biochemical pathways in the fungus *Neurospora* established the one gene, one enzyme hypothesis, the idea that each gene encodes a separate enzyme. This hypothesis was later modified to become the one gene, one polypeptide hypothesis.

(a)



(b)



(c)



15.4 Proteins serve a number of biological functions and are central to all living processes.

(a) The light produced by fireflies is the result of a light-producing reaction between luciferin and ATP catalyzed by the enzyme luciferase. (b) The protein fibroin is the major structural component of spider webs. (c) Castor beans contain a highly toxic protein called ricin. (Part a, Gregory K. Scott/Photo Researchers; part b, A. Shay/Animals Animals; part c, Gerald & Buff Corsi/Visuals Unlimited).

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Further information about the use of *Neurospora* in genetics

The Structure and Function of Proteins

Proteins are central to all living processes (FIGURE 15.4). Many proteins are enzymes, the biological catalysts that drive the chemical reactions of the cell; others are structural components, providing scaffolding and support for membranes, filaments, bone, and hair. Some proteins help transport substances; others have a regulatory, communication, or defense function.

All proteins are composed of **amino acids**, linked end to end. There are 20 common amino acids found in proteins; these amino acids are shown in FIGURE 15.5 with both their three- and one-letter abbreviations. (Other amino acids that are sometimes found in proteins are modified forms of the common amino acids.) The 20 common amino acids are similar in structure, differing only in the structures of the R (radical) groups. The amino acids in proteins are joined by **peptide bonds** (FIGURE 15.6) to form **polypeptide** chains, and a protein consists of one or more polypeptide chains. Like nucleic acids, polypeptides have polarity with one end having a free amino group (NH_2) and the other end possessing a free carboxyl group (CO_2H). Some proteins consist of only a few amino acids, whereas others may have thousands.

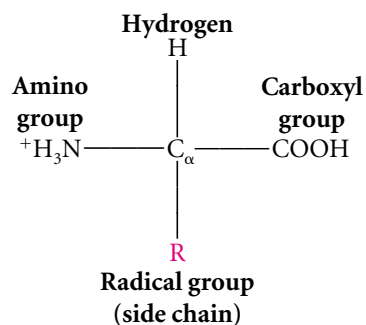
Like that of nucleic acids, the molecular structure of proteins has several levels of organization. The *primary structure* of a protein is its sequence of amino acids (FIGURE 15.7a). Through interactions between neighboring amino acids, a polypeptide chain folds and twists into a *secondary structure* (FIGURE 15.7b); two common secondary structures found in proteins are the beta (β) pleated sheet and the alpha (α) helix. Secondary structures interact and fold further to form a *tertiary structure* (FIGURE 15.7c), which is the overall, three-dimensional shape of the protein. The secondary and tertiary structures of a protein are ultimately determined by the primary structure—the amino acid sequence—of the protein. Finally, some proteins consist of two or more polypeptide chains that associate to produce a *quaternary structure* (FIGURE 15.7d).

Concepts

The product of many genes is a protein, whose action produces the trait encoded by that gene. Proteins are polymers, consisting of amino acids linked by peptide bonds. The amino acid sequence of a protein is its primary structure. This structure folds to create the secondary and tertiary structures; two or more polypeptide chains may associate to create a quaternary structure.

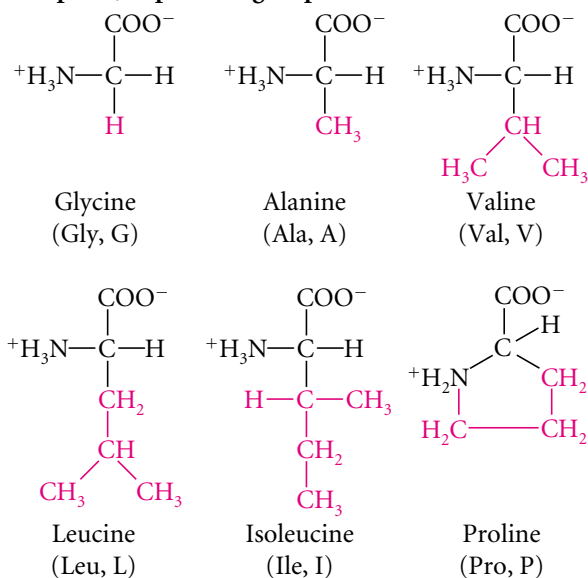
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More information on protein structure

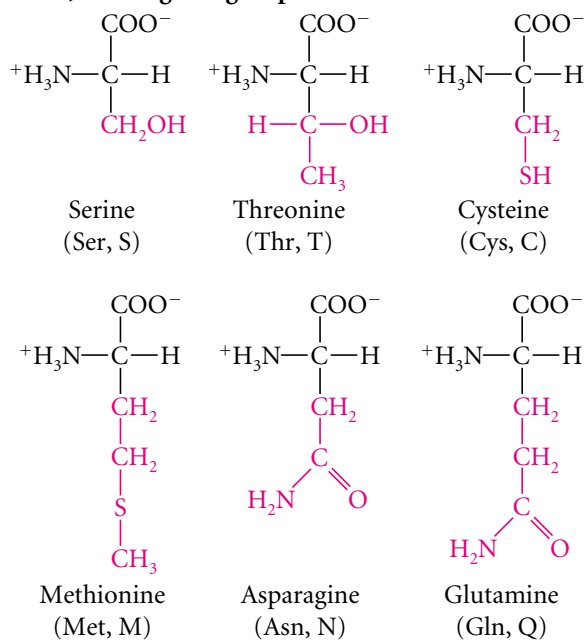


15.5 The common amino acids have similar structures. Each amino acid consists of a central (α) carbon atom attached to: (1) an amino group (NH_3^+); (2) a carboxyl group (COO^-); (3) a hydrogen atom (H); and (4) a radical group, designated R. In the structures of the 20 common amino acids, the parts in black are common to all amino acids and the parts in red are the R groups.

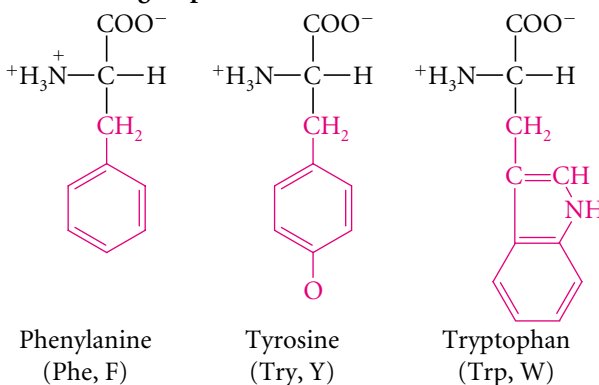
Nonpolar, aliphatic R groups



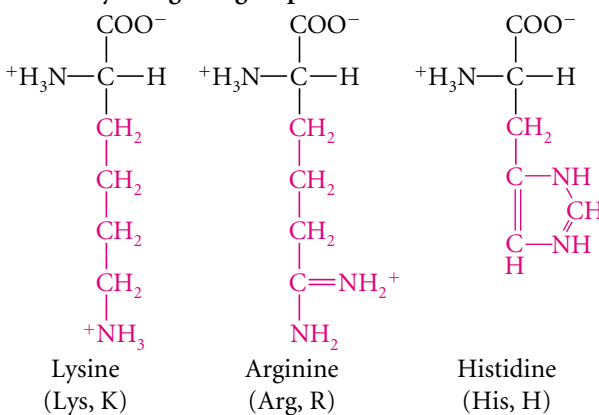
Polar, uncharged R groups



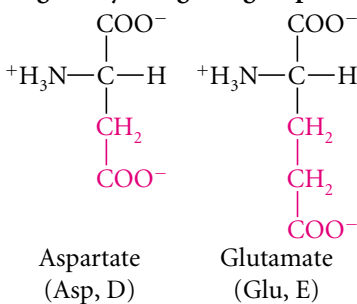
Aromatic R groups

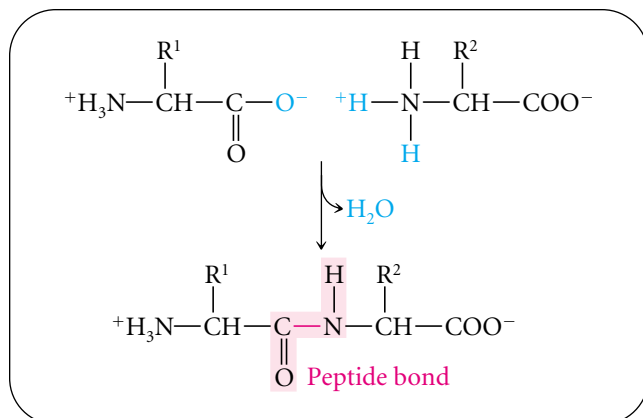


Positively charged R groups

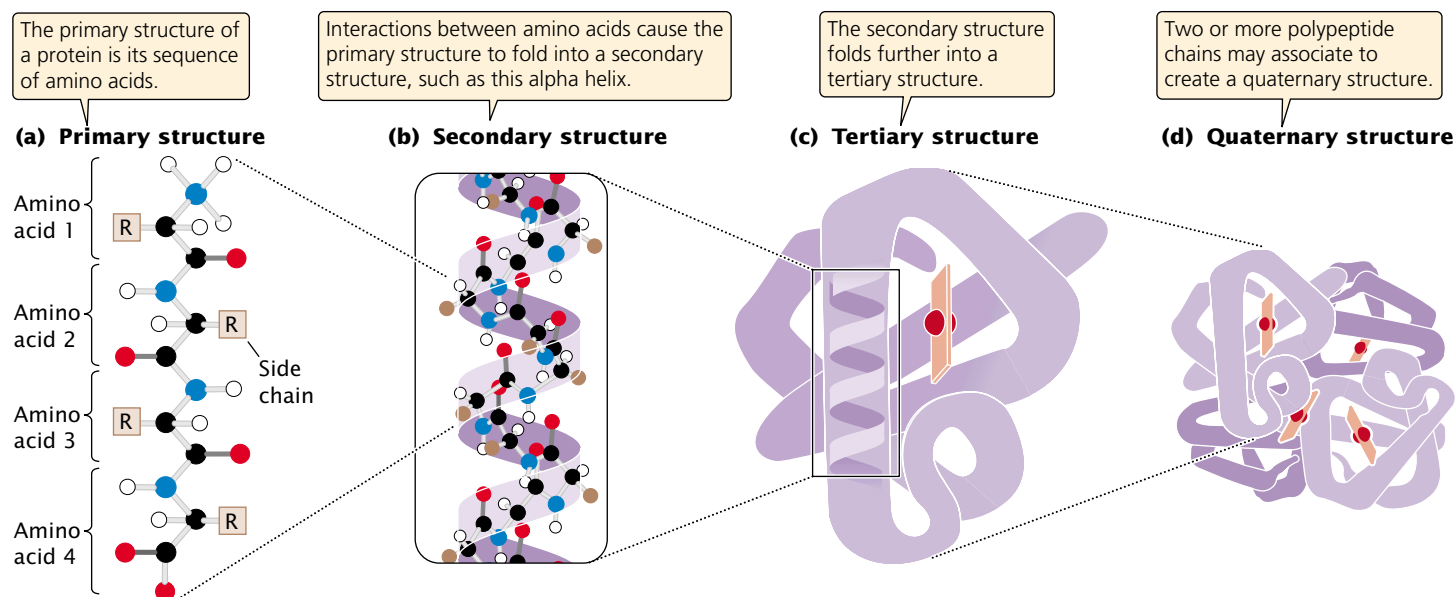


Negatively charged R groups





15.6 Amino acids are joined together by peptide bonds. In a peptide bond, the carboxyl group of one amino acid is covalently attached to the amino group of another amino acid.



15.7 Proteins have several levels of structural organization.

The Genetic Code

In 1953, Watson and Crick solved the structure of DNA and identified the base sequence as the carrier of genetic information. However, the way in which the base sequence of DNA specified the amino acid sequences of proteins (the genetic code) was not immediately obvious and remained elusive for another 10 years.

One of the first questions about the genetic code to be addressed was: *How many nucleotides are necessary to specify a single amino acid?* This basic unit of the genetic code—the set of bases that encode a single amino acid—is a *codon* (Chapter 14). Many early investigators recognized that codons must contain a minimum of three nucleotides. Each nucleotide position in mRNA can be occupied by one of four bases: A, G, C, or U. If a codon consisted of a single nucleotide, only four different codons (A, G, C, and U) would be possible, which is not enough to code for the 20 different

amino acids commonly found in proteins. If codons were made up of two nucleotides each (i.e., GU, AC, etc.) there would be $4 \times 4 = 16$ possible codons—still not enough to code for all 20 amino acids. With three nucleotides per codon, there are $4 \times 4 \times 4 = 64$ possible codons, which is more than enough to specify 20 different amino acids. Therefore, a *triplet code* requiring three nucleotides per codon is the most efficient way to encode all 20 amino acids. Using mutations in bacteriophage, Francis Crick and his colleagues confirmed in 1961 that the genetic code is indeed a triplet code.

Concepts

The genetic code is a triplet code, in which three nucleotides code for each amino acid in a protein.

www.whfreeman.com/pierce An electronic table of codons and the amino acids they specify

Breaking the Genetic Code

When it had been firmly established that the genetic code consists of codons that are three nucleotides in length, the next step was to determine which groups of three nucleotides specify which amino acids. This task required the development of a cell-free system for protein synthesis (FIGURE 15.8), which would make it possible to study the translation of a known mRNA.

Logically, the easiest way to break the code would have been to determine the base sequence of a piece of RNA, add it to a cell-free protein-synthesizing system, and allow it to direct the synthesis of a protein. The amino acid sequence

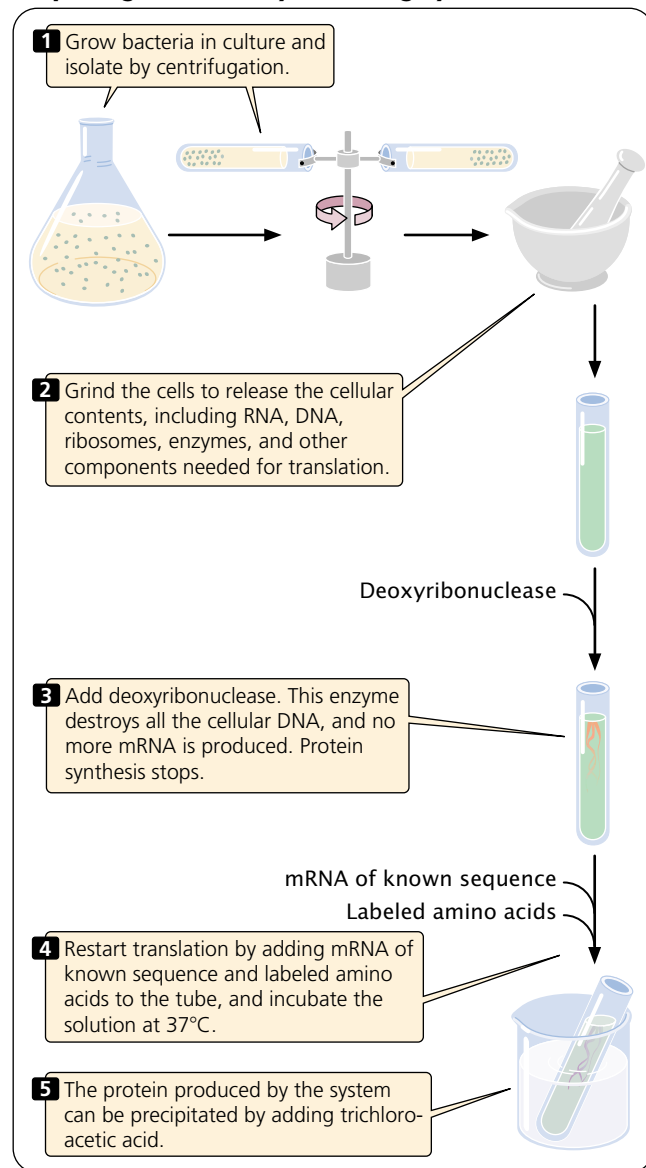
of the newly synthesized protein could then be determined, and its sequence could be compared with that of the RNA. Unfortunately, there was no way at that time to determine the nucleotide sequence of a piece of RNA; so indirect methods were necessary to break the code.

The first clues to the genetic code came in 1961, from the work of Marshall Nirenberg and Johann Heinrich Matthaei. These investigators created synthetic RNAs by using an enzyme called polynucleotide phosphorylase. Unlike RNA polymerase, polynucleotide phosphorylase does not require a template; it randomly links together any RNA nucleotides that happen to be available. The first synthetic mRNAs used by Nirenberg and Matthaei were homopolymers, RNA molecules consisting of a single type of nucleotide. For example, by adding polynucleotide phosphorylase to a solution of uracil nucleotides, they generated RNA molecules that consisted entirely of uracil nucleotides and thus contained only UUU codons (FIGURE 15.9). These poly(U) RNAs were then added to 20 tubes, each containing a cell-free protein-synthesizing system and the 20 different amino acids, one of which was radioactively labeled. Translation took place in all 20 tubes, but radioactive protein appeared in only one of the tubes—the one containing labeled phenylalanine (see Figure 15.9). This result showed that the codon UUU specifies the amino acid phenylalanine. The results of similar experiments using poly(C) and poly(A) RNA demonstrated that CCC codes for proline and AAA codes for lysine; for technical reasons, the results from poly(G) were uninterpretable.

To gain information about additional codons, Nirenberg and his colleagues created synthetic RNAs containing two or three different bases. Because polynucleotide phosphorylase incorporates nucleotides randomly, these RNAs contained random mixtures of the bases and are thus called random copolymers. For example, when adenine and cytosine nucleotides are mixed with polynucleotide phosphorylase, the RNA molecules produced have eight different codons: AAA, AAC, ACC, ACA, CAA, CCA, CAC, and CCC. In cell-free protein-synthesizing systems, these poly(AC) RNAs produced proteins containing six different amino acids: asparagine, glutamine, histidine, lysine, proline, and threonine.

The proportions of the different amino acids in the proteins depended on the ratio of the two nucleotides used in creating the synthetic mRNA, and the theoretical probability of finding a particular codon could be calculated from the ratios of the bases. If a 4:1 ratio of C to A were used in making the RNA, then the probability of C occurring at any given position in a codon is $\frac{4}{5}$ and the probability of A being in it is $\frac{1}{5}$. With random incorporation of bases, the probability of any one of the codons with two Cs and one A (CCA, CAC, or ACC) should be $\frac{4}{5} \times \frac{4}{5} \times \frac{1}{5} = \frac{16}{125} = 0.13$, or 13%, and the probability of any codon with two As and one C (AAC, ACA, or CAA) should be $\frac{1}{5} \times \frac{1}{5} \times \frac{4}{5} = \frac{4}{125} = 0.032$, or about 3%. Therefore, an amino acid encoded by

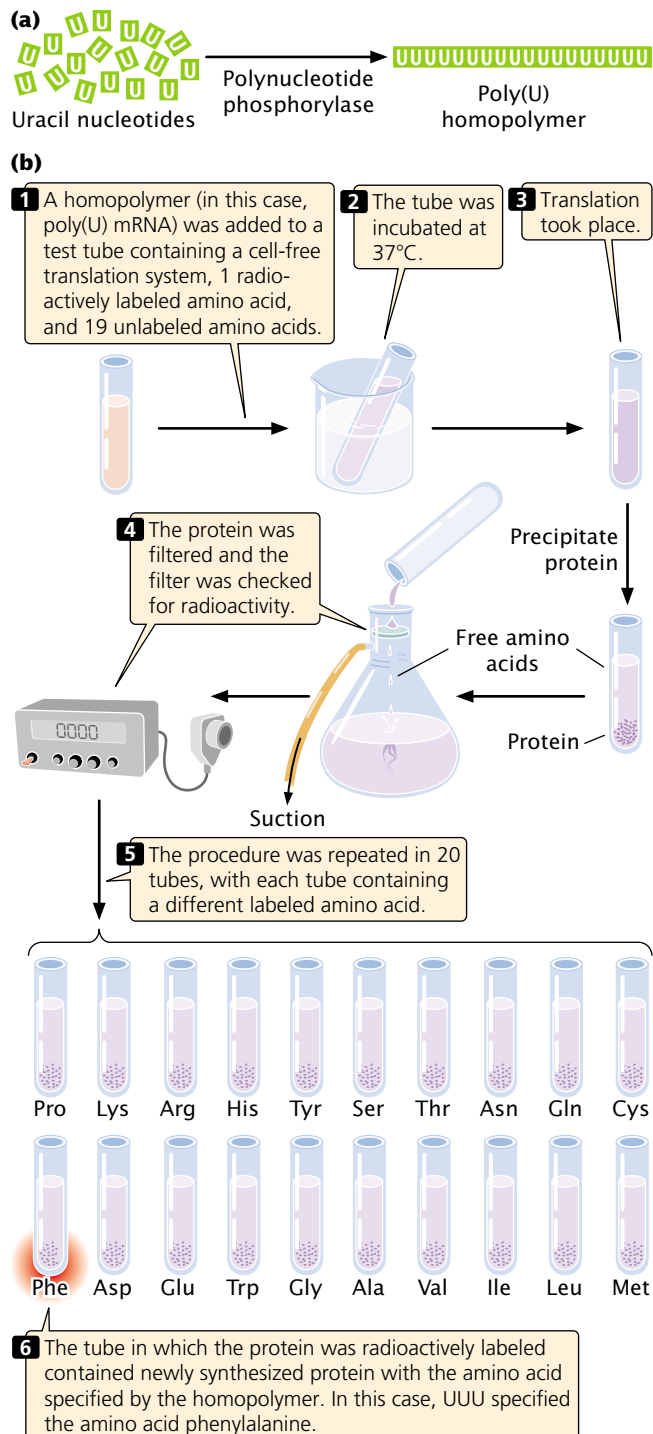
Preparing a cell-free synthesizing system



15.8 Breaking the genetic code required a cell-free protein-synthesizing system.

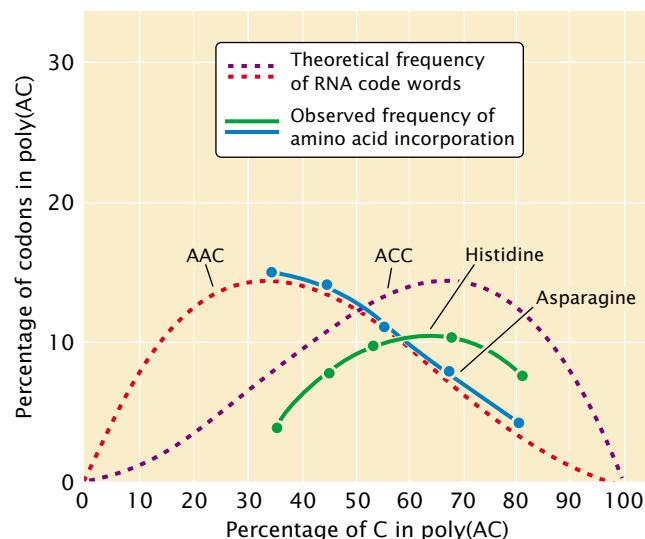
Experiment

Question: What amino acids are specified by codons composed of only one type of base?



Conclusion: UUU codes for phenylalanine; in other experiments, AAA was found to code for alanine, and CCC for proline.

15.9 Nirenberg and Matthaei developed a method for identifying the amino acid specified by a homopolymer.



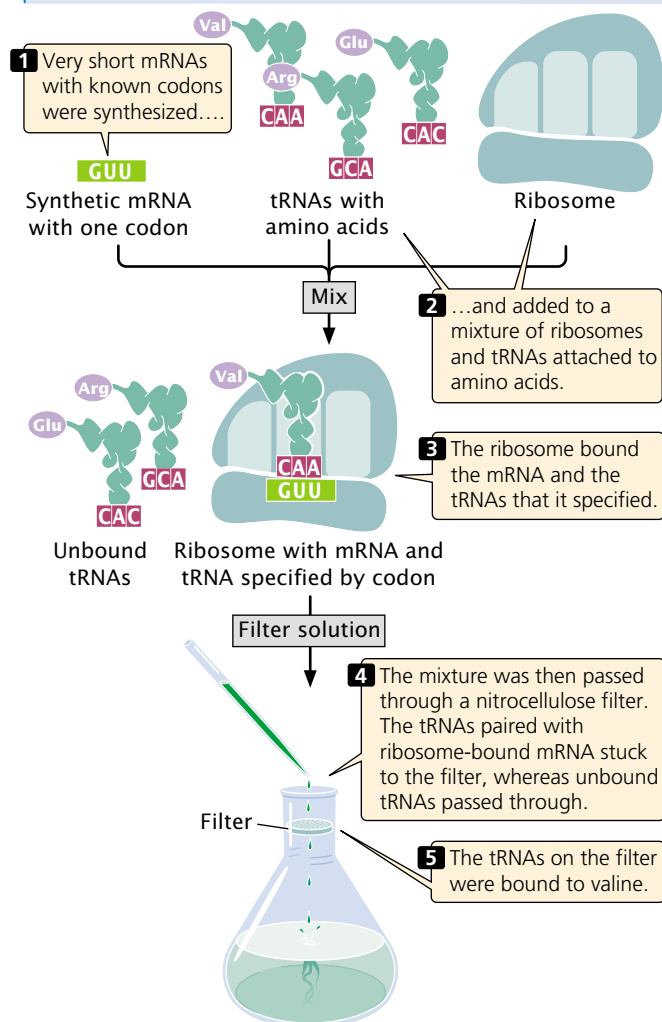
15.10 Nirenberg and Matthaei's use of random copolymers provided information about the genetic code. The theoretical percentage of codons (vertical axis) is plotted against various percentages of cytosine in random AC copolymers. Notice that the distribution of histidine approximates the theoretical percentage of a codon with two Cs and one A, whereas the distribution of asparagine approximates the percentage expected of a codon with two As and one C (Credit to come-allowed 1 line).

two Cs and one A should be more common than an amino acid encoded by two As and one C. By comparing the percentages of amino acids in proteins produced by random copolymers with the theoretical frequencies expected for the codons (FIGURE 15.10), information about the base composition of the codons was derived. Findings from these experiments revealed nothing, however, about the codon base sequence; histidine was clearly encoded by a codon with two Cs and one A (see Figure 15.10), but whether that codon was ACC, CAC, or CCA was unknown. There were other problems with this method: the theoretical calculations depended on the random incorporation of bases, which did not always occur, and, because the genetic code is redundant, sometimes several different codons specify the same amino acid.

To overcome the limitations of random copolymers, Nirenberg and Philip Leder developed another technique in 1964 that used ribosome-bound tRNAs. They found that a very short sequence of mRNA—even one consisting of a single codon—would bind to a ribosome. The codon on the short mRNA would then base pair with the matching anticodon on a tRNA that carries the amino acid specified by the codon (FIGURE 15.11). The ribosome-bound mRNA was mixed with tRNAs and amino acids, and this mixture was passed through a nitrocellulose filter. The tRNAs paired with the ribosome-bound mRNA stuck to the filter, whereas unbound tRNAs passed through. The advan-

Experiment

Question: With the use of tRNAs, what other matches between codon and amino acid could be determined?



Conclusion: The codon GUU specifies valine; many other codons were determined by using this method.

15.11 Nirenberg and Leder developed a technique for using ribosome-bound tRNAs to provide additional information about the genetic code.

stage of this system was that it could be used with very short synthetic mRNA molecules that could be synthesized with a known sequence. Nirenberg and Leder synthesized over 50 short mRNAs with known codons and added them individually to a mixture of ribosomes and tRNAs. They then isolated the ribosome-bound tRNAs and determined which amino acids were present on the bound tRNAs. For example, synthetic RNA with the codon GUU retained a tRNA to which valine was attached, whereas RNAs with the codons UGU and UUG did not. Using this method, Nirenberg and his colleagues were able to determine the amino acids encoded by more than 50 codons.

A third method provided additional information about the genetic code. Gobind Khorana and his colleagues used chemical techniques to synthesize RNA molecules that contained known repeating sequences. They hypothesized that an mRNA that contained, for instance, alternating uracil and guanine nucleotides (UGUG UGUG) would be read during translation as two alternating codons, UGU GUG UGU GUG, producing a protein composed of two alternating amino acids. When Khorana and his colleagues placed this synthetic mRNA in a cell-free protein-synthesizing system, it produced a protein made of alternating cysteine and valine residues. This technique could not determine which of the two codons (UGU or GUG) specified cysteine, but, combined with other methods, it made a crucial contribution to cracking the genetic code. The genetic code was fully understood by 1968 ( **FIGURE 15.12**). In the next section, we will examine some of the features of the code, which is so important to modern biology that Francis Crick has compared its place to that of the periodic table of the elements in chemistry.

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A brief biography of Marshall

Nirenberg

		Second base				
		U	C	A	G	
First base	U	UUU Phe	UCU	UAU Tyr	UGU Cys	U
		UUC	UCC Ser	UAC	UGC	C
		UUA Leu	UCA	UAA Stop	UGA Stop	A
		UUG	UCG	UAG Stop	UGG Trp	G
	C	CUU	CCU	CAU His	CGU	U
		CUC Leu	CCC Pro	CAC	CGC Arg	C
		CUA	CCA	CAA Gln	CGA	A
		CUG	CCG	CAG	CGG	G
	A	AUU	ACU	AAU Asn	AGU Ser	U
		AUC Ile	ACC Thr	AAC	AGC	C
		AUA	ACA	AAA Lys	AGA Arg	A
		AUG Met	ACG	AAG	AGG	G
	G	GUU	GCU	GAU Asp	GGU	U
		GUC Val	GCC Ala	GAC	GGC Gly	C
		GUA	GCA	GAA Glu	GGA	A
		GUG	GCG	GAG	GGG	G

15.12 The genetic code consists of 64 codons and the amino acids specified by these codons.

The codons are written 5'→3', as they appear in the mRNA. AUG is an initiation codon; UAA, UAG, and UGA are termination codons.

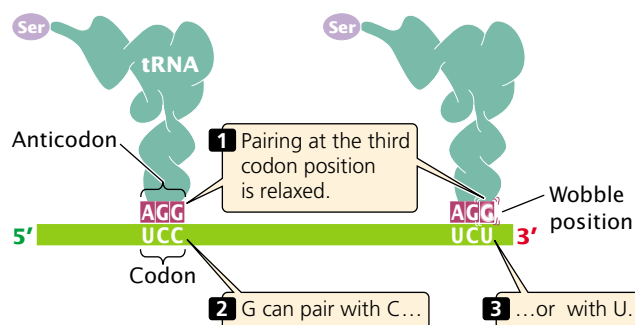
The Degeneracy of the Code

One amino acid is encoded by three consecutive nucleotides in mRNA, and each nucleotide can have one of four possible bases (A, G, C, and U) at each nucleotide position thus permitting $4^3 = 64$ possible codons (see Figure 15.12). Three of these codons are stop codons, specifying the end of translation. Thus, 61 codons, called **sense codons**, code for amino acids. Because there are 61 sense codons and only 20 different amino acids commonly found in proteins, the code contains more information than is needed to specify the amino acids and is said to be a **degenerate code**. This expression does not mean that the genetic code is depraved; *degenerate* is a term that Francis Crick borrowed from quantum physics, where it describes multiple physical states that have equivalent meaning. The degeneracy of the genetic code means that amino acids may be specified by more than one codon. Only tryptophan and methionine are encoded by a single codon (see Figure 15.12). Others amino acids are specified by two codons, and some, such as leucine, are specified by six different codons. Codons that specify the same amino acid are said to be **synonymous**, just as synonymous words are different words that have the same meaning.

Isoaccepting tRNAs As we learned in Chapter 14, tRNAs serve as adapter molecules, binding particular amino acids and delivering them to a ribosome, where the amino acids are then assembled into polypeptide chains. Each type of tRNA attaches to a single type of amino acid. The cells of most organisms possess from about 30 to 50 different tRNAs, and yet there are only 20 different amino acids in proteins. Thus, some amino acids are carried by more than one tRNA. Different tRNAs that accept the same amino acid but have different anticodons are called **isoaccepting tRNAs**. Some synonymous codons code for different isoacceptors.

Wobble Many synonymous codons differ only in the third position (see Figure 15.12). For example, alanine is encoded by the codons GCU, GCC, GCA, and GCG, all of which begin with GC. When the codon on the mRNA and the anticodon of the tRNA join (▶ **FIGURE 15.13**), the first (5') base of the codon pairs with the third base (3') of the anticodon, strictly according to Watson and Crick rules: A with U; C with G. Next, the middle bases of codon and anticodon pair, also strictly following the Watson and Crick rules. After these pairs have hydrogen bonded, the third bases pair weakly—there may be flexibility, or **wobble**, in their pairing.

In 1966, Francis Crick developed the wobble hypothesis, which proposed that some nonstandard pairings of bases could occur at the third position of a codon. For example, a G in the anticodon may pair with either a C or a U in the third position of the codon (Table 15.2). The important thing to remember about wobble is that it allows



15.13 Wobble may exist in the pairing of a codon on mRNA with an anticodon on tRNA. The mRNA and tRNA pair in an antiparallel fashion. Pairing at the first and second codon positions is in accord with the Watson and Crick pairing rules (A with T, G with C); however, pairing rules are relaxed at the third position of the codon, and G on the anticodon can pair with either U or C on the codon in this example.

some tRNAs to pair with more than one codon on an mRNA; thus from 30 to 50 tRNAs can pair with 61 sense codons. Some codons are synonymous through wobble.

Concepts

The genetic code consists of 61 sense codons that specify the 20 common amino acids; the code is degenerate and some amino acids are encoded by more than one codon. Isoaccepting tRNAs are different tRNAs with different anticodons that specify the same amino acid. Wobble exists when more than one codon can pair with the same anticodon.

The Reading Frame and Initiation Codons

Findings from early studies of the genetic code indicated that it is generally **nonoverlapping**. An overlapping code is one in which a single nucleotide is included in more than one codon, as shown in ▶ **FIGURE 15.14**. Usually, however, each nucleotide sequence of an mRNA specifies a single amino acid. A few overlapping codes are found in viruses; in these cases, two different proteins may be encoded within the same sequence of mRNA.

For any sequence of nucleotides, there are three potential sets of codons—three ways that the sequence can be read in groups of three. Each different way of reading the sequence is called a **reading frame**, and any sequence of nucleotides has three potential reading frames. The three reading frames have completely different sets of codons and therefore will specify proteins with entirely different amino acid sequences. Thus, it is essential for the

Table 15.2 The wobble rules, indicating which bases in the third position (3' end) of the mRNA codon can pair with bases at the first (5' end) of the anticodon of the tRNA

First Position of Anticodon	Third Position of Codon	Pairing
C	G	Anticodon 3'-X-Y-C-5' 5'-Y-X-G-3' Codon
G	U or C	Anticodon 3'-X-Y-G-5' 5'-Y-X-U-3' C Codon
A	U	Anticodon 3'-X-Y-A-5' 5'-Y-X-U-3' Codon
U	A or G	Anticodon 3'-X-Y-U-5' 5'-Y-X-A-3' G Codon
I (inosine)	A, U, or C	Anticodon 3'-X-Y-I-5' 5'-Y-X-A-3' U C Codon

translational machinery to use the correct reading frame. How is the correct reading frame established? The reading frame is set by the **initiation codon**, which is the first codon of the mRNA to specify an amino acid. After the initiation codon, the other codons are read as successive groups of three nucleotides. No bases are skipped between the codons; so there are no punctuation marks to separate the codons.

The initiation codon is usually AUG, although GUG and UUG are used on rare occasions. The initiation codon is not just a punctuation mark; it specifies an amino acid. In bacterial cells, AUG encodes a modified type of methionine, *N*-formylmethionine; all proteins in bacteria begin with this amino acid, but the formyl group (or, in some cases, the entire amino acid) may be removed after the protein has been synthesized. When the codon AUG is at an internal

Nucleotide sequence	A U A C G A G U C
Nonoverlapping code	A U A C G A G U C 1 2 3 . . .
Overlapping code	A U A C G A G U C 1 U A C 2 A C G 3

15.14 The genetic code is generally nonoverlapping. In a nonoverlapping code, each nucleotide belongs to only one codon. In an overlapping code, some nucleotides belong to more than one codon. The genetic code used in almost all living organisms is nonoverlapping.

position in a gene, it codes for unformylated methionine. In archaeal and eukaryotic cells, AUG specifies unformylated methionine both at the initiation position and at internal positions.

Termination Codons

Three codons—UAA, UAG, and UGA—do not encode amino acids. These codons signal the end of the protein in both bacterial and eukaryotic cells and are called **stop codons**, **termination codons**, or **nonsense codons**. No tRNA molecules have anticodons that pair with termination codons.

The Universality of the Code

For many years the genetic code was assumed to be **universal**, meaning that each codon specifies the same amino acid in all organisms. We now know that the genetic code is almost, but not completely, universal; a few exceptions have been found. Most of these exceptions are termination codons, but there are a few cases in which one sense codon substitutes for another. The majority of exceptions are found in mitochondrial genes; a few nonuniversal codons have also been detected in nuclear genes of protozoans (Table 15.3).

Concepts

Each sequence of nucleotides possesses three potential reading frames. The correct reading frame is set by the initiation codon. The end of a protein-encoding sequence is marked by a termination codon. With a few exceptions, all organisms use the same genetic code.

Connecting Concepts

Characteristics of the Genetic Code

We have now considered a number of characteristics of the genetic code. Let's pause for a moment and review these characteristics.

1. The genetic code consists of a sequence of nucleotides in DNA or RNA. There are four letters in the code, corresponding to the four bases—A, G, C, and U (T in DNA).
2. The genetic code is a triplet code. Each amino acid is encoded by a sequence of three consecutive nucleotides, called a codon.
3. The genetic code is degenerate—there are 64 codons but only 20 amino acids in proteins. Some codons are synonymous, specifying the same amino acid.
4. Isoaccepting tRNAs are tRNAs with different anticodons that accept the same amino acid; wobble allows the anticodon on one type of tRNA to pair with more than one type of codon on mRNA.
5. The code is generally nonoverlapping; each nucleotide in an mRNA sequence belongs to a single reading frame.
6. The reading frame is set by an initiation codon, which is usually AUG.
7. When a reading frame has been set, codons are read as successive groups of three nucleotides.
8. Any one of three termination codons (UAA, UAG, and UGA) can signal the end of a protein; no amino acids are encoded by the termination codons.
9. The code is almost universal.

The Process of Translation

Now that we are familiar with the genetic code, we can begin to study the mechanism by which amino acids are assembled into proteins. Because more is known about translation in bacteria, we will focus primarily on bacterial translation. In most respects, eukaryotic translation is similar, although there are some significant differences that will be noted as we proceed through the stages of translation.

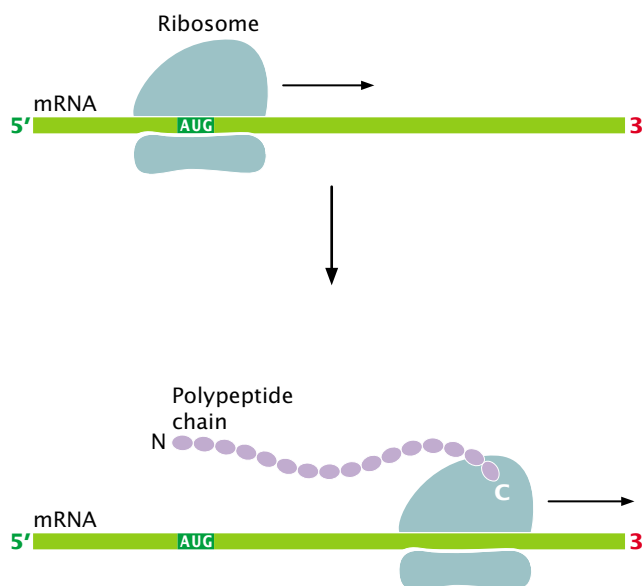
Translation takes place on ribosomes; indeed, ribosomes can be thought of as moving protein-synthesizing machines. Through a variety of techniques, a detailed view of the structure of the ribosome has been produced in recent years, which has greatly improved our understanding of the translational process. A ribosome attaches near the 5' end of an mRNA strand and moves toward the 3' end, translating the codons as it goes (FIGURE 15.15). Synthesis

Table 15.3 Some exceptions to the universal genetic code

Genome	Codon	Universal Code	Altered Code
Bacterial DNA			
<i>Mycoplasma capricolum</i>	UGA	Stop	Trp
Mitochondrial DNA			
Human	UGA	Stop	Trp
Human	AUA	Ile	Met
Human	AGA, AGG	Arg	Stop
Yeast	UGA	Stop	Trp
Trypanosomes	UGA	Stop	Trp
Plants	CGG	Arg	Trp
Nuclear DNA			
<i>Tetrahymena</i>	UAA	Stop	Gln
<i>Paramecium</i>	UAG	Stop	Gln

begins at the amino end of the protein, and the protein is elongated by the addition of new amino acids to the carboxyl end.

Protein synthesis can be conveniently divided into four stages: (1) the binding of amino acids to the tRNAs; (2) initiation, in which the components necessary for translation



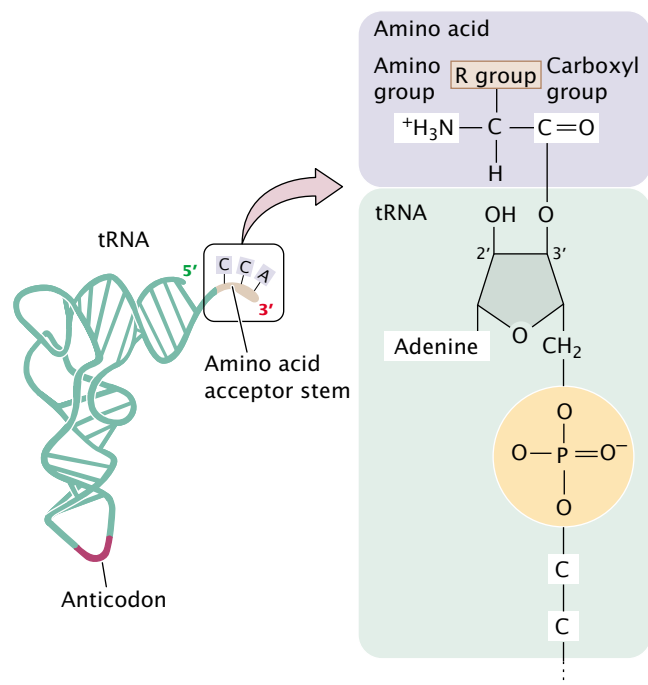
15.15 The translation of an mRNA molecule takes place on a ribosome.

are assembled at the ribosome; (3) elongation, in which amino acids are joined, one at a time, to the growing polypeptide chain; and (4) termination, in which protein synthesis halts at the termination codon and the translation components are released from the ribosome.

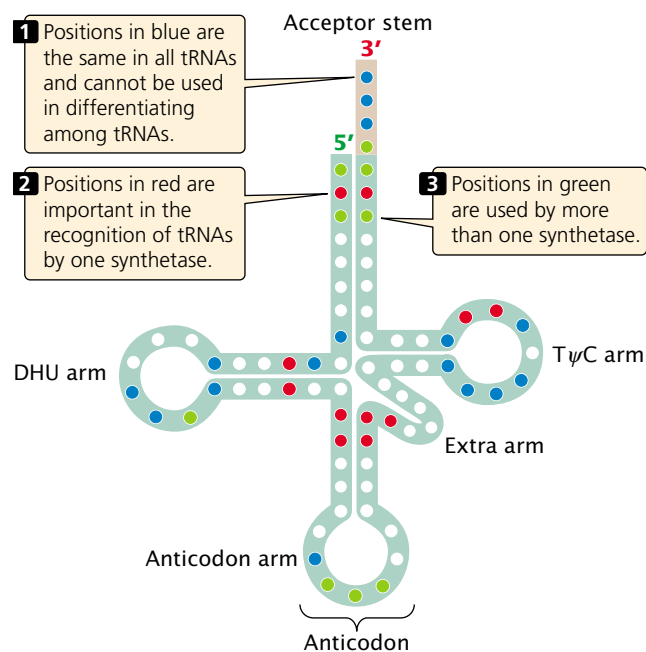
The Binding of Amino Acids to Transfer RNAs

The first stage of translation is the binding of tRNA molecules to their appropriate amino acids. When linked to its amino acid, a tRNA delivers that amino acid to the ribosome, where the tRNA's anticodon pairs with a codon on mRNA. This process enables the amino acids to be joined in the order specified by the mRNA. Proper translation, then, first requires the correct binding of tRNA and amino acid.

As already mentioned, a cell typically possesses from 30 to 50 different tRNAs, and, collectively, these tRNAs are attached to the 20 different amino acids. Each tRNA is specific for a particular kind of amino acid. All tRNAs have the sequence CCA at the 3' end, and the carboxyl group (COO^-) of the amino acid is attached to the 2'- or 3'-hydroxyl group of the adenine nucleotide at the end of the tRNA, (FIGURE 15.16). If each tRNA is specific for a particular amino acid but all amino acids are attached to the same nucleotide (A) at the 3' end of a tRNA, how does a tRNA link up with its appropriate amino acid?



15.16 An amino acid attaches to the 3' end of a tRNA. The carboxyl group (COO^-) of the amino acid attaches to the hydroxyl group of the 2'- or 3'-carbon atom of the final nucleotide at the 3' end of the tRNA, in which the base is always an adenine.

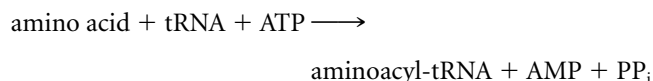


15.17 Certain positions on tRNA molecules are recognized by the appropriate aminoacyl-tRNA synthetase.

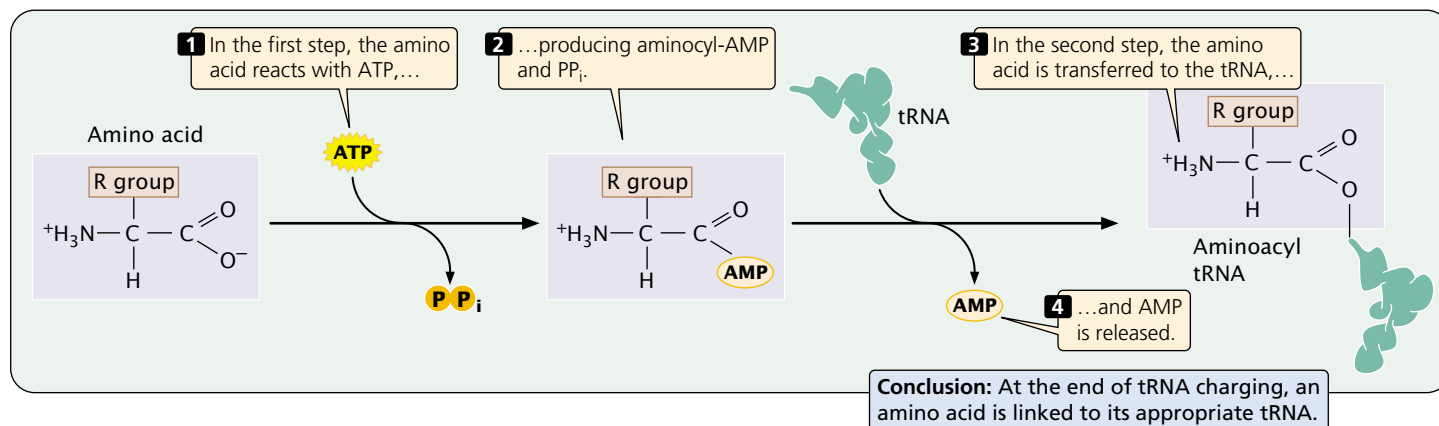
The key to specificity between an amino acid and its tRNA is a set of enzymes called **aminoacyl-tRNA synthetases**. A cell has 20 different aminoacyl-tRNA synthetases, one for each of the 20 amino acids. Each synthetase recognizes a particular amino acid, as well as all the tRNAs that accept that amino acid. Recognition of the appropriate amino acid by a synthetase is based on the different sizes, charges, and R groups of the amino acids. The tRNAs, however, are all similar in tertiary structure. How does a synthetase distinguish among tRNAs?

The recognition of tRNAs by a synthetase depends on the differing nucleotide sequences of tRNAs. Researchers have identified which nucleotides are important in recognition by altering different nucleotides in a particular tRNA and determining whether the altered tRNA is still recognized by its synthetase. The results of these studies revealed that the anticodon loop, the DHU-loop, and the acceptor stem are particularly critical for the identification of most tRNAs (FIGURE 15.17).

The attachment of a tRNA to its appropriate amino acid (termed **tRNA charging**) requires energy, which is supplied by adenosine triphosphate (ATP):



Two phosphates are cleaved from ATP, producing adenosine monophosphate (AMP) and pyrophosphate (PP_i), as well as the aminoacylated tRNA (the tRNA with its attached



15.18 An amino acid becomes attached to the appropriate tRNA in a two-step reaction.

amino acid). This reaction takes place in two steps (FIGURE 15.18). To identify the resulting aminoacylated tRNA, we write the three-letter abbreviation for the amino acid in front of the tRNA; for example, the amino acid alanine (Ala) attaches to its tRNA (tRNA^{Ala}), giving rise to its aminoacyl-tRNA ($\text{Ala-tRNA}^{\text{Ala}}$).

Errors in tRNA charging are rare; they occur in only about 1 in 10,000 to 1 in 100,000 reactions. This fidelity is due to the presence of proofreading activity in the synthetases, which detects and removes incorrectly paired amino acids from the tRNAs.

Concepts

Amino acids are attached to specific tRNAs by aminoacyl-tRNA synthetases in a two-step reaction that requires ATP.

The Initiation of Translation

The second stage in the process of protein synthesis is initiation. During initiation, all the components necessary for protein synthesis assemble: (1) mRNA; (2) the small and large subunits of the ribosome; (3) a set of three proteins called initiation factors; (4) initiator tRNA with *N*-formylmethionine attached ($\text{fMet-tRNA}^{\text{fMet}}$); and (5) guanosine triphosphate (GTP). Initiation comprises three major steps. First, mRNA binds to the small subunit of the ribosome. Second, initiator tRNA binds to the mRNA through base pairing between the codon and anticodon. Third, the large ribosome joins the initiation complex. Let's look at each of these steps more closely.

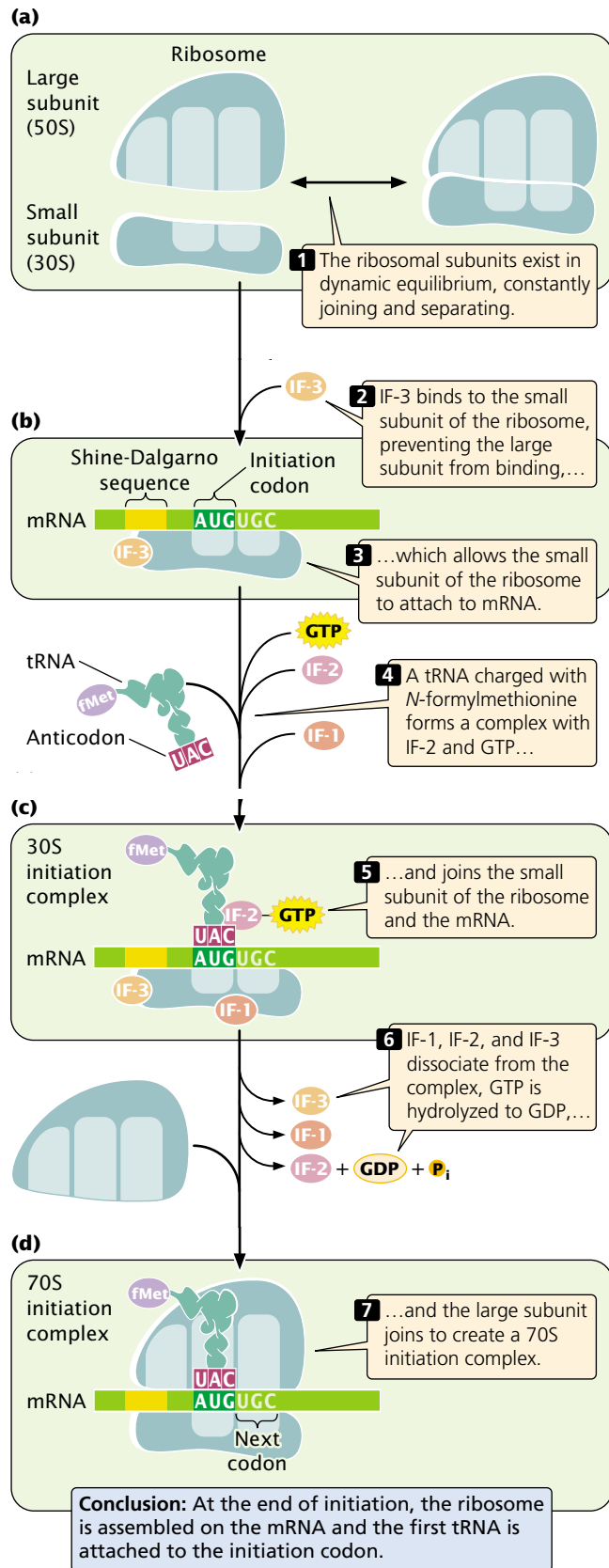
A functional ribosome exists as two subunits, the small 30S subunit and the large 50S subunit (in bacterial cells). When not actively translating, the two subunits exist in dynamic equilibrium, in which they are constantly joining

and separating (FIGURE 15.19). An mRNA molecule can bind to the small ribosome subunit only when the subunits are separate. **Initiation factor 3** (IF-3) binds to the small subunit of the ribosome and prevents the large subunit from binding during initiation (see Figure 15.19b).

Key sequences on the mRNA required for ribosome binding have been identified in experiments in which the ribosome is allowed to bind to mRNA under conditions that allow initiation but prevent later stages of protein synthesis, thereby stalling the ribosome at the initiation site. After the ribosome has attached to the mRNA in these experiments, ribonuclease is added, which degrades all the mRNA except the region covered by the ribosome. The remaining mRNA can be separated from the ribosome and studied. The sequence covered by the ribosome during initiation is from 30 to 40 nucleotides long and includes the AUG initiation codon. Within the ribosome-binding site is the Shine-Dalgarno consensus sequence (FIGURE 15.20) (see Chapter 14), which is complementary to a sequence of nucleotides at the 3' end of 16S rRNA (part of the small subunit of the ribosome). During initiation, the nucleotides in the Shine-Dalgarno sequence pair with their complementary nucleotides in the 16S rRNA, allowing the small subunit of the ribosome to attach to the mRNA and positioning the ribosome directly over the initiation codon.

Next, the initiator $\text{fMet-tRNA}^{\text{fMet}}$ attaches to the initiation codon (see Figure 15.19c). This step requires **initiation factor 2** (IF-2), which forms a complex with GTP. A third factor, **initiation factor 1** (IF-1), enhances the dissociation of the large and small ribosomal subunits.

At this point, the initiation complex consists of (1) the small subunit of the ribosome; (2) the mRNA; (3) the initiator tRNA with its amino acid ($\text{fMet-tRNA}^{\text{fMet}}$); (4) one molecule of GTP; and (5) IF-3, IF-2, and IF-1. These components are collectively known as the **30S initiation complex** (see Figure 15.19c). In the final step of initiation, IF-3 dissociates from the small subunit, allowing the large



15.19 The initiation of translation in bacterial cells requires several initiation factors and GTP.

E. coli trpA gene 5'AGCACGAGGGGAAAUUCG AUGGAACGCUAC3'
E. coli araB gene UUUUGGAUGGAGUGAAACG AUGGCGAUUGCA
E. coli lacI gene CAAUUCAGGGUGGUGAAUCUGAAACCAGUA
 λ phage *CRO* gene AUGUACUAAGGAGGUUGUAUGGAACAAGCG

Shine-Dalgarno sequence Initiation codon; pairs with fMet-tRNA^{fMet}

mRNA 5'AUGUACUAAGGAGGUUGUAUGGAACAAGCG3'
 16S rRNA 3'AUUCCUCCA5'

15.20 Shine-Dalgarno consensus sequences in mRNA are required for the attachment of the small subunit of the ribosome. The Shine-Dalgarno sequences are complementary to a sequence of nucleotides found near the 3' end of 16S rRNA in the small subunit of the ribosome. These complementary nucleotides base pair during the initiation of translation.

subunit of the ribosome to join the initiation complex. The molecule of GTP (provided by IF-2) is hydrolyzed to guanosine diphosphate (GDP), and IF-1 and IF-2 depart (see Figure 15.19d). When the large subunit has joined the initiation complex, it is called the **70S initiation complex**.

Similar events take place in the initiation of translation in eukaryotic cells, but there are some important differences. In bacterial cells, sequences in 16S rRNA of the small subunit of the ribosome bind to the Shine-Dalgarno sequence in mRNA; this binding positions the ribosome over the start codon. No analogous consensus sequence exists in eukaryotic mRNA. Instead, the cap at the 5' end of eukaryotic mRNA plays a critical role in the initiation of translation. The small subunit of the eukaryotic ribosome, with the help of initiation factors, recognizes the cap and binds there; the small subunit then migrates along (scans) the mRNA until it locates the first AUG codon. The identification of the start codon is facilitated by the presence of a consensus sequence (called the Kozak sequence) that surrounds the start codon:

Kozak sequence
 5'-ACCAUGG-3'
 Start codon

Another important difference is that eukaryotic initiation requires more initiation factors. Some factors keep the ribosomal subunits separated, just as IF-3 does in bacterial cells. Others recognize the 5' cap on mRNA and allow the small subunit of the ribosome to bind there. Still others possess RNA helicase activity, which is used to unwind secondary structures that may exist in the 5' untranslated region of mRNA, allowing the small subunit to move down

the mRNA until the initiation codon is reached. Other initiation factors help bring the initiator tRNA and methionine (Met-tRNA^{fMet}) to the initiation complex.

The poly(A) tail at the 3' end of eukaryotic mRNA also plays a role in the initiation of translation. Proteins that attach to the poly(A) tail interact with proteins that bind to the 5' cap, enhancing the binding of the small subunit of the ribosome to the 5' end of the mRNA. This interaction between the 5' cap and the 3' tail suggests that the mRNA bends backward during the initiation of translation, forming a circular structure (FIGURE 15.21). A few eukaryotic mRNAs contain internal ribosome entry sites, where ribosomes can bind directly without first attaching to the 5' cap.

Concepts

In the initiation of translation in bacterial cells, the small ribosomal subunit attaches to mRNA, and initiator tRNA attaches to the initiation codon. This process requires several initiation factors (IF-1, IF-2, and IF-3) and GTP. In the final step, the large ribosomal subunit joins the initiation complex.

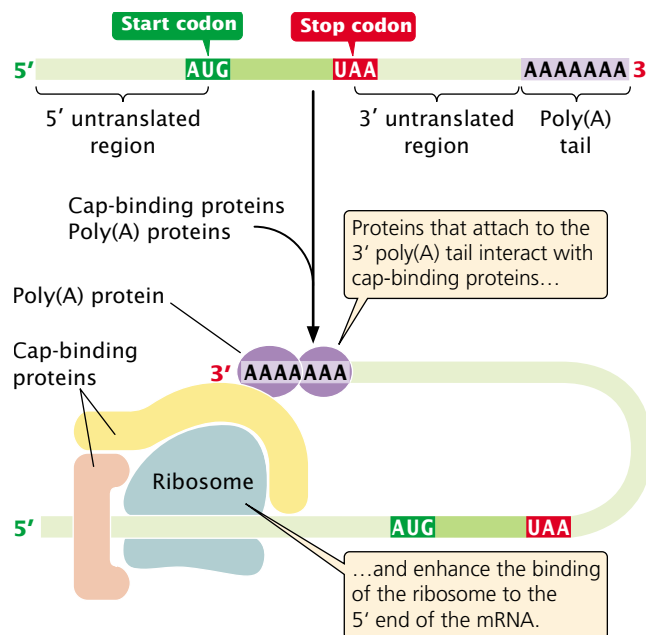
Elongation

The next stage in protein synthesis is elongation, in which amino acids are joined to create a polypeptide chain. Elongation requires (1) the 70S complex just described; (2) tRNAs charged with their amino acids; (3) several elongation factors (EF-Ts, EF-Tu, and EF-G); and (4) GTP.

A ribosome has three sites that can be occupied by tRNAs; the **aminoacyl**, or **A, site**, the **peptidyl**, or **P, site**, and the **exit**, or **E, site** (FIGURE 15.22a). The initiator tRNA immediately occupies the P site (the only site to which the fMet-tRNA^{fMet} is capable of binding), but all other tRNAs first enter the A site. After initiation, the ribosome is attached to the mRNA, and fMet-tRNA^{fMet} is positioned over the AUG start codon in the P site; the adjacent A site is unoccupied (see Figure 15.22a).

Elongation occurs in three steps. The first step (FIGURE 15.22b) is the delivery of a charged tRNA (tRNA with its amino acid attached) to the A site. This requires the presence of **elongation factor Tu** (EF-Tu), **elongation factor Ts** (EF-Ts), and GTP. EF-Tu first joins with GTP and then binds to a charged tRNA to form a three-part complex. This three-part complex enters the A site of the ribosome, where the anticodon on the tRNA pairs with the codon on the mRNA. After the charged tRNA is in the A site, GTP is cleaved to GDP, and the EF-Tu-GDP complex is released (FIGURE 15.22c). Factor EF-Ts regenerates EF-Tu-GDP to EF-Tu-GTP. In eukaryotic cells, a similar set of reactions delivers the charged tRNA to the A site.

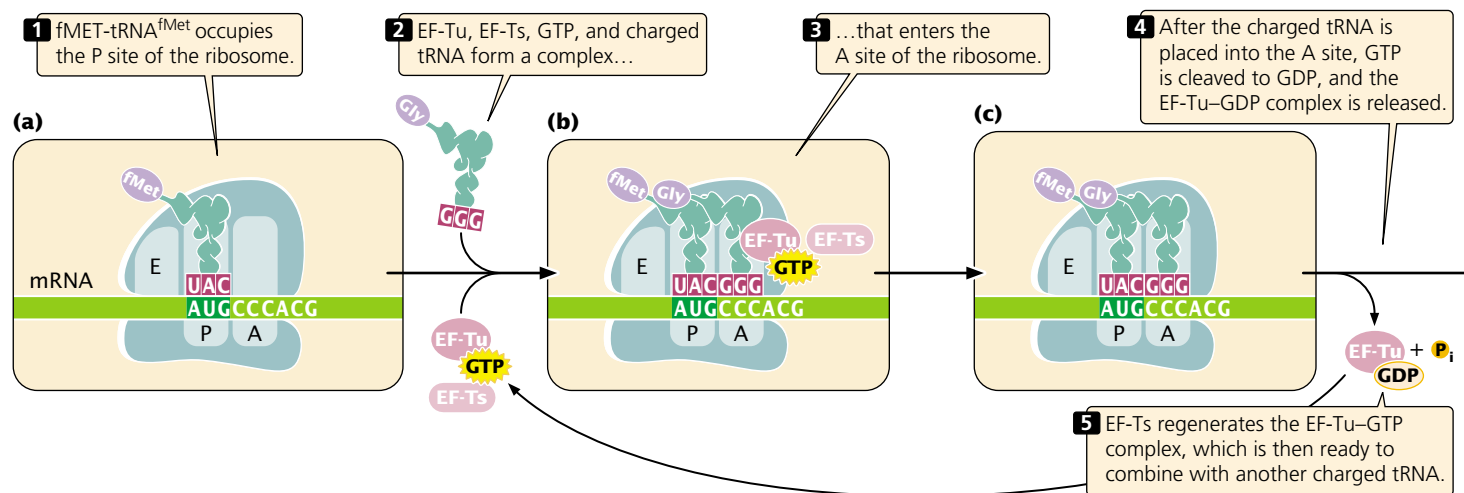
The second step of elongation is the creation of a peptide bond between the amino acids that are attached to



15.21 The poly(A) tail at the 3' end of eukaryotic mRNA plays a role in the initiation of translation.

tRNAs in the P and A sites (FIGURE 15.22d). The formation of this peptide bond releases the amino acid in the P site from its tRNA. The activity responsible for peptide-bond formation in the ribosome is referred to as **peptidyl transferase**. For many years, the assumption was that this activity is carried out by one of the proteins in the large subunit of the ribosome. Evidence, however, now indicates that the catalytic activity is a property of the rRNA in the large subunit of the ribosome; this rRNA acts as a ribozyme (see pp. 000 in Chapter 14).

The third step in elongation is **translocation**, (FIGURE 15.22e), the movement of the ribosome down the mRNA in the 5'→3' direction. This step positions the ribosome over the next codon and requires **elongation factor G** (EF-G) and the hydrolysis of GTP to GDP. Because the tRNAs in the P and A site are still attached to the mRNA through codon-anticodon pairing, they do not move with the ribosome as it translocates. Consequently, the ribosome shifts so that the tRNA that previously occupied the P site now occupies the E site, from which it moves into the cytoplasm where it may be recharged with another amino acid. Translocation also causes the tRNA that occupied the A site (which is attached to the growing polypeptide chain) to be in the P site, leaving the A site open. Thus, the progress of each tRNA through the ribosome during elongation can be summarized as follows: cytoplasm → A site → P site → E site → cytoplasm. As discussed earlier, the initiator tRNA is an exception: it attaches directly to the P site and never occupies the A site.



15.22 The elongation of translation comprises three steps.

After translocation, the A site of the ribosome is empty and ready to receive the tRNA specified by the next codon. The elongation cycle (Figure 15.22a through d) repeats itself: a charged tRNA and its amino acid occupy the A site, a peptide bond is formed between the amino acids in the A and P sites, and the ribosome translocates to the next codon. Throughout the cycle, the polypeptide chain remains attached to the tRNA in the P site. The ribosome moves down the mRNA in the 5'→3' direction, adding amino acids one at a time according to the order specified by the mRNA's codon sequence. Elongation in eukaryotic cells takes place in a similar manner.

Concepts

Elongation consists of three steps: (1) a charged tRNA enters the A site, (2) a peptide bond is created between amino acids in the A and P sites, and (3) the ribosome translocates to the next codon. Elongation requires several elongation factors (EF-Tu, EF-Ts, and EF-G) and GTP.

Termination

Protein synthesis terminates when the ribosome translocates to a termination codon. Because there are no tRNAs with anticodons complementary to the termination codons, no tRNA enters the A site of the ribosome when a termination codon is encountered (FIGURE 15.23a). Instead, proteins called **release factors** bind to the ribosome (FIGURE 15.23b). *E. coli* has three release factors—RF₁, RF₂, and RF₃. Release factor 1 recognizes the termination codons UAA and UAG, and RF₂ recognizes UGA and UAA. Release factor 3 forms a complex with GTP and binds to the ribosome. The release factors then promote the cleavage of the tRNA in the P site from the polypeptide chain; in the

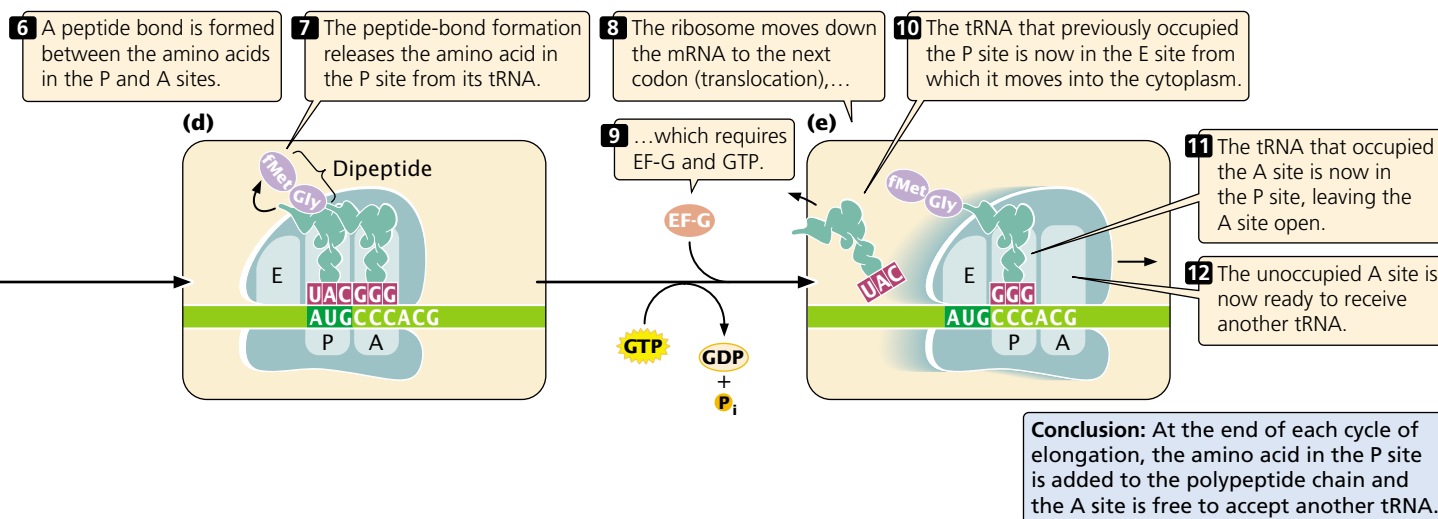
process, the GTP that is complexed to RF₃ is hydrolyzed to GDP. Additional factors help bring about the release of the tRNA from the P site, the release of the mRNA from the ribosome, and the dissociation of the ribosome (FIGURE 15.23c). Translation in eukaryotic cells terminates in a similar way, except that there are two release factors: eRF1, which recognizes all three termination codons, and eRF2, which binds GTP and stimulates the release of the polypeptide from the ribosome.

Findings from recent studies suggest that the release factors bring about the termination of translation by completing a final elongation cycle of protein synthesis. In this model, RF₁ and RF₂ are similar in size and shape to tRNAs and occupy the A site of the ribosome, just as the amino acid–tRNA–EF–Tu–GTP complex does during an elongation cycle. Release factor 3 is structurally similar to EF-G; it then translocates RF₁ and RF₂ to the P site, as well as the last tRNA to the E site, in a way similar to that in which EF-G brings about translocation. When both the A site and the P site of the ribosome are cleared of tRNAs, the ribosome can dissociate. Research findings also indicate that some of the sequences in the rRNA play a role in the recognition of termination codons.

Concepts

Termination takes place when the ribosome reaches a termination codon. Release factors bind to the termination codon, causing the release of the polypeptide from the last tRNA, the tRNA from the ribosome, and the mRNA from the ribosome.

The overall process of protein synthesis, including tRNA charging, initiation, elongation, and termination, is summarized in FIGURE 15.24, and the components taking part in this process are listed in Table 15.4 (See page 00).



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A brief overview of translation and how it fits into the central dogma of genetics

RNA–RNA Interactions in Translation

The process of translation is rich in RNA–RNA interactions (which were discussed in Chapter 14 in the context of RNA processing). For example, in bacterial translation, the Shine-Dalgarno consensus sequence at the 5' end of the mRNA pairs with the 3' end of the 16S rRNA (see Figure 15.20), which ensures the binding of the ribosome to mRNA. Mutations that alter the Shine-Dalgarno sequence, so that the mRNA and rRNA are no longer complementary, inhibit translation. Corresponding mutations affecting the rRNA that restore complementarity allow translation to proceed. RNA–RNA interactions also take place between the tRNAs in the A and P sites and the rRNAs found in both the large and the small subunits of the ribosome. Furthermore, association of the large and small subunits of the ribosome may require interactions between the 16S rRNA and the 23S rRNA, although whether ribosomal proteins are implicated is not yet clear. Finally, tRNAs and mRNAs interact through their codon–anticodon pairing.

Polyribosomes

In both prokaryotic and eukaryotic cells, mRNA molecules are translated simultaneously by multiple ribosomes; see page 000 (FIGURE 15.25). The resulting structure—an mRNA with several ribosomes attached—is called a **polyribosome**. Each ribosome successively attaches to the ribosome-binding site at the 5' end of the mRNA and moves toward the 3' end; the polypeptide associated with each ribosome becomes progressively longer as the ribosome moves along the mRNA.

In prokaryotic cells, transcription and translation are simultaneous; so multiple ribosomes may be attached to the 5' end of the mRNA while transcription is still taking place at the 3' end, as shown in FIGURE 15.26; see page 000. Until recently, transcription and translation were thought *not* to be simultaneous in eukaryotes, because transcription takes place in the nucleus and all translation was assumed to take place in the cytoplasm. However, research findings have now demonstrated that some translation takes place within the eukaryotic nucleus, and evidence suggests that, when the nucleus is the site of translation, transcription and translation may be simultaneous, much as in prokaryotes.

Concepts

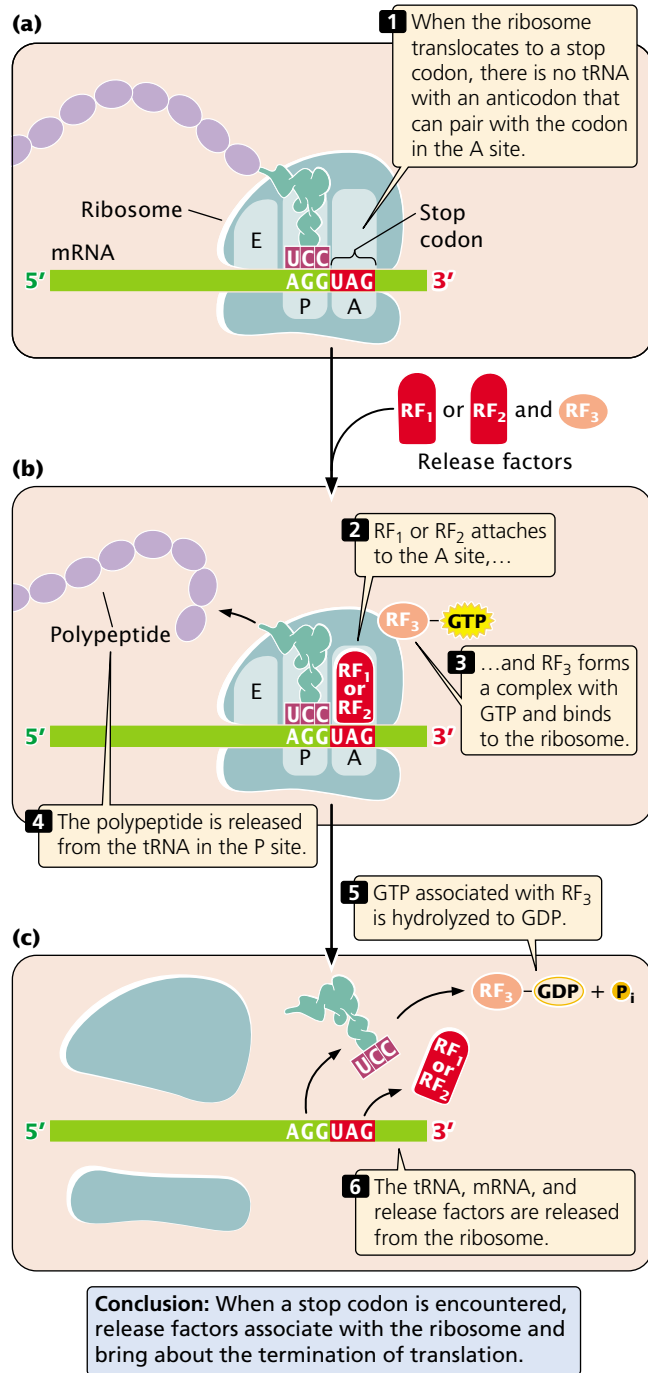
In both prokaryotic and eukaryotic cells, multiple ribosomes may be attached to a single mRNA, generating a structure called a polyribosome.

Connecting Concepts

A Comparison of Bacterial and Eukaryotic Translation

We have now considered the process of translation in bacterial cells and noted some distinctive differences that exist in eukaryotic cells. Let's take a few minutes to reflect on some of the important similarities and differences of protein synthesis in bacterial and eukaryotic cells.

First, we should emphasize that the genetic code of bacterial and eukaryotic cells is virtually identical; the only difference is in the amino acid specified by the initiation codon. In bacterial cells, AUG codes for a modified type of methionine, *N*-formylmethionine, whereas, in eukaryotic



15.23 Translation ends when a stop codon is encountered.

15.24 The four steps involved in translation are **tRNA charging** (the binding of amino acids to tRNAs), **initiation**, **elongation**, and **termination**. In this process, amino acids are linked together in the order specified by the mRNA to create a polypeptide chain. A number of initiation, elongation, and release factors take part in the process, and energy is supplied by ATP and GTP.

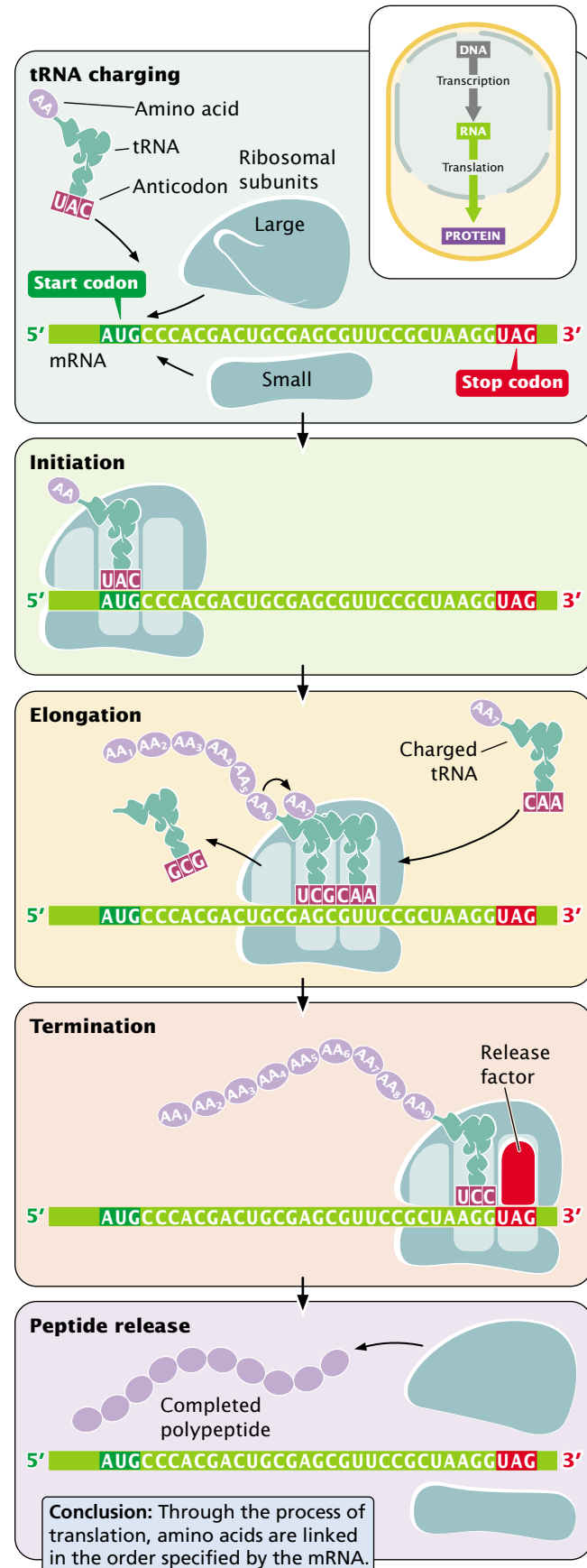


Table 15.4 Components required for protein synthesis in bacterial cells

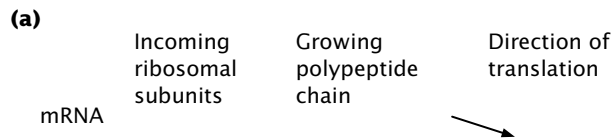
Stage	Component	Function
Binding of amino acid to tRNA	Amino acids	Building blocks of proteins
	tRNAs	Deliver amino acids to ribosomes
	aminoacyl-tRNA synthetase	Attaches amino acids to tRNAs
	ATP	Provides energy for binding amino acid to tRNA
Initiation	mRNA	Carries coding instructions
	fMet-tRNA ^{fMet}	Provides first amino acid in peptide
	30S ribosomal subunit	Attaches to mRNA
	50S ribosomal subunit	Stabilizes tRNAs and amino acids
	Initiation factor 1	Enhances dissociation of large and small subunits of ribosome
	Initiation factor 2	Binds GTP; delivers fMet-tRNA ^{fMet} to initiation codon
	Initiation factor 3	Binds to 30S subunit and prevents association with 50S subunit
Elongation	70S initiation complex	Functional ribosome with A, P, and E sites and peptidyl transferase activity where protein synthesis takes place
	Charged tRNAs	Bring amino acids to ribosome and help assemble them in order specified by mRNA
	Elongation factor Tu	Binds GTP and charged tRNA; delivers charged tRNA to A site
	Elongation factor Ts	Generates active elongation factor Tu
	Elongation factor G	Stimulates movement of ribosome to next codon
	GTP	Provides energy
	Peptidyl transferase	Creates peptide bond between amino acids in A site and P site
Termination	Release factors 1, 2, and 3	Bind to ribosome when stop codon is reached and terminate translation

cells, AUG codes for unformylated methionine. One consequence of the fact that bacteria and eukaryotes use the same code is that eukaryotic genes can be translated in bacterial systems, and vice versa; this feature makes genetic engineering possible, as we will see in Chapter 18.

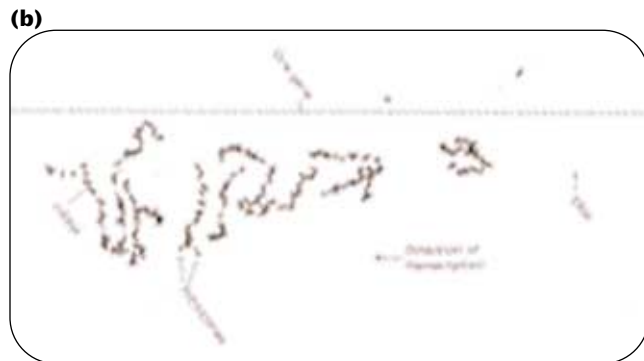
Another difference is that transcription and translation take place simultaneously in bacterial cells, but the nuclear envelope may separate these processes in eukaryotic cells. The physical separation of transcription and translation has important implications for the control of gene expression, which we will consider in Chapter 16, and it allows for extensive modification of eukaryotic mRNAs, as discussed in Chapter 14. However, it is now evident that some translation does take place in the eukaryotic nucleus and, there, transcription and translation may be simultaneous. The extent of nuclear translation and how it may affect gene regulation are not yet clear.

Yet another difference is that mRNA in bacterial cells is short lived, typically lasting only a few minutes, but the longevity of mRNA in eukaryotic cells is highly variable and is frequently hours or days. Thus the synthesis of a particular bacterial protein ceases very quickly after transcription of the corresponding mRNA stops, but protein synthesis in eukaryotic cells may continue long after transcription has ended.

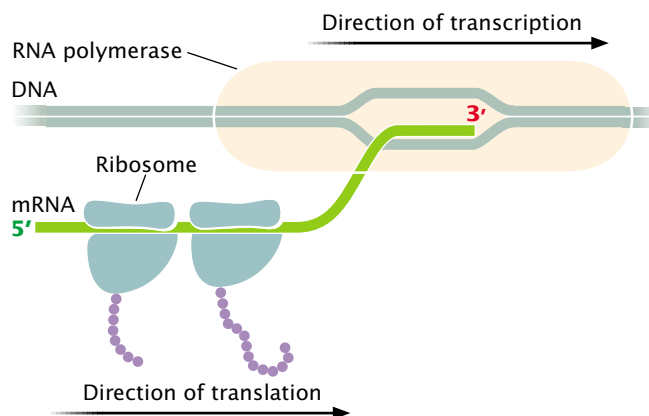
In both bacterial and eukaryotic cells, aminoacyl-tRNA synthetases attach amino acids to their appropriate tRNAs; the chemical reaction employed is the same. There are significant differences in the sizes and compositions of bacterial and eukaryotic ribosomal subunits. For example, the large subunit of the eukaryotic ribosome contains three rRNAs, whereas the bacterial ribosome contains only two. These differences allow antibiotics and other substances to inhibit bacterial translation while having no effect on the transla-



Drawing to be rendered from actual micrograph being used in part (b)



15.25 An mRNA molecule may be transcribed simultaneously by several ribosomes. (a) Four ribosomes are translating a eukaryotic mRNA molecule, moving from the 5' end to the 3' end. (b) In this electron micrograph of a polyribosome from the silkworm, the dark staining spheres are ribosomes, and the long, thin filament connecting the ribosomes is mRNA. The 5' end of the mRNA is toward the upper left-hand corner of the micrograph. The twisted filament coming out of each ribosome is a polypeptide chain. The polypeptide chains become longer as the ribosomes move toward the 3' end of the mRNA. (Part b, O. L. Miller, Jr., and Barbara A. Hamaklo).



15.26 In prokaryotic cells, transcription and translation take place simultaneously. While mRNA is being transcribed from the DNA template at mRNA's 3' end, translation is taking place simultaneously at mRNA's 5' end.

tion of eukaryotic nuclear genes, as will be discussed near the end of this chapter.

Other fundamental differences lie in the process of initiation. In bacterial cells, the small subunit of the ribosome attaches directly to the region surrounding the start codon through hydrogen bonding between the Shine-Dalgarno consensus sequence in the 5' untranslated region of the mRNA and a sequence at the 3' end of the 16S rRNA. In contrast, the small subunit of a eukaryotic ribosome first binds to proteins attached to the 5' cap on mRNA and then migrates down the mRNA, scanning the sequence until it encounters the first AUG initiation codon. (A few eukaryotic mRNAs have internal ribosome-binding sites that utilize a specialized initiation mechanism similar to that seen in bacterial cells.) Additionally, more initiation factors take part in eukaryotic initiation than in bacterial initiation.

Elongation and termination are similar in bacterial and eukaryotic cells, although different elongation and termination factors are used. In both types of organisms, mRNAs are translated multiple times and are simultaneously attached to several ribosomes, forming polyribosomes.

What about translation in archaea, which are prokaryotic in structure (see Chapter 2) but are similar to eukaryotes in other genetic processes such as transcription? Much less is known about the process of translation in archaea, but available evidence suggests that they possess a mixture of eubacterial and eukaryotic features. Because archaea lack nuclear membranes, transcription and translation take place simultaneously, just as they do in eubacterial cells. As mentioned earlier, archaea utilize unformylated methionine as the initiator amino acid, a characteristic of eukaryotic translation. Findings from recent studies of DNA sequences that code for initiation and elongation factors in archaea suggest that some of them are similar to those found in eubacteria, whereas others are similar to those found in eukaryotes. Finally, some of the antibiotics that inhibit translation in eubacteria have no effect on translation in archaea.

The Posttranslational Modifications of Proteins

After translation, proteins in both prokaryotic and eukaryotic cells may undergo alterations termed posttranslational modifications. A number of different types of modifications are possible. As mentioned earlier, the formyl group or the entire methionine residue may be removed from the amino end of a protein. Some proteins are synthesized as larger precursor proteins and must be cleaved and trimmed by enzymes before the proteins can become functional. For others, the attachment of carbohydrates may be required for activation. The functions of many proteins depend critically on the proper folding of the polypeptide chain; some pro-

teins spontaneously fold into their correct shapes, but, for others, correct folding may initially require the participation of other molecules called **molecular chaperones**.

In eukaryotic cells, the amino end of a protein is often acetylated after translation. Another modification of some proteins is the removal of 15 to 30 amino acids, called the **signal sequence**, at the amino end of the protein. The signal sequence helps direct a protein to a specific location within the cell, after which the sequence is removed by special enzymes. Amino acids within a protein may also be modified: phosphates, carboxyl groups, and methyl groups are added to some amino acids.

Concepts

Many proteins undergo posttranslational modifications after their synthesis.



Translation and Antibiotics

Antibiotics are drugs that kill microorganisms. To make an effective antibiotic—not just any poison will do—the trick is to kill the microbe without harming the patient. Antibiotics must be carefully chosen so that they destroy bacterial cells but not the eukaryotic cells of their host.

Translation is frequently the target of antibiotics because translation is essential to all living organisms and differs significantly between bacterial and eukaryotic cells. For example, bacterial and eukaryotic ribosomes differ in size and composition. A number of antibiotics bind selectively to bacterial ribosomes and inhibit various steps in translation, but they do not affect eukaryotic ribosomes. Tetracyclines, for instance, are a class of antibiotics that bind to the A site of bacterial ribosomes and block the entry of charged tRNAs, yet they have no effect on eukaryotic ribosomes. Neomycin binds to the ribosome near the A site and induces translational errors, probably by causing mistakes in the binding of charged tRNAs to the A site. Chloramphenicol binds to the large subunit of the ribosome and blocks peptide-bond formation. Streptomycin binds to the small subunit of the ribosome and inhibits initiation, and erythromycin blocks translocation. Although chloramphenicol and streptomycin are potent inhibitors of translation in bacteria, they do not inhibit translation in archaeobacteria.

The three-dimensional structure of puromycin resembles the 3' end of a charged tRNA, permitting puromycin to enter the A site of a ribosome efficiently and inhibit the entry of tRNAs. A peptide bond can form between the puromycin molecule in the A site and an amino acid on the tRNA in the P site of the ribosome, but puromycin cannot bind to the P site and translocation does not take place, blocking further elongation of the protein. Because tRNA structure is similar in all organisms, puromycin inhibits translation in both bacterial and eukaryotic cells; consequently, puromycin kills eukaryotic cells along with

bacteria and is sometimes used in cancer therapy to destroy tumor cells.

Many antibiotics act by blocking specific steps in translation, and different antibiotics affect different steps in protein synthesis. Because of this specificity, antibiotics are frequently used to study the process of protein synthesis.

Connecting Concepts Across Chapters



This chapter has focused on the process by which genetic information in an mRNA molecule is transferred to the amino acid sequence of a protein. This process is termed translation because information contained in the language of nucleotides must be “translated” into the language of amino acids.

The link between genotype and phenotype is usually a protein: most genes affect phenotypes by encoding proteins. How the presence of a protein produces a particular anatomical, physiological, or behavioral trait, however, is often far from clear, as was illustrated by the story of Lesch-Nyhan disease. The relation between genes and traits is the subject of much current research and will be explored further in Chapters 16 and 21.

In this chapter, we have examined the nature of the genetic code. It is a very concise code, with each codon consisting of three nucleotides, the minimum number capable of specifying all 20 common amino acids. Breaking the genetic code required great ingenuity and hard work on the part of a number of geneticists.

Much of this chapter has centered on protein synthesis. We learned that translation is a highly complex process: rRNAs, ribosomal proteins, tRNAs, mRNA, initiation factors, elongation factors, release factors, and aminoacyl-tRNA synthetases all help to assemble amino acids into a protein. This complexity might seem surprising, because the peptide bonds that hold amino acids together are simple covalent bonds. Translation is complex not because of any special property of the peptide bond, but rather because the amino acids must be linked in a highly precise order. The amino acid sequence determines the secondary and tertiary structures of a protein, which are critical to its function; so the genetic information in a mRNA molecule must be accurately translated. The complexity of translation has evolved to ensure that few mistakes are made in the course of protein synthesis.

An important theme in protein synthesis is RNA–RNA interaction, which takes place between tRNAs and mRNA, between mRNA and rRNAs, and between tRNAs and rRNAs. The prominence of these RNA–RNA interactions in translation reinforces the proposal that life first evolved in an RNA world, where flexible and versatile RNA molecules carried out many life processes (Chapter 13).

This chapter has built on our understanding of other processes of information transfer covered earlier in the book: replication (Chapter 12), transcription (Chapter 13), and RNA processing (Chapter 14). It also provides a critical foundation for later discussions of

gene regulation (Chapter 16), gene mutations (Chapter 17), and the advanced topics of developmental genetics, cancer genetics, and immunological genetics (Chapter 21).

CONCEPTS SUMMARY

- Genes code for phenotypes by specifying the amino acid sequences of proteins.
- The relation between genes and proteins was first suggested by Archibald Garrod.
- Beadle and Tatum developed the one gene, one enzyme hypothesis, which proposed that each gene specifies one enzyme; this hypothesis was later modified to become the one gene, one polypeptide hypothesis.
- Proteins are composed of 20 different amino acids, several or many of which are linked together by peptide bonds. Chains of amino acids fold and associate to produce the secondary, tertiary, and quaternary structures of proteins.
- The genetic code is the way in which genetic information is stored in the nucleotide sequence of a gene.
- Solving the genetic code required several different approaches: the use of synthetic mRNAs with random sequences; short mRNAs that bind tRNAs with their amino acids; and long synthetic mRNAs with regularly repeating sequences.
- The genetic code is a triplet code: three nucleotides specify a single amino acid. It is also degenerate, nonoverlapping, and universal (almost).
- The degeneracy of the code means that more than one codon may specify an amino acid. Different tRNAs (isoaccepting tRNAs) may accept the same amino acid, and different anticodons may pair with the same codon through wobble, which can exist at the third position of the codon and which allows some nonstandard pairing of bases in this position.
- The reading frame is set by the initiation codon.
- The end of the protein-coding section of an mRNA is marked by one of three termination codons.
- Protein synthesis comprises four steps: (1) the binding of amino acids to the appropriate tRNAs, (2) initiation, (3) elongation, and (4) termination.
- The binding of an amino acid to a tRNA requires the presence of a specific aminoacyl-tRNA synthetase and ATP. The amino acid is attached by its carboxyl end to the 3' end of the tRNA.
- In bacterial translation initiation, the small subunit of the ribosome attaches to the mRNA and is positioned over the initiation codon. It is joined by the first tRNA and its associated amino acid (*N*-formylmethionine in bacterial cells) and, later, by the large subunit of the ribosome. Initiation requires several initiation factors and GTP.
- In elongation, a charged tRNA enters the A site of a ribosome, a peptide-bond is formed between amino acids in the A and P sites, and the ribosome moves (translocates) along the mRNA to the next codon. Elongation requires several elongation factors and GTP.
- Translation is terminated when the ribosome encounters one of the three termination codons. Release factors and GTP are required to bring about termination.
- Like RNA processing, translation requires a number of RNA–RNA interactions.
- Each mRNA may be simultaneously translated by several ribosomes, producing a structure called a polyribosome.
- Many proteins undergo posttranslational modification.

IMPORTANT TERMS

auxotroph (p. 000)	isoaccepting tRNAs (p. 000)	initiation factors (IF-1, IF-2, IF-3) (p. 000)	elongation factor Ts (EF-Ts) (p. 000)
one gene, one enzyme hypothesis (p. 000)	wobble (p. 000)	30S initiation complex (p. 000)	peptidyl transferase (p. 000)
one gene, one polypeptide hypothesis (p. 000)	nonoverlapping genetic code (p. 000)	70S initiation complex (p. 000)	translocation (p. 000)
amino acid (p. 000)	reading frame (p. 000)	aminoacyl (A) site (p. 000)	elongation factor G (EF-G) (p. 000)
peptide bond (p. 000)	initiation codon (p. 000)	peptidyl (P) site (p. 000)	release factors (RF ₁ , RF ₂ , RF ₃) (p. 000)
polypeptide (p. 000)	stop (termination or nonsense) codon (p. 000)	exit (E) site (p. 000)	polyribosome (p. 000)
sense codon (p. 000)	universal genetic code (p. 000)	elongation factor Tu (EF-Tu) (p. 000)	molecular chaperone (p. 000)
degenerate genetic code (p. 000)	aminoacyl-tRNA synthetase (p. 000)		signal sequence (p. 000)
synonymous codons (p. 000)	tRNA charging (p. 000)		

Worked Problems

1. A series of auxotrophic mutants were isolated in *Neurospora*. Examination of fungi containing these mutations revealed that they grew on minimal medium to which various compounds (A, B, C, D) were added; growth responses to each of the four compounds are presented in the following table. Give the order of compounds A, B, C, and D in a biochemical pathway. Outline a biochemical pathway that includes these four compounds and indicate which step in the pathway is affected by each of the mutations.

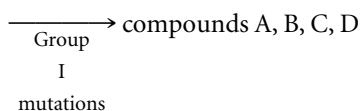
Mutation number	Compound			
	A	B	C	D
134	+	+	—	+
276	+	+	+	+
987	—	—	—	+
773	+	+	+	+
772	—	—	—	+
146	+	+	—	+
333	+	+	—	+
123	—	+	—	+

• Solution

To solve this problem, we should first group the mutations for which compounds allow growth, as follows.

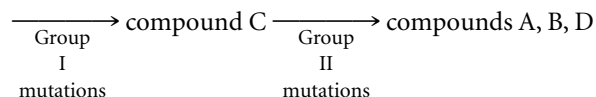
Mutation		Compound			
Group	Number	A	B	C	D
I	276	+	+	+	+
	773	+	+	+	+
	772	—	—	—	+
II	134	+	+	—	+
	146	+	+	—	+
	333	+	+	—	+
III	123	—	+	—	+
IV	987	—	—	—	+
	772	—	—	—	+

The underlying principle used to determine the order of the compounds in the pathway is as follows: If a compound is added after the block, it will allow the mutant to grow, whereas, if a compound is added before the block, it will have no effect. Applying this principle to the data in the table, we see that mutants in group I will grow if compound A, B, C, or D is added to the medium; so these mutations must affect a step before the production of all four compounds:

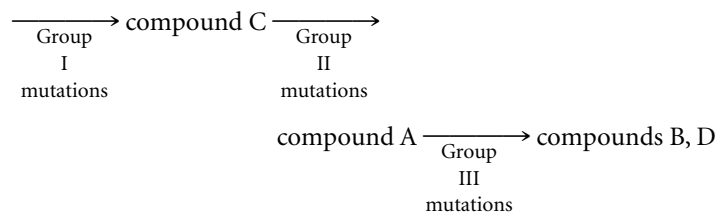


Group II mutants will grow if compound A, B, or D is added but not if compound C is added. Thus compound C comes

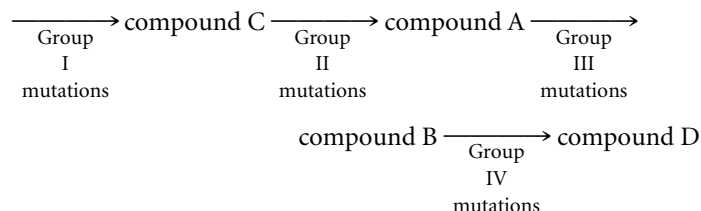
before A, B, and D; and group II mutations affect the conversion of compound C into one of the other compounds:



Group III mutants allow growth if compound B or D is added but not if compound A or C is added. Thus group III mutations affect steps that follow the production of A and C; we have already determined that compound C precedes A in the pathway; so A must be the next compound in the pathway:



Finally, mutants in group IV will grow if compound D is added, but not if compound A, B, or C are added. Thus compound D is the fourth compound in the pathway, and mutations in group IV block the conversion of B into D:



2. If there were five different types of bases in mRNA instead of four, what would be the minimum codon size (number of nucleotides) required to specify the following numbers of different amino acid types: (a) 4, (b) 20, (c) 30?

• Solution

To answer this question, we must determine the number of combinations (codons) possible when there are different numbers of bases and different codon lengths. In general, the number of different codons possible will be equal to:

$$b^{lg} = \text{number of codons}$$

where, b equals the number of different types of bases and lg equals the number of nucleotides in each codon (codon length). If there are five different types of bases, then:

$$5^1 = 5 \text{ possible codons}$$

$$5^2 = 25 \text{ possible codons}$$

$$5^3 = 125 \text{ possible codons}$$

The number of possible codons must be greater than or equal to the number of amino acids specified. Therefore, a codon length of one nucleotide could specify 4 different amino acids, a codon length of 2 nucleotides could specify 20 different amino acids, and a codon length of 3 nucleotides could specify 30 different amino acids: (a) 1, (b) 2, (c) 3.

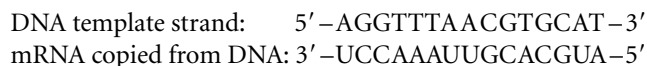
3. A template strand in bacterial DNA has the following base sequence:



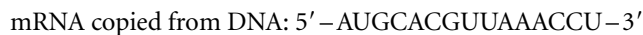
What amino acids would be encoded by this sequence?

• Solution

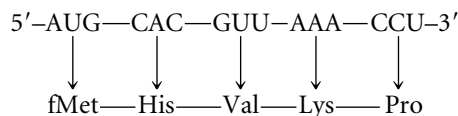
To answer this question, we must first work out the mRNA sequence that will be transcribed from this DNA sequence. The mRNA must be antiparallel and complementary to the DNA template strand:



An mRNA is translated $5' \rightarrow 3'$; so it will be helpful if we turn the RNA molecule around with the 5' end on the left:



The codons consist of groups of three nucleotides that are read successively after the first AUG codon; using Figure 15.14, we can determine that the amino acids are:

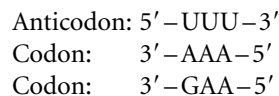


4. The following triplets constitute anticodons found on a series of tRNAs. Give the amino acid carried by each of these tRNAs.

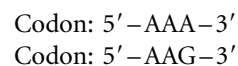
- (a) $5' - \text{UUU} - 3'$
- (b) $5' - \text{GAC} - 3'$
- (c) $5' - \text{UUG} - 3'$
- (d) $5' - \text{CAG} - 3'$

• Solution

To solve this problem, we first determine the codons with which these anticodons pair and then look up the amino acid specified by the codon in Figure 15.14. The codons are antiparallel and complementary to the anticodons. For part *a*, the anticodon is $5' - \text{UUU} - 3'$. According to the wobble rules in Table 15.2, U in the first position of the anticodon can pair with either A or G in the third position of the anticodon, so there are two codons that can pair with this anticodon:



Listing these codons in the conventional manner, with the 5' end on the right, we have:



According to Figure 15.14, both codons specify the amino acid lysine (Lys). Recall that the wobble in the third position allows more than one codon to specify the same amino acid; so any wobble that exists should produce the same amino acid as the standard base pairings would, and we do not need to figure the wobble to answer this question. The answers for parts *b*, *c*, and *d* are:

- (b) Anticodon: $5' - \text{GAC} - 3'$
Codon: $3' - \text{CUG} - 5'$
 $5' - \text{GUC} - 3'$ codes for Val
- (c) Anticodon: $5' - \text{UUG} - 3'$
Codon: $3' - \text{AAC} - 5'$
 $5' - \text{CAA} - 3'$ codes for Gln
- (d) Anticodon: $5' - \text{CAG} - 3'$
Codon: $3' - \text{GUC} - 5'$
 $5' - \text{CUG} - 3'$ codes for Leu

The New Genetics

MINING GENOMES

THREE DIMENSIONAL PROTEIN STRUCTURE

The function of proteins and other biological macromolecules is directly related to their shape, and understanding the three-dimensional shape of proteins is an important developing field in bioinformatics. This exercise uses tools available at the

Biology Workbench, managed by the San Diego Supercomputing Center at the University of California, San Diego, to visualize the three-dimensional shape of some interesting proteins.

COMPREHENSION QUESTIONS

1. What is the one gene, one enzyme hypothesis? Why was this hypothesis an important advance in our understanding of genetics?
2. What three different methods were used to help break the genetic code? What did each reveal and what were the advantages and disadvantages of each?
3. What are isoaccepting tRNAs?
4. What is the significance of the fact that many synonymous codons differ only in the third nucleotide position?
5. Define the following terms as they apply to the genetic code:
 - (a) reading frame
 - (b) overlapping code
 - (c) nonoverlapping code
 - (d) initiation codon
 - (e) termination codon
 - (f) sense codon
 - (g) nonsense codon
 - (h) universal code
 - (i) nonuniversal codons
6. How is the reading frame of a nucleotide sequence set?
7. How are tRNAs linked to their corresponding amino acids?
8. What role do the initiation factors play in protein synthesis?
9. How does the process of initiation differ in bacterial and eukaryotic cells?
10. Give the elongation factors used in bacterial translation and explain the role played by each factor in translation.
11. What events bring about the termination of translation?
12. Give several examples of RNA–RNA interactions that take place in protein synthesis.
13. What are some types of posttranslational modification of proteins?
14. Explain how some antibiotics work by affecting the process of protein synthesis.
15. Compare and contrast the process of protein synthesis in bacterial and eukaryotic cells, giving similarities and differences in the process of translation in these two types of cells.

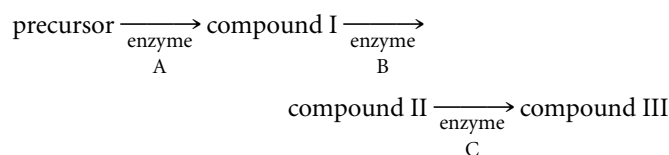
APPLICATION QUESTIONS AND PROBLEMS

16. Sydney Brenner isolated *Salmonella typhimurium* mutants that were implicated in the biosynthesis of tryptophan and would not grow on minimal medium. When these mutants were tested on minimal medium to which one of four compounds (indole glycerol phosphate, indole, anthranilic acid, and tryptophan) had been added, the growth responses shown in the following table were obtained.

Mutant	Minimal medium	Anthranilic acid	Indole glycerol phosphate	Indole	Tryptophan
<i>trp</i> -1	—	—	—	—	+
<i>trp</i> -2	—	—	+	+	+
<i>trp</i> -3	—	—	—	+	+
<i>trp</i> -4	—	—	+	+	+
<i>trp</i> -6	—	—	—	—	+
<i>trp</i> -7	—	—	—	—	+
<i>trp</i> -8	—	+	+	+	+
<i>trp</i> -9	—	—	—	—	+
<i>trp</i> -10	—	—	—	—	+
<i>trp</i> -11	—	—	—	—	+

Give the order of indole glycerol phosphate, indole, anthranilic acid, and tryptophan in a biochemical pathway leading to the synthesis of tryptophan. Indicate which step in the pathway is affected by each of the mutations.

17. The addition of a series of compounds yielded in the following biochemical pathway:



Mutation *a* inactivates enzyme A, mutation *b* inactivates enzyme B, and mutation *c* inactivates enzyme C. Mutants, each having one of these defects, were tested on minimal medium to which compound I, II, or III was added. Fill in the results expected of these tests by placing a plus sign (+) for growth or a minus sign (–) for no growth in the following table:

Strain with mutation	Minimal medium to which is added		
	Compound I	Compound II	Compound III
<i>a</i>			
<i>b</i>			
<i>c</i>			

18. Assume that the number of different types of bases in RNA is four. What would be the minimum codon size (number of nucleotides) required if the number of different types of amino acids in proteins were:
(a) 2, (b) 8, (c) 17, (d) 45, (e) 75.
19. How many codons would be possible in a triplet code if only three bases (A, C, and U) were used?

20. Using the genetic code given in Figure 15.14, give the amino acids specified by the following bacterial mRNA sequences, and indicate the amino and carboxyl ends of the polypeptide produced.
- 5'–AUGUUUAAAUUUAAAUUUUGA–3'
 - 5'–AUGUAUAUAUAUAUGA–3'
 - 5'–AUGGAUGAAAGAUUUCUCGCUUGA–3'
 - 5'–AUGGGUUAGGGGACAUCAUUUUGA–3'
21. A nontemplate strand on DNA has the following base sequence. What amino acid sequence would be encoded by this sequence?
- 5'–ATGATACTAAGGCCC–3'
22. The following amino acid sequence is found in a tripeptide: Met-Trp-His. Give all possible nucleotide sequences on the mRNA, on the template strand of DNA, and on the nontemplate strand of DNA that could encode this tripeptide.
23. How many different mRNA sequences can code for a polypeptide chain with the amino acid sequence Met-Leu-Arg? (Be sure to include the stop codon.)
24. A series of tRNAs have the following anticodons. Consider the wobble rules given in Table 14.2 and give all possible codons with which each tRNA can pair.
- 5'–GGC–3'
 - 5'–AAG–3'
 - 5'–IAA–3'
 - 5'–UGG–3'
 - 5'–CAG–3'
25. An anticodon on a tRNA has the sequence 5'–GCA–3'.
- What amino acid is carried by this tRNA?
 - What would be the effect if the G in the anticodon were mutated to a U?
26. Which of the following amino acid changes could result from a mutation that changed a single base? For each change that could result from the alteration of a single base, determine which position of the codon (first, second, or third nucleotide) in the mRNA must be altered for the change to result.
- Leu → Gln
 - Phe → Ser
 - Phe → Ile
 - Pro → Ala
 - Asn → Lys
 - Ile → Asn
27. A synthetic mRNA added to a cell-free protein-synthesizing system produces a peptide with the following amino acid sequence: Met-Pro-Ile-Ser-Ala. What would be the effect on translation if the following components were omitted from the cell-free protein-synthesizing system? What, if any, type of protein would be produced? Explain your reasoning.
- initiation factor 1
 - initiation factor 2
 - elongation factor Tu
 - elongation factor G
 - release factors R₁, R₂, and R₃
 - ATP
 - GTP

CHALLENGE QUESTIONS

28. In what ways are spliceosomes and ribosomes similar? In what ways are they different? Can you suggest some possible reasons for their similarities.
29. Several experiments were conducted to obtain information about how the eukaryotic ribosome recognizes the AUG start codon. In one experiment, the gene that codes for methionine initiator tRNA (tRNA_i^{Met}) was located and changed. The nucleotides that specify the anticodon on tRNA_i^{Met} were mutated so that the anticodon in the tRNA was 5'–CCA–3' instead of 5'–CAU–3'. When this mutated gene was placed into a eukaryotic cell, protein synthesis took place, but the proteins produced were abnormal. Some of the proteins produced contained extra amino acids, and others contained fewer amino acids.
- What do these results indicate about how the ribosome recognizes the starting point for translation in eukaryotic cells? Explain your reasoning.
 - If the same experiment had been conducted on bacterial cells, what results would you expect?

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A report of the structure of the complete ribosome with mRNA and bound tRNAs at high resolution.

16

Control of Gene Expression



The giant transgenic mouse on the left was produced by injecting a rat gene for growth hormone into a mouse embryo; a normal-size mouse is on the right. To ensure expression, the rat gene was linked to a DNA sequence that stimulates the transcription of mouse DNA whenever heavy metals are present. Zinc was provided in the food for the transgenic mouse; some transgenic mice produced 800 times the normal levels of growth hormone. (Courtesy of Dr. Ralph L. Brinster, School of Veterinary Medicine, University of Pennsylvania.)

Creating Giant Mice Through Gene Regulation

In 1982, a group of molecular geneticists led by Richard Palmiter at the University of Washington produced gigantic mice that grew to almost twice the size of normal mice. Palmiter and his colleagues created these large mice through genetic engineering, by injecting the rat gene for growth hormone into the nuclei of fertilized mouse embryos and then implanting these embryos into surrogate mouse mothers. In a few embryos, the rat gene became incorporated into the mouse chromosome and, after birth, these *trans-*

genic mice produced growth hormone encoded by the rat gene. Some of the transgenic mice produced from 100 to 800 times the amount of growth hormone found in normal mice, which caused them to grow rapidly into giants.

Inserting foreign genes into bacteria, plants, mice, and even humans is now a routine procedure for molecular geneticists (see Chapter 18). However, simply putting a gene into a cell does not guarantee that the gene will be transcribed or produce a protein; indeed, most foreign genes are never transcribed or translated, which isn't surprising. Organisms have evolved complex systems to ensure that genes are expressed at the appropriate time and in the

- Creating Giant Mice Through Gene Regulation
- General Principles of Gene Regulation
 - Levels of Gene Control
 - Genes and Regulatory Elements
 - DNA-Binding Proteins
- Gene Regulation in Bacterial Cells
 - Operon Structure
 - Negative and Positive Control: Inducible and Repressible Operons
 - The *lac* Operon of *E. coli*
 - lac* Mutations
 - Positive Control and Catabolite Repression
 - The *trp* Operon of *E. coli*
 - Attenuation: The Premature Termination of Transcription
 - Antisense RNA in Gene Regulation
 - Transcriptional Control in Bacteriophage Lambda
- Eukaryotic Gene Regulation
 - Chromatin Structure and Gene Regulation
 - Transcriptional Control in Eukaryotic Cells
 - Gene Control Through Messenger RNA Processing
 - Gene Control Through RNA Stability
 - RNA Silencing
 - Translational and Posttranslational Control

appropriate amounts, and sequences other than the gene itself are required to ensure transcription and translation. In this chapter, we will learn more about these sequences and other mechanisms that control gene expression.

If foreign genes are rarely expressed, why did the transgenic mice with the gene for rat growth hormone grow so big? Palmiter and his colleagues, aware of the need to provide sequences that control gene expression, linked the rat gene with the mouse metallothionein I promoter sequence, a DNA sequence normally found upstream of the mouse metallothionein I gene. When heavy metals such as zinc are present, they activate the metallothionein promoter sequence, thereby stimulating transcription of the metallothionein I gene. By connecting the rat growth-hormone gene to this promoter, Palmiter and his colleagues provided a means of turning on the transcription of the gene, simply by putting extra zinc in the food for the transgenic mice.

This chapter is about **gene regulation**, the mechanisms and systems that control the expression of genes. We begin by discussing why gene regulation is necessary; the levels at which gene expression is controlled; and the difference between genes and regulatory elements. We then examine gene regulation in bacterial cells. In the second half of the chapter, we turn to gene regulation in eukaryotic cells, which is often more complex than in bacterial cells.

General Principles of Gene Regulation

One of the major themes of molecular genetics is the central dogma, which stated that genetic information flows from DNA to RNA to proteins (see Figure 10.17a) and provided a molecular basis for the connection between genotype and phenotype. Although the central dogma brought coherence to early research in molecular genetics, it failed to address a critical issue: How is the flow of information along the molecular pathway *regulated*?

Consider *E. coli*, a bacterium that resides in your large intestine. Your eating habits completely determine the nutrients available to this bacteria: it can't seek out nourishment when nutrients are scarce; nor can it move away when confronted with unpleasant changes. *E. coli* makes up for its inability to alter the external environment by being internally flexible. For example, if glucose is present, *E. coli* uses it to generate ATP; if there's no glucose, it utilizes lactose, arabinose, maltose, xylose, or any of a number of other sugars. When amino acids are available, *E. coli* uses them to synthesize proteins; if a particular amino acid is absent, *E. coli* produces the enzymes needed to synthesize that amino acid. Thus, *E. coli* responds to environmental changes by rapidly altering its biochemistry. This biochemical flexibility, however, has a high price. Producing all the enzymes necessary for every environmental condition would be energetically expensive. So how does *E. coli* maintain biochemical flexibility while optimizing energy efficiency?

The answer is through gene regulation. Bacteria carry the genetic information for many proteins, but only a subset of this genetic information is expressed at any time. When the environment changes, new genes are expressed, and proteins appropriate for the new environment are synthesized. For example, if a carbon source appears in the environment, genes encoding enzymes that take up and metabolize this carbon source are quickly transcribed and translated. When this carbon source disappears, the genes that encode them are shut off. This type of response, the synthesis of an enzyme stimulated by a specific substrate, is called **induction**.

Multicellular eukaryotic organisms face a different dilemma. Individual cells in a multicellular organism are specialized for particular tasks. The proteins produced by a nerve cell, for example, are quite different from those produced by a white blood cell. The problem that a eukaryotic cell faces is how to specialize. Although they are quite different in shape and function, a nerve cell and a blood cell still carry the same genetic instructions.

A multicellular organism's challenge is to bring about the specialization of cells that have a common set of genetic instructions. This challenge is met through gene regulation: all of an organism's cells carry the same genetic information, but only a subset of genes are expressed in each cell type. Genes needed for other cell types are not expressed. Gene regulation is therefore the key to both unicellular flexibility and multicellular specialization, and it is critical to the success of all living organisms.

Concepts

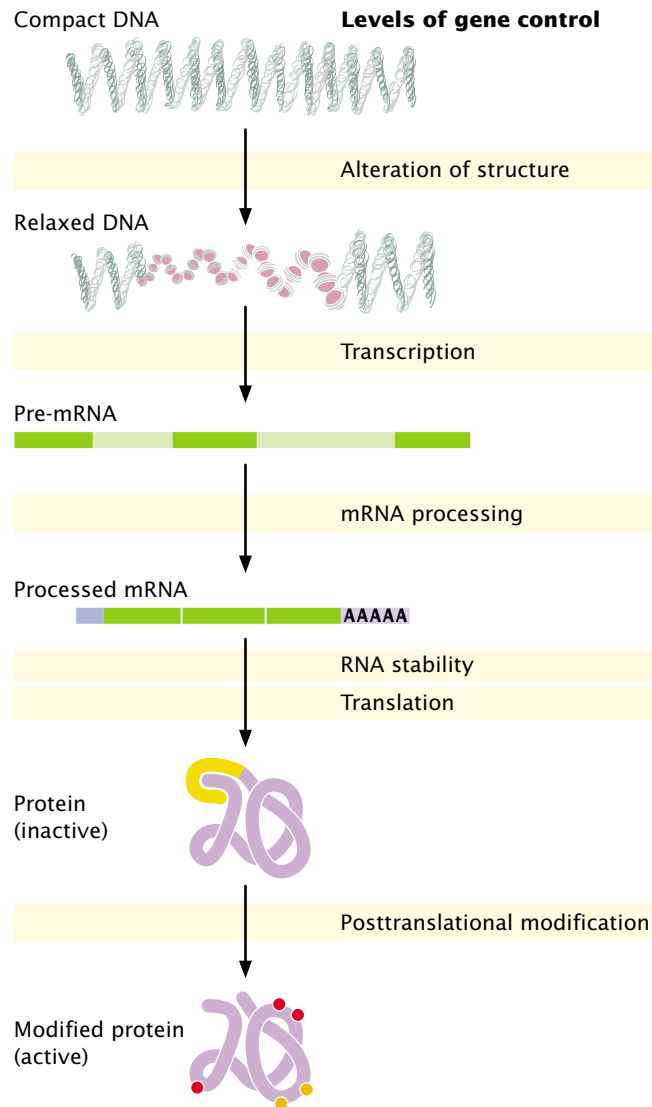
In bacteria, gene regulation maintains internal flexibility, turning genes on and off in response to environmental changes. In multicellular eukaryotic organisms, gene regulation brings about cellular differentiation.



Levels of Gene Control

A gene may be regulated at a number of points along the pathway of information flow from genotype to phenotype (FIGURE 16.1). First, regulation may be through the alteration of gene structure. Modifications to DNA or its packaging may influence which sequences are available for transcription or the rate at which sequences are transcribed. DNA methylation and changes in chromatin are two processes that play a pivotal role in gene regulation.

A second point at which a gene can be regulated is at the level of transcription. For the sake of cellular economy, it makes sense to limit protein production early in the transfer of information from DNA to protein, and transcription is an important point of gene regulation in both bacterial and eukaryotic cells. A third potential point of gene regulation is mRNA processing. Eukaryotic mRNA is



16.1 Gene expression may be controlled at multiple levels.

extensively modified before it is translated; a 5' cap is added, the 3' end is cleaved and polyadenylated, and introns are removed (see Chapter 14). These modifications determine the stability of the mRNA, whether mRNA can be translated, the rate of translation, and the amino acid sequence of the protein produced. There is growing evidence that a number of regulatory mechanisms in eukaryotic cells operate at the level of mRNA processing.

A fourth point for the control of gene expression is the regulation of RNA stability. The amount of protein produced depends not only on the amount of mRNA synthesized, but also on the rate at which the mRNA is degraded; so RNA stability plays an important role in gene expression. A fifth point of gene regulation is at the level of translation, a complex process requiring a large number of enzymes, protein factors, and RNA molecules (Chapter 15).

All of these factors, as well as the availability of amino acids and sequences in mRNA, influence the rate at which proteins are produced and therefore provide points at which gene expression may be controlled.

Finally, many proteins are modified after translation (Chapter 13), and these modifications affect whether the proteins become active; so genes can be regulated through processes that affect posttranscriptional modification. Gene expression may be affected by regulatory activities at any or all of these points.

Concepts

Gene expression may be controlled at any of a number of points along the molecular pathway from DNA to protein, including gene structure, transcription, mRNA processing, RNA stability, translation, and posttranslational modification.

Genes and Regulatory Elements

In our consideration of gene regulation, it will be necessary to distinguish between the DNA sequences that are transcribed and the DNA sequences that regulate the expression of other sequences. We will refer to any DNA sequence that is transcribed into an RNA molecule as a *gene*. According to this definition, genes include DNA sequences that encode proteins, as well as sequences that encode rRNA, tRNA, snRNA, and other types of RNA. **Structural genes** encode proteins that are used in metabolism or biosynthesis or that play a structural role in the cell. **Regulatory genes** are genes whose products, either RNA or proteins, interact with other sequences and affect their transcription or translation. In many cases, the products of regulatory genes are DNA-binding proteins.

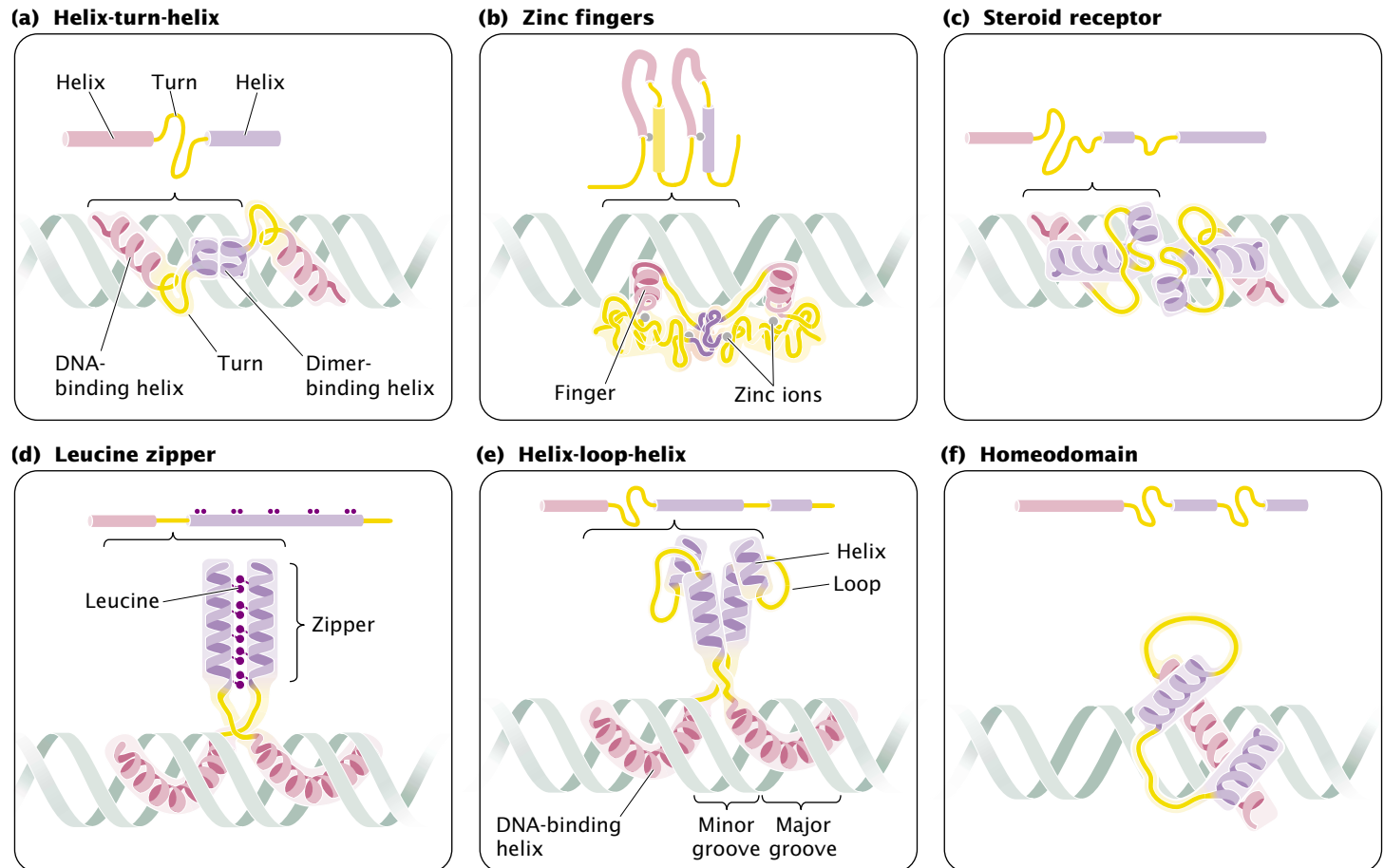
We will also encounter DNA sequences that are not transcribed at all but still play a role in regulating other nucleotide sequences. These **regulatory elements** affect the expression of sequences to which they are physically linked. Much of gene regulation takes place through the action of proteins produced by regulatory genes that recognize and bind to regulatory elements.

Concepts

Genes are DNA sequences that are transcribed into RNA. Regulatory elements are DNA sequences that are not transcribed but affect the expression of genes.

DNA-Binding Proteins

Much of gene regulation is accomplished by proteins that bind to DNA sequences and influence their expression. These regulatory proteins generally have discrete functional



16.2 DNA-binding proteins can be grouped into several types on the basis of their structure, or motif.

(a) The helix-turn-helix DNA motif consists of two alpha helices connected by a turn. (b) The zinc-finger motif consists of a loop of amino acids containing a single zinc ion. Most proteins containing zinc fingers have several repeats of the zinc-finger motif. Each zinc finger fits into the major groove of DNA and forms hydrogen bonds with bases in the DNA. (c) The steroid receptor binding motif has two alpha helices, each with a zinc ion surrounded by four cysteine residues. The two alpha helices are perpendicular to one another: one fits into the major groove of the double helix, whereas the other is parallel to the DNA. (d) The leucine-zipper motif consists of a helix of leucine nucleotides and an arm of basic amino acids. DNA-binding proteins usually have two polypeptides; the leucine nucleotides of the two polypeptides face one another, whereas the basic amino acids bind to the DNA. (e) The helix-loop-helix binding motif consists of two alpha helices separated by a loop of amino acids. Two polypeptide chains with this motif join to form a functional DNA-binding protein. A highly basic set of amino acids in one of the helices binds to the DNA. (f) The homeodomain motif consists of three alpha helices; the third helix fits in a major groove of DNA.

parts—called **domains**, typically consisting of 60 to 90 amino acids—that are responsible for binding to DNA. Within a domain, only a few amino acids actually make contact with the DNA. These amino acids (most commonly asparagine, glutamine, glycine, lysine, and arginine) often form hydrogen bonds with the bases or interact with the sugar–phosphate backbone of the DNA. Many regulatory proteins have additional domains that can bind other molecules such as other regulatory proteins.

DNA-binding proteins can be grouped into several distinct types on the basis of a characteristic structure, called a motif, found within the binding domain. Motifs are simple structures, such as alpha helices, that can fit into the major groove of the DNA. Some common DNA-binding motifs are illustrated in **FIGURE 16.2** and are summarized in Table 16.1.

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Molecular images of several DNA-binding proteins

Molecular images of several DNA-binding proteins

Table 16.1 Common DNA-binding motifs

Motif	Location	Characteristics	Binding Site in DNA
Helix-turn-helix	Bacterial regulatory proteins; related motifs in eukaryotic proteins	Two alpha helices	Major groove
Zinc-finger	Eukaryotic regulatory and other proteins	Loop of amino acids with zinc at base	Major groove
Steroid receptor	Eukaryotic proteins	Two perpendicular alpha helices with zinc surrounded by four cysteine residues	Major groove and DNA backbone
Leucine-zipper	Eukaryotic transcription factors	Helix of leucine residues and a basic arm; two leucine residues interdigitate	Two adjacent major grooves
Helix-loop-helix	Eukaryotic proteins	Two alpha helices separated by a loop of amino acids	Major groove
Homeodomain	Eukaryotic regulatory proteins	Three alpha helices	Major groove

Gene Regulation in Bacterial Cells

The mechanisms of gene regulation were first investigated in bacterial cells, where the availability of mutants and the ease of laboratory manipulation made it possible to unravel the mechanisms. When the study of these mechanisms in eukaryotic cells began, it seemed clear that bacterial and eukaryotic gene regulation were quite different. As more and more information has accumulated about gene regulation, however, a number of common themes have emerged, and today many aspects of gene regulation in bacterial and eukaryotic cells are recognized to be similar. Although we will look at gene regulation in these two cell types separately, the emphasis will be on the common themes that apply to all cells.

Operon Structure

One significant difference in prokaryotic and eukaryotic gene control lies in the organization of functionally related genes. Many bacterial genes that have related functions are clustered and are under the control of a single promoter. These genes are often transcribed together into a single mRNA. Eukaryotic genes, in contrast, are dispersed, and typically, each is transcribed into a separate mRNA. A group of bacterial structural genes that are transcribed together (along with their promoter and additional sequences that control transcription) is called an **operon**.

The organization of a typical operon is illustrated in **FIGURE 16.3**. At one end of the operon is a set of structural genes, shown in Figure 16.3 as gene *a*, gene *b*, and gene *c*. These structural genes are transcribed into a single mRNA, which is translated to produce enzymes A, B, and C. These

enzymes carry out a series of biochemical reactions that convert precursor molecule X into product Y. The transcription of structural genes *a*, *b*, and *c* is under the control of a promoter, which lies upstream of the first structural gene. RNA polymerase binds to the promoter and then moves downstream, transcribing the structural genes.

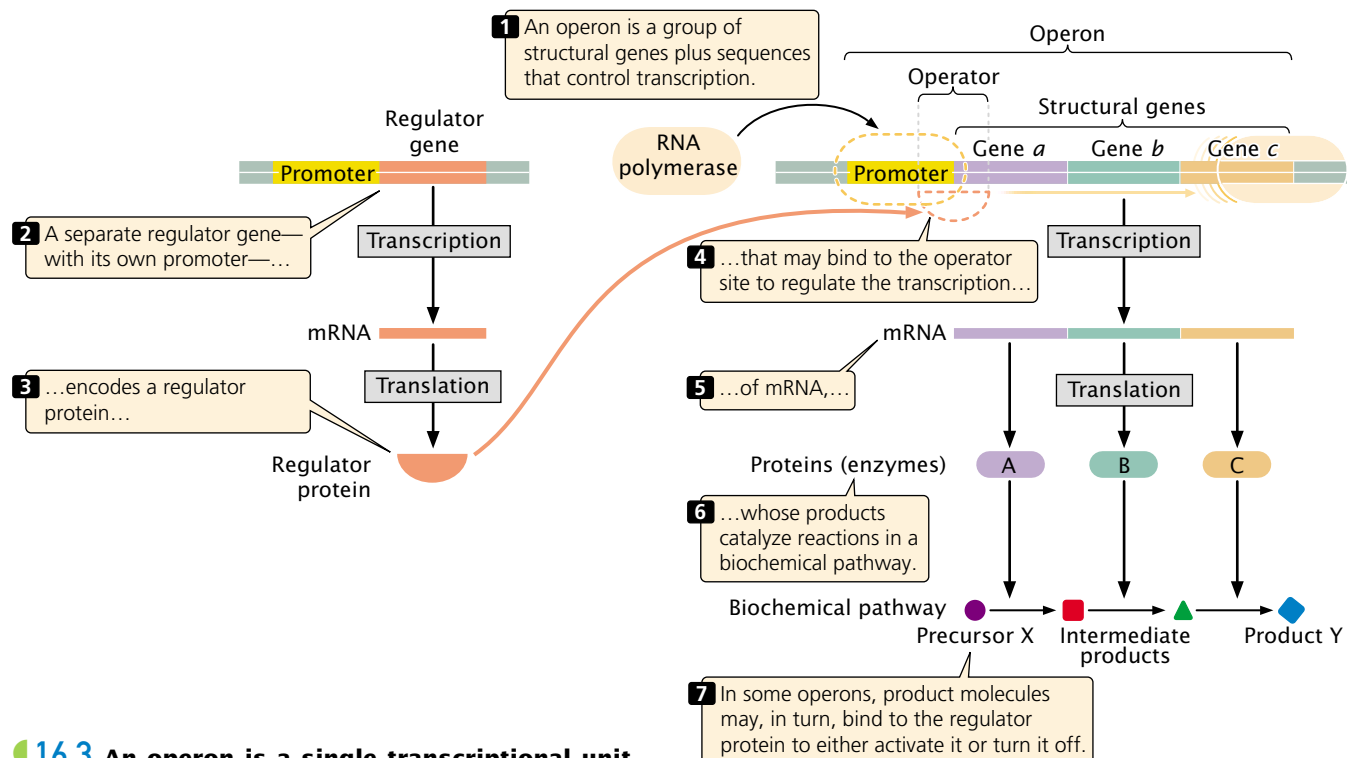
A **regulator gene** helps to regulate the transcription of the structural genes of the operon. The regulator gene is not considered part of the operon, although it affects operon function. The regulator gene has its own promoter and is transcribed into a relatively short mRNA, which is translated into a small protein. This **regulator protein** may bind to a region of DNA called the **operator** and affect whether transcription can take place. The operator usually overlaps the 3' end of the promoter and sometimes the 5' end of the first structural gene (see Figure 16.3).

Concepts

Functionally related genes in bacterial cells are frequently clustered together as a single transcriptional unit termed an operon. A typical operon includes several structural genes, a promoter for the structural genes, and an operator site where the product of a regulator gene binds.

Negative and Positive Control: Inducible and Repressible Operons

There are two types of transcriptional control: **negative control**, in which a regulatory protein acts as a repressor, binding to DNA and inhibiting transcription; and **positive control**, in



16.3 An operon is a single transcriptional unit that includes a series of structural genes, a promoter, and an operator.

which a regulatory protein acts as an activator, stimulating transcription. In the next sections, we will consider several varieties of these two basic control mechanisms.

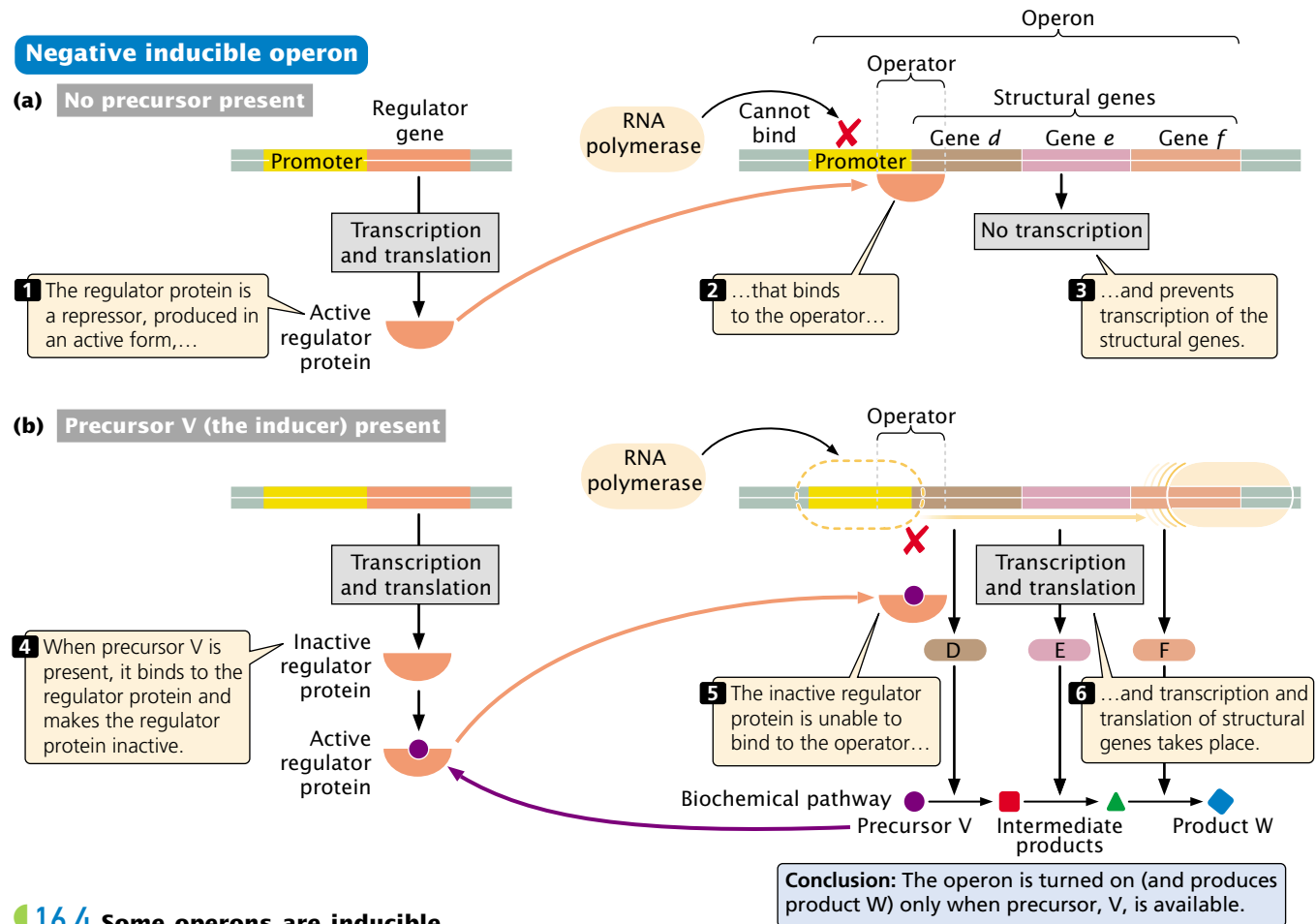
Negative inducible operons In an operon with negative control at the operator site, the regulatory protein is a repressor—the binding of the regulator protein to the operator inhibits transcription. In a *negative inducible operon*, transcription and translation of the regulator gene produce an *active* repressor that readily binds to the operator (FIGURE 16.4a). Because the operator site overlaps with the promoter site, the binding of this protein to the operator physically blocks the binding of RNA polymerase to the promoter and prevents transcription. For transcription to take place, something must happen to prevent the binding of the repressor at the operator site. This type of system is said to be inducible, because transcription is normally off (inhibited) and must be turned on (induced).

Transcription is turned on when a small molecule, an **inducer**, binds to the repressor. FIGURE 16.4b shows that, when precursor V (acting as the inducer) binds to the repressor, the repressor can no longer bind to the operator. Regulatory proteins frequently have two binding sites: one that binds to DNA and another that binds to a small molecule such as an inducer. Binding of the inducer alters the shape of the repressor, preventing it from binding to DNA. Proteins of this type, which change shape on binding to another molecule, are called **allosteric proteins**.

When the inducer is absent, the repressor binds to the operator, the structural genes are not transcribed, and enzymes D, E, and F (which metabolize precursor V) are not synthesized (see Figure 16.4a). This is an adaptive mechanism: because no precursor V is available, it would be wasteful for the cell to synthesize the enzymes when they have no substrate to metabolize. As soon as precursor V becomes available, some of it binds to the repressor, rendering the repressor inactive and unable to bind to the operator site. Now RNA polymerase can bind to the promoter and transcribe the structural genes. The resulting mRNA is then translated into enzymes D, E, and F, which convert substrate V into product W (see Figure 16.4b). So, an operon with negative inducible control regulates the synthesis of the enzymes economically: the enzymes are synthesized only when their substrate (V) is available.

Negative repressible operons Some operons with negative control are **repressible**, meaning that transcription *normally* takes place and must be turned off, or repressed. The regulator protein in this type of operon also is a repressor but is synthesized in an inactive form that cannot by itself bind to the operator. Because there is no repressor bound to the operator, RNA polymerase readily binds to the promoter and transcription of the structural genes takes place (FIGURE 16.5a).

To turn transcription off, something must happen to make the repressor active. A small molecule called a **core-**



16.4 Some operons are inducible.

pressor binds to the repressor and makes it capable of binding to the operator. In the example illustrated (see Figure 16.5a), the product (U) of the metabolic reaction is the corepressor. As long as the level of product U is high, it is available to bind to and activate the repressor, preventing transcription (FIGURE 16.5b). With the operon repressed, enzymes G, H, and I are not synthesized, and no more U is produced from precursor T. However, when all of product U is used up, the repressor is no longer activated by U and cannot bind to the operator. The inactivation of the repressor allows the transcription of the structural genes and the synthesis of enzymes G, H, and I, resulting in the conversion of precursor T into product U.

As with inducible operons, repressible operons are economical: the enzymes are synthesized only as needed. Note that both the inducible and the repressible systems that we have considered are forms of negative control, in which the regulatory protein is a repressor. We will now consider positive control, in which a regulator protein stimulates transcription.

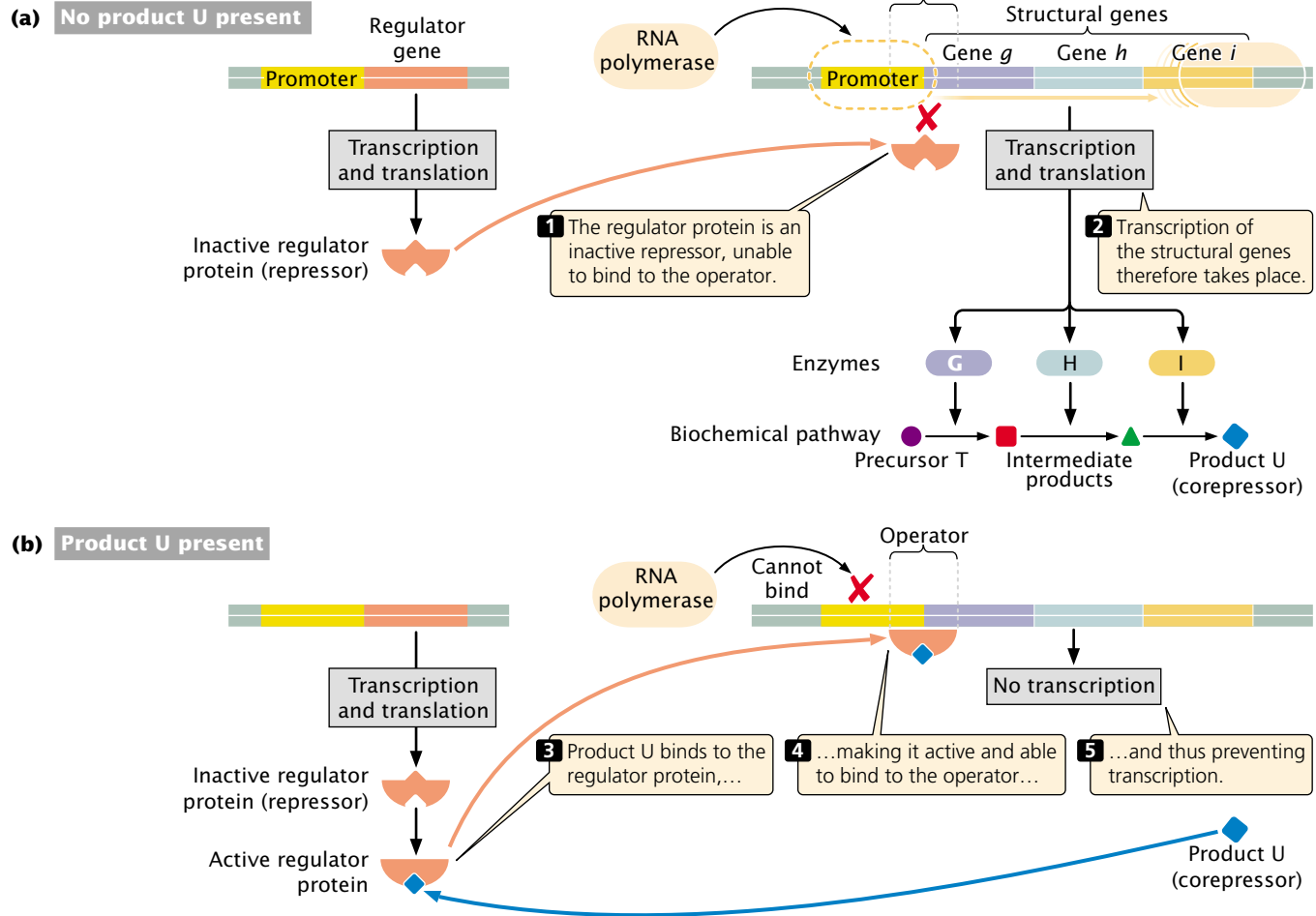
Positive control With positive control, a regulatory protein binds to DNA (usually at a site other than the operator)

and stimulates transcription. Theoretically, positive control could be inducible or repressible.

In a positive *inducible* operon, transcription would normally be turned off because the regulator protein would be produced in an inactive form. Transcription would take place when an inducer became attached to the regulatory protein, rendering the regulator active. Logically, the inducer should be the precursor of the reaction controlled by the operon so that the necessary enzymes would be synthesized only when the substrate for their reaction was present.

A positive operon could also be repressible; transcription would normally take place and would have to be repressed. In this case, the regulator protein would be produced in a form that readily binds to DNA and stimulates transcription. Transcription would be inhibited when a substance became attached to the activator and rendered it unable to bind to the DNA so that transcription was no longer stimulated. Here, the product (P) of the reaction controlled by the operon would logically be the repressing substance, because it would be economical for the cell to prevent the transcription of genes that allow the synthesis of P when plenty of P is already available.

Negative repressible operon



16.5 Some operons are repressible.

Putting it all together Theoretically, operons might exhibit positive or negative control and be either inducible or repressible. Try sketching out all possible types—negative inducible, negative repressible, positive inducible, and positive repressible. To do so, learn the meanings of *positive* and *negative control* and *inducible* and *repressible*; then use logic to work out the details of whether the regulatory protein is a repressor or an activator and whether it is produced in an active or inactive form. You can check your answers against Table 16.2, where the important features of these four types of operons are summarized. Another useful exercise is to think about the effects of mutations at various sites in different types of operon systems.

Although it is a useful learning device to think of operons as either positive or negative and either inducible or repressible, in reality both positive and negative controls often exist in the same operon.

Concepts

There are two basic types of transcriptional control: negative and positive. In negative control, when a regulatory protein (repressor) binds to DNA, transcription is inhibited; in positive control, when a regulatory protein (activator) binds to DNA, transcription is stimulated. Some operons are inducible; transcription is normally off and must be turned on. Other operons are repressible; transcription is normally on and must be turned off.

The *lac* Operon of *E. coli*

In 1961, François Jacob and Jacques Monod described the “operon model” for the genetic control of lactose metabolism in *E. coli*. This work and subsequent research on the genetics of lactose metabolism established the operon as

Table 16.2 Features of inducible and repressible operons with positive and negative control

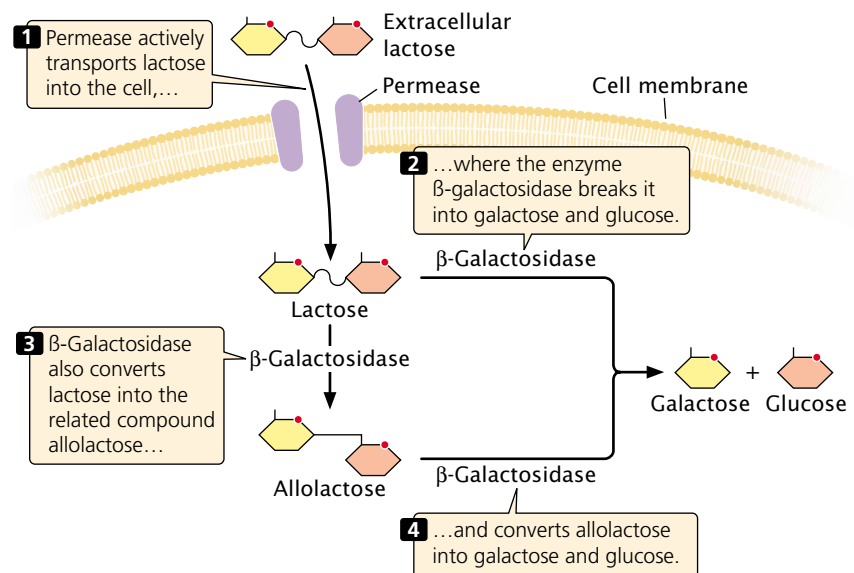
Type of Control	Transcription Normally	Regulator Protein	Effect of Regulatory Protein	Action of Modulator
Negative inducible	Off	Active repressor	Inhibits transcription	Substrate makes repressor inactive
Negative repressible	On	Inactive repressor	Inhibits transcription	Product makes repressor active
Positive inducible	Off	Inactive activator	Stimulates transcription	Substrate makes activator active
Positive repressible	On	Active activator	Stimulates transcription	Product makes activator inactive

the basic unit of transcriptional control in bacteria. Despite the fact that, at the time, no methods were available for determining nucleotide sequences, Jacob and Monod deduced the structure of the operon *genetically* by analyzing the interactions of mutations that interfered with the normal regulation of lactose metabolism. We will examine the effects of some of these mutations after seeing how the *lac* operon regulates lactose metabolism.

Lactose (a disaccharide) is one of the major carbohydrates found in milk; it can be metabolized by *E. coli* bacteria that reside in the gut of mammals. Lactose does not easily diffuse across the *E. coli* cell membrane and must be actively transported into the cell by the enzyme permease (FIGURE 16.6). To utilize lactose as an energy source, *E. coli* must first break it into glucose and galactose, a reaction catalyzed by the enzyme β -galactosidase. This enzyme can also convert lactose into allolactose, a compound that plays an

important role in regulating lactose metabolism. A third enzyme, thiogalactoside transacetylase, also is produced by the *lac* operon, but its function in lactose metabolism is not yet known.

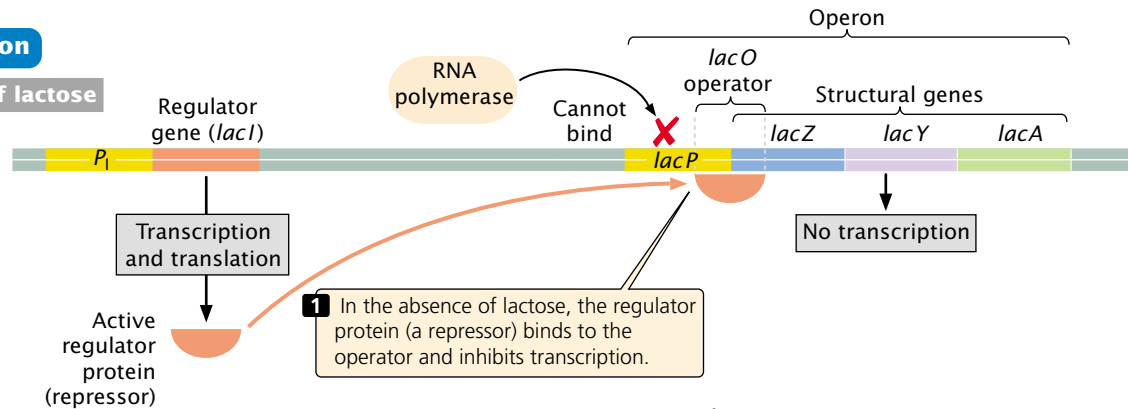
The enzymes β -galactosidase, permease, and transacetylase are encoded by adjacent structural genes in the *lac* operon of *E. coli*. β -Galactosidase is encoded by the *lacZ* gene, permease by the *lacY* gene, and transacetylase by the *lacA* gene (FIGURE 16.7a). When lactose is absent from the medium in which *E. coli* grows, only a few molecules of each enzyme are produced. If lactose is added to the medium and glucose is absent, the rate of synthesis of all three enzymes simultaneously increases about a thousandfold within 2 to 3 minutes. This boost in enzyme synthesis results from transcription of *lacZ*, *lacY*, and *lacA* and exemplifies **coordinate induction**, the simultaneous synthesis of several enzymes, stimulated by a specific molecule, the inducer (FIGURE 16.7b). Although



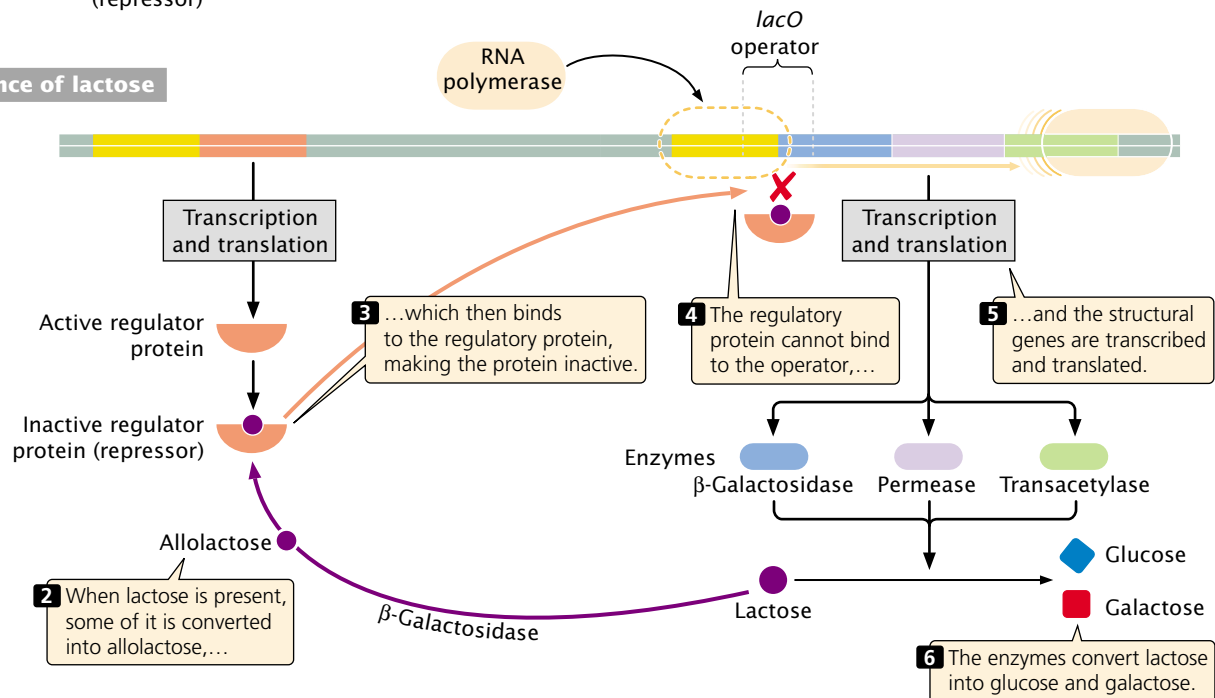
16.6 Lactose, a major carbohydrate found in milk, consists of 2 six-carbon sugars linked together.

The *lac* operon

(a) Absence of lactose



(b) Presence of lactose



16.7 The *lac* operon regulates lactose metabolism.

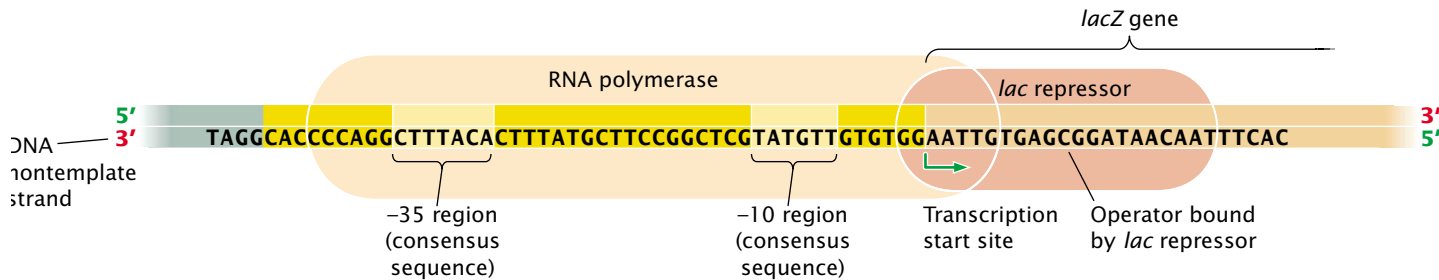
lactose appears to be the inducer here, allolactose is actually responsible for induction.

In the *lac* operon, the *lacZ*, *lacY*, and *lacA* genes have a common promoter (*lacP* in Figure 16.7a) and are transcribed together. Upstream of the promoter is a regulator gene, *lacI*, which has its own promoter (P_1). The *lacI* gene is transcribed into a short mRNA that is translated into a repressor. Each repressor consists of four identical polypeptides and has two binding sites; one site binds to allolactose and the other binds to DNA. In the absence of lactose (and, therefore, allolactose), the repressor binds to the *lac* operator site *lacO* (see Figure 16.7a). Jacob and Monod mapped the operator to a position adjacent to the *lacZ* gene; more recent nucleotide sequencing has demonstrated that the operator actually overlaps the 3' end of the promoter and the 5' end of *lacZ* (FIGURE 16.8).

Immediately upstream of the structural genes is the *lac* promoter. RNA polymerase binds to the promoter and moves

down the DNA molecule, transcribing the structural genes. When the repressor is bound to the operator, the binding of RNA polymerase is blocked, and transcription is prevented. When lactose is present, some of it is converted into allolactose, which binds to the repressor and causes the repressor to be released from the DNA. In the presence of lactose, then, the repressor is inactivated, the binding of RNA polymerase is no longer blocked, the transcription of *lacZ*, *lacY*, and *lacA* takes place, and the *lac* enzymes are produced.

Have you spotted the flaw in the explanation just given for the induction of the *lac* enzymes? You might recall that permease is required to transport lactose into the cell. If the *lac* operon is repressed and no permease is being produced, how does lactose get into the cell to inactivate the repressor and turn on transcription? Furthermore, the inducer is actually allolactose, which must be produced from lactose by β -galactosidase. If β -galactosidase production is repressed, how can lactose metabolism be induced?



16.8 In the *lac* operon, the operator overlaps the promoter and the 5' end of the first structural gene.

The answer is that repression never *completely* shuts down transcription of the *lac* operon. Even with active repressor bound to the operator, there is a low level of transcription and a few molecules of β -galactosidase, permease, and transacetylase are synthesized. When lactose appears in the medium, the permease that is present transports a small amount of lactose into the cell. There, the few molecules of β -galactosidase that are present convert some of the lactose into allolactose. The allolactose then attaches to the repressor and alters its shape so that the repressor no longer binds to the operator. When the operator site is clear, RNA polymerase can bind and transcribe the structural genes of the *lac* operon.

Several compounds related to allolactose also can bind to the *lac* repressor and induce transcription of the *lac* operon. One such inducer is isopropylthiogalactoside (IPTG). Although IPTG inactivates the repressor and allows the transcription of *lacZ*, *lacY*, and *lacA*, IPTG is not metabolized by β -galactosidase; for this reason it is often used in research to examine the effects of induction, independent of metabolism.

Concepts

The *lac* operon of *E. coli* controls the transcription of three genes in lactose metabolism: the *lacZ* gene, which encodes β -galactosidase; the *lacY* gene, which encodes permease; and the *lacA* gene, which encodes thiogalactoside transacetylase. The *lac* operon is inducible: a regulator gene produces a repressor that binds to the operator site and prevents the transcription of the structural genes. The presence of allolactose inactivates the repressor and allows the transcription of the *lac* operon.

lac Mutations

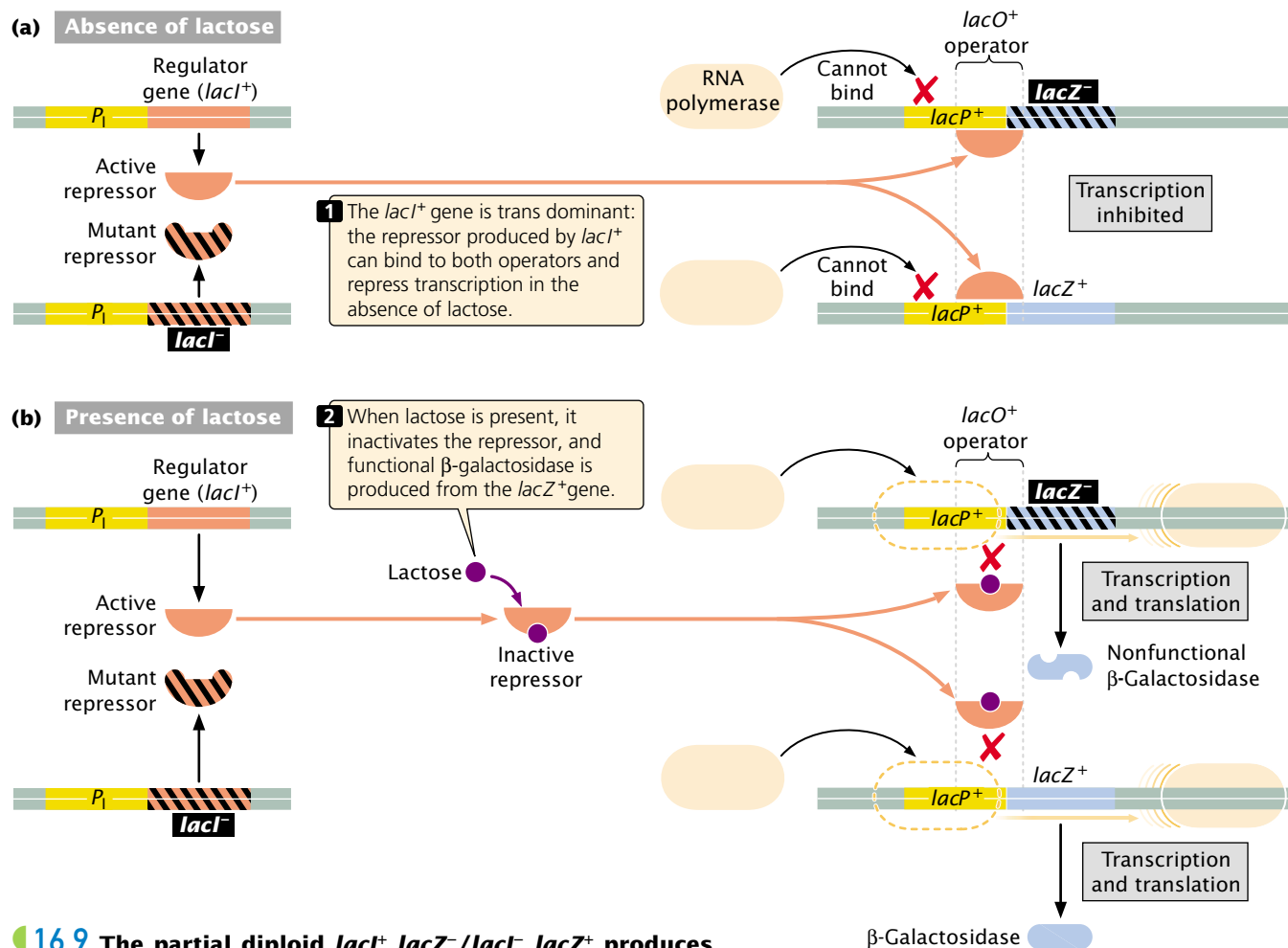
Jacob and Monod worked out the structure and function of the *lac* operon by analyzing mutations that affected lactose metabolism. To help define the roles of the different components of the operon, they used **partial diploid** strains of

E. coli. The cells of these strains possessed two different DNA molecules: the full bacterial chromosome and an extra piece of DNA. Jacob and Monod created these strains by allowing conjugation to take place between two bacteria (see Chapter 8). In conjugation, a small circular piece of DNA (a plasmid) is transferred from one bacterium to another. The plasmid used by Jacob and Monod contained the *lac* operon; so the recipient bacterium became partly diploid, possessing two copies of the *lac* operon. By using different combinations of mutations on the bacterial and plasmid DNA, Jacob and Monod determined that parts of the *lac* operon were *cis* acting (able to control the expression of genes on the same piece of DNA only) or *trans* acting (able to control the expression of genes on other DNA molecules).

Structural-gene mutations Jacob and Monod first discovered some mutant strains that had lost the ability to synthesize either β -galactosidase or permease. (They did not study in detail the effects of mutations on the transacetylase enzyme, so it will not be considered here.) These mutations mapped to the *lacZ* or *lacY* structural genes and altered the amino acid sequence of the enzymes encoded by the genes. These mutations clearly affected the *structure* of the enzymes and not the regulation of their synthesis.

Through the use of partial diploids, Jacob and Monod were able to establish that mutations at the *lacZ* and *lacY* genes were independent and usually affected only the product of the gene in which they occurred. Partial diploids with *lacZ*⁺ *lacY*[−] on the bacterial chromosome and *lacZ*[−] *lacY*⁺ on the plasmid functioned normally, producing β -galactosidase and permease in the presence of lactose. (The genotype of a partial diploid is written by separating the genes on each DNA molecule with a slash: *lacZ*⁺ *lacY*[−]/*lacZ*[−] *lacY*⁺.) In this partial diploid, a single functional β -galactosidase gene (*lacZ*⁺) is sufficient to produce β -galactosidase; it makes no difference whether the functional β -galactosidase gene is coupled to a functional (*lacY*⁺) or a defective (*lacY*[−]) permease gene. The same is true of the *lacY*⁺ gene.

Regulator-gene mutations Jacob and Monod also isolated mutations that affected the *regulation* of enzyme production.



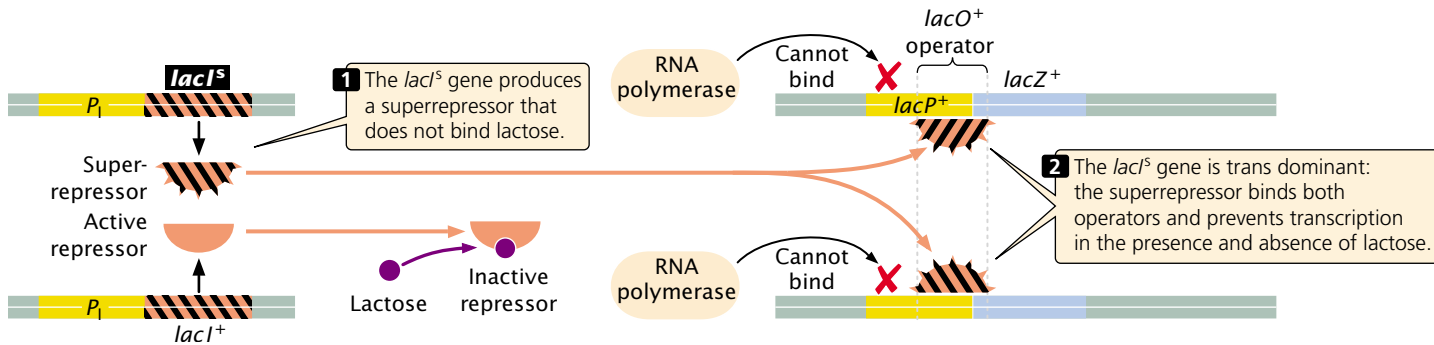
16.9 The partial diploid *lacI*⁺ *lacZ*⁻/*lacI*⁻ *lacZ*⁺ produces β -galactosidase only in the presence of lactose because the *lacI* gene is trans dominant.

These mutations affected the production of both β -galactosidase and permease, because genes for both enzymes are in the same operon and are regulated coordinately.

Some of these mutations were **constitutive**, causing the *lac* enzymes to be produced all the time, whether lactose was present or not, and these mutations fell into two classes: regulator and operator. Jacob and Monod mapped one class to a site upstream of the structural genes; these mutations occurred in the regulator gene and were designated *lacI*⁻. The construction of partial diploids demonstrated that a *lacI*⁺ gene was dominant over a *lacI*⁻ gene; a single copy of *lacI*⁺ (genotype *lacI*⁺/*lacI*⁻) was sufficient to bring about normal regulation of enzyme production. Furthermore, *lacI*⁺ restored normal control to an operon even if the operon was located on a different DNA molecule, showing that *lacI* was able to act in trans. A partial diploid with genotype *lacI*⁺ *lacZ*⁻/*lacI*⁻ *lacZ*⁺ functioned normally, synthesizing β -galactosidase only when lactose was present (FIGURE 16.9). In this strain, the *lacI*⁺ gene on the bacterial chromosome was functional, but the *lacZ*⁻ gene was defec-

tive; on the plasmid, the *lacZ*⁻ gene was defective, but the *lacI*⁺ gene was functional. The fact that a *lacI*⁺ gene could regulate a *lacZ*⁺ gene located on a different DNA molecule indicated to Jacob and Monod that the *lacI*⁺ gene product was able to diffuse to either the plasmid or the chromosome.

Some *lacI* mutations isolated by Jacob and Monod prevented transcription from taking place even in the presence of lactose and other inducers such as IPTG. These mutations were referred to as superrepressors (*lacI*^s), because they produced repressors that could not be inactivated by an inducer. Recall that the repressor has two binding sites, one for the inducer and one for DNA. The *lacI*^s mutations produced a repressor with an altered inducer-binding site, which made the inducer unable to bind to the repressor; consequently, the repressor was always able to attach to the operator site and prevent transcription of the *lac* genes. Superrepressor mutations were dominant over *lacI*⁺; partial diploids with genotype *lacI*^s *lacZ*⁺/*lacI*⁺ *lacZ*⁺ were unable to synthesize either β -galactosidase or permease, whether or not lactose was present (FIGURE 16.10).



16.10 The partial diploid $lacI^s lacZ^+/lacI^+ lacZ^+$ fails to produce β -galactosidase in the presence *and* absence of lactose, because the $lacI^s$ gene encodes a superrepressor.

Operator mutations Jacob and Monod mapped the other class of constitutive mutants to a site adjacent to $lacZ$. These mutations occurred at the operator site, and were labeled $lacO^c$ (O stands for operator and *c* for constitutive). The $lacO^c$ mutations altered the sequence of DNA at the operator so that the repressor protein was no longer able to bind. A partial diploid with genotype $lacI^+ lacO^c lacZ^+/lacI^+ lacO^+ lacZ^+$ exhibited constitutive synthesis of β -galactosidase, indicating that $lacO^c$ was dominant over $lacO^+$.

Analysis of other partial diploids showed that the $lacO$ gene was *cis* acting, affecting only genes on the same DNA molecule. For example, a partial diploid with genotype $lacI^+ lacO^+ lacZ^-/lacI^+ lacO^c lacZ^+$ was constitutive, producing β -galactosidase in the presence or absence of lactose (FIGURE 16.11a), but a partial diploid with genotype $lacI^+ lacO^+ lacZ^+/lacI^+ lacO^c lacZ^-$ produced β -galactosidase only in the presence of lactose (FIGURE 16.11b). In the constitutive partial diploid ($lacI^+ lacO^+ lacZ^-/lacI^+ lacO^c lacZ^+$; see Figure 16.11a), the $lacO^c$ mutation and the functional $lacZ^+$ gene are present on the same DNA molecule; but in $lacI^+ lacO^+ lacZ^+/lacI^+ lacO^c lacZ^-$ (see Figure 16.11b), the $lacO^c$ mutation and the functional $lacZ^+$ gene are on different molecules. The $lacO$ mutation affects only genes to which it is physically connected, as is true of all operator mutations. They prevent the binding of a repressor protein to the operator and thereby allow RNA polymerase to transcribe genes on the same DNA molecule. However, they cannot prevent a repressor from binding to normal operators on other DNA molecules.

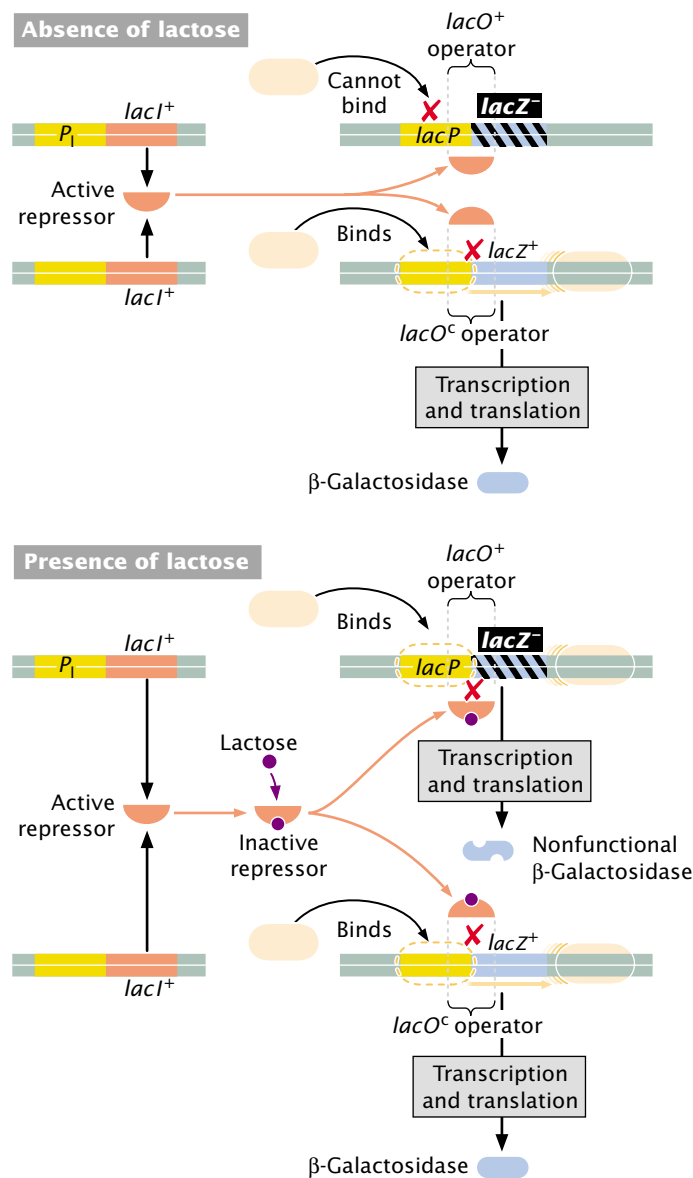
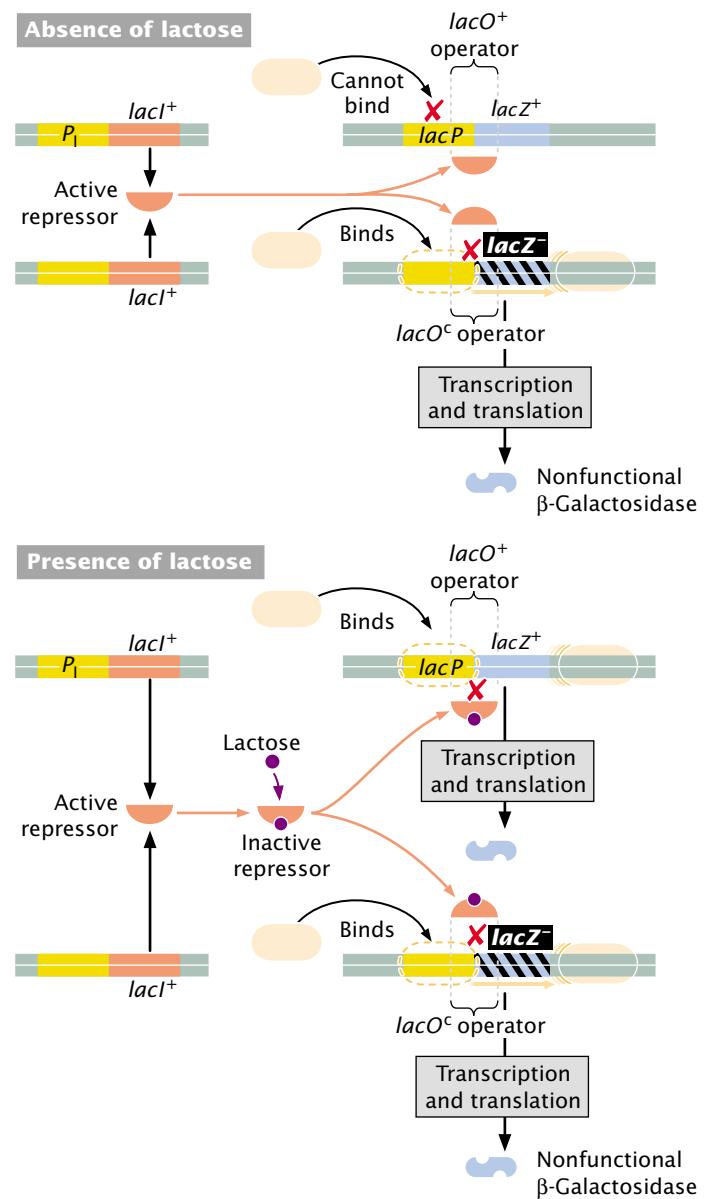
Promoter mutations Mutations affecting lactose metabolism have also been isolated at the promoter site; these mutations are designated $lacP^-$, and they interfere with the binding of RNA polymerase to the promoter. Because this binding is essential for the transcription of the structural genes, *E. coli* strains with $lacP^-$ mutations don't produce lac enzymes either in the presence or in the absence of lactose. Like operator mutations, $lacP^-$ mutations are *cis*

acting and affect only genes on the same DNA molecule. The partial diploid $lacI^+ lacP^+ lacZ^+/lacI^+ lacP^- lacZ^+$ exhibits normal synthesis of β -galactosidase, whereas the $lacI^+ lacP^- lacZ^+/lacI^+ lacP^+ lacZ^-$ fails to produce β -galactosidase whether or not lactose is present.

Positive Control and Catabolite Repression

E. coli and many other bacteria will metabolize glucose preferentially in the presence of lactose and other sugars. They do so because glucose enters glycolysis without further modification and therefore requires less energy to metabolize than do other sugars. When glucose is available, genes that participate in the metabolism of other sugars are repressed, in a phenomenon known as **catabolite repression**. For example, the efficient transcription of the lac operon takes place only if lactose is present and glucose is absent. But how is the expression of the lac operon influenced by glucose? What brings about catabolite repression?

Catabolite repression results from positive control in response to glucose. (This regulation is in addition to the negative control brought about by the repressor binding at the operator site of the lac operon when lactose is absent.) Positive control is accomplished through the binding of a dimeric protein called the **catabolite activator protein** (CAP) to a site that is about 22 nucleotides long and is located within or slightly upstream of the promoter of the lac genes (FIGURE 16.12). RNA polymerase does not bind efficiently to many promoters unless CAP is first bound to the DNA. Before CAP can bind to DNA, it must form a complex with a modified nucleotide called **adenosine-3', 5'-cyclic monophosphate** (cyclic AMP or cAMP), which is important in cellular signaling processes in both bacterial and eukaryotic cells. In *E. coli*, the concentration of cAMP is inversely proportional to the level of available glucose. A high concentration of glucose within the cell lowers the amount of cAMP, and so little cAMP–CAP complex is available to bind to the DNA. Subsequently, RNA polymerase has poor affinity for the lac promoter, and little

(a) Partial diploid $lacI^+ lacO^+ lacZ^- / lacI^+ lacO^c lacZ^+$ **(b) Partial diploid $lacI^+ lacO^+ lacZ^+ / lacI^+ lacO^c lacZ^-$** 

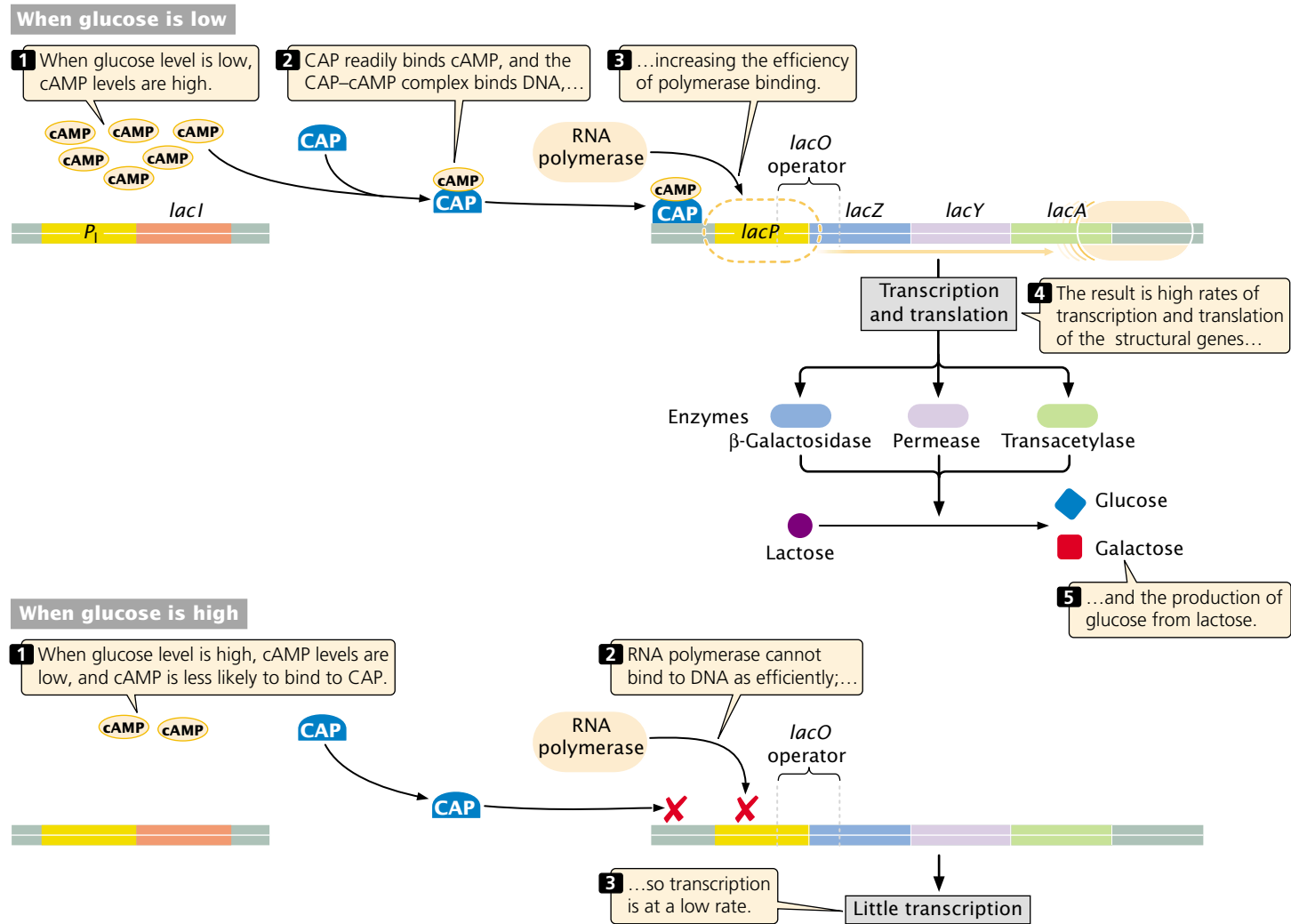
16.11 Mutations in $lacO$ are constitutive and cis acting.

- (a) The partial diploid $lacI^+ lacO^+ lacZ^- / lacI^+ lacO^c lacZ^+$ is constitutive, producing β -galactosidase in the presence and absence of lactose.
- (b) The partial diploid $lacI^+ lacO^+ lacZ^+ / lacI^+ lacO^c lacZ^-$ is inducible (produces β -galactosidase only when lactose is present), demonstrating that the $lacO$ gene is cis acting.

transcription of the lac operon takes place. Low concentrations of glucose stimulate high levels of cAMP, resulting in increased cAMP–CAP binding to DNA. This increase enhances the binding of RNA polymerase to the promoter and increases transcription of the lac genes by some 50-fold.

The catabolite activator protein exerts positive control in more than 20 operons of *E. coli*. The response to

CAP varies among these promoters; some operons are activated by low levels of CAP, whereas others require high levels. CAP contains a helix–turn–helix DNA-binding motif and, when it binds at the CAP site, it causes the DNA helix to bend (FIGURE 16.13). The bent helix enables CAP to interact directly with the RNA polymerase enzyme bound to the promoter and facilitate the initiation of transcription.



16.12 The catabolite activator protein (CAP) binds to the promoter of the *lac* operon and stimulates transcription.

CAP must complex with cAMP before binding to the promoter of the *lac* operon. The binding of cAMP–CAP to the promoter activates transcription by facilitating the binding of RNA polymerase. Levels of cAMP are inversely related to glucose: low glucose stimulates high cAMP; high glucose stimulates low cAMP.

Concepts

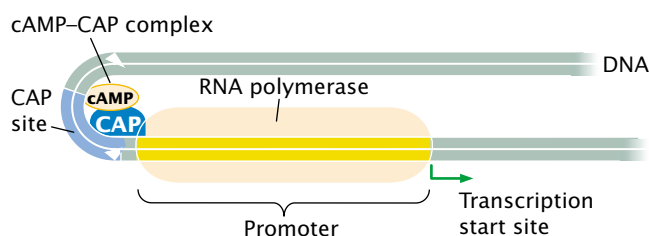
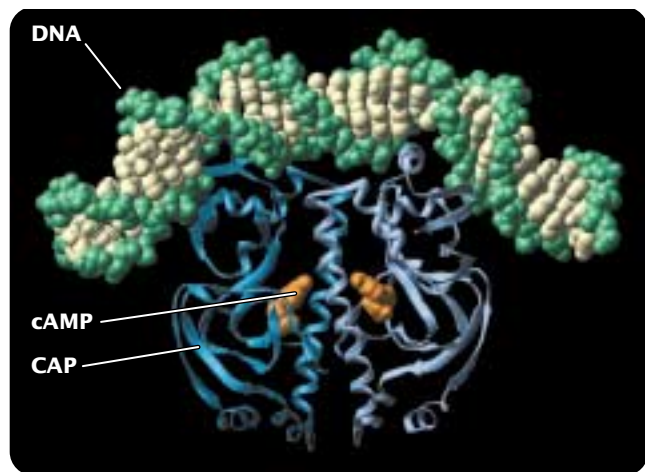
In spite of its name, catabolite repression is a type of positive control in the *lac* operon. CAP, complexed with cAMP, binds to a site near the promoter and stimulates the binding of RNA polymerase. Cellular levels of cAMP in the cell are controlled by glucose; a low glucose level increases the abundance of cAMP and enhances the transcription of the *lac* structural genes.

www.whfreeman.com/pierce Information on CAP control and the binding of CAP to DNA

The *trp* Operon of *E. coli*

The *lac* operon just discussed is an inducible operon, one in which transcription does not normally take place and must be turned on. Other operons are repressible; transcription in these operons is normally turned on and must be repressed. The tryptophan (*trp*) operon in *E. coli*, which controls the biosynthesis of the amino acid tryptophan, is an example of a repressible operon.

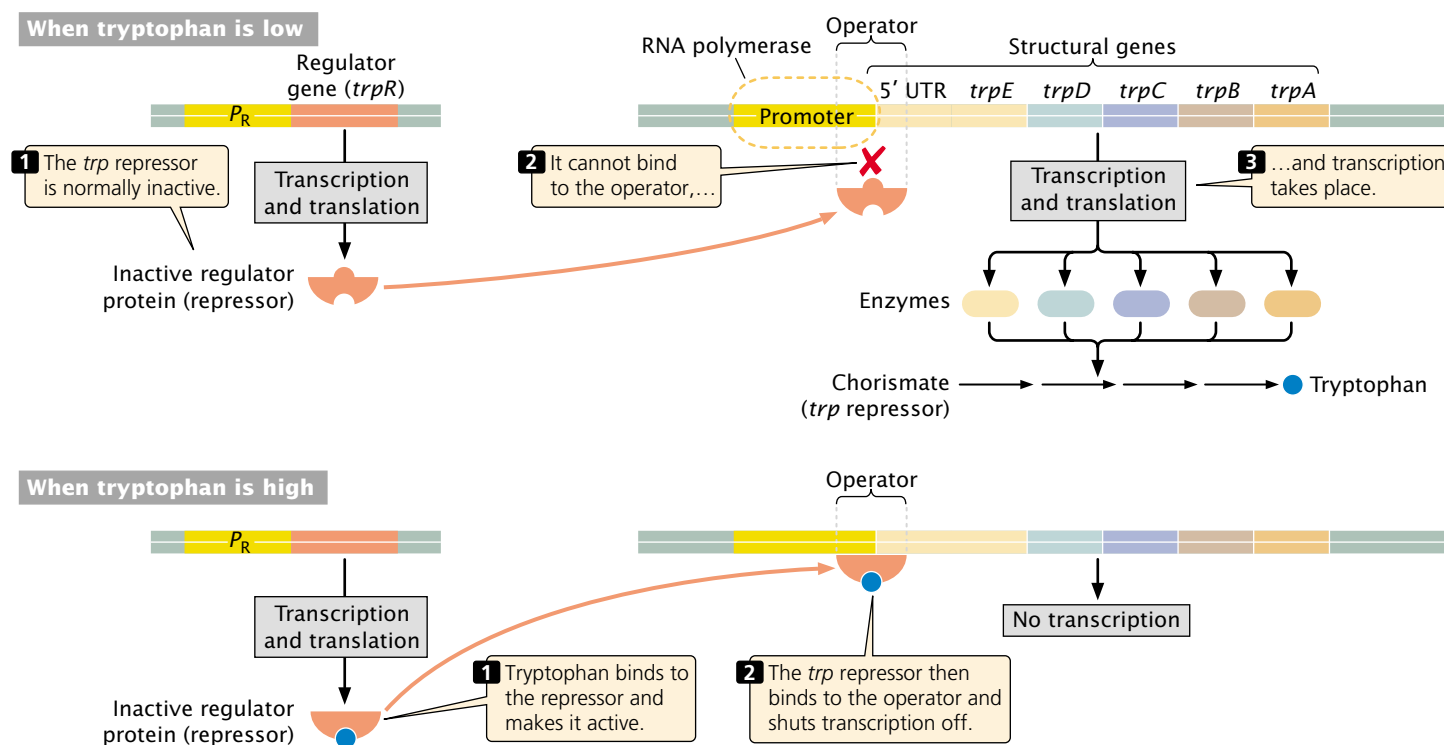
The *trp* operon contains five structural genes (*trpE*, *trpD*, *trpC*, *trpB*, and *trpA*), which produce components of three enzymes (two of the enzymes consist of two polypeptide chains). These enzymes convert chorismate into tryptophan (FIGURE 16.14). The first structural



16.13 Binding of the cAMP–CAP complex to DNA produces a sharp bend in DNA that activates transcription.

gene, *trpE*, contains a long 5' untranslated region (5' UTR) that is transcribed but does not encode any of these enzymes. Instead, this 5' UTR plays an important role in another regulatory mechanism, discussed in the next section. Upstream of the structural genes is the *trp* promoter. When tryptophan levels are low, RNA polymerase binds to the promoter and transcribes the five structural genes into a single mRNA, which is then translated into enzymes that convert chorismate into tryptophan.

Some distance from the *trp* operon is a regulator gene, *trpR*, which encodes a repressor that alone cannot bind DNA (see Figure 16.14). Like the *lac* repressor, the tryptophan repressor has two binding sites, one that binds to DNA at the operator site and another that binds to tryptophan (the activator). Binding with tryptophan causes a conformational change in the repressor that makes it capable of binding to DNA at the operator site, which overlaps the promoter (see Figure 16.14). When the operator is occupied by the tryptophan repressor, RNA polymerase cannot bind to the promoter and the structural genes cannot be transcribed. Thus, when cellular levels of tryptophan are low, transcription of the *trp* operon takes place and more tryptophan is synthesized; when cellular levels of tryptophan are high, transcription of the *trp* operon is inhibited and the synthesis of more tryptophan does not take place.



16.14 The *trp* operon controls the biosynthesis of the amino acid tryptophan in *E. coli*.

Concepts

The *trp* operon is a repressible operon that controls the biosynthesis of tryptophan. In a repressible operon, transcription is normally turned on and must be repressed. Repression is accomplished through the binding of tryptophan to the repressor, which renders the repressor active. The active repressor binds to the operator and prevents RNA polymerase from transcribing the structural genes.

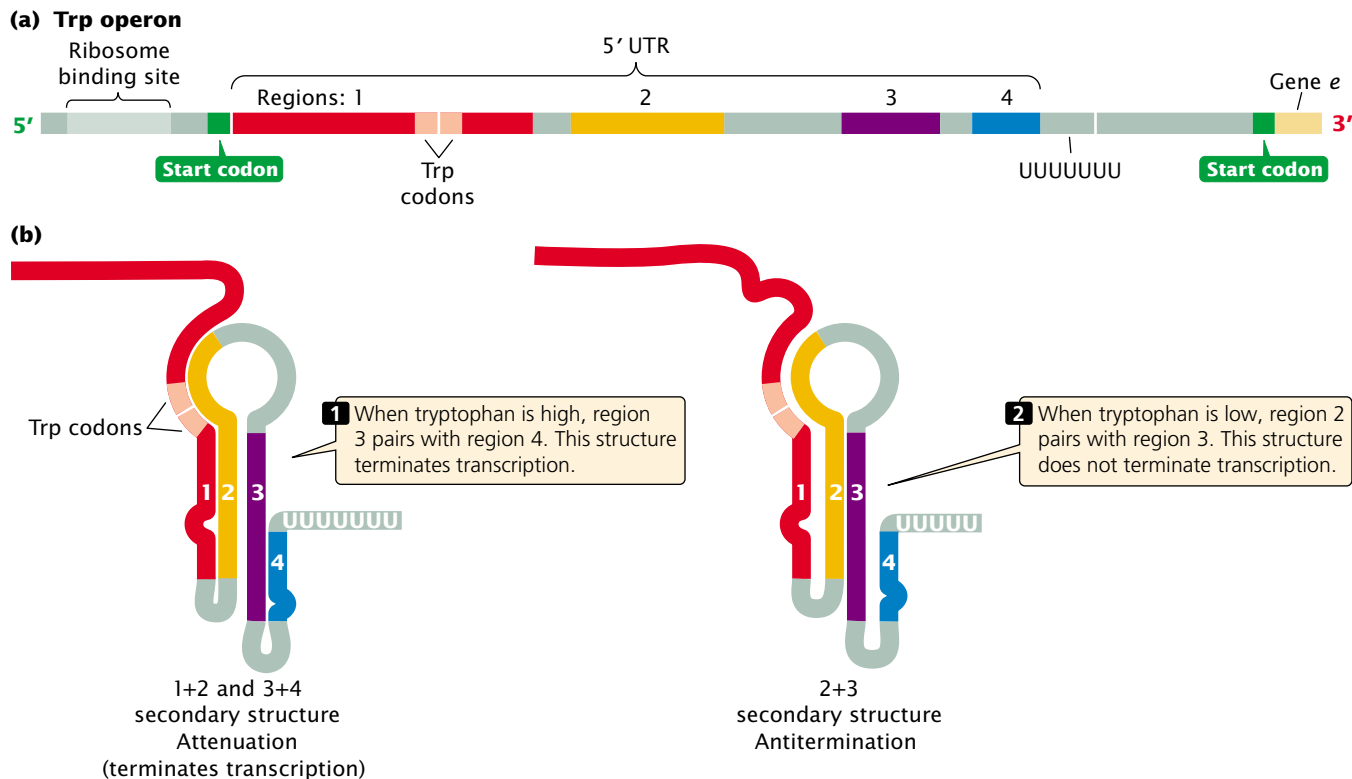
Attenuation: The Premature Termination of Transcription

We've now seen how both positive and negative control regulate the initiation of transcription in an operon. Some operons have an additional level of control that affects the *continuation* of transcription rather than its initiation. In **attenuation**, transcription begins at the start site, but termination takes place prematurely, before the RNA polymerase even reaches the structural genes. Attenuation occurs in a number of operons that code for enzymes participating in the biosynthesis of amino acids.

We can understand the process of attenuation most easily by looking at one of the best-studied examples,

which is found in the *trp* operon of *E. coli*. Several observations by Charles Yanofsky and his colleagues in the early 1970s indicated that repression at the operator site is not the only method of regulation in the *trp* operon. They isolated a series of mutants that possessed deletions in the transcribed region of the operon. Some of these mutants exhibited increased levels of transcription, yet control at the operator site was unaffected. Furthermore, they observed that two mRNAs of different sizes were transcribed from the *trp* operon: a long mRNA containing sequences for the structural genes and a much shorter mRNA of only 140 nucleotides. These observations led Yanofsky to propose that another mechanism—one that caused premature termination of transcription—also regulates transcription in the *trp* operon.

Close examination of the *trp* operon reveals a region of 162 nucleotides that corresponds to the long 5' UTR of the mRNA (mentioned earlier) transcribed from the *trp* operon (FIGURE 16.15a). The 5' UTR (also called a leader) contains four regions: region 1 is complementary to region 2, region 2 is complementary to region 3, and region 3 is complementary to region 4. These complementarities allow the 5' UTR to fold into two different secondary structures (FIGURE 16.15b). Which secondary structure is assumed determines whether attenuation will occur.



16.15 Two different secondary structures may be formed by the 5' UTR of the mRNA transcript of the *trp* operon.

One of the secondary structures contains one hairpin produced by the base pairing of regions 1 and 2 and another hairpin produced by the base pairing of regions 3 and 4. Notice that a string of uracil nucleotides follows the 3+4 hairpin. Not coincidentally, the structure of a bacterial intrinsic terminator (Chapter 13) includes a hairpin followed by a string of uracil nucleotides; this secondary structure in the 5' UTR of the *trp* operon is indeed a terminator and is called an **attenuator**. When cellular levels of tryptophan are high, regions 3 and 4 of the 5' UTR base pair, producing the attenuator structure; this base pairing causes transcription to be terminated before the *trp* structural genes can be transcribed.

The alternative secondary structure of the 5' UTR is produced by the base pairing of regions 2 and 3 (see Figure 16.15b). This base pairing also produces a hairpin, but this hairpin is not followed by a string of uracil nucleotides; so this structure does *not* function as a terminator. When cellular levels of tryptophan are low, regions 2 and 3 base pair, and transcription of the *trp* structural genes is not terminated. RNA polymerase continues past the 5' UTR into the coding section of the structural genes, and the enzymes that synthesize tryptophan are produced. Because it prevents the termination of transcription, the 2+3 structure is called an **antiterminator**.

To summarize, the 5' UTR of the *trp* operon can fold into one of two structures. When tryptophan is high, the 3+4 structure forms, transcription is terminated within the 5' UTR, and no additional tryptophan is synthesized. When tryptophan is low, the 2+3 structure forms, transcription continues through the structural genes, and tryptophan is synthesized. The critical question, then, is, Why does the 3+4 structure arise when tryptophan is high and the 2+3 structure when tryptophan is low?

To answer this question, we must take a closer look at the nucleotide sequence of the 5' UTR. At the 5' end, upstream of region 1, is a ribosome-binding site. Region 1 actually encodes a small protein (see Figure 16.15b). Within the coding sequence for this protein are two UGG codons, which specify the amino acid tryptophan; so tryptophan is required for the translation of this 5' UTR sequence. The protein encoded by the 5' UTR has not been isolated and is presumed to be unstable; its only apparent function is to control attenuation. Although it was stated in Chapter 14 that a 5' UTR is not translated into a protein, the 5' UTR of operons subject to attenuation are exceptions to this rule.

The formation of hairpins in the 5' UTR of the *trp* operon is controlled by the interplay of transcription and translation that takes place near the 5' end of the mRNA. Recall that, in prokaryotic cells, transcription and translation are coupled: while transcription is taking place at the 3' end of the mRNA, translation is initiated at the 5' end. The precise timing and interaction of these two processes in the 5' UTR determine whether attenuation occurs.

Transcription when tryptophan levels are high Let's first consider what happens when intracellular levels of tryptophan are high. RNA polymerase begins transcribing the DNA, producing region 1 of the 5' UTR (● [FIGURE 16.16a](#)). Following RNA polymerase closely, a ribosome binds to the 5' UTR (at the Shine-Dalgarno sequence, see Chapter 14) and begins to translate the coding region. Meanwhile, RNA polymerase is transcribing region 2 (● [FIGURE 16.16b](#)). Region 2 is complementary to region 1 but, because the ribosome is translating region 1, the nucleotides in regions 1 and 2 cannot base pair. As RNA polymerase begins to transcribe region 3, the ribosome is continuing to translate region 1 (● [FIGURE 16.16c](#)). When the ribosome reaches the two UGG tryptophan codons, it doesn't slow or stall, because tryptophan is abundant and tRNAs charged with tryptophan are readily available. This point is critical to note: because tryptophan is abundant, translation can keep up with transcription.

As it moves past region 1 to the stop codon, the ribosome partly covers region 2; (● [FIGURE 16.16d](#)); meanwhile, RNA polymerase completes the transcription of region 3. Although regions 2 and 3 are complementary, region 2 is partly covered by the ribosome; so it can't base pair with 3.

RNA polymerase continues to move along the DNA, eventually transcribing regions 4 of the 5' UTR. Region 4 is complementary to region 3, and, because region 3 cannot base pair with region 2, it pairs with region 4. The pairing of regions 3 and 4 (see Figure 16.16d) produces the attenuator—a hairpin followed by a string of uracil nucleotides—and transcription terminates just beyond region 4. The structural genes are not transcribed, no tryptophan-producing enzymes are translated, and no additional tryptophan is synthesized.

Transcription when tryptophan levels are low What happens when tryptophan levels are low? Once again, RNA polymerase begins transcribing region 1 of the 5' UTR (● [FIGURE 16.16e](#)), and the ribosome binds to the 5' end of the 5' UTR and begins to translate region 1 while RNA polymerase continues transcribing region 2 (● [FIGURE 16.16f](#)). When the ribosome reaches the UGG tryptophan codons, it stalls (● [FIGURE 16.16g](#)) because the level of tryptophan is low, and tRNAs charged with tryptophan are scarce or even unavailable. The ribosome sits at the tryptophan codons, awaiting the arrival of a tRNA charged with tryptophan. Stalling of the ribosome does not, however, hinder transcription; RNA polymerase continues to move along the DNA, and transcription gets ahead of translation.

Because the ribosome is stalled at the tryptophan codons in region 1, region 2 is *not* covered by the ribosome when region 3 has been transcribed. Therefore, nucleotides in region 2 and region 3 base pair, forming the 2+3 hairpin (● [FIGURE 16.16h](#)). This hairpin does not cause termination, and so transcription continues. Because region 3 is already paired with region 2, the 3+4 hairpin (the attenuator)

When tryptophan is high**(a)**

- 1 RNA polymerase begins transcribing DNA, producing region 1 of the 5' UTR.

(b)

- 2 A ribosome binds to the 5' end of the 5' UTR and begins to translate region 1, while region 2 is being transcribed.

(c)

- 3 The ribosome translates region 1 while RNA polymerase transcribes region 3.

(4)

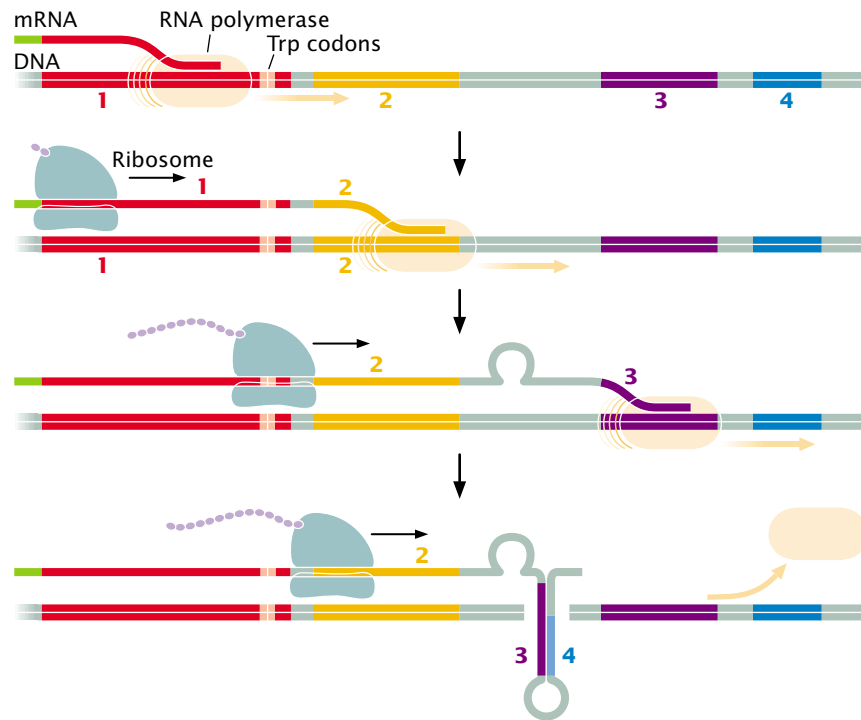
- The ribosome does not stall at the Trp codons, because tryptophan is abundant.

(d)

- 5 The leading edge of the ribosome covers part of region 2, preventing it from pairing with region 3.

(6)

- Region 4 is transcribed and pairs with region 3. The pairing of regions 3 and 4 produces the attenuator that terminates transcription.

**When tryptophan is low****(e)**

- 1 RNA polymerase begins transcribing the DNA, producing region 1 of the 5' UTR.

(f)

- 2 A ribosome attaches to the 5' end of the 5' UTR and begins to translate region 1 while region 2 is being transcribed.

(g)

- 3 The ribosome stalls at the Trp codons in region 1 because tryptophan is low.

(4)

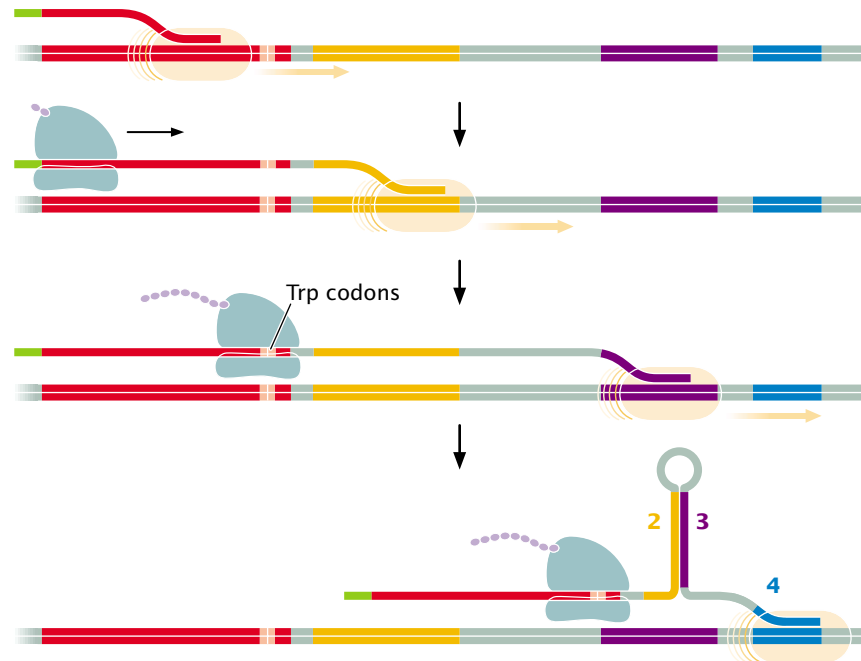
- Because the ribosome is stalled, region 2 is not covered by the ribosome when region 3 is transcribed.

(h)

- 5 When region 3 is transcribed, it pairs with region 2.

(6)

- When region 4 is transcribed, it cannot pair with region 3, because region 3 is already paired with region 2; the attenuator never forms, and transcription continues.



16.16 The premature termination of transcription (attenuation) takes place in the *trp* operon, depending on the cellular level of tryptophan.

never forms, and so attenuation does not occur. RNA polymerase continues along the DNA, past the 5' UTR, transcribing all the structural genes into mRNA, which is translated into the enzymes encoded by the *trp* operon. These enzymes then synthesize more tryptophan. Important

events in the process of attenuation are summarized in Table 16.3.

Several additional points about attenuation need clarification. The key factor controlling attenuation is the number of tRNA molecules charged with tryptophan, because

Table 16.3 Events in the process of attenuation

Intracellular Level of Tryptophan	Ribosome Stalls at Trp Codons	Position of Ribosome When Region 3 Is Transcribed	Secondary Structure of 5' UTR	Termination of Transcription of <i>trp</i> Operon
High	No	Covers region 2	3+4 hairpin	Yes
Low	Yes	Covers region 1	2+3 hairpin	No

their availability is what determines whether the ribosome stalls at the tryptophan codons. A second point concerns the synchronization of transcription and translation, which is critical to attenuation. Synchronization is achieved through a pause site located in region 1 of the 5' UTR. After initiating transcription, RNA polymerase stops temporarily at this site, which allows time for a ribosome to bind to the 5' end of the mRNA so that translation can closely follow transcription. A third point is that ribosomes do not traverse the convoluted hairpins of the 5' UTR to translate the structural genes. Ribosomes that attach to the ribosome-binding site at the 5' end of the mRNA encounter a stop codon at the end of region 1. Ribosomes translating the structural genes attach to a different ribosome-binding site located near the beginning of the *trpE* gene.

Why does attenuation occur? Why do bacteria need attenuation in the *trp* operon? Shouldn't repression at the operator site prevent transcription from taking place when tryptophan levels in the cell are high? Why does the cell have two types of control? Part of the answer is that repression is never complete; some transcription is initiated even when the *trp* repressor is active; repression reduces transcription only as much as 70-fold. Attenuation can further reduce transcription another 8- to 10-fold; so together the two processes are capable of reducing transcription of the *trp* operon more than 600-fold. Both mechanisms provide *E. coli* with a much finer degree of control over tryptophan synthesis than either could achieve alone.

Another reason for the dual control is that attenuation and repression respond to different signals: repression responds to the cellular levels of tryptophan, whereas attenuation responds to the number of tRNAs charged with tryptophan. There may be times when it is advantageous for the cell to be able to respond to these different signals. Finally, the *trp* repressor affects several operons other than the *trp* operon. It's possible that at an earlier stage in the evolution of *E. coli*, the *trp* operon was controlled only by attenuation. The *trp* repressor may have evolved primarily to control the other operons and only incidentally affects the *trp* operon.

Attenuation is a complex process to grasp because you must simultaneously visualize how two dynamic processes—transcription and translation—interact, and it's easy to get the two processes confused. Remember that attenuation

entails the early termination of *transcription*, not translation (although events in translation bring about the termination of transcription). Attenuation often causes confusion because we know that transcription must precede translation. We're comfortable with the idea that transcription might affect translation, but it's harder to imagine that the effects of translation could influence transcription, as it does in attenuation. The reality is that transcription and translation are closely coupled in prokaryotic cells, and events in one process can easily affect the other.

Concepts

In attenuation, transcription is initiated but terminates prematurely. When tryptophan levels are low, the ribosome stalls at the tryptophan codons and transcription continues. When tryptophan levels are high, the ribosome does not stall at the tryptophan codons, and the 5' UTR adopts a secondary structure that terminates transcription before the structural genes can be copied into RNA (attenuation).

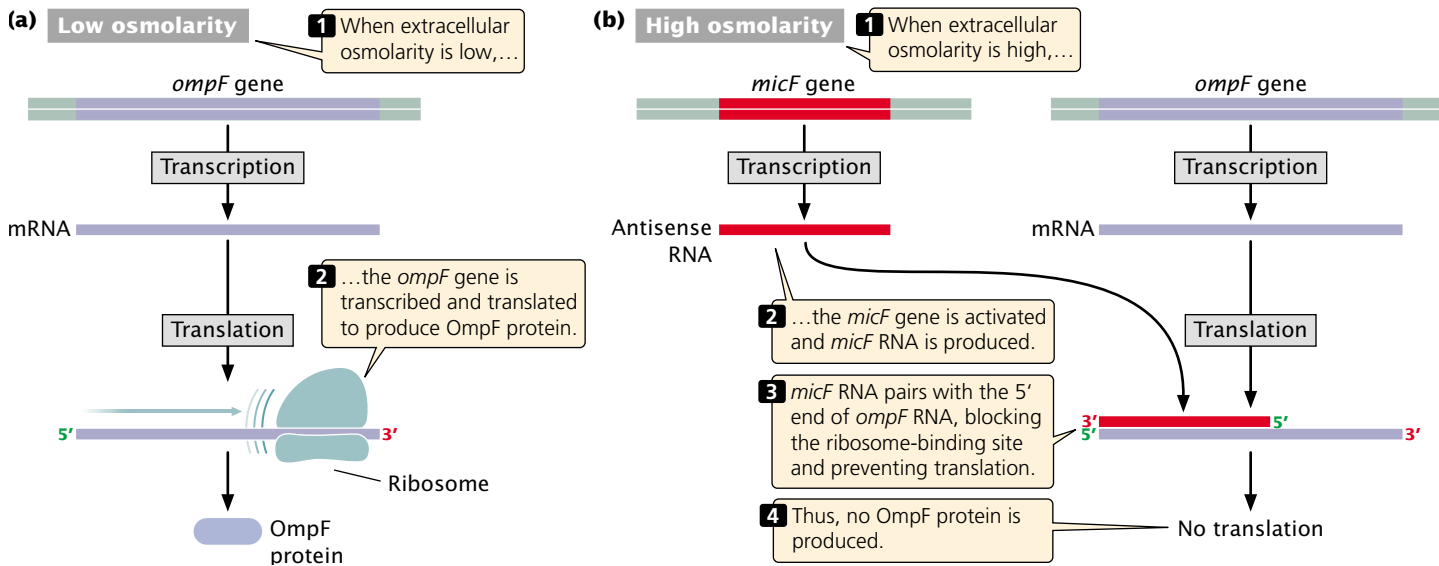
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More information on attenuation

Antisense RNA in Gene Regulation

All the regulators of gene expression that we have considered so far have been proteins. Several examples of RNA regulators have also been discovered. These small RNA molecules are complementary to particular sequences on mRNAs and are called **antisense RNA**. They control gene expression by binding to sequences on mRNA and inhibiting translation.

Translational control by antisense RNA is seen in the regulation of the *ompF* gene of *E. coli* (FIGURE 16.17a). Two *E. coli* genes, *ompF* and *ompC*, produce outer-membrane proteins that function as diffusion pores, allowing bacteria to adapt to external osmolarities (the tendency of water to move across a membrane owing to different ion concentrations). Under most conditions, both the *ompF* and the *ompC* genes are transcribed and translated. When the osmolarity of the medium increases, a regulator gene named *micF*—for



16.17 Antisense RNA can regulate translation.

mRNA-interfering complementary RNA—is activated and *micF* RNA is produced (FIGURE 16.17b). The *micF* RNA, an antisense RNA, binds to a complementary sequence in the 5' UTR of the *ompF* mRNA and inhibits the binding of the ribosome. This inhibition reduces the amount of translation (see Figure 16.17b), which results in fewer OmpF proteins in the outer membrane and thus reduces the detrimental movement of substances across the membrane owing to the changes in osmolarity. A number of examples of antisense RNA controlling gene expression have now been identified in bacteria and bacteriophages.

Concepts

Antisense RNA is complementary to other RNA or DNA sequences. In bacterial cells, it may inhibit translation by binding to sequences in the 5' UTR of mRNA and preventing the attachment of the ribosome.

Transcriptional Control in Bacteriophage Lambda

Bacteriophage λ is a virus that infects the bacterium *E. coli* (Chapter 8). Bacteriophage λ possesses a single DNA chromosome consisting of 48,502 nucleotides surrounded by a protein coat. A bacteriophage infects a bacterial cell by attaching to the cell wall and injecting its DNA into the cell. Inside the cell, λ phage undergoes either of two life cycles.

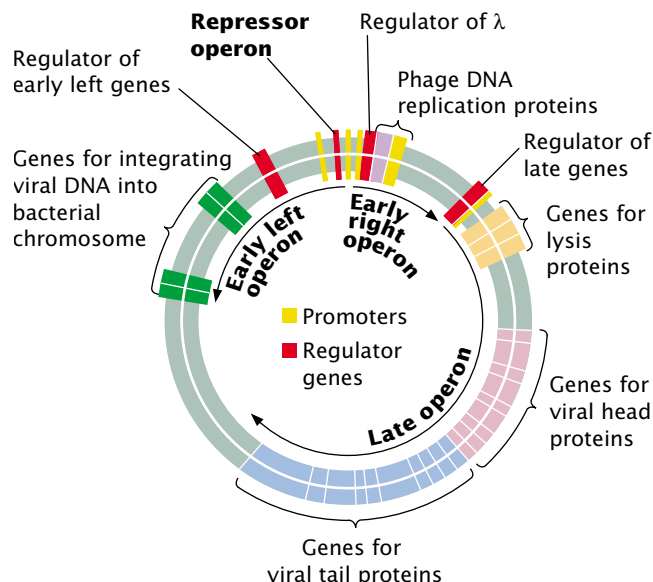
In the *lytic cycle* (see Chapter 8), phage genes are transcribed and translated to produce phage coat proteins and enzymes that synthesize from 100 to 200 copies of the phage DNA. The viral components are assembled to produce phage

particles, and the phage produces a protein that causes the cell to lyse. The released phage can then infect other bacterial cells. In the *lysogenic cycle*, phage genes that encode replication enzymes and phage proteins are not immediately transcribed. Instead, the phage DNA integrates into the bacterial chromosome as a prophage. When the bacterial chromosome replicates, the prophage is duplicated along with the bacterial genes and is passed to the daughter cells in bacterial reproduction. The prophage may later excise from the bacterial chromosome and enter the lytic cycle.

Whether a λ phage enters the lytic or the lysogenic cycle depends on the regulation of the phage genes. In the lytic cycle, the genes that encode replication enzymes, phage proteins, and bacterial cell lysis are transcribed; but, in the lysogenic cycle, these genes are repressed.

Like bacterial genes, functionally related phage genes are clustered together into operons. There are four major operons in the phage λ chromosome (FIGURE 16.18). The early right operon contains genes that are required for DNA replication and are transcribed early in the lytic cycle. The early left operon contains genes necessary for recombination and the integration of phage DNA into the bacterial chromosome as a part of the lysogenic cycle. A third operon, the late operon, contains genes that encode the protein coat of the phage, produced late in the lytic cycle. The fourth operon is the repressor operon, which produces the λ repressor responsible for maintaining the prophage DNA in a dormant state. Although there are several additional promoters on the λ chromosome that may be activated at special times, here the emphasis is on three general features of transcriptional control in bacteriophage λ .

First, both positive control and negative control are seen in λ gene regulation. Several proteins act as repressors, inhibiting transcription, whereas others act as activators,



16.18 The bacteriophage λ chromosome contains four major operons: the early left operon, the early right operon, the late operon, and the repressor operon.

stimulating transcription. The λ repressor, which plays a major role in λ gene regulation, can act as either an activator or a repressor.

A second feature is that transcription is accomplished through a cascade of reactions. As one operon is transcribed, it produces a protein that regulates the transcription of a second operon, which produces a protein that affects the transcription of a third operon. Thus, the operons are activated and repressed in a particular order, with the use of several different promoters, each with an affinity for specific activators and repressors. As each promoter is activated, only the genes under its control are transcribed;

this controlled transcription ensures that genes appropriate to each stage of the lytic or lysogenic cycle are expressed.

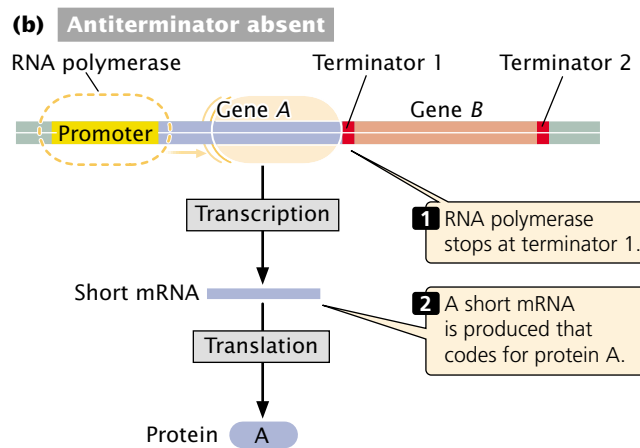
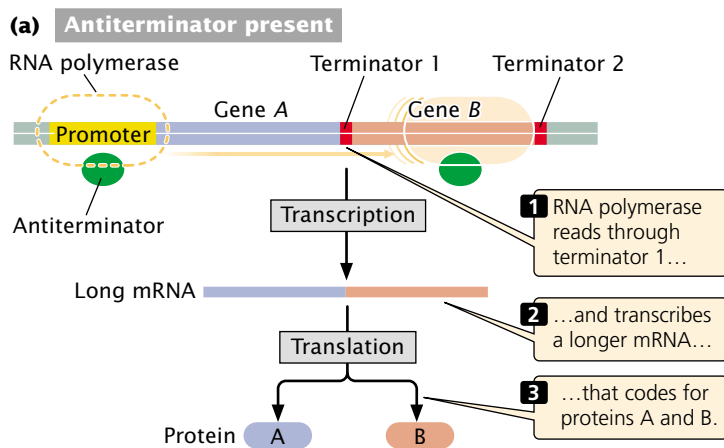
A third feature of λ gene regulation is the use of **transcriptional antiterminator proteins**, which bind to RNA polymerase and alter its structure, allowing it to ignore certain terminators (FIGURE 16.19a). In the absence of the antiterminator protein, RNA polymerase stops at a terminator located early in the operon (FIGURE 16.19b), and so only some of the genes in the operon are transcribed and translated.

Concepts

The entry of bacteriophage λ into lysis or lysogeny is controlled by a cascade of reactions, in which the transcription of operons is turned on and off in a specific sequence. The expression of the operons is controlled by the affinity of different promoters for repressor and activator proteins and through transcriptional antiterminators.

Eukaryotic Gene Regulation

Many features of gene regulation are common to both bacterial and eukaryotic cells. For example, in both types of cells, DNA-binding proteins influence the ability of RNA polymerase to initiate transcription. However, there are also some differences, although these differences are often a matter of degree. First, eukaryotic genes are not organized into operons and are rarely transcribed together into a single mRNA molecule; instead, each structural gene typically has its own promoter and is transcribed separately. Second, chromatin structure affects gene expression in eukaryotic cells; DNA must unwind



16.19 Antiterminator proteins bind to RNA polymerase and alter its structure so that it ignores certain terminators.

from the histone proteins before transcription can take place. Third, although both repressors and activators function in eukaryotic and bacterial gene regulation, activators seem to be more common in eukaryotic cells. Finally, the regulation of gene expression in eukaryotic cells is characterized by a greater diversity of mechanisms that act at different points in the transfer of information from DNA to protein.

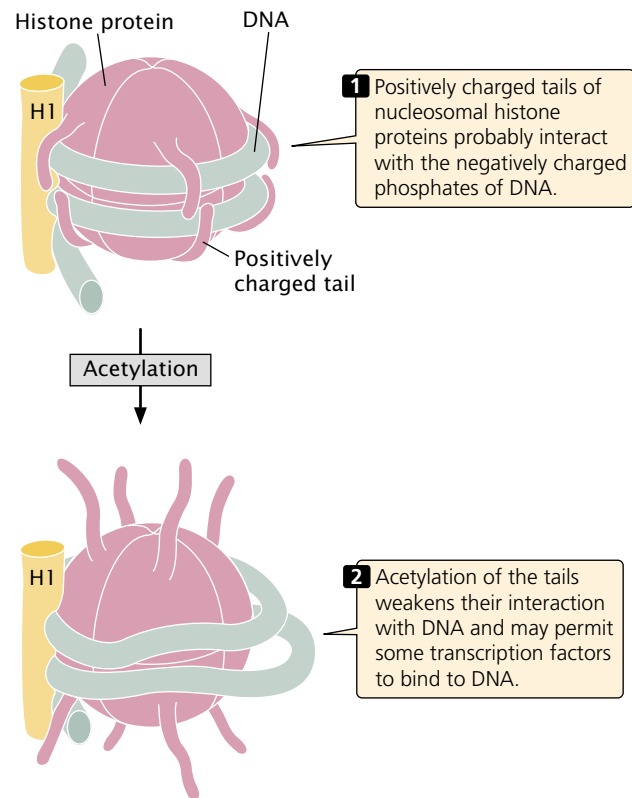
Eukaryotic gene regulation is less well understood than bacterial regulation, partly owing to the larger genomes in eukaryotes, their greater sequence complexity, and the difficulty of isolating and manipulating mutations that can be used in the study of gene regulation. Nevertheless, great advances in our understanding of the regulation of eukaryotic genes have been made in recent years, and eukaryotic regulation continues to be one of the cutting-edge areas of research in genetics.

Chromatin Structure and Gene Regulation

One type of gene control in eukaryotic cells is accomplished through the modification of gene structure. In the nucleus, histone proteins associate to form octamers, around which helical DNA tightly coils to create chromatin (see Figure 11.5). In a general sense, this chromatin structure represses gene expression. For a gene to be transcribed, transcription factors, activators, and RNA polymerase must bind to the DNA. How can these events take place with DNA wrapped tightly around histone proteins? The answer is that before transcription, chromatin structure changes, and the DNA becomes more accessible to the transcriptional machinery.

DNase I hypersensitivity Several types of changes are observed in chromatin structure when genes become transcriptionally active. One type is an increase in the sensitivity of chromatin to degradation by DNase I, an enzyme that digests DNA. When tightly bound by histone proteins, DNA is resistant to DNase I digestion because the enzyme cannot gain access to the DNA. When DNA is less tightly bound by histones, it becomes sensitive to DNase I degradation. Thus, the ability of DNase I to digest DNA provides an indication of the DNA–histone association.

As genes become transcriptionally *active*, regions around the genes become highly sensitive to the action of DNase I (see Chapter 11). These regions, called **DNase I hypersensitive sites**, frequently develop about 1000 nucleotides upstream of the start site of transcription, suggesting that the chromatin in these regions adopts a more open configuration during transcription. This relaxation of the chromatin structure may allow regulatory proteins access to binding sites on the DNA. Indeed, many DNase I hypersensitive sites correspond to known binding sites for regulatory proteins.



16.20 The acetylation of histone proteins alters chromatin structure and permits some transcription factors to bind to DNA.

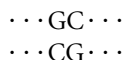
Histone acetylation One factor affecting chromatin structure is acetylation, the addition of acetyl groups (CH_3CO) to histone proteins. Histones in the octamer core of the nucleosome have two domains: (1) a globular domain that associates with other histones and the DNA and (2) a positively charged tail domain that probably interacts with the negatively charged phosphates on the backbone of DNA (FIGURE 16.20).

Acetyl groups are added to histone proteins by acetyltransferase enzymes; the acetyl groups destabilize the nucleosome structure, perhaps by neutralizing the positive charges on the histone tails and allowing the DNA to separate from the histones. Other enzymes called deacetylases strip acetyl groups from histones and restore chromatin repression. Certain transcription factors (see Chapter 13) and other proteins that regulate transcription either have acetyltransferase activity or attract acetyltransferases to the DNA.

Some transcription factors and other regulatory proteins are known to alter chromatin structure without acetylating histone proteins. These **chromatin-remodeling complexes** bind directly to particular sites on DNA and reposition the nucleosomes, allowing transcription factors to bind to promoters and initiate transcription.

DNA methylation Another change in chromatin structure associated with transcription is the methylation of cytosine bases, which yields 5-methylcytosine (see Figure 10.19). Heavily methylated DNA is associated with the repression of transcription in vertebrates and plants, whereas transcriptionally active DNA is usually unmethylated in these organisms.

DNA methylation is most common on cytosine bases adjacent to guanine nucleotides on the same strand (CpG); so two methylated cytosines sit diagonally across from each other on opposing strands:



DNA regions with many CpG sequences are called **CpG islands** and are commonly found near transcription start sites. While genes are not being transcribed, these CpG islands are often methylated, but the methyl groups are removed before the initiation of transcription. CpG methylation is also associated with long-term gene repression, such as on the inactivated X chromosome of female mammals (see Chapter 4).

Recent evidence suggests an association between DNA methylation and the deacetylation of histones, both of which repress transcription. Certain proteins that bind tightly to methylated CpG sequences form complexes with other proteins that act as histone deacetylases. In other words, methylation appears to attract deacetylases, which remove acetyl groups from the histone tails, stabilizing the nucleosome structure and repressing transcription. Demethylation of DNA would allow acetyltransferases to remove these acetyl groups, disrupting nucleosome structure and permitting transcription.

Concepts

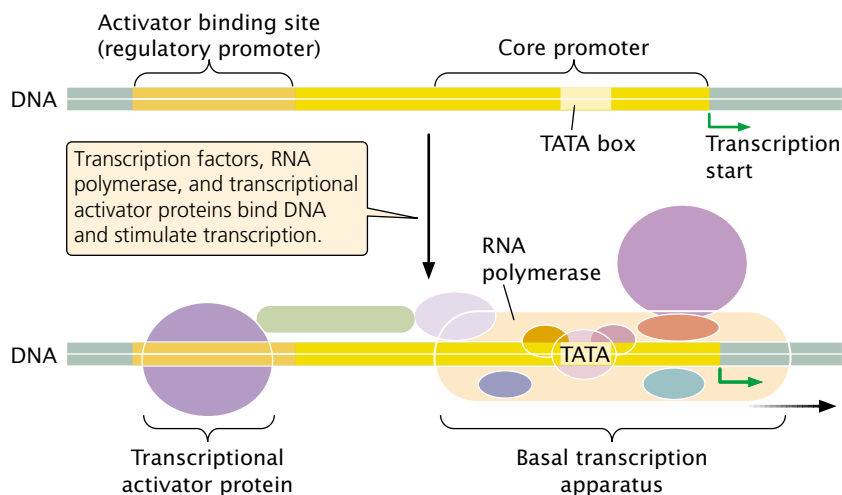
Sensitivity to DNase I digestion suggests that transcribed DNA assumes an open configuration before transcription. The acetylation of histone proteins disrupts nucleosome structure and may facilitate transcription. The activation of transcription is often preceded by demethylation of DNA; methylated sequences may attract deacetylases, which remove acetyl groups from histone proteins, stabilizing chromatin structure and repressing transcription.

Transcriptional Control in Eukaryotic Cells

Transcription is an important level of control in eukaryotic cells, and this control requires a number of different types of proteins and regulatory elements. The initiation of eukaryotic transcription was discussed in detail in Chapter 13. Recall that general transcription factors and RNA polymerase assemble into a *basal transcription apparatus*, which binds to a *core promoter* located immediately upstream of a gene. The basal transcription apparatus is capable of minimal levels of transcription; *transcriptional activator proteins* are required to bring about normal levels of transcription. These proteins bind to a regulatory promoter, which is located upstream of the core promoter, and to *enhancers*, which may be located some distance from the gene (FIGURE 16.21).

Transcriptional activators, coactivators and repressors

Transcriptional activator proteins stimulate transcription by facilitating the assembly or action of the basal transcription apparatus at the core promoter; the activators may interact directly with the basal transcription apparatus or indirectly



16.21 Transcriptional activator proteins bind to sites on DNA and stimulate transcription. Most act by stimulating or stabilizing the assembly of the basal transcription apparatus.

through protein **coactivators**. Some activators and coactivators, as well as the general transcription factors, also have acetyltransferase activity and facilitate transcription further by altering chromatin structure (see earlier subsection on histone acetylation).

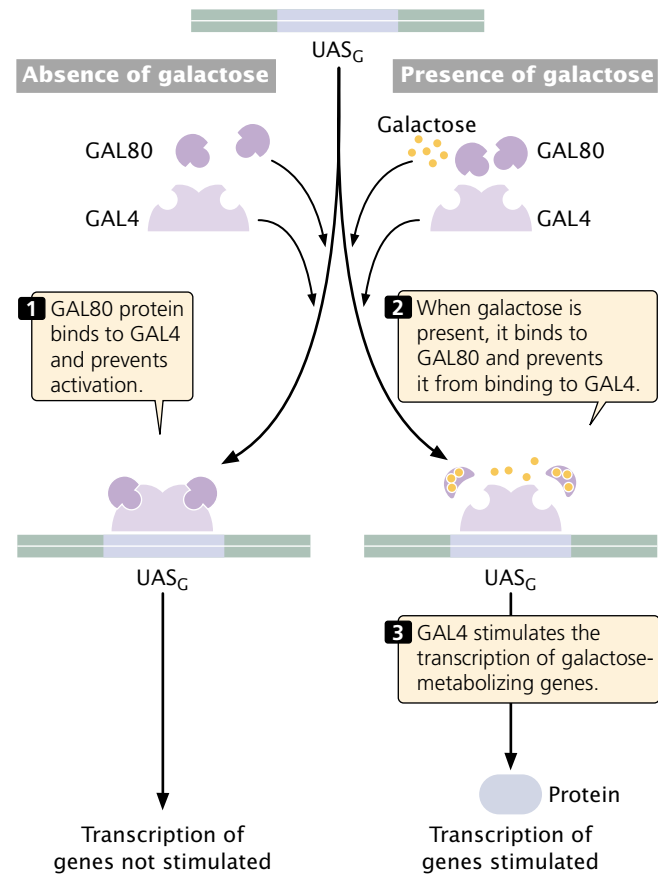
Transcriptional activator proteins have two distinct functions (see Figure 16.21). First, they are capable of binding DNA at a specific base sequence, usually a consensus sequence in a regulatory promoter or enhancer; for this function, most transcriptional activator proteins contain one or more of the DNA-binding motifs discussed at the beginning of this chapter. A second function is the ability to interact with other components of the transcriptional apparatus and influence the rate of transcription. Most do so by either stabilizing or stimulating the assembly of the basal transcription apparatus.

GAL4 is a transcription activator protein that regulates the transcription of several yeast genes in galactose metabolism. GAL4 contains several zinc fingers and binds to a DNA sequence called UAS_G (upstream activating sequence for GAL4). UAS_G exhibits the properties of an enhancer—a regulatory sequence that may be some distance from the regulated gene and is independent of the gene in position and orientation (see Chapter 13). When bound to UAS_G, GAL4 stimulates the transcription of yeast genes needed for metabolizing galactose.

A particular region of GAL4 binds another protein called GAL80, which regulates the activity of GAL4 in the presence of galactose. When galactose is absent, GAL80 binds to GAL4 (two molecules of GAL80 bind to each molecule of GAL4), preventing GAL4 from activating transcription (FIGURE 16.22). When galactose is present, however, it binds to GAL80, causing a conformational change in the protein so that it can no longer bind GAL4. The GAL4 protein is then available to activate the transcription of the genes whose products metabolize galactose.

GAL4 and a number of other transcriptional activator proteins contain multiple amino acids with negative charges that form an *acidic activation domain*. These acidic activators stimulate transcription by enhancing the ability of TFIIB (see Chapter 13), one of the general transcription factors, to join the basal transcription apparatus. Without the activator, the binding of TFIIB is a slow process; the activator helps “recruit” TFIIB to the initiation complex, thereby stimulating the binding of RNA polymerase and the initiation of transcription. Acidic activators may also enhance other steps in the assembly of the basal transcription apparatus.

Some regulatory proteins in eukaryotic cells act as repressors, inhibiting transcription. These repressors may bind to sequences in the regulatory promoter or to distant sequences called *silencers*, which, like enhancers, are position and orientation independent. Unlike repressors in bacteria, most eukaryotic repressors do not directly block RNA polymerase. These repressors may compete with activators for DNA binding sites: when a site is occupied by an activa-



16.22 Transcription is activated by GAL4 in response to galactose. GAL4 binds to the UAS_G site and controls the transcription of genes in galactose metabolism.

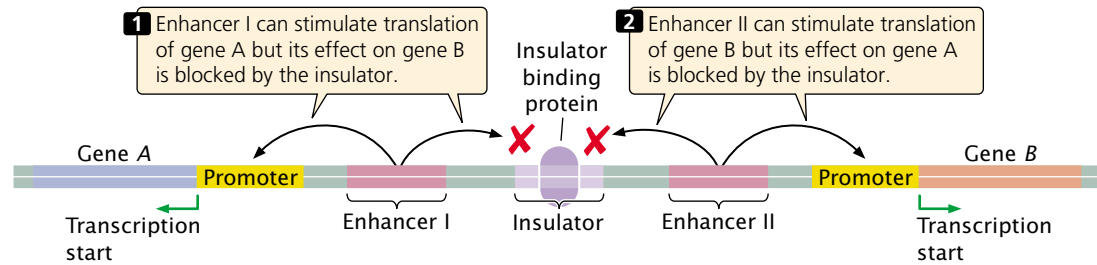
tor, transcription is stimulated, but, if a repressor occupies that site, no activation occurs. Alternatively, a repressor may bind to sites near an activator site and prevent the activator from contacting the basal transcription apparatus. A third possible mechanism of repressor action is direct interference with the assembly of the basal transcription apparatus, thereby blocking the initiation of transcription.

Concepts

Transcriptional regulatory proteins in eukaryotic cells can influence the initiation of transcription by affecting the stability or assembly of the basal transcription apparatus. Some regulatory proteins are activators and stimulate transcription; others are repressors and inhibit transcription.

Enhancers and insulators Enhancers are capable of affecting transcription at distant promoters. For example, an enhancer that regulates the gene encoding the alpha chain of the T-cell receptor is located 69,000 bp down-

16.23 An insulator blocks the action of an enhancer on a promoter when the insulator lies between the enhancer and the promoter.



stream of the gene's promoter. Furthermore, the exact position and orientation of an enhancer relative to the promoter can vary. How can an enhancer affect the initiation of transcription taking place at a promoter that is tens of thousands of base pairs away? The mechanism of action of many enhancers is not known, but evidence suggest that, in some cases, activator proteins bind to the enhancer and cause the DNA between the enhancer and the promoter to loop out, bringing the promoter and enhancer close to one another, so that the transcriptional activator proteins are able to directly interact with the basal transcription apparatus at the core promoter.

Most enhancers are capable of stimulating any promoter in their vicinities. Their effects are limited, however, by **insulators** (also called boundary elements), which are DNA sequences that block or insulate the effect of enhancers in a position-dependent manner. If the insulator lies between the enhancer and the promoter, it blocks the action of the enhancer; but, if the insulator lies outside the region between the two, it has no effect (FIGURE 16.23). Specific proteins bind to insulators and play a role in their blocking activity, but exactly how this takes place is poorly understood. Some insulators also limit the spread of changes in chromatin structure that affect transcription.

Concepts

Some activator proteins bind to enhancers, which are regulatory elements that are distant from the gene whose transcription they stimulate. Insulators are DNA sequences that block the action of enhancers.

Coordinated gene regulation Although eukaryotic cells do not possess operons, several eukaryotic genes may be activated by the same stimulus. For example, many eukaryotic cells respond to extreme heat and other stresses by producing **heat-shock proteins** that help to prevent damage from such stressing agents. Heat-shock proteins are produced by approximately 20 different genes. During times of environmental stress, the transcription of all the heat-shock genes is greatly elevated. Groups of bacterial genes are often coordinately expressed (turned on and off together) because they are physically clustered as an operon and have the same promoter, but coordinately expressed genes in eukaryotic cells are not clustered. How, then, is the transcription of eukaryotic genes coordinately controlled if they are not organized into an operon?

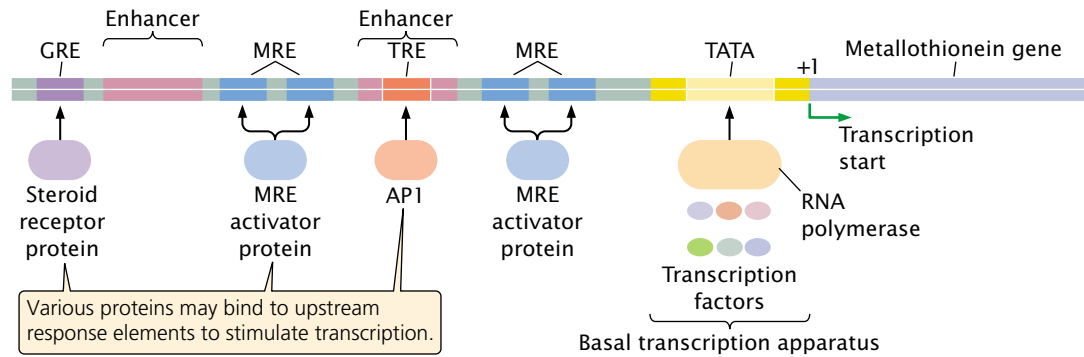
Genes that are coordinately expressed in eukaryotic cells are able to respond to the same stimulus because they have regulatory sequences in common in their promoters or enhancers. For example, different eukaryotic heat-shock genes possess a common regulatory element upstream of their start sites. A transcriptional activator protein binds to this regulatory element during stress and elevates transcription. Such common DNA regulatory sequences are called **response elements**; they typically contain short consensus sequences (Table 16.4) at varying distances from the gene being regulated.

A single eukaryotic gene may be regulated by several different response elements. The metallothionein gene protects cells from the toxicity of heavy metals by encoding a protein that binds to heavy metals and removes them from cells. The basal transcription apparatus assembles around the TATA box, just upstream of the transcription start site for the metallothionein gene, but the apparatus alone is capable of only low rates of transcription. The presence of heavy metals stimulates much higher rates of transcription.

Table 16.4 A few response elements found in eukaryotic cells

Response Element	Responds to	Consensus Sequence
Heat-shock element	Heat and other stress	CNNGAANNTCCNNG
Glucocorticoid response element	Glucocorticoids	TGGTACAAATGTTCT
Phorbol ester response element	Phorbol esters	TGACTCA
Serum response element	Serum	CCATATTAGG

Source: Adapted from B. Lewin, *Genes IV* (Oxford: Oxford University Press, 1994), p. 880.



16.24 Multiple response elements (MREs) are found in the upstream region of the metallothionein gene. The basal transcription apparatus binds near the TATA box. In response to heavy metals, activator proteins bind to several MRE elements and stimulate transcription. The TRE response element is the binding site for transcription factor AP1. In response to glucocorticoid hormones, steroid receptors bind to the GRE response element located approximately 250 nucleotides upstream of the metallothionein gene and stimulate transcription.

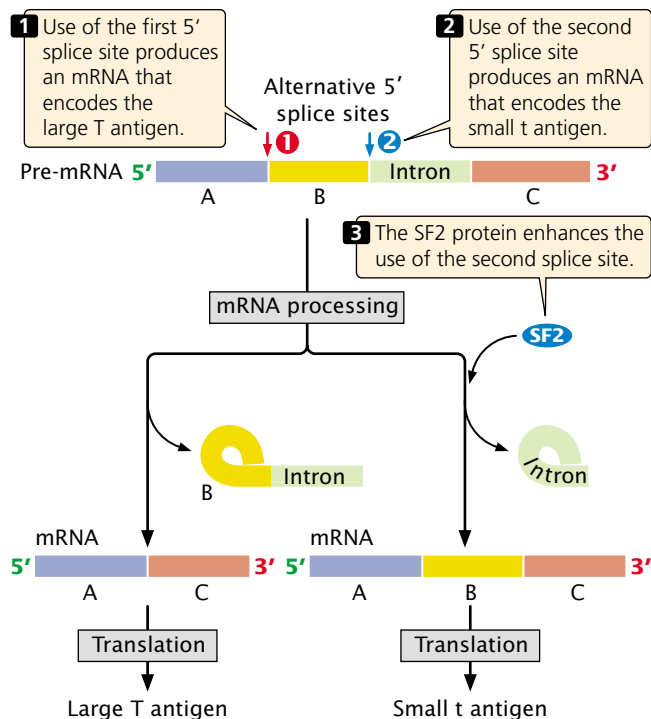
Other response elements found upstream of the metallothionein gene also contribute to increasing its rate of transcription. For example, several copies of a metal response element (MRE) are upstream of the metallothionein gene (FIGURE 16.24). Heavy metals stimulate the binding of an activator protein to MREs, which elevates the rate of transcription of the metallothionein gene. The presence of multiple copies of this response element permits high rates of transcription to be induced by metals. Two enhancers also are located in the upstream region of the metallothionein gene; one enhancer contains a response element known as TRE, which stimulates transcription in the presence of phorbol esters. A third response element called GRE is located approximately 250 nucleotides upstream of the metallothionein gene and stimulates transcription in response to glucocorticoid hormones.

This example illustrates a common feature of eukaryotic transcriptional control: a single gene may be activated by several different response elements, found in both promoters and enhancers. Multiple response elements allow the same gene to be activated by different stimuli. At the same time, the presence of the same response element in different genes allows a single stimulus to activate multiple genes. In this way, response elements allow complex biochemical responses in eukaryotic cells.

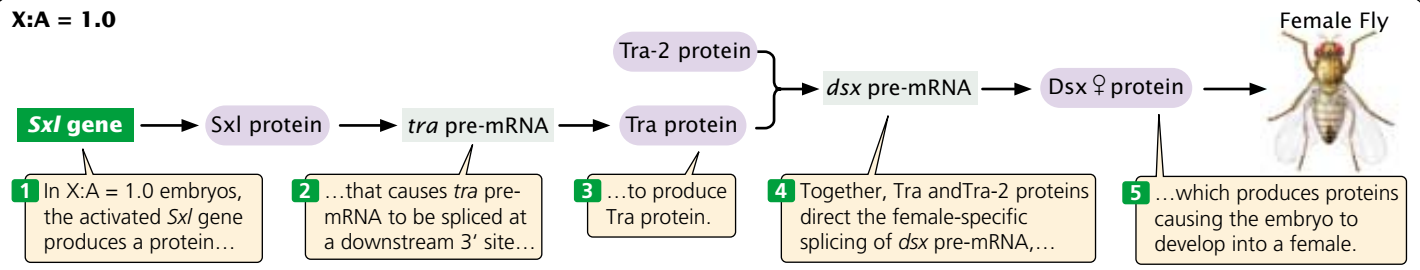
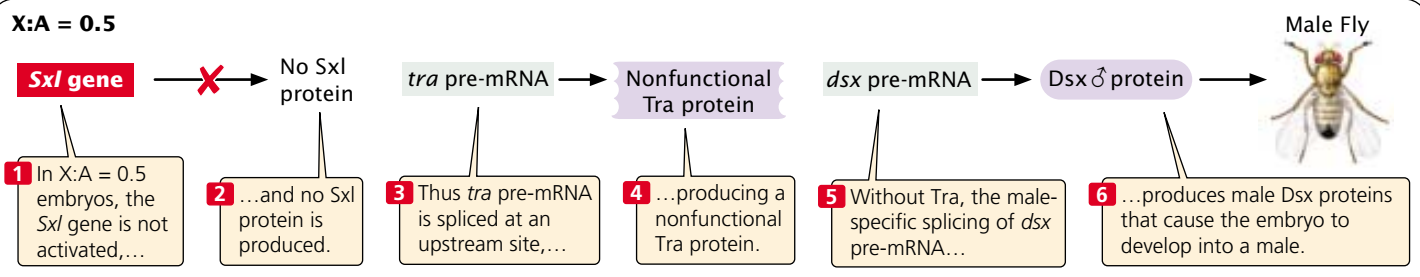
Gene Control Through Messenger RNA Processing

Alternative splicing allows a pre-mRNA to be spliced in multiple ways, generating different proteins in different tissues or at different times in development (see Chapter 14). Many eukaryotic genes undergo alternative splicing, and the regulation of splicing is probably an important means of controlling gene expression in eukaryotic cells.

The T-antigen gene of the mammalian virus SV40 serves as a well-studied example of alternative splicing. This gene is capable of encoding two different proteins, the large T and small t antigens. Which of the two proteins is produced depends on which of two alternative 5' splice sites is used during RNA splicing (FIGURE 16.25). The use of one 5' splice site produces mRNA that encodes the large T



16.25 Alternative splicing leads to the production of the small t antigen and the large T antigen in the mammalian virus SV40.

XX genotype**X:A = 1.0****XY genotype****X:A = 0.5**

16.26 Alternative splicing controls sex determination in *Drosophila*.

antigen, whereas the use of the other 5' splice site (which is farther downstream) produces an mRNA encoding the small t antigen.

A protein called splicing factor 2 (SF2) enhances the production of mRNA encoding the small t antigen (see Figure 16.25). Splicing factor 2 has two binding domains: one is an RNA-binding region and the other has alternating serine and arginine amino acids. These two domains are typical of **SR proteins**, which often play a role in regulating splicing. Splicing factor 2 stimulates the binding of U1 snRNP to the 5' splice site, one of the earliest steps in RNA splicing (see Chapter 14). The precise mechanism by which SR proteins influence the choice of splice sites is poorly understood. One model suggests that SF2 and other SR proteins bind to specific splice sites on mRNA and stimulate the attachment of snRNPs, which then commit the site to splicing.

Another example of alternative mRNA splicing that regulates the expression of genes controls whether a fruit fly develops as male or female. Sex differentiation in *Drosophila* arises from a cascade of gene regulation (see Figure 16.26). When the ratio of X chromosomes to the number of haploid sets of autosomes (the X:A ratio; see Chapter 4) is 1, a female-specific promoter is activated early in development and stimulates the transcription of the *sex-lethal* (*Sxl*) gene. The protein encoded by *Sxl* regulates the splicing of the pre-mRNA transcribed from another gene called *transformer* (*tra*). The splicing of *tra* pre-mRNA results in the production of Tra protein. Together with another protein (Tra-2), Tra stimulates the female-specific splicing of

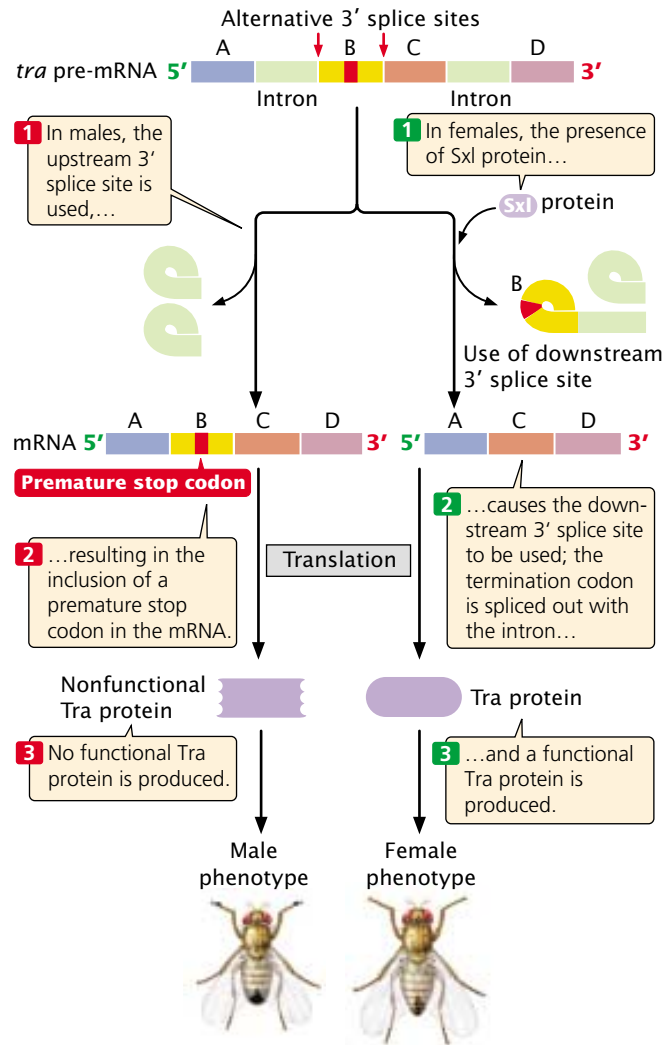
pre-mRNA from yet another gene called *doublesex* (*dsx*). This event produces a female-specific Dsx protein, which causes the embryo to develop female characteristics.

In male embryos, which have an X:A ratio of 0.5 (see Figure 16.26), the promoter that transcribes the *Sxl* gene in females is inactive; so no Sxl protein is produced. In the absence of Sxl protein, *Tra* pre-mRNA is spliced at a different 3' splice site to produce a nonfunctional form of Tra protein (see Figure 16.27). In turn, the presence of this nonfunctional Tra in males causes *Dsx* pre-mRNAs to be spliced differently (see Figure 16.26), and a male-specific Dsx protein is produced. This event causes the development of male-specific traits.

In summary, the Tra, Tra-2, and Sxl proteins regulate alternative splicing that produces male and female phenotypes in *Drosophila*. Exactly how these proteins regulate alternative splicing is not yet known, but it's possible that the Sxl protein (produced only in females) may block the upstream splice site on the *tra* pre-mRNA. This blockage would force the spliceosome to use the downstream 3' splice site, which causes the production of Tra protein and eventually results in female traits (see Figure 16.27).

Concepts

Eukaryotic genes may be regulated through the control of mRNA processing. The selection of alternative splice sites leads to the production of different proteins.



16.27 Alternative splicing of *tra* pre-mRNA. Two alternative 3' splice sites are present.

Gene Control Through RNA Stability

The amount of a protein that is synthesized depends on the amount of corresponding mRNA available for translation. The amount of available mRNA, in turn, depends on both the rate of mRNA synthesis and the rate of mRNA degradation. Eukaryotic mRNAs are generally more stable than bacterial mRNAs, which typically last only a few minutes before being degraded, but nonetheless there is great variability in the stability of eukaryotic mRNA: some persist for only a few minutes; others last for hours, days, or even months. These variations can result in large differences in the amount of protein that is synthesized.

Cellular RNA is degraded by ribonucleases, enzymes that specifically break down RNA. Most eukaryotic cells contain 10 or more types of ribonucleases, and there are several different pathways of mRNA degradation. In one pathway, the 5' cap is first removed, followed by 5'→3'

removal of nucleotides. A second pathway begins at the 3' end of the mRNA and removes nucleotides in the 3'→5' direction. In a third pathway, the mRNA can be cleaved at internal sites.

Messenger RNA degradation from the 5' end is most common and begins with the removal of the 5' cap. This pathway is usually preceded by the shortening of the poly(A) tail. Poly(A)-binding proteins (PABPs) normally bind to the poly(A) tail and contribute to its stability-enhancing effect. The presence of these proteins at the 3' end of the mRNA protects the 5' cap. When the poly(A) tail has been shortened below a critical limit, the 5' cap is removed, and nucleases then degrade the mRNA by removing nucleotides from the 5' end. These observations suggest that the 5' cap and 3' poly(A) tail of eukaryotic mRNA physically interact with each other, most likely by the poly(A) tail bending around so that the PABPs make contact with the 5' cap (see Chapter 14). Other parts of eukaryotic mRNA, including sequences in the 5' UTR, the coding region, and the 3' UTR, also affect mRNA stability.

Poly(A) tails are added to the 3' ends of some bacterial mRNAs, but they are shorter than those typically associated with eukaryotic mRNA and have the opposite effect; they appear to destabilize most prokaryotic mRNAs.

Concepts

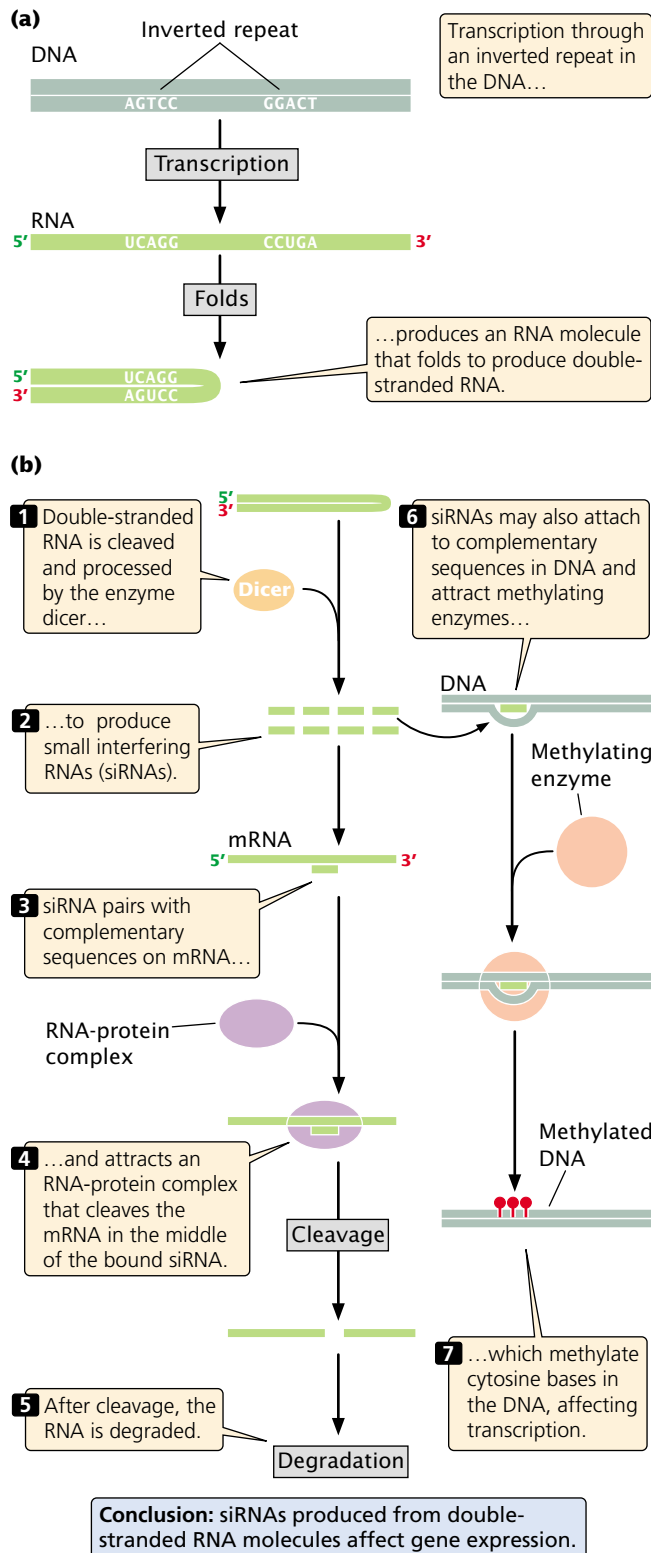
The stability of mRNA influences gene expression by affecting the amount of mRNA available to be translated. The stability of mRNA is affected by the 5' cap, the poly(A) tail, the 5' UTR, the coding section, and the 3' UTR.

RNA Silencing

Recent evidence indicates that the expression of some genes may be suppressed through **RNA silencing**, also known as RNA interference and posttranscriptional gene silencing. Although many of the details of this mechanism are still poorly understood, it appears to be widespread, existing in fungi, plants, and animals. It may also prove to be a powerful tool for artificially regulating gene expression in genetically engineered organisms.

RNA silencing is initiated by the presence of double-stranded RNA, which may arise in several ways: by the transcription of inverted repeats in DNA into a single RNA molecule that base pairs with itself; by the simultaneous transcription of two different RNA molecules that are complementary to one another and pair; or by the replication of double-stranded RNA viruses (FIGURE 16.28a). In *Drosophila*, an enzyme called Dicer cleaves and processes the double-stranded RNA to produce small pieces of single-stranded RNA that range in length from 21 to 25 nucleotides (FIGURE 16.28b). These **small interfering**

RNAs (siRNAs) then pair with complementary sequences in mRNA and attract an RNA–protein complex that cleaves the mRNA approximately in the middle of the bound siRNA. After cleavage, the mRNA is further degraded. In



the nucleus, siRNAs serve as guides for the methylation of complementary sequences in DNA, which then affects transcription. Some related RNA molecules produced through the cleavage of double-stranded RNA bind to complementary sequences in the 3' UTR of mRNA and inhibit their translation.

RNA silencing is thought to have evolved as a defense against RNA viruses and transposable elements that move through an RNA intermediate (see Chapter 20). The extent to which it contributes to normal gene regulation is uncertain, but dramatic phenotypic effects result from some mutations that occur in the enzymes that carry out RNA silencing.

Concepts

RNA silencing is initiated by double-stranded RNA molecules that are cleaved and processed. The resulting small interfering RNAs bind to complementary sequences in mRNA and bring about their cleavage and degradation. Small interfering RNAs may also stimulate the methylation of complementary sequences in DNA.

Translational and Posttranslational Control

Ribosomes, aminoacyl tRNAs, initiation factors, and elongation factors are all required for the translation of mRNA molecules. The availability of these components affects the rate of translation and therefore influences gene expression. The initiation of translation in some mRNAs is regulated by proteins that bind to the mRNA's 5' UTR and inhibit the binding of ribosomes, similar to the way in which repressor proteins bind to operators and prevent the transcription of structural genes.

Many eukaryotic proteins are extensively modified after translation by the selective cleavage and trimming of amino acids from the ends, by acetylation, or by the addition of phosphates, carboxyl groups, methyl groups, and carbohydrates to the protein). These modifications affect the transport, function, and activity of the proteins and have the capacity to affect gene expression.

Concepts

The initiation of translation may be affected by proteins that bind to specific sequences at the 5' end of mRNA. The availability of ribosomes, tRNAs, initiation and elongation factors, and other components of the translational apparatus may affect the rate of translation.

16.28 RNA silencing leads to the degradation of mRNA and the methylation of DNA.

Connecting Concepts



A Comparison of Bacterial and Eukaryotic Gene Control

Now that we have considered the major types of gene regulation, let's review some of the similarities and differences of bacterial and eukaryotic gene control.

1. Much of gene regulation in both bacterial and eukaryotic cells is accomplished through proteins that bind to specific sequences in DNA. Regulatory proteins come in a variety of types, but most can be characterized according to a small set of DNA-binding motifs.
2. Regulatory proteins that affect transcription exhibit two basic types of control: *repressors* inhibit transcription (negative control); *activators* stimulate transcription (positive control). Both negative control and positive control are found in bacterial and eukaryotic cells.
3. Complex biochemical and developmental events in bacterial and eukaryotic cells may require a cascade of gene regulation, in which the activation of one set of genes stimulates the activation of another set.
4. Most gene regulation in bacterial cells is at the level of transcription (although it does exist at other levels). Gene regulation in eukaryotic cells often takes place at multiple levels, including chromatin structure, transcription, mRNA processing, and RNA stability.
5. In bacterial cells, genes are often clustered in operons and are coordinately expressed by transcription into a single mRNA molecule. In contrast, each eukaryotic gene typically has its own promoter and is transcribed independently. Coordinate regulation in eukaryotic cells takes place through common response elements, present in the promoters and enhancers of the genes. Different genes that have the same response element in common are influenced by the same regulatory protein.
6. Chromatin structure plays a role in eukaryotic (but not bacterial) gene regulation. In general, condensed chromatin represses gene expression; chromatin structure must be altered before transcription. Acetylation of the histone proteins, which may be influenced by the degree of DNA methylation, appears to be important in bringing about these changes in chromatin structure.
7. The initiation of transcription is a relatively simple process in bacterial cells, and regulatory proteins function by blocking or stimulating the binding of RNA polymerase to DNA. Eukaryotic transcription requires complex machinery that includes RNA polymerase, general transcription factors, and transcriptional activators, which allows transcription to be influenced by multiple factors.

8. Some eukaryotic transcriptional activator proteins function at a distance from the gene by binding to enhancers, causing a loop in the DNA, and bringing the promoter and enhancer into close proximity. Some distant-acting sequences analogous to enhancers have been described in bacterial cells, but they appear to be less common.
9. The greater time lag between transcription and translation in eukaryotic cells than in bacterial cells allows mRNA stability and mRNA processing to play larger roles in eukaryotic gene regulation.

Connecting Concepts Across Chapters



The focus of this chapter has been on how the flow of information from genotype to phenotype is controlled. We have seen that there are a number of potential points of control in this pathway of information flow, including changes in gene structure, transcription, mRNA processing, mRNA stability, translation, and posttranslational modifications.

Gene regulation is critically important from a number of perspectives. It is essential to the survival of cells, which cannot afford to simultaneously transcribe and translate all of their genes. The evolution of complex genomes consisting of thousands of genes would not have been possible without some mechanism to selectively control gene expression. Gene regulation is also important from a practical point of view. A number of human diseases are caused by the breakdown of gene regulation, which produces proteins at inappropriate times or places. Gene regulation is also important to genetic engineering, where the key to success is often not getting genes into a cell, which is relatively easy, but getting them expressed at useful levels. For all of these reasons, there is tremendous interest in how gene expression is controlled, and understanding gene regulation is one of the frontiers of genetic research.

Information presented in this chapter builds on the foundation of molecular genetics developed in Chapters 10 through 15. The mechanisms of gene regulation provide important links to several topics in subsequent chapters. Gene regulation is important to the success of recombinant DNA, which is discussed in Chapter 18. Gene regulation also plays an important role in the genetics of development and cancer, which are discussed in Chapter 21.

CONCEPTS SUMMARY

- Gene expression may be controlled at different levels, including the alteration of gene structure, transcription, mRNA processing, RNA stability, translation, and posttranslational modification. Much of gene regulation is through the action of regulatory proteins binding to specific sequences in DNA.
- Genes in bacterial cells are typically clustered into operons—groups of functionally related structural genes and the sequences that control their transcription. Structural genes in an operon are transcribed together as a single mRNA.
- In negative control, a repressor protein binds to DNA and inhibits transcription. In positive control, an activator protein binds to DNA and stimulates transcription. In inducible operons, transcription is normally off and must be turned on; in repressible operons, transcription is normally on and must be turned off.
- The *lac* operon of *E. coli* is a negative inducible operon that controls the metabolism of lactose. In the absence of lactose, a repressor binds to the operator and prevents transcription of the structural genes that encode β -galactosidase, permease, and transacetylase. When lactose is present, some of it is converted into allolactose, which binds to the repressor and makes it inactive, allowing the structural genes to be transcribed and lactose to be metabolized. When all the lactose has been metabolized, the repressor once again binds to the operator and blocks transcription.
- Positive control in the *lac* operon and other operons is through catabolite repression. When complexed with cAMP, the catabolite activator protein (CAP) binds to a site in or near the promoter and stimulates the transcription of the structural genes. Levels of cAMP are indirectly correlated with glucose; so low levels of glucose stimulate transcription and high levels inhibit transcription.
- The *trp* operon of *E. coli* is a negative repressible operon that controls the biosynthesis of tryptophan.
- Attenuation is another level of control that allows transcription to be stopped before RNA polymerase has reached the structural genes. It takes place through the close coupling of transcription and translation and depends on the secondary structure of the 5' UTR sequence.
- Small RNA molecules, called antisense RNA, are complementary to sequences in mRNA and may inhibit translation by binding to these sequences, thereby preventing the attachment or progress of the ribosome.
- Transcriptional control regulates the lytic and lysogenic cycles of bacteriophage λ . The transcription of certain operons stimulates the transcription of some operons and represses the transcription of others. Which operons are stimulated and which are repressed depends on the affinity of promoters for repressor and activator proteins.
- Like gene regulation in bacterial cells, much of eukaryotic regulation is accomplished through the binding of regulatory proteins to DNA. However, there are no operons in eukaryotic cells, and gene regulation is characterized by a greater diversity of mechanisms acting at different levels.
- In eukaryotic cells, chromatin structure represses gene expression. During transcription, chromatin structure may be altered by the acetylation of histone proteins and demethylation.
- The initiation of eukaryotic transcription is controlled by general transcription factors that assemble into the basal transcription apparatus and by transcriptional activator proteins that stimulate normal levels of transcription by binding to regulatory promoters and enhancers.
- Some DNA sequences limit the action of enhancers by blocking their action in a position-dependent manner.
- Coordinately controlled genes in eukaryotic cells respond to the same factors because they have common response elements that are stimulated by the same transcriptional activator.
- Gene expression in eukaryotic cells may be influenced by RNA processing.
- Gene expression may be regulated by changes in RNA stability. The 5' cap, the coding sequence, the 3' UTR, and the poly(A) tail are important in controlling the stability of eukaryotic mRNAs. Proteins binding to the 5' end of eukaryotic mRNA may affect its translation.
- RNA silencing takes place when double-stranded RNA is cleaved and processed to produce small interfering RNAs that bind to complementary mRNAs and bring about their cleavage and degradation.
- Control of the posttranslational modification of proteins also may play a role in gene expression.

IMPORTANT TERMS

gene regulation (p. 000)	domain (p. 000)	negative control (p. 000)	repressible operon (p. 000)
induction (p. 000)	operon (p. 000)	positive control (p. 000)	corepressor (p. 000)
structural gene (p. 000)	regulator gene (p. 000)	inducible operon (p. 000)	coordinate induction (p. 000)
regulatory gene (p. 000)	regulator protein (p. 000)	inducer (p. 000)	partial diploid (p. 000)
regulatory element (p. 000)	operator (p. 000)	allosteric protein (p. 000)	constitutive mutation (p. 000)

catabolite repression (p. 000)	attenuator (p. 000)	chromatin-remodeling complex (p. 000)	SR protein (p. 000)
catabolite activator protein (CAP) (p. 000)	antiterminator (p. 000)	CpG island (p. 000)	RNA silencing (p. 000)
adenosine-3', 5'-cyclic monophosphate (cAMP) (p. 000)	antisense RNA (p. 000)	coactivator (p. 000)	small interfering RNAs (siRNAs) (p. 000)
attenuation (p. 000)	transcriptional antiterminator protein (p. 000)	insulator (p. 000)	
	DNase I hypersensitive site (p. 000)	heat-shock protein (p. 000)	
		response element (p. 000)	

Worked Problems

1. A regulator gene produces a repressor in an inducible operon. A geneticist isolates several constitutive mutations affecting this operon. Where might these constitutive mutations occur? How would the mutations cause the operon to be constitutive?

• Solution

An inducible operon is normally not being transcribed, meaning that the repressor is active and binds to the operator, inhibiting transcription. Transcription takes place when the inducer binds to the repressor, making it unable to bind to the operator.

Constitutive mutations cause transcription to take place at all times, whether the inducer is present or not. Constitutive mutations might occur in the regulator gene, altering the repressor so that it is never able to bind to the operator. Alternatively, constitutive mutations might occur in the operator, altering the binding site for the repressor so that the repressor is unable to bind under any conditions.

2. For *E. coli* strains with the *lac* genotypes, use a plus sign (+) to indicate the synthesis of β -galactosidase and permease and a minus sign (–) to indicate no synthesis of the enzymes.

Genotype of strain	Lactose absent		Lactose present	
	β -Galactosidase	Permease	β -Galactosidase	Permease
(a) <i>lacI⁺ lacP⁺ lacO⁺ lacZ⁺ lacY⁺</i>				
(b) <i>lacI⁺ lacP⁺ lacO^c lacZ[–] lacY⁺</i>				
(c) <i>lacI⁺ lacP[–] lacO⁺ lacZ⁺ lacY[–]</i>				
(d) <i>lacI⁺ lacP⁺ lacO⁺ lacZ[–] lacY[–] / lacI[–] lacP⁺ lacO⁺ lacZ⁺ lacY⁺</i>				

• Solution

Genotype of strain	Lactose absent		Lactose present	
	β -Galactosidase	Permease	β -Galactosidase	Permease
(a) <i>lacI⁺ lacP⁺ lacO⁺ lacZ⁺ lacY⁺</i>	–	–	+	+
(b) <i>lacI⁺ lacP⁺ lacO^c lacZ[–] lacY⁺</i>	–	+	–	+
(c) <i>lacI⁺ lacP[–] lacO⁺ lacZ[–] lacY⁺</i>	–	–	–	–
(d) <i>lacI⁺ lacP⁺ lacO⁺ lacZ[–] lacY[–] / lacI[–] lacP⁺ lacO⁺ lacZ⁺ lacY⁺</i>	–	–	+	+

- (a) All the genes possess normal sequences, and so the *lac* operon functions normally: when lactose is absent, the regulator protein binds to the operator and inhibits the transcription of the structural genes, and so β -galactosidase and permease are not produced. When lactose is present, some of it is converted into allolactose, which binds to the repressor and makes it inactive; the repressor does not bind to the operator, and so the structural genes are transcribed, and β -galactosidase and permease are produced.
- (b) The structural *lacZ* gene is mutated; so β -galactosidase will not be produced under any conditions. The *lacO* gene has a constitutive mutation, which means that the repressor is

- unable to bind to it, and so transcription takes place at all times. Therefore, permease will be produced in both the presence and the absence of lactose.
- (c) In this strain, the promoter is mutated, and so RNA polymerase is unable to bind and transcription does not take place. Therefore β -galactosidase and permease are not produced under any conditions.
- (d) This strain is a partial diploid, which consists of two copies of the *lac* operon—one on the bacterial chromosome and another on a plasmid. The *lac* operon represented in the upper part of the genotype has mutations in both the *lacZ* and *lacY* genes, and so it

is not capable of encoding β -galactosidase or permease under any conditions. The *lac* operon in the lower part of the genotype has a defective regulator gene, but the normal regulator gene in the upper operon produces a diffusible repressor (trans acting) that binds to the lower operon in the absence of lactose and inhibits transcription. Therefore no β -galactosidase or permease is produced when lactose is absent. In the presence of lactose, the repressor cannot bind to the operator, and so the lower operon is transcribed and β -galactosidase and permease are produced.

3. The *fox* operon, which has sequences A, B, C, and D, encodes enzymes 1 and 2. Mutations in sequences A, B, C, and D have the following effects, where a plus sign (+) = enzyme synthesized and a minus sign (–) = enzyme not synthesized.

Mutation in sequence	Fox absent		Fox present	
	Enzyme 1	Enzyme 2	Enzyme 1	Enzyme 2
No mutation	–	–	+	+
A	–	–	–	+
B	–	–	–	–
C	–	–	+	–
D	+	+	+	+

(a) Is the *fox* operon inducible or repressible?

(b) Indicate which sequence (A, B, C, or D) is part of the following components of the operon:

Regulator gene	_____
Promoter	_____
Structural gene for enzyme 1	_____
Structural gene for enzyme 2	_____

• Solution

Because the structural genes in an operon are coordinately expressed, mutations that affect only one enzyme are likely to occur in the structural genes; mutations that affect both enzymes must occur in the promoter or regulator.

(a) When no mutations are present, enzymes 1 and 2 are produced in the presence of Fox but not in its absence, indicating that the operon is inducible and Fox is the inducer.

(b) Mutation A allows the production of enzyme 2 in the presence of Fox, but enzyme 1 is not produced in the presence or absence of Fox, and so A must have a mutation in the structural gene for enzyme 1. With B, neither enzyme is produced under any conditions, and so this mutation most likely occurs in the promoter and prevents RNA polymerase from binding. Mutation C affects only enzyme 2, which is not produced in the presence or absence of lactose; enzyme 1 is produced normally (only in the presence of Fox), and so mutation C most likely occurs in the structural gene for enzyme 2. Mutation D is constitutive, allowing the production of enzymes 1 and 2 whether or not Fox is present. This mutation most likely occurs in the regulator gene, producing a defective repressor that is unable to bind to the operator under any conditions.

Regulator gene	<u>D</u>
Promoter	<u>B</u>
Structural gene for enzyme 1	<u>A</u>
Structural gene for enzyme 2	<u>C</u>

4. A mutation occurs in the 5' UTR of the *trp* operon that reduces the ability of region 2 to pair with region 3. What would be the effect of this mutation when the tryptophan level is high and when the tryptophan level is low?

• Solution

When the tryptophan level is high, regions 2 and 3 do not normally pair, and therefore the mutation will have no effect. When the tryptophan level is low, however, the ribosome normally stalls at the Trp codons in region 1 and does not cover region 2, and so regions 2 and 3 are free to pair, which prevents regions 3 and 4 from pairing and forming a terminator, ending transcription. If regions 2 and 3 cannot pair, then regions 3 and 4 will pair even when tryptophan is low and attenuation will always occur. Therefore, no more tryptophan will be synthesized even in the absence of tryptophan.

The New Genetics

MINING GENOMES

MICROARRAY ANALYSIS AND THE ANALYSIS OF GENE EXPRESSION

This exercise introduces the powerful technique of microarray analysis, one of the most potent tools in bioinformatics. After

a general introduction to microarrays, you will explore the use of microarrays in studies of gene expression. You will use SAGE (Serial Analysis of Gene Expression) to try to identify which genes are important in the development of specific diseases.

COMPREHENSION QUESTIONS

- * 1. Name six different levels at which gene expression might be controlled.
- * 2. Draw a picture illustrating the general structure of an operon and identify its parts.

3. What is the difference between positive and negative control? What is the difference between inducible and repressible operons?
- * 4. Briefly describe the *lac* operon and how it controls the metabolism of lactose.
5. What is catabolite repression? How does it allow a bacterial cell to use glucose in preference to other sugars?
- * 6. What is attenuation? What is the mechanism by which the attenuator forms when tryptophan levels are high and the antiterminator forms when tryptophan levels are low?
- * 7. What is antisense RNA? How does it control gene expression?
8. What general features of transcriptional control are found in bacteriophage λ ?
- * 9. What changes take place in chromatin structure and what role do these changes play in eukaryotic gene regulation?
10. Briefly explain how transcriptional activator proteins and repressors affect the level of transcription of eukaryotic genes.
11. What is an insulator?
12. What is a response element? How do response elements bring about the coordinated expression of eukaryotic genes?
13. Outline the role of alternative splicing in the control of sex differentiation in *Drosophila*.
- *14. What role does RNA stability play in gene regulation? What controls RNA stability in eukaryotic cells?
15. Define RNA silencing. Explain how siRNAs arise and how they potentially affect gene expression.
- *16. Compare and contrast bacterial and eukaryotic gene regulation. How are they similar? How are they different?

APPLICATION QUESTIONS AND PROBLEMS

- *17. For each of the following types of transcriptional control, indicate whether the protein produced by the regulator gene will be synthesized initially as an active repressor, inactive repressor, active activator, or inactive activator.
 - (a) Negative control in a repressible operon
 - (b) Positive control in a repressible operon
 - (c) Negative control in an inducible operon
 - (d) Positive control in an inducible operon
- *18. A mutation occurs at the operator site that prevents the regulator protein from binding. What effect will this mutation have in the following types of operons?
 - (a) Regulator protein is a repressor in a repressible operon.
 - (b) Regulator protein is a repressor in an inducible operon.
19. The *blob* operon produces enzymes that convert compound A into compound B. The operon is controlled by a regulatory gene *S*. Normally the enzymes are synthesized only in the absence of compound B. If gene *S* is mutated, the enzymes are synthesized in the presence *and* in the absence of compound B. Does gene *S* produce a repressor or an activator? Is this operon inducible or repressible?
- *20. A mutation prevents the catabolite activator protein (CAP) from binding to the promoter in the *lac* operon. What will be the effect of this mutation on transcription of the operon?
21. Under which of the following conditions would a *lac* operon produce the greatest amount of β -galactosidase? The least? Explain your reasoning.
22. A mutant strain of *E. coli* produces β -galactosidase in the presence *and* in the absence of lactose. Where in the operon might the mutation in this strain occur?
- *23. For *E. coli* strains with the following *lac* genotypes, use a plus sign (+) to indicate the synthesis of β -galactosidase and permease and a minus sign (−) to indicate no synthesis of the enzymes.

	Lactose present	Glucose present
Condition 1	Yes	No
Condition 2	No	Yes
Condition 3	Yes	Yes
Condition 4	No	No

Lactose absent		Lactose present	
β -Galactosidase	Permease	β -Galactosidase	Permease
Genotype of strain <i>lacI</i> ⁺ <i>lacP</i> ⁺ <i>lacO</i> ⁺ <i>lacZ</i> ⁺ <i>lacY</i> ⁺ <i>lacI</i> [−] <i>lacP</i> ⁺ <i>lacO</i> ⁺ <i>lacZ</i> ⁺ <i>lacY</i> ⁺ <i>lacI</i> ⁺ <i>lacP</i> ⁺ <i>lacO</i> ^c <i>lacZ</i> ⁺ <i>lacY</i> ⁺ <i>lacI</i> [−] <i>lacP</i> ⁺ <i>lacO</i> ⁺ <i>lacZ</i> ⁺ <i>lacY</i> [−] <i>lacI</i> [−] <i>lacP</i> [−] <i>lacO</i> ⁺ <i>lacZ</i> ⁺ <i>lacY</i> ⁺ <i>lacI</i> ⁺ <i>lacP</i> ⁺ <i>lacO</i> ⁺ <i>lacZ</i> [−] <i>lacY</i> ⁺ / <i>lacI</i> [−] <i>lacP</i> ⁺ <i>lacO</i> ⁺ <i>lacZ</i> ⁺ <i>lacY</i> [−]			

(continued on p. 469)

*23. (continued)

Genotype of strain	Lactose absent		Lactose present	
	β -Galactosidase	Permease	β -Galactosidase	Permease
$lacI^- lacP^+ lacO^c lacZ^+ lacY^+ /$				
$lacI^+ lacP^+ lacO^+ lacZ^- lacY^-$				
$lacI^- lacP^+ lacO^+ lacZ^+ lacY^- /$				
$lacI^+ lacP^- lacO^+ lacZ^- lacY^+$				
$lacI^+ lacP^- lacO^c lacZ^- lacY^+ /$				
$lacI^- lacP^+ lacO^+ lacZ^+ lacY^-$				
$lacI^+ lacP^+ lacO^+ lacZ^+ lacY^+ /$				
$lacI^+ lacP^+ lacO^+ lacZ^+ lacY^+$				
$lacI^s lacP^+ lacO^+ lacZ^+ lacY^- /$				
$lacI^+ lacP^+ lacO^+ lacZ^- lacY^+$				
$lacI^s lacP^- lacO^+ lacZ^- lacY^+ /$				
$lacI^+ lacP^+ lacO^+ lacZ^+ lacY^+$				

24. Give all possible genotypes of a *lac* operon that produces β -galactosidase and permease under the following conditions. Do not give partial diploid genotypes.

	Lactose absent		Lactose present	
	β -Galactosidase	Permease	β -Galactosidase	Permease
(a)	—	—	+	+
(b)	—	—	—	+
(c)	—	—	+	—
(d)	+	+	+	+
(e)	—	—	—	—
(f)	+	—	+	—
(g)	—	+	—	+

- *25. Explain why mutations in the *lacI* gene are trans in their effects, but mutations in the *lacO* gene are cis in their effects.

- *26. The *mmm* operon, which has sequences A, B, C, and D, encodes enzymes 1 and 2. Mutations in sequences A, B, C, and D have the following effects, where a plus sign (+) = enzyme synthesized and a minus sign (—) = enzyme not synthesized.

Mutation in sequence	Mmm absent		Mmm present	
	Enzyme 1	Enzyme 2	Enzyme 1	Enzyme 2
No mutation	+	+	—	—
A	—	+	—	—
B	+	+	+	+
C	+	—	—	—
D	—	—	—	—

- (a) Is the *mmm* operon inducible or repressible?

- (b) Indicate which sequence (A, B, C, or D) is part of the following components of the operon:

Regulator gene _____
 Promoter _____
 Structural gene for enzyme 1 _____
 Structural gene for enzyme 2 _____

- *27. Listed in parts a through g are some mutations that were found in the 5' UTR region of the *trp* operon of *E. coli*. What would the most likely effect of each of these mutations be on the transcription of the *trp* structural genes?

- (a) A mutation that prevented the binding of the ribosome to the 5' end of the mRNA 5' UTR
 (b) A mutation that changed the tryptophan codons in region 1 of the mRNA 5' UTR into codons for alanine
 (c) A mutation that created a stop codon early in region 1 of the mRNA 5' UTR
 (d) Deletions in region 2 of the mRNA 5' UTR
 (e) Deletions in region 3 of the mRNA 5' UTR

- (f) Deletions in region 4 of the mRNA 5' UTR
 (g) Deletion of the string of adenine nucleotides that follows region 4 in the 5' UTR
28. Some mutations in the *trp* 5' UTR region increase termination by the attenuator. Where might these mutations occur and how might they affect the attenuator?
29. Some of the mutations mentioned in Question 28 have an interesting property. They prevent the formation of the antiterminator that normally takes place when the tryptophan level is low. In one of the mutations, the AUG start codon for the 5' UTR peptide has been deleted. How might this mutation prevent antitermination from occurring?
30. Several examples of antisense RNA regulating translation in bacterial cells have been discovered. Molecular geneticists have also used antisense RNA to artificially control transcription in both bacterial and eukaryotic genes. If you wanted to inhibit the transcription of a bacterial gene with antisense RNA, what sequences might the antisense RNA contain?
- *31. What would be the effect of deleting the *Sxl* gene in a newly fertilized *Drosophila* embryo?
32. What would be the effect of a mutation that destroyed the ability of poly(A)-binding protein (PABP) to attach to a poly(A) tail?

CHALLENGE QUESTIONS

33. Would you expect to see attenuation in the *lac* operon and other operons that control the metabolism of sugars? Why or why not?
34. A common feature of many eukaryotic mRNAs is the presence of a rather long 3' UTR, which often contains consensus sequences. Creatine kinase B (CK-B) is an enzyme important in cellular metabolism. Certain cells—termed U937D cells—have lots of CK-B mRNA, but no CK-B enzyme is present. In these cells, the 5' end of the CK-B mRNA is bound to ribosomes, but the mRNA is apparently not translated. Something inhibits the translation of the CK-B mRNA in these cells.

In recent experiments, numerous short segments of RNA containing only 3' UTR sequences were introduced into U937D cells. As a result, the U937D cells began to synthesize the CK-B enzyme, but the total amount of CK-B mRNA did not increase. Short segments of other RNA sequences did not stimulate the synthesis of CK-B; only the 3' UTR sequences turned on the translation of the enzyme.

On the basis of these experiments, propose a mechanism for how CK-B translation is inhibited in the U937D cells. Explain how the introduction of short segments of RNA containing the 3' UTR sequences might remove the inhibition.

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17

Gene Mutations and DNA Repair



This is photo legend x 26 picas width for opening chapter photo for Chapter 17. This is legend copy area for Chapter opening photo for Chapter Seventeen allowing 4lines, if more space is needed crop photo at top to allow for deeper legend here. (Volodymyr Repik/AP).

The Genetic Legacy of Chernobyl

Early on the morning of April 26, 1986, unit 4 of the Chernobyl nuclear power plant in northern Ukraine exploded, creating the worst nuclear disaster in history. The explosion blew off the 2000-ton metal plate that sealed the top of the reactor and ignited hundreds of tons of graphite, which burned uncontrollably for 10 days. The exact amount of radiation released in the explosion and ensuing fire is still unknown, but a minimum estimate is 100 mil-

lion curies, equal to a medium-sized nuclear strike. A plume of radioactive particles blew west and north from the crippled reactor, raining dangerous levels of radiation down on thousands of square kilometers. Regions as far away as Germany and Norway were affected; even Japan and the United States received measurable increases in radiation.

Immediately after the accident, 31 people, mostly firefighters who heroically battled the blaze, died of acute radiation sickness. More than 400,000 workers later toiled to

- The Genetic Legacy of Chernobyl
- The Nature of Mutation
 - The Importance of Mutations
 - Categories of Mutations
 - Types of Gene Mutations
 - Mutation Rates
- Causes of Mutations
 - Spontaneous Replication Errors
 - Spontaneous Chemical Changes
 - Chemically Induced Mutations
 - Radiation
- The Study of Mutations
 - The Analysis of Reverse Mutations
 - Detecting Mutations with the Ames Test
 - Radiation Exposure in Humans
- DNA Repair
 - Mismatch Repair
 - Direct Repair
 - Base-Excision Repair
 - Nucleotide-Excision Repair
 - Other Types of DNA Repair
 - Genetic Diseases and Faulty DNA Repair